

prone to pulmonary arteriovenous fistulas with associated risk of paradoxical cerebral embolism which predisposes to embolic stroke and cerebral abscess formation (p.320).

82.5.2 Epidemiology

Rare autosomal dominant genetic disorder of blood vessels affecting ≈ 1 in 5,000 people. 95% have recurrent epistaxis.

82.5.3 Imaging

CT

May show a well demarcated homogeneous or mottled high density³⁵ (high density due to hematoma, calcification, thrombosis, hemosiderin deposition, alterations in BBB, and/or increased blood volume³¹) with some form of contrast enhancement (around or within lesion) in 17 of 24 patients.³⁵ Surrounding edema or mass effect is rare (except in cases that have recently hemorrhaged).

MRI

May demonstrate previous hemorrhage(s),³⁷ (may be important when the presence of multiple occurrences affects therapeutic choices). T2WI finding: reticulated core of increased and decreased intensity, a prominent surrounding rim of reduced intensity may be present (due to hemosiderin laden macrophages from previous hemorrhages). GRASS image demonstrates flow related enhancement in $\approx 60\%$ of cases, which allows signal dropout from flowing blood on other sequences to be differentiated from that due to calcium (and thus, bone) or air (limitations: hemosiderin causes signal dropout, and slow in-plane flow does not enhance).³⁸

82.5.4 Treatment

Surgery is indicated mainly for evacuation of hematoma or diagnosis, especially when favorably located. Also consider surgery for recurrent hemorrhages (rupture has been reported even after normal angiography) or medically intractable seizures. Stereotactic radiosurgery has not had a satisfactorily high enough benefit to risk ratio to justify its use.³⁹

82.6 Cavernous malformation

82.6.1 General information

Key concepts

- usually angiographically occult. May show up on MRI (open channels $\rightarrow$ flow void on T2WI, previous hemorrhage $\rightarrow$ “popcorn” pattern especially on T2* gradient echo) or contrast CT
- low-flow. No intervening neural parenchyma, no arteries. Associated with venous anomaly (represents venous outflow and should be preserved)
- XRT is a risk factor for developing cavernous malformation
- presentation: usually seizures. Hemorrhage: rare, risk is difficult to predict
- treatment:
 - a) $\blacklozenge$ surgery is the treatment of choice for symptomatic accessible lesions
 - b) $\square$ radiosurgery should not be considered as a treatment option

AKA: cavernous hemangioma, cavernoma, cavernous angioma, angioma, and in medical jargon “cav-mal.” A well-circumscribed, benign vascular hamartoma consisting of irregular thick and thin walled sinusoidal vascular channels located within the brain but lacking intervening neural parenchyma,²⁹ large feeding arteries, or large draining veins. Usually 1–5 cm in size. May hemorrhage, calcify, or thrombose. Occur rarely in the spinal cord.⁴⁰ Caverns are filled with blood in various stages of thrombus formation/organization/dissolution. Frequently associated with venous angiomas (p.1245). Capillary telangiectasias may be found adjacent to lesions and may represent a precursor. Stain positive for angiogenesis factor.⁴¹ Lesions may arise de novo,⁴² and may grow (although slower than hemangioblastomas), shrink, or remain unchanged with time.⁴³

82.6.2 Pathology

Gross appearance resembles a mulberry (facetiously dubbed a “hemorrhoid of the brain”). Light microscopy: stains for von Willebrand’s factor. Smooth muscle layer is absent (except for some tiny portions). EM: shows abnormal gapping of the tight junctions between endothelial cells⁴⁴ (may permit leakage of blood) and sparse or poorly characterized subendothelial smooth muscle cells.⁴⁴

82.6.3 Epidemiology

Cerebral cavernous malformations (CM) comprise 5–13% of CNS vascular malformations, and develop in 0.02–0.13% of the population (based on large autopsy⁴⁵ and MRI⁴⁶ series). 48–86% are supratentorial, 4–35% brainstem, 5–10% basal ganglia.⁴⁷ Multiple in 23%⁴⁸ to 50%⁴⁹ of cases, and multiplicity may be more common in hereditary cases.⁵⁰

Spinal CMs: CMs rarely may occur in spinal cord. XRT appears to be a risk factor,⁵¹ (e.g. following craniospinal XRT⁵² for medulloblastoma) especially for spinal CMs. 42% of patients with spinal CMs also harbor ≥ 1 intracranial CM.⁵³

82.6.4 Genetics

Two types: sporadic and hereditary. The latter may be inherited in a Mendelian autosomal dominant pattern with variable expressivity.⁵⁴ There appears to be at least 3 gene loci (see □ Table 82.8. Multiple lesions are more common in the familial form.⁴⁶

82.6.5 Presentation/natural history

See references.^{55,56,57}

General information

Seizures (60%), progressive neurologic deficit (50%), hemorrhage (20%, usually intraparenchymal; note: here, hemorrhage is defined as symptomatic, radiologically proven extralesional bleeding), hydrocephalus, or as in incidental finding (over 50% in one series).

Hemorrhage

Risk is not well delineated. Even the definition of hemorrhage is controversial since, by definition, all CMs have surrounding hemosiderin indicative of small leaks. The Angioma Alliance⁵⁸ definition of hemorrhage: acute or subacute symptoms (any of: headache, seizure, impaired consciousness, or new/worsened focal neurological deficit referable to the anatomic location of the CM) accompanied by radiological, pathological, surgical, or rarely only cerebrospinal fluid evidence of recent extra- or intralesional hemorrhage. This definition does not include either an increase in CM diameter without other evidence of recent hemorrhage, nor the presence of a hemosiderin halo. Risk of significant hemorrhage is much less than with AVMs. CMs are prone to recurrent small hemorrhages that are rarely devastating. Hemorrhage rate tends to be low in cohort studies ≈ 2.6–3.1%/yr; appears higher in females (4.2%/yr) than males (0.9%/yr).⁴⁶ Bleeding risk is not related to the size of the CM. Controversial if hemorrhage increases the risk of future bleeding: it did not in one study⁴⁶ whereas another study⁵⁹ found only a 0.6%/yr risk of bleeding in lesions without prior hemorrhage. Some CMs behave benignly after an initial hemorrhage. Others behave more malignantly with (>2 hemorrhages) with increasingly more detrimental outcome. Pregnancy and parturition are not known to be risk factors for hemorrhage.⁵⁴

Table 82.8 Subtypes of CCM			
	CCM1	CCM2	CCM3
locus	7q11-q22	7p15-13	3q25.2-q27
gene	KRIT1	MGC4607 (malcavernin)	PDCD10
feature	more common in Hispanics		

Σ

The bleeding rate is variable & the criteria for what constitutes “bleeding” has not been uniform. Each patient appears to have their own natural history □ it is difficult to assign a risk of hemorrhage for any individual patient.

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Seizures

The rate of new-seizure onset is 2.4%/yr.⁴⁶

82.6.6 Evaluation

CT

Not sensitive: CT misses many small lesions, some large ones, and even some that have bled. Not specific: CT findings may overlap with low grade tumors, hemorrhages, granulomas.

MRI

Gradient-echo T2WI MRI is the most sensitive test due to high sensitivity to susceptibility artifact. Findings are similar to AOVM in general (mixed signal core with low signal rim – sometimes described as “popcorn” pattern; see above). The diagnosis is strongly suggested by finding multiple lesions with these characteristics and a positive family history.⁴⁹ A venous malformation may be seen adjacent to a solitary CM, but not with multiple CMs.⁶⁰ Diffusion tensor imaging/white matter tractography⁶¹ and pre-op 3D-constructive interference in steady-state (CISS) MRI⁶² may improve localization, approach, and post-op outcomes.

Angiography

Does not demonstrate lesion. MRI appearance is nearly pathognomonic, and angiography is not necessary in classically appearing cases. Angiography may be needed to R/O other diagnoses in questionable cases.

Familial considerations

First degree relatives of patients with more than one family member having a cavernous malformation should have MRI screening and appropriate genetic counselling.

82.6.7 Treatment/management

Overview

Options:

1. observe
2. surgical excision
3. XRT or stereotactic radiosurgery.^{63,64,65,66} Controversial: results appear comparable to natural history

No randomized prospective study has been done. Determining treatment response is difficult since no imaging study can prove elimination of the lesion. Therefore it has been suggested that recurrent hemorrhage rate be followed as an endpoint.

Recommendations

Incidental lesions

Asymptomatic, incidentally discovered CMs should be managed expectantly with serial imaging studies for about 2–3 years (to rule-out frequent subclinical bleeds); additional studies thereafter based on clinical grounds. However, some experts recommend removal for single, easily accessible incidental CMs in non-eloquent brain.⁶⁷

□ Since the radiographic appearance is almost pathognomonic, biopsy or excision solely to verify the diagnosis is rarely appropriate.

Brainstem CMs

Surgery is almost never indicated for brainstem CMs that have not bled. With a bleed rate of 2–6%, Gross et. al⁶⁸ suggest operative management for a history of >2 prior hemorrhages and “pial/ependymal representation” on T1WI MRI.

Bleeds that do not come to the surface cannot be removed without creating neurologic deficit (worsening of neurologic outcome was 9% vs. 29% in superficial vs. deep brainstem CM resections, respectively⁶⁹). The approach is chosen to expose the site where the bleed comes closest to the surface. Spetzler says brainstem CMs are almost always associated with a venous angioma (p.1245) (which, again, must be preserved since it provides the venous outflow). Outcome was worse with surgery through the floor of the 4th ventricle than with a lateral approach. Significant short-term neurologic deficit is expected with brainstem CM resection.⁶⁸

Spinal cord CMs

Managed essentially the same as brainstem CMs.

Cranial nerve CMs

Many case reports and reviews document CMs of cranial nerves (rarely extra-axial) with various presentations.^{70,71,72} Case reports suggest patients may benefit from early surgical decompression from hemorrhagic chiasmal cavernomas since they are at risk for recurrent micro-hemorrhages.⁷³

Surgery

General information

Indications for surgery for intracranial CMs:

1. accessible lesions with
 - a) focal deficit
 - b) or symptomatic hemorrhage
 - c) or seizures:
 - new onset seizures (p.461): there is a suggestion that removing CMs before “kindling” occurs may have a better chance of preventing future seizures
 - difficult to manage seizures
2. less accessible lesions that repeatedly bleed with progressive neurologic deterioration may be considered for excision, even in delicate regions such as the brain stem^{74,75,76} or spinal cord

Surgical technique

Goal of surgery: complete removal of the malformation. Since CMs are not particularly bloody, piecemeal excision is an option; especially important in brainstem lesions.

Stereotactic localization or intraoperative ultrasound may be particularly helpful in localizing. When operating on CMs that have bled, one usually encounters a cavity containing the CM and blood degradation products.⁷⁷ Initial dissection is directed at separating the lesion from the adjacent brain. Although bleeding is usually not a problem, it occasionally may be brisk if the CM is entered before the dissection and devascularization is complete. Once the dissection is complete, the contents of the CM capsule may be removed piecemeal to minimize the parenchymal opening (especially important in the brainstem). For supratentorial CMs presenting with seizures, it is desirable to also remove the hemosiderin-stained brain immediately surrounding the CM.

Keep in mind the relatively common association of CMs with venous angiomas (p.1245), which if encountered should not be removed as they represent the venous drainage of the area.

Brainstem CMs

The use of retractors is to be avoided; cottonoids and exploitation of hematoma cavity may be used to gain access. Brainstem CMs may be extremely adherent to brain parenchyma⁷⁶ unlike supratentorial CMs. Bipolar cautery: use on low power with constant irrigation to reduce thermal injury. Unlike supratentorial CMs with seizures (where you want to remove adjacent hemosiderin-stained brain), just remove the CM itself.

Post-op follow-up

Follow-up MRI $\approx$ 3 months post-op is recommended. It never looks “normal” but can determine if removal was complete.

Stereotactic radiosurgery (SRS)

Some non-controlled studies have shown a possible reduction in recurrent hemorrhage rate following a 2 year latency period after SRS,⁶⁶ however, radiation induced morbidity was significant.^{78,79} Other series have failed to show reduction.⁸⁰ Findings may reflect the natural history of CMs with temporal clustering of hemorrhagic events with decrease in hemorrhage rates after 28 months.⁸¹

Σ

SRS is not an alternative to surgery and should not be considered for the treatment of CMs.

82.6.8 Prognosis

When CMs can be completely removed, the risk of further growth or hemorrhage is essentially permanently eliminated⁷⁷ (however, recurrence of symptoms has been reported after partial and even seemingly-complete removal^{76,82}).
For CMs treated surgically, patients need to be aware that post-op neurologic worsening is very common, especially with brainstem CMs.⁸³ Worsening may be transient,⁸⁴ but may take months to resolve.

82.7 Dural arteriovenous fistulae (DAVF)

82.7.1 General information

AKA dural AVMs (DAVM). Vascular abnormality in which an arteriovenous shunt is contained within the leaflets of the dura mater, exclusively supplied by branches of the internal/external carotid or vertebral arteries.⁸⁵ Because they are considered acquired rather than congenital lesions, the term fistula is preferred over malformation, although the latter term has also been used in the literature. Multiple fistulas may be found in up to 8%of cases.
Usually found adjacent to dural venous sinuses. Common locations:

1. transverse/sigmoid: the most common⁸⁶ (63%of cases) with a slight left-sided predominance,⁸⁷ with the epicenter of these almost invariably at the junction of the transverse and sigmoid sinuses
2. tentorial/petrosal
3. anterior fossa/ethmoidal
4. middle fossa/Sylvian
5. cavernous sinus (carotid-cavernous fistula – CCF)
6. superior sagittal sinus
7. foramen magnum

82.7.2 Etiology

Evidence suggests that most DAVFs are acquired, idiopathic lesions, and they have a well-recognized association with venous sinus thrombosis, although their exact pathogenesis is not fully understood. Theories include:

1. venous sinus occlusion awakens dormant embryonic dural arteriovenous channels⁸⁶
2. venous hypertension/thrombosis promotes local angiogenesis and the de novo formation of DAVF⁸⁸
3. the DAVF may arise first and itself result in venous sinus thrombosis⁸⁹

82.7.3 Epidemiology

DAVFs comprise 10–15%of all intracranial AVMs.⁸⁷ 61–66%occur in females, and patients are usually in their 40 s or 50 s. They occur rarely in children, and when they do they tend to be complex, bilateral dural sinus malformations.⁹⁰

82.7.4 Presentation

Common findings are listed in □ Table 82.9. Pulsatile tinnitus is the most common presenting symptom of a DAVF. Cortical venous drainage with resultant venous hypertension can produce IC-HTN,

Table 82.9 Clinical findings in 27 patients with dural AVMs ⁹¹	
Sign/symptom	No. (%)
pulsatile tinnitus	25 (92%)
occipital bruit	24 (89%)
headache	11 (41%)
visual impairment	9 (33%)
papilledema	7 (26%)

Table 82.10 Borden classification	
Type	Features
I	DAVF drainage into a dural venous sinus or meningeal veins, with normal anterograde flow. Usually clinically benign.
II	DAVF draining anterograde into dural venous sinus, but with retrograde flow into cortical veins.
III	DAVF with direct retrograde flow from fistula into cortical veins causing venous hypertension.
Above further subclassified into a: with single hole and b: multiple holes	

and this is the most common cause of morbidity and mortality and thus the strongest indication for DAVF treatment. DAVFs may also cause global cerebral edema or hydrocephalus due to poor cerebral venous drainage or by impairing the function of the arachnoid granulations, respectively. Other DAVF symptoms/signs include headaches, seizures, cranial nerve palsies, and orbital venous congestion. a.

82.7.5 Evaluation

General information

Brain CT or MRI without contrast are often normal. CTA may reveal dilated tortuous vessels corresponding to enlarged arterial feeders or ectatic draining veins. MRA may reveal dilated pial vessels, early prominent venous sinus filling, sinus enlargement or occlusion, and white matter edema related to venous hypertension. Full 6 vessel cerebral angiography (bilateral ICAs, bilateral ECAs, bilateral vertebral arteries) is required to establish the diagnosis and to plan treatment.

Angiographic classification

Several classification systems have been described to characterize DAVFs. The Borden⁹² (□ Table 82.10) and the Cognard⁹³ (□ Table 82.11) systems have emerged as the most commonly utilized contemporary grading schemes. See Tables. Cortical venous drainage is the defining angiographic feature that distinguishes benign (low-grade) from aggressive (high-grade) fistulas. (Borden I, Cognard I, and Cognard IIa are low-grade, all others are high-grade.)

Borden classification system

The Borden classification⁹² is shown in □ Table 82.10.

Cognard angiographic classification

The Cognard system⁹³ is shown in □ Table 82.11. This system is generally most applicable to DAVFs involving the transverse sinus.

Cognard found 54% had no cortical venous reflux (Types I and IIa) and usually exhibit benign behavior.

□ Key determinant: in the Cognard system, the pattern of venous drainage is the most critical factor. As a general rule, lesions with retrograde flow in the cortical veins (IIb, IIa+b, III & IV – red frames in □ Table 82.11) are high risk (for bleeding or intracranial hypertension...).

82.7.6 Natural history

The concept of benign vs aggressive DAVF behavior based on the absence or presence, respectively, of cortical venous drainage was validated by data reported by University of Toronto group. Over a 3 year period, 98% of benign lesions (no cortical venous drainage) remained benign.⁹⁴ On the other hand, over a 4 year period, annual rates of hemorrhage, non-hemorrhagic neurologic deficit, and mortality were 8.1%, 6.9% and 10.4% for aggressive lesions (with cortical venous drainage).⁹⁵

In a meta-analysis of 377 cases,⁹⁶ three DAVF locations were associated with particularly aggressive behavior (aggressive:benign ratio) – tentorial (31:1), middle fossa/sylvian (2.5:1), anterior fossa/ethmoidal 2.1:1).

82.7.7 Management

General information

Lesions with cortical venous drainage should generally be treated. Lesions without cortical venous drainage should be followed radiographically and clinically (2% may evolve to develop cortical venous drainage). A change in a bruit (either worsening, or disappearance) should prompt restudy.

Indications for intervention:

1. presence of cortical venous drainage
2. neurologic dysfunction
3. hemorrhage
4. orbital venous congestion
5. refractory symptoms (headache, pulsatile tinnitus)

Manual carotid self compression

Advocated by some, the thrombosis rate of $\approx 22\%$ and clinical improvement rate of 33% ⁹⁷ may mimic the natural course. Patients are advised to compress with the hand that would be affected by ischemia if it were to occur (e.g. with a left-sided DAVF, the right hand should be used to compress the left carotid artery). That way, the hand would fall away if ischemia develops. Recommendations vary, one option: start with 10 minutes once a day, gradually increase frequency and duration.

Endovascular embolization

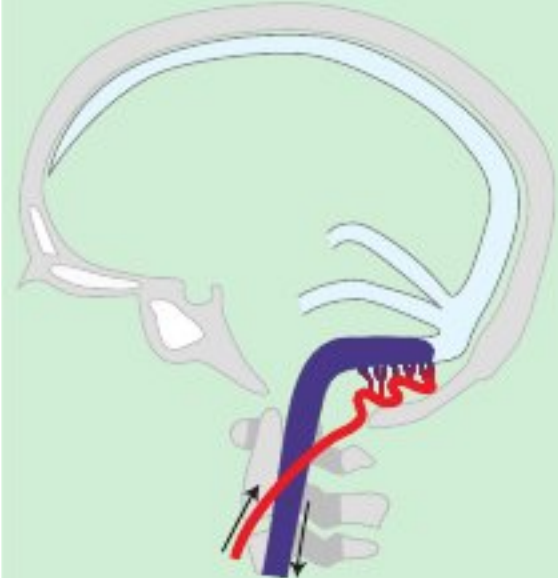
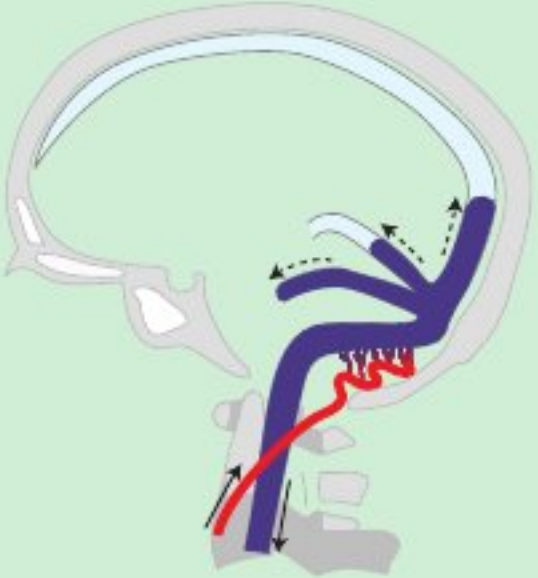
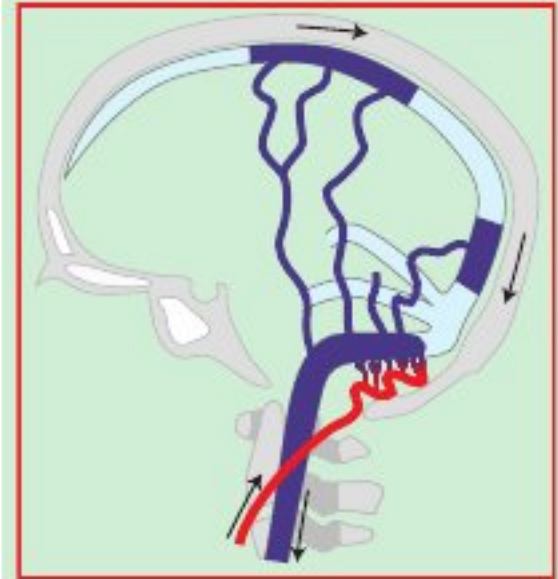
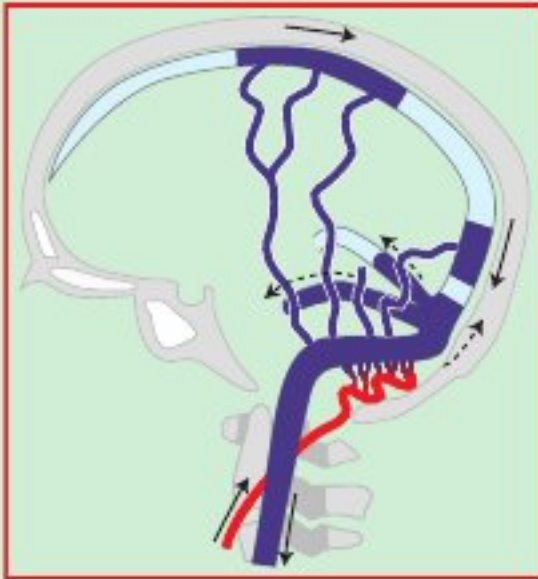
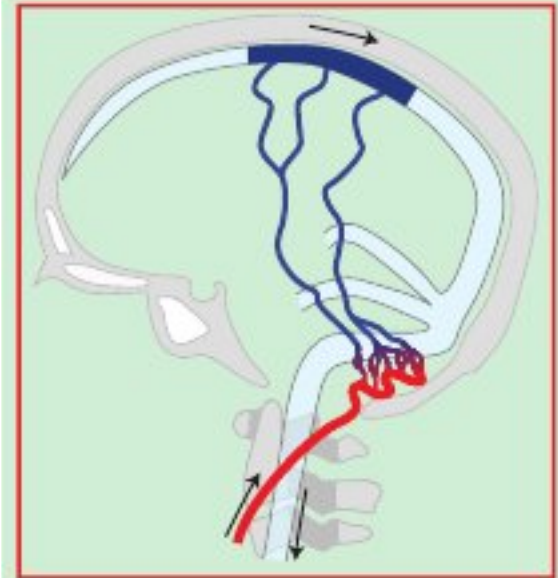
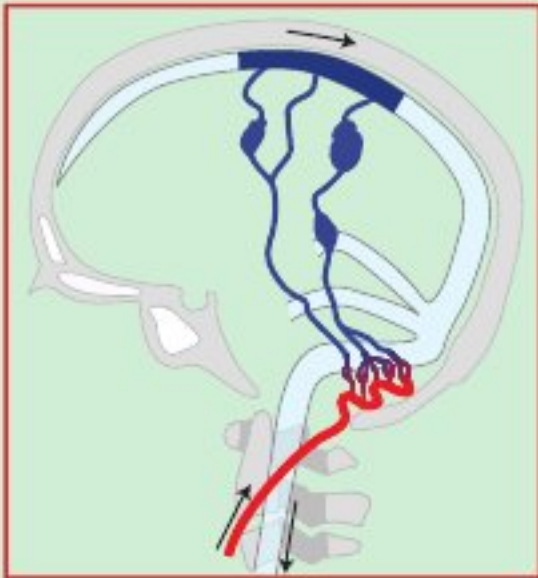
May be performed transarterial or transvenous. Before the availability of liquid embolic agents (Onyx and NBCA), treatment was directed at the venous drainage (unlike pial AVMs) which had higher success, because the coils could be deployed to sacrifice the venous drainage very close to point of arteriovenous shunting, resulting in thrombosis of the fistula. It is more difficult to deploy coils across the point of arteriovenous shunting from the arterial side, whereas the liquid embolic agents, particularly Onyx, can be injected at somewhat of a distance and pushed forward across the fistulous point. Whether a transarterial, transvenous, or combined approach is utilized depends on the unique angioarchitecture of the fistula.

Surgery

While endovascular approaches have emerged as the primary treatment for most DAVFs, certain fistula types are still best dealt with via open surgery as the first line strategy.⁹⁸ Furthermore, surgery has been used to successfully treat DAVFs after previous partial, incomplete, or failed endovascular treatment. Finally, surgery can be used adjunctively in a combined approach to provide direct access for embolization of DAVFs that are inaccessible by a purely endovascular route.

Preoperative embolization may facilitate surgical treatment⁹⁹ by lessening the risk of catastrophic hemorrhage, which may occur simply during the performance of the craniotomy.⁹¹ The use of the craniotome is discouraged, as a sinus or venous laceration could produce a fatal hemorrhage. Contingencies for the rapid administration of blood products must be made (large bore central lines). The scalp incision, craniotomy flap, and dural incision should be planned in a strategic manner to control and sequentially eliminate the blood supply to the lesion at each step, while maximizing the exposure as needed. Surgical options for the treatment of DAVFs include the following techniques⁹⁸:

1. radical fistula excision
2. sinus skeletonization

Table 82.11 Angiographic classification of dural AVMS ^{93a}	
Venous drainage: sinus	
Type I  Normal antegrade flow into a dural venous sinus Course: benign^b	Type IIa  Drainage into a sinus with retrograde flow within the sinus^c Course: sinus reflux caused IC-HTN in 20%
Type IIb  Drainage into a sinus with retrograde flow into cortical vein(s) Course: reflux into veins induced hemorrhage in 10%	Type IIa + b  Drainage into a sinus with retrograde flow within the sinus^c and cortical vein(s) Course: aggressive in 66% with bleeding and/or IC-HTN
Venous drainage: directly into cortical veins	
Type III  Direct drainage into a cortical vein without venous ectasia Description: direct cortical venous drainage without venous ectasia Course: hemorrhage occurs in 40%	Type IV  Direct drainage into a cortical vein with venous ectasia Course: hemorrhage occurs in 65%

Venous drainage: spinal in addition to all of the above

Type V



Direct drainage into spinal perimedullary veins in addition to all of the above
Course: progressive myelopathy in 50%

^athose in red boxes are high risk for bleeding or intracranial hypertension

^bdespite a usually good prognosis, $\approx 2\%$ will progress and therefore follow-up studies may be warranted

^cdashed arrows signify retrograde flow

- 3. disconnection of cortical venous drainage
- 4. ligation of the fistulous point and/or outflow vein
- 5. sinus packing
- 6. coagulation of arterial feeders to the lesion

While surgery vs. endovascular treatment can be considered for all DAVF locations, two locations generally remain more favorable for surgery:

- 1. anterior fossa/ethmoidal
- 2. tentorial DAVFs

The endovascular approach to these fistulas is difficult whereas the surgical approach is often straightforward. Surgically-assisted embolization, whereby a craniotomy is performed followed by direct puncture for embolization of the target vessel, may be utilized in select cases.

Stereotactic radiosurgery

May be used post-embolization.¹⁰⁰ Pan et al¹⁰¹ reported a complete obliteration rate of 58% of transverse/sigmoid fistulae treated with only radiosurgery (1650–1900 cGy) or with radiosurgery after surgery/embolization had failed to produce complete obliteration. 71% of the patients were cured of their symptoms.

With the continued improvement of endovascular technology over the past two decades, the role of radiosurgery for DAVF treatment has steadily decreased; however, it remains an option for those difficult lesions in which endovascular/surgical options have been exhausted.

82.8 Vein of Galen malformation

82.8.1 General information

Enlargement of the great cerebral vein of Galen (VOG) may occur in “vein of Galen malformations” (VOGM) (some refer to these as vein of Galen aneurysms). Congenital, develop before the 3-month embryo stage, and probably consists not of the vein of Galen but rather of the medial vein of the prosencephalon) or secondarily to high flow from adjacent deep parenchymatous AVMs or pial fistulae. Parenchymatous AVMs can be distinguished from true VOG malformations by retrograde filling of the of the internal cerebral vein in the former.¹⁰²

True VOG malformations are predictably fed from the medial and lateral choroidal, circumferential, mesencephalic, anterior choroidal, pericallosal and meningeal arteries.^{102,103} Agensis of the straight sinus may be an associated finding.

82.8.2 Presentation

Newborns tend to present with congestive heart failure in first few weeks of life (due to high blood flow)¹⁰⁴ and a cranial bruit. Hydrocephalus may result from obstruction of the sylvian aqueduct by

the enlarged VOG, or it may be caused by the increased venous pressure (which can also produce prominence of the scalp veins¹⁰⁵).

Parenchymatous AVMs are usually diagnosed later in life due to neurological manifestations¹⁰⁶ including focal neurologic deficit and hemorrhage.

82.8.3 Classification

Classified based on the location of the fistula^{107,108}:

1. pure internal fistulae: single or multiple
2. fistulae between thalamoperforators and the VOG
3. mixed form: the most common
4. plexiform AVMs

82.8.4 Natural history

Untreated VOG malformations have a poor prognosis, with neonates having nearly 100% mortality, and 1–12 month olds having $\approx$ 60% mortality, 7% major morbidity, and 21% being normal.¹⁰⁹

Parenchymatous AVMs behave similar to other AVMs.

82.8.5 Treatment

Hydrocephalus

Hydrocephalus associated with VOGM is obstructive, due to the varix. Although admonitions about shunting were common due to fear of precipitating a hemorrhage, when hydrocephalus is present the patient needs a shunt.

Vein of Galen malformations

Pediatric patients are often in poor medical condition, limiting the efficacy of operative treatment. Treatment options for these include embolization of the main feeding arteries. Prognosis is poor. Those presenting with hydrocephalus from aqueductal obstruction often do so at the end of the first year of life. Neurosurgical excision may be considered here, and the prognosis is better.

Repeated embolization while monitoring the venous drainage is employed.

Parenchymal AVM with enlarged VOG

The AVM is treated by the same methods as other AVMs (embolization, resection or radiosurgery).

82.9 Carotid-cavernous fistula

82.9.1 General information

Key concepts

- direct (high flow, from ICA) or indirect (low flow, from meningeal branches)
- classic triad (more common with direct CCF): chemosis, pulsatile proptosis, ocular bruit
- risk of SAH is low. Major risk is to vision
- natural history of low flow CCF is up to 50% spontaneous thrombosis

See also anatomy and venous inflow and outflow of the cavernous sinus (p. 86).

Carotid-cavernous fistula (CCF): divided into direct (Type A) and indirect (Types B-D)¹¹⁰:

1. Type A: direct high-flow shunts between the internal carotid artery and cavernous sinus:
 - a) traumatic (including iatrogenic): occur in 0.2% of patients with craniocerebral trauma. Iatrogenic: may follow percutaneous trigeminal rhizotomy,¹¹¹ endovascular procedures...
 - b) spontaneous: usually due to ruptured cavernous sinus ICA aneurysm. May also occur in patients with connective tissue disorders
2. indirect (dural): most are shunts: from dural arteries that are branches of the external carotid (not from ICA) (exception: Type B) – low flow

- a) Type B: from meningeal branches of the internal carotid artery (ICA)
- b) Type C: from meningeal branches of the external carotid artery (ECA)
- c) Type D: from meningeal branches of both the ICA and ECA

82.9.2 Presentation

1. orbital and/or retro-orbital pain
2. chemosis (arteriolization of conjunctiva)
3. pulsatile proptosis
4. ocular and/or cranial bruit
5. deterioration of visual acuity: may be due to hypoxic retinopathy as a result of reduced arterial pressure and increased venous pressure and increased intraocular pressure
6. diplopia: abducens (VI) palsy is the most common
7. pupillary dilatation
8. ophthalmoplegia (usually unilateral, but may present initially as bilateral or may progress to bilateral)
9. increased intraocular pressure
10. neo-vascularization of the iris or retina
11. rarely: SAH

Indirect CCFs generally have a more gradual onset and milder presentation than direct.

82.9.3 Evaluation

CT or MRI: usually demonstrates proptosis. Serpiginous and engorged intraocular vessels including the superior ophthalmic vein (best seen on T2WI coronals – helps to differentiate from rectus muscles) and convexity of lateral wall of cavernous sinus.

Angiography: shunting of blood from ICA into cavernous sinus. Rapid opacification of petrosal sinus and/or ophthalmic vein may be seen.

1. Huber maneuver: lateral view, inject VA and manually compress affected carotid. Helps identify upper extent of fistula, multiple fistulous openings, and complete transection of ICA
2. Mehringer-Hieshima maneuver: inject contrast at a rate of 2–3 ml/s into affected carotid while compressing the carotid in the neck (below the catheter tip) to control flow to help demonstrate the fistula

82.9.4 Treatment

General information

20–50% of low flow CCF spontaneously thrombose, therefore one may observe these as long as visual acuity is stable and intra-ocular pressure is $< \approx 25$. Symptomatic (e.g. progressive visual deterioration) high-flow CCFs rarely resolve spontaneously, and urgent treatment is usually indicated. Treatment is usually in the form of embolization by an interventional neuroradiologist or trapping between surgically placed clips.

Even if normal ocular motility cannot be achieved in affected eye, preservation of vision is desirable because:

1. for some ocular motility abnormalities, surgical treatment may reduce diplopia
2. patient may be provided with frosted eyeglass lens which will eliminate diplopia but will maintain peripheral vision
3. in the rare event of injury to contralateral eye (trauma, central retinal artery occlusion...) there would be “reserve” vision in the eye with reduced motility (with loss of the other eye, there would not be diplopia)

Indications for treatment

1. proptosis
2. visual loss
3. cranial nerve VI palsy
4. intractable bruit
5. severely elevated intraocular pressure
6. increased filling of cortical veins on angiography

Endovascular treatment

Options include:

- 1. Electrolytically detachable coils
- 2. Amplatzer vascular plug

Routes available include:

- 1. transarterial through internal carotid. If this fails (e.g. wide aneurysm neck), the carotid artery may be occluded on either side of fistula to trap it (sacrifices carotid artery, therefore test occlusion must be done first to determine if patient can tolerate this; however note: test occlusion with an open fistula may give false positive result because steal through the fistula may reduce CBF and cause neurologic symptoms not related to the occlusion acting alone). The distal occlusion needs to be proximal to the ophthalmic artery
- 2. transarterial through external carotid: useful only for dural fistulas
- 3. transvenous:
 - a) traversing heart to enter jugular vein, then through petrosal sinus to cavernous sinus. Lower success rate ($\approx 20\%$) than transarterial route
 - b) via superior ophthalmic vein: entered where supra-optic vein enters orbit to become superior ophthalmic vein. If possible, it is best to wait for the vein to become arterialized by the high flow pressure. Reports of “disasters” due to injury to the fragile vein performed before arterio- lization took place may have been due to more primitive balloon catheters that were standard before current commercially produced versions were available (which are softer than origi- nal). Must avoid lacerating the vein inside the orbit, and avoid distal ligation of the vein with- out proximal occlusion (shunts even more blood into eye)

Choice of technique

With indirect fistulas, it is mandatory to place coils on the venous side (otherwise new feeders will be recruited).

Coils or clips may be used to occlude direct fistulas.

82.10 Sigmoid sinus diverticulum

Sigmoid sinus diverticulum (SSD, see □ Fig. 82.1) or sigmoid sinus dehiscence is found in 1.2% of asymptomatic patients.¹¹² However, these abnormalities may be found ipsilaterally in up to 23% of patients with pulsatile tinnitus, presumably due to turbulent flow which may occur in these abnor- malities.¹¹³ SSD are more common in women.



Fig. 82.1 Axial contrast head CT (left) and AP venous phase angiogram (right) showing sigmoid sinus diverticulum on the patient’s left side.

When treatments such as masking noise generators fail, surgical intervention can be considered. Surgical treatment options include:

- Endovascular coiling/stenting
- Transmastoid “resurfacing” (see below)
- Craniectomy with clip reconstruction

Transmastoid “resurfacing” consists of partial mastoidectomy, and subtotal obliteration of the area of the diverticulum (so-called sinus wall resurfacing^[13] Otto, 2007 #7186^[14]) with e.g. bone chips, fibrin glue, or muscle.

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Part XIX

Stroke and Occlusive
Cerebrovascular Disease

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83 General Information and Stroke Physiology

83.1 Definitions

83

Stroke AKA cerebral infarction. Obsolete term: cerebrovascular accident (CVA).

□ TIA. (Transient ischemic attack): transient neuronal dysfunction secondary to focal ischemia (of brain, spinal cord, or retina) without (permanent) acute infarction¹ (note: obsolete operational definitions used an arbitrary 24 hour cutoff for duration of symptoms).

10–15%of patients with TIA have a stroke within 3 months, 50%of which occur within 48 hours.

□ Stroke. Permanent (i.e. irreversible) death of neurons caused by inadequate perfusion of a region of brain or brain stem.

□ Watershed infarct. Ischemic infarction in a territory located at the periphery of two bordering arterial distributions due to a disturbance in flow in one or both of the arteries.

83.2 Cerebrovascular hemodynamics

83.2.1 Cerebral blood flow (CBF) and oxygen utilization

□ Table 83.1 shows typical CBF values and the corresponding neurophysiologic state. CBF<20 is generally associated with ischemia and if prolonged will produce cell death.² However, this assumes normal metabolic rate and may be more applicable to global cerebral hypoperfusion.³ There is a higher CBF threshold for loss of electrical excitability than that for cell death – this has led to the concept of the ischemic penumbra – nonfunctioning cells that are still viable.²

CBF is related to blood pressure as shown in Eq (83.1).

$$CBF \propto \frac{CPP}{CVR} \propto \frac{MAP - ICP}{CVR}$$

83:1P

where CPP=cerebral perfusion pressure (p.856), CVR=cerebrovascular resistance (see below), and MAP=mean arterial pressure.

83.2.2 Cerebrovascular resistance (CVR)

CVR is affected by the PaCO₂ such that there is a linear increase in CBF with increasing PaCO₂ within the range of 20–80 mm Hg.

CVR is also affected by changes in CPP which produce changes in blood vessel tone via a myogenic mechanism. In the range of CPP=50–150 mm Hg the CVR of normal brain tissue varies linearly to

Table 83.1 Correlates of CBF	
CBF (ml per 100 gm tissue/min)	Condition
>60 (approx)	hyperemia (CBF>tissue demand)
45–60	normal brain at rest
75–80	gray matter
20–30	white matter
<20: Ischemia	
16–18	EEG becomes flatline
15	physiologic paralysis
12	brainstem auditory evoked response (BAER) changes
10	alterations in cell membrane transport (cell death; stroke)

maintain an almost constant CBF. This phenomenon is called (cerebral) autoregulation, which is altered in pathologic states.

83.2.3 Cerebral metabolic rate of oxygen consumption (CMRO₂)

CMRO₂ averages 3.0–3.8 ml/100 gm tissue/min. The ratio of CBF to CMRO₂ (the coupling ratio⁴) in the quiescent brain is 14–18. With focal cortical activity, local CBF increases $\approx 30\%$ while CMRO₂ increases $\approx 5\%$ ⁵ CMRO₂ can be manipulated to some degree.

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83.2.4 Cerebrovascular reserve and reactivity

May be evaluated with xenon-enhanced CT, CTP (p.228), TCD, SPECT, or MRI.^{6,7,8,9} Response of CBF to a vasodilator challenge with 1000 mg of IV acetazolamide (ACZ) (Diamox®) is classified as^{8,9}:

Type I: normal baseline CBF with 30–60% increase following ACZ challenge

Type II: decreased baseline CBF with blunted response of $<10\%$ increase or <10 ml/100 g/min absolute increase after ACZ challenge

Type III: decreased baseline CBF with paradoxical decrease of regional CBF following ACZ challenge, suggesting a steal phenomenon in regions with maximally dilated vasculature at baseline

83.3 Collateral circulation

83.3.1 Collateral circulation for ICA stenosis/occlusion

The effects of ICA stenosis/occlusion may be ameliorated by collateral blood flow. Potential alternate routes for blood to reach brain tissue include:

1. flow through the circle of Willis
 - a) from contralateral ICA through anterior communicating a.
 - b) from forward flow through the ipsilateral posterior communicating a.
2. retrograde flow through ophthalmic a. parasitizing blood from both ECAs via:
 - a) facial a. $\rightarrow$ angular a. $\rightarrow$ dorsal nasal a. & medial palpebral a.
 - b) maxillary a.
 - middle meningeal a. $\rightarrow$ lacrimal a.
 - vidian a. (a. of the pterygoid canal)
 - c) transverse facial a. $\rightarrow$ lateral palpebral a.
 - d) superficial temporal a. $\rightarrow$ supraorbital a.
3. proximal maxillary a. $\rightarrow$ anterior tympanic a. $\rightarrow$ caroticotympanic branch of ICA
4. cortical-cortical anastomoses
5. dural-leptomeningeal anastomoses

83.3.2 Collateral circulation for vertebrobasilar stenosis/occlusion

Available collaterals depend on the site of occlusion.

Basilar artery occlusion. Collateral flow via:

1. posterior communicating aa.
2. anastomoses between SCA and PICA

Proximal vertebral artery (VA) occlusion. Collateral flow via:

1. ECA $\rightarrow$ occipital a. $\rightarrow$ muscular branches of VA $\rightarrow$ VA
2. thyrocervical trunk $\rightarrow$ ascending cervical a. $\rightarrow$ direct connection or spinal radicular aa. $\rightarrow$ VA
3. contralateral VA and/or ascending cervical a. via spinal radicular branches and anterior spinal artery

83.4 “Occlusion” syndromes

83.4.1 Occlusion of major vessels organized by vascular territories

See □ Fig. 2.1 for the distribution territories of the major cerebral arteries. To indicate lateralization of findings, {CL}=contralateral, {IL}=ipsilateral.

Internal carotid artery and its branches

Risk and extent of stroke is influenced by suddenness of occlusion, location of occlusion, and collateral circulation (see above)

- 1. statistics:
 - a) acute ICA occlusion (all comers): 26–49%risk of stroke¹⁰ (not all of these strokes are severe)
 - b) annual stroke risk in 1261 patients with symptomatic ICA occlusion: 7%overall, 5.9%ipsilateral to the occlusion (mean follow-up =45.5 mos) (even with anticoagulation or antiplatelets drugs) (12 prospective studies¹¹)
 - c) St. Louis Carotid Occlusion Study¹²: 2-year ipsilateral ischemic stroke rate in patients with symptomatic ICA occlusion =5%in patients with normal O₂ extraction fraction (OEF) by PET scan, and 26%in patients with increased OEF
 - d) stroke risk is less when one includes asymptomatic ICA occlusions (i.e. there are people walking around with ICA occlusion and no symptoms)
 - e) in patients presenting with ICA territory stroke or TIA, complete ICA occlusion is found in 10–15%²
- 2. worst-case scenario of total ICA occlusion with no a-comm or p-comm flow and no collateral rescue: stroke in ACA and MCA territories (□ Table 83.2)
- 3. anterior cerebral artery: {CL} weakness of LE>UE
- 4. posterior cerebral artery
 - a) unilateral occipital lobe infarction →homonymous hemianopsia with macular sparing (visual cortex of the macula receives dual blood supply from MCA and PCA)
 - b) Balint syndrome
 - c) cortical blindness (Anton syndrome)
 - d) Weber syndrome
 - e) alexia without agraphia
 - f) thalamic pain syndrome (Dejerine-Roussy syndrome)
- 5. artery of Percheron (p.83): bilateral thalamic and mesencephalic infarctions¹³

Posterior circulation

- 1. vertebral artery
 - a) medial medullary syndrome (Dejerine syndrome)
 - b) lateral medullary syndrome (Wallenberg syndrome): see below
- 2. basilar artery
- 3. AICA: lateral pontine syndrome (Marie-Foix syndrome)

Table 83.2 Total ICA occlusion			
Deficit ^a	complete (M1 occlusion)	superior division	inferior division
{CL} weakness of UE>LE	X	X	
{CL} weakness of lower face	X	X	
{CL} hemisensory loss (UE & LE)	X	X	
{CL} hemisensory loss face (all modalities)	X	X	
{CL} neglect ^b	X	X	
{IL} gaze preference	X		
{CL} homonymous hemianopsia	X		X ^c
receptive aphasia ^d (Wernicke’s area)	X		X
expressive aphasia ^d (Broca’s area)	X	X	
Gerstmann syndrome (p.98): with dominant parietal lobe infarct			
^a [CL]=contralateral, [IL]=ipsilateral. An “X” indicates that the deficit is present			
^b with involvement on side of nondominant hemisphere			
^c plus {CL} upper quadrantanopsia			
^d with involvement on the side of the dominant hemisphere			

- 4. PICA: sometimes lateral medullary (Wallenberg) syndrome: see below
- 5. SCA: infarction of superior cerebellar vermis and superior cerebellum
- 6. anterior spinal artery
- 7. recurrent medial striate artery (of Heubner): expressive aphasia + mild hemiparesis (UE>LE, proximal muscles weaker than distal)
- 8. anterior choroidal artery (AChA) syndrome: first described by Foix et al. in 1925. The complete triad consists of {CL} hemiplegia, hemihypesthesia and homonymous hemianopsia (mnemonic: 3 H's), however, incomplete forms are more common.¹⁴ Occlusion is usually due to small vessel disease and CT or MRI usually shows infarct in posterior limb of IC (just above temporal horn of lateral vent)¹⁵ and white matter posterior and lateral to it. Occlusion is usually tolerated fairly well, and ligation of this artery was actually utilized in treatment of Parkinsonism sometimes without ill effect^{16 (p 540)} – see Surgical treatment of Parkinson's disease (p.1524) – but internal capsule infarct occurred in ≈ 15%

Lateral medullary syndrome (LMS)

AKA Wallenberg's syndrome, AKA PICA syndrome. Classically attributed to PICA occlusion, but in 80–85% of cases the vertebral artery is also involved.¹⁷ No cases have been reported arising from brainstem hemorrhage. Onset is usually acute. The findings are listed in □ Table 83.3 (NB: absence of pyramidal tract findings, and no change in sensorium). The location of the lesion and medullary structures are shown in □ Fig. 83.1.

□ Note. This is essentially the only location where a lesion will produce sensory loss on one side of the face (ipsilateral to the lesion) and contralateral sensory loss in the body. All in the absence of pyramidal tract findings (i.e. overt weakness).

These patients sometimes develop severe cerebellar swelling that responds to neurosurgical decompression (the tissue aspirates easily).

In a patient presenting with LMS, one needs to rule-out vertebral dissection (p.1325) since this would be treated with heparin. MRI including fat-suppressed T1WI and MRA would detect dissection in most cases.

Prognosis: 12% of 43 patients died during the acute phase from respiratory and cardiovascular complications and 2 new posterior-fossa strokes occurred.¹⁸ Recurrent vertebrobasilar territory stroke rate was 1.9% per year.¹⁸

83.4.2 Lacunar strokes

General information

Small infarcts in deep noncortical cerebrum or brainstem (□ Table 83.4) resulting from occlusion of penetrating branches of cerebral arteries. Size of infarcts ranges from 3–20 mm (CT detects larger ones; better sensitivity in white matter).

Table 83.3 Findings in lateral medullary syndrome ^{16 (p 547)}	
GENERALIZED symptoms	Responsible lesion
• vertigo, N/V, nystagmus, diplopia, oscillopsia	vestibular nuclei & connections
• hiccups	?
IPSILATERAL to lesion	Responsible lesion
• facial pain, paresthesias, & impaired sensation	descending tract and nucleus V over half of face
• ataxia of limbs	(restiform body?)
• Horner's syndrome	descending sympathetic tract
• dysphagia, diminished gag, hoarseness	exiting fibers of IX & X
• numbness of arm, trunk, or leg	cuneate & gracile nuclei
CONTRALATERAL to lesion	Responsible lesion
• impaired pain & temp sense over half of body	spinothalamic tract

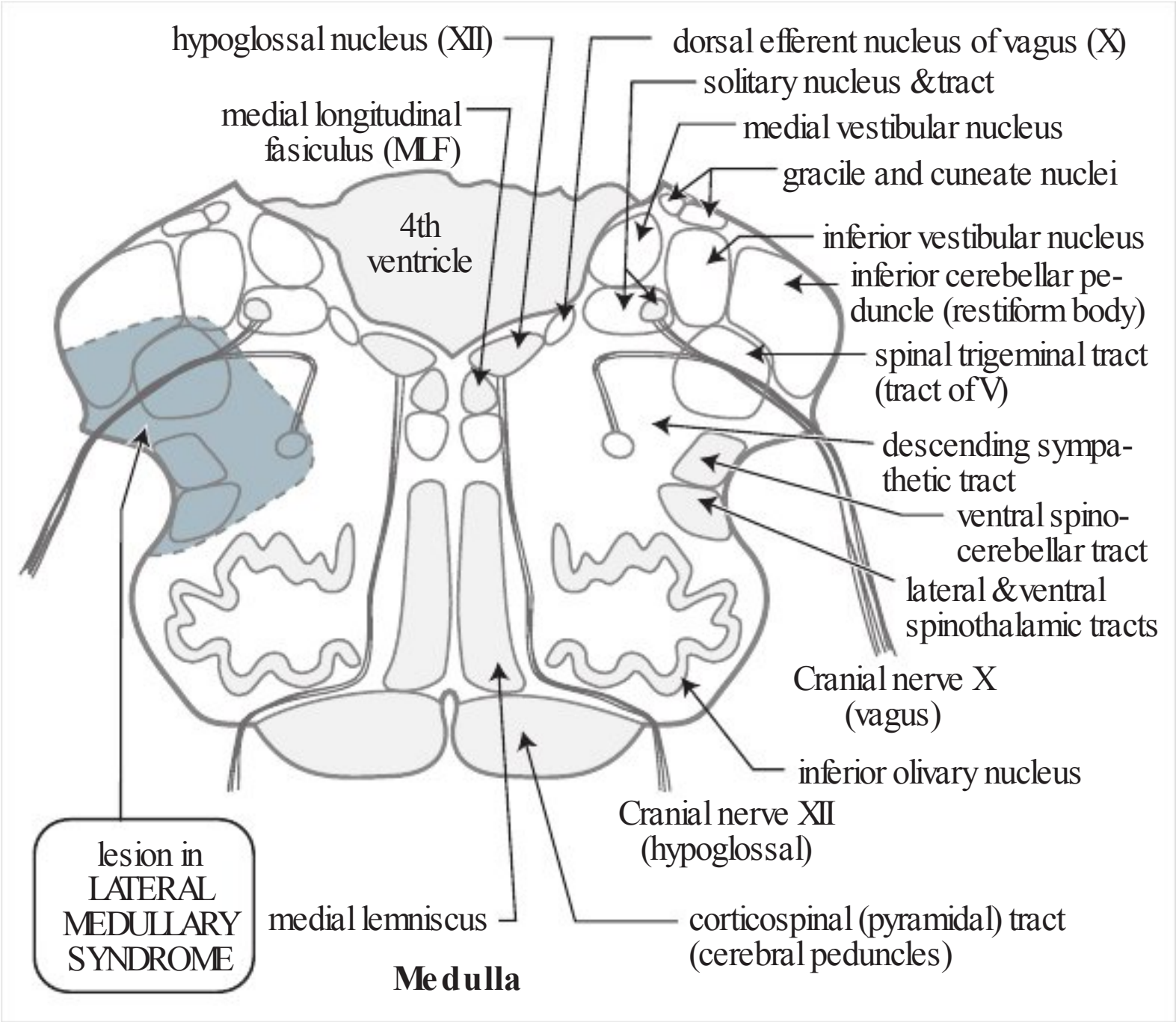


Fig. 83.1 Typical lesion in lateral medullary syndrome (indicated as shaded area)

Table 83.4 Typical locations for lacunar strokes (in descending frequency)
• putamen • caudate • thalamus • pons • internal capsule (IC) • convolutional white matter

Small (3–7 mm) lacunes may be due to lipohyalinosis (vasculopathy due to HTN) of arteries <200 microns (may also be cause of many ICHs); this vasculopathy is indicative of small vessel disease, unlikely to be prevented by carotid endarterectomy.

Clinically, diagnosis virtually excluded by: aphasia, apractagnosia, sensorimotor stroke, monoplegia, homonymous hemianopsia (HH), severe isolated memory impairment, stupor, coma, LOC, or seizures.

L'etat lacunaire: multiple lacunes → chronic progressive neuro decline with one or more episodes of hemiparesis; results in invalidism, dysarthria, small-step gait (marche á petits pas), imbalance, incontinence, pseudobulbar signs, dementia. Many signs and symptoms are possibly due to NPH (unrecognized originally).

Lacunar syndromes

Major syndromes (see reference¹⁹ for others):

1. pure sensory stroke or TIA: (the most common lacunar manifestation) usually isolated unilateral numbness of face, arm, and leg. Only 10% of TIA go on to stroke. Lacune in sensory (posteroventral) thalamus → CT detection is poor. Dejerine-Roussy = rare thalamic pain syndrome that may develop late

2. pure motor hemiparesis (PMH): (2nd most common lacunar manifestation) pure unilateral motor deficit of face, arm and leg without sensory deficit, HH, etc. Lacune in posterior limb of IC, or in lower basis pontis where corticospinal (CS) tracts coalesce, or rarely in mid-cerebral peduncle
3. ataxic hemiparesis: contralateral PMH + cerebellar ataxia of affected limbs (if they can move). Lacune in basis pontis at junction of upper third and lower two-thirds → dysarthria, nystagmus and unidirectional toppling possible. Differential severity in face, arm and leg possible because CS fibers dispersed by nuclei pontis (unlike compact pyramids and peduncle)
 - a) variant: dysarthria-clumsy hand syndrome: lesion in same location or genu of IC. May be mimicked by a cortical infarct, but latter will have numb lips
4. PMH sparing the face: lacune in medullary pyramid; at onset, there may be vertigo and nystagmus (approaching lateral medullary syndrome)
 - a) variant: thalamic dementia: central region of one thalamus + adjacent subthalamus → abulia, memory impairment + partial Horner's (miosis + anhydrosis)
5. mesencephalothalamic syndrome: "top o' the basilar syndrome." Usually caused by embolus. Infarct typically butterfly shaped & bilateral involving rostral brainstem and cerebral hemisphere regions fed by the distal basilar artery. Clinical: III palsy, Parinaud's syndrome & abulia, may have amnesia, hallucinations and somnolence, usually without significant motor dysfunction
6. Weber's syndrome: Cr. N. III palsy with contralateral PMH (no sensory loss). Usually due to occlusion of interpeduncular branches of basilar artery → central midbrain infarction, disrupting cerebral peduncle and issuing fibers of III. May also be due to aneurysm of basilar bifurcation or BA-SCA junction
7. PMH with crossed VI palsy: lacune in paramedian inferior pons
8. cerebellar ataxia with crossed III palsy (Claude syndrome): lacune in dentatorubral tract (superior cerebellar peduncle)
9. hemiballism: classically, infarct or hemorrhage in subthalamic semilunar nucleus of Luys
10. lateral medullary syndrome: see below
11. locked-in syndrome: bilateral PMH from infarct at IC, pons, pyramid or (rarely) cerebral peduncles

83.5 Stroke in young adults

83.5.1 General information

Only 3% of ischemic strokes occur in patients <40 yrs age.²⁰ Over 10% of ischemic strokes occur in patients ≤55 yrs.²¹ Incidence: 10 per 100,000 persons age 35–44 yrs,²² 73 per 100,000 for age <55 yrs.²¹

83.5.2 Etiologies

The differential diagnosis is lengthy,²⁰ with trauma being the most common cause of strokes (22%) in patients under 45 yrs.²³ Most of the rest are covered by the small number of etiologies listed below (excludes: trauma, post-op stroke, SAH, and intracerebral hemorrhage).

1. atherosclerosis: 20%—less common than in older population (all 18 patients in one series had either ID-DM, or were males >35 yrs with ≥1 risk factors (see below), most had TIAs earlier)
2. embolism with recognized source: 20%
 - a) cardiac origin is the most common (above), most have previously known cardiac disease:
 - rheumatic heart disease
 - prosthetic valve
 - endocarditis
 - mitral valve prolapse (MVP): present in 5–10% of young adults, in 20–40% of young adults with stroke (although one series found MVP in only 2% of stroke in young adults²²)
 - A-fib
 - left-atrial myxoma
 - b) fat embolism syndrome: neurologic manifestation is usually global neurologic dysfunction; see Fat embolism syndrome (p.835)
 - c) paradoxical embolism: e.g. ASD, pulmonary AVM including Osler-Weber-Rendu syndrome, patent foramen ovale (above)
 - d) amniotic fluid embolism: may occur typically in the post-partum period
3. vasculopathy: 10%
 - a) inflammatory
 - Takayasu's
 - infective: TB, syphilis, ophthalmic zoster

- amphetamine abuse
- herpes zoster ophthalmicus (HZO): usually presents with delayed contralateral hemiplegia with a mean of ≈ 8 weeks following HZO²⁴
- mucormycosis: a nasal and orbital fungal infection primarily in diabetics and immunocompromised patients that causes an arteritis which may thrombose the orbital veins and ICA or ACA. Produces proptosis, ocular palsy, and hemiplegia
- associated with systemic disease such as: SLE (lupus), (also see below under Coagulopathy); arteritis (especially periarteritis nodosa (p.199)) when confined to CNS is usually multifocal and progressive, but may mimic stroke early; multiple sclerosis (MS); cancer; rheumatoid arthritis
- b) non-inflammatory
 - fibromuscular dysplasia (p.200)
 - carotid or vertebral artery dissections (including posttraumatic)
 - moyamoya disease (p.1313)
 - homocystinuria: a genetic defect in methionine metabolism that produces intimal thickening and fibrosis in almost all vessels with associated thromboembolic events (arterial and venous, including dural venous sinuses). Estimated risk of stroke is 10–16% Patients have a Marfan syndrome-like physical appearance, malar blotches, mental retardation, and elevated levels of urinary homocysteine
 - pseudoxanthoma elasticum
- 4. coagulopathy: 10% The following are associated with hypercoagulable states
 - a) SLE: lupus anticoagulant $\rightarrow$ prolonged PTT incompletely corrects with 50/50 mix. Collagen vascular disease only rarely presents initially with stroke
 - b) polycythemia or thrombocytosis
 - c) sickle cell disease
 - d) TTP (thrombotic thrombocytopenic purpura)
 - e) antithrombin III deficiency (controversial – not seen in large series of young adults with stroke)
 - f) protein C or protein S deficiency (familial): protein C attenuates hemostatic reactions, homozygous deficiency is fatal in the neonatal period. Heterozygous deficiency is associated with thrombotic strokes. A rare complication during initial therapy with warfarin is a drop in protein C before other coagulation factors resulting in a hypercoagulable state
 - g) antiphospholipid-antibody syndrome (APLAS)^{25,26}: causes venous and/or arterial thrombosis. The two best known antiphospholipid-antibodies are anticardiolipin antibodies (ACLA), and lupus anticoagulant (LAC). Once they become symptomatic, treatment is high-intensity warfarin therapy to an $\text{INR} \geq 3$.²⁷ There is a dramatic increase in thrombotic events after discontinuing warfarin. Aspirin is useless
 - h) following use of the drug 3,4-methylenedioxymethamphetamine (MDMA, known on the street as ecstasy),²⁸ possibly independent of the hypercoagulable state that occurs with hyperthermia when insufficient fluids are consumed in conjunction with use of the drug
- 5. peripartum: 5%(usually within 2 wks of parturition)
- 6. miscellaneous causes: 35%
 - a) uncertain etiology
 - b) oral contraceptives (BCP): associated with ninefold increased risk for stroke, many with prior migraine history
 - c) venous thrombosis (including dural sinus thrombosis): incidence may be increased with use of BCP
 - d) migraine²⁹: widely accepted, but difficult to assess objectively (incidence of stroke in these patients may be same as general population). Rare. Usually occurs in women, with a benign long-term course; recurs in $<3\%$ Possible mechanisms include: vasospasm, platelet dysfunction and arteriopathy.³⁰ strokes often occur during a migrainous attack³¹ or shortly thereafter
 - e) cocaine abuse³²: stroke may result from vasoconstriction, or from HTN in the presence of aneurysms or AVMs (frank vasculitis occurs³³ but is rare with cocaine, unlike amphetamines); strokes with alkaloidal cocaine (“crack”) are $\approx$ equally divided between ischemic and hemorrhagic
 - f) posterior reversible encephalopathy syndrome (PRES) (p.194)

83.5.3 Risk factors

In a retrospective “neighborhood control” study of 201 Australian patients aged 15–55 (mean = 45.5) with first-time strokes, the following risk factors were identified²¹:

1. diabetes: odds ratio = 12
2. HTN: odds ratio = 6.8

3. current cigarette smoking: odds ratio = 2.5
4. long-term heavy alcohol consumption: odds ratio = 1.5 (heavy alcohol ingestion within 24 hrs preceding the stroke was not a risk factor)

83.5.4 Evaluation

1. history & physical exam directed at uncovering systemic disease (see above) and modifiable risk factors (see above)
2. cardiology work-up including EKG and echocardiogram
3. bloodwork (include as appropriate):
 - a) routine: electrolytes, CBC, platelet count and/or function, ESR (elevation may suggest SLE, arteritis, atrial myxoma... but a normal ESR does not rule-out vasculitis), PT/PTT, VDRL (should be obtained in all young adults with stroke), fasting lipid profile
 - b) for unexplained stroke: ANA, antithrombin III, protein C, protein S, homocysteine, factor V Leiden, PPD, sickle-cell screen, toxicology screen (blood and urine, to R/O drugs such as cocaine), SPEP, lupus anticoagulant, serum amino acid, tissue plasminogen-activator and -inhibitor
4. miscellaneous tests: U/A, CXR, CSF exam when indicated
5. cerebral angiography: not always necessary for patients with obvious systemic disease or strong evidence for cardiac embolism; may occasionally diagnose cerebral embolism if performed within 48 hrs of ictus

83.6 Atherosclerotic carotid artery disease

83.6.1 General information

Atherosclerotic plaques begin to form in the carotid artery at 20 yrs of age. In the extracranial cerebral circulation, plaques typically start on the back wall of the common carotid artery (CCA). As they enlarge, they encroach on the lumen of the ICA. Calcified hard plaques may not change with time. The risk of stroke correlates with the degree of stenosis and with certain types of plaque morphology, and is also increased in hypercoagulable states and with increased blood viscosity.

Plaque morphology

“Vulnerable” plaques are atherosclerotic plaques likely to cause thrombotic complications, or those that tend to progress rapidly. Criteria for vulnerable plaques include: intimal thickening, plaque fissure, lipid/necrotic core with thin fibrous cap, calcification, thrombus, intraplaque hemorrhage, and outward remodeling. Some of these features can be identified with high-resolution MRI.^{34,35,36,37}

83.6.2 Presentation

General information

Carotid artery lesions are considered symptomatic if there is one or more lateralizing ischemic episodes appropriate to the distribution of the lesion. A lesion is considered to be asymptomatic if the patient only has non-specific visual complaints, dizziness, or syncope not associated with TIA or stroke.³⁸ The majority (80%) of carotid atherothrombotic strokes occur without warning symptoms.³⁹

Asymptomatic carotid stenosis

Usually discovered as a carotid bruit. Asymptomatic bruit: prevalence increases with age (2.3% in ages 45–54 yrs, 8.2% at ≥ 75).⁴⁰ Accuracy of a bruit in predicting ICA stenosis: 50–83% (depending on cohort, criteria for stenosis...). Sensitivity is as low as 24%.⁴¹

Symptomatic carotid disease

May present as a TIA, RIND or stroke with any of the following findings; see also ICA occlusion syndromes (p. 1265):

1. retinal insufficiency or infarction (central retinal artery is a branch of the ophthalmic artery):
 - ipsilateral monocular blindness
 - a) may be temporary: amaurosis fugax, AKA transient monocular blindness (TMB). Four types:

Type I: embolic. Described “like a black curtain coming down” in one eye. Complete loss of vision, usually lasts 1–2 minutes

Type II: flow related. Retinal hypoperfusion → desaturation of color, usually described as a graying of vision

Type III: vasospastic. May occur with migraines

Type IV: miscellaneous. May occur with anticardiolipin antibodies

a) blindness may be permanent

1. middle cerebral artery symptoms:

a) contralateral motor or sensory TIA (arm and face worse than leg) with hyperreflexia and upgoing toe

b) language deficits if dominant hemisphere involved

83.6.3 Evaluation of the extent of carotid disease

Overview

Symptomatic patients will usually be assessed as part of a stroke/TIA protocol.

Check CBC with platelet count, fibrinogen, PT/PTT/INR (to R/O hypercoagulable state).

Funduscopy exam may show Hollenhorst plaques (cholesterol crystal emboli) in the retina.

Classification of patients based on the hemodynamics and also the embolic propensity of carotid lesions has thus far been too complex to be utilized in large studies. The tests described below place a great deal of emphasis on the greatest degree of stenosis which is probably an oversimplification. Plaque composition and morphology is probably important.

Recommendations for screening for carotid stenosis

- 1. the U.S. Preventive Services Task Force (USPSTF) currently recommends against screening for carotid stenosis in the adult general population (grade D recommendation: moderate or high certainty that the service has no net benefit or that the harm outweighs the benefit)⁴²
- 2. the AHA Primary Prevention of Stroke Guidelines does not recommend screening for asymptomatic carotid stenosis⁴³
- 3. the American Society of Neuroimaging advised that screening should be considered only for age ≥ 65 years with 3 or more cardiovascular risk factors⁴⁴
- 4. the Society of Vascular Surgery recommends ultrasonography screening for age ≥ 55 years with cardiovascular risk factors, such as a HTN, diabetes, smoking, hypercholesterolemia, or known cardiovascular disease⁴⁵

Assessment options

See also recommendations for which tests to use (p. 1274).

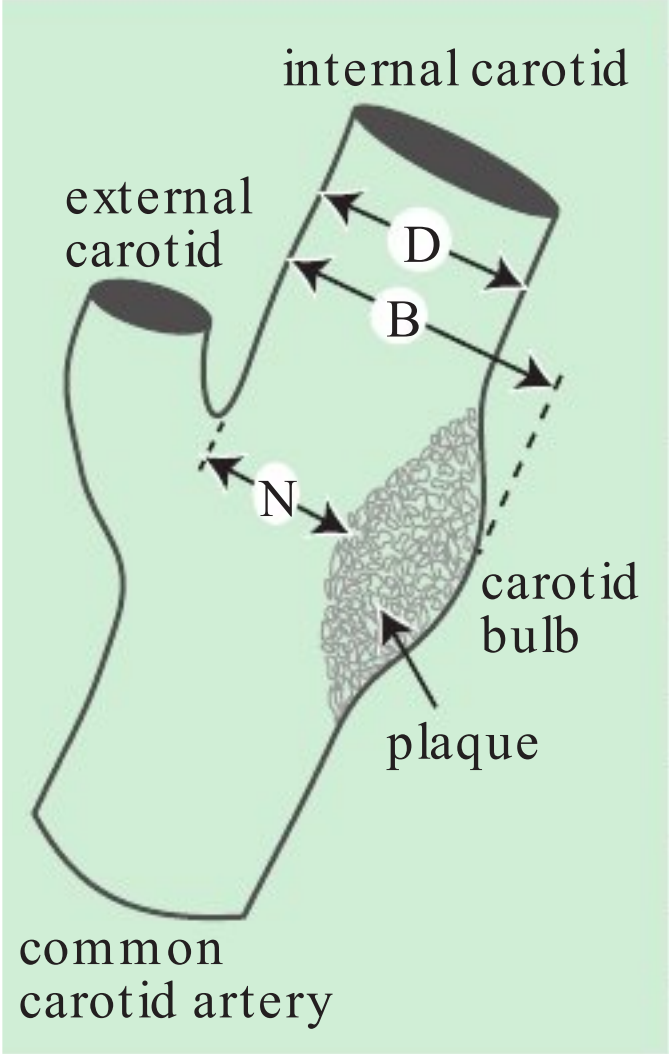
Angiography

The “gold standard” test is a catheter arteriogram. It cannot be justified as a screening test because it is invasive, and too costly and risky (recent data show <1% risk of transient or permanent deficit (risk is 2–3 times higher in symptomatic patients than in asymptomatic)^{46,47,48} in good hands). Also, unlike duplex doppler and MRA, it does not provide any information about the thickness of the plaque. Different definitions of the degree of stenosis are employed, □ Table 83.5 compares the definitions used by the NASCET study⁴⁹ to that of the ECST.⁵⁰ For both, N is the linear diameter of the carotid artery at the site of greatest narrowing. The studies differ in the denominator, NASCET uses D (the diameter of the normal artery distal to the carotid bulb – taken at the first point at which the arterial walls become parallel), whereas the ECST uses B (the estimated carotid bulb diameter).

For example, using the NASCET definition, the degree of stenosis is shown in Eq (83.2).

% stenosis $\delta_{\text{NASCET}} = \left(1 - \frac{N}{D}\right) \hat{=} 100$ 83:2b

The relationship between the degree of narrowing based on the NASCET definition vs. that of the ECST has also been estimated by equation⁵¹ as shown in Eq (83.3).

Table 83.5 Comparison of NASCET and ECST measurements of ICA stenosis ^a		
	NASCET	ECST
	$1 - \frac{N}{D}$	$1 - \frac{N}{B}$
	Approximate equivalent degrees of ICA stenosis based on direct comparison (%)	
	30 ^b	65 ^b
	40 ^b	70
	50	75
	60	80
	70	85
	80	91
	90	97
^a adapted from Donnan G A, Davis S M, Chambers B R, et al.: Surgery for the Prevention of Stroke. Lancet 351: 1372, 1998, with permission ^b indicates degrees of stenosis for which surgery was NOT of clear benefit for symptomatic stenosis (p. 1290)		

% stenosis by ECST = 0.6 × % stenosis by NASCET + 40%

83.3%

Angiography also affords the opportunity to perform endovascular intervention if indicated.

Duplex doppler ultrasound

B-mode image evaluates the artery in cross-sectional plane, and spectrum analysis shows blood flow. Performs poorly with a “string sign.” Cannot scan above the angle of the mandible. Lower frequencies give greater depth of penetration, but signal definition is sacrificed (used in transcranial doppler). Sensitivity: 88% specificity: 76%⁵²

Magnetic resonance angiography (MRA)

May obviate the need for angiography in some cases of carotid stenosis, specifically in symptomatic patients with a focal “gap” of signal intensity loss with distal reappearance of signal.^{53,54} Sometimes overestimates the degree of stenosis.⁵⁵ Sensitivity: 91% specificity: 88%for extracranial carotid disease.⁵⁶ 2D TOF-MRA is adequate (contrast-MRA shows more, but is not necessary for surgical lesions⁵⁷).
Can be performed at the time as MRI with stroke protocol in TIA/stroke patients, and also detects thrombus or dissection. As with Doppler, has difficulties distinguishing very severe stenosis from occlusion. Less operator dependent than Doppler, but is more expensive and time-consuming. MRA is more difficult to perform if the patient is critically ill, unable to lie supine, or has claustrophobia, a pacemaker or ferromagnetic implants. High-resolution MRI may also detect vulnerable plaques (p.1271).

Computed tomography angiography (CTA)

CTA involves ionizing radiation (x-rays) and IV iodinated contrast, limiting its use in patients with dye allergies and renal dysfunction. Results are comparable to MRA and Doppler. CTA

can be performed within a few seconds and yields high-resolution images of all vessels from the aortic arch through the intracranial/extracranial vessels as well as the surrounding soft tissues. In a metaanalysis, sensitivity and specificity for detection of a 70% to 99% stenosis were 85% and 93%, respectively.⁵⁸ CTA is still evolving and may help detect vulnerable plaques (p.1271). Another potential advantage: ability to obtain CT-perfusion (p.228) studies at the same time.

Choice of imaging test/management decisions

Despite a great deal of research on the subject, there are no data to support a particular testing algorithm.¹ Doppler, CTA, or MRA are acceptable initial screening tests. In patients with an abnormal screening test, a common strategy is to obtain a second confirmatory noninvasive test to evaluate the carotid bifurcation before intervention. The combination of carotid ultrasound and MRA has proved cost effective with good interobserver reliability.⁵⁹ If 2 noninvasive tests are discordant, catheter angiography should be considered before intervention.

83.6.4 Treatment

Treatment alternatives are primarily between the following.

- 1. “best medical management”: see below
- 2. carotid endarterectomy (p.1290)
- 3. endovascular techniques: combined angioplasty and stenting (± distal embolus protection)

Medical treatment

General information

What constitutes “best medical management” has not been precisely determined, and recommendations are constantly changing. Some or all of the following are utilized:

- 1. antiplatelet therapy (p.164):
 - a) usually aspirin (ASA) (see below)
 - b) clopidogrel, either alone or in combination with ASA (see below)
 - c) combination of extended release dipyridamole and ASA (Aggrenox®) (no benefit from dipyridamole (Persantine®) alone)
- 2. antihypertensive therapy as appropriate
- 3. good control of diabetes if present
- 4. patients with asymptomatic A-fib should be treated with anticoagulation; see Cardiogenic brain embolism (p.1304)
- 5. antilipid therapy if needed – Statins
- 6. intervention to help patients to quit smoking

Antiplatelet therapy

Drug info: Aspirin

Irreversibly inhibits cyclooxygenase preventing synthesis of vascular prostacyclin (a vasodilator and platelet inhibitor) and platelet thromboxane A2 (a vasoconstrictor and platelet activator). Platelets, lacking cellular organelles, cannot resynthesize cyclooxygenase whereas the vascular tissues do so rapidly.⁶⁰ NB: <1000 mg ASA per day probably does not help with high grade stenosis where there is perfusion failure or flow failure. Some (but not all) studies show less effectiveness in women,⁶¹ and no large study has shown that ASA prevents a second stroke in patients that have already had one.

□: For angina, a bolus dose of 160–325 mg PO is followed by maintenance doses of 80–160 mg/d (lower doses appear to be as effective as higher doses).⁶² Optimal dose for cerebrovascular ischemia continues to be debated. 325 mg PO qd reduces risk of stroke following TIA by 25–30% Daily doses of 81 or 325 mg when compared to higher doses were associated with a lower rate of stroke, MI and death (6.2%vs. 8.4%) following carotid endarterectomy.⁶³

Drug info: Aspirin/ER-Dipyridamole (Aggrenox®)

Combination of extended release dipyridamole and ASA (Aggrenox) is more effective than ASA alone for prevention of TIA, stroke, and myocardial infarction.^{64,65,66} Aggrenox was not superior to clopidogrel, with increased hemorrhage with Aggrenox.⁶⁷ Side effects: H/A with initial therapy.

□: 1 capsule PO BID. Supplied: fixed dose capsules of aspirin 25 mg/extended-release with dipyridamole 200 mg.

Drug info: Clopidogrel (Plavix®)

A thienopyridine. Incidence of severe neutropenia (0.04%) is close to that of ASA ($\approx$ 0.02%).⁶⁸ Interferes with platelet membrane function by inhibiting ADP-induced platelet fibrinogen binding and release of platelet granule contents, as well as subsequent platelet-platelet interactions. Produces a time and dose dependent irreversible inhibition of platelet aggregation and prolongation of bleeding time. May replace ASA if intolerance or resistance. Used in combination with ASA for some endovascular procedures. Although clopidogrel plus aspirin is recommended over aspirin for acute coronary syndromes, the MATCH⁶⁹ results do not suggest a similar benefit for stroke and TIA. Combination therapy significantly increased risk of hemorrhage.⁶⁹

Pharmacokinetics: Dosed once daily. Requires several days to reach maximal effect (□a loading dose may be used, e.g. after an acute event such as an MI, or before stenting). Takes $\approx$ 5 days off the drug for platelet inhibition to reverse.

□: 75 mg PO qd. Loading dose: 225 mg (3 pills) the first day of therapy. Supplied: 75 mg film-coated tablet.

Choice of antiplatelet agents

Individualization is recommended for antiplatelet agents for secondary stroke prevention. ASA is effective, and its low cost may help compliance. A small reduction of vascular events with Aggrenox may justify its expense from a broader healthcare perspective. Clopidogrel is appropriate for those intolerant or resistant to ASA. Clopidogrel plus ASA may be indicated in patients with recent cardiac ischemia or vascular stenting.⁷⁰

Asymptomatic carotid artery stenosis

Key concepts

- natural history: reveals low stroke rate (2%/yr) half of which are not disabling
- large randomized trials have revealed moderate surgical benefit versus medical management for: asymptomatic stenosis >60%
- treatment selection criteria depend on patient's age, gender and comorbidities (and therefore life expectancy), and on perioperative complication rate

Practice guideline: Asymptomatic carotid stenosis

- Level I⁷¹: CEA is reasonable in asymptomatic patients with >70%ICA stenosis if risk of perioperative stroke, MI and death is low
- Level II⁷¹: It is reasonable to choose CEA over CAS when revascularization is indicated in older patients especially when the anatomy is unfavorable for endovascular intervention
- Level II⁷¹: It is reasonable to choose CAS over CEA when revascularization is indicated in patients with anatomy unfavorable for surgery
- Level II⁷¹: prophylactic CAS may be considered in highly selected patients with asymptomatic ICA stenosis ($\geq$ 60% by angiography, >70% by validated Doppler ultrasound), but the effectiveness compared to medical therapy alone is not well established

- Level II⁷¹: In patients with high risk of complications by either CEA or CAS (includes: age >80 years, NYHA heart failure class II or IV, LVEF <30% class III or OV angina pectoris, left main or multivessel CAD, need for cardiac surgery within 30 days, MI within 4 weeks, and severe chronic lung disease), the effectiveness of revascularization over medical therapy alone is not well established

Abbreviations: CEA=carotid endarterectomy, CAS=carotid artery stenting

Natural history

Prevalence of carotid stenosis >50% in men and women >65 years of age is 5–10% with 1% having stenosis >80%^{72,73,74}

Natural history studies reflect an annual stroke risk of 1–3.4% with asymptomatic carotid artery stenosis of 50–99% at 2–3 years.^{75,76,77,78,79,80} A cohort study found similar cumulative rates of ipsilateral stroke over 10 years (9.3% or 0.9%/year) and 15 years (16.6% or 1.1%/year).⁸¹

Attempts to identify subgroups of patients with asymptomatic carotid stenosis at elevated stroke risk suggest that the rate of unheralded stroke ipsilateral to a hemodynamically significant extracranial carotid artery stenosis is 1–2% annually, with some data suggesting that the stroke rate may be higher with progressing stenosis or with more severe stenosis. Asymptomatic carotid stenosis is an important marker of concomitant ischemic cardiac disease.^{75,76,77,80,81} In the REACH Study,⁸² patients with asymptomatic carotid stenosis (n=3164) had statistically significantly higher age- and sex-adjusted 1-year rates of transient ischemic attack, non-fatal stroke, fatal stroke, and cardiovascular death compared with patients without asymptomatic carotid stenosis (n=30 329).

Surgery vs. medical management: the studies

ACST

See reference.⁸³

Σ

The largest multicenter randomized trial to date⁸³ revealed a moderate benefit for immediate CEA vs. medical management in patients age <75 with asymptomatic stenosis ≥60%

Details: 3,120 patients with ≥60% stenosis by duplex ultrasound were randomized to immediate CEA (50% had CEA within 1 month, 88% within 1 year) or medical therapy at the discretion of the treating physician. Mean follow-up: 3.4 years. Exclusion criteria included: poor surgical risk, prior ipsilateral CEA, and probable cardiac emboli. Surgeons were required to have a perioperative morbidity and mortality rate of <6%

Net five-year risk for all stroke or perioperative stroke or death: 6.4% in the CEA group, vs. 11.8% in the medical group (p <0.0001). Fatal or disabling stroke: 3.5 vs. 6.1% Fatal stroke alone: 2.1 vs. 4.2% Although men and women benefited, men benefited more. CEA did not demonstrate a statistically significant benefit for patients over the age of 75. Statistical benefit was not seen in the immediate CEA group until nearly two years after surgery, despite a relatively low perioperative morbidity and mortality rate of 3.1% (in contrast to patients with symptomatic stenosis (NASCET⁸⁴) where benefit was seen much earlier).

ACAS

See reference.⁸⁵

Σ

Large trial that randomized patients in good health with asymptomatic stenosis (calculated in the same manner as the NASCET study) ≥60% to CEA plus aspirin, or aspirin alone⁸⁵ found a reduced 5-year risk of ipsilateral stroke if CEA was performed with <3% perioperative morbidity and mortality and is added to aggressive management of modifiable risk factors.

Details: CEA reduced 5-year stroke risk 66% in males, 17% in females (not statistically significant), and 53% overall (males & females lumped together). CEA did not significantly protect against major stroke or death ($P=0.16$) (half of the strokes were not disabling), and was somewhat protective against any stroke or death ($P=0.08$). The study group was 95% caucasian, and 66% were male. Excluded patients (age > 79 yrs, unstable CAD, uncontrolled HTN) may have been higher risk. Surgeons were carefully selected and the surgical morbidity (1.5%) and mortality (0.1%) was very low. Surprisingly, $\approx$ half of the total morbidity (1.2%) was related to angiography. The implication is that for a generally healthy white male with ACAS > 60% management with CEA (when performed by a surgeon with a low complication rate, as described) reduces his annual risk of all strokes from 0.5% to 0.17% (the reduction of risk for severe stroke is less). The benefit from CEA is realized within less than one year after the CEA. This is in contrast to the ACST trial (see above) and is most likely due to the lower perioperative event rate. The risk from mortality from other causes (including MI) is $\approx$ 3.9% per year. Combined stroke and death rates in community hospitals⁸⁶ while improved over the last 20 yrs, remains higher at $\approx$ 6.3% than at centers used in this study.

Veteran's Administration Cooperative Study (VACS)

See reference.⁸⁴

CEA reduces ipsilateral neurologic events, but did not reduce the rate of ipsilateral strokes nor death (most deaths were secondary to MI). This trial did not include women and was not powered to detect differences in outcome subgroups.

CASANOVA Study

See reference.⁸⁷

No difference in outcome between CEA vs. aspirin (new stroke or death), but an unusual protocol lessened its statistical validity.⁸⁸

Mayo Clinic Asymptomatic Carotid Endarterectomy (MACE) Study

See reference.⁸⁹

There were no major strokes or deaths in either the medical or the endarterectomy group. Surgically treated patients were not given aspirin, and 26% had an MI compared to 9% in the aspirin-treated medical arm, reflecting the high incidence of concomitant CAD in patients with an asymptomatic carotid artery stenosis.

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84 Evaluation and Treatment for Stroke

84.1 Rationale for acute stroke treatment

84.1.1 General information

In the complete absence of blood flow, neuronal death occurs within 2–3 minutes from exhaustion of energy stores. However, in most strokes, there is a salvageable penumbra (tissue at risk) that retains viability for a period of time through suboptimal perfusion from collaterals. Progression of local cerebral edema from the injury results in compromise of these collaterals and progression of ischemic penumbra to infarction if flow is not restored and maintained. Prevention of this secondary neuronal injury drives the treatment of stroke and has led to the creation of designated Primary Stroke Centers that offer appropriate and timely triage and treatment of all potential stroke patients.

Current standard of care requires the administration of IV tPA to all eligible patients. Documentation is necessary to justify deviation from this standard of care in the current medico-legal environment.

In centers with advanced capabilities (Comprehensive Stroke Centers), other treatment modalities are also offered.

84.2 Evaluation

84.2.1 History – key components

1. time last seen normal (stroke on awakening being increasingly evaluated by perfusion studies to ensure the presence of viable tissue)
2. current deficit and clinical presentation
3. NIH Stroke Scale score (p.1282) should be assessed and recorded
4. reasons for not administering IV tPA (if any) should be documented

84.2.2 CATscan (emergent)

General information

Upon presentation with symptoms of a potential stroke, a noncontrast brain CT scan should be done immediately to rule-out hemorrhage (intraparenchymal or SAH), hematoma, early signs of ischemia, old infarcts or injuries, and other lesions (e.g. tumor).

CATscan findings with ischemic stroke (“pale” infarcts)

General information

NB: These principles do not apply to small lacunar infarcts, nor to hemorrhagic strokes. NB: CT is normal in 8–69% of MCA strokes in the first 24 hours.¹

Findings at various times after ischemic stroke

□ Hyperacute (<6 hours after stroke). Early signs of infarction involving large areas of the MCA territory correlate with poor outcome.² Early findings may include³:

1. hyperdense artery sign (see below): low sensitivity, but helpful if present
2. focal low attenuation within the gray matter*
3. loss of the gray-white interface*
4. attenuation of the lentiform nucleus
5. mass effect*
 - a) early: effacement of the cerebral sulci (often subtle)⁴
 - b) late: midline shift in large territory infarction
6. loss of the insular ribbon (hypodensity involving the insular region)
7. enhancement with IV contrast: occurs in only 33% Stroke becomes isodense (called “masking” effect) or hyperdense with normal brain, and, rarely, may be the only indication of infarction⁴

* These findings are probably due to increased water content resulting from the following: cellular edema arising from altered cell permeability which produces a shift of sodium and water from the extracellular to the intracellular compartment, which also increases the extracellular osmotic pressure causing transudation of water from capillaries into the interstitium.⁵

- 24 hrs. Most strokes can be identified as a low density by this time.
- 1–2 wks. Strokes are sharply demarcated. In 5–10% there may be a short window (at around day 7–10) where the stroke becomes isodense, called “fogging effect.” IV contrast will usually demonstrate these.
- 3 wks. Stroke approaches CSF density.
- Mass effect. Common between day 1 to 25. Then atrophy is usually seen by ≈ 5 wks (2 wks at the earliest). Serial CT scans have shown that midline shift increases after ischemic stroke and reaches a maximum 2–4 days after the insult.
- Calcifications. Over a long period of time (months to years) approximately 1–2% of strokes calcify (in adults, it is probably a much smaller fraction than this; and in peds it is a higher percentage than this). Therefore, in an adult, calcifications almost rule-out a stroke (consider AVM, low grade tumor...).

Hyperdense artery sign

First described in the MCA in 1983.⁶ The cerebral vessel (usually the MCA) appears as a high density on unenhanced CT, indicating intra-arterial clot (thrombus or embolus).⁷ Seen in 12% of 50 patients scanned within 24 hrs of stroke, and in 34% of 23 very early CTs done to R/O hemorrhage. Sensitivity for MCA occlusion is low, but specificity is high (although it may also be seen with carotid dissection, or (usually bilaterally) with calcific atherosclerosis or high hematocrit⁷). Does not have independent prognostic significance.⁸

Enhancement

CT enhancement with IV contrast in stroke:

1. many enhance by day 6, most by day 10, some will enhance up to 5 wks
2. rule of 2's: 2% enhance at 2 days, 2% enhance at 2 mos
3. gyral enhancement: AKA called “ribbon” enhancement. Common. Usually seen by 1 week (grey matter enhances > white). DDx includes inflammatory infiltrating lesions such as lymphoma, neurosarcoidosis... (due to breakdown of BBB)
4. rule of thumb: there should not be enhancement at the same time there is mass effect

84.2.3 CT angio (CTA)

CTA (p.227) is useful for assessing the location and extent of vascular occlusion in acute ischemic stroke,⁹ and may identify the bleeding source in subarachnoid hemorrhage. Findings can direct treatment towards endovascular options when a proximal or significant large vessel occlusion is seen.

84.2.4 CT perfusion

Theoretically identifies salvageable penumbra as a region of mismatch between CBF and CBV. Assumption: the infarcted core (with no salvageable tissue) has decreased CBF within a region of decreased CBV (CBF/CBV match). A mismatched area (decreased CBV without a decrease in CBF) represents potentially salvageable penumbra.¹⁰ Implication: thrombolytics and interventional treatment modalities without mismatch will likely increase morbidity and mortality without clinical benefit.

84.2.5 MRI

With newer, faster acquisition times, and with gradient echo sequences that are highly sensitive to hemorrhage, MRI is increasingly being utilized in the hyperacute setting and is at times replacing CT as the initial evaluation. More sensitive than CT (especially DWI-MRI (p.232) – and particularly in the 1st 24 hrs after stroke), and especially with brainstem or cerebellar infarction. More contraindications (p.230) than CT.

Contrast MRI: not often used. 4 enhancement patterns¹¹:

1. intravascular enhancement: occurs in $\approx 75\%$ of 1–3 day-old cortical infarcts, and is probably due to sluggish flow and vasodilatation (thus, it is not seen with complete occlusion). May indicate areas of brain at risk of infarction
2. meningeal enhancement: especially involving the dura. Seen in 35% of cortical strokes 1–3 days old (not seen in deep cerebral or brainstem strokes). No angiographic nor CT equivalent

- 3. transitional enhancement: above two types of enhancement coexist with early evidence of BBB breakdown; usually seen on days 3–6
- 4. parenchymal enhancement: classically appears as a cortical or subcortical gyral ribbon enhancement. May not be apparent for the first 1–2 days, and gradually approaches 100%by 1 week. Enhancement may eliminate “fogging effect” (as on CT) which may obscure some strokes at ≈ 2 weeks on unenhanced T2WI

84.2.6 MRI perfusion

Similar to CT perfusion (p.228), areas of matched DWI and PWI abnormality are thought to represent infarcted tissue. PWI abnormalities that do not have a DWI correlate are thought to represent potentially salvageable penumbra.¹²

84.2.7 Emergency cerebral catheter angiography

Indications:

- 1. early stroke in carotid distribution +history of amaurosis fugax or bruit or retinal emboli, etc. suggesting increasing carotid stenosis, thrombogenic ulcerated plaque, or carotid dissection
- 2. if diagnosis still questionable (e.g. aneurysm, vasculitis)
- 3. with rapid recovery, suggesting carotid TIA in face of increasing stenosis
- 4. AVOID angio if unstable or if severe disabling neuro deficit

Findings:

- 1. cutoff sign: vessel ends abruptly at the point of obstruction
- 2. string sign: narrow strand of contrast in a vessel with high grade stenosis
- 3. “luxury perfusion”: reactive hyperemia is a recognized response of cerebral tissue to injury (trauma, infarction, epileptogenic focus...). Luxury perfusion is blood flow in excess of demand due to abolition of CBF autoregulation due to acidosis.¹³ The exact nature of the perfusion is not known (i.e. if it is capillary, arteriole...). On angiography it shows up as accelerated circulation adjacent to the infarct with a stain or blush and early venous drainage

84.2.8 NIH stroke scale (NIHSS)

Administer in order shown (□ Table 84.1). Record initial performance only (do not go back).

84.3 Management of TIA or stroke

84.3.1 Treatment options timeline

See □ Fig. 84.1.

Σ
1. within 4.5 hours of onset of symptoms: a) patients may be candidates for IV tPA (p. 1286) b) failures to respond to IV tPA who are in good clinical grade)NIHSS (p. 1282)>8–10) may be candidates for • intraarterial tPA (IA tPA) or • mechanical embolectomy/clot disruption 2. 4.5–6 hours after onset: a) intraarterial tPA (IA tPA) or b) mechanical embolectomy/clot disruption 3. 6–8 hours, check perfusion with CTP or MRI-DWI before mechanical embolectomy (studied up to 8 hours after onset). □ Embolectomy contraindicated if stroke > 1/3 of MCA distribution (risk of ICH with reperfusion)

These times are more applicable to anterior circulation strokes. Posterior circulation occlusions may be treated more aggressively, e.g. IA tPA has been used up to 12 hrs.

Table 84.1 NIH Stroke scale ^a	
Scale	Finding
1a. Level of consciousness (LOC)	
0	alert; keenly responsive
1	not alert, but arousable by minor stimulation to obey, answer or respond
2	not alert, requires repeated stimulation to attend, or is obtunded and requires strong painful stimulation to make movements (not stereotyped)
3	comatose: responds only with reflex motor (posturing) or autonomic effects, or totally unresponsive, flaccid and areflexic
1b. Level of consciousness questions	
Patient is asked the month and their age.	
0	answers both questions correctly: must be correct (no credit for being close)
1	answers one question correctly, or cannot answer because of: ET tube, orotracheal trauma, severe dysarthria, language barrier, or any other problem not secondary to aphasia.
2	answers neither question correctly, or is: aphasic, stuporous, or does not comprehend the questions
1c. Level of consciousness commands	
Patient is asked to open and close the eyes, and then to grip and release the non-paretic hand. Substitute another 1-step command if both hands cannot be used. Credit is given for an unequivocal attempt even if it cannot be completed due to weakness. If there is no response to commands, demonstrate (pantomime) the task. Record only first attempt.	
0	performs both tasks correctly
1	performs one task correctly
2	performs neither task correctly
2. Best gaze	
Test only horizontal eye movement. Use motion to attract attention of aphasic patients.	
0	normal
1	partial gaze palsy (gaze abnormal in one or both eyes, but forced deviation or total gaze paresis are not present) or patient has an isolated cranial nerve III, IV or VI paresis
2	forced deviation or total gaze paresis not overcome by oculocephalic (Doll's eyes) maneuver (do not do caloric testing)
3. Visual	
Visual fields (upper and lower quadrants) are tested by confrontation. May be scored as normal if patient looks at side of finger movement. Use ocular threat where consciousness or comprehension limits testing. Then test with double sided simultaneous stimulation (DSSS).	
0	no visual loss
1	partial hemianopia (clear cut asymmetry), or extinction to DSSS
2	complete hemianopia
3	bilateral hemianopia (blind, including cortical blindness)
4. Facial palsy	
Ask patient (or pantomime) to show their teeth, or raise eyebrows and close eyes. Use painful stimulus and grade grimace response in poorly responsive or non-comprehending patients.	
0	normal symmetrical movement
1	minor paralysis (flattened nasolabial fold, asymmetry on smiling)

Table 84.1 continued	
Scale	Finding
2	partial paralysis (total or near total paralysis of lower face)
3	complete paralysis of one or both sides (absent facial movement in upper and lower face)
5. Motor Arm (5a = left, 5b = right)	
Instruct patient to hold the arms outstretched, palms down (at 90° if sitting, or 45° if supine). If consciousness or comprehension impaired, cue patient by actively lifting arms into position while verbally instructing patient to maintain position.	
0	no drift (holds arm at 90° or 45° for full 10 seconds)
1	drift (holds limbs at 90° or 45° position, but drifts before full 10 seconds but does not hit bed or other support)
2	some effort against gravity (cannot get to or hold initial position, drifts down to bed)
3	no effort against gravity, limb falls
4	no movement
9	amputation or joint fusion: explain
6. Motor leg (6a = left, 6b = right)	
While supine, instruct patient to maintain the non-paretic leg at 30°. If consciousness or comprehension impaired, cue patient by actively lifting leg into position while verbally instructing patient to maintain position. Then repeat in paretic leg.	
0	no drift (holds leg at 30° full 5 seconds)
1	drift (leg falls before 5 seconds, but does not hit bed)
2	some effort against gravity (leg falls to bed by 5 seconds)
3	no effort against gravity (leg falls to bed immediately)
4	no movement
9	amputation or joint fusion: explain
7. Limb ataxia	
Looking for unilateral cerebellar lesion). Finger-nose-finger and heel-knee-shin tests are performed on both sides. Ataxia is scored only if clearly out of proportion to weakness. Ataxia is absent in the patient who cannot comprehend or is paralyzed.	
0	absent
1	present in one limb
2	present in two limbs
9	amputation or joint fusion: explain
8. Sensory	
Test with pin. When consciousness or comprehension impaired, score sensation normal unless deficit clearly recognized (e. g. clear-cut asymmetry of grimace or withdrawal). Only hemisensory losses attributed to stroke are counted as abnormal.	
0	normal, no sensory loss
1	mild to moderate sensory loss (pinprick dull or less sharp on the affected side, or loss of superficial pain to pinprick but patient aware of being touched)
2	severe to total (patient unaware of being touched in the face, arm and leg)
9. Best language	
In addition to judging comprehension of commands in the preceding neurologic exam, the patient is asked to describe a standard picture, to name common items, and to read and interpret the standard text in the box below. The intubated patient should be asked to write:	

Table 84.1 continued	
Scale	Finding
• You know how. • Down to earth. • I got home from work. • Near the table in the dining room. • They heard him speak on the radio last night.	
0	normal, no aphasia
1	mild to moderate aphasia (some loss of fluency, word finding errors, naming errors, paraphasias and/or impairment of communication by either comprehension or expression disability)
2	severe aphasia (great need for inference, questioning and guessing by listener; limited range of information can be exchanged)
3	mute or global aphasia (no usable speech or auditory comprehension) or patient in coma (item 1a=3)
10. Dysarthria	
Patient may be graded based on information already gleaned during evaluation. If patient is thought to be normal, have them read (or repeat) the standard text shown in this box. • MAMA • TIP-TOP • FIFTY-FIFTY • THANKS • HUCKLEBERRY • BASEBALL PLAYER • CATERPILLAR	
0	normal speech
1	mild to moderate (slurs some words, can be understood with some difficulty)
2	severe (unintelligible slurred speech in the absence of, or out of proportion to any dysphasia, or is mute/anarthric)
9	intubated or other physical barrier
11. Extinction and inattention (formerly neglect)	
Sufficient information to identify neglect may already be gleaned during evaluation. If the patient has severe visual loss preventing visual DSSS, and the cutaneous stimuli are normal, the score is normal. Scored as abnormal only if present.	
0	normal, no sensory loss
1	visual, tactile, auditory, spatial or personal inattention or extinction to DSSS in one of the sensory modalities
2	profound hemi-inattention or hemi-inattention to more than one modality. Does not recognize own hand or orients to only one side of space.
A. Distal motor function (not part of NIHSS) (a = left arm, b = right)	
Patients hand is held up at the forearm by the examiner, and is asked to extend the fingers as much as possible. If patient cannot do so, the examiner does it for them. Do not repeat the command.	
0	normal (no finger flexion after 5 seconds)
1	at least some extension after 5 seconds (any finger movement is scored)
2	no voluntary extension after 5 seconds
Higher NIHSS scores correlate with more proximal vascular lesions (larger vessel occlusion causes more widespread deficit).	
^a Revised 1/24/91. Based on Cincinnati stroke scale. ¹⁴ Contact the Public Health Service, National Institutes of Health, National Institute of Neurologic Disorders and Stroke, Bethesda, Maryland, U.S.A. for copies of a grading form (which has more details on some aspects of grading) and for training information. ¹⁵	

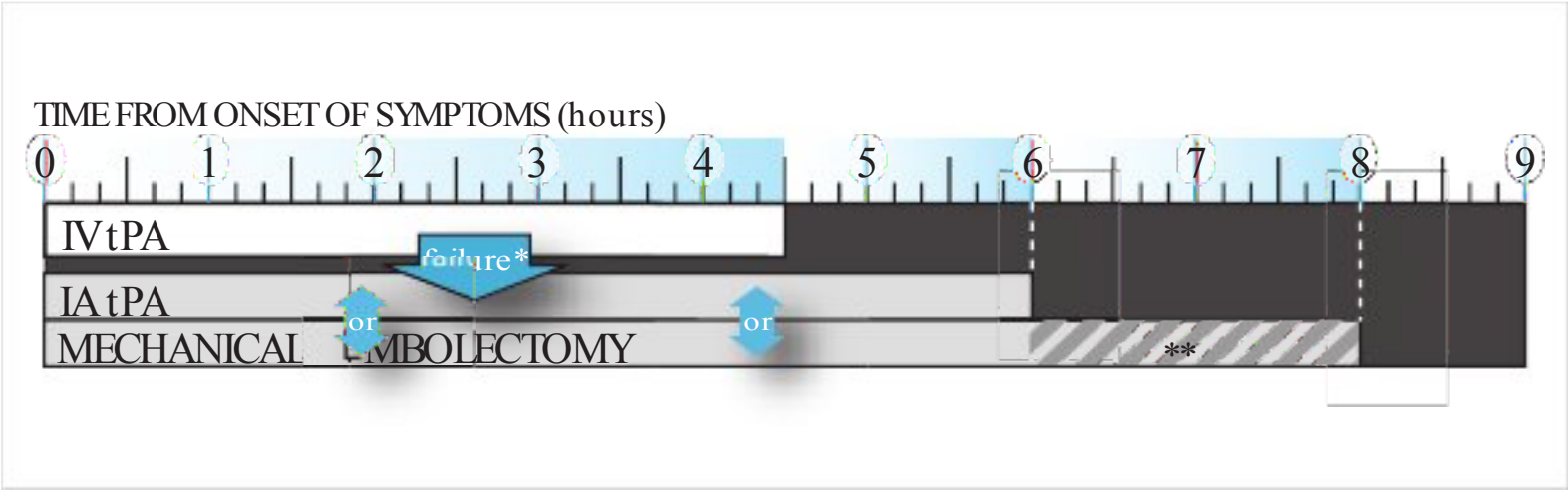


Fig. 84.1 Treatment options timeline.
* option for failures with NIHSS > 8-10
** from 6-8 hrs, check perfusion before mechanical embolectomy

84.3.2 Thrombolytic therapy

General information

Plasminogen activators catalyze the conversion of plasminogen to the fibrinolytic compound plasmin. The primary agent used is alteplase (recombinant tissue plasminogen activator (rtPA, or just tPA)) (Activase®) which is FDA approved for the IV treatment of acute ischemic stroke (see below).

Tissue plasminogen activator

IV tPA

For intraarterial tPA see endovascular therapy.

A randomized double-blind NINDS study of 624 patients with an ischemic stroke having a clearly defined time of onset and a CT scan prior to drug administration, found improved neurologic outcome at 3 months in patients who received alteplase (these patients were 30% more likely to have minimal or no disability)¹⁶ which persisted at 6 and 12 mos¹⁷ in all subgroups of ischemic stroke. The recurrent stroke rate in control and tPA patients was similar (5%). In contrast, statistical benefit at 90 days could not be confirmed in the second European Cooperative Acute Stroke Study (ECASS II).¹⁸

Early data indicated tPA had to be given ≤ 3 hrs after the onset of symptoms, however, this window was extended to 4.5 hours after the ECASS-3¹⁹ looked at 821 stroke patients randomized between placebo or tPA in the 3- to 4.5-hour time-window. Compared to placebo, tPA-treated patients experienced a 7.2% absolute increase in the rate of excellent recovery at 90-day follow-up (P=0.04). Although tPA therapy was associated with an increased rate of symptomatic intracerebral hemorrhage (7.9% for tPA versus 3.5% for placebo, P<0.001), it was not associated with an increased rate of death (7.7% for tPA versus 8.4% for placebo, P=0.68). For every 100 acute ischemic stroke patients given tPA in accordance with the NINDS protocols, 32 will benefit and 3 will be harmed.²⁰

Guidelines for the administration of IV tPA

Eligibility:

1. age ≥ 18 years (although use in childhood stroke is increasing²¹)
2. time since last seen normal < 3 hrs prior to administration (ECASS III extends window to 4.5 hours in select patients²²; ECASS III did not include: patients ≥ 80 years of age, patients with baseline NIHSS (p. 1282) scores > 25, and prior stroke in diabetics. These patients are not excluded from treatment with IV tPA in the 0–3-hour window by the regulatory authorities in the United States and Canada.)
3. “wake-up” stroke (seen in 25% of ischemic stroke patients) may also be safe to treat in select circumstances²³

Contraindications (see reference¹⁶):

1. intracerebral hemorrhage (ICH): on admitting CT, or history of prior ICH
2. clinical presentation of SAH (even with negative CT)
3. known intracranial aneurysm or AVM
4. active internal bleeding
5. known bleeding diathesis, including but not limited to:
 - a) patients on anticoagulants, or those who received heparin in past 48 hrs
 - b) platelet count < 100,000/mm³

6. serious head trauma, serious stroke, or intracranial surgery within past 3 months
7. SBP > 185 mm Hg, or DBP > 110 mm Hg that cannot be controlled despite use of nicardipine infusion or IV labetalol

□ Cautions:

1. seizure witnessed at the time of onset of stroke symptoms
2. major surgery within the last 14 days
3. arterial puncture at non-compressible site within past 7 days
4. recent lumbar puncture
5. rapidly improving or minor symptoms
6. blood glucose > 400 mg/dl or < 50 mg/dl
7. history of GI or urinary tract hemorrhage within past 21 days
8. post myocardial infarction pericarditis

Treatment protocol: Also, see Contraindications (p.1286).

□ alteplase (Activase®): initiate < 4.5 hrs from onset of deficit. NINDS protocol: 0.09 mg/kg IV bolus over 1 min, followed by 0.81 mg/kg constant infusion over 60 minutes (up to a maximum of 90 mg total, including the bolus).¹⁶

HTN is aggressively controlled.

Anticoagulants and antiplatelet drugs are held for 24 hrs after treatment. If there is an indication for anticoagulation, obtain a non-contrast CT 24 hours prior to starting anticoagulation since there is a risk of subclinical intracerebral hemorrhage.

ICH following IV tPA

There is an increased risk of symptomatic intracerebral hemorrhage (ICH) with the use of tPA (NINDS study: 6.4% vs. 0.6% with placebo; ECASS II: 8.8% vs. 3.4%). In spite of this, the NINDS study found that mortality in the tPA group was similar to controls at 3 mos (17% vs. 21%). The following factors were associated with an increased risk of symptomatic ICH (with only a 57% efficiency rate of predicting ICH): severity of NIHSS score, or pre-treatment CT showing brain edema or mass effect. In one study, ICH did not influence outcome except in the rare instance when a massive hematoma occurred.²⁴ Outcomes were still better in the treated group, and the conclusion is that these patients are still reasonable candidates for tPA.²⁵ Since then, multicenter analyses have demonstrated that size of infarction and elevated blood sugar are independent risk factors for symptomatic ICH.²⁶

Management of post-tPA ICH:

1. discontinue tPA infusion of and obtain STAT head CT
2. send labs: PT, aPTT, platelet count, fibrinogen, and type & cross
3. prepare to administer 6–8 units cryoprecipitate containing Factor VIII
4. prepare to administer 6–8 units of platelets
5. if emergent EVD placement or other interventional procedure is needed, consider the use of recombinant Factor VIIa (40–80 mg/kg) immediately beforehand (NB: this is only a temporizing measure and cryoprecipitate needs to still be given)

84.3.3 Endovascular therapy for stroke

Recent trials favor rapid endovascular intervention in acute ischemic stroke with proximal vessel occlusion, small infarct core and moderate to good collateral circulation.^{27,28,29,30} Techniques include intraarterial tPA, mechanical clot retrieval. See Endovascular neurosurgery section (p.1595) for details.

84.3.4 Management of patients not undergoing therapy directed at thrombus

Admitting orders

These guidelines are for TIA or stroke, but not SAH (p.1163) nor intracerebral hemorrhage (ICH) (p.1339). See reference³¹ for rationale for these recommendations. The following guidelines for initial management should be maintained 48 hrs after last neuro deterioration.

1. frequent VS with crani checks (q 1 hr × 12 hrs, then q 2 hrs)
2. activity: bed rest

3. labs:
 - a) routine: CBC+platelet count, electrolytes, PT/PTT, U/A, EKG, CXR, ABG
 - b) “special” (when appropriate): RPR (to rule-out neurosyphilis), ESR (to rule-out giant cell arteritis), hepatic profile, cardiac profile
 - c) at 24 hrs: CBC, platelet count, cardiac profile, lipid profile, EKG
4. O₂ at 2 L per NC; repeat ABG on 2 L O₂
5. monitor cardiac rhythm × 24 hrs (literature quotes 5–10% prevalence of EKG changes, and 2–3% acute MIs in patients with stroke)
6. diet: NPO
7. nursing care
 - a) indwelling Foley (urinary) catheter if consciousness impaired or if unable to use urinal or bedpan; intermittent catheterization q 4–6 hrs PRN no void if Foley not used
 - b) accurate I’s & O’s; notify M.D. for urine output <20 cc/hr × 2 hrs by Foley, or <160 cc in 8 hrs if no Foley
8. IV fluids: NS or 1/2 NS at 75–125 cc/hr for most patients (to eliminate dehydration if present)
 - a) avoid glucose: hyperglycemia may extend ischemic zone (penumbra).³² Although hyperglycemia may be a stress response and may not be neurotoxic,³³ recommendations are to strive for normoglycemia³⁴
 - b) avoid overhydration in cases of ICH, CHF, or SBP > 180. It had been suggested that an optimal Hct for compromise between O₂ delivery and decreased viscosity was ≈ 33% and that fluid management should strive for this, however the early promise of this theory has not been borne out
9. treat CHF and arrhythmias (check CXR & EKG). MI or myocardial ischemia may present with neuro deficit, these patients should be admitted to CCU
10. avoid diuretics unless volume overloaded
11. blood pressure (BP) management:
 - a) for patients presenting with HTN: management must take baseline BP into account: see Hypertension in stroke patients below for management
 - b) for patients presenting with hypotension (SBP < 110 or DBP < 70):
 - unless contraindicated (viz.: ICH, cerebellar infarct, or decreased cardiac output) give 250 cc NS over 1 hr, then 500 cc over 4 hrs, then 500 cc over 8 hrs
 - if fluid ineffective or contraindicated: consider pressors
12. medications
 - a) ASA 325 mg PO q d (unless hemorrhagic stroke proven or suspected)
 - b) stool softener
13. see following sections for discussion of anticoagulation (p. 1289), steroids (p. 1289), and mannitol (p. 1290)

Hypertension in stroke patients

General information

HTN may actually be needed to maintain CBF in the face of elevated ICP, and it usually resolves spontaneously. Therefore treat HTN cautiously and slowly to avoid rapid reduction and overshooting the target. Avoid treating mild HTN. Indications to treat HTN emergently include:

1. acute LV failure (rare)
2. acute aortic dissection (rare)
3. acute hypertensive renal failure (rare)
4. neurologic complications of HTN
 - a) hypertensive encephalopathy
 - b) converting a massive pale (ischemic) infarct into a hemorrhagic infarct
 - c) patients with ICH; some HTN is needed to maintain CBF, see Initial management of ICH (p. 1339)

Hypertension treatment algorithm (modified)

See reference.³¹

Recommended lower limits for treatment endpoints are shown in □ Table 84.2.

1. If DBP > 140 (malignant hypertension): ≈ 20–30% reduction is desirable. Cardene infusion or IV labetalol are agents of choice; arterial-line monitor recommended; sympatholytics (e.g. trimethaphan) contraindicated (they reduce CBF)
2. SBP > 230 or DBP 120–140 × 20 mins: labetalol (p. 126) (unless contraindicated): start at 10 mg slow IVP over 2 mins, then double q 10 min (20, 40, 80, then 160 mg slow IVP) until controlled or

Table 84.2 Guidelines for lower limits of treatment endpoints for HTN in strokes		
	No prior history of HTN	Prior history of HTN
do not lower SBP below	160–170 mm Hg	180–185 mm Hg
do not lower DBP below	95–105 mm Hg	105–110 mm Hg

total of 300 mg given. Maintenance: effective dose (from above) q 6–8 hrs PRN SBP > 180 or DBP > 110

3. SBP 180–230 or DBP 105–120: defer emergency treatment unless there is evidence of LV failure or if readings persist × 60 mins
- a) oral labetalol (p.126) (unless contraindicated) dosed as follows:
- for SBP > 210 or DBP > 110: 300 mg PO BID

• for SBP 180–210 or DBP 100–110: 200 mg PO BID
- b) if labetalol contraindicated:
- nicardipine (p.126)

Anticoagulants

Heparin

A prospective trial³⁵ administering continuous IV infusion of unfractionated heparin titrated to keep APTT 1.5–2.5 × control found no significant improvement in outcome.³⁶ The recurrent stroke rate in the 7 days following a stroke was only 0.6–2.2% per week.^{35,37} Effectiveness is unproven in strokes and TIAs except with cardiogenic brain embolism (p.1304). Anticoagulation may also be hazardous,³⁸ however the complication rate has not been assessed prospectively (small, nonrandomized studies have found symptomatic ICH in 1–8% and other bleeding complications in 3–12%³⁵). Conversion rate of pale → hemorrhagic stroke is 2–5% (dog studies suggest the risk is increased only when HTN not well controlled). Conclusion: the risk of heparin therapy for acute focal cerebral ischemia exceeds any proven benefit,³⁵ and is not justified in most cases (especially when used just to placate the frustrated clinician).^{39,40} The American Heart Association has recommended: “Until more data are available, the use of heparin remains a matter of preference of the treating physician”.³⁵ A small but significant reduction in recurrent stroke has been shown with ASA.

Warfarin

High-intensity warfarin therapy has proven helpful for the antiphospholipid antibody syndrome (APLAS) (p.1270).

For the rare indication for anticoagulation therapy:

1. first, R/O hemorrhage by CT before beginning therapy
2. ASA 325 mg PO q d in all patients with non-hemorrhagic stroke where anticoagulants or surgery not indicated (NB: angiography may be slightly more difficult in patients on ASA)
3. anticoagulants (heparin/warfarin):
- a) indications (rare)
- probably effective for cardiogenic emboli (below)

• shown ineffective for stroke in evolution (neuro deficit that begins, recurs, fluctuates, or worsens while patient in hospital), crescendo TIA or completed stroke. Note: in 74 patients with recent TIAs, elevating PTT 1.5–2.5 x normal with heparin did not reduce recurrent TIAs nor strokes. Bleeding occurred in 9 (12.2%). Additional risk: hemorrhage from heparin induced thrombocytopenia.⁴¹

• unproven, but generally used for carotid dissection
- b) contraindicated with large cardiac embolism, large stroke (risk of hemorrhagic conversion), peptic-ulcer disease that has bled in past 6 mos, uncontrolled severe HTN
- c) start IV heparin and simultaneous warfarin (Coumadin®). Maintain heparin during first ≈ 3 days of warfarin because of initial hypercoagulability, see Anticoagulation (p.156) for target APTT and INR
- d) stop warfarin after 6 months (benefits decline, risks rise)

Steroids (including Dexamethasone (Decadron®))

Indications:

1. steroid responsive vasculitis, e.g. giant cell arteritis (temporal arteritis)
2. cerebellar infarct/bleed with mass effect

Mannitol

- 1. indicated for cerebellar infarct/bleed, prior to surgery, or if mass effect
- 2. contraindicated in hypotension
- 3. initial dose: 50 to 100 gm IV over 20 minutes

Emergency surgery

Possible indications:

- 1. herniation from subdural hematoma
- 2. suboccipital craniectomy for progressive neurologic deterioration due to brainstem compression from cerebellar hemorrhage/infarction (see below)
- 3. decompressive craniectomy for malignant MCA territory stroke (see below)
- 4. carotid endarterectomy for high grade carotid stenosis ipsilateral to fluctuating neuro deficit; see Emergency carotid endarterectomy (p.1295)

84

84.4 Carotid endarterectomy

84.4.1 Indications

Trials and results

□ Table 84.3 shows the status of current studies for the surgical treatment of carotid stenosis (NB: some of the results may be contradictory).

The North American Symptomatic Carotid Endarterectomy Trial⁴³ (NASCET) found that for patients with a hemispheric or retinal TIA or a mild (non-disabling) stroke within 120 days and ipsilateral high-grade stenosis (>70%), that carotid endarterectomy (CEA) reduced the rate of fatal and non-fatal strokes (by 17%at 18 months) and death from any cause (by 7%at 18 months) when compared to best medical management (when surgery was performed with perioperative risk of stroke

Table 84.3 Summary of study findings for carotid endarterectomy (CEA) ^a (modified ⁴²)			
Stenosis	Relevant study	Recommendation	Risk reduction ^b
Symptomatic Narrowing			
70–99%	NASCET ⁴³	CEA	16.5 @ 2 yrs
>60%	ECST ⁴⁴	CEA	11.6 @ 3 yrs
50–69%	NASCET ⁴⁵	CEA ^c	10.1 @ 5 yrs
<30%	NASCET ⁴⁵	BMM	0.8 @ 5 yrs
<40%	ECST ⁴⁶	BMM	CEA worse @ 3 yrs
Asymptomatic Narrowing (p. 1275)			
>60%	ACST ⁴⁷	CEA if age <75 yrs	5.4% @ 5 yrs
>60%	ACAS, ⁴⁸ ACST ^d	CEA ^d	6.3 @ 5 yrs
>50%	VACS	± CEA ^e	
<90%	CASANOVA	BMM ^e	
^a abbreviations: NASCET=North American Symptomatic Carotid Endarterectomy Trial; ECST=European Carotid Surgery Trial; CASANOVA=Carotid Artery Stenosis with Asymptomatic Narrowing Operation Versus Aspirin; ACAS=Asymptomatic Carotid Atherosclerosis Study; ACST=Asymptomatic Carotid Atherosclerosis Study; VACS=Veteran’s Administration Cooperative Study; CEA=carotid endarterectomy; BMM=best medical management			
^b reduction in risk of all nonfatal strokes and death from any cause with CEA vs. BMM (e.g. with an absolute risk reduction of 16.5 at 2 yrs, for every 100 patients treated, 16.5 nonfatal strokes or deaths were prevented over a 2 year period)			
^c surgery moderately beneficial (requires low complication rate)			
^d the overall health of the patient is critical			
^e results equivocal			

or death of 5.8%). Results were twice as good for patients with stenosis from 90–99% than for those with 70–79%. Furthermore, with CEA the frequency of major functional impairment was reduced at 2 years.⁴⁹ NB: see differences in techniques for measuring stenosis between NASCET and ECST □ Table 83.5.
See also asymptomatic patients (p.1275).

Unresolved controversies

- Include:
1. progressive STROKE (“stroke in evolution”): see Emergency carotid endarterectomy (p.1295)
 2. abrupt occlusion: see Emergency endarterectomy (p.1295)
 3. tandem lesions (e.g. carotid siphon and bifurcation stenosis): although this topic remains controversial, CEA in patients with tandem lesions has not been associated with increased postoperative stroke rates.^{50,51} Recent case series also report success with endovascular treatment
 4. progressive retinal ischemia

84.4.2 Timing with respect to acute stroke

For patients with small fixed deficits or small infarcts on CT or MRI, the risk of early CEA is not increased.^{50,52} In the pooled analysis of the three symptomatic CEA studies, patients randomized in the trials within 2 weeks of the last symptomatic event had greater benefit from CEA.⁵³ Data from Sundt (see below) indicates that a stroke is a risk factor for a complication only if it occurred within 7 days pre-op.
Since the introduction of tPA for the treatment of acute ischemic stroke, there have been reports on the successful treatment of residual critical ICA stenosis following tPA recanalization as early as 24 hours after administration of tPA in patients with small fixed deficit or small ischemic areas on MRI.^{53,54}

84.4.3 Pre-op risk factors for CEA

Σ

The characteristics of patients who are high risk for complications from CEA has not been well defined, despite the perception that this group exists.

Identifying patients at high risk for complications after CEA has proven challenging. Typically, the exclusion criteria from studies is cited, but in most cases these are simply patients that were not included in the study because it was the investigators perception these patients might be “high risk.” Therefore these risk factors are not validated. They are included here for completeness.
NASCET and ACAS: age >80 years, prior ipsilateral CEA, prior contralateral CEA within 4 months, prior neck XRT, tandem lesion larger than target lesion, other conditions that could cause symptoms (atrial fibrillation, prior stroke with persistent major deficit, valvular heart disease), major organ failure, uncontrolled hypertension or diabetes mellitus, and significant coronary artery disease^{55,56}.
The SAPHIRE Trial (Stenting and Angioplasty with Protection in Patients at High-Risk for Endarterectomy): patients with clinically significant cardiac disease (CHF, abnormal stress test, or need for open-heart surgery), severe pulmonary disease, contralateral carotid occlusion, contralateral laryngeal-nerve palsy, previous radical neck surgery or neck XRT, recurrent stenosis after endarterectomy, and age >80 years.⁵⁷ The ARChER Trial (ACCULINK for Revascularization of Carotids in High-Risk patients) also included patients with tracheostomy, spinal immobility, and dialysis-dependant renal failure.⁵⁸

84.4.4 Carotid endarterectomy – surgical considerations

Perioperative management

- Pre-op management (carotid endarterectomy)
1. ASA 325 mg TID for at least 2 days, preferably 5 days pre-op⁵⁹ (NB: patients should be kept on their ASA for surgery, and if not on ASA they should be started, in order to reduce risks of MI and TIA⁶⁰)

Post-op management (carotid endarterectomy)

1. patient monitored in ICU with A-line
2. keep patient well hydrated (run IVF ≥ 100 cc/hr for most adults)
3. SBP ideally 110 – 150 mm Hg (higher pressures are permitted in patients with chronic severe HTN)
 - a) BP frequently labile in 1st 24 hrs post-op, may be due to “new” pressure in carotid bulb; to prevent rebound hyper- or hypo-tension, avoid long acting agents
 - b) hypotension
 - check EKG – R/O cardiogenic shock
 - if mild, start with fluids (crystalloid or colloid)
 - phenylephrine (Neo-Synephrine®) for resistant hypotension
 - c) hypertension: nicardipine (Cardene®) (p. 126) is the agent of choice. Avoid rebound hypotension
4. avoid antiplatelet drugs for 24–48 hrs post-op (causes oozing); may start these 24–72 hrs post-op (note: ASA 325 mg + dipyridamole 75 mg TID have been shown not to reduce the rate of restenosis after endarterectomy⁶¹)
5. optional: reverse half of heparin with protamine 10 minutes after closing arteriotomy

Post-op check (carotid endarterectomy)

In addition to routine, the following should be checked:

1. change in neurologic status due to cerebral dysfunction, including:
 - a) pronator drift (R/O new hemiparesis)
 - b) signs of dysphasia (especially for left sided surgery)
 - c) mimetic muscle symmetry (assesses facial nerve function)
2. pupil diameter and reaction (R/O stroke, Horner’s syndrome)
3. severe H/A (especially unilateral) > may indicate hyperperfusion syndrome
4. STA pulses (R/O external carotid occlusion)
5. tongue deviation (R/O hypoglossal nerve injury)
6. symmetry of lips (R/O weakness of lower lip depressors due to retraction of marginal mandibular branch of facial nerve against mandible, usually resolves in 6–12 wks, must differentiate from central VII palsy due to stroke)
7. check for hoarseness (R/O recurrent laryngeal nerve injury)
8. assess for hematoma in operative site: note any tracheal deviation, dysphagia

Post-op complications (carotid endarterectomy)

To justify CEA, the absolute upper limit of (significant) complication rate should be $\leq 3\%$

1. overall in-hospital mortality: 1%⁶²
2. disruption of arteriotomy closure: rare, but emergent (see below)
 - a) evidenced by:
 - swelling of neck: rupture may produce a pseudoaneurysm
 - tracheal deviation (visible, palpable, or on CXR)
 - symptoms: dysphagia, air hunger or worsening hoarseness, difficulty swallowing
 - b) dangers:
 - asphyxiation: most immediate danger
 - stroke
 - exsanguination (unlikely, unless skin closure is also disrupted)
 - c) late (often delayed weeks to months): false aneurysm.⁶³ Risk = 0.33% Presents as neck mass. Risk is increased with wound infection and possibly with patch graft as compared to endarterectomy alone^{63,64,65}
3. stroke (cerebral infarction) intra-op or post-op rate⁶⁶: 5%
 - a) embolic (the most common cause of minor post-op neurologic deficit): source may be denuded media of endarterectomy
 - b) intracerebral hemorrhagic (ICH) (breakthrough bleeding): occurs in $<0.6\%$ ⁶⁷ Related to cerebral hyperperfusion in most^{68,69} (see below). Usually occurs within first 2 weeks, often in basal ganglion 3–4 days post-op with hypertensive episode. Patients at greatest risk are those with severe stenosis and limited hemispheric collateral flow
 - c) post-op ICA occlusion
 - most common cause of major post-op stroke, but may be asymptomatic
 - risk is reduced by attention to technical details at surgery⁷⁰ (p 249)
 - some may be due to hypercoagulable state induced by heparin (predictable in patients whose platelet count drops while on heparin. No known therapy for this condition⁷⁰ (p 249–50))

- the endarterectomized surface is highly thrombogenic for 4 hrs following endarterectomy (Sundt recommends not reversing heparin)
 - in Sundt's series using patch graft⁷⁰ (p 229): 0.8% incidence, associated with major stroke in 33% and minor stroke in 20%
 - occlusion rate with primary closure: 4% in Sundt's experience, 2–5% in literature⁷⁰ (p 249)
4. post-op TIAs: most due to ICA occlusion. Some may be due to microemboli. Hyperperfusion syndrome produces a 1% incidence of post-op TIAs⁷⁰ (p 229)
 5. seizures⁷¹: usually focal in onset with possible generalization, most occur late (post-op day 5–13) with an incidence of $\approx 0.4\%$ ⁷ to 1% ⁷² May be due to cerebral hyperperfusion,⁶⁷ emboli,⁷³ and/or intracerebral hemorrhage. Usually difficult to control initially, lorazepam and phenytoin are recommended (p.205)
 6. late restenosis: identifiable restenosis occurs in $\approx 25\%$ by 1 yr, and half of these reduce luminal diameters by $>50\%$ ⁷⁴ Restenosis within 2 yrs is usually due to fibrous hyperplasia, after 2 yrs it is typically due to atherosclerosis⁷⁵
 7. cerebral hyperperfusion syndrome (AKA normal pressure hyperperfusion breakthrough): classically thought to result from return of blood flow to an area that has lost autoregulation due to chronic cerebral ischemia typically from high-grade stenosis. Controversial.⁶⁹ Usually presents as ipsilateral vascular H/A or eye pain that subsides within several days⁷⁶ or with seizures ($\pm$ PLEDs on EEG, more common with Halothane®, due to petechial hemorrhages⁶⁷). May cause ICH.⁷⁷ Most complications occur several days post-op
 8. hoarseness: the most common cause is laryngeal edema and not superior nor recurrent laryngeal nerve injury
 9. cranial nerve injury: the most common complication after CEA with an incidence of up to 8–10%⁷⁸
 - a) hypoglossal nerve $\rightarrow$ tongue deviation towards the side of injury: incidence $\approx 1\%$ (with mobilizing XII to allow displacement). Unilateral injury may cause speaking, chewing and swallowing difficulties. Bilateral injuries can cause upper airway obstruction.⁷⁹ The presence of a unilateral palsy is a contraindication to doing contralateral endarterectomy until the first side recovers. May last as long as four months
 - b) vagus or recurrent laryngeal nerve $\rightarrow$ unilateral vocal cord paralysis: 1% risk
 - c) mandibular branch of facial nerve $\rightarrow$ loss of ipsilateral lip depressor
 10. headache⁶⁷
 11. hypertension^{80,81}: may develop 5–7 days post-op. Longstanding HTN may occur as a result of the loss of the carotid sinus baroreceptor reflex

Complication management

1. post-op TIAs
 - a) if TIA occurs in recovery room, emergency CT (to R/O hemorrhage) and then angiogram recommended to assess for ICA or CCA occlusion (vs. emboli)
 - b) if TIA occurs later, consider emergent OPG; if abnormal $\rightarrow$ emergent surgery (if neurologically intact, pre-op angiogram is appropriate)⁷⁰
2. fixed post-op deficit in distribution of endarterectomized carotid
 - a) if deficit occurs immediately post-op (i.e. in PACU), recommend immediate re-exploration without delay for CT or angiogram⁸² (case reports of no deficit when flow re-established in ≤ 45 mins). For later onset, workup is indicated. Technical considerations for emergency re-operation⁷⁰ (p 255):
 - isolate the 3 arteries (CCA, ECA, & ICA)
 - occlude CCA 1st, then ECA, and ICA last (to minimize emboli)
 - open arteriotomy, check backflow; if none, pass a No. 4 Fogarty catheter into ICA, gently inflate and withdraw (avoid intimal tears)
 - if good backflow established, close with patch graft
 - remove tortuous vessel loops and kinks before closing
 - b) immediate management (unless ICH or SDH are likely) includes
 - fluids (e.g. Plasmanate®) to improve rheology and to elevate BP
 - pressors (e.g. phenylephrine) to elevate SBP to ≈ 180 mm Hg
 - oxygen
 - heparinization (may be controversial)
 - c) theoretical benefits of radiographic evaluation include:
 - CT: identifies ICH or SDH that might require treatment other than re-exploration of the surgical site, elevating BP, etc.

- angiogram: identifies whether ICA is occluded, or if deficit is from another cause (e.g. emboli from endarterectomy site) that would not benefit from re-exploration or possibly endovascular treatment
- 3. disruption of arteriotomy closure, management
 - a) OPEN WOUND – if there is any stridor, it is critical to do this before trying to intubate (although ideally performed in O.R., the delay may be decisive). Evacuate clot (start with a sterile gloved finger) and stop bleeding, preferably without traumatizing the artery, a DeBakey clamp is optimal
 - b) INTUBATION – high priority, may be difficult or impossible if trachea is deviated (open wound immediately). Preferably done by anesthesiologist in controlled setting (i.e. O.R.) unless there is acute airway obstruction
 - c) call O.R. and have them prepare set-up for endarterectomy, and take patient to O.R.

84.4.5 Operative technique

Anesthesia and monitoring

Most (but not all) surgeons monitor some parameter of neurologic function during carotid endarterectomy, and will alter technique (e.g. insert a vascular shunt) if there is evidence of hemodynamic intolerance of carotid clamping (only occurs in $\approx 1\text{--}4\%$).

1. local/regional anesthesia: permits “clinical” monitoring of patient’s neurologic function.^{83,84} Disadvantages: patient movement during procedure (often exacerbated by sedation and alterations in CBF), lack of cerebral protection from anesthetic and adjunctive agents. The only prospective randomized study found no difference between local and general anesthesia.⁸⁵ The multicenter, randomized controlled General Anesthesia versus Local Anesthesia (GALA) Trial⁸⁶ found no significant differences in the prevention of stroke, MI, or death for either anesthetic technique. Subgroup analysis showed trends (not statistically significant) favoring local anesthesia for perioperative death, event-free survival at 1 year, and patients with contralateral occlusion. Local anesthesia was associated with a significant reduction of shunt insertion.⁸⁶ A Cochrane Database Review found no evidence from randomized trials to favor either anesthetic technique⁸⁷
2. general anesthesia, possibly including barbiturates (thiopental boluses of 125–250 mg until 15–30 second burst suppression on EEG, followed by small bolus injections or constant infusion to maintain burst suppression⁵⁹)
 - a) EEG monitoring
 - b) SSEP monitoring
 - c) measurement of distal stump pressure after CCA occlusion (unreliable), e.g. using a shunt if stump pressure < 25 mm Hg
 - d) transcranial Doppler
 - e) near-infrared spectroscopy

Position and incision

1. supine, neck slightly extended and rotated slightly ($\approx 30^\circ$) away from the operative side
2. the incision curves gently and follows the anterior border of the sternocleidomastoid muscle, and curves posteriorly at the rostral end
3. keep the horizontal portion of the incision ≈ 1 cm away from the mandible to avoid injury to marginal mandibular branch of facial nerve (which lies in the inferior parotid gland and supplies lip depressor) due to retraction against mandible
4. retractors should not be placed deeper than the platysma to avoid injury to recurrent laryngeal nerve which runs between the esophagus and trachea. Blunt retractors are used to avoid internal jugular vein injury

Dissection

1. the common facial vein (CFV) usually crosses the field over the carotid bifurcation, it is doubly ligated and divided. It leads to the internal jugular vein (IJV)
2. identifying the IJV is key, dissection is carried down between the carotid artery and the IJV
3. the ansa hypoglossi runs superficial to the ICA and serves as a useful guide to the hypoglossal nerve (XII) which should be identified since it is at greater risk when it is not seen. XII can arise anywhere from the carotid bifurcation to the angle of the mandible, although it is usually in the vicinity of the CFV. Mobilization can be facilitated by dividing the small artery (sternocleidomastoid branch of the ECA) and vein that cross over it⁷⁹

4. the ansa hypoglossi can usually be spared, and if mobilized, allows medial retraction of the hypoglossal nerve out of harm's way. If it is necessary to divide the ansa it is done close to the hypoglossal nerve to be certain it is not a branch of the vagus and to minimize neurologic deficit (the ansa has an anterior cervical limb from the cervical plexus)
5. the superior thyroid artery is the first branch of the ECA, and helps differentiate ECA from the ICA (the ICA is located posterior to the ECA)
6. the carotid bulb may be anesthetized with $\approx 2-3$ ml of 1% plain lidocaine using a 27 Ga needle. This may be done routinely, or, as some prefer only if hypotension and/or bradycardia occur during dissection (indicating IX nerve stimulation)
7. the ICA must be exposed beyond the extent of the plaque which can be determined by gentle palpation with a moistened finger and by visualization as the area where the artery turns from yellowish to its normal pinker color

Occlusion and arteriotomy

1. a vessel loop is placed around the ECA at least 2 cm above the bifurcation
2. a vessel loop is also placed around the ICA but is looped only once
3. umbilical tape with a choke is placed around the CCA 2–3 cm below the bifurcation
4. IV heparin (usually 5,000 IU) is given 1 minute prior to cross clamping
5. a temporary aneurysm clip is placed on the superior thyroid artery
6. the order of occlusion of the vessels is as follows (mnemonic: "ICE"):
 - a) ICA (e.g. with temporary aneurysm clip)
 - b) CCA (e.g. with a small DeBakey clamp)
 - c) ECA (e.g. with temporary aneurysm clip)
7. during ICA clamping, mild hypertension is maintained by the anesthesiologist
8. shunt: some surgeons use some form of monitoring (EEG, BSAER, etc.) to determine if a shunt is needed – see Anesthesia and monitoring (p. 1294) -, yet others routinely use a shunt whenever possible without assessing the need
9. the arteriotomy is begun in the CCA with a #11 scalpel, and once the lumen is entered, a Potts' scissors carries the incision through to the ICA beyond the plaque. Stay in the midline to facilitate arteriotomy closure

Plaque removal

1. the plaque usually cannot be completely removed from the CCA, and thus it is usually transected with a Potts' scissors taking care not to inadvertently incise the artery wall and to leave as smooth an edge as possible
2. in the ICA, great care must be made to avoid leaving an intimal flap which could become a nidus for an arterial dissection. If necessary the intima may be tacked down by suturing from the lumen out on both ends (using double armed suture) and tying the knot outside the vessel

Arteriotomy closure and vessel release

1. arteriotomy may be performed with a running Prolene suture using either
 - a) primary closure
 - b) or with a patch graft to increase the caliber of the vessel and reduce the risk of re-stenosis
 - c) limited evidence suggests that carotid patch angioplasty may reduce the risk of perioperative arterial occlusion and re-stenosis. Synthetic patches (Dacron, PTFE) are preferred to autologous vein (risk of aneurysmal dilatation, thrombogenic surface)^{88,89}
2. the order of releasing the vessels (reverse that of the clamping order):
 - a) ECA
 - b) CCA (allows air and debris to be washed into the ECA)
 - c) ICA

84.4.6 Emergency carotid endarterectomy

General information

Emergency CEA indications include crescendo TIAs and stroke in evolution. The treatment paradigm of these conditions has shifted towards the use of interventional methods, such as thrombolysis and stenting, although there are no randomized controlled trial data to support that approach. A recent meta-analysis of emergent CEA has shown that the pooled stroke and stroke/death rates after CEA

for crescendo TIA in 176 patients were 6.5%and 9.0% respectively. For those with stroke in evolution, the overall stroke and stroke/death rates in 114 patients were 16.9%and 20.0% respectively.⁹⁰

After retrospective analysis of 64 emergency endarterectomies⁹¹ the guidelines given below were suggested. However, the efficacy of immediate surgical removal of obstruction is controversial and unproven. In one early study, over 50%of patients suffered fatal intracranial hemorrhage within 72 hours ofemergency carotid endarterectomy.

Initial management of patient presenting with acute neuro deficit

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1. obtain history directed at determining presence of previous stroke and other serious medical illness, and to try to differentiate from seizure
2. baseline neurological assessment including evaluation of STA pulses and carotid bruits
3. during evaluation: close control of BP. O₂ per NC. Labs +EKG; see Management of TIA or stroke (p.1282). Consider hemodilution with LMD
4. CT to R/O ICH or infarction (early stroke will not be visible)
5. when carotid disease is suspected, and CT is negative for ICH or acute infarct, emergency angiography, MRI/MRA or CTA is performed

Indications for emergency carotid endarterectomy

General information

In patients with acute neurological deficits, the need for rapid decision making often does not allow differentiating between TIA, stroke in evolution and acute stroke, nor in assessing the stability or fluctuating nature of the deficit.

Indications

1. stroke in evolution
2. crescendo TIAs: TIAs that abruptly increase in frequency to ≥several per day
3. following intra-arterial thrombolysis, emergent/urgent CEA is indicated for residual critical carotid stenosis^{54,92}

Contraindications

See also more details (p.1286). Patients with depressed levels of consciousness or acute fixed deficits.

Surgical management

Again, most cases would now be managed initially with endovascular thrombolysis and stenting. Surgery would considered if this is not an option.

1. for emergency surgery, it is essential that blood pressure be stable
2. in patients with complete occlusion, ICA is not occluded intra-op (to avoid breaking up thrombus, if present)
3. if thrombus present
 - a) attempt spontaneous extrusion using back pressure
 - b) if this fails, attempt to remove with smoothened suction catheter
 - c) if this fails, pass balloon embolectomy catheter as far as base of skull (caution: avoid injury to distal ICA that could cause CCF)
 - d) obtain intra-op angiogram unless thrombus emerges and backflow is excellent
 - e) plicate ICA (avoid creating a blind pouch at origin) if there is good backflow or if satisfactory angiography cannot be obtained

Surgical results

Highest correlation was with presenting neurologic status (□ Table 84.4).

Table 84.4 Surgical results		
Presenting deficit	Same or improved	Deaths
intact or mild	92%	0
moderate	80%	1 (7%)
severe	77%	3 (13%)

84.5 Carotid angioplasty/stenting

84.5.1 General information

Σ

There are no well-designed studies that convincingly show superiority of angioplasty/stenting over CEA in average risk symptomatic patients, and the recommendation in these patients is to continue with the time-tested technique of CEA.

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There is a paucity of randomized control trials^{57,58,93,94,95,96} comparing carotid angioplasty/stenting with CEA, and many nonrandomized registries.^{58,97,98,99,100,101,102,103,104,105}

However, data from multicenter randomized trials showing that carotid angioplasty/stenting is as safe over the short term or as efficacious over the long term as CEA in average-risk symptomatic patients are lacking. Published trials are heterogeneous (clinically and methodologically), too small to provide robust and convincing data, and limited in long-term follow-up. Only the SAPHIRE study⁵⁷ comparing CEA with stenting (using a distal embolic protection device) for moderate to severe carotid stenosis with comorbidities that might increase the risk of CEA (high-risk patients), found that angioplasty/stenting was not inferior (risk within 3% $P=0.004$) to CEA (based on a composite primary end point of stroke, death, or MI within 30 days, or death from neurologic causes or ipsilateral stroke between 31 days and 1 year).⁵⁷ However, the study methodology has been criticized.^{106,107,108}

A 2007 Cochrane review concluded that available data on carotid angioplasty/stenting are difficult to interpret and does not support a change in clinical practice away from recommending CEA as the treatment of choice for suitable carotid artery stenosis.¹⁰⁹

84.5.2 Indications for angioplasty/stenting

Carotid stenting performed with adequate procedural quality levels, should be considered instead of CEA in the presence of¹¹⁰:

1. severe vascular and cardiac comorbidities:
 - a) congestive heart failure (New York Heart Association class III/IV) and/or known severe left ventricular dysfunction
 - b) open heart surgery needed within 6 weeks
 - c) recent myocardial infarction (<24 hours and >4 weeks)
 - d) unstable angina (Canadian Cardiovascular Society class III/IV)
 - e) contralateral carotid occlusion
2. specific conditions:
 - a) contralateral laryngeal nerve palsy
 - b) radiation therapy to the neck
 - c) previous CEA with recurrent restenosis
 - d) high cervical internal carotid/below the clavicle common carotid lesions
 - e) severe tandem lesions
 - f) age >80 years
 - g) severe pulmonary disease

The 2009 European Society for Vascular Surgery (ESVS) Guidelines state that carotid angioplasty/stenting is indicated in cases of: contralateral laryngeal nerve palsy, previous radical neck dissection or cervical XRT, prior CEA (re-stenosis), high bifurcation or intracranial extension of a carotid lesion, provided that the peri-interventional stroke or death rate is not higher than that accepted for CEA (Class C recommendation).⁸⁸

AHA Guidelines state that angioplasty/stenting might be a reasonable alternative to CEA in asymptomatic high risk patients. However, they stress that it remains uncertain whether this group of patients should have either procedure.¹¹¹

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85 Special Conditions

85.1 Totally occluded internal carotid artery

85.1.1 General information

10–15% of patients presenting with carotid territory stroke or transient ischemic attacks (TIA) are found to have carotid occlusion. This amounts to an estimated 61,000 first ever strokes and 19,000 TIAs per year in the United States. Prevention of subsequent stroke in symptomatic patients with carotid artery occlusion remains a difficult challenge. The overall rate of subsequent stroke is 7% per year for all stroke and 5.9% per year for ischemic stroke ipsilateral to the occluded carotid artery.¹ These risks persist even despite treatment with antiaggregants and anticoagulants.² The prevalence of asymptomatic carotid occlusion is not known, and the incidence of ipsilateral stroke in never-symptomatic carotid occlusion is negligible.³

85

85.1.2 Presentation

3 patterns of stroke with acute carotid artery occlusion:

1. stump emboli: produces cortical infarcts. Emboli usually go up the external carotid (higher flow, and reverse flow that may occur through ICA initially prevents emboli from ICA). Later, ICA emboli may occur.
2. whole hemisphere stroke
3. watershed infarct

In symptomatic patients⁴: hemiparetic TIA 53%, dysphasic TIA 34%, fixed neuro deficit 21%, crescendo TIAs 21%, amaurosis fugax 17%, acute hemiplegia 6%. One series had 27% asymptomatic.⁵ Patients may have the so-called “slow carotid stroke” of carotid occlusion which is a stuttering progressive stroke.

85.1.3 Natural history

See reference.⁶

Patients with mild deficit and angiographically proven ICA occlusion have a stroke rate (in two series) of 3 or 5% per year (2 or 3.3% related to occluded side). In patients with acute ICA occlusion and profound neurological deficit, 2–12% make good recovery, 40–69% will have profound deficit, and 16–55% will have died by the time of follow-up.

85.1.4 Endovascular thrombolysis and stenting for acute carotid occlusion

Case reports and series of endovascular treatment of internal carotid artery occlusion have confirmed the feasibility of this technique. Intra-arterial thrombolysis within 6 hours of stroke onset may increase recanalization rates to 37%–100% and clinical improvement to 53%–94% without significant increase in hemorrhagic transformation when compared with intravenous thrombolytic therapy alone.^{7,8,9,10,11,12} Although results appear promising, randomized controlled trials on cervical carotid thrombolysis and/or stenting are lacking.

85.1.5 Surgery

Options include: endarterectomy, Fogarty balloon catheter embolectomy (utilizing a No. 2 French catheter with 0.2 ml balloon gently passed 10–12 cm up ICA from small arteriotomy made distal to atheromatous plaque¹³), extracranial-intracranial bypass. Restored patency rate is inversely related to suspected duration of occlusion. Chronically occluded ICA has poor patency rate and little gain from re-opening.

Determining the exact time of occlusion is frequently impossible. One must often rely on clinical grounds, therefore an occasional chronic occlusion will be included.

Retrograde filling of ICA to petrous or cavernous segment from ECA (e.g. via ophthalmic) or from contralateral ICA is a good sign of operability.⁴

Surgical results⁴

32% (15/47 cases) immediate surgical failures (no or minimal back bleeding), at least 3 deaths. Among immediate successes no strokes and no TIAs. If operated <2 days reported patency rate 70–100% from 3–7 days 50–100% 8–14 days 27–58% 15–30 days 4–61% over 1 month (2 series) 20–50%

85.1.6 Guidelines

Emergency operations for acute neuro deficit associated with total occlusion should not be performed after about 2 hrs. Extremely poor neuro status (lethargy/coma) is a contraindication to surgery. Patients without persistent neuro deficit: operate ASAP. If the patient has recurrent TIAs (despite maximal medical therapy) following recent carotid occlusion, and no definite infarct on MRI, consider by-pass surgery.

85

85.2 Cerebellar infarction

85.2.1 General information

Relatively rare (seen on only 0.6% of all CTs obtained for any reason¹⁴). Cerebellar infarcts may be classified as involving the PICA distribution (cerebellar tonsil and/or inferior vermis), superior cerebellar artery distribution (superior hemisphere or superior vermis), or other indeterminate patterns.¹⁵ 80% of patients developing signs of brainstem compression will die, usually within hours to days.

85.2.2 Early clinical findings

In most cases the onset is sudden, without premonitory symptoms.¹⁶ The first 12 hrs after onset were characterized by lack of progression. Early findings are due to the intrinsic cerebellar lesion (ischemic infarction or hemorrhage):

1. symptoms
 - a) dizziness or vertigo
 - b) nausea/vomiting
 - c) loss of balance, often with a fall and inability to get up
 - d) headache (infrequent in one series¹⁶)
2. signs
 - a) truncal and appendicular ataxia
 - b) nystagmus
 - c) dysarthria

85.2.3 Later clinical findings

Patients with cerebellar infarction may subsequently develop increased pressure within the posterior fossa (due to cerebellar edema or mass effect from clot), with brainstem compression (particularly posterior pons). Clinical findings generally increase between 12–96 hrs following onset. Compression the sylvian aqueduct can cause acute hydrocephalus with attendant increased ICP.

85.2.4 Imaging studies

CT scan: may be normal very early in these patients. There may be subtle findings of a tight posterior fossa: compression or obliteration of basal cisterns or 4th ventricle or hydrocephalus.

MRI: (including DWI) more sensitive for ischemia, especially in the posterior fossa.

85.2.5 Surgical indications

Surgical decompression (see below) should probably be done as soon as any of the following signs develop if there is no response to medical therapy.¹⁷ It is important to recognize a lateral medullary syndrome (LMS) (p. 1267) which may often accompany a cerebellar infarct. With LMS, the signs are usually present from the onset (dysphagia, dysarthria, Horner's syndrome, ipsilateral facial numbness, crossed sensory loss...), and are not accompanied by a change in sensorium. There is no place for surgical decompression in LMS since it represents primary brainstem ischemia and not compression.

Findings proceed in the approximate following sequence if there is no intervention:

1. abducens (VI) nerve palsy
2. loss of ipsilateral gaze (compression of VI nucleus and lateral gaze center)
3. peripheral facial nerve paresis (compression of facial colliculus)
4. confusion and somnolence (may be partly due to developing hydrocephalus)
5. Babinski sign
6. hemiparesis
7. lethargy
8. small but reactive pupils
9. coma
10. posturing → flaccidity
11. ataxic respirations

85.2.6 Suboccipital craniectomy for cerebellar infarction

Unlike the situation with supratentorial masses causing herniation, there are several reports of patients in deep coma from direct brainstem compression who were operated upon quickly who made useful recovery.^{17,18,19} See also Guidelines for patients with cerebellar hemorrhage (p.1344).

The operation of choice is a suboccipital decompression to include enlargement of the foramen magnum. The dura is then opened and the infarcted cerebellar tissue usually exudes “like tooth-paste” and is easily aspirated. Avoid using ventricular drainage alone as this may cause upward cerebellar herniation (p.303) and does not relieve the direct brainstem compression.

85.3 Malignant middle cerebral artery territory infarction

85.3.1 General information

A distinct syndrome that occurs in up to 10% of stroke patients,^{20,21} which carries a mortality of up to 80% (mostly due to severe postischemic cerebral edema → increased ICP → herniation).²¹

Patients usually present with findings of severe hemispheric stroke (hemiplegia, forced eye and head deviation) often with CT findings of major infarct within the first 12 hours. Most develop drowsiness shortly after admission. There is progressive deterioration during the first 2 days, and subsequent transtentorial herniation usually within 2–4 days of stroke. Fatalities are often associated with: severe drowsiness, dense hemiplegia, age >45–50 yrs,²² early parenchymal hypodensity involving >50% of the MCA distribution on CT scan,²³ midline shift >8–10 mm, early sulci effacement and hyperdense artery sign (p.1281)²² in MCA.

Neurosurgeons may become involved in caring for these patients because aggressive therapies in these patients may reduce morbidity and mortality. Options include:

1. conventional measures to control ICP (with or without ICP monitor): mortality is still high in this group, and elevated ICP is not a common cause of initial neurologic deterioration in large hemispheric stroke
2. hemicraniectomy (decompressive craniectomy): see below
3. □ to date, the following treatments have not improved outcome: agents to lyse clot, hyperventilation, mannitol, or barbiturate coma

85.3.2 Hemicraniectomy for malignant MCA territory infarction

May reduce mortality to as low as 32% in nondominant hemisphere strokes²⁴ (37% in all comers²⁵) with surprising reduction of hemiplegia, and in dominant hemisphere strokes, with only mild-moderate aphasia (better results occur with early surgery, especially if surgery is performed before any changes associated with herniation occur). Meta-analysis²⁶ of 3 randomized controlled trials found that hemicraniectomy within 48 hours after stroke onset resulted in decreased mortality and increased the number of patients with a favorable functional outcome.

Indications: No firm indications. Guidelines:

1. age <70 years
2. more strongly considered in nondominant hemisphere (usually right)
3. clinical & CT evidence of acute, complete ICA or MCA infarcts and direct signs of impending or complete severe hemispheric brain swelling (severe post-admission neurologic deterioration is the usual event that triggers surgical intervention)

See also Technique (p.1467).

85.4 Cardiogenic brain embolism

85.4.1 General information

About one stroke in six is cardioembolic. Emboli may be composed of fibrin-rich thrombi (e.g. mural thrombi due to segmental myocardial hypokinesis following MI or ventricular aneurysm), platelets (e.g. nonbacterial thrombotic endocarditis), calcified material (e.g. in aortic stenosis), or tumor particles (e.g. atrial myxoma).

85.4.2 Following acute myocardial infarction (AMI)

2.5% of patients will have a stroke within 1–2 weeks of an AMI (the period when most emboli occur). The risk is higher with anterior wall MI ($\approx 6\%$) vs. inferior wall MI ($\approx 1\%$).

85.4.3 Atrial fibrillation (A-fib)

Nonrheumatic patients with a-fib have a 3–5 fold increased risk of stroke,²⁷ with a 4.5% rate of stroke per year without treatment.²⁸ The incidence of a-fib in the U.S. is 2.2 million. About 75% of strokes in patients with A-fib are due to left atrial thrombi.²⁹ Independent risk factors for stroke in patients with A-fib are: advanced age, prior embolism (stroke or TIA), HTN, DM, and echocardiographic evidence of left atrial enlargement or left ventricular dysfunction.²⁷

The CHADS2 scoring system for patients with a-fib has been widely validated³⁰ and is shown in □ Table 85.1. The points are totaled and risk assessment is shown in □ Table 85.2. For patients with a CHADS2 score ≥ 2 , warfarin therapy was significantly protective for out-of-hospital death or hospitalization for stroke, MI or hemorrhage (CI=0.61–0.91).³¹

85.4.4 Prosthetic heart valves

Patients with mechanical prosthetic heart valves on long-term anticoagulation have an embolism rate of 3%/year for mitral and 1.5%/year for aortic valves. With bioprosthetic heart valves and no anticoagulation the risk is 2–4%/year.

Table 85.1 CHADS2 scoring items	
Item	Points
CHF (any history)	1
HTN (prior history)	1
Age >75 yrs	1
Diabetes mellitus	1
Secondary prevention: in patients with prior ischemic stroke or TIA; most also include systemic embolic events	2

Table 85.2 Risk based on CHADS2 score	
CHADS 2 score	Annual stroke risk (%/year)
0	1.9
1	2.8
2	4
3	5.9
4	8.5
5	12.5
6	18.2

85.4.5 Paradoxical embolism

Paradoxical embolism can occur with a patent foramen ovale which is present in 10–18% of the general population, but in up to 56% of young adults with unexplained stroke.³²

85.4.6 Endocarditis

Diagnosis

Blood cultures and TEE help evaluate.

No specific neurologic features can distinguish these patients. The diagnosis is suggested in imaging studies showing multiple intracranial ischemic strokes in different arterial distributions, the differential diagnosis includes: vasculitis, intracranial atherosclerosis (focal plaques, more common in Asian populations that consume Western diets), and intravascular lymphomatosis.

The diagnosis of cardiogenic brain embolism (CBE) as a cause of a stroke relies on demonstrating a potential cardiac source, the absence of cerebrovascular disease, and non-lacunar stroke.

Large areas of hemorrhagic transformation within an ischemic infarct may be more indicative of CBE due to thrombolysis of the clot and reperfusion of infarcted brain with subsequent hemorrhagic conversion. Hemorrhagic transformation most often occurs within 48 hrs of a CBE stroke, and is more common with larger strokes.

Detection of cardiac source

Most centers rely on echocardiography (without transesophageal ability). Using restricted criteria (i.e. excluding mitral valve prolapse), about 10% of patients with ischemic stroke will have potential cardiac source detected by echo, and most of these patients have other manifestations of cardiac disease. In stroke patients without clinical heart disease, only 1.5% will have a positive echo; the yield is higher in younger patients without cerebrovascular disease.³³

EKG may detect atrial fibrillation which may be seen in 6–24% of ischemic strokes, and may be associated with a 5-fold increased risk of stroke (see below).

Treatment

CBE is essentially the only condition for which anticoagulation has been shown to significantly reduce the rate of further strokes.

One must balance the risk of recurrent emboli (12% of patients with a cardioembolic stroke will have a second embolic stroke within 2 weeks) against that of converting a pale infarct into a hemorrhagic one. No study has shown a clear benefit of early anticoagulation.

Recommendations for anticoagulation:

1. if anticoagulation is to be used, it should not be instituted within the first 48 hrs of a probable CBE stroke
2. CT should be obtained after 48 hrs following a CBE stroke and before starting anticoagulation (to R/O hemorrhage)
3. anticoagulation should not be used in the face of large infarcts
4. start heparin and warfarin simultaneously. Continue heparin for 3 days into warfarin therapy, see Anticoagulation (p. 156)
5. optimal range of oral anticoagulation to minimize subsequent embolism and/or hemorrhage has not been determined, but pending further data, an INR of 2–3 appears satisfactory
6. patients with asymptomatic A-fib have 66–86% reduction in stroke risk with warfarin (Coumadin®).^{27,34} ASA is only about half as effective, but may be sufficient for those without associated risk factors (p. 1304)²⁷

85.5 Vertebrobasilar insufficiency

85.5.1 General information

Signs and symptoms resulting from inadequate blood flow through the posterior cerebral circulation (vertebral arteries, basilar artery and their branches).

85.5.2 Symptoms

□ Table 85.3 shows a mnemonic of the symptoms of vertebrobasilar insufficiency (VBI). Predicting the site of the lesion based only on clinical evaluation is very unreliable.

Diagnostic criteria for VBI are shown in □ Table 85.3.

VBI may also be suspected in a patient with transient episodes of “dizziness” (vertigo that is otherwise unexplained, e.g. absence of orthostatic hypotension or benign positional vertigo) that is initiated by positional changes. VBI may sometimes be due to compression of the VA at the C1-C2 level with:

- 1. head turning (below)
- 2. os odontoideum (p.981)
- 3. anterior atlantoaxial subluxation: e.g. in rheumatoid arthritis (p.1134)
- 4. with rotatory atlantoaxial subluxation (p.968)

85.5.3 Pathophysiology

Atheromatous and stenotic lesions of the posterior circulation occur most frequently at the VA origin.

VBI symptoms may be due to:

- 1. hemodynamic insufficiency (may be the most common etiology), including:
 - a) subclavian steal: reversed flow in VA due to proximal stenosis of subclavian artery
 - b) stenosis of both VAs or of one VA where the other is hypofunctional (e.g. hypoplastic, occluded, or terminates in PICA) causing reduced distal flow in face of inadequate collaterals (below)
- 2. embolism from ulcerations
- 3. atherosclerotic occlusion of brainstem perforators
- 4. Vertebrobasilar hypoplasia: reported as a possible etiology for cerebellar stroke.

85.5.4 Natural history

No clinical study accurately defines the natural history. The estimated stroke rate is 22–35% over 5 years, or 4.5–7% per year³⁵ (one study estimating 35% stroke rate in 5 years did not use angiography).

Risk of stroke after first VBI-TIA has been estimated as 22% for first year.³⁶

85.5.5 Evaluation

Adequate investigation usually requires selective four-vessel angiography,³⁷ sometimes with provocative maneuvers (see e.g. Bow Hunter’s Syndrome below). CTA may also be useful.

85.5.6 Treatment

Anticoagulation is the mainstay of medical management. Alternatives include anti-platelet drugs such as ASA (efficacy of either remains unproven^{35,37}).

Table 85.3 Mnemonic: “The 5 D’s of VBI”
• “drop attack” • diplopia • dysarthria • defect (visual field) • dizziness

Table 85.4 Criteria for clinical diagnosis of VBI
Clinical diagnosis requires 2 or more of the following • motor or sensory symptoms or both, occurring bilaterally in the same event • diplopia: ischemia of upper brainstem (midbrain) near ocular nuclei • dysarthria: ischemia of lower brainstem • homonymous hemianopsia: ischemia of occipital cortex (NB: this is binocular, in contrast to amaurosis fugax which is monocular)

Surgical treatment includes:

1. vertebral endarterectomy
2. transposition of VA to ICA (with or without carotid endarterectomy, with or without saphenous vein patch graft) or to thyrocervical trunk or to subclavian artery³⁸
3. bypass grafting (e.g. occipital artery to PICA)
4. C1–2 posterior arthrodesis (p. 1479) may prevent potentially life-threatening stroke in cases of os odontoideum (p. 982)

85.6 Bow hunter's stroke

85.6.1 General information

A special subset of VBI. The term was coined in 1978 by Sorensen.³⁹ Bow hunter's stroke (BHS): hemodynamic VBI induced by intermittent VA occlusion resulting from head rotation⁴⁰ (ischemic sequelae range from TIA (bow hunter's sign) to completed stroke). May occur with forced (e.g. with chiropractic neck manipulation⁴¹) or voluntary⁴² head rotation.

Occlusion usually involves the VA contralateral to the direction of rotation, and usually occurs at the C1–C2 junction (due to the immobility of the VA at this location).⁴³ However, other sites have also been reported.^{44,45}

VA occlusion does not produce symptoms in most individuals due to collateral flow through the contralateral VA and/or the circle of Willis. Symptomatic occlusion usually involves the dominant VA,⁴⁶ however, may also occur with non-dominant VA⁴²). Most cases of BHS occur in patients with an isolated posterior circulation (incompetent posterior communicating arteries).

BHS has also been postulated as one possible cause of SIDS.⁴⁷

85.6.2 Contributing factors

1. external VA compression⁴⁵
 - a) spondylotic bone spurs: particularly in the foramen transversarium⁴⁸
 - b) tumors
 - c) fibrous bands (e.g. proximal to entrance of VA into C6 foramen transversarium⁴⁴)
 - d) infectious processes
 - e) trauma
2. tethering of the VA
 - a) at the transverse foramina of C1 & C2
 - b) along the sulcus arteriosus proximal to where the VA enters the dura
3. defect in odontoid process⁴⁹
4. atherosclerotic vascular disease

85.6.3 Diagnosis

General information

BHS should be suspected in patient with symptoms of VBI precipitated by head movement. This may be very difficult to differentiate from vertigo and nausea due to vestibular dysfunction (which can also be induced by head movement) – (rotation of the body keeping the head motionless should not cause symptoms from vestibular symptoms and might help distinguish these conditions⁵⁰).

Dynamic cerebral angiography (DCA)

□ NB: significant consequences can be precipitated during DCA in patients with BHS.⁴³ The involved VA shows loss of flow as the head is rotated from the neutral position to the contralateral side. Carotid injections demonstrate patency of P-comms and the presence of any persistent fetal anastomoses.

CT angiogram (CTA)

Same precautions as with DCA (see above). Probably not the initial diagnostic study of choice. If the DCA is negative, CTA is not needed. If DCA is positive, CTA may be helpful to demonstrate the arterial relationship to the bony anatomy.

85.6.4 Treatment

- Options include:
- 1. anticoagulation⁵⁰
 - 2. cervical collar: to remind patient not to turn their head
 - 3. for VA compression at C1–2 (see □ Table 85.5 for a comparison):
 - a) C1–2 fusion (p.1481)
 - b) VA decompression: C1 “hemilaminectomy” via a posterior approach⁵¹
 - 4. for compression at other sites: elimination of the source of compression where possible (e.g. sectioning of offending fibrous band,⁴⁴ removal of osteophytic spurs⁴⁸...)

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Management recommendations: For compression at C1–2, it is suggested that VA decompression be performed as the initial treatment. This should be followed by DCA to verify maintenance of patency with head turning. Patients who fail clinically or on DCA should undergo C1–2 fusion.⁴³ Patients need to know pros and cons of each option.

85.7 Cerebrovascular venous thrombosis

85.7.1 General information

- There are 3 types of cerebrovascular venous thrombosis (CVVT) (any of which may produce venous infarctions):
- 1. dural sinus thrombosis (DST)
 - 2. cortical venous thrombosis
 - 3. deep venous thrombosis

85.7.2 Etiologies

Partial list of etiologies

- Many conditions have been incriminated with CVVT. Some common ones are listed here⁵³ (p 1301):
- 1. infection
 - a) usually local, e.g. otitis media^{54,55} (leading to the now obsolete term otitic hydrocephalus), sinusitis, peritonsillar abscess, paranasal sinusitis⁵⁶: in the pre-antibiotic era, CVVT was most commonly associated with chronic suppurative infection
 - b) meningitis
 - 2. pregnancy & puerperium: see below
 - 3. birth control pills (BCP) (oral contraceptives)⁵⁷
 - 4. dehydration and cachexia (marantic thrombosis): includes burns and cachexia of neoplastic disease
 - 5. cardiac disease (including CHF)
 - 6. ulcerative colitis (UC): 1% of UC patients have some thrombotic complication (not necessarily all intracranial), and this is the cause of ≈ 33% of deaths (usually pulmonary embolism, PE)
 - 7. periarteritis nodosa
 - 8. sickle cell trait
 - 9. trauma: including closed head injury (see below)
 - 10. iatrogenic: e.g. S/P radical neck surgery,⁵⁸ transvenous pacemaker placement, post-craniotomy
 - 11. malignancy: including myeloproliferative disorders
 - 12. hypercoagulable state (AKA thrombophilia)
 - a) protein C deficiency or resistance to activated protein C: hereditary factor V Leiden mutation may produce resistance to activated protein C.⁵⁹ Apparent protein C deficiency may be an artifact of dehydration in some cases
 - b) antithrombin III deficiency
 - c) protein S deficiency

Table 85.5 Comparison of surgical treatment for positional VA occlusion at C1–2		
Procedure	Advantages	Disadvantages
C1–2 fusion	high success rate in eliminating symptoms	loss of 50–70% of neck rotation with possible discomfort
VA decompression	no loss of motion	33% continue to have symptoms ⁵²

- d) antiphospholipid antibodies: associated with a variety of clinical syndromes including ischemic stroke, DVTs, thrombocytopenia, systemic lupus erythematosus SLE. The best known antibodies include
 - anticardiolipin antibodies
 - lupus anticoagulant
- e) paroxysmal nocturnal hemoglobinuria (PNH)
- f) plasminogen deficiency
- g) systemic lupus erythematosus⁶⁰
- h) factor VIII elevation⁶¹: may explain some cases of CVVT in pregnancy (see below)
- 13. diabetes mellitus: especially with ketoacidosis
- 14. homocystinuria (p.200)
- 15. Behçet's syndrome (p.200)⁶²
- 16. rarely associated with lumbar puncture, associated with hereditary activated protein C resistance due to the factor V R506Q mutation (FV Leiden) in one report⁶³

□ In the absence of factors such as BCP use, CVVT is highly suggestive of myeloproliferative disorder.

Pregnancy/puerperium

Highest risk is in first 2 wks post-partum. One series⁶⁴ found no case of CVVT occurred later than 16 days post-partum. Incidence $\approx$ 1/10,000 births. Etiology may be related to elevation of clotting factors (VII, X and especially factor VIII⁶⁵).

Trauma

A rare sequelae of closed head injury.⁶⁶ CVVT occurs in $\approx$ 10% of combat injuries involving the brain. May occur in absence of skull fracture. CVVT should be suspected in patients with fractures or missiles crossing sinus.

85.7.3 Relative frequency of involvement

The relative frequency of involvement of dural sinuses and other veins with thrombotic issues

1. sinuses
 - a) superior sagittal sinus (SSS) and left transverse sinus (TS) (70% each)
 - b) multiple sinuses in 71%
 - c) isolated inferior sagittal sinus: rare, first case report in 1997⁶⁷
 - d) straight sinus⁶⁸
2. superficial cortical veins
3. deep venous system (e.g. internal cerebral vein)
4. cavernous sinus^{69,70}: rare. Thrombophlebitis of the cavernous sinus may be caused by sphenoid sinusitis. MRI may show enlargement and abnormal enhancement of the cavernous sinus, increased signal of the petrous apex and clivus on T2WI, and narrowing of the cavernous portion of the ICA⁷⁰

85.7.4 Pathophysiology

Venous thrombosis reduces venous outflow from the brain and diminishes effective blood flow to the involved area. This venous engorgement causes white matter edema. The increased venous pressure may also lead to infarction and/or hemorrhage. These processes may all elevate ICP. Thus, clinical findings may be due to elevated ICP, and focal findings may be due to edema and/or hemorrhage. Cerebral infarction due to venous stasis is called venous infarction.

85.7.5 Clinical

Clinical presentations of DST are shown in □ Table 85.6. There are no pathognomonic findings. Many signs and symptoms are due to elevated ICP. May present as a syndrome clinically indistinguishable from idiopathic intracranial hypertension; see pseudotumor cerebri (p.768).

There is a high association of concurrent thromboembolic disease in other organs.

The anterior 1/3 of the SSS may occlude often without sequelae. Posterior to this, venous infarction is more likely to develop. Midportion SSS occlusion usually $\rightarrow$ increased muscle tone ranging from spastic hemi- or quadri-paresis to decerebration. Posterior SSS thrombosis $\rightarrow$ field cuts or cortical blindness, or massive stroke with cerebral edema and death. Occlusion of the TS may occur

Table 85.6 Presentation of dural sinus thrombosis		
Sign/symptom	Series A ^a	Series B ^a
H/A	100%	74%
N/V	75%	–
seizures	70%	29%
hemiparesis	70%	34%
papilledema	70%	45%
blurred vision	60%	–
altered consciousness	35%	26%
^a series A: 20 young females ⁶⁴ ; series B: 38 cases from France ⁷²		

without deficit unless the contralateral TS is hypoplastic, in which cases presentation is similar to posterior SSS occlusion.

SSS occlusion alone will not cause cranial nerve findings except perhaps for visual obscuration and abducens (VI) nerve palsy from elevated of ICP. Thrombosis in the jugular bulb may compress the nerves in the jugular foramen pars nervosa causing hoarseness, aphonia, difficulty swallowing and breathlessness; see Vernet’s syndrome (p.100).⁷¹

85.7.6 Diagnosis of DST

General information

Catheter angiography is better at demonstrating the presence of residual flow, and can identify areas of reversal of flow, and can sometimes demonstrate clot as a filling defect. However, CT (especially CTA) & MRI are better for identifying areas of clot. Angiography is often used as a complementary test⁷³ when the diagnosis is suggested by CT or MRI.

CTscan

Non-contrast CT

May be normal in 10–20%of cases of DST. Findings include:

- 1. hyperdense sinuses and veins (high density clots in cortical veins has been dubbed the cord sign which is pathognomonic for cerebral venous thrombosis; seen in only 2/30 patients)
- 2. petechial “flame” hemorrhages (intraparenchymal): seen in 20%(suspect sinus thrombosis with intracerebral hemorrhages in unusual locations for aneurysm or “hypertensive” hemorrhage)
- 3. small ventricles: seen in 50%
- 4. thrombosis of superior sagittal sinus may produce a triangular-shaped high density within the sinus posteriorly near the Torcular herophili on axial CT images (some refer to this as the delta sign, but this causes confusion with the “empty delta sign”, see below) (there is also confusion when an apparent “empty delta sign” is seen without contrast, this may occur when there is blood surrounding the SSS, e.g. following subarachnoid hemorrhage, this has been called a “false delta sign” or pseudodelta sign⁷⁴). Recommendation: avoid the confusion of the variations on “delta signs” and describe the findings
- 5. white matter edema
- 6. above changes occurring bilaterally

IVcontrast CT

Findings of DST include:

- 1. with contrast, the dura around the sinus may enhance and become denser than clot in 35%of cases.⁷⁵ Near the Torcular herophili this produces what has been called the empty delta sign,⁷⁶ but sometimes this, too, is called the delta sign
- 2. gyral enhancement occurs in 32%
- 3. dense deep (white matter) veins (collateral flow)
- 4. intense tentorial enhancement (common)

MRI

MRI excels for diagnosis and follow-up. Shows absence of flow and clot burden, also demonstrates parenchymal changes. Can differentiate occluded sinus from congenital absence. Shows cerebral edema and non-acute hemorrhagic changes to better advantage than CT. Also may help estimate age of clots (□ Table 85.7). MR-angiography may increase the utility. MR-venography (MRV) tends to overestimate the degree of occlusion.

Angiography for DST

Accuracy is close to MRI, and some still consider it to be the standard of diagnosis. MRI has some advantages over angiography (e.g. on angiography a hypoplastic transverse sinus may not visualize, or non-opacified blood entering a sinus may mimic a filling defect).

Findings include:

- 1. non-filling of segments of sinuses, or filling defects in segments that are visualized
- 2. prolonged circulation time: present in 50%of cases (may need delayed films to see veins)
- 3. stumps and abnormal collateral pathways

LP

OP usually increased. CSF may be bloody or xanthochromic.

Bloodwork

To detect predisposing conditions when the etiology is unknown. Some tests that may be useful include evaluation for thrombophilia (protein C and S levels, antiphospholipid antibodies=anticardiolipin antibodies and lupus anticoagulant) as well as tests for specific predisposing conditions (CBC, Factor II level, serum homocysteine level, paroxysmal nocturnal hemoglobinuria (PNH) panel, leukocyte alkaline phosphatase).

Ultrasound for DST

May be used in diagnosis of superior sagittal sinus thrombosis in the neonate.⁷⁷

Evaluation for underlying disorder

At the time of presentation, work-up is difficult because the acute process will cause numerous abnormalities in the clotting system. The best time to work these patients up is ≈ 3 months after they recover from the acute phase.

85.7.7 Treatment

General information

Driving principle: treat the underlying abnormality (if possible).
Should be aggressive because recoverability of brain is probably greater than with arterial occlusive stroke. Management is complicated because measures that counteract thrombosis (e.g. anticoagulation) tend to increase the risk of hemorrhagic infarct (the risk of which is already increased), and measures that lower ICP tend to increase blood viscosity →increased coagulability.

Table 85.7 MRI appearance of thrombosed sinuses at various stages		
Age of clot in sinus	Appearance of clotted sinus	
	T1 WI	T2 WI
acute	iso-intense	decreased (black): can mimic flow void
subacute	increased (1st)	increased (2nd)
late (>10 d, recanalized)	black (flow void)	black (flow void)

Specific measures

1. correct underlying abnormality when possible (e.g. antibiotics for infection)
2. heparin (systemic); see dosing information (p. 164) especially if patient is in DIC. Several studies show a lower mortality rate with heparin than without.^{78,79,80} It remains the best treatment even when there is evidence of intracerebral hemorrhage ICH with the attendant risk of increasing the size of the hemorrhage.⁷³ There is no consensus on duration of treatment or if warfarin should be used afterwards. Success rate may be higher if administered before patient becomes moribund
3. avoid steroids (reduces fibrinolysis, increases coagulation)
4. control HTN
5. anticonvulsants to control seizures
6. monitor ICP if patient continues to deteriorate: ventriculostomy preferred, but caution must be used if patient is on heparin
 - a) hydrate aggressively as ICP tolerates
 - b) measures to lower ICP: in general, order is almost reverse of that for traumatic intracranial hypertension because diuretics → hypertonicity → ↑ viscosity → ↑ coagulation:
 - elevate HOB
 - hyperventilate
 - drain CSF
 - pentobarbital coma
 - reserve hyperosmotic and/or loop diuretics for last (replace fluid loss with isotonic IV fluids to prevent dehydration; i.e. goal is hypertonic euvolemia)
7. thrombolytic therapy: either systemically or infused directly into clotted sinus,^{73,81} may be followed with heparin. Many reports, but no controlled trials have been published
 - a) urokinase^{68,81} or streptokinase
 - b) intravenous tissue plasminogen activator (tPA): promising animal evidence,⁸² not yet reported in humans
8. when above measures fail, either
 - a) decompressive craniectomy (± decompressive lobectomy): this decreases ICP, but may not improve outcome
 - b) OR direct “attack” on clotted sinus: direct surgical treatment when deficit progresses in spite of above measures, or ICP not manageable (i.e. failure of medical therapy) (see below)
9. interventional neuroradiology: effectiveness is much reduced with chronic clot
 - a) Penumbra® system: sucks out clot, only takes out a small amount of clot
 - b) AngioJet®: not FDA approved for this. Not used for arterial clots because of injury to cerebral tissue, but may have some efficacy with venous clot
10. visual loss with papilledema may be treated with optic nerve sheath fenestration/decompression⁸³
11. long-term treatment after resolution of acute phase with heparin and/or warfarin × 3–6 months

Direct surgical treatment for DST

Rarely indicated. Thrombectomy and sinus reconstruction are technically possible, but rethrombosis is common. Surgery may be indicated for abscess requiring excision.

Surgical technique for direct treatment of SSS thrombosis

Have available: blood for massive transfusion, large bore IV access, sinus shunt (pre-fabricated, or improvise with high-vacuum silicone grease inside & outside pediatric anode endotracheal tube with cuff on both ends, gas sterilize, prepared pre-op), tissue for sinus reconstruction (e.g. 20 cm of saphenous vein (arteries have a high rate of fibrosis, and there is no published experience with synthetics grafts). Vein grafts are dilated with heparinized saline and should be oriented correctly in case of valves).

Expose a wide portion of the sinus.

Consider ligation when lesion in non-critical location (SSS anterior to rolandic vein, non-dominant transverse or sigmoid, minor sinuses on skull base).

Hemorrhage is controlled by digital pressure, Fogarty catheter (insert directly, or insert a no. 7 Fogarty thru tiny sinotomy proximal to bleeding which allows repair of bleeding site), and/or insertion of shunt.

85.7.8 Prognosis

Mortality: approximately 30% (range: 5–70%) (10% in French series⁷²).

Poor prognosticators:

1. clinical status:
 - a) coma⁸⁴:
 - b) rapid neurologic deterioration,⁸⁴ focal signs.
2. demographics
 - a) age: extremes of age (infancy or elderly)⁸⁴ and age >37 years
 - b) male gender
3. radiographic findings:
 - a) hemorrhages, especially larger hemorrhages
 - b) venous infarcts
4. deep venous involvement

85.8 Moyamoya disease

85.8.1 General information

Key concepts

- progressive bilateral spontaneous occlusion of ICAs with compensatory capillary collaterals that look like a “puff of smoke” (Japanese: moyamoya) on angio
- typical presentation: juvenile form → ischemic infarcts/TIAs (suspect diagnosis in any child presenting with TIAs). Adult form → hemorrhage
- pathology: intimal thickening w/o inflammation, may also involve heart, kidneys. Associated aneurysms may be source of bleeding
- evaluation: cerebral angiography is necessary to delineate degree of stenosis as well as to evaluate potential extracranial donor vessels for revascularization. Also identifies aneurysms
- treatment:
 - a) medical treatment (antiplatelet drugs, anticoagulation, vasodilators...): not shown to be effective although antiplatelet/anticoagulation is often used
 - b) surgical revascularization: reduces the incidence of strokes and TIAs, but benefit on reducing the rate of hemorrhage is unproven

Progressive spontaneous occlusion of one or usually both ICAs (usually at the level of the carotid siphon) and their major branches, with secondary formation of an anastomotic collateral capillary network at the base of the brain which has been termed “moyamoya”, the Japanese word for something hazy like a “puff of cigarette smoke”⁸⁵ (which it fancifully resembles on angiography) (first described in 1957,⁸⁶ and named in 1969⁸⁵). With progression, involvement includes the proximal MCAs and ACAs and on rare occasion the vertebrobasilar system. Associated aneurysms (see below) and rarely AVMs^{87,88} may be observed.

Eventually the dilated capillary (moyamoya) vessels disappear with the development of collaterals from the ECA (meningeal collaterals are called “rete mirabile”).

85.8.2 Pathophysiology

Primary moyamoya disease

The most common pathology is stenosis of the proximal anterior and middle cerebral arteries that is neither atherosclerotic nor inflammatory in origin. Exact etiology is unknown but some studies show elevated basic fibroblast growth factor in the dura and scalp arteries in patients with moyamoya.⁸⁹ The internal elastic lamina of affected vessels may be thinned or duplicated. Similar vascular changes may also occur in the heart, kidney and other organs, suggesting it may be a systemic vascular disease.

Secondary moyamoya disease

AKA “quasi-moyamoya disease” or “moyamoya syndrome”.⁹⁰ Angiographic findings of moyamoya associated with e.g.:

1. Graves’ disease/thyrotoxicosis
2. history of cerebral inflammatory disease, including meningitis (especially tubercular (TB) meningitis and leptospirosis)

3. retinitis pigmentosa
4. vascular disorders: atherosclerosis, fibromuscular dysplasia, pseudoxanthoma elasticum
5. congenital disorders: Down syndrome, Marfan syndrome, Turner syndrome, neurofibromatosis type 1, tuberous sclerosis, Apert syndrome
6. hematologic disorders: Fanconi anemia, sickle cell anemia (in the U.S. one of the more common associations) and sickle cell trait
7. following radiation therapy for skull base glioma in children⁹¹
8. head trauma
9. systemic lupus erythematosus (SLE)

Associated aneurysms

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Intracranial aneurysms are frequently associated with moyamoya disease (MMD). This may be a result of the increased flow through dilated collaterals, or it may be that patients with moyamoya may also have a congenital defect in the arterial wall that predisposes them to aneurysms. 3 types: 1) usual sites of aneurysms in the Circle of Willis, 2) in peripheral portions of cerebral arteries, e.g. posterior/anterior choroidal, Heubner's, and 3) within moyamoya vessels. The frequency of aneurysms in the vertebrobasilar system is $\approx 62\%$ which is much higher than in the general population.⁹² Aneurysmal SAH may be the actual cause of some hemorrhages that were erroneously attributed to moyamoya vessels.

85.8.3 Epidemiology

Risk factors

A history of inflammation in the head & neck region has been implicated.

Demographics

Incidence in Japan is higher (0.35/100,000/yr) than in North America. Two peaks (may not be same disease): juvenile (highest peak), age <10 yrs (mean 3); adult, 3rd & 4th decade. Slight female predominance (1.8:1). Some evidence for familial tendency (some Asian families have an incidence as high as 7%), genetics appears autosomal dominant with low penetrance. Associated with some HLA antigens (B40 in juvenile form; B54(20) in adult) and anti-double-stranded DNA antibody.

85.8.4 Presentation

Juvenile form

Moyamoya is associated with 6% of childhood strokes.⁸⁹ Ischemic presentation more common (81%); includes TIAs (41%) which may alternate sides (alternating hemiplegia is a suggestive clinical finding), RINDs, or infarct (40%). Neurologic events are often provoked by straining or by hyperventilation (e.g. during crying or blowing a wind instrument) which is thought to produce hypocapnea with reactive vasoconstriction.

Headache is the most common presenting symptom, but seizures, focal neurologic deficits, choreoathetotic movements and hemorrhages can also be presenting symptoms. The risk of hemorrhage is increased in stages 5 & 6 of MMD.

Adult form

Hemorrhage has been described as being more common (60%), but in a Stanford series⁹³ 89% presented with ischemia. Rupture of the fragile moyamoya vessels produces bleeding in the basal ganglia (BG), thalamus or ventricles (from the ventricular wall) in 70–80% of hemorrhages. SAH may occur, usually due to rupture of associated aneurysms (see above). In the pre-CT era, the most common form of hemorrhage was thought to be SAH from the rupture of moyamoya vessels, but most cases were probably intraventricular blood or SAH from associated aneurysms.⁹⁴

85.8.5 Natural history

Incidence of disease progression in one study was 20% in adult patients with MMD.⁹⁵ Female patients had a higher risk of disease progression than males.

Prognosis of untreated MMD is poor, with 73% rate of major deficit or death within 2 years of diagnosis in children, and similarly poor outlook in adults.⁹³

85.8.6 Evaluation and diagnosis

Diagnostic criteria

Diagnosis of moyamoya requires bilateral symmetrical stenosis or occlusion of the terminal portion of the ICAs as well as the presence of dilated collateral vessels at the base of the brain.⁸⁹ (If unilateral, the diagnosis is considered questionable,⁹⁶ and these cases may progress to bilateral involvement).

Other characteristic findings include:

- 1. stenosis/occlusion starting at termination of ICA and at origins of ACA and MCA
- 2. abnormal vascular network in region of BG (intraparenchymal anastamosis)
- 3. transdural anastomosis (rete mirabile), AKA “vault moyamoya.” Contributing arteries: anterior falcial, middle meningeal, ethmoidal, occipital, tentorial, STA
- 4. moyamoya collaterals may also form from internal maxillary artery via ethmoid sinus to fore-brain in frontobasal region

CT

Work-up in suspected cases typically begins with a non-enhanced head CT. Up to 40%of ischemic cases have normal CT. Low density areas (LDAs) may be seen, usually confined to cortical and sub-cortical areas (unlike atherosclerotic disease or acute infantile hemiplegia which tend to have LDAs in basal ganglia as well). LDAs tend to be multiple and bilateral, especially in the PCA distribution (poor collaterals), and are more common in children.

MRI and MRA

MRA usually discloses the stenosis or occlusion of the ICA. Moyamoya vessels appear as flow voids on MRI (especially in basal ganglia) and a fine network of vessels on MRA, and are demonstrated better in children than adults. Parenchymal ischemic changes are commonly shown, usually in watershed areas.

Angiography

In addition to helping to establish the diagnosis, angiography also identifies suitable vessels for revascularization procedures and unearths associated aneurysms. The angiography-related complication rate is higher than with atherosclerotic occlusive disease. Avoid dehydration prior to and hypotension during the procedure. Six angiographic stages of MMD are described in □ Table 85.8 ⁸⁵ that tend to progress up until adolescence and stabilize by age 20.

EEG

Non-specific in the adult. Juvenile cases: high-voltage slow waves may be seen at rest, predominantly in the occipital and frontal lobes. Hyperventilation produces a normal buildup of monophasic slow waves (delta-bursts) that return to normal 20–60 seconds after hyperventilation. In >50%of cases, after or sometimes continuous with buildup is a second phase of slow waves (this characteristic finding is called “rebuildup”) which are more irregular and slower than the earlier waves, and usually normalize in ≤10 minutes.⁹⁷

Table 85.8 Six angiographic stages of MMD ⁸⁵	
Stage	Finding
1	stenosis of suprasellar ICA, usually bilateral
2	development of moyamoya vessels at base of brain; ACA MCA & PCA dilated
3	increasing ICA stenosis & prominence of moya-moya vessels (most cases diagnosed at this stage); maximal basal moyamoya
4	entire circle of Willis and PCAs occluded, extracranial collaterals start to appear, moyamoya vessels begin to diminish
5	further progression of stage 4
6	complete absence of moyamoya vessels and major cerebral arteries

Cerebral blood flow (CBF) studies

CBF is decreased in children with MMD, but relatively normal in adults. There is a shift of CBF from the frontal to the occipital lobes⁹⁸ probably reflecting the increasing dependency of CBF on the posterior circulation. Children with MMD have impaired autoregulation of CBF to blood pressure and CO₂ (with more impairment of vasodilatation in response to hypercapnia or hypotension than vasoconstriction in response to hypocapnia or hypertension).⁹⁹

Xenon (Xe-133) CT can identify areas of low perfusion. Repeating the study after an acetazolamide challenge (which causes vasodilatation) evaluates reserve capacity of CBF and can identify areas of “steal” which are at high risk of future infarction.

85.8.7 Treatment

General information

No medical or surgical treatment has been proven effective in reducing the rate of hemorrhage in the adult with MMD. However, multiple large case series have supported the efficacy of cerebral revascularization for reducing the incidence of ischemic strokes and TIAs.⁹⁰

Asymptomatic moyamoya disease

Guidelines for management of asymptomatic moyamoya disease have not yet been established. A multi-center, nation-wide survey in Japan focusing on asymptomatic moyamoya disease provided the following findings¹⁰⁰: subtle findings of cerebral infarction and disturbed cerebral hemodynamics were detected in 20% and 40% of the involved hemispheres, respectively. Angiographic stage was more advanced in elderly patients. Of 34 medically-treated patients, 7 experienced TIA, ischemic stroke or hemorrhage during a mean follow-up period of 43.7 months. Cerebral infarction or hemorrhage did not occur in the 6 patients who underwent surgical revascularization.

Medical treatment

Medical treatment with platelet inhibitors, anticoagulants, calcium channel blockers,⁹³ steroids, mannitol, low-molecular-weight dextran and antibiotics have not proven to be of benefit. Steroids may be considered for involuntary movements and acutely during recurrent TIAs.

Surgical treatment

General information

Patients with mass effect from clot may be candidates for urgent decompression. Revascularization procedures, however, should be performed when the patient is stable under nonemergent conditions.

Perioperative management

During any surgical procedure:

1. avoid hyperventilation: due to increased sensitivity of collaterals, keep PaCO₂ 40–50 mm Hg to avoid ischemic infarction
2. avoid hypotension: maintain BP at normotensive levels
3. avoid alpha-adrenergic agents because of vasoconstrictive effects
4. cerebral protection: mild hypothermia (32–43° C) and barbiturates are routinely used
5. papaverine helps prevent vascular spasm

Postoperatively following STA-MAC bypass procedures:

1. avoid hypertension: may cause bleeding at anastomotic site and in areas of increased perfusion within the brain
2. avoid hypotension: may result in graft occlusion
3. aspirin is started on the post-op day # 1
4. watch for evidence of CSF leak
5. monitor coag studies and correct abnormalities
6. cerebral arteriogram is recommended 2–6 months post-op

Suggested criteria for revascularization procedures

See reference.⁹⁰

1. patients presenting with infarction or hemorrhage but are in good neurologic condition

2. infarction <2 cm maximal diameter on CT, and all previous hemorrhages have completely resolved
3. angiographic stage is II-IV (see □ Table 85.8)
4. timing of operation: ≥ 2 months after most recent attack

Surgical revascularization options

Various methods to revascularize the ischemic brain, used primarily in children, include:

1. direct revascularization procedures:
 - a) results are superior to indirect revascularization procedures^{101,102} if a donor and recipient vessel of sufficient caliber (≥ 1 mm outer dia) can be identified (may be difficult in the pediatric age group who are the most likely to benefit¹⁰³). Otherwise, indirect revascularization procedures (see below) are options.
 - b) Among direct revascularization procedures, STA-MCA bypass¹⁰⁴ is the procedure of choice
2. indirect revascularization procedures: usually reserved for younger patients (suggested cutoff age ≈ 15 years). May be combined with STA-MCA bypass. Includes:
 - a) encephalomyosynangiosis (EMS): laying the temporalis muscle on the surface of the brain (may cause problems with muscle contractions during talking and chewing, and neural impulses on surface of brain)
 - b) encephaloduroarteriosynangiosis (EDAS)^{105,106}: suturing the STA with a galeal cuff to a linear defect created in the dura. Variations on this technique include splitting the dura¹⁰⁷
 - c) omental transposition¹⁰⁸: either as a pedicle graft or as a vascularized free flap. Felt to have higher potential to revascularize ischemic tissue than above procedures, but there is greater risk of mass effect from the thickness of the omentum
3. the above indirect revascularization procedures improve blood flow in the MCA distribution, but not ACA circulation. This may be rectified by:
 - a) simple placement of frontal burr holes with opening of the underlying dura and arachnoid¹⁰⁹
 - b) “ribbon EDAS” where a pedicle of galea is inserted into the interhemispheric fissure on both sides¹¹⁰
4. stellate ganglionectomy and perivascular sympathectomy: unproven that this increases CBF permanently

Outcome with surgical treatment

Neurologic status at time of treatment generally predicts long-term outcome.⁸⁹ The mortality rate in adults ($\approx 10\%$) is higher than for juveniles ($\approx 4.3\%$).⁹⁶ The cause of death was bleeding in 56% of 9 children and 63% of 30 adults. With treatment the prognosis is good in 58%⁹⁴

85.9 Extracranial-intracranial (EC/IC) bypass

85.9.1 EC/IC bypass for atherosclerotic occlusive disease

The EC/IC bypass study

The EC/IC bypass, pioneered by Donaghy and Yasargil in 1967,¹¹¹ plummeted in popularity¹¹² after publication of the international cooperative EC/IC bypass study¹¹³ in 1985. The EC/IC trial randomized 1377 patients with symptomatic ICA or MCA stenosis to either STA-MCA bypass or medical therapy with ASA. Despite a graft patency rate of 96%, surgical patients suffered more and earlier fatal and nonfatal strokes. Patients with severe MCA stenosis and those with persistent symptoms following ICA occlusion fared especially worse with bypass. During the 55.8 months mean follow-up, the percentage of patients experiencing 1 or more strokes in the medical group compared to the surgical group was 29% vs. 31%

Critics highlight the failure of the study's inclusion criteria to distinguish between hemodynamic vs. thromboembolic causes of stroke^{2,114,115} (ischemia secondary to thromboembolic events would not be expected to improve with flow augmentation, and inclusion of such patients in the surgical arm could therefore artificially lower the apparent efficacy of the procedure).

Current state of affairs

Imaging technologies introduced since the EC/IC trial can identify flow-dependent ischemia. Xenon-CT, TCD, SPECT, and MRI may be used in combination with acetazolamide challenge to evaluate cerebrovascular reserve and reactivity (p.1265).

As cerebral perfusion pressure decreases in severe atherosclerotic occlusive disease, cerebral autoregulation is unable to maintain adequate CBF to meet metabolic demands. In this state of “misery perfusion”, oxygen extraction fraction (OEF) of available blood flow will increase.^{116,117} Abnormal OEF, as quantified by PET, is an independent predictor of subsequent stroke.² Patients with abnormal response to acetazolamide challenge (p.228) and/or with elevated OEF are therefore potential candidates for cerebral revascularization.^{2,115,118,119,120}

Indications for EC/IC bypass

1. Patients with misery perfusion (see above)
2. aneurysms: certain aneurysms are not amenable to either direct microsurgical clipping or endovascular coiling due to extreme size, location, calcification or atherosclerosis, dissection, or the incorporation of perforators or major arteries. EC/IC bypass remains a highly viable adjunctive measure in patients requiring Hunterian occlusion of parent vessel or prolonged temporary occlusion for definite treatment.^{121,122,123,124,125} Cerebrovascular reserve and need for bypass can be assessed preoperatively using balloon test occlusion (BTO) with hypotensive challenge
3. tumors encasing or invading major arteries
4. moyamoya disease (p.1313)

Bypass types

The type of graft used depends on preoperative determination of amount of flow augmentation necessary, the size of the recipient graft and the availability of donor vessel¹²⁶:

1. pedicled arterial grafts: STA, occipital artery
 - a) low-flow (15 – 25 ml/min)
 - b) only one anastomosis required
 - c) 95%graft patency in superficial temporal artery-middle cerebral artery (STA-MCA) bypasses
2. radial artery graft
 - a) moderate to high flow (40 – 70 ml/min)
 - b) advantages: physiological conduit for arterial blood; constant location makes it easy to harvest; lumen size closely approximates that of M2 or P1 and reduces flow mismatch with subsequent flow turbulence and graft thrombosis
 - c) disadvantages: risk of vasospasm (reduced with pressure distension technique)
 - d) >90%graft patency at 5 years
3. saphenous vein graft
 - a) high flow (70 – 140 ml/min)
 - b) advantages: easy accessibility; longer length
 - c) disadvantages: risk of thrombosis at distal anastomosis due to flow mismatch and turbulence; lower graft patency rates
 - d) 82%graft patency at 5 years

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86 Cerebral Arterial Dissections

86.1 General information

Key concepts

- hemorrhage into the tunica media of an artery
- may be spontaneous, post-traumatic, or iatrogenic [e.g. angiography-related], may be intracranial or extracranial
- may present with pain (usually ipsilateral H/A or carotidynia), Horner's syndrome
- (in carotid dissections), TIA/stroke, or SAH
- extracranial dissections are usually treated medically (anticoagulation), intracranial dissections with SAH are treated surgically

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This section primarily discusses “spontaneous” dissections. ICA dissection following blunt cervical trauma is much more common (p.851).

86.2 Nomenclature

Some confusion has arisen because of inconsistent terminology in the literature. Although by no means standard, Yamaura¹ has suggested the following:

- Dissection. Extravasation of blood between the intima and media, creating luminal narrowing or occlusion.
- Dissecting aneurysm. Dissection of blood between the media and adventitia, or at the media, causing aneurysmal dilatation, which may rupture into the subarachnoid space.
- Pseudoaneurysm. Rupture of artery with subsequent encapsulation of the extravascular hematoma, may or may not produce luminal narrowing.

86.3 Pathophysiology

The lesion common to all dissections is hemorrhage outside of the vascular lumen due to pathological trans-intimal extravasation of blood from the true lumen into the vessel wall. The hematoma may either dissect the internal elastic membrane from the intima² causing narrowing of the true lumen, or it may dissect into the subadventitial plane producing an adventitial outpouching from the vessel wall (pseudoaneurysm). Rupture through the vessel wall producing SAH occurs occasionally.

Subintimal dissection is more common with intracranial dissections, whereas extracranial vessels (including the aorta) usually dissect at the media or between media and adventitia.

“Spontaneous” dissections have been associated with a large number of conditions, oftentimes the association is unproven. These conditions include:

- fibromuscular dysplasia (FMD): found in $\approx 15\%$ of cases³
- cystic medial necrosis (or degeneration): originally thought to be a common finding, now thought to perhaps be linked to a higher likelihood of fatal dissection
- saccular aneurysm
- Marfan syndrome: autosomal dominant inherited disorder of connective tissue. Phenotypic manifestations are due to production of abnormal fibrillin, the main component of extracellular microfibrils, a component in the media of certain blood vessels, encoded by the FBN1 gene on chromosome 15q21
- Ehlers-Danlos syndrome
- atherosclerosis: only rarely implicated as an etiology. More likely to be a factor with subintimal dissection of extracranial arteries
- Takayasu's disease
- medial degeneration
- syphilitic arteritis (more common in the past, associated with 60% of dissections before 1950)

- autosomal dominant polycystic kidney disease (p.1193): associated with a higher incidence of cerebral aneurysms
- variant periarteritis nodosa
- allergic arteritis
- homocystinuria
- moyamoya disease (p.1313) ⁴
- strenuous physical activity

86.4 Epidemiology

Occurs primarily in middle aged patients, with a mean age of ≈ 45 yrs (average age of traumatic dissections is slightly younger). More frequent in men.^{1,3} Incidence is unknown, since often times the condition produces mild, transient symptoms. Increased awareness of the condition has resulted in an increased rate of diagnosis. ICA dissection accounts for 1-2.5% of first strokes.⁵ However, in middle aged and young adults it comprises 10-25% of strokes.⁶

86.5 Sites of dissection

A review of 260 cases¹ (literature review + new cases) found the incidence by location shown in Table 86.1. The vertebral artery was the most common intracranial site. Previously, it was believed that the ICA was the most common site. This change may be due to the recent increased recognition of arterial dissections as a source of SAH (and vertebral dissections most often present as SAH). Multiple dissections occur in $\approx 10\%$ (the most common: bilateral vertebrobasilar lesions).

86.6 Clinical

Cerebral arterial dissections may cause symptoms by:

1. embolization secondary to:
 - a) platelet aggregation stimulated by the exposed surfaces
 - b) dislodged thrombus (formation of which is enhanced by reduced flow)
2. reduced distal flow secondary to:
 - a) thrombosis due to reduced flow
 - b) occlusion of the true lumen by the expansion of the mural hematoma
3. subarachnoid hemorrhage (atypical presentation, may be more common with posterior circulation dissection than with anterior circulation)⁷

The most common presentation in patients <30 yrs of age was due to internal carotid dissection without SAH. In patients >30 yrs, vertebrobasilar artery (VBA) dissection with SAH was the most common.¹

Headache, usually severe, often predates neurologic deficit by days or weeks (p.1228). See sections ICA (p.1324) and vertebrobasilar (p.1325) for specifics.

Table 86.1 Spontaneous intracranial dissections by site			
Location	Left	Right	Total
vertebral	122	82	204
basilar	35		35
internal carotid	17	13	30
middle cerebral	16	10	26
anterior cerebral	10	3	13
posterior cerebral	7	9	16
PICA	4	10	14
Total	176	127	338

86.7 Evaluation

- CT. More useful for evaluating brain for infarction. Dissection can sometimes be visualized directly.⁸
- CT angiogram (CTA). Often obviates the need for cerebral angiography since CTA scanners with ≥ 16 detectors are equal in predictive value and have an accuracy near 99%⁹
- Angiography. The definitive diagnostic study. However, diagnosis may be delayed if the dissection is misinterpreted as:
 1. an unusual saccular aneurysm (the most common error)
 2. atherosclerotic lesions: with dissections, the location is unusual, the lesion may be isolated, the age is usually younger, and the stenosis is smooth. Cervical ICA dissection typically spares the carotid bulb whereas cervical ICA atherosclerosis tends to involve the bulb.
 3. vasospasm following SAH: however, the narrowing with vasospasm is delayed in onset vs. the changes with dissection which are present from the beginning

Angiographic findings may include:

1. luminal stenosis: irregular stenosis over long segments of the artery often with focal areas of near total stenosis (“string sign”)
2. fusiform dilation with proximal or distal narrowing (string and pearl sign)
3. occlusion: artery usually tapers to a point
4. intimal flap: when seen, usually found at proximal end of dissection
5. may see proximal beading (“string of beads” configuration, indicative of FMD)
6. “double lumen sign”: true vessel lumen and an intramural false lumen with an intimal flap. Usually with retention of contrast within the false lumen well into the venous phase. The only pathognomonic sign
7. wavy “ripple” appearance
8. severe kinking (frequently bilateral). With VBA lesions: dolichoectasia

A characteristic of arterial dissections is that they often change configuration on repeat angiography¹⁰ (some resolve, and some worsen). NB: forceful intra-arterial contrast injection during the performance of angiography carries a potential for worsening the dissection.

□ MRI. Probably not as accurate as CTA or angiography. Optimal MRI study is source T1WI axial images with fat suppression (“fat sat”), look for loss of visualization over several slices, with good visualization above and below. May visualize intimal flap and distinguish a dissection from a fusiform aneurysm.

Crescent sign: bright signal in wall of ICA on T2WI axial images (hematoma in vessel wall).

86.8 Overall outcome

An early review of the literature found an 83% mortality within a few weeks of presentation with vertebrobasilar artery (VBA) dissection.¹¹ A later report tempered that grim prognosis.¹²

Based on a review of 260 cases,¹ an overall mortality of 26% was found. 70% had a favorable outcome (assessed using Glasgow Outcome scale), 5% poor. Mortality was higher in ICA lesions (49%) than VBA lesions (22%). Mortality was 24% in the SAH group, and 29% in non-SAH cases.

86.9 Vessel specific information

86.9.1 Internal carotid dissection

See above for general information. Posttraumatic ICA dissection (p.1323) is much more common than spontaneous.

Some cases considered “spontaneous” may actually be due to trivial trauma, including violent coughing, nose blowing, and simple neck turning. Usually seen in young women.

In spontaneous dissection, the most common initial symptom is ipsilateral headache. Most of these (60%) are orbital or periorbital, but they may also be auricular or mastoid (39%), frontal (36%), temporal (27%). May also produce sudden onset of severe pain over carotid artery (carotidynia).¹³

Incomplete Horner’s syndrome (oculosympathetic palsy): ptosis and miosis without anhidrosis (due to involvement of plexus around the ICA, sparing the ECA plexus which innervates facial sweat

glands) may occur. Bruits may be heard by the examiner or by the patient. These and other clinical features are shown in □ Table 86.2.

May be a cause of some cases of infantile and childhood hemiplegia and hemiparesis.¹⁴

86.9.2 Vertebrobasilar system artery dissection

Vertebral artery dissections

General information

See general information on cerebral arterial dissections (p.1322). See also posttraumatic vertebrobasilar dissection (p.852).

Less common than carotid. Extracranial lesions outnumber intracranial.

Traumatic dissections tend to occur where the VA crosses bony prominences, e.g. at the C1–2 junction or where it enters the foramen transversarium (usually at C6). Spontaneous dissections tend to be intracranial and commonly occur on the dominant VA. Unlike cervical ICA dissections, which tend not to propagate intracranially through the carotid canal, high cervical VA dissections can readily propagate intracranially through the foramen magnum.

Spontaneous VA dissections have been associated with FMD, migraine, and oral contraceptives.¹⁵ Unrecognized or forgotten trauma or sudden head motion may have occurred in some cases reported as spontaneous. Commonly occurs in young adults (mean age: 48 yrs). With spontaneous dissections, 36% of patients have dissections at other sites, 21% of cases have bilateral VA dissections.¹⁶

Dissecting aneurysms of the VA (possibly a distinct entity) are also described.^{17,18,19} They tend to be fusiform, and may be amenable to clipping, and were associated with vertebral dissections in 5 of 7 cases reported in one series.²⁰ As of 1984, only ≈ 50 cases of dissecting aneurysms were published.²⁰

Presentation

In spontaneous extradural dissections, neck pain is a prominent early finding in most patients, and is commonly located over the occiput and posterior cervical region. Generalized severe headache is also common. TIAs or stroke (usually lateral medullary syndrome (p.1267) ²¹ or cerebellar infarction, especially in patients with occlusion of the third or fourth portion of the VA²²). None of 5 patients developed new neurologic symptoms after the original stroke in an average of 21 months follow-up.²² In 3 of these 5, VA dissection was bilateral.

Dissecting aneurysms may present with altered consciousness, and may cause SAH (seen in 6 of 30 cases of vertebrobasilar complex dissections).²⁰ Rebleeding occurs in 24–30% of those cases presenting with SAH,¹⁶ making these lesions treacherous, with a very high mortality.^{23,24}

Traumatic extradural dissections or pseudoaneurysms may have a similar presentation, but can also produce massive external hemorrhage or neck hematomas.¹⁶

Evaluation

See section under Cerebral arterial dissections, Evaluation (p.1324).

Table 86.2 Clinical features of spontaneous ICA dissection ³	
Feature	%
focal cerebral ischemia	76%
headache	59%
oculosympathetic palsy	30%
bruit	25%
amaurosis fugax	10%
neck pain	9%
syncope	4%
scalp tenderness	2%
neck swelling	2%

□ **Angiography.** Diagnosis by angiography may be difficult in many cases (the most common misdiagnosis is ruptured saccular aneurysm of unusual shape²⁵).

In post-traumatic dissections, the most common finding is irregular stenosis of horizontal loops of distal extracranial VAs as they pass behind C1, often bilateral.

In 14 of 15 post-traumatic VA dissections, the lesion was located posterior to the atlas (distal extracranial 3rd segment), the single exception being a patient with direct trauma causing proximal VA involvement. This predilection is possibly explained by the fact that the first and third portions of the VA are movable, whereas the second and fourth are relatively immobilized by bone.

Treatment

Except for cases presenting with hemorrhage or large ischemic stroke, medical therapy should be started emergently. Classically consists of anticoagulation, with heparin acutely, followed by oral agents (e.g. Coumadin) probably for a total of 6 months. Recent preliminary study showed antiplatelet therapy was equally as effective.²⁶

As with traumatic dissections, endovascular techniques are now assuming more prominent role in management.

□ **Indications for intervention.** Surgery or endovascular techniques (mostly stents, but also occlusion, angioplasty¹⁶) are required for dissections presenting with SAH (due to their propensity to rebleed) and is recommended for most intradural dissections. For extradural lesions it is indicated for dissections that progress (angiographically) or for persistent symptoms in spite of adequate medical therapy. Some less malignant lesions may be amenable to endovascular stenting.

□ **Endovascular treatment.** Balloon-mounted, self-expandable, or covered stents have been used relatively infrequently to treat dissections of the internal carotid or vertebral arteries, with good technical results and low procedure-related complication rates²⁷ 2011]. Given the small number of patients treated and the fact that medical therapy is generally effective, the role of stenting for dissection remains to be defined. It should be reserved for patients in whom medical therapy is ineffective or contra-indicated or when the dissection causes symptomatic flow-limiting stenosis.

□ **Surgical treatment.** At the time of surgery, the site of dissection may be recognized by fusiform or tubular enlargement of the artery with discoloration due to blood within the arterial wall (the discoloration has been described as black, bluish, purple, purple red, or brown²⁵).

Surgical treatment of intradural dissection when endovascular techniques are not an option includes the following alternatives:

1. non-clippable aneurysms may be candidates for Hunterian occlusion of the VA proximal to the BA (either by microsurgical technique, or by endovascular techniques which may not be as precise). Some may not tolerate clipping the dominant VA, especially if the contralateral VA is hypoplastic. Conversely, some may tolerate bilateral VA occlusion.²⁸ Balloon test occlusion¹⁶ is recommended
 - a) if the dissection involves the PICA origin, then clip proximal to dissection. PICA then fills from retrograde flow, and the reversal of flow across the site of dissection should push the intima back against the wall
 - b) if the dissection is proximal to PICA and doesn't involve PICA, then trap the aneurysm between clips. PICA fills by retrograde flow
 - c) if the aneurysm begins distal to the PICA origin, occlude the VA⁷ distal to the PICA takeoff²⁹
2. combining VA clipping (non-clippable aneurysms may be candidates for Hunterian occlusion of the VA proximal to the aneurysm) with vascular bypass, options:
 - a) side-to-side PICA-PICA anastomosis
 - b) transplantation of the PICA origin to the VA outside the aneurysm
 - c) occipital artery-to-PICA bypass
3. resection accompanied by autogenous interposition vein graft
4. non occlusive surgical techniques
 - a) clipping with specially designed clips for fusiform aneurysms (e.g. Sundt-Kees clip)
 - b) wrapping: of dubious benefit

86.9.3 Vertebrobasilar system dissections excluding the VA

Basilar artery dissections tend to present with brain stem infarction and more rarely with SAH.²⁴ The prognosis is generally regarded as poor. Endovascular techniques may be able to treat some.

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Part XX

Intracerebral Hemorrhage

87 Intracerebral
Hemorrhage

1330



87 Intracerebral Hemorrhage

87.1 General Information

Intracerebral hemorrhage (ICH) is a hemorrhage within the brain parenchyma. Formerly commonly referred to as “hypertensive hemorrhage”, but hypertension is a debatable etiology in many cases; see Hypertension as a cause? (p.1334).

87.2 Intracerebral hemorrhage in adults

Key concepts

- the second most common form of stroke (15–30% of strokes), but the most deadly
- unlike ischemic infarct: smooth progressive onset over minutes to hours, often with severe headache, vomiting and alterations in level of consciousness
- unenhanced CT scan of the brain is the initial diagnostic study of choice
- the volume of the hematoma correlates highly with morbidity and mortality
- the clot enlarges in at least 33% of cases within the first 3 hours of onset
- angiography is recommended (as long as it doesn’t delay emergent treatment) except for patients >45 yrs of age with preexisting hypertension and ICH in thalamus, putamen or posterior fossa
- treatment
 - a) still controversial. The initial promise of rFVIIa has not been actualized
 - b) the usefulness of surgery is still controversial, but seems limited to some cerebellar hemorrhages and select supratentorial hemorrhages that come within 1 cm of the cortical surface

87.3 Epidemiology

87.3.1 Incidence

The second most common form of stroke ($\approx$ 15–30% of all strokes) (earlier estimates: 10%¹), and the most deadly. Approximately 12–15 cases per 100,000/yr. Early studies estimated an incidence equal to SAH, but more recent studies in the CT era show approximately twice the incidence as SAH² (pre-CT studies may have misclassified some ICH as ischemic stroke, and some cases of ICH that rupture into the subarachnoid space (occurs in $\approx$ 7%) may have been misclassified as SAH). After a decline in the 1970’s, the incidence increased in the 1980s for those age $\geq$ 65 years.³ Onset is usually during activity (rarely during sleep), which may be related to elevation of BP or increased CBF; see Etiologies (p.1332).

87.3.2 Risk factors

The following are epidemiologic risk factors; see also others (p.1332).

1. age: the incidence increases significantly after age 55 years and doubles with each decade of age until age >80 yrs where incidence is 25 times that during previous decade. Relative risk for age >70 yrs is 7
2. gender: more common in men
3. race: in the U.S., ICH affects blacks more than whites. May be related to higher prevalence of HTN in blacks. Incidence may also be higher in Asians⁴
4. previous stroke (any type) increases risk to 23:1
5. alcohol consumption^{4,5}:
 - a) recent use: moderate or heavy alcohol consumption both within the 24 hours and the week preceding the ICH were independent risk factors for ICH⁶ as shown in □ Table 87.1
 - b) chronic use: one study suggests that consuming >3 drinks a day increases the risk of ICH by $\approx$ 7 times⁷ (p 15)
 - c) ICH in patients with high ethanol consumption were more commonly lobar than the typical “hypertensive hemorrhages” in the basal ganglia⁸

Table 87.1 Relative risk of ICH with EtOH consumption		
Period prior to ICH	Amount ^a (g EtOH)	Relative risk
24 hours	41–120	4.6
	>120	11.3
1 week	1–150	2.0
	151–300	4.3
	>300	6.5
^a 1 standard drink = 12 g EtOH		

Table 87.2 Common sites for ICH (modified ¹³)	
%	Location
50%	striate body (basal ganglia); putamen most common; also includes: lenticular nucleus, internal capsule, globus pallidus
15%	thalamus
10–15%	pons (≈ 90% of these are genuinely hypertensive)
10%	cerebellum
10–20%	cerebral white matter
1–6%	brain stem

6. cigarette smoking: increases the risk of SAH and ischemic infarction but probably does not increase the risk of ICH,^{9,10} further clarification is needed
7. street drugs: cocaine, amphetamines, phencyclidine¹¹
8. liver dysfunction: hemostasis may be impaired on the basis of thrombocytopenia, reduced coagulation factors, and hyperfibrinolysis¹² (may be responsible for the increased risk of ICH with chronic EtOH consumption)

87.4 Locations of hemorrhage within the brain

87.4.1 General information

Common sites of ICH are shown in □ Table 87.2. Common arterial feeders of ICHs:

1. lenticulostriates: the source of putaminal hemorrhages (possibly secondary to microaneurysms of Charcot-Bouchard, see below)
2. thalamoperforators
3. paramedian branches of BA

87.4.2 Lobar hemorrhage

This term was popularized in 1980 after a report delineating 4 clinical syndromes associated with hemorrhage in each of the cerebral lobes.¹⁴ It incorporates primary hemorrhages into the occipital, temporal, frontal and parietal lobes (including ICH arising from cortex and subcortical white matter), as opposed to hemorrhage of deep structures (e.g. basal ganglion, thalamus, and infratentorial structures).¹⁴ Accounts for 10–32% of nontraumatic ICHs.¹⁴ With large hemorrhages, it may be difficult to make a distinction between lobar and deep ICH.

Lobar hemorrhages are more likely to be associated with structural abnormalities than deep hemorrhages (see below). They may also be more common in patients with high alcohol consumption (see above). Lobar hemorrhages may also have a more benign outcome than ganglionic-thalamic hemorrhages.¹⁴

Etiologies of lobar hemorrhage: Although many causes of ICH can produce lobar hemorrhages (see below for a detailed list), those that are more likely to produce lobar hemorrhages include:

1. extension of a deep hemorrhage
2. cerebral amyloid angiopathy (p.1334): the most common cause of lobar ICH in elderly normotensive patients
3. trauma
4. hemorrhagic transformation of an ischemic infarct: see below
5. hemorrhagic tumor (p.1335). Multiple lobar hemorrhages may occur with metastases
6. cerebrovascular malformation (especially AVM) (p.1246)
7. rupture of an aneurysm: see below for circumstances likely to produce this
8. idiopathic

87.4.3 Internal capsule hemorrhages

There may be prognostic significance with regard to contralateral motor function if the hemorrhage is medial to and/or extending through the internal capsule (IC), or lateral to the IC and merely compressing it, making the clot more accessible to surgical treatment without damaging the IC.

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87.5 Etiologies

87.5.1 History check list

Based on information in this chapter, the following check-list is presented to assist in the gathering of historical information important in evaluating the adult with ICH:

1. hypertension
2. drugs:
 - a) sympathomimetics:
 - amphetamines, cocaine
 - appetite suppressants or nasal decongestants (phenylpropanolamine, pseudoephedrine)
 - b) dietary supplements: especially ephedra alkaloids (ma huang)
 - c) anticoagulants: warfarin in particular
 - d) antiplatelet drugs: aspirin (patients often forget taking low dose 81 mg), Plavix, NSAIDS
 - e) oral contraceptive (birth control pills): questionable association
3. history of alcohol abuse
4. coagulopathies
5. leukemia
6. previous stroke
7. history of known vascular abnormalities (AVM, venous angioma...)
8. tumor: known history of cancer, especially those that tend to go to brain (lung, breast, GI, renal, melanoma...)
9. recent surgery: especially carotid endarterectomy, procedures requiring heparin...
10. recent childbirth and/or eclampsia or preeclampsia
11. history of recent trauma

87.5.2 Etiology list

1. “hypertension” (debatable as a cause or effect, see below) but is a risk factor
 - a) acute hypertension (HTN): as may occur in eclampsia (see below) or with use of certain drugs, e.g. cocaine, phenylpropanolamine... (p.1334)
 - b) chronic HTN: possibly causes degenerative changes within blood vessels
2. possibly associated with acutely increased CBF (globally or focally),¹⁵ especially to areas previously rendered ischemic:
 - a) following carotid endarterectomy^{16,17}
 - b) following repair of congenital heart defects in children¹⁸
 - c) previous stroke (embolic¹⁹ or otherwise): hemorrhagic transformation may occur in up to 43% of strokes during the first month.²⁰ May follow dislodgment or recanalization of an arterial occlusion, although it has been demonstrated with persistent occlusion.²¹ May occur as early as ≤ 24 hrs after a stroke in patients with a negative CT done within 6 hours.²² Two types^{20,23}:
 - type 1: diffuse or multifocal. Heterogeneous or mottled appearance within the boundaries of the stroke. Less hyperdense than primary ICH

- type 2: extensive hematoma. Probably unifocal source. As hyperdense as primary ICH and may extend outside the original stroke boundaries. Unlike type 1, classically associated with anticoagulation therapy, and tends to occur in initial few days after stroke and is often associated with clinical worsening. May be difficult to distinguish from primary ICH, and may be frequently misdiagnosed as such²²
- migraine: during²⁴ or following²⁵ a migraine attack (probably an exceedingly rare event)
 - a) following surgery to remove an AVM: “normal perfusion pressure breakthrough.” Some cases may be due to incomplete AVM excision
 - b) physical factors: following strenuous physical exertion,²⁶ exposure to cold²⁷...
- 3. vascular anomalies
 - a) AVM: rupture; see Arteriovenous malformation (p.1239)
 - b) aneurysm rupture
 - saccular (“berry”) aneurysms: (i) aneurysms of the circle of Willis (COW): aneurysms that have become adherent to brain surface by fibrosis as a result of inflammation or previous hemorrhage may produce ICH when they rupture instead of the usual SAH; (ii) aneurysms distal to the COW (e.g. MCA aneurysms)
 - microaneurysms of Charcot-Bouchard (p.1334)
- 4. venous angioma rupture: significant ICH from these common lesions is a very rare event
 - a) “arteriopathies”
 - amyloid angiopathy: usually → repeated lobar hemorrhages (see below)
 - fibrinoid necrosis^{1,28} (sometimes seen in cases of amyloid angiopathy)
 - lipohyalinosis: subintimal lipid-rich hyaline material²⁹
 - cerebral arteritis (including necrotizing angiitis)
- 5. brain tumor (primary or met): below
- 6. coagulation or clotting disorders
 - a) leukemia
 - b) thrombocytopenia:
 - thrombotic thrombocytopenic purpura
 - aplastic anemia
 - c) patients receiving anticoagulation therapy (p.1336)
 - d) patients receiving thrombolytic therapy:
 - for acute ischemic stroke: incidence of symptomatic ICH within 36 hrs of treatment with rtPA is 6.4%(vs. 0.6%in the placebo treated group)³⁰
 - for acute MI or other thrombosis: incidence is $\approx 0.36\text{--}2\%$ ^{31,32,33} Risk is increased with higher doses than the recommended 100 mg of alteplase (Activase®, recombinant tissue plasminogen activator (rt-PA)),³⁴ in older patients, in those with anterior MI or higher Killip class, and with bolus administration (vs. infusion).³⁵ When heparin was used adjunctively, higher doses were associated with higher risk of ICH.³⁶ ICH is thought to occur in those patients with some preexisting underlying vascular abnormality.³⁷ Immediate coronary angioplasty is safer than rt-PA when available³³
 - e) aspirin therapy:
 - one ASA qod was associated with increased risk of ICH,³⁸ with a rate of 0.2–0.8%per year³⁹
 - ASA 100 mg/d did not increase the risk of significant ICH in patients >60 yrs with mild to moderate head injury (GCS $\geq$ 9)⁴⁰
 - Vitamin E supplements⁴¹: associated with reduction of 1 ischemic stroke in 476 individuals, and increase of 1 ICH in 1250 patients taking vitamin E
- 7. CNS infection:
 - a) especially fungal, which attack blood vessels
 - b) granulomas
 - c) herpes simplex encephalitis: may initially produce low density lesions that progress to hemorrhagic ones
- 8. venous or dural sinus thrombosis (p.1308)
- 9. drug related
 - a) substance abuse
 - alcohol: consumption of >3 drinks/day increases the risk of ICH $\approx$ 7-fold (p.1330)
 - drug abuse: especially sympathomimetics (cocaine,^{42,43} amphetamine⁴⁴)
 - b) drugs that raise BP:
 - alpha-adrenergic agonists (sympathomimetics): phenylpropanolamine^{45,46} (may also cause ischemic stroke (p.1287)) which was removed on the order of the FDA from OTC nasal decongestants and appetite suppressants, but other OTC alpha agonists (including phenylephrine, ephedrine,⁴⁷ and pseudoephedrine⁴⁸) are also problematic⁴⁹
 - ephedra alkaloids: sold as a dietary supplement (ma huang) to suppress appetite and increase energy. Associated in case reports with HTN, SAH, ICH, seizures and death⁵⁰

10. post-traumatic: often in a delayed fashion^{51,52}; see Hemorrhagic contusion (p.891)
11. pregnancy related: the risk of ICH in pregnancy and puerperium (up to 6 weeks post partum) is ≈ 1 in 9,500 births⁵³
 - a) most commonly associated with eclampsia or preeclampsia: the mortality of eclampsia is $\approx 6\%$ with ICH being the most frequent direct cause⁵⁴; also see Pregnancy & intracranial hemorrhage (p.1169)
 - b) postpartum ICH (median 8 days, range 3–35 days) in the absence of eclampsia has been reported⁵⁵; when associated with vasculopathy the term postpartum cerebral angiopathy has been used
 - c) vascular findings:
 - some cases associated with isolated cerebral vasculopathy in the absence of systemic vasculitis⁵⁶
 - some cases demonstrate vasospasm
 - some cases show findings (e.g. patchy enhancement in occipital lobes) suggestive of cerebrovascular dysautoregulation (p.1264)
 - some cases show no vascular-related abnormalities
12. post-operative:
 - a) following carotid endarterectomy (see above)
 - b) following craniotomy:
 - at site of craniotomy⁵⁷: risk factors identified: within residual astrocytoma after subtotal resection, following craniotomy for AVM (see above)
 - at site remote from craniotomy. In a series of 37 patients, unlike hematomas at craniotomy site, the following were identified as not being related to risk of hemorrhage: HTN, coagulopathy, CSF drainage, underlying occult lesion, following drainage of chronic SDH (p.899) cerebellar hemorrhage following pterional craniotomy⁵⁸ (this author incriminated possibly rapid overdrainage of CSF), or following temporal lobectomy⁵⁹
13. idiopathic¹⁴

87.5.3 Cerebellar hemorrhage etiologies

Etiologies are similar to ICH of any location, however, some nuances:

1. HTN is a factor in up to two-thirds of cerebellar hemorrhages
2. AVM is a consideration, aneurysm is very rare (possibly AICA aneurysm, but usually only in association with other high-flow lesion, e.g. AVM⁶⁰)
3. may be related to recent previous spinal or supratentorial surgery

87.5.4 Hypertension as a cause?

Hypertension (HTN) is controversial as cause of ICH since the incidence of both ICH and HTN increases with age (66% of patients >65 yrs have HTN). The relative risk for ICH with HTN is 3.9–5.4, depending on the definition of HTN used.⁶¹ Many patients with ICH are dramatically hypertensive on presentation, however, acute elevations of ICP from the hemorrhage may actually precipitate HTN (part of Cushing's triad, see □ Table 56.2). HTN is probably a risk factor primarily for pontine/cerebellar ICH and is probably not a factor in at least 35% of basal ganglion hemorrhages.

87.5.5 Microaneurysms of Charcot-Bouchard

AKA miliary aneurysms.⁶² Occur primarily at bifurcation of small (<300 μ m) perforating branches of lateral lenticulostriate arteries in basal ganglia (found in 46% of hypertensive patients over age 66, but only in 7% of controls⁶³). Possibly the origin of some “hypertensive” ganglionic (putaminal) hemorrhages,⁶⁴ but this is controversial.

87.5.6 (Cerebral) amyloid angiopathy

Cerebral amyloid angiopathy (CAA) AKA congophilic angiopathy. Pathologic deposition of beta amyloid protein (appears as birefringent “apple-green” under polarized light when stained with congo red) within the media of small meningeal and cortical vessels (especially those in white matter) without evidence of systemic amyloidosis.⁶⁵ Some vessels may show fibrinoid necrosis of vessel wall.^{66,67}

CAA should be suspected in patients with recurrent hemorrhages (uncommon with “hypertensive hemorrhages” (p.1330)⁶⁸) that are lobar in location. Gradient-echo MRI may identify petechial hemorrhages or hemosiderin deposits from small cortical hemorrhages which may be associated with CAA.⁶⁹ Less likely in the case of basal ganglion or brain stem hemorrhages.¹⁴

Incidence increases with age: CAA is present in $\approx 50\%$ of those over 70 years of age,⁷⁰ however, most do not hemorrhage. CAA is probably responsible for $\approx 10\%$ of cases of ICH. May be associated with genetic factors (including the apolipoprotein E $\epsilon 4$ allele⁷¹), and may be more prevalent in patients with Down syndrome. Although they are distinct diseases, there is some overlap between CAA and Alzheimer’s disease; the amyloid in CAA is identical to that found in senile plaques of Alzheimer’s disease. CAA may increase the risk of ICH by potentiating plasminogen⁷² (may be of special relevance to patients receiving tissue plasminogen activator (t-PA) to treat MI or stroke).

Patients with CAA may present with a TIA-like prodrome (see below).

Among patients with lobar hemorrhage, those with the apoE $\epsilon 4$ allele typically have their first hemorrhage > 5 yrs earlier than noncarriers (73 ± 8 yrs vs. 79 ± 7 yrs).⁷¹

Diagnostic tests are useful mainly to rule-out other conditions. The definitive diagnosis of CAA requires pathologic evaluation of brain tissue. Criteria for the diagnosis of CAA are shown in □ Table 87.3.⁷³

87.5.7 Hemorrhagic brain tumors

Although any brain tumor can hemorrhage, tumoral ICH is usually associated with malignancies. Tumors on occasion can also produce SAH or subdural hematomas.

Malignant tumors most commonly associated with ICH:

- 1. glioblastoma
- 2. lymphoma
- 3. metastatic tumors
 - a) melanoma^{74,75}: $\approx 40\%$ hemorrhage
 - b) choriocarcinoma^{74,76,77}: $\approx 60\%$ hemorrhage
 - c) renal cell carcinoma
 - d) bronchogenic carcinoma: although only $\approx 9\%$ hemorrhage, this tumor is such a frequent source of cerebral mets that it therefore is a more common source of tumoral ICH

Malignant tumors that hemorrhage less commonly include:

- 1. medulloblastoma^{78,79,80,81} (most commonly in children)
- 2. gliomas^{82,83}

Some benign brain tumors that have been associated with ICH include:

- 1. meningiomas have been associated with intratumoral, subdural, and nearby parenchymal hemorrhage.^{84,85,86,87} Tendency to bleed is similar for angioblastic variety as for other highly vascular meningiomas

Table 87.3 Criteria for the diagnosis of cerebral amyloid angiopathy (CAA) ⁷³	
Diagnosis	Criteria
Definite CAA	Full postmortem exam showing all 3 of the following: a) lobar, cortical, or corticosubcortical hemorrhage b) severe CAA c) absence of another diagnostic lesion
Probable CAA with supporting pathological evidence	Clinical data & pathological tissue showing all 3 of the following: a) lobar, cortical, or corticosubcortical hemorrhage b) some degree of vascular amyloid deposition in specimen c) absence of another diagnostic lesion
Probable CAA	Clinical data and MRI findings showing all 3 of the following: a) age ≥ 60 yrs b) multiple hemorrhages restricted to the lobar, cortical, or corticosubcortical region c) absence of another cause of hemorrhage^a
Possible CAA	Clinical data and MRI findings: a) age ≥ 60 yrs b) single lobar, cortical, or corticosubcortical hemorrhage without another cause^a, or multiple hemorrhages with a possible but not a definite cause^a or with some hemorrhages in an atypical location (e.g. brain stem)
^a e.g. excessive anticoagulation (INR >3.0), head trauma, ischemic stroke, CNS tumor, cerebrovascular malformation, vasculitis or blood dyscrasia	

2. pituitary adenoma, see Pituitary apoplexy (p.720)
3. oligodendroglioma (relatively benign): rarely presents with hemorrhage,⁸⁸ classically after years of causing seizures
4. hemangioblastoma⁸⁹
5. vestibular schwannoma^{90,91,92}
6. cerebellar astrocytoma⁹³

87.5.8 Anticoagulation preceding ICH

10% of patients on warfarin (Coumadin®) develop a significant bleeding complication per year (not all are intracranial), including ICH (65% mortality in this group). The risk of ICH in patients treated with warfarin for A-fib varies between 0–0.3% per year³⁹ (historically, this was as high as $\approx 1.8\%$ in older studies⁹⁴ from the 1960s and 1970s), but when an elderly subgroup (mean age 80 yrs) was analyzed, this rate was 1.8% per year.³⁹ ICH was the only cause of fatal bleeding complications of warfarin therapy in one series where the cumulative risk of a fatal hemorrhage was 1% at 1 year and 2% at 3 yrs.⁹⁵

The risk of hemorrhagic complications was increased with the length and also the variability of the PT, and during the first three months of anticoagulation.⁹⁵ Patients with cerebral amyloid angiopathy (CAA) (see above) are also at increased risk of ICH following administration of antiplatelet drugs or anticoagulants.⁷³

87.6 Clinical

87.6.1 General information

In general, the neurologic deficit with ICH is characterized by a smooth progressive onset over minutes to hours, unlike embolic/ischemic stroke where deficit is maximal at onset. With ICH, severe headache, vomiting and alterations in level of consciousness may be more common (H/A may not be more prevalent than in embolic stroke, but it is often a first and prominent symptom¹⁴).

87.6.2 Prodrome

TIA-like symptoms may precede lobar hemorrhages^{96,97} in patients with CAA, and may occur in up to $\approx 50\%$ of patients for whom a complete history is obtainable. Unlike typical TIAs, these usually consist of numbness, tingling or weakness (corresponding to the area where the hemorrhage will subsequently occur) that gradually spreads in a manner reminiscent of a Jacksonian-march and may spill-over vascular territories (probably an electrical phenomenon rather than an ischemic event). This is suggestive of but not pathognomonic for the subsequent development of lobar ICH.

87.6.3 Concomitants of specific lesions in ICH

Putaminal hemorrhage

The most common site for ICH. Smooth gradual deterioration in 62% (maximal deficit at onset in 30%); never fluctuating. Contralateral hemiparesis, may progress to hemiplegia or even coma or death. H/A in 14% at onset. No H/A at any time in 72%. Papilledema and subhyaloid preretinal hemorrhage are rare.

Thalamic hemorrhage

Classically, contralateral hemisensory loss. Also hemiparesis when the internal capsule is involved. Extension into upper brain stem $\rightarrow$ vertical gaze palsy, retraction nystagmus, skew deviation, loss of convergence, ptosis, miosis, anisocoria, $\pm$ unreactive pupils. H/A in 20–40%. Motor deficit similar to putaminal hemorrhage, but contralateral sensory deficit widespread and striking. Hydrocephalus may occur from compression of CSF pathways.

In 41 patients, when hemorrhage > 3.3 cm on CT, all died. Smaller hematomas usually caused permanent disability.

Cerebellar hemorrhage

May include any combination of the following:

1. symptoms of increased ICP (lethargy, N/V, HTN with bradycardia...) due to hydrocephalus which may occur as a result of:
 - a) compression of the 4th ventricle → obstruction of CSF
 - b) extension of the hemorrhage into the ventricular system
2. direct compression of brain stem may produce:
 - a) facial palsy: due to pressure on the facial colliculus
 - b) these patients classically become comatose without first having hemiparesis, unlike many supratentorial etiologies

Lobar hemorrhage

Syndromes associated with hemorrhage in the 4 cerebral lobes¹⁴ (≈ 50% have H/A as a first and prominent symptom):

1. frontal lobe (the most distinctive of the syndromes): frontal H/A with contralateral hemiparesis, usually in the arm with mild leg and facial weakness
2. parietal lobe: contralateral hemisensory deficit and mild hemiparesis
3. occipital lobe: ipsilateral eye pain and contralateral homonymous hemianopsia, some may spare superior quadrant
4. temporal lobe: on dominant side, produces fluent dysphasia with poor auditory comprehension but relatively good repetition

87.6.4 Delayed deterioration

General information

Deterioration after the initial hemorrhage is usually due to any combination of the following:

1. rebleeding: see below
2. edema: see below
3. hydrocephalus: higher risk with intraventricular extension or posterior fossa ICH
4. seizures

Rebleeding or extension of bleed

Early rebleeding: Rebleeding (more so in basal ganglion hemorrhages than in lobar hemorrhages) has been documented during the first hour by “ultra-early” scanning and repeating CT scans. Rebleeding is usually accompanied by clinical deterioration.⁹⁸ The incidence of hematoma enlargement decreases with time, 33–38% in 1–3 hours,⁹⁹ 16% in 3–6 hrs, and 14% between 24 hrs of onset and a second CT within 24 hrs of the first.¹⁰⁰ Patients with enlarging hematomas were more likely to have larger hematomas and/or coagulopathy, and had a worse outcome.¹⁰⁰ Rebleeding may still occur following surgical evacuation of clot even with satisfactory intraoperative hemostasis. Hemostatic agents (e.g. NovoSeven®) may reduce this risk (p.1339). The “spot sign”¹⁰¹ on CTA (small enhancing foci within acute ICH) correlated with increased risk of hematoma expansion.

Late rebleeding: Quoted rates for late rebleeding from ICH range from 1.8–5.3% (depending on length of follow-up).¹⁰² Diastolic BP was significantly higher in the group with recurrent hemorrhage, with a 10%/yr risk for DBP > 90 mm Hg vs. < 1.5% for DBP ≤ 90 (mean F/U of 67 months).¹⁰² Other risk factors include diabetes and tobacco and alcohol abuse.¹⁰³ Recurrent hemorrhages may indicate underlying vascular malformations or amyloid angiopathy (lobar rebleeding is likely to be due to amyloid angiopathy¹⁰³).

Edema

Edema and ischemic necrosis around the hemorrhage may cause delayed deterioration.¹ Although necrosis from mass effect of the clot contributes a small part to the edema, experiments indicate that by itself, the mass effect is insufficient to account for the amount of edema that occurs. It is believed that an edemogenic toxin is released from the clot. Experiments with various components of blood clots has disclosed that thrombin concentrations that could be released from the clot causes increased permeability of the blood-brain-barrier, and is also a potent vasoconstrictor. This is the leading suspect as the major cause of delayed edema and deterioration. Also see Cerebral edema (p.90).

87.7 Evaluation

87.7.1 CTscan

CT scan is rapid, and easily demonstrates blood as high density within the brain parenchyma immediately after hemorrhage. Although mass effect is common, the tendency for the hemorrhage to dissect through brain tissue often results in less mass effect than would be anticipated from the size of the clot.

Clot volume carries prognostic significance (p.1343). It can be measured volumetrically using computer algorithms available on some CT scanners, or it can very simply be approximated by the ellipsoid method¹⁰⁴ (originally developed for AVMs, based on the principal that the volume of an ellipsoid is approximately half of that of a parallelepiped into which it is placed)¹⁰⁵ and is simpler than other slightly more accurate estimation methods¹⁰⁶ as shown in Eq (87.1), where AP, LAT and HT are the diameters of the clot in each of the 3 dimensions (antero-posterior, lateral, and height). To estimate height of a lesion when only axial images are available (as on most initial CTs), count the number of images on which the lesion is seen, and multiply by the slice thickness of the CT cuts^{104,106,107}(this information is usually printed on the CT), or, subtract the table position of the highest cut that shows the clot from the table position of the lowest cut showing clot.

ellipsoid volume %
$$\frac{AP \times LAT \times HT}{2}$$

87:1P

On the average, the size of the clot decreases ≈ 0.75 mm/day, and the density decreases by ≈ 2 CT units/day, with little change for 1st 2 wks.

87.7.2 MRI

Usually not the procedure of choice for initial study. Does not show blood well within the first few hours. Difficult to ventilate or access patient during the study. Slower and more expensive than CT. May be useful later, e.g. to help diagnose cerebral amyloid angiopathy (CAA) (p.1334).

The appearance of ICH on MRI is very complicated. It is highly dependent on the age of the clot¹⁰⁸ with 5 stages identified (□ Table 87.4).

87.7.3 Cerebral angiography

For making the diagnosis of the ICH itself, angiography cannot reliably differentiate the mass effect from an ICH from that due to an ischemic infarct or tumor.¹⁰⁹ May demonstrate AVMs and aneurysms when they are associated with the ICH. The yield may be increased by delaying the study.¹⁴

Table 87.4 Variation of MRI appearance of ICH with time since hemorrhage^{a 108}

Stage	Age	Condition of hemoglobin	T1 WI	T2 WI
hyperacute	<24 hrs	oxy-Hgb (intracellular)	iso	sl. ↑
acute	1–3 d	deoxy-Hgb (intracellular)	sl. ↓	very ↓
subacute				
• early	>3 d	met-Hgb (intracellular)	very ↑	very ↓
• late	>7 d	met-Hgb (extracellular ^b)	very ↑	very ↑
chronic				
• center	>14 d	hemichromes ^c (extracellular)	iso	sl. ↑
• rim		hemosiderin (intracellular)	sl. ↓	very ↓

^aAbbreviations: oxy-Hgb =oxyhemoglobin, deoxy-Hgb =deoxyhemoglobin, met-Hgb =methemoglobin, iso = iso-intense to brain, ↓ =hypo-intense, ↑ =hyperintense, sl=slightly
^bwhen the RBCs lyse, the Hgb becomes extracellular
^cdiamagnetic (non-paramagnetic) heme derivatives

May demonstrate vascular blush in some cases of tumor. Normal arteriography cannot eliminate cerebral amyloid angiopathy as the etiology of ICH in the elderly.¹¹⁰
For indications for cerebral angiography in ICH, see below.

87.7.4 ICH score

The system of Hemphill et al.¹¹¹ assigns points based on 5 features as indicated in □ Table 87.5. The points are then summed for the “ICH score.” The associated 30 day mortality is tabulated in □ Table 87.6.

87.8 Initial management of ICH

87.8.1 Outline

(The following assumes the diagnosis has already been made, usually on CT scan.) There is not uniform agreement on almost all aspects of the management of ICH from the optimal BP to the indications for surgery. The following is offered as a guide.

Most aspects of management are controversial. The following is offered as a guide.

- 87
1. patients should be managed in an ICU
 2. HTN: controversial. Issues: HTN may contribute to further bleeding, especially within the first hour.⁹⁸ However, some HTN may be needed to maintain perfusion. Some say reduce MAP to pre-morbid level if known, or by $\approx 20\%$ if unknown. Note: A study of 8 ICHs showed autoregulation was maintained, but with an elevated lower limit. However, CBF fell when MAP was lowered pharmacologically below the usual MAP, which averaged 80% of the admission MAP (admission HTN followed the ICH).¹¹²
 3. intubate if stuporous or comatose
 4. maintain euglycemia
 5. maintain normothermia
 6. anticonvulsants
 - a) seizures are treated with appropriate AEDs
 - b) prophylactic AEDs: optional. May decrease risk of early seizures in patients with lobar hemorrhages
 - c) AED options
 - Keppra has a very favorable therapeutic/toxic profile. Dose 500 mg BID
 - OR phenytoin: load with 17 mg/kg slow IV over 1 hour, follow with 100 mg q 8 hrs; see phenytoin (PHT, Dilantin®) (p.446)

Table 87.5 ICH Score ¹¹¹		
Feature	Finding	Points
GCS score (□ Table 18.1)	3–4	2
	5–12	1
	13–15	0
Age ^a	≥ 80 years	1
	<80	0
Location	infratentorial	1
	supratentorial	0
ICH volume see (87.1)	≥ 30 cc	1
	<30 cc	0
Intraventricular blood	yes	1
	no	0
“ICH Score” = Total Points		0–6
^a possible bias since treatment decisions in elderly patients may have differed from younger patients		

Table 87.6 Mortality based on ICH Score		
ICH Score ^a	30 day mortality	
0	0%	(26 pts)
1	13%	(32 pts)
2	26%	(27 pts)
3	72%	(32 pts)
4	97%	(29 pts)
5	100%	(6 pts)
6	? 100% ^b	(0 pts)
^a from □ Table 87.5		
^b no pt. in the study had a score of 6, but “it is expected this would be associated with high rate of mortality”		

- 7. hemostatic issues:
 - a) check INR (or PT), PTT & platelet count (PC), platelet function assay (PFA)
 - correct coagulopathies, see Correction of coagulopathies or reversal of anticoagulants (p. 166)
 - platelets: correct thrombocytopenia or platelet inhibiting drugs as discussed below
 - b) bleeding time: not generally helpful
 - c) □ hemostatic agents: NovoSeven® (recombinant activated coagulation factor VII (rFVIIa)) given IV within 4 hours of onset,¹¹³ see below
- 8. steroids: controversial. No benefit from dexamethasone in ICH, with significantly more complications (primarily infectious, GI bleeding and diabetogenic).¹¹⁴ Consider use if significant perihemorrhage edema on imaging (suggested dosage¹¹⁵: 4 mg dexamethasone IV q 6 hrs, tapered over 7–14 days)
- 9. treat intracranial hypertension presumptively: mannitol and/or furosemide as tolerated, also helps with HTN; for more, see Treatment measures for elevated ICP (p. 866). If significant problems from suspected increased ICP, consider ICP monitor
- 10. external ventricular drain (EVD): for hydrocephalus, some cases of intraventricular blood, or to manage ICP (see below). □ R/O coagulopathy before placing
- 11. follow electrolytes and osmolarity
 - a) aggressively treat hyperglycemia (insulin drip if problematic)
 - b) watch for SIADH (p. 114)
- 12. angiography: primarily to R/O underlying vascular malformation, but also to R/O aneurysm (a less common cause of ICH), and tumor (which is usually better diagnosed on contrast CT or MRI)
 - a) if urgent surgery is indicated (e.g. for herniation), the delay in obtaining an angiogram may be detrimental and it may be best deferred to post-op
 - b) □ indications: angiography is recommended except for patients >45 yrs of age with preexisting hypertension and ICH in thalamus, putamen or posterior fossa because there was a 0% yield out of 29 patients in this group¹¹⁶ and low yield in all patients with isolated deep ICH¹¹⁷
 - patients >45 yrs with a history of HTN and a lobar ICH: angiography had a 10% yield,¹¹⁶ with the ratio of AVM:aneurysm ≈ 4.3:1
 - patients with intraventricular hemorrhage (without parenchymal hematoma): the yield of angiography was ≈ 65%¹¹⁶ primarily AVM
 - c) an underlying lesion may be obliterated by ICH, especially acutely. If initial angio is negative, repeat after CT shows resorption of clot (≈ in 2–3 mos). If still negative, follow CT or MRI q 4–6 mos for ≈ 1 year to R/O tumor.¹ Delaying the initial angiogram for several weeks may increase the yield and is also an option¹⁴
 - d) the literature indicates that MRI/MRA has only ≈ 90% sensitivity for detecting structural abnormalities in this setting, and so a negative study cannot completely exclude this possibility¹¹⁶
 - e) the yield of angiography in ICH would be expected to be lower in patients at increased risk of ICH: patients on warfarin (Coumadin®), chronic alcoholics, patients with amyloid angiopathy...

Σ

Treat HTN. Suggested target BP $\approx$ 140/90. Avoid overcorrection (relative or absolute hypotension)

87.8.2 Thrombocytopenia or platelet inhibiting drugs

1. thrombocytopenia: although platelet transfusions are generally recommended only for PC<50K, ICH is so serious that a suggestion is ideally to keep PC>100K (if this is difficult to attain, aim for platelet count >75K)
2. patients on platelet inhibiting drugs (e.g. aspirin or Plavix®) should receive platelets
3. when needed: start with 6 units of platelets; see Platelets (p.154)

87.8.3 NovoSeven® (recombinant activated coagulation factor VII (rFVIIa))

At the site of a tissue factor (TF) bearing cell, rFVIIa forms a complex with TF resulting in thrombin production. It also converts factor X to its active form, Xa on the surface of activated platelets resulting in a “thrombin burst” at the site of damage.¹¹⁸ Half life: 2.6 hrs. Expensive ($\approx$ \$10,000 per dose).

FDA approved for various bleeding diatheses (including hemophiliacs with antibodies to factor VIII or IX). Phase II “off label” Factor Seven for Acute Hemorrhagic Stroke (FAST) study for ICH¹¹³ appeared promising, however, preliminary results of the phase 3 trial showed no difference in death or major disability at 90 days.

□ for ICH. Doses studied: 40, 80 & 160 mcg/kg IV over 1–2 minutes given IV within 4 hours of symptom onset reduces 90 day morbidity & mortality, with a dose-related reduction in mean increase of ICH volume at 24 hrs, and a small increase in thrombotic complications (studied in patients with GCS>5, with no plan for surgical evacuation within 24 hours and no history of thrombotic or vaso-occlusive disease). Side effects: thrombotic events (MI, stroke...) primarily with higher doses ($\geq$ 120 mcg/kg),¹¹⁹ risk may be increased in presence of DIC, predisposing coagulopathy, advanced atherosclerotic disease, crush injury, septicemia, or concomitant treatment with activated or nonactivated prothrombin complex concentrates (aPCC/PCCs) due to increased levels of circulating TF.

87.8.4 Anticoagulation following ICH

Patients with ICH who subsequently require anticoagulation (e.g. for embolic ischemic stroke or for mechanical heart valve) pose a management dilemma. In the case of embolic disease, the fear of converting an ischemic infarct to a hematoma or increasing the size a small ICH with continued anticoagulation has traditionally outweighed the possible benefit of protection from further embolization. However, an anecdotal (retrospective uncontrolled) report of 12 such patients found no incidence of increased intracranial bleeding with either continued anticoagulation (6 patients) or resumption of anticoagulation after an hiatus (several days in 4 patients, 5 days in 1, and 14 days in 1).¹²⁰ In another study¹²¹ none of 35 patients who had resumption of warfarin had recurrent intracranial hemorrhage (ICH, SAH or subdural hematoma). While this does not prove that anticoagulation is safe after ICH, it does demonstrate that if there is a strong indication for anticoagulation, and if there is not an acceptable alternative (e.g. Greenfield filter for DVT (p.170)), that anticoagulation in this setting is not always met with disastrous results.

The probability of having an ischemic stroke at 30 days following cessation of warfarin for a median of 10 days using Kaplan-Meier survival estimates are approximately 2.9% for patients who had originally been treated with warfarin for prosthetic heart valves, 2.6% for those treated for atrial fibrillation, and 4.8% for those treated for cardioembolic stroke.¹²¹ These numbers may be gross underestimates as many patients died within 2 weeks, and follow-up imaging was scant¹²²; another study¹²³ showed a much higher rate of 20% see Cardiogenic brain embolism (p.1304) for more details.

Antiplatelet therapy after ICH is not associated with a substantially increased risk of recurrent ICH¹²⁴ (prospective cohort study).

Recommendations

A-fib: long-term anticoagulation should be avoided after ICH.¹²⁵
Mechanical heart valves: 1–2 weeks off anticoagulation (to observe ICH, or to evacuate a SDH or clip an aneurysm).^{121,126} Patients with deep hemispheric ICH at high-risk for thromboembolic stroke may benefit from resumption of long-term anticoagulation).¹²⁵
Patients requiring hemodialysis after ICH: heparin-free dialysis may be used.

87.8.5 Ventriculostomy (IVC) AKA external ventricular drainage (EVD)

- Indications:
1. intraventricular extension of blood causing acute obstruction of the third ventricular outlet. In these cases, the IVC is usually placed in the lateral ventricle contralateral to the hemorrhage (to avoid putting the catheter directly in clot, which may obstruct the inlets). The prognosis for patients with a significant volume of intraventricular blood is poor. It may be difficult to maintain the patency of the catheter due to occlusion by clot, tissue plasminogen activator may help (see below)
 2. acute hydrocephalus
 3. fICP management

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87.9 Surgical treatment

87.9.1 General information

The first successful evacuation of an intracerebral hematoma was reported by MacEwan in 1888.¹²⁷ The patient recovered completely from an upper extremity monoplegia.

Booking the case: Craniotomy for ICH

Also see defaults & disclaimers (p. 27).

1. position: (depends on location of bleed)
2. equipment:
 - a) microscope (not used for all cases)
 - b) image guided navigation (not typically used)
3. post-op: ICU
4. consent (in lay terms for the patient – not all-inclusive):
 - a) procedure: surgery through the skull to remove blood clot, stop any bleeding identified, possible placement of external (ventricular) drain
 - b) alternatives: nonsurgical management
 - c) complications: usual craniotomy complications (p.28) plus further bleeding which may cause problems (especially in patients taking blood thinners, antiplatelet drugs including aspirin, or those with coagulation abnormalities or previous bleeds) and may require further surgery, areas of the brain that have already been damaged by the bleeding are not likely to recover, hydrocephalus

87.9.2 Indications for surgery

General information

Amazingly, after repeated attempts to resolve this dilemma, considerable controversy persists regarding indications for surgery. Surgery may lower morbidity from rebleeding (especially if an aneurysm or AVM is identified as the cause of the ICH), edema, or necrosis from mass effect of hematoma (unproven), but rarely causes neurologic improvement. Meta-analyses^{128,129} yield inconclusive or conflicting results and could not identify whether there was a favorable effect of surgery, the types of ICH and patients that are likely to benefit, and the relative effectiveness of the various available surgical options.

Randomized prospective studies (RPS) in the current CT/surgical era

One RPS¹³⁰ found lower mortality for patients with GCS 7–10 treated surgically (note: only 20% of these patients were operated on <8 hrs from the bleed, and the mean time for all patients to

operation was 14.5 hours (range: 6–48 hrs), which may be long). However, survivors in this group were all severely disabled (none were independent).

Another¹¹⁵ found no benefit from surgery for putaminal hemorrhages, also with poor outcomes in all patients.

International STICH¹³¹: enrolled 1,033 patient. Study shortcomings: possible selection bias (the responsible neurosurgeon had to be uncertain of the benefits of medical vs. surgical treatment), “early surgery” had a somewhat long median time to treatment of 30 hours, and 26% of medically treated patients crossed over and had surgery at a mean of 60 hours (late). Given these limitations, the conclusion was that for supratentorial ICH there was no benefit of early surgery (although there may have been some benefit in the subgroup with a hematoma within 1 cm of the cortical surface). This trial may be more accurately considered to be a comparison of early vs. delayed surgery in patients subjectively judged to need surgery by the investigator.

Conclusion

The decision to operate therefore must be individualized based on patient’s neurologic condition, size and location of hematoma, patient’s age, and the patient’s expressed preferences (e.g. by a “living will”) and the family’s wishes concerning “heroic” measures in the face of catastrophic illness.

Guidelines for considering surgery vs. medical management

(for separate indications for surgery for cerebellar hemorrhage, see below)

1. NON-SURGICAL: factors that favor medical management
 - a) minimally symptomatic lesions: e.g. alert patient with subtle hemiparesis (especially patients with $GCS > 10$ ¹³⁰)
 - b) situations with little chance of good outcome
 - high ICH score (p. 1339), which overlaps with the following
 - massive hemorrhage with significant neuronal destruction (see below)
 - large hemorrhage in dominant hemisphere
 - poor neurologic condition: e.g. comatose with posturing (i.e. $GCS \leq 5$), loss of brain stem function (fixed pupils, posturing...)
 - $\approx$ age > 75 yrs: do not do well with surgery for this
 - c) severe coagulopathy or other significant underlying medical disorder(s): in the event of herniation, rapid decompression surgery may be considered in spite of the risks
 - d) basal ganglion (putaminal) or thalamic hemorrhage: surgery is no better than medical management, and both have little to offer^{115,132} (see below)
2. SURGICAL: factors that favor rapid surgical removal of the blood clot
 - a) lesions with marked mass effect, edema, or midline shift on imaging (removal is considered due to the potential for herniation)
 - b) lesions where the symptoms (e.g. hemiparesis/plegia, aphasia, or sometimes just confusion or agitation...) appear to be due to increased ICP or to mass effect (i.e. compression) from the clot or surrounding edema. Symptoms attributable directly to brain injury from the hemorrhage are unlikely to be reversed by surgical evacuation
 - c) volume: surgery for moderate volume hematomas (i.e. ≈ 10 –30 cc, Eq (87.1)) may be more appropriate than with:
 - ☐ small clot (< 10 cc): mass effect from clot + edema is usually not significant enough to require surgery
 - ☐ large clot: > 30 cc: associated with poor outcome (only 1 of 71 patients could function independently at 30 days¹³³)
 - ☐ massive hemorrhage > 60 cc with $GCS \leq 8$: 91% 30-day mortality¹³³
 - ☐ massive hemorrhage > 85 cc (the volume of a sphere with a diameter of 5.5 cm): no patient survived, regardless of treatment in one series¹³⁴
 - d) persistent elevated ICP in spite of therapy (failure of medical management). Evacuating clot definitely lowers ICP, but the effect on outcome is uncertain
 - e) rapid deterioration (especially with signs of brain stem compression) regardless of location in a patient considered to be salvageable
 - f) favorable location, for example:
 - lobar (as opposed to deep hemispheric): in spite of optimistic results in a non-randomized study done in 1983 indicating good outcomes in patients with deep hemorrhages treated with early surgery,⁶⁴ a later randomized study failed to confirm this benefit¹¹⁵
 - cerebellar: see below

- external capsule
 - non-dominant hemisphere
- g) young patient (especially age ≤ 50 yrs): they tolerate surgery better than elderly patients, and, unlike elderly patients with brain atrophy, they also have less room in the head to accommodate the mass effect of clot + edema
- h) early intervention following hemorrhage: surgery after 24 hrs from onset of symptoms or deterioration may be of less benefit¹³⁰

Management of cerebellar hemorrhage

Recommendations¹³⁵:

1. patients with a Glasgow Coma Scale (GCS) score ≥ 14 and hematoma < 4 cm diameter: treat conservatively
2. patients with GCS ≤ 13 or with a hematoma ≥ 4 cm: surgical evacuation
3. patients with absent brain stem reflexes and flaccid quadriplegia: intensive therapy is not indicated. Note: some authors contend that the loss of brain stem reflexes from direct compression may not be irreversible,¹³⁶ and that cerebellar hemorrhage represents a surgical emergency (and that the above criteria would thus deny potentially helpful surgery to some, see discussion of cerebellar infarction and decompression (p.1302)).
4. patients with hydrocephalus: ventricular catheter (if no coagulopathy). Caution: do not overdrain to avoid upward cerebellar herniation (p.303). Most cases with hydrocephalus also require evacuation of the clot

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87.9.3 Surgical considerations

General recommendations

1. send specimens (hematoma, abnormal looking tangle of blood vessels if present, and possibly biopsy walls of hematoma cavity) to pathology for analysis¹³⁷ (to rule-out tumor, AVM, amyloid angiopathy...)
2. surgical options:
 - a) “standard approach”: craniotomy with evacuation of the clot under direct vision (with or without microscope)
 - b) stereotactic aspiration with thrombolytic agents has also been used; see Stereotactic surgery (p.1441), evacuation of intracerebral hemorrhage
 - c) endoscopic surgery¹³⁸

Surgical techniques for cerebellar hemorrhage

1. position: lateral oblique (p.1446) with the involved side up
2. if rapidity is crucial, a midline skin incision is preferred because it can be taken down quickly with little fear of encountering a vertebral artery
3. craniectomy (without bone replacement) is preferred over craniotomy to accommodate post-op swelling
4. a prophylactic Frazier burr hole is recommended to allow rapid treatment if post-op hydrocephalus develops – see placement (p.1450) and use (p.1452) -, or a ventricular catheter may be placed to monitor ICP and allow CSF drainage post-op
5. in cases where there has been rupture into the ventricular system, the surgical microscope should be used to follow the clot to the fourth ventricle which is then cleared of clot

Intraventricular tissue plasminogen activator (rt-PA)

Intraventricular rt-PA may help lyse clot and maintain catheter patency or reopen a clotted catheter. No well-designed randomized study has been done; but anecdotal evidence suggests it is relatively safe. □ In cases of suspected aneurysm, AVM or other vascular malformation, it cannot be used until the source of bleeding has been corrected.^{139,140}

□: 2–5 mg of rt-PA^{139,141,142} in NS is administered through an intraventricular catheter (IVC). The IVC is closed for 2 hours after injection.¹⁴² In the low dose CLEAR-IVH (Clot Lysis: Evaluating Accelerated Resolution of Intraventricular Hemorrhage) trial (a phase II trial with 52 patients), 1 mg tPA intrathecally via a ventricular catheter every 8 hours up to a maximum of 4 days, was associated with a 30 d mortality of 15% (compared to an expected 80–85%).¹⁴³ Hemorrhagic complication rate

was 6% A phase III trial<http://clinicaltrials.gov/ct2/show/NCT00784134?term=clear+III+stroke&rank=1> is underway to confirm this.

87.10 Outcome

Thalamic hemorrhages that tend to destroy the internal capsule (IC) are more likely to produce hemiplegia than hemorrhages lateral to the IC that compress but do not disrupt the IC.

Mortality: The chief cause of death (in a series testing the effects of dexamethasone) is cerebral herniation,¹¹⁴ occurring mainly during the first week and mostly in patients with initial Glasgow Coma Scale scores ≤ 7 . The in-hospital death rate decreased overall during the 1980s but increased for patients ≥ 65 years of age.³

Quoted mortality rates vary widely, and depend on size and location of clot, age and medical condition of the patient, and etiology of the hemorrhage. Overall, the 30-day mortality rate is $\approx 44\%$ for ICH,² which is similar to that for SAH ($\approx 46\%$). Patients with lobar hemorrhages (p.1336) tend to fare better than deep ICH (basal ganglion, thalamus...) with only $\approx 11\%$ mortality in 26 patients.¹⁴

87.11 ICH in young adults

87.11.1 General information

In a review of 72 patients age 15–45 yrs suffering nontraumatic ICH,¹⁴⁴ a presumed cause was found in 76% (□ Table 87.7). 3 patients had labor or post-partum hemorrhages (p.1334); see also Pregnancy & intracranial hemorrhage (p.1169).

AVM: lobar hemorrhages in this age group are highly suggestive of AVM. Of 40 lobar hemorrhages, 37.5% were determined to be from AVMs.¹⁴⁴

Herpes simplex encephalitis: may produce hemorrhagic changes on CT, especially in the temporal lobes; see Herpes simplex encephalitis (p.364).

Drug abuse: especially with sympathomimetics such as cocaine (p.1333) should also be considered in young adults.

Leukemia: ICH may be the initial presentation of leukemia in a young adult (may be due to metastases (chloroma) or to thrombocytopenia).

87.11.2 Outcome

Overall in-hospital survival (including those treated medically) was 87.5%

Table 87.7 Causes of spontaneous ICH in young adults ¹⁴⁴	
Etiology	%
ruptured AVM	29.1%
arterial hypertension	15.3%
ruptured saccular aneurysm	9.7%
sympathomimetic drug abuse	6.9%
tumor ^a	4.2%
acute EtOH intoxication	2.8%
pre-eclampsia/eclampsia	2.8%
superior sagittal sinus thrombosis	1.4%
moyamoya	1.4%
cryoglobulinemia	1.4%
undetermined	23.6%
^a hemangioma, ependymoma, metastatic choriocarcinoma...see Hemorrhagic brain tumors (p.1335)	

87.12 Intracerebral hemorrhage in the newborn

87.12.1 General information

Occurs primarily in premature infants. Alternate terms: subependymal hemorrhage (SEH), germinal matrix hemorrhage (GMH), periventricular-intraventricular hemorrhage (PIVH). Intraventricular hemorrhage (IVH) arises from extension of SEH through ependymal lining of ventricle and occurs in 80% of cases of SEH.¹⁴⁵

87.12.2 Etiology

The highly vascular germinal matrix is part of the primordial tissue of the developing brain and is the source of future neurons and glial cells. It is located just beneath the ependymal lining of the lateral ventricles, and undergoes progressive involution until 36 weeks gestational age (GA). Thus, the matrix may persist out of utero in premature infants. A disproportionate amount of the total CBF perfuses the periventricular circulation through these capillaries which are immature and fragile and have impaired autoregulation.^{146,147} The site of hemorrhage is age dependent. Between 24–28 weeks GA they occur over the body of the caudate nucleus and at 29 weeks GA or greater they arise over the head of the caudate nucleus.¹⁴⁸

87.12.3 Pathogenesis of PIVH in the pre-term infant

The metabolically active GM is susceptible to hypotension and hypoperfusion which can lead to infarction. The GM is a vulnerable watershed zone supplied by Heubner's artery (from the anterior cerebral artery), terminal branches of the lateral striate arteries (off the middle cerebral artery) and the anterior choroidal artery (off the internal carotid or middle cerebral artery).

1. postnatal hypoxia due to respiratory distress syndrome related to hyaline membrane disease, pneumothorax and/or anemia can deprive the metabolically active GM of oxygen. This ischemia to the endothelial cells lining the capillaries makes them vulnerable to infarction and then disruption
2. hypercapnia maximally dilates the thin walled vessels of the GM. If this is followed by sudden increases in perfusion the result can be rupture of the vessels
3. increased venous pressure from any cause (labor and delivery, positive pressure ventilation, stimulation, endotracheal suctioning, myocardial failure from ischemia) can result in increased venous pressure in the GM leading to hemorrhage
4. dehydration followed by rapid resuscitation with hyperosmolar solutions increases the intravascular volume by osmotically encouraging the movement of fluid from tissues into the intravascular space. With associated increases in systemic blood pressure the GM capillaries are at increased risk of rupture

87.12.4 Risk factors for PIVH

Increased cerebral perfusion pressure (CPP) with the associated increased cerebral blood flow (CBF) and hypoxia are the common denominators for most risk factors for PIVH. The elevated pressure may cause the hemorrhage by rupturing the fragile vessels of the germinal matrix, possibly already damaged by previous insults of high or fluctuating CBF and hypoxia.

Risk factors for PIVH include¹⁴⁹:

1. those associated primarily with increased CBF or CPP:
 - a) asphyxia: including hypercapnia (see above)
 - b) rapid volume expansion
 - c) seizures
 - d) pneumothorax
 - e) cyanotic heart disease (including PDA)
 - f) infants being mechanically ventilated having RDS and fluctuating CBF velocity documented by Doppler flow meter¹⁵⁰
 - g) anemia
 - h) decreased blood glucose
 - i) arterial catheterization
 - j) blood pressure fluctuations
2. younger gestational age (GA)
3. low birth weight
4. acute amnionitis

- 5. failure to give antenatal steroids (p.1347) during the 48 hours prior to pre-term delivery¹⁵¹ (i.e. to women at risk of delivering low birth-weight infants):
- 6. APGAR's <4 at 1 minute and <8 at 5 minutes
- 7. acidosis
- 8. coagulopathies
- 9. general anesthesia for C-section
- 10. extracorporeal membrane oxygenation (ECMO): due to heparinization in addition to increased CPP
- 11. maternal cocaine abuse¹⁵²
- 12. maternal aspirin use

87.12.5 Epidemiology

Incidence

Depends on the method used for detection (many PIVHs are asymptomatic) and the population being evaluated. 540,000 pre-term infants are born in the United States annually. 85,000 are very pre-term (<32 weeks GA) and 385,000 are late pre-term (34–36 weeks GA). 63,000 very low birth weight (<1500 grams) infants are born each year. Of the preemies weighing <1500 gm birth weight, 20–25% will suffer from a PIVH.^{153,154}

In a 1978 study, PIVH was found by CT in 43% (20/46) of preemies with birth-weight <1500 gm.¹⁵⁵ Mortality in infants with PIVH was 55% compared to 23% in those without PIVH.¹⁵⁵ Ultra-sound (U/S) detected PIVH in 90% of 113 preemies <34 weeks gestation¹⁵⁶ (49% were grade III or IV, see □ Table 87.8 for grading).

Timing

The timing of PIVH has a bimodal distribution. A substantial number occur within 6 hours of birth with 50% occurring within 12 hours of birth.^{157,158} At postnatal days 3–4, a second peak occurs. Only 5% of bleeds will develop after postnatal day 4. Progression of hemorrhage has been documented in 10–20% of infants.¹⁵⁸ Early onset PIVH is more likely to progress and has a higher mortality.¹⁵⁹

87.12.6 Prevention

Numerous studies have been conducted to find a method of directly reducing the incidence of PIVH among premature infants. Many are controversial. Optimal resuscitation and neonatal care, with an emphasis on measures which minimize cerebral blood flow fluctuations are key.

- 1. good prenatal care and avoiding pre-term labor
- 2. antenatal corticosteroids: administration of one course of antenatal corticosteroids to women at risk of having premature birth infants reduces neonatal mortality, respiratory distress syndrome and PIVH.¹⁶⁰ Multiple courses of antenatal corticosteroids did not improve outcomes and were associated with decreased head circumference, weight and length at birth¹⁶¹
- 3. indomethacin: results in cerebral vasoconstriction and reduces the responsiveness of CBF to changes in CO₂, lowers CBF and increases arterial oxygenation reducing patent ductus arteriosus (PDA). However, use is possibly associated with increased risk of intestinal perforation
- 4. antenatal vitamin K given IM >4 hrs prior to delivery decreases PIVH from 33% to 5%
- 5. sluicing umbilical cord blood and delaying umbilical cord clamping by 30–120 seconds in premature babies increased hematocrit and decreased PIVH in 5 of 7 studies¹⁶²
- 6. using surfactant to reduce RDS
- 7. minimizing external stimulation (some centers use fentanyl drips)
- 8. steroids to stabilize the GM vessels

Table 87.8 Grading subependymal hemorrhage ¹⁵⁵	
Grade	Description
I	subependymal
II	IVH without ventricular dilatation
III	IVH with ventricular dilatation
IV	IVH with parenchymal hemorrhage

87.12.7 Clinical

Grading

The most commonly used grading system of Papile et al. based on CT or U/S findings is shown in □ Table 87.8. PIVH may present acutely, subacutely. Most commonly, it is discovered incidentally on surveillance U/S.

There is a direct correlation between younger gestational age (GA) and the severity of PIVH. In infants 24–26 weeks GA, 32% will have a Grade III PIVH and 19% will have a Grade IV PIVH compared with infants 31–32 weeks GA, 11% will have a Grade III PIVH and 5% will have a Grade IV PIVH.¹⁶³

Presentation

Asymptomatic PIVH

Most PIVHs will be clinically unsuspected, usually with smaller hemorrhages. Retrospectively, these PIVHs may have been suggested by a fall in Hct or delays in neurologic development. These have a 78% 6-month survival, vs. 20% for PIVH showing signs.

Subacute presentation

Usually smaller or more slowly developing hemorrhages. Clinically may present as irritability, reduced motor activity, or abnormal eye movements.

Acute presentation

1. changes in muscle tone or activity: usually decerebrate or decorticate posturing, sometimes flaccid paralysis
2. seizures: often subclinical
3. tense fontanelle
4. hypotension
5. respiratory and cardiac irregularities: apnea & bradycardia (“A’s and B’s”)
6. unreactive pupils and/or loss of extraocular muscle movements
7. Hct drop > 10%

Hydrocephalus

General information

20–50% of infants with PIVH will develop either transient or progressive hydrocephalus (HCP). Grades III and IV are more often associated with progressive ventricular dilatation than are lower grades (however, HCP may develop even after low grade PIVH¹⁶⁴). Younger gestational age infants may be at lower risk.

Post PIVH hydrocephalus usually occurs 1–3 weeks after the hemorrhage. Probably caused by cellular debris and/or the toxic effects of blood breakdown products on the arachnoid granulations (communicating HCP), or by an adhesive arachnoiditis in the posterior fossa or rarely by compression or blockage of critical pathways, e.g. at the sylvian aqueduct (obstructive HCP). In a case of HCP following intra-uterine PIVH, aqueductal gliosis was found at autopsy.¹⁶⁵

Differential diagnosis of ventriculomegaly in PIVH

When ventriculomegaly is detected, it needs to be differentiated from the following:

1. transient ventriculomegaly: occurs in the first few days after PIVH. This may not cause elevated ICP. As implied, it is self limited
2. progressive ventriculomegaly: occurs in 20–50% of cases (true hydrocephalus)
3. “hydrocephalus ex vacuo”: due to loss of brain tissue or maldevelopment. Is not progressive on serial U/S. OFCs may fall below normal due to lack of growing brain as stimulus for head growth

Possible presentations

Abnormally increasing OFC (crossing percentile curves faster than body weight), lethargy, apnea and bradycardia, vomiting. There is progressive dilatation of the ventricular system on serial U/S or CT or MRI evaluations.

87.12.8 Pathophysiologic effects of PIVH

Deleterious effects of PIVH on the brain are due to¹⁶⁶:

1. destruction of the germinal matrix and glial precursors
2. direct injury to neural tissue from hematoma: once hemorrhage resorbs may leave patient with porencephaly or cystic lesions
3. pressure of hematoma on nearby brain tissue reducing CBF even to parts of the same hemisphere distant from the hemorrhage¹⁶⁷
4. diffuse decreased CBF following the hemorrhage¹⁶⁸ due to elevated ICP
5. injury from the same hypoxic event that precipitated the PIVH
6. decreased CPP leads to periventricular leukomalacia (PVL) and cerebral infarction
7. periventricular hemorrhagic infarction
8. hydrocephalus (see above): numerous deleterious effects on the CNS
9. seizures: repeated or prolonged seizures may be deleterious to neuronal function

87.12.9 Diagnosis

Ultrasound (U/S)

Performed through the open fontanelles.¹⁵⁶ Accuracy $\approx$ 88%(91%sensitivity, 85%specificity).¹⁶⁹ U/S is invaluable because:

1. it demonstrates the size of the ventricles, the location and size of the hematoma, and the thickness of the cortical mantle
2. it may be brought to the infant's bedside (obviating transportation)
3. it is non-invasive
4. it is not adversely affected by occasional infant movements (eliminating the need for sedation)
5. there is no exposure to ionizing radiation (radiation from diagnostic imaging in children has long-term risks for cancer¹⁷⁰ and damage to the lens)
6. it may be followed serially with relative ease

CTscan

Sometimes necessary when U/S is not readily available, or in complicated cases where anatomy is difficult to deduce from U/S images. Many ICUs have portable CTscans available which obviates need for patient transport.

Rapid sequence MRI

Pros: Eliminates the risk of ionizing radiation associated with CTscan.

Cons: Requires moving the infant from the neonatal ICU to the radiology suite.

87.12.10 Treatment

General measures

General measures are directed at optimizing CPP without further excessive elevation of CBF by carefully maintaining normal MAP and normalizing pCO₂, and by treating active hydrocephalus as needed (see above).

While daily LPs can control the deleterious effects of posthemorrhagic HCP, they do not reduce the frequency of long-term HCP (requiring permanent shunting). Ventricular size must be monitored with serial U/S.

Medical treatment

1. not very effective. Treated patients fared worse in several studies
2. osmotic agents: isosorbide, glycerol. Effects are short-lived
3. □ diuretic therapy: has been used, but a large study showed increased nephrocalcinosis and biochemical abnormalities, resulting in a borderline increase in the risk for motor impairment at one year.¹⁷¹ The results were so compelling, the data-monitoring committee terminated the study prematurely. Furosemide and acetazolamide therapy was deemed neither safe nor effective in treating post-hemorrhagic ventricular dilatation and cannot therefore be recommended¹⁷²

Surgical/interventional treatment for the clot

Due to poor operative results, surgical evacuation of an intracerebral hemorrhage in the newborn is not indicated with the possible exception of a posterior fossa hemorrhage causing brain stem compression that does not respond to medical treatment.¹⁷³ Supportive measures are usually in order.

Intervention for intraventricular blood

General information

34% of infants <1500 g require shunt/reservoir drainage after failed medical management. Grade III and IV PIVH: >70% of cases develop progressive ventricular dilatation, and 32–47% of this subset will ultimately require shunting.¹⁷⁴

Indications for intervention

Intervention for intraventricular blood is indicated in the setting of progressive ventriculomegaly with the OFC crossing percentile curves and clinical evidence of increased ICP (split sutures, tense fontanelle...).

Serial lumbar punctures

Used at many facilities for hemorrhages with intraventricular extension and communicating hydrocephalus (the usual type of HCP that occurs with PIVH).¹⁷⁵

This should be undertaken with the knowledge that meta-analysis¹⁷⁶ showed sequential lumbar or ventricular taps of ≈ 10 ml/kg/tap for prophylaxis or treatment of progressive hydrocephalus offers no clear benefit over conservative treatment, and had an infection rate of 5–9%. In rare cases, LPs may succeed in temporizing progressive HCP for a few weeks until the infant is large enough for shunt placement.

Infants <800 gm may not tolerate LPs because of desaturation when lying on their side, or the LP itself may be difficult. In these patients, consider 1–2 ventricular taps to at least obtain fluid for analysis (in some cases nothing further needs to be done).

Serial ventricular taps

May be a viable short-term option for those infants who cannot tolerate LPs or in whom there is obstruction to CSF flow in the lumbar subarachnoid space (e.g. due to spinal subdural hematoma from previous LP). However it is not desirable for long-term use because of repeated trauma to brain (risk of porencephaly) and risk of intracerebral, intraventricular, or subdural hemorrhage.

If continued taps are likely (i.e. large hemorrhage, or rapid recurrence of intracranial hypertension as determined by palpation of fullness of anterior fontanelle (AF) following several taps) the acceptable options include:

1. continuing serial LPs (see below)
2. percutaneous ventricular taps: not recommended for more than a few treatments as it causes porencephaly
3. placement of a temporary ventricular access device (TVAD) – a ventricular catheter connected to a subgaleal reservoir (either a Rickham reservoir, or a low profile McComb reservoir¹⁷⁷). These can be inserted safely at the bedside, obviating the need for transport to the O.R.¹⁷⁸
 - a) temporary ventricular access: the reservoir can be used for serial percutaneous taps. Usually tapped QD or QOD (see below). Use a 27 Ga butterfly needle, clean with at least 3 betadine stick swabs, withdraw ≈ 10 ml and send for culture. Reported infection rate: 8–12%⁷⁹
 - b) ventricular-subgaleal shunt: the side-port of the reservoir is left uncapped. A subgaleal pocket must be created at the time of surgery. Fluid is reabsorbed from this potential space. First performed in 1893 by Mikulicz-Radecki (1850–1905). Use has been reported up to 35 days.¹⁸⁰ Infection rate: $\approx 6\%$
 - c) the reservoir may be converted to VP shunt if and when appropriate. Not recommended in infants <1100 gms due to very high infection rate
4. external ventricular drainage (EVD): similar to reservoir placement, but with possibility of inadvertent dislodgment (13%) and comparable infection rate (6%)
5. early VP shunting: high infection rate, peritoneal cavity not suitable in many cases, e.g. due to necrotizing enterocolitis (NEC), paucity of subcutaneous tissue through which to pass shunt tube... Not recommended for infants <2000 gms

Temporary ventricular access device (TVAD)

Advantages of TVAD

- 1. avoids shunt in unhealthy children at risk of infection, skin breakdown or other operative/anes-
thetic complications
- 2. clears protein and cellular debris (more favorable for subsequent shunting)
- 3. avoids repeated penetration of brain with risk of porencephaly
- 4. provides port for infusion of medication (e.g. antibiotics) PRN
- 5. avoids cumbersome, easily dislodged EVD with infection risk 6%on average of 13 days of EVD
- 6. up to 25%of patients will recover and avoid permanent shunt placement^{181,182}

Disadvantages of TVAD

- 1. requires services of a neurosurgeon (not always available)
- 2. increases risk of infection of subsequent permanent shunt from 5%to 13%¹⁸³
- 3. inherent risks of surgery including hemorrhage, infection, ventriculitis, meningitis, CSF leak
- 4. risks of overdrainage including subdural hematoma, impaired skull growth

Technical considerations for serial taps (via ventricular reservoir or LP)

8–20 cc of fluid are removed initially, and this is repeated daily (or more often if AF become very tense before 24 hours elapse) for several days, and then usually varies from 5–20 cc qod to 15 cc TID depending on response. The frequency and volume of the taps are modified based on:

- 1. fullness of AF: attempt to keep AF from becoming tense
- 2. appearance of ventricles on serial U/S: strive to prevent progressive enlargement, reduction in size can usually be achieved
- 3. follow OFC: should not cross percentile curves (need to differentiate from the so-called “catch-up phase” of brain growth which may occur once the infant overcomes their overall medical prob-
lems and is able to adequately utilize nutrition^{184,185}; serial U/S will show rapid brain growth without progressive ventriculomegaly in cases of catch-up brain growth)
- 4. CSF protein concentration: controversial. Diminishes with serial taps. Some feel that as long as it is ≥ 100 mg/dl it is unlikely that significant spontaneous resorption will occur and continued seri-
al taps will probably be needed
- 5. NB: removal of this volume of fluid may cause electrolyte disturbances, primarily hyponatremia;
□ follow serum electrolytes on regular basis

Follow with serial U/S on day 3–5, and then weekly for several weeks, and then bi-weekly. A baseline CT scan is often obtained prior to placement of a permanent shunt.

Insertion of VP shunt or conversion of sub-Q reservoir to VP shunt

Indications and requirements:

- 1. symptomatic hydrocephalus (p.1348) and/or progressive ventriculomegaly
- 2. infant is extubated (and thus off ventilator)
- 3. infant weighs ≥ 2000 grams (some prefer ≥ 2500 grams)
- 4. no evidence of NEC (might create problems with peritoneal end of catheter)
- 5. CSF protein ideally < 100 mg/dl (because of concerns about plugging of the shunt, or causing ileus or malabsorption of the fluid – which was not seen with high protein fluid shunted from the sub-
dural space¹⁸⁶ – and also to see if patient will start reabsorbing CSF on their own)

Technical recommendations:

- 1. do not tap reservoir for at least 24 hrs before inserting a new ventricular catheter (allows ven-
tricles to expand to facilitate catheterization)
- 2. obtain U/S the day prior to conversion
- 3. use a low or very-low pressure system (if CSF protein is high, consider a valveless system),
upgrade later in infancy if necessary
- 4. avoid placing shunt hardware in areas on which these debilitated infants tend to lay (to prevent
skin breakdown with hardware exposure)

87.12.11 Outcome

Short-term

Preemies with PIVH have higher mortality than matched preemies without PIVH.

The incidence of mortality and progression of hemorrhage is higher the earlier the hemorrhage occurs. The more severe the hemorrhage, the higher the mortality and the higher the risk of HCP (□ Table 87.9).

Table 87.9 Short-term outcome of PIVH (≈ 250 cases ¹⁴⁵)		
Severity of hemorrhage	Deaths (%)	Progressive hydrocephalus (%)
mild	0	0–10
moderate	5–15	15–25
severe	50–65	65–100

Long-term

The effect of low grade PIVH on long-term neurodevelopment has not been studied well. Most investigators feel that higher grades of PIVH are associated with greater degrees of handicaps than matched controls.

In one study of 12 infants with Grade II PIVH treated with serial LPs and in the 7 with progressive ventriculomegaly with VP shunt followed for a mean of 4.5 years found all were ambulatory and 75% had IQ within normal range.¹⁸⁷

A recent study of very low birth weight infants showed that children 18–22 months of age with severe PIVH and shunts had significantly lower scores on the Bayley Scales of Infant Development IIR compared with children with no PIVH and children with equal grades of PIVH who did not require a shunt.¹⁸⁸

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87.13 Other causes of intracerebral hemorrhage in the newborn

1. birth trauma may result in subdural hemorrhage, tentorial hemorrhage, parenchymal hemorrhage and/or subarachnoid blood. This is usually detected by imaging (U/S or CT) when an infant develops seizures, apnea, bradycardia or rarely focal neurological deficits. It rarely requires surgical intervention
2. choroid plexus hemorrhage can result in IVH. In some cases HCP can develop and require shunt placement
3. hemorrhagic stroke has been identified in 6.2 per 100,000 live births.¹⁸⁹ The usual presentation was with encephalopathy (100%) and seizures (65%). 75%of the strokes were idiopathic. Other identified etiologies were thrombocytopenia and a single case of a cavernous malformation. Risk factors for perinatal hemorrhagic stroke include: male gender, fetal distress, emergent c-section, prematurity and post-maturity
4. tumors in the neonate can present with hemorrhage
5. vascular malformations of any form can present in the neonate with hemorrhage, although this is uncommon. Vein of Galen malformations are diagnosed in the neonate in about 40%of cases.¹⁹⁰ Most of these infants present with fulminant congestive heart failure and 50%have ventriculomegaly

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Part XXI

Outcome Assessment

88 Outcome
Assessment

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XXI

88 Outcome Assessment

88.1 Cancer

□ Karnofsky performance scale (KPS). □ Table 88.1, (after David A. Karnofsky) often used for grading functional status in patients with cancer. A KPS score <70 (particularly with brain tumors) often identifies patients with a worse prognosis for any given treatment.

WHO performance score. □ Table 88.2, the WHO Performance score³ (AKA The Eastern Cooperative Oncology Group (ECOG) score, AKA Zubrod score (after C. Gordon Zubrod), ranges from 0 to 5, with 0 indicating perfect health and 5 death. The advantage over the Karnofsky scale is its simplicity.

88.2 Head injury

The Ranchos Los Amigos scale (□ Table 88.3) is often used in rating disability following head injury. The Glasgow outcome scale (□ Table 88.4) is frequently employed in outcome assessment.

88.3 Cerebrovascular events

88.3.1 General information

Several outcome grading scales have come to be favored for use following strokes or SAH. Each emphasizes different aspects of outcome. The Barthel Index (□ Table 88.6) places weight on activities of daily living (ADLs), while others, such as the modified Rankin scale⁵ (□ Table 88.5) assess levels of independence and includes a comparison to previous activity levels and shows fairly good interobserver consistency.⁶ While it does measure functional status, the modified Rankin is not sensitive to subtle neurologic deficits such as dysphasia or visual field defects.

Table 88.1 Karnofsky performance status scale (modified ^{1,2})		
Score	Criteria	General category
100	normal: no complaints, no evidence of disease	Able to carry on normal activity and work. No special care is needed
90	able to carry on normal activity: minor signs or symptoms	
80	normal activity with effort: some signs or symptoms	
70	cares for self: unable to carry on normal activity or to do active work	Unable to work. Able to live at home, care for most personal needs. Variable assistance is required
60	requires occasional assistance: cares for most of needs	
50	requires considerable assistance and frequent care	
40	disabled: requires special care and assistance	Unable to care for self. Requires equivalent of institutional or hospital care. Disease may be rapidly progressing
30	severely disabled: hospitalized; death not imminent	
20	very sick: hospitalized; active supportive care needed	
10	moribund: fatal processes are progressing rapidly	
0	dead	

Table 88.2 WHO performance scale	
Grade	Description
0	Fully active, no performance restriction as a result of disease.
1	Restricted in physically strenuous activity. Ambulatory. Able to do light work, e.g. light house work, desk work.
2	Unable to perform any work activities. Ambulatory. Up & about >50% of waking hours.
3	Only able to perform limited self care. Wheelchair confined >50% of waking hours.
4	Completely disabled. Unable to perform self care. Totally bed or chair confined.
5	Dead

Table 88.3 Ranchos Los Amigos cognitive scale	
Level	Meaning
I	No response to pain, touch, sight or sound.
II	Generalized reflex responses to pain.
III	Localized response. Blinks to strong light, turns towards/away from sound, responds to physical discomfort, inconsistent responses to commands.
IV	Confused – Agitated Alert, very active, agitated, aggressive, or bizarre behaviors. Performs motor activities but behavior is non-purposeful, extremely short attention span.
V	Confused – Non agitated Gross attention to environment, easily distracted, requires continual redirection, difficulty learning new tasks, agitated by excess stimulation. May converse socially but with inappropriate verbalizations.
VI	Confused – Appropriate Inconsistent orientation to time and place. Retention span and recent memory impaired. Begins to recall past, consistently follows simple commands, goal directed behavior with assistance.
VII	Automatic – Appropriate Performs daily routine in highly familiar environment in a non-confused but automatic “robot-like” fashion. Skills deteriorate in unfamiliar environment. Lacks realistic planning for future.
VIII	Purposeful – Appropriate

Table 88.4 Glasgow outcome scale ⁴	
Score	Meaning
5	good recovery – resumption of normal life despite minor deficits (“return to work” not reliable)
4	moderate disability (disabled but independent) – travel by public transportation, can work in sheltered setting (exceeds mere ability to perform “activities of daily living”)
3	severe disability (conscious but disabled) – dependent for daily support (may be institutionalized, but this is not a criteria)
2	persistent vegetative state – unresponsive & speechless; after 2–3 weeks, may open eyes & have sleep/wake cycles
1	death – most deaths ascribable to primary head injury occur within 48 hrs

Table 88.5 The modified* Rankin scale	
Grade	Description
0	no symptoms at all
1	no significant disability despite symptoms: able to carry out all usual duties & activities
2	slight disability: unable to carry out all previous activities. Able to look after own affairs without assistance
3	moderate disability: requiring some help, but able to walk without assistance
4	moderately severe disability: unable to walk without assistance, and unable to attend to own bodily needs without assistance
5	severe disability: bedridden, incontinent, and requiring constant nursing care and attention
* the original Rankin scale ⁷ : did not have Grade 0, Grade 1 did not include the words “despite symptoms” and “& activities”, and it defined Grade 2 as “unable to carry out some of previous activities...”	

Table 88.6 The Barthel index								
Item	Original Barthel Index			Modified Barthel Index				
	Unable to perform task	Needs assistance	Fully independent	CODE 1	CODE 2	CODE 3	CODE 4	CODE 5
				Unable to perform task	Attempts task but unsafe	Moderate help required	Minimal help required	Fully independent
Personal hygiene	0	0	5	0	1	3	4	5
Self bathing	0	0	5	0	1	3	4	5
Feeding	0	5	10	0	2	5	8	10
Toilet	0	5	10	0	2	5	8	10
Stair climbing	0	5	10	0	2	5	8	10
Dressing	0	5	10	0	2	5	8	10
Bowel control	0	5	10	0	2	5	8	10
Bladder control	0	5	10	0	2	5	8	10
Ambulation	0	5–10	15	0	3	8	12	15
Wheelchair ^a	0	0	5	0	1	3	4	5
Chair/bed transfers	0	5–10	15	0	3	8	12	15
Total (range)	0	→→	100	0	→→→→→			100
^a score only if unable to walk and patient trained in wheelchair management								

88.3.2 Scales

- Modified Rankin scale (□ Table 88.5)
- Barthel index (□ Table 88.6). The original Barthel index^{8,9} assigns one of three scores to 10 ratable ADLs, and then the individual scores are summed. The modified Barthel index (MBI) with a 5-step scoring system appears to have greater sensitivity.¹⁰ The total ranges from 0 to 100 (a score of 100 implies functional independence, not necessarily normality).

Table 88.7 The Functional Independence Measure™(FIM)	
Classification	Item
Motor	
Self-care	Eating
	Grooming
	Bathing
	Dressing – upper body
	Dressing – lower body
	Toileting
Sphincter control	Bladder management
	Bowel management
Mobility	Bed, chair, wheelchair
	Toilet
	Tub, shower
Locomotion	Walk or wheelchair
	Stairs
Cognitive	
Communication	Comprehension
	Expression
Social cognition	Social interaction
	Problem solving
	Memory

Table 88.8 The 7 FIM™rating levels of disability		
Degree of dependency	Level of function	Score
No helper	Complete independence	7
	Modified independence	6
Modified dependence on a helper	Supervision	5
	Minimal assist (≥ 75% independent)	4
	Moderate assist (≥ 50% independent)	3
Complete dependence on a helper	Maximal assist (≥ 25% independence)	2
	Total assist (<25% independence)	1

Of all the factors, independence in bathing was the most difficult. Abilities on the Barthel index tend to return in a fairly consistent order, and so most patients with the same score will have similar patterns of disability.

88.4 Spinal cord injury

□ Functional Independence Measure™ (FIM™) ^{11,12,13} (FIM™). Developed to provide uniform evaluation of disability for spinal cord injuries. Rates 18 items shown in □ Table 88.7 (13 motor, 5 cognitive) on the 7 level scale shown in □ Table 88.8.

The FIM™ has high internal consistency and is a good indicator of burden of care.^{14,15}

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Part XXII

Differential Diagnosis

89	Differential Diagnosis by Location or Radiographic Finding – Intracranial	1364
90	Differential Diagnosis by Location or Radiographic Finding – Spine	1390
91	Differential Diagnosis (DDx) by Signs and Symptoms – Primarily Intracranial	1395
92	Differential Diagnosis (DDx) by Signs and Symptoms – Primarily Spine and Other	1407

89 Differential Diagnosis by Location or Radiographic Finding – Intracranial

89.1 Diagnoses covered outside this chapter

Table 89.1 Differential diagnoses by location or radiographic finding, intracranial – covered outside this chapter	
DDx	
chordomas (p. 778)	
extra-axial fluid (peds) (p. 903)	
gyral enhancement (p. 1281)	
hydrocephalus (p. 399)	
pineal region tumors (p. 658)	
pneumocephalus (p. 887)	
schizencephaly (p. 288)	

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89.2 Posterior fossa lesions

89.2.1 Cerebellar lesions

General information

The following addresses intra-axial p-fossa abnormalities (for extra-axial lesions, below).

Adult

Single lesion

□ Note. Rule of thumb: “the differential diagnosis of a solitary intraparenchymal lesion in an adult p-fossa is metastasis, metastasis, metastasis, until proven otherwise.”

- tumors:
 - metastasis
 - hemangioblastoma (p. 701): the most common PRIMARY intra-axial p-fossa tumor in adults (7–12% of p-fossa tumors). Very vascular nodule, often has cyst. Almost all p-fossa tumors are relatively avascular on angiography except these (look for serpentine signal voids especially in the periphery of the lesion on MRI,¹ much less common in cavernous hemangioma)
 - cerebellar (pilocytic) astrocytoma (p. 630): may be solid or cystic, tends to occurs in younger adults
 - brainstem glioma: an isolated glioblastoma in the posterior fossa of an adult is a reportable rarity
 - choroid plexus tumor: usually infratentorial in adults (p. 1380)
 - cerebellar liponeurocytoma (p. 646)
- infectious: abscess
- vascular
 - cavernous hemangioma
 - hemorrhage
 - infarction: cerebellar stroke may be associated with H/A and/or pain in suboccipital region or upper neck
 - embolic
 - thrombotic/plaque related
 - vertebral artery dissection: much less common than carotid dissection (p. 1325)
 - vertebrobasilar hypoplasia (p. 1306)
- Lhermitte-Duclos (p. 647): Focal or diffuse. Nonenhancing. Characteristic tiger stripes. Widens folia (c.f. most neoplasms which destroy folial pattern)

Multiple lesions

1. metastases
2. hemangioblastoma (possibly as part of von Hippel-Lindau) (p.703)
3. abscesses
4. cavernous hemangiomas

Pediatric

Also see Pediatric brain tumors (p.593).

Early data: 67% of childhood brain tumors occur in p-fossa, and astrocytomas were the most common there. Currently: p-fossa tumors comprise 54–60% of childhood brain tumors (breakdown listed below). 4 types account for $\approx 95\%$ of infratentorial tumors in patients ≤ 18 yrs age.² The 3 most common are equal in incidence (expressed as percent of p-fossa tumors pooled from 1350 pediatric brain tumors³):

1. PNET, including medulloblastoma (p.663): 27%
 - a) most start in roof of 4th ventricle (fastigium), and most are solid
 - b) differentiating medulloblastoma (MB) from ependymoma:
 - 4th ventricle drapes around medulloblastoma (“banana sign”) from the anterior aspect, c.f. ependymoma which tends to grow into 4th ventricle from floor. Ependymoma may grow through foramen of Luschka and/or Magendie
 - ependymomas tend to be inhomogeneous on T1WI MRI (unlike MB)
 - the exophytic component of ependymomas tends to be high signal on T2WI MRI (with MB this is only mildly hyperintense)
 - calcifications: common in ependymomas, but only in $<10\%$ of MB
2. cerebellar (pilocytic) astrocytoma (p.630): 27% Most start in cerebellar hemisphere. Often cystic with enhancing mural nodule
3. brainstem gliomas (p.633): 28% Usually present with multiple cranial nerve palsies and long tract findings
4. ependymoma (p.642): usually arise in floor of 4th ventricle
5. choroid plexus papilloma (p.648): majority of patients are <2 yrs old
6. atypical teratoid/rhabdoid tumor (AT/RT) (p.666)
7. metastasis: neuroblastoma, rhabdomyosarcoma, Wilm’s tumor...
8. PHACES syndrome: acronym for a group of findings including Posterior fossa malformations, cervicofacial Hemangioma, Arterial anomalies of the head and neck, Coarctation of the aorta and cardiac defects, Eye anomalies and Sternal cleft. Ratio girls:boys=9:1. Thought to begin during gestation weeks 8–10

89.2.2 Cerebellopontine angle (CPA) lesions

Lesions in general

Vestibular schwannoma, meningioma, and epidermoid account for most. For those lesions that may be cystic, see below.

1. vestibular schwannoma: (80–90% of CPA lesions); see below for differentiating from meningioma (p.1366)
2. meningioma: (5–10%); see below for differentiating from vestibular schwannoma (p.1366)
3. ectodermal inclusion tumors (p.761)
 - a) epidermoid (cholesteatoma): 5–7% High signal on DW MRI (p.761). Tumor passing from the posterior fossa to the middle fossa though the incisura is highly suggestive of epidermoid
 - b) dermoid
4. metastases
5. neuroma from cranial nerves other than VIII (also see below for some differentiating features)
 - a) trigeminal neuroma: expands towards Meckel’s cave
 - b) facial nerve neuroma⁴: may arise in any portion of the VII nerve, with a predilection for the geniculate ganglion.⁵ Even in these tumors, hearing loss tends to precede facial paresis. Hearing loss may be sensorineural from VIII nerve compression from tumors arising in the proximal portion of VII (cisternal or internal auditory canal (IAC) segment), or it may be conductive from erosion of the ossicles by tumors arising in the second (tympanic, or horizontal) segment of VII. Facial palsy (peripheral) (p.576) may also develop, usually late⁴
 - c) neurinoma of lowest 4 cranial nerves (IX, X, XI, XII)
6. arachnoid cyst (p.248)

7. neurenteric cyst (p.290): rare.⁶ May secrete mucin
8. cholesterol granuloma (distinct from epidermoid) (p.761)
9. lipoma
10. aneurysm: PICA, AICA, vertebralbasilar
11. dolichobasilar ectasia
12. cysticercosis
13. extensions of:
 - a) brain stem or cerebellar glioma
 - b) pituitary adenoma
 - c) craniopharyngioma
 - d) chordoma & tumors of skull base
 - e) fourth ventricle tumors (ependymoma, medulloblastoma)
 - f) choroid plexus papilloma: from 4th ventricle through foramen of Luschka
 - g) glomus tumor
 - glomus jugulare
 - glomus tympanicum
 - h) primary tumors of temporal bone (e.g. sarcoma or carcinoma)

Cystic lesions of the CPA

CPA lesions from the above list that may be cystic or have a cystic component⁶:

1. arachnoid cyst: same intensity as CSF on all MRI sequences, homogeneous
2. epidermoid cyst (p.270): □ high signal on DWMRI differentiates this from arachnoid cyst
3. dermoid cyst: high intensity areas on T1WI similar to fat; usually midline
4. cystic schwannoma
5. cholesterol granuloma: □ ≈ only lesion that is high signal on T1WI (due to blood breakdown products; exception: the rare “white” epidermoid). Also high signal on T2WI. Usually extradural, especially near petrous apex. Bone destruction is common
6. neurenteric cyst: nonenhancing. Low intensity on DWMRI
7. choroidal cyst
8. cysticercosis: enhancing nodule (scolex)

Differentiating neuromas of V, VII and VIII cranial nerves

All 3 of these tumors may present in the CPA and may cross from posterior fossa to middle fossa, but they tend to do so in different manners. Vestibular schwannomas show “transhiatal” extension by passing through the tentorial hiatus medially. Most trigeminal neuromas show “transapicopetrosal” extension by crossing into the middle fossa via the petrous apex (although some show transhiatal extension). When facial neuromas cross, they tend to spread across the midpetrosal bone, which is characteristic for facial neuromas spread across the midpetrosal bone is characteristic for facial neuromas.⁴ When a facial neuroma enlarges the IAC, unlike a vestibular schwannoma, it tends to erode the anterosuperior aspect of the IAC.

Differentiating vestibular schwannoma from CPA meningioma

1. vestibular schwannoma (VS) (AKA acoustic neuroma):
 - a) clinical: progressive unilateral hearing loss, usually with tinnitus. Progression results in unsteadiness, with true vertigo being rare. The facial nerve is more resistant to stretching, thus facial nerve signs and symptoms occur late. Trigeminal nerve involvement may occur with tumors > 3 cm (check corneal reflex), with tic douloureux-like symptoms being unusual
 - b) imaging: often heterogeneous signal and nonuniform enhancement. Medium size tumors look like ice cream in a cone (IAC is the cone). Rarely calcified. Except for very small tumors, IAC is frequently enlarged. Look for an acute angle between the tumor and the petrous bone (meningiomas usually have an obtuse angle)
2. meningiomas: may mimic VSs with these differences:
 - a) clinical: since they often arise from the superior anterior edge of the IAC, early facial nerve involvement is more common, and hearing loss is usually late. Trigeminal neuralgia-like pain is more common than with VSs
 - b) imaging: homogeneous signal and enhancement. The tumor may enter the IAC but it tends not to enlarge it. IAC often eccentric in tumor. Tumor is flat against petrous bone with an obtuse angle to the bone. Calcification and bony hypertrophy may occur (which occasionally narrows the IAC).

89.2.3 Petrous apex lesions

1. infection/inflammatory:
 - a) osteomyelitis: may produce Gradenigo's syndrome (p.570)
 - b) cholesterol granuloma (bright on T1WI epidermoid cyst are bright on DWI, neither enhance)
2. vascular lesions: aneurysm
3. neoplastic:
 - a) squamous cell cancer
 - b) glomus tumor
 - c) chondrosarcoma: will displace the carotid from medial to lateral (almost every other tumor in this region encases the carotid)

89.2.4 Foramen magnum lesions

Differential diagnosis

See Foramen magnum lesions (p.1367) for nonneoplastic lesions. Most foramen magnum (FM) region tumors are extra-axial. This includes:

1. Extra-axial tumors
 - a) meningioma: the anterior lip of the foramen magnum is the second most common site of origin of posterior fossa meningiomas. Meningiomas (p.698) comprise 38–46% of FM tumors^{7,8} and most are intradural
 - b) Chordoma (p.778): a mass behind the dens compressing the spinal cord is a chordoma until proven otherwise
 - c) neurilemmoma
 - d) epidermoid
 - e) chondroma
 - f) chondrosarcoma
 - g) metastases
2. Exophytic component of a brainstem tumor
3. Non-neoplastic lesions
 - a) aneurysms or ectasia of the vertebral artery
 - b) odontoid process in cases of basilar invagination (p.278)
 - c) pannus from involvement of the odontoid with rheumatoid arthritis or old nonunion of fracture
 - d) synovial cyst of the quadrate ligament of the odontoid⁹

Presentation

In the pre-imaging era (i.e. before CT & MRI) these lesions were often diagnosed relatively late due to the unusual associated clinical syndromes and the rarity of visualizing this region on myelography.

Clinical findings

Symptoms:

1. sensory
 - a) craniocervical pain: usually an early symptom, commonly in neck and occiput. Aching in nature. ↑ with head movement
 - b) sensory findings: usually occur later. Numbness and tingling of the fingers
2. motor
 - a) spastic weakness of the extremities: weakness usually starts in the ipsilateral UE, then the ipsilateral LE, then contralateral LE, and finally contralateral UE (“rotating paralysis”)

Signs:

1. sensory
 - a) dissociated sensory loss: loss of pain and temperature contralateral to lesion with preservation of tactile sensation
 - b) loss of position and vibratory sense, greater in the upper than the lower extremities
2. motor
 - a) spastic weakness of the extremities
 - b) atrophy of the intrinsic hand muscles: a lower motor nerve finding
 - c) cerebellar findings may rarely be present with extensive intracranial extension
3. long tract findings
 - a) brisk muscle stretch reflexes (hyperreflexia, spasticity)

- b) loss of abdominal cutaneous reflexes
- c) neurogenic bladder: usually a very late finding
- 4. ipsilateral Horner's syndrome: due to compression of cervical sympathetics
- 5. nystagmus: classically downbeat (p.558) but other types can occur

It had been postulated that long tract findings were due to direct compression at the cervicomedullary junction, and that lower motor nerve findings in the upper extremities were due to central necrosis of the grey matter as a result of compression of arterial blood supply. Anatomic study suggests that it is actually venous infarction at lower cervical levels (C8-T1) that is responsible for the lower motor neuron findings.

89.3 Multiple intracranial lesions on CT or MRI

1. neoplastic
 - a) primary
 - multicentric gliomas; $\approx 6\%$ of gliomas are multicentric, more common in neurofibromatosis, see Multiple gliomas (p.619)
 - tuberous sclerosis (including giant cell astrocytomas); (usually periventricular)
 - multiple meningiomas
 - lymphoma
 - PNET
 - multiple neuromas (usually in neurofibromatosis, including bilateral vestibular schwannomas)
 - b) metastatic: usually cortical or subcortical, surrounded by prominent vasogenic edema (p.803). More common tumors include:
 - lung
 - breast
 - melanoma: may be higher density than brain on unenhanced CT
 - renal cell
 - gastrointestinal tumors
 - genitourinary tract tumors
 - choriocarcinoma
 - testicular
 - atrial myxoma
 - leukemia
2. infection: mostly abscess or cerebritis. Most commonly due to:
 - a) pyogenic bacteria
 - b) toxoplasmosis: common in AIDS patients (p.371)
 - c) fungal
 - cryptococcus
 - mycoplasma
 - coccidiomycosis
 - aspergillosis
 - candidiasis
 - d) echinococcus
 - e) schistosomiasis
 - f) paragonimiasis
 - g) herpes simplex encephalitis (HSE): usually temporal lobe (p.364)
3. inflammatory
 - a) demyelinating disease
 - MS: usually in white matter, periventricular, with little mass effect, margins are usually very sharp. Ring enhancing lesions can occur with tumefactive demyelinating lesions (p.179)
 - progressive multifocal leukoencephalopathy (PML): primarily in white matter. No enhancement. Patients are usually very sick
 - b) gummas
 - c) granulomas
 - d) amyloidosis
 - e) sarcoidosis
 - f) vasculitis or arteritis
 - g) collagen vascular disease, including:
 - periarteritis nodosa (PAN) (p.199)

- systemic lupus erythematosus (SLE)
 - granulomatous arteritis
4. vascular
 - a) multiple aneurysms (congenital or atherosclerotic)
 - b) multiple hemorrhages, e.g. associated with DIC or other coagulopathies (including anticoagulant therapy)
 - c) venous infarctions, especially in dural sinus thrombosis (p.1308)
 - d) moyamoya disease (p.1313)
 - e) subacute hypertension (as in malignant HTN, eclampsia...) →symmetric confluent lesions with mild mass effect and patchy enhancement usually in occipital subcortical white matter
 - f) multiple strokes
 - lacunar strokes (l'état lacunaire)
 - multiple emboli (e.g. in atrial fibrillation, mitral valve prolapse, SBE, air emboli)
 - sickle cell disease
 - vasculitis
 - intravascular lymphomatosis (p.711)
 5. hematomas and contusions
 - a) traumatic (multiple hemorrhagic contusions, multiple SDH)
 - b) multiple “hypertensive” hemorrhages (amyloid angiopathy, etc.)
 6. intracranial calcifications (p.1380)
 7. miscellaneous
 - a) radiation necrosis
 - b) foreign bodies (e.g. post gunshot wound)
 - c) periventricular low densities
 - Binswanger's disease
 - transependymal absorption of CSF (e.g. in active hydrocephalus)

□ Evaluation. Deciding which of the following tests are needed to evaluate a patient with multiple intracranial lesions must be individualized for the appropriate clinical setting.

1. cardiac echo: to R/O SBE that could shed septic emboli
2. “metastatic workup” (p.806) including:
 - a) CT of chest/abdomen/pelvis with and without contrast: has become a relatively standard part of the metastatic workup. It has largely supplanted CXR, lower GI (barium enema) and IVP.
Rationale:
 - Chest: R/O primary bronchogenic Ca or pulmonary metastases of another Ca. Can demonstrate mediastinal lymphadenopathy. Also to R/O pulmonary abscess that could shed septic emboli.
 - Assesses for possible primary lesions: e.g. kidneys, GI, prostate
 - Evaluates for metastases to liver, adrenal, and even spine.
 - b) mammogram in women
 - c) PSA in men

89.4 Ring-enhancing lesions on CT/MRI

89.4.1 Abscess vs. tumor

See □ Fig. 89.1 and □ Fig. 89.2. Tumor: the enhancing ring may be incomplete and irregular. Abscess: ring is usually complete, often thinner and smoother than with tumor. Abscess: usually brighter than tumor on DWI MRI.

MRS should theoretically be ideal for differentiating tumor from abscess (abscess should show reduced NAA, Cr and choline, and “atypical peaks” may be present), but in practice is often not conclusive.

89.4.2 Short list

Multiple lesions: metastases or abscess are much more likely than astrocytoma.

In adults, the main differential (short list) is:

1. High grade glioma (glioblastoma)
2. Metastasis
3. Abscess
4. Lymphoma should also always be tacked on as a possibility

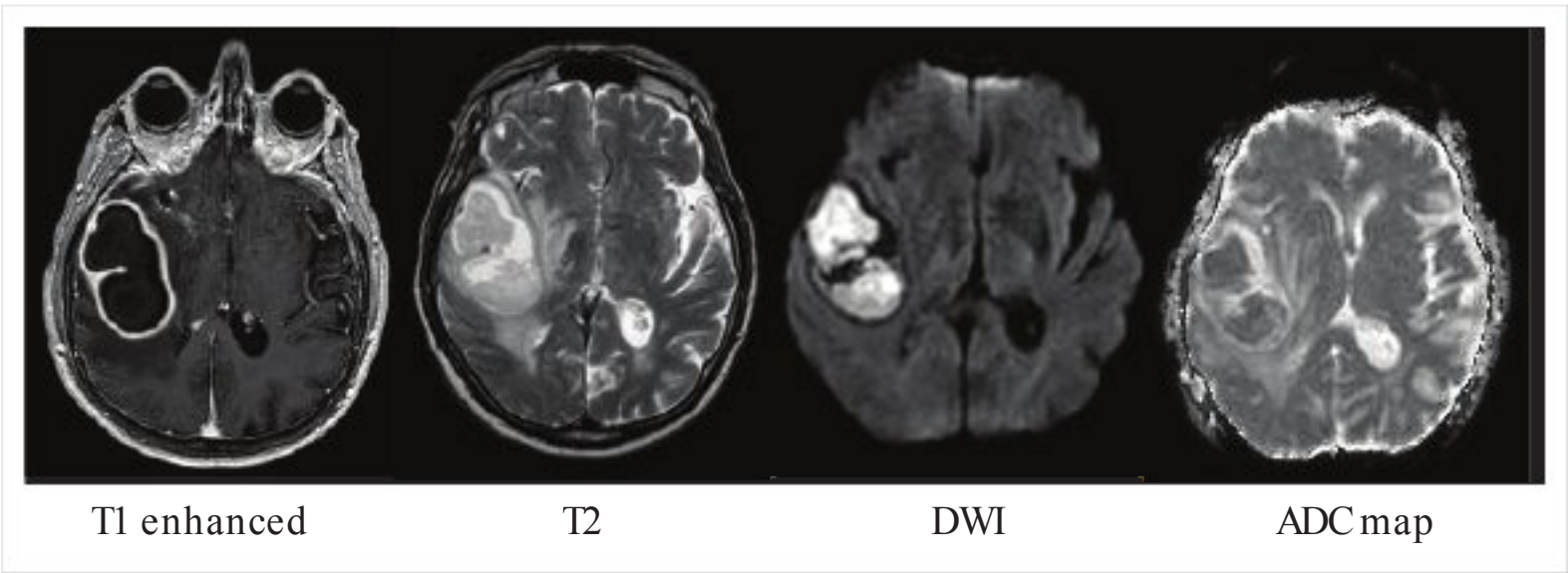


Fig. 89.1 MRI of right hemispheric cerebral abscess (bright on DWI)

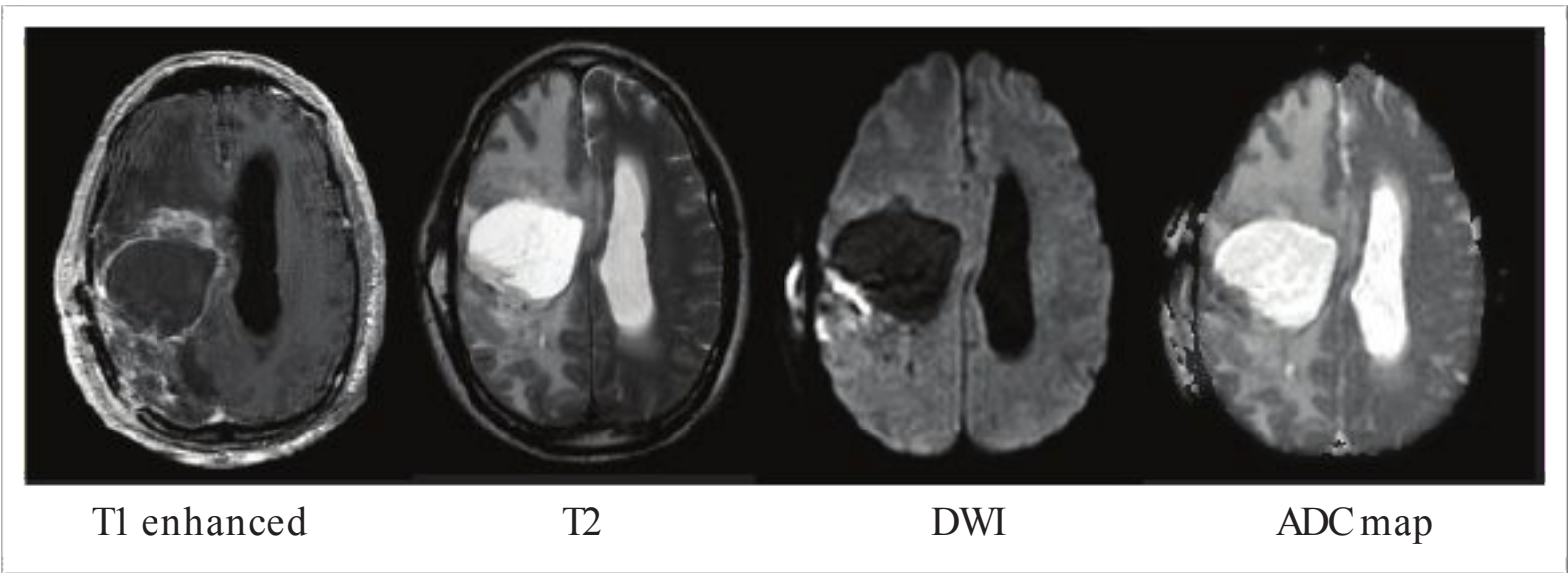


Fig. 89.2 MRI of right hemispheric glioblastoma (dark on DWI)

89.4.3 Long list

Mnemonic: “Magic Dr” (metastasis (including lymphoma), abscess, glioma, infarct, contusion, demyelination, radiation).

1. astrocytoma: usually glioblastoma multiforme
2. metastases (p.800): especially lung
3. abscess (p.320):
 - a) may see visible growth over several days on serial imaging
 - b) pyogenic abscesses are often (but not always) associated with fever and rapidly progressing neurologic deficit
 - c) Nocardia abscesses (p.335) are often multiloculated and are usually associated with a lung lesion
4. others
 - a) lymphoma (primary brain lymphoma or metastatic systemic lymphoma): wall is thicker than abscess.¹⁰ Incidence is increasing (p.712)
 - b) radiation necrosis
 - c) resolving intracerebral hematoma: on T1 gradient echo sequence, a continuous ring suggests hematoma, an interrupted ring suggests malignancy
 - d) cystic lesions with enhancing wall or mural nodule (see also intracranial cysts):
 - cysticercosis cyst, see Neurocysticercosis (p.371)
 - hemangioblastoma
 - pilocytic astrocytoma
 - cystic acoustic neuroma
 - e) trauma
 - f) recent infarct
 - g) thrombosed giant aneurysm

89.5 White matter lesions

89.5.1 Leukoencephalopathy

Disease largely confined to the white matter. Demyelinating disease cause most.

Appear as white matter low density on CT or low signal on T1WI MRI, and high-intensity on T2WI. Usually does not enhance. Unlike a stroke, changes tend to spare the cortex. Conditions such as metabolic derangements, leuko-araiosis, etc. tend to produce fairly symmetric findings.

Differential diagnosis:

1. anoxia/ischemia
2. demyelinating disease
 - a) MS
 - b) ADEM (p.182)
3. intoxication: cyanide, organic solvents, carbon monoxide
4. vitamin deficiencies: B12 with subacute combined degeneration
5. infectious, especially viral:
 - a) progressive multifocal leukoencephalopathy (PML) (p.331)
 - b) herpes varicella-zoster leukoencephalitis (p.366)
 - c) HIV infection (AIDS): perivascular pattern of demyelination
 - d) cytomegalovirus infection
 - e) Creutzfeldt-Jakob disease: small and perivascular demyelination
6. metabolic derangements: hyponatremia (p.110), excessively rapid correction of hyponatremia (causing osmotic myelinolysis)
7. hereditary: metachromatic leukodystrophy, adult-onset Schilder's disease
8. leuko-araiosis (p.1384)
9. multiple myeloma (p.714)
10. low grade (WHO grade II infiltrating) glioma

89.5.2 Corpus callosum lesions

1. lymphoma
2. MS plaque
3. tumefactive demyelinating lesions (p.181)
4. lipoma
5. diffuse axonal injury from trauma

89.6 Sellar, suprasellar and parasellar lesions

89.6.1 General information

May enlarge, erode or destroy the sella turcica. Considerations in adults (adenoma is the most common enhancing pituitary lesion) are different than for children (adenomas are rare, craniopharyngioma and germinoma are more common). Includes (modified¹¹):

89.6.2 Tumors/pseudotumors

Tumors having epicenter within the sella

□ Pituitary tumor:

1. adenohypophyseal tumors
 - a) adenoma
 - microadenoma: <1 cm diameter (p.718)
 - macroadenoma: ≥1 cm diameter
 - invasive adenoma (p.721): Includes aggressive tumors of Nelson's syndrome (p.724)
 - b) pituitary carcinoma or carcinosarcoma (p.718)
2. neurohypophyseal tumors
 - a) metastases: the most common tumor found in the posterior pituitary (presumably due to rich blood supply): breast and lung are most common primaries¹²
 - b) pituicytoma (p.728): the most common tumor arising from neurohypophysis/pituitary stalk (i.e. primary)
 - c) astrocytoma: arising from stalk or posterior pituitary

□ Pituitary “pseudotumor”:

1. hyperplasia (enlargement)
 - a) thyrotroph hyperplasia due to primary hypothyroidism¹³ (see □ Table 46.2) causing chronic pituitary stimulation by TRH. Typically: free T4 low or normal, TSH ↑ ↑, symmetrical sellar mass on MRI
 - b) gonadotroph hyperplasia: due to primary hypogonadism
 - c) somatotroph hyperplasia: due to ectopic GH-RH secretion
 - d) lactotroph hyperplasia: in pregnancy
2. pituitary enlargement may occur in intracranial hypotension (p.391)
3. the pituitary gland of young women of childbearing potential is normally slightly enlarged

Juxtapellar or suprasellar tumors or masses: any of these lesions may extend into the sella

1. craniopharyngioma (p.763): in this region, these account for 20% of tumors in adult, 54% in peds
2. Rathke cleft cyst (p.756)
3. meningioma (parasellar, tuberculum sellae, or diaphragma sellae): to differentiate tuberculum sellae meningioma from pituitary macroadenoma on MRI (□ Fig. 89.3), 3 characteristics of meningioma are: 1) bright homogeneous enhancement with gadolinium (c.f. heterogeneous, poor enhancement with macroadenoma), 2) suprasellar epicenter (vs. sellar), 3) tapered extension of intracranial dural base¹⁴ (dural tail). Also, the sella is usually not enlarged, and even large suprasellar meningiomas rarely produce endocrine disturbances.¹⁵ The pituitary stalk is sometimes seen being pushed posteriorly by a meningioma. Tuberculum sellae meningiomas may be associated with sphenoid pneumosinus dilatans¹⁶ (enlargement of the underlying sphenoid sinus without bone erosion)
4. pituitary tumor (mostly adenomas) with extrasellar extension: tends to push carotids laterally (unlike meningioma which may encase carotid), more symmetric than meningioma
5. germ cell tumors (GCT) (p.659): choriocarcinoma, germinoma, teratoma, embryonal carcinoma, endodermal sinus tumor. In females, suprasellar GCTs are more common; in males pineal region is more common
 - a) suprasellar GCT: triad of diabetes insipidus, visual deficit and panhypopituitarism.¹⁷ May also present with obstructive hydrocephalus
 - b) simultaneous suprasellar and pineal lesions is diagnostic of GCT (so-called synchronous germ cell tumors (p.659))
6. glioma
7. hypothalamic glioma
8. optic nerve or chiasm (optic glioma) (p.631)
9. metastasis
10. chordoma
11. parasitic infections: cysticercosis
12. epidermoid cyst

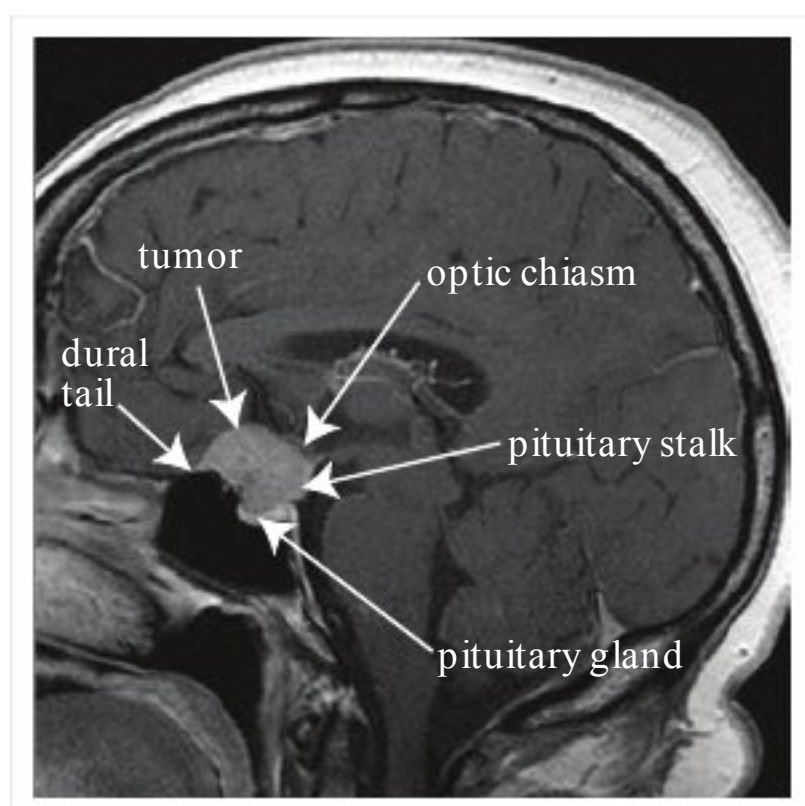


Fig. 89.3 Tuberculum sellae meningioma that could be mistaken for a pituitary adenoma. Contrast enhanced T1 WI sagittal MRI

- 13. suprasellar arachnoid cyst: see Arachnoid cysts (p.265)
- 14. sarcoidosis (p.189): hypothalamic involvement is a more likely site as a cause of anterior and/or posterior pituitary insufficiency
- 15. bone abnormalities
 - a) giant cell tumor (p.794)
 - b) chondromyxoid fibroma
 - c) brown tumor of hyperparathyroidism
 - d) bone spur
 - e) extramedullary hematopoiesis¹⁸

89.6.3 Vascular lesions

- a) aneurysm: ACoA, ICA (p.1334) (cavernous carotid or suprasellar variant of superior hypophyseal artery aneurysm), ophthalmic, basilar bifurcation. Giant aneurysms may produce mass effect
- b) carotid cavernous fistula (CCF) (p.1256)

89.6.4 Inflammatory

- a) (autoimmune) hypophysitis (see below):
 - distinguishing imaging characteristics are shown in □ Table 89.2
 - the most important clinical feature: pregnancy
 - the most important laboratory feature: diabetes insipidus (if DI is present, it is unlikely to be an adenoma)
- b) pituitary granuloma¹⁹

89.6.5 Empty sella syndrome

- 1. primary (p.773)
- 2. secondary: following pituitary tumor resection (p.773)

89.6.6 Hypophysitis

- AKA autoimmune hypophysitis (AH)
- Two main forms:
- 1. lymphocytic (adeno)hypophysitis²⁰ AKA lymphoid adenohypophysitis: the more commonly encountered form. Inflammation of the pituitary stalk with lymphocytic infiltrate. Well established autoimmune etiology, although the antigens have not been identified. Primarily affects women in late pregnancy or early postpartum period
 - 2. granulomatous hypophysitis: more aggressive. No gender bias. No association with pregnancy. May be autoimmune, but pathogenesis not definitely known

Table 89.2 Imaging characteristics of hypophysitis vs. adenoma ²⁰		
Feature	Hypophysitis	Adenoma
Enlargement	symmetric	asymmetric
Pituitary stalk	thickened, nontapering	not thickened, tapering, deviated
Sellar floor ^a	spared	may be eroded
Enhancement	intense, may be heterogeneous	less intense, usually homogeneous
Mean size at time of presentation	3 cm ³	10 cm ³
Posterior pituitary bright spot ^b	lost	preserved in 97%
^a on CT scan		
^b the normal hyperintensity of the posterior pituitary on T1 WI MRI (p.737) ²¹		

Because AH often mimics a nonsecretory pituitary macroadenoma (enhancing sellar mass on imaging, with negative endocrine tests), these lesions often undergo surgical resection instead of what may be more appropriate medical therapy (e.g. steroids,²² or discontinuing possible offending agents such as ipilimumab²³).

For distinguishing imaging characteristics, see □ Table 89.2.

89.7 Intracranial cysts

89.7.1 In general

Modified²⁴:

1. arachnoid cysts (p.265): Typically lined with meningotheelial cells
2. suprasellar cyst from dilated third ventricle
3. interhemispheric cyst from porencephaly
4. neuroectodermal cysts (glioependymal cysts): intraparenchymal, located near ventricles
5. old infarct: if it communicates with ventricle it is called a porencephalic cyst
6. tumor cysts (the solid portion may sometimes be isodense to brain on CT):
 - a) ganglioglioma (p.652): usually solid but may appear cystic on CT
 - b) pilocytic astrocytoma (p.629): usually has enhancing mural nodule
 - c) neurilemmomas may be cystic
 - d) supratentorial ependymomas are often cystic (p.643)
7. infectious
 - a) abscess
 - b) cysticercosis: see Neurocysticercosis (p.371)
 - c) hydatid cyst: see Echinococcosis (p.375)
8. pineal cysts (p.658)
9. colloid cyst (p.756)
10. Rathke’s cleft cyst (p.756)
11. giant aneurysm
12. on CT, a low density non-enhancing tumor can mimic a cyst
13. chronic subdural hematoma or hygroma may mimic a cyst
14. posterior fossa: (for cysts of the CPA) (p.1366). Includes:
 - a) cyst associated with Dandy-Walker malformation (p.256)
 - b) epidermoid (p.760)
 - c) enlarged cisterna magna may mimic a cyst
 - d) cerebellar hemangioblastoma: often has an enhancing mural nodule (p.704)
 - e) arachnoid cyst of posterior fossa
 - f) neurenteric cyst (p.290)
 - g) pilocytic astrocytoma of the cerebellum (p.629): usually has an enhancing mural nodule

89.7.2 Midline cavities

Three potential supratentorial midline cavities in the center of the brain and differentiating features are shown in □ Table 89.3.

Table 89.3 Features of midline brain cavities ²⁵			
Cavity	Anatomy	Frequency	Clinical significance
cavum septum pellucidum (CSP) (see text)	located between leaflets of septum pellucidum	100% of preemies, 97% of newborns, 10% of adults	no known association with pathologic conditions
cavum vergae	directly posterior to, and often communicating with CSP	relatively uncommon	possible association with neurologic abnormalities ^a
cavum velum interpositum	due to separation of crura of fornix between thalami above the 3rd ventricle	present in 60% of children <1 yr age, and in 30% between 1 and 10 yrs	no known association with pathologic conditions
^a including developmental delay, macrocephaly, Apert’s syndrome, abnormal EE			

89.7.3 Cavum septum pellucidum (CSP)

AKA fifth ventricle, among others. A variable slit-like fluid-filled space between the leaflets of the left and right septum pellucidum. The compartment is usually isolated, although some communicate with third ventricle. The CSP is part of normal development, and persists until shortly after birth. Thus, it is present in $\approx$ all preemies. It is found in $\approx$ 10% of the adult population, usually representing an asymptomatic developmental anomaly. However, it is also commonly seen in boxers suffering from chronic traumatic encephalopathy (p.924).

89.8 Orbital lesions

89.8.1 General information

4 compartments of the orbit:

1. ocular (AKA globe, AKA bulbar)
2. optic nerve sheath
3. intraconal
4. extraconal

CT remains a strong imaging modality within the orbit (less susceptible to motion artifact than MRI because of the speed, images bony structures to good advantage).

89.8.2 Orbital lesions in adults

Orbital pseudotumor is the most common.

1. neoplastic
 - a) discrete tumors that may occur adjacent to but not envelope the optic nerve sheath
 - cavernous hemangioma: the most common benign primary intra-orbital neoplasm. Choroidal hemangioma is seen in Sturge-Weber syndrome
 - fibrohistiocytoma
 - hemangiopericytoma
 - b) capillary hemangioma: produces infantile proptosis. Regresses spontaneously
 - c) lymphangioma: produces infantile proptosis. Does not regress
 - d) melanoma: the most common primary ocular malignancy of adulthood
 - e) retinoblastoma: congenital, malignant primary retinal tumor. 40% are bilateral, 90% are calcified (often a key differentiating feature; does not portend benignity as with other lesions). CT may show retinal detachment
 - f) lymphoma of the orbit: causes painless proptosis. The 3rd most common cause of proptosis
 - g) intraorbital meningioma
 - h) primary optic nerve tumors
 - optic glioma
 - optic nerve sheath tumor (schwannoma)
2. congenital
 - a) Coats disease: telangiectatic vascular malformation of retina which leaks a lipid exudate causing retinal detachment. May mimic retinoblastoma. Vitreous is hyperintense on MRI on both T1WI and T2WI due to lipid
 - b) persistent hyperplastic primary vitreous
 - c) retinopathy of prematurity (retrolental fibroplasia)
3. infectious
 - a) toxocara endophthalmitis
4. inflammatory/collagen vascular disease: usually bilateral
 - a) scleritis
 - b) pseudotumor of the orbit: the most common intraconal lesion. Usually unilateral (p.569)
 - c) sarcoidosis: usually affects the conjunctiva and lacrimal gland and spares connective tissues and intraorbital muscles
 - d) Sjögren's syndrome
5. vascular
 - a) enlargement of the superior orbital vein: may occur in thrombosis of cavernous sinus or in carotid-cavernous fistula
 - b) dural AVM
6. miscellaneous
 - a) drusen: degenerated retinal pigment cells in the posterior globe that may resemble calcified masses on CT

- b) thyroid ophthalmopathy: Graves' disease (hyperthyroidism & swelling of EOMs → painless proptosis). 80% of cases are bilateral. The ophthalmopathy is independent of the level of thyroid hormone (possibly an autoimmune process). NB: a swollen inferior rectus muscle may resemble an orbital tumor if seen only on lower CT cut through the orbit
- c) EOM enlargement can also occur with steroid use or occasionally with obesity
- d) fibrous dysplasia

89.8.3 Orbital tumors in pediatrics

1. dermoid cyst: 37% The most common lesion in children
2. hemangioma: 12% Most regress spontaneously without surgery
3. rhabdomyosarcoma: 9% The most common malignant tumor of the orbit
4. optic nerve glioma: 6%
5. lymphangioma: <7% Imaging resembles hemangioma. But will not regress spontaneously, requires surgery. Proptosis may worsen after a URI. May bleed into itself (chocolate cysts)

89.9 Cavernous sinus lesions

Modified²⁶:

1. primary tumors (rare)
 - a) meningiomas²⁷
 - b) neurinomas
2. tumors from adjacent areas that may extend into cavernous sinus (head and neck cancers may track intracranially along cranial nerves, especially V)
 - a) meningiomas
 - b) neurinomas
 - c) chordomas
 - d) chondromas
 - e) Chondrosarcomas
 - f) pituitary tumors²⁸
 - g) nasopharyngeal carcinomas
 - h) esthesioneuroblastomas
 - i) nasopharyngeal angiofibromas
 - j) metastatic tumors
3. inflammation: e.g. Tolosa-Hunt (p.569)
4. infection: mucormycosis (phycomycosis) (p.568). Usually in diabetics
5. vascular
 - a) cavernous carotid aneurysm
 - b) carotid-cavernous fistula (p.1256)
 - c) cavernous sinus thrombosis

89.10 Skull lesions

89.10.1 General information

The most common benign tumors of skull are osteoma and hemangioma. Osteogenic sarcoma is the most common malignancy. See also specific skull tumors (p.775).

Evaluating roentgenographic skull lucencies

There is enough overlap of features to prevent any systematic means of determining the etiology of all or even most radiographic skull lucencies. The following features should be noted for any lucency, some are more helpful than others (modified²⁹):

1. multiplicity (single or multiple?): except for multiple venous lakes, the presence of 6 or more defects is usually indicative of a malignancy
2. origin (intradiploic, full thickness, inner or outer table only):
 - a) most vault lesions originate intradiploically, so limitation to this space may merely signify early recognition of a lesion
 - b) expansion of the diploë with bulging of one or both tables almost always signifies a benign lesion
 - c) full thickness lesions affecting both tables congruently usually indicates malignancy, whereas non-congruent erosion is more common with benign lesions

3. edges (smooth or ragged):
 - a) smooth edges, whether regular, distinct or indistinct: no predictive value
 - b) irregular margins (especially ragged undermined edges): more suggestive of infection (osteomyelitis) or malignancy
 - c) sharply demarcated, full thickness punched out defects: suggest myeloma
4. presence of peripheral sclerosis: circumferential bony sclerosis suggests benignity (may indicate slow expansion and longstanding nature). The ring of sclerosis is generally narrow except in fibrous dysplasia
5. presence or absence of peripheral vascular channels: presence is highly suggestive of benign lesions (seen in $\approx 66\%$ of venous lakes and $\approx 50\%$ of hemangiomas)
6. pattern within the lucency:
 - a) ☐ hemangiomas classically show honeycomb or trabecular pattern (seen in $\approx 50\%$ of cases) or sunburst pattern (seen in $\approx 11\%$ of cases)
 - b) fibrous dysplasia may show well defined islands of bone, or a grossly mottled appearance with randomly arranged cystic and dense areas
7. location on the cranial vault (high vs. low): poor correlation with benign vs. malignant lesions
8. pain: Langerhans cell histiocytosis lesions are often tender

NB: keep in mind that skull lesions may have an intracranial component. CT scanning is good for assessing bone (MRI is poor for this), however, CT may miss small intracranial lesions tucked within the convexity of the calvaria due to bone hardening artifact (MRI has better sensitivity in this setting).

Nuclear bone scan may be a helpful adjunctive test (see specific lesion for findings).

Biopsy: indicated for questionable skull lesions. If the bone has not been destroyed by soft tissue, biopsy may be accomplished with a Craig needle, and the specimen may need decalcification by the pathologist before histologic evaluation can be completed.

89.10.2 Radiolucent lesion or bone defect in skull (AKA lytic lesions)

1. congenital or developmental
 - a) epidermoid (cholesteatoma): sclerotic edge
 - b) congenital: encephalocele, meningoencephalocele, dermal sinus
 - c) fibrous dysplasia (p.779). A benign condition in which normal bone is replaced by fibrous connective tissue. Tends to occur higher in calvaria. 3 types:
 - cystic: widening of the diploë usually with thinning of the outer table and little involvement of the inner table. Typically involves calvaria
 - sclerotic: usually involves skull base (especially sphenoid bone) and facial bones
 - mixed: appearance is similar to cystic type with patches of increased density within the lucent lesions
 - d) hemangioma or AVM of bone or scalp
 - e) pacchionian depression: arachnoid granulations (older terms: Pacchioni's granulation (after Italian anatomist Antonio Pacchioni) or pacchionian bodies) resorb CSF into vascular system and occasionally cause a bony lucency, usually near the superior sagittal sinus
 - f) Albright's syndrome
 - g) congenital foramina: "holes" in skull traversed by emissary veins
 - h) parietal thinning: usually a bilateral process
 - i) frontal fenestrae
 - j) venous lakes
 - k) cerebral herniations: AKA occipital pacchionian granulations
2. traumatic
 - a) surgical defect: burr hole, craniectomy
 - b) fracture
 - c) post-traumatic leptomeningeal cyst (p.915)
 - d) following trauma in children³⁰
3. inflammatory
 - a) osteomyelitis: including tuberculosis³¹
 - b) sarcoidosis
 - c) syphilis
4. neoplastic
 - a) hemangioma: fine, honeycombed matrix. Classic x-ray finding: "starburst" appearance due to radiating bone spicules (may occur in as few as $\approx 11\%$ of cases²⁹)
 - b) intracranial tumor with erosion

- c) lymphoma, leukemia
 - d) meningioma
 - e) metastasis: usually hot on bone scan
 - f) multiple myeloma, plasmacytoma (p.714). Usually cool on bone scan
 - g) sarcoma or fibrosarcoma of bone
 - h) skin tumor with invasion (rodent ulcer)
 - i) neuroblastoma
 - j) lipoma
 - k) epidermoid (may also be considered congenital, thus also see above)
5. miscellaneous
- a) Langerhans cell histiocytosis (p.777). Perfectly round non-sclerotic punched out lesion, may be single (formerly called eosinophilic granuloma) or multiple, tender
 - b) Paget's disease (when seen as a zone of osteolysis without osteoblastic sclerosis on skull films, this is defined as osteoporosis circumscripta). Usually "hot" on bone scan
 - c) aneurysmal bone cyst: rare. Arises in diploë and expands both tables which become thin but remain intact
 - d) brown tumor of hyperparathyroidism

89.10.3 Diffuse demineralization or destruction of the skull

Includes "salt and pepper skull."

- 1. common
 - a) hyperparathyroidism, primary or secondary
 - b) metastatic carcinoma or neuroblastoma
 - c) multiple myeloma
 - d) osteoporosis
- 2. uncommon
 - a) Paget's disease (osteoporosis circumscripta)

89.10.4 "Hair-On-End" appearance in skull

- 1. common
 - a) congenital hemolytic anemia (e.g. thalassemia, sickle cell, hereditary spherocytosis, pyruvate kinase deficiency)
- 2. uncommon
 - a) hemangioma
 - b) cyanotic congenital heart disease (with secondary polycythemia)
 - c) iron deficiency anemia
 - d) metastases: especially neuroblastoma, thyroid carcinoma
 - e) multiple myeloma
 - f) meningioma
 - g) osteosarcoma
 - h) polycythemia vera

89.10.5 Diffuse increased density, hyperostosis, or calvarial thickening

- 1. common
 - a) anemia (sickle cell, iron deficiency, thalassemia, hereditary spherocytosis)
 - b) fibrous dysplasia
 - leontiasis ossea ("lion-like facies"): a form of polyostotic fibrous dysplasia
 - c) hyperostosis interna generalisata
 - d) osteoblastic metastases (especially prostate, breast)
 - e) Paget's disease (begins with lytic zone and diploic thickening)
 - f) treated hydrocephalus
- 2. uncommon
 - a) chronic phenytoin therapy
 - b) Engelmann's disease (progressive diaphyseal dysplasia)
 - c) fluorosis
 - d) hypervitaminosis D
 - e) hypoparathyroidism, pseudohypoparathyroidism
 - f) meningioma

- g) osteogenesis imperfecta
- h) osteopetrosis (p. 1401)
- i) secondary polycythemia
- j) syphilitic osteitis
- k) tuberous sclerosis

89.10.6 Focal increased density of skull base

1. common
 - a) fibrous dysplasia
 - b) meningioma
2. uncommon
 - a) mastoiditis
 - b) nasopharyngeal carcinoma
 - c) osteoblastic metastasis
 - d) osteoma of the outer table or diploe
 - e) chondroma
 - f) sarcoma of bone (e.g. osteosarcoma, chondrosarcoma)
 - g) sphenoid sinusitis

89.10.7 Generalized increased density of skull base

1. common
 - a) fibrous dysplasia
 - b) Paget's disease
2. uncommon
 - a) severe anemia (e.g. thalassemia, sickle cell)
 - b) Engelmann's disease (progressive diaphyseal dysplasia)
 - c) fluorosis
 - d) hyperparathyroidism, primary or secondary (treated)
 - e) hypervitaminosis D
 - f) idiopathic hypercalcemia
 - g) meningioma
 - h) osteopetrosis (p. 1401)

89.10.8 Localized increased density or hyperostosis of the calvaria

1. common
 - a) anatomic variation (e.g. sutural sclerosis)
 - b) fibrous dysplasia
 - c) osteoma (p. 775)
 - d) meningioma
 - e) hyperostosis frontalis interna (p. 779)
 - f) osteoblastic metastases (especially: prostate, breast)
 - g) Paget's disease (begins with lytic zone and diploic thickening)
 - h) cephalhematoma
 - i) depressed skull fracture
2. uncommon
 - a) osteosarcoma
 - b) chronic osteomyelitis, tuberculosis
 - c) tuberous sclerosis
 - d) osteoid osteomas: radiolucent nidus with surrounding zone of dense sclerosis
 - e) osteblastoma:
 - f) ossifying fibromas: predilection for frontotemporal region
 - g) radiation necrosis

89.10.9 Pneumocele

Pneumocele: enlargement of an air sinus often with bone erosion. Pneumosinus dilatans (p. 1372) generally denotes enlargement of an air sinus without bone erosion, as may occur with tuberculum sellae or planum sphenoidale meningiomas.

Pneumoceles occur primarily in the maxillary antrum. Involvement of frontal sinus. usually involves pneumosinus dilatans. Etiology is unknown, and may involve a trap-valve mechanism, ruptured mucocele, or possibly congenital. May occur with fibrous dysplasia.

Presentation of pneumocele or pneumosinus dilatans:

1. headache
2. neuralgia
3. facial asymmetry
4. frontal bossing (with frontal pneumosinus dilatans)
5. exophthalmous
6. CSF fistula (leak)
7. Treatment for maxillary pneumocele: opening the sinus into the nasal cavity via endoscopic approach. Watch for encephalocele.

89.11 Combined intracranial/extracranial lesions

Lesion causing mass outside skull with intracranial component.

1. intra-axial: rule of thumb – “there is no intra-axial lesion that grows out of skull”, however untreated fungating malignant gliomas may do this
2. extra-axial:
 - a) meningioma
 - may arise in diploe, grows outward and inward
 - intracranial meningioma can grow through bone by destroying it
 - intracranial meningioma can induce hyperostosis that causes extracranial mass
 - b) metastatic disease (e.g. GI carcinoma, and especially prostate Ca)
 - c) bone (skull) lesion:
 - hemangioma
 - epidermoid
 - fibrous dysplasia (rare)
 - giant cell tumor (rare)
 - Ewing’s sarcoma (rare in skull)
 - aneurysmal bone cyst (5% occur in skull, occipital bone most common)

89.12 Intracranial hyperdensities

I.e. Differential diagnosis of an intracranial structure that is hyperdense with respect to brain (i.e. appear “whiter” than brain) on non-contrast CT):

1. acute blood
2. calcium
3. vessels with low flow
4. melanoma: may be slightly hyperdense to brain due to melanin

89.13 Intracranial calcifications

89.13.1 Single intracranial calcifications

1. physiologic
 - a) choroid plexus: calcifications usually bilateral (see below)
 - b) arachnoid granulation
 - c) diaphragma sellae
 - d) dural (falcine, tentorial, sagittal sinus)
 - e) habenular commissure
 - f) petroclinoid or interclinoid ligaments
 - g) pineal: 55% of patients > 20 yrs age have a calcified pineal gland visible on plain skull x-ray
2. infection
 - a) cysticercosis cyst: single or multiple, see Neurocysticercosis (p.371)
 - b) encephalitis, meningitis, cerebral abscess (acute and healed)
 - c) granuloma (torulosis and other fungi)
 - d) hydatid cyst
 - e) tuberculoma
 - f) paragonimiasis
 - g) rubella
 - h) syphilitic gumma

3. vascular
 - a) aneurysm, including:
 - vein of Galen aneurysm
 - giant aneurysm
 - b) arteriosclerosis (especially carotid artery in siphon region)
 - c) hemangioma, AVM, Sturge-Weber syndrome
4. neoplastic: calcifications usually suggest a more benign process
 - a) meningioma (p.690)
 - b) craniopharyngioma
 - c) choroid plexus papilloma
 - d) ependymoma
 - e) glioma (especially oligodendroglioma, also astrocytoma)
 - f) ganglioglioma
 - g) lipoma of corpus callosum
 - h) pinealoma
 - i) hamartoma of tuber cinereum
5. miscellaneous
 - a) hematoma: ICH, EDH or SDH. Calcifications usually only when chronic
 - b) idiopathic
 - c) tuberous sclerosis (p.606)

89.13.2 Multiple intracranial calcifications

1. common
 - a) choroid plexus: the most common site for physiologic calcification (in lateral ventricles where it is usually bilateral and symmetric; rare in 3rd & 4th ventricles). Increases in frequency and extent with age (prevalence: 75% by 5th decade). Rare under age 3. Under age 10, consider possible choroid plexus papilloma. Involvement in the temporal horns is often associated with neurofibromatosis
 - b) basal ganglia (BG): slight bilateral BG calcifications on CT are common, especially in the elderly. Considered a normal radiographic variant by some. They may be idiopathic, secondary to conditions such as hypoparathyroidism or long-term anticonvulsant use, or part of rare conditions such as Fahr's disease (see below). BG calcifications >0.5 cm dia are possibly associated with cognitive impairment and a high prevalence of psychiatric symptoms (including bipolar and obsessive-compulsive disorders, but no patients had schizophreniform disorders)³²
2. uncommon
 - a) Fahr's disease: progressive idiopathic calcification of medial portions of basal ganglia, sulcal depths of cerebral cortex, and dentate nuclei³³
 - b) hemangioma, AVM, Sturge-Weber syndrome, von Hippel-Lindau disease
 - c) basal cell nevus syndrome (falx, tentorium)
 - d) Gorlin's syndrome. Associated findings: mandibular cysts, rib and vertebral deformities, short metacarpals. Medulloblastoma seen in several patients
 - e) deposition of calcium in the media of medium-sized blood vessels without compromise of the lumen. Usually asymptomatic. May become symptomatic by the time the involvement is significant enough to be visible on plain x-ray in a young person
 - f) cytomegalic inclusion disease
 - g) encephalitis (e.g. measles, chickenpox, neonatal herpes simplex)
 - h) hematomas (SDH or EDH, chronic)
 - i) neurofibromatosis (choroid plexi)
 - j) toxoplasmosis
 - k) tuberculomas; tuberculous meningitis (treated)
 - l) tuberous sclerosis
 - m) hypoparathyroidism (including post-thyroidectomy cases³⁴) and pseudohypoparathyroidism
 - n) multiple tumors (e.g. meningiomas, gliomas, metastases)
 - o) cysticercosis cyst: may be single or multiple, see Neurocysticercosis (p.371)

89.14 Intraventricular lesions

89.14.1 General information

Intraventricular tumors represent only $\approx 10\%$ of CNS neoplasms. A clue to differentiating a tumor located within the ventricle from an intraparenchymal tumor invaginating into the ventricle is a "cap" of CSF surrounding an intraventricular tumor on CT or MRI.

89.14.2 Differential diagnosis

(Percentages quoted here are from a series of 73 patients with an intraventricular lesion on CT seen at UCSF³⁵).

1. astrocytoma: (20%) the most common lesion. Hydrocephalus (HCP) is present in 73% Hyperdense on non-contrast CT (NCCT) in 77%
Locations in descending order of frequency:
 - a) frontal horn
 - b) third ventricle
 - c) atrium (AKA trigone)
 - d) fourth ventricle
2. colloid cyst: (14%) essentially seen only in anterior third ventricle at foramen of Monro (other sites have been described but are exceedingly rare). 50%are hyperdense on NCCT. MRI appearance is variable, and may occasionally be missed. Little or no enhancement on CT/MRI (p.756). DDx includes xanthogranuloma
3. meningioma: (12%) most in atrium, rarely in frontal horn. All hyperdense with dense uniform enhancement. May be calcified. Most have dense tumor blush on angiogram, most supplied from anterior choroidal artery, posterior choroidal less common. Thought to arise from arachnoidal cells within the choroid plexus
4. ependymoma: (10%) most in 4th ventricle, may occur in body of lateral ventricle. Often hyperdense on CT because of high cellularity
5. craniopharyngioma: (7%) primarily in 3rd ventricle. Most have punctate calcification. Squamous epithelial rests in region of lamina terminalis are felt to give rise to this uncommon variety of craniopharyngioma
6. medulloblastoma: (5%) often fill 4th ventricle. Hyperdense on CT with homogeneous enhancement
7. cysticercosis: (5%) may involve any ventricle or may be panventricular (NB: incidence related to geographic location),
8. choroid plexus papilloma: (5%) most common in lateral ventricle (may be bilateral), but also may be seen in 4th and occasionally in 3rd. Non-obstructive HCP may occur (possible CSF overproduction). Intense blush on angiogram
9. epidermoid: (4%) mostly in 4th ventricle. Hypodense on CT with no enhancement (tend to follow CSF signal). The most common 4th ventricular low density lesion in the U.S.
10. dermoid: (3%) common in 4th ventricle. May see free floating fat in ventricles suggestive of cyst rupture. Tendency to form in midline
11. choroid plexus carcinoma: (3%) common in atrium of lateral ventricle. May extended into adjacent brain parenchyma with edema and shift. Intense blush on angio. NB: very rare lesion
12. subependymoma: (3%) 4th ventricle or frontal horn. Typically isodense on CT with minimal enhancement. May have calcification or cystic degeneration (more common in ependymoma). Most commonly in floor of 4th ventricle near obex
13. ependymal cyst: (3%) common in lateral ventricle. Absence of communication demonstrated by water soluble contrast cisternography
14. arachnoid cyst: (1%) lateral ventricle. Absence of communication demonstrated by water soluble contrast cisternography
15. arteriovenous malformation (AVM): (3%)
16. teratoma: (1%) Located in anterior 3rd ventricle. Partially calcified with foci of fat density. Marked enhancement
17. central neurocytoma (p.645)
18. metastases: breast and lung reported³⁶
19. chordoid glioma of the 3rd ventricle³⁷

89.14.3 Features to help identify type of intraventricular lesions

By location within ventricular system

□ Table 89.4 shows the breakdown of lesion type by location within the ventricular system.

By location and age within lateral ventricle

See reference.³⁸

See □ Table 89.5. This study excluded tumors that were clearly arising in the third ventricle or were predominantly parenchymal with intraventricular extension.

Table 89.4 Type of intraventricular lesions by location ³⁵ (numbers are patients out of 73 ^a)									
3rd ventricle		4th ventricle		Lateral ventricle					
				Atrium		Body		Frontal horn	
colloid cyst	10	medulloblast.	4	meningioma	8	ependymoma	3	astrocytoma	7
craniopharyng	5	ependymoma	4	astrocytoma	3	ch. plexus papil. ^b	1	meningioma	1
astrocytoma	4	epidermoid	3	ch. plexus papil.	1	ch. plexus carc.	1	subependym.	1
teratoma	1	cysticercosis	2	ch. plexus carc.	1	ependym. cyst	1	dermoid	1
ch. plexus papil.	1	astrocytoma	1	arachnoid cyst	1	AVM	1		
cysticercosis	1	subependym.	1	ependym. cyst	1				
dermoid	1								
ch. plexus carc.	1								
AVM	1								
^a 1 patient had cysticercosis diffusely throughout ventricles									
^b 1 patient with bilateral lateral ventricle papillomas									

Table 89.5 Lateral ventricle tumor type by location & age			
Age (yrs)	Location within lateral ventricle ^a		
	Foramen of Monroe region	Trigone	Body
0–5	0	8 CPP	2 PNETs 1 teratoma
6–30	5 SEGAs 2 pilocytic astrocytomas 1 CPP 1 meningioma 1 oligodendroglioma	1 ependymoma 1 oligodendroglioma	1 mixed glioma 1 ependymoma 1 pilocytic astrocytoma
>30	2 metastases	8 meningiomas	2 glioblastomas 1 lymphoma 1 metastasis 6 subependymomas
^a abbreviations: CPP=choroid plexus papillomas, PNET=primitive neuroectodermal tumor, SEGA=subependymal giant cell astrocytoma			

The teratoma and both PNETs occurred in age <1 year, and all showed calcifications. Only one CPP occurred above age 5 years.

In adults >30 years age, the only tumors found in the trigone were meningiomas. Subependyomas were the only nonenhancing tumor in this age group.

By location within third ventricle

1. anterior third ventricle
- a) colloid cyst
- b) sellar mass

- c) sarcoidosis
- d) aneurysm
- e) hypothalamic glioma
- f) histiocytosis
- g) meningioma
- h) optic glioma
- 2. posterior third ventricle
 - a) pinealoma (dysgerminoma)
 - b) meningioma
 - c) arachnoid cyst
 - d) vein of Galen aneurysm

By enhancement

All lesions enhanced except: cysts (ependymal and arachnoid), dermoids and epidermoids. There are differences of opinion of the tendency for subependymomas to enhance, Jelinek et al.³⁸ found that they did not.

By multiplicity

Multiple lesions are more suggestive of: neurocysticercosis, metastases, or ruptured epidermoid cyst.

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89.15 Periventricular lesions

89.15.1 Periventricular solid enhancing lesions (in decreasing frequency)

1. lymphoma; CNS involvement from systemic, or rarely primary brain (p.710): must be included in differential diagnosis of any solid enhancing periventricular brain tumor. Very radiosensitive.
2. ependymoma (usually invaginates)
3. metastatic Ca: especially malignant melanoma or choriocarcinoma
4. ventriculitis
5. medulloblastoma (in peds), AKA cerebellar sarcoma in adults
6. pineal tumor (dysgerminoma type): usually midline, young patient
7. occasionally, glioblastoma can present like this

89.15.2 Periventricular low density on CT, or high signal on T2WI MRI

1. increased extracellular or intracellular water content (edema)
 - a) in hydrocephalus: transependymal CSF absorption (p.399)
 - b) necrosis from infarction
 - c) edema from tumor
2. uncommon late variants of adrenoleukodystrophy
3. vascular disorders
 - a) subacute arteriosclerotic encephalopathy (Binswanger's disease)^{39,40}
 - b) cerebral embolism
 - c) vasculitis
 - d) amyloid angiopathy
 - e) low flow states
4. demyelination: including multiple sclerosis
5. leukoaraiosis⁴¹: a term coined to describe white matter disease with symmetric (or nearly so) periventricular white matter changes on CT or MRI. May be asymptomatic or may present with findings including dementia. May be related to:
 - a) Binswanger's encephalopathy
 - b) watershed infarction⁴²
 - c) normal aging⁴³: increases each decade after age 60, usually patchy
 - d) hypoxia
 - e) hypoglycemia⁴⁴
6. heterotopias: islands of grey matter in abnormal locations
7. following radiation therapy (XRT)

89.16 Meningeal thickening/enhancement

89.16.1 Dural enhancement

See reference.⁴⁵

Visible beneath the inner table of the skull. Unlike leptomeningeal enhancement, does not follow the gyral convolutions. May be either focal or diffuse:

1. focal
 - a) adjacent to meningioma: so called “dural tail”
 - b) pleomorphic xanthoastrocytoma: also can have “dural tail”
2. diffuse dural enhancement⁴⁶: associated with extraaxial neoplastic processes in $\approx 65\%$ Clinically: H/A, multiple cranial nerve palsies, seizures; may be indistinguishable from leptomeningeal metastases
 - a) intracranial hypotension: diffuse pachymeningeal enhancement on cerebral MRI (p.389)
 - b) bacterial meningitis
 - c) primary CNS tumors: medulloblastoma, malignant meningioma
 - d) sarcoidosis
 - e) following craniotomy
 - f) metastases (mostly carcinomas):
 - bony mets to skull: present in 10 of 13 patients
 - dural metastases
 - leptomeningeal
 - g) following subdural hemorrhage⁴⁷

89.16.2 Leptomeningeal enhancement

See reference.⁴⁵

1. thin linear enhancement that closely follows the gyri
2. small nodules attached to the brain

89.17 Ependymal and subependymal enhancement

Some overlap with periventricular enhancement. Ependymal enhancement often heralds a serious condition.⁴⁸ Main DDx is tumor vs. infectious process.

1. ventriculitis or ependymitis: ependymal enhancement occurs in 64% of cases of pyogenic ventriculitis.⁴⁹
 - a) Infection may occur in the following settings
 - following shunt surgery
 - after intraventricular surgery
 - with indwelling prosthetic devices (e.g. Ommaya reservoir)
 - with use of intrathecal chemotherapy
 - with meningitis
 - with viral ependymitis
 - in some cases of CMV encephalitis in immunocompromised patients
 - granulomatous involvement: esp. in immunocompromised patients; e.g. tuberculosis, mycobacterium, syphilis
 - b) Infections may be⁴⁸
 - bacterial (pyogenic) ventriculitis
 - tuberculous ventriculitis
 - cystic lesions suggest cysticercosis
2. carcinomatous meningitis: typically also produces meningeal enhancement (p.812)
3. multiple sclerosis: usually more periventricular (in the white matter)
4. tumors
 - a) lymphoproliferative disorders
 - CNS lymphoma (p.710)
 - leukemia
 - b) ependymoma
 - with tumor spread
 - transient enhancement reported in a child with ependymoma in the absence of tumor spread⁵⁰
 - c) Metastasis
 - d) Germ cell tumors

5. tuberous sclerosis: subependymal hamartomas appear as nodules which occasionally enhance (p.606). These gradually calcify with age
6. in the presence of appropriate constitutional symptoms: rare causes of linear enhancement include: neurosarcoidosis, Whipple's disease, metastatic multiple myeloma (usually nodular)

In immunocompromised patients, the enhancement pattern may help distinguish between the following (which tend to occur in this population⁴⁸):

1. thin linear enhancement: suggests virus (CMV or varicella-zoster)
2. nodular enhancement: suggests CNS lymphoma
3. band enhancement: less specific (may occur with virus, lymphoma, or tuberculosis (TB))

89.18 Intraventricular hemorrhage

Etiologies:

1. most occur as a result of extension of intraparenchymal hemorrhages
 - a) in the adult:
 - spontaneous ICH: especially thalamic or putaminal hemorrhages (p.1336)
 - associated with AVM
 - b) in newborns: extension of subependymal hemorrhage (p.1346)
2. pure intraventricular hemorrhage (IVH) is usually the result of a rupture of
 - a) aneurysm: accounts for $\approx 25\%$ of IVH in adults, and is second only to extension of intracerebral hemorrhage as the most common cause. IVH occurs in 13–28% of ruptured aneurysms in clinical series.⁵¹ More common with the following aneurysms: a-comm, distal basilar artery or carotid terminus, VA or distal PICA (p.1192) (for patterns)
 - b) vertebral artery dissection (or dissecting aneurysms) (p.1323)
 - c) intraventricular AVM
 - d) intraventricular tumor
 - e) SAH outside the ventricles refluxing into foramina of Luschka and/or Magendie

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89.19 Medial temporal lobe lesions

May be responsible for seizures, especially “uncal fits” (temporal lobe seizures).

1. hamartoma
2. mesial temporal sclerosis (p.441): should see atrophy of the parenchyma in this area with dilatation of the temporal horn of the lateral ventricle
3. glioma: may be low grade. Look for mass effect and possibly enhancement

89.20 Basal ganglion abnormalities

1. generally symmetric abnormalities
 - a) calcification (p.1381)
 - b) Wilson's disease (hepatolenticular degeneration): autosomal recessive disease causing accumulation of copper in tissues
 - c) Huntington's disease (or chorea): caused by >40 trinucleotide CAG repeats in the Huntington gene (4p16.3) which leads to the production of the protein huntingtin. Cell loss in caudate nucleus can be seen on CT or MRI
 - d) manganese: symmetrical high signal abnormalities on T1WI primarily in the globus pallidus with essentially no findings on T2WI or GRASS (almost pathognomonic) (p.177)
 - e) globus pallidus (low density on CT):
 - severe carbon monoxide intoxication
 - cyanide poisoning
 - hypoxia
 - f) putamen
 - hypoglycemia: affects corpus striatum (caudate and putamen)
2. stroke

89.21 Thalamic lesions

Astrocytomas are the most common tumors.

1. Common neoplasms
 - a) Adults
 - anaplastic astrocytoma
 - glioblastoma multiforme
 - metastasis
 - primary CNS lymphoma
 - b) Pediatrics
 - anaplastic astrocytoma
 - astrocytoma (WHO grade II)
 - glioblastoma multiforme
 - pilocytic astrocytoma
2. Uncommon neoplasms
 - a) Adults
 - astrocytoma (WHO grade II)
 - neurocytoma
 - oligodendroglioma
 - pilocytic astrocytoma
 - hamartoma
 - b) Pediatrics
 - germinoma
 - oligodendroglioma
 - PNET
 - subependymal giant cell tumor
3. non-neoplastic (pediatric and adult)
 - a) cavernous angioma
 - b) granuloma
 - c) heterotopias
 - d) AVM
 - e) infarct

89.22 Intranasal/intracranial lesions

Lesions within the nose that may communicate with the intracranial cavity:

1. infectious
 - a) tuberculosis
 - b) syphilis
 - c) Hansen's disease (leprosy)
 - d) fungal infections, especially:
 - aspergillosis
 - mucormycosis: seen primarily in diabetics or immunocompromised patients (p.568)
 - *Sporothrix schenckii*
 - *Coccidioides*
 - e) Wegener's granulomatosis (p.199): necrotizing granulomatous vasculitis of the upper and lower respiratory tracts with glomerulonephritis and nasal destruction⁵²
 - f) lethal midline granuloma (p.199): a locally destructive lymphomatoid infiltrative disease that may not have true granulomas, and may also cause local nasal destruction. However, renal and tracheal involvement do not occur as in Wegener's granulomatosis
 - g) polymorphic reticulosis: may be a nasal lymphoma. Possibly the same disease as lethal midline granuloma (see above)
2. mucocele: a retention cyst of an air sinus that results from an occluded ostium and may cause expansive erosion of the involved sinus. Often enhances with IV contrast (MRI or CT), and may contain mucus or pus
3. neoplasms
 - a) carcinoma of the nasal sinus
 - squamous cell
 - glandular
 - nasopharyngeal carcinomas: may be related to Epstein-Barr Virus (EBV) infection
 - sinonasal undifferentiated carcinoma (SNUC)⁵³: distinct from lymphoepithelioma (less keratinizing). Rare, aggressive carcinoma (more lethal variant of squamous cell carcinoma) with poor prognosis. Incidence may be higher with prior XRT and in woodworkers and nickel factory workers. May invade adjacent structures, those relevant to neurosurgeons: frontal fossa, and cavernous sinus. No relation to EBV. Treatment: tri-modal therapy (XRT, chemotherapy and salvage surgery)

Table 89.6 Encephalocele vs. nasal glioma		
Finding	Encephalocele	Nasal glioma
pulsatile?	frequently (may not be if small)	no
changes with Valsalva maneuver	swells (Furstenberg sign)	no change
presence of hypertelorism	suggests encephalocele	does not correlate
attachment to CNS	stalk	none, or minimal
probe	can be passed lateral	cannot be passed lateral

- b) esthesioneuroblastoma⁵⁴ or aesthesioneuroblastoma AKA olfactory neuroblastoma: named for the stem cell of the olfactory epithelium (esthesioneuroblast). A malignant tumor arising from crest cells of the nasal vault, often with intracranial invasion. Very rare (≈ 200 reported cases). Presents with epistaxis (76%), nasal obstruction (71%), tearing (14%), pain (11%), diplopia, proptosis, anosmia and endocrinopathies.⁵⁵ Treatment: surgical resection followed by XRT, ± chemotherapy
- c) metastatic tumors: very rare, possibly with renal cell carcinoma
- d) benign tumors
 - frontal meningioma: rarely erodes into nasal cavity
 - rhabdomyoma
 - benign hemangiopericytoma
 - cholesteatoma
 - chordoma
4. congenital lesions
- a) encephalocele : a nasal polypoid mass in a newborn should be considered an encephalocele until proven otherwise. Classifications:
 - cranial vault
 - frontal ethmoidal
 - basal
 - posterior fossa
- b) nasal glioma: non-neoplastic glial tissue located within the nose, often conceptually and diagnostically confused with an encephalocele (□ Table 89.6). The term “glioma” is a misnomer, and nasal glial heterotopia is preferred. Does not communicate with the subarachnoid space

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90 Differential Diagnosis by Location or Radiographic Finding – Spine

90.1 Diagnoses covered outside this chapter

Table 90.1 Differential diagnoses by location or radiographic finding, spine – covered outside this chapter	
DDx	
chordomas (p. 778)	
lateral disc herniation (p. 1058)	
spinal cord tumors (p. 783)	
spinal epidural abscess (p. 350)	
spinal stenosis	
• lumbar (p. 1101)	
synovial cyst (spinal) (p. 1144)	
thoracic outlet syndrome (p. 554)	

90.2 Atlantoaxial subluxation

- 90
- incompetence of the transverse atlantal ligament (TAL): results in increased atlanto-dental interval (ADI) (p.213)
 - rheumatoid arthritis: erosion of insertion points of the TAL (p.1134)
 - traumatic
 - disruption (tear) of the TAL (rare)
 - avulsion of the insertion points of the TAL (as in comminuted C1 fx)
 - congenital laxity of the TAL:
 - Down syndrome: 20% incidence (p.1138) ¹
 - may be associated with neurofibromatosis
 - retropharyngeal infections: chronic tonsillitis (p.968), Grisel syndrome
 - chronic steroid use
 - incompetence of the odontoid process: ADI is normal
 - odontoid fractures (p.978)
 - os odontoideum (p.981)
 - erosion of the odontoid due to rheumatoid arthritis (RA) (p.1134)
 - neoplastic erosion of the odontoid:
 - metastases to the upper cervical spine (p.815)
 - other tumors of the axis
 - Morquio syndrome: hypoplasia of the dens (p.1151)
 - congenital absence/dysplasia of the odontoid
 - following transoral odontoidectomy: creates severe ligamentous instability (p.1472)
 - local infection

□ Note. Chronic AAS seen in conditions such as rheumatoid arthritis or Down syndrome may be significant yet asymptomatic. Treatment decisions in this group are difficult. Acute AAS is more commonly symptomatic and may be life threatening.

90.3 Abnormalities in vertebral bodies

- See also lesions unique to the craniocervical junction & upper cervical spine (p.1151).
For abnormalities unique to the axis (C2), see below.
- neoplasms; see more extensive list (p.542)
 - metastases: prostate, breast, lung, renal cell, thyroid, lymphoma & myeloma commonly go to bone. Four patterns (≈ all are low intensity on T1WI):
 - focal lytic (most common): T1WI=hypointense, T2WI=hyperintense

- focal sclerotic: hypointense on T1WI and T2WI
- diffuse homogeneous: T1WI=hypointense, T2WI=hyperintense or heterogeneous
- diffuse heterogeneous: mixed signal intensities on T1WI & T2WI
- b) primary bone tumors; see more extensive discussion (p.792)
 - vertebral hemangioma
 - osteoblastoma
- 2. infection: osteomyelitis/discitis
- 3. fatty infiltrate or replacement of bone marrow: with age, hematopoietic red marrow of VBs is gradually replaced by yellow marrow in a splotchy pattern at a slower rate than in many other locations, e.g. distal appendicular bones.² T1WI: yellow marrow (MRI characteristics similar to subcutaneous fat) is hyperintense to red marrow (caution: bright areas on T1WI may be fat, or may be a normal area next to a low intensity met). T2WI: yellow marrow is bright
- 4. degenerative changes (Modic changes), see □ Table 68.2
- 5. metabolic
 - a) Paget's disease: plain x-rays → enlargement of VBs with cortical thickening usually involving several contiguous levels (p.1121)
 - b) osteoporosis: reduced bone density. Vertebral compression fractures may be seen
 - c) ankylosing spondylitis (p.1123): osteoporotic VBs, calcified intervertebral discs (sparing the nucleus pulposus), and ossified ligaments, → square VBs with bridging syndesmophytes ("bamboo spine"). Starts in sacroiliac joints & lumbar spine

90.4 Axis (C2) vertebra lesions

1. tumors: rare. Possibilities include those that involve the spine at any location. Some factors pertinent to this location³:
 - a) primary bone
 - chondroma
 - chondrosarcoma: rare in the craniovertebral junction. Lobulated tumors with calcified areas
 - chordoma: slow-growing radioresistant malignancy (p.778)
 - osteochondroma (chondroma)
 - osteoblastoma (p.792)
 - osteoid osteoma (p.792): more common in posterior elements than VB⁴
 - giant-cell tumors of bone: typically arise in adolescence. Lytic with bony collapse⁵
 - b) metastatic: including
 - typical metastases that spread hematogenously to bone, including: breast cancer, prostate cancer, malignant melanoma, paraganglioma, renal cell carcinoma
 - extension of regional tumors: nasopharyngeal tumors, craniopharyngioma
 - c) miscellaneous
 - plasmacytoma
 - multiple myeloma
 - Langerhans cell histiocytosis: osteolytic defect with progressive vertebral collapse. Occasionally occur in C2⁶
 - Ewing's sarcoma: malignant. Peak incidence during 2nd decade of life
 - aneurysmal bone cyst⁷
2. infection: osteomyelitis of the axis
3. pannus from old nonunion of fracture or from rheumatoid arthritis (RA)
4. erosive changes in the odontoid process with RA (p.1134)

90.5 Pathologic fractures of the spine

90.5.1 General information

Fractures due to metastatic involvement are hypointense on T1WI and hyperintense on T2WI. Benign VB collapse should be isointense to normal VBs on all sequences^{8,9} and the VB should look homogeneous. On T2WI or STIR images, the cortex of the VB (which should be dark border around the VB due to low water content of cortical bone) should be intact.

90.5.2 Etiologies

1. osteoporosis
2. neoplasm: short list
 - a) metastases: common sources of spine mets: lung, breast, prostate, myeloma

- b) Langerhans cell histiocytosis (p.777): may cause vertebra plana (see below)
- c) lymphoma
- d) hemangioma (p.794)
- 3. infection
- 4. avascular necrosis of the vertebral body
 - a) Calve-Kummel-Verneuil disease (see below)
 - b) with steroid use

90.5.3 Vertebra plana

Criteria:

1. uniform collapse of vertebral body into flat thin disc
2. increased density of vertebra
3. spares neural arches
4. normal disc and intervertebral disc space
5. intervertebral vacuum cleft sign (pathognomonic)
6. no kyphosis

Etiologies include:

1. Langerhans cell histiocytosis
2. Calve-Kummel-Verneuil disease: avascular necrosis of the vertebral body. Occurs in 2–15 year olds
3. hemangioma

90.6 Spinal epidural masses

See items marked with a dagger (†) under Myelopathy (p.1407).

90.7 Destructive lesions of the spine

90.7.1 Etiologies

1. neoplastic; see Differential diagnosis: spine & spinal cord tumors (p.783) for more:
 - a) metastatic tumors with a predilection for bone: prostate, breast, renal cell, lymphoma, thyroid, lung...; see Spinal epidural metastases (p.814)
 - b) primary bone tumors: chordomas (p.778), osteoid osteoma (p.792), hemangioma (p.794)
2. infection:
 - a) vertebral osteomyelitis: occurs mostly in IV drug abusers, patients with diabetes mellitus, and hemodialysis patients. May have associated spinal epidural abscess. Also see Vertebral osteomyelitis (p.353)
 - b) Discitis (p.356)
3. chronic renal failure: some patients develop a destructive spondyloarthropathy that resembles infection^{10,11}
4. ankylosing spondylitis (p.1123): bamboo spine (square VBs with bridging syndesmophytes)
5. lesions producing posterior scalloping of VB (mnemonic: AMEN)

A acromegaly or achondroplasia

M Marfan syndrome or mucopolysaccharidosis

E Ehlers-Danols

N neurofibromatosis

also: dural ectasia

1. lesions producing anterior scalloping of VB
 - a) aortic aneurysm
 - b) lymphoma
 - c) spinal TB

90.7.2 Differentiating factors

Of the many lytic or destructive lesions that involve the vertebra, destruction of the disc space is highly suggestive of infection which often involves at least two adjacent vertebral levels. Although tumors may involve adjacent vertebral levels and cause collapse of disc height, the disc space is usually not destroyed¹² (possible exceptions include: some vertebral

plasmacytomas, a reported metastatic cervical carcinoma, and there may occasionally be destruction of the disc in ankylosing spondylitis¹³). Unlike pyogenic infections, the disc may be relatively resistant to tuberculous involvement in Pott's disease.¹⁴ Also, since metastatic tumor involvement usually produces widespread bony involvement, it is less likely with involvement of a single bone.

90.8 Vertebral hyperostosis

1. Paget's disease (p. 1120): classic "ivory bone" with cortical thickening ("picture frame" appearance on plain x-rays). Consider Paget's with a dense vertebra on x-ray in an older patient, commonly involving several contiguous vertebrae
2. osteoblastic metastases
 - a) in men: prostate
 - b) in women: breast
 - c) lymphoma

90.9 Sacral lesions

Tumors:

1. metastases: the most common sacral neoplasm
2. primary neoplasms of the sacrum are uncommon and include:
 - a) giant cell tumor (p. 1412)
 - b) chordoma
 - c) teratoma:
 - adults: pre-sacral or sacro-coccygeal teratomas may arise from cells sequestered from Hensen's node in the caudal embryo. Rarely cause neurologic involvement (distinguishing this from chordoma). Sacrum may be normal in up to 50% (abnormal in almost all chordomas). Treatment is complete removal usually by general surgeon
 - peds: malignant pre-sacral teratoma is a rare tumor seen primarily in female children

Infection:

Most infections of the sacrum or sacroiliac joint are due to contiguous spread from a suppurative focus

Arthritic disorders:

1. ankylosing spondylitis (p. 1124): involves SI joint almost by definition
2. osteoarthritis

Sacral fractures, may be due to:

1. trauma
2. repetitive stress
3. sacral insufficiency (p. 1415)

Congenital:

sacral agenesis (caudal regression syndrome): rare (prevalence: 0.005–0.01% higher (0.1–0.2% in children of diabetic mothers (16–20% of children with sacral agenesis have diabetic mothers)). Increased incidence of associated spinal abnormalities including: syrinx, tethered cord, lipoma, and lipomyelomeningocele.

1. Four types:
 - a) Type 1: partial unilateral agenesis, localized to the sacrum or coccyx
 - b) Type 2: partial bilaterally symmetric defects in the sacrum. Iliac bones articulate with S1, and distal segments of the sacrum and coccyx fail to develop
 - c) Type 3: total sacral agenesis + iliac bones articulate with the lowest segment of the lumbar spine present
 - d) Type 4: total sacral agenesis + iliac bones fused posteriorly along the midline
2. in cases of total sacral agenesis (types 3 & 4), MR findings include: absence of the sacrum and coccyx and variable absence of a portion of the lumbar spine, with a characteristic club-shaped configuration of the conus medullaris

Miscellaneous:

osteitis condensans ilii: increased density in ilium, usually asymptomatic (incidental) finding. Occasionally may produce low back pain or tenderness

90.10 Enhancing nerve roots

- 1. tumor
 - a) meningeal carcinomatosis
 - b) lymphoma
- 2. infection: especially CMV (often seen in AIDS patients)
- 3. inflammatory
 - a) Guillain-Barre
 - b) arachnoiditis
 - c) sarcoid

90.11 Nodular enhancing lesions in the spinal canal

- 1. neurofibromatosis (NFT)
- 2. tumor
 - a) drop mets
 - b) neurofibroma
 - c) schwannoma

90.12 Intraspinal cysts

- 1. spinal meningeal cysts (p.1142)
- 2. cystic neurofibroma
- 3. ependymoma: may be cystic. In filum terminale: myxopapillary ependymoma (p.789)
- 4. syringomyelia (p.1144)
- 5. dilated central canal (p.1144)

90

90.13 Diffuse enhancement of nerve roots/cauda equina

- (As distinct from nodular enhancement, see above)
- 1. Guillain-Barre (p.184)
 - 2. meningitis
 - 3. cytomegalovirus (CMV) (especially in AIDS)
 - 4. lymphoma
 - 5. sarcoid (look for hilar adenopathy)

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91 Differential Diagnosis (DDx) by Signs and Symptoms – Primarily Intracranial

91.1 Diagnoses covered outside this chapter

Table 91.1 Differential diagnoses by signs and symptoms, primarily intracranial – covered outside this chapter	
DDx	
abducens palsy (p. 567)	
anisocoria (p. 561)	
chordomas (p. 778)	
chronic meningitis (p. 319)	
coma (p. 297)	
Creutzfeldt-Jakob disease (p. 370)	
diabetes insipidus (p. 120)	
dizziness (p. 572)	
facial nerve palsy (Bell’s palsy) (p. 576)	
giant cell arteritis (p. 195)	
gyral enhancement (p. 1281)	
hemiplegia/hemiparesis – see spine section (p. 1414)	
internuclear ophthalmoplegia (p. 565)	
Meniere’s disease (p. 573)	
multiple sclerosis (p. 180)	
ophthalmoplegia	
• painful (p. 568)	
• painless (p. 569)	
papilledema (p. 558)	
Parinaud’s syndrome (p. 99)	
Parkinson’s disease (p. 177)	
pneumocephalus (p. 887)	
prolactin elevation (□ Table 46.4)	
pseudotumor cerebri (p. 770)	
retinal hemorrhage (p. 916)	
sarcoidosis (p. 191)	
seizures	
• new onset, adult (p. 461)	
• new onset, peds (p. 462)	
• nonepileptic (p. 464)	

Table 91.1 continued
DDx
• status epilepticus (p.470)
schizencephaly (p.288)
torticollis (p.1533)
trigeminal neuralgia (p.481)
vertigo (p.572)

91.2 Encephalopathy

Many etiologies are similar to that for coma (p.297). EEG may be helpful in distinguishing some etiologies (p.238).

- 1. a rare cause may be (spontaneous) intracranial hypotension (p.389)
- 2. hypertensive encephalopathy from malignant hypertension

91.3 Syncope and apoplexy

91.3.1 General information

Syncope may be defined as one or more episodes of brief loss of consciousness (LOC) with prompt recovery (this term is considered by many to signify a vasovagal episode). The uncommonly used term lipothymia may be less likely to imply an etiology. Prevalence may be as high as ≈ 50%(higher in the elderly). Apoplexy is traditionally considered a form of hemorrhage, usually intracerebral. The recovery from apoplexy would therefore usually be slower than for syncope.

91.3.2 Etiologies

Adapted^{1,2}). NB: in a large number of cases no cause can be determined.

- 1. vascular: a few myotonic jerks may be seen in cerebral ischemia
 - a) cerebrovascular
 - subarachnoid hemorrhage (most commonly aneurysmal)
 - intracerebral hemorrhage
 - brain stem infarction
 - pituitary apoplexy (p.720) (rare)
 - vertebrobasilar insufficiency (VBI) (p.1305)
 - rarely with migraine
 - b) cardiovascular
 - Stokes-Adams attacks: disorder of AV-node conduction in the heart resulting in syncope with bradycardia
 - carotid sinus syncope: minimal stimulation (e.g. tight shirt collar, syncope while shaving...) causes reflex bradycardia with hypotension, more common in patients with carotid vascular disease. Bedside carotid massage with ECG and BP monitor may diagnose²
 - cardiac standstill: seen rarely in patients with glossopharyngeal neuralgia (p.492)
 - vasodepressor syncope (the common faint), AKA vasovagal response, and recently AKA neurocardiogenic syncope³: the most common cause of transient LOC. Hypotension usually with any of the following autonomic manifestations: pallor, nausea, heavy perspiration, pupillary dilatation, bradycardia, hyperventilation, salivation. Usually benign. Most common in age <35 yrs
 - orthostatic hypotension: drop in SBP ≥25 mm Hg on standing
 - triggered syncope: includes micturition syncope, tussive syncope, weight lifting syncope... (most involve elevation of intrathoracic pressure)
- 2. infectious
 - a) meningitis
 - b) encephalitis
- 3. seizure (p.443): in general, there are involuntary movements and confusion afterwards, lasts at least several minutes. Todd’s paralysis may follow and usually resolves slowly over a period of a few hours. There may be irritative special-sense phenomena (visual, auditory, or olfactory hallucinations)
 - a) generalized

- b) complex partial
- c) akinetic seizure
- d) drop attack (loss of posture without LOC): seen in Lennox-Gastaut
- 4. metabolic: hypoglycemia (may produce seizure, usually generalized)
- 5. miscellaneous
 - a) intermittent ventricular obstruction: the classic example is a colloid cyst of the third ventricle (p.756), but this mechanism is questionable
 - b) narcoleptic cataplexy: narcolepsy is characterized by somnolence and sudden attacks of weakness (cataplexy) when awake. Easy arousal and lack of post-ictal drowsiness distinguishes cataplexy from a seizure. The somnolence is treated with CNS stimulants (such as amphetamines or modafinil (Provigil®) 200 mg PO q AM), and cataplexy is treated with antidepressants
 - c) psychogenic
- 6. intracranial hypotension: usually with CSF shunt when upright (p.389)
- 7. unknown: in $\approx 40\%$ of cases no cause can be diagnosed

91.3.3 Practical approach to syncope

Introduction

The core of diagnosis and management are the H&P, orthostatic vital signs, and the ECG, with a combined diagnostic yield of 50%⁴ covering:

- 1. reflex mediated such as vasovagal or valsalva/stress-induced: 36 -62%
- 2. cardiac valvular etiology or arrhythmia: 10–30%
- 3. orthostatic due to autonomic dysregulation, dehydration or polypharmacy: 2–24%
- 4. cerebrovascular due to stroke: $\approx 1\%$
- 5. seizure

Evaluation

- 1. history: includes
 - a) medication list: look for drugs that may cause orthostatic hypotension, especially blood pressure medication, beta blockers
 - b) precipitating factors: e.g. change in position, sensitivity to tight collars...
 - c) premonitory factors: e.g. sweating and tremulousness may signify hypoglycemia, bradycardia is associated with vasovagal events, tonic-clonic movements may occur with a seizure
 - d) post-ictal emergence: usually rapid after a simple faint, slower after seizure which may also exhibit Todd's paralysis (p.443)
- 2. cardiovascular etiologies: Testing is also guided by H&P, vital signs & ECG:
 - a) cardiac arrhythmia evaluation: 12-lead ECG & 24-hour Holter monitor, and may lead to electrophysiologic (EP) testing/intervention^{4,5}
 - b) abnormal orthostatics warrants a formal tilt-table test
 - c) history of cardiomyopathy or CAD merits an echocardiogram and formal stress testing. These results determine the need for cardiac catheterization
- 3. neurologic etiologies: comprise $< 1\%$ of cases.⁶ In the absence of clinical evidence of a neurologic etiology, neurodiagnostic testing (EEG, CT scan, MRI/MRA, carotid doppler) have a diagnostic yield of 2–6% □ These tests are warranted only when clinically indicated⁵ (seizures, altered consciousness, gradually resolving Todd's paralysis, known history of cerebrovascular compromise). Tests include:
 - a) unenhanced brain CT: rules out most acute neurosurgical etiologies (bleed, hydrocephalus, edema which may be associated with tumor)
 - b) MRI without and with enhancement in cases with unexplained CT findings, or with a negative CT but high suspicion of a CNS etiology
 - c) seizure evaluation: when symptoms suggest possible seizure:
 - EEG: usually a sleep deprived EEG. Not very sensitive
 - 24 hour video EEG monitoring: in cases with high index of suspicion of seizures or nonepileptic seizures

Management

Admission and inpatient management is warranted for patients with diagnosed cardiac or neurologic syncope, either by suggestive history (family history of sudden death, syncope during exertion, witnessed seizure) or diagnostic testing (arrhythmia, severe orthostatic changes, hemodynamic instability).^{4,7}

91.4 Transient neurologic deficit

For apoplexy, etc., see Syncope and apoplexy (p.1396).

The first three etiologies listed below cover most cases of transient neurologic deficit:

1. transient ischemic attack (TIA) (p.1264): temporary neurologic dysfunction as a result of ischemia. Maximum deficit usually at onset. Most resolve in <20 mins
2. migraine: unlike TIA, tends to progress in a march-like fashion over several minutes. May or may not be followed by headache; see Migraine (p.175)
3. seizure: may be followed by a Todd's paralysis (p.443)
4. TIA-like syndrome
 - a) "tumor TIA": a transient deficit in a patient with a tumor, may be clinically indistinguishable from an ischemic TIA. Intravascular lymphomatosis may mimic TIAs (p.711)
 - b) TIA-like symptoms may occur as a prodrome (p.1336) to a lobar intracerebral hemorrhage^{8,9} in cases of cerebral amyloid angiopathy (CAA). Unlike typical TIAs, these usually consist of numbness, tingling or weakness that gradually spreads in a manner reminiscent of a Jacksonian-march and may cross-over vascular territories. Caution: antiplatelet drugs and anticoagulation may increase the risk of hemorrhage in patients with CAA (p.1334)
 - c) chronic subdural hematoma: may cause recurrent TIA-like symptoms of the involved hemisphere¹⁰ (including transient aphasia with dominant hemisphere involvement, hemisensory or motor abnormalities). The duration of symptoms tends to be longer than the typical TIA.¹⁰ Postulated mechanisms include:
 - electrical basis: the possibility of epileptic activity due (e.g. due to irritation of the cortex by blood breakdown products) has not been supported in the literature; however, spreading depression of Leao has been considered¹¹
 - impairment of venous outflow by compression of surface veins
 - compromised regional cerebral perfusion by indirect shifting of the anterior and posterior cerebral arteries¹²
 - transient ICP elevations → variations in cerebral perfusion pressure

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91.5 Ataxia/balance difficulties

1. cerebellar origin: usually with involvement of UEs in addition to LEs
 - a) cerebellar tumors
 - b) cerebellar hemorrhage
 - c) acute cerebellar ataxia: usually follows viral infection in a child <3 years. Usually self-limited with good prognosis for complete recovery
2. spinal cord: usually worse with eyes closed (loss of proprioceptive input)
 - a) spinal stenosis
 - b) neoplastic cord compression
 - c) syringomyelia (may be part of Chiari malformation)
3. degenerative
 - a) ataxia-telangiectasia syndrome
 - b) ataxia oculomotor apraxia
 - c) Friedreich's ataxia
 - d) spinocerebellar degeneration
4. metabolic/nutritional
 - a) vitamin B12 deficiency
 - b) drugs
 - AEDs (especially phenytoin or carbamazepine)
 - alcohol: acutely with intoxication and chronic
 - heavy metal poisoning
5. conditions that may mimic ataxia
 - a) weakness
 - b) peripheral neuropathy
 - c) dizziness: including orthostatic hypotension; see Dizziness and vertigo (p.572)
6. peripheral neuropathy:
 - a) ataxia can occur with Guillain-Barré syndrome (p.185), especially Miller Fisher variant (p.186)
 - b) balance difficulties are common with chronic immune demyelinating polyradiculoneuropathy (CIDP) (p.186)

91.6 Diplopia

1. cranial nerve palsy of any one or combination of III, IV (rare), or VI
 - a) for multiple cranial nerve palsies, below
 - b) VI palsy (p.567): can occur with increased intracranial pressure, e.g. in idiopathic intracranial hypertension (pseudotumor cerebri) (p.766), sphenoid sinusitis...; see other causes of abducens palsy (p.321)
 - c) isolated muscle paresis of III suggests nuclear lesion or myasthenia gravis
2. intraorbital mass compressing extraocular muscles
 - a) orbital pseudotumor (p.569)
 - b) meningioma
3. Graves' disease: hyperthyroidism + ophthalmopathy (p.1375)
4. myasthenia gravis
5. giant cell arteritis (p.196)
6. botulism: due to toxin from *Clostridium botulinum* (in adults: ingested or in wound). N/V, abdominal cramps, and diarrhea often precede neurologic symptoms. Neurologic involvement is typically symmetric. Dry mouth & cranial nerves palsies (diplopia, ptosis, loss of accommodation and pupillary light reflex) are followed by descending weakness. Bulbar paresis (dysarthria, dysphagia, dysphonia, flaccid facial muscles) follows. Muscles of the trunk/extremities and respiration progressively weaken in a descending fashion. Sensory disturbances are absent. Sensorium usually remains clear
7. following head trauma: includes injury to EOMs, orbital hematoma, VI palsy from increased ICP

91.7 Anosmia

1. abrupt onset of anosmia
 - a) severe upper respiratory infection with damage to the neuroepithelium: the most common cause
 - b) head trauma: second most common cause. Anosmia occurs in 7–15% of patients with significant head trauma
2. gradual onset of anosmia
 - a) allergic rhinitis and sinus disease¹³: third most common cause of anosmia (anosmia in this setting may be intermittent)
 - b) intracranial neoplasms: olfactory groove meningioma, see Foster Kennedy syndrome (p.99), esthesioneuroblastoma (p.1387)
 - c) may also be associated with Alzheimer's disease
 - d) olfactory sense diminishes with age: $\approx 50\%$ of patients 65–85 years of age have some loss of sense of smell
 - e) metabolic abnormalities: vitamin deficiency
 - f) physical blockage of nasal passages: nasal polyps...
 - g) endocrine abnormalities: diabetes...
 - h) chemical: alcohol abuse, exposure to solvents,¹⁴ cocaine (ischemic infarction of olfactory mucosa from vasoconstriction)
3. congenital anosmia: Kallmann syndrome (anosmia with hypogonadotropic hypogonadism¹⁵)

91.8 Multiple cranial nerve palsies (cranial neuropathies)

91.8.1 Framework

The differential diagnosis is legion. The following is a framework (modified¹⁶):

1. congenital
 - a) Möbius syndrome: AKA congenital facial diplegia. Facial plegia is complete in $\approx 35\%$ (in rest, affects upper face more than lower face, unlike central or peripheral facial palsy), associated with abducens palsy in 70% external ophthalmoplegia in 25% ptosis in 10% lingual palsy in 18%
 - b) congenital facial diplegia may be part of facioscapulohumeral or myotonic muscular dystrophy
2. infectious
 - a) chronic meningitis:
 - spirochetal, fungal, mycoplasma, viral (including AIDS)
 - mycobacterial AKA tuberculous (TB) meningitis: 6th nerve involved first and most frequently. CSF shows lymphocytic pleocytosis and hypoglycorrhachia. Smears are usually negative and multiple cultures are needed to diagnose

- b) stage II Lyme disease (p.576). Facial nerve weakness is common, sometimes bilateral (Lyme disease is the most common cause of facial diplegia in endemic areas). Other cranial nerve involvement is rare
- c) neurosyphilis: rare nowadays except with AIDS. Diagnosed by serologic testing
- d) fungal infection
 - cryptococcal meningitis (p.376): CSF analysis for cryptococcal antigen and India ink prep can detect
 - aspergillosis: may extend to the orbit from sinuses and involve cranial nerves
 - mucormycosis (phycomycosis) (p.1269): produces cavernous sinus syndrome, usually occurs in diabetics
- e) cysticercosis: especially with basal form; see Neurocysticercosis (p.371)
- 3. traumatic: especially with basal skull fractures. Lower cranial nerve palsies may occur (sometimes delayed in onset) with occipital condyle fractures (p.966) or atlanto-occipital dislocation (p.963)
- 4. neoplastic (brain stem compression and intrinsic lesions usually also produce long tract findings early). Also see Jugular foramen syndromes (p.100)
 - a) chordoma (p.778)
 - b) sphenoid-ridge meningioma
 - c) neoplasms of the temporal bone (often in conjunction with chronic otitis media and otalgia): adenoid cystic carcinoma, adenocarcinoma, mucoepidermoid carcinoma
 - d) glomus jugulare tumors: often affects nerves IX, X, and XI. May cause pulsatile tinnitus; see Paraganglioma (p.652)
 - e) carcinomatous or lymphomatous meningitis (p.811): CSF pleocytosis and elevated protein. Palsies are painless or associated with diffuse headache. Sensory palsies are common, resulting in deafness and blindness
 - f) invasive pituitary adenomas involving the cavernous sinus (p.720): extraocular cranial neuropathies tend to develop after visual field deficits in these tumors, and are less common when compared to other intracavernous solid tumors¹⁷
 - g) primary CNS lymphoma (p.712)
 - h) multiple myeloma involving the skull base (p.714)
 - i) intrinsic brain stem tumors: gliomas, ependymoma, metastases...
- 5. vascular
 - a) aneurysm: intracranial or cavernous sinus (p.1225)
 - b) brain stem stroke: usually also produces long tract findings (p.99)
 - Weber's syndrome: Cr N III (usually pupil sparing) + contralateral hemiparesis
 - Millard-Gubler syndrome: Cr N VI + VII + contralateral hemiparesis
 - c) vasculitis: Wegener's granulomatosis usually affects eighth nerve in addition to others
- 6. granulomatous
 - a) sarcoidosis: ≈ 5% have CNS involvement, usually as fluctuating single or multiple cranial neuropathies (facial nerve is most common, and may be indistinguishable from Bell's palsy). CSF pleocytosis (p.200) is common
- 7. inflammatory
- 8. neuropathies
 - a) Guillain-Barré syndrome (GBS) (p.184): cranial nerve involvement includes facial diplegia, oropharyngeal paresis. Peripheral neuropathy usually presents with ascending weakness, proximal muscle weakness > distal, and absent deep tendon reflexes
 - b) Miller-Fisher variant GBS: ataxia, areflexia & ophthalmoplegia. Serum marker: anti-GQ1b antibodies
 - c) idiopathic cranial polyneuropathy: subacute onset of constant facial pain, usually retro-orbital. Frequently precedes sudden onset of cranial-nerve palsies usually involving III, IV & VI, less frequently V, VII, and lower nerves (IX through XII). Olfactory and auditory nerves usually spared. Acute and chronic inflammation of unknown etiology similar to Tolosa-Hunt and orbital pseudotumor. Steroids reduce pain and expedite recovery
- 9. entrapment in abnormal bone
 - a) hyperostosis cranialis interna: a rare autosomal dominant abnormality of the bone of the base of the skull causing recurrent facial palsy and other cranial nerve palsies¹⁸
 - b) osteopetrosis: see below
 - c) Paget's disease (p.1120) involving the skull: 8th nerve involvement (deafness) is most common. Optic nerve atrophy, and palsies of oculomotor, facial, IX, XI, olfactory nerves and others may also occur¹⁹
 - d) fibrous dysplasia (p.779)

91.8.2 Specific syndromes

Facial diplegia

Items culled from the above list that have facial diplegia (p.576) as a prominent finding:

1. congenital: Möbius syndrome, congenital facial diplegia
2. infectious: Lyme disease
3. neuropathies: Guillain-Barré syndrome
4. isolated 4th ventricle (p.576): compression at the facial colliculus
5. granulomatous: sarcoidosis

Cavernous sinus syndrome

Multiple cranial nerve palsies (involving any of the cavernous sinus cranial nerves: III, IV, VI, V2, VI) which primarily produce diplopia (due to ophthalmoplegia). Classically the third nerve palsy (e.g. from an enlarging cavernous carotid artery aneurysm) will not produce a dilated pupil because the sympathetics which dilate the pupil are also paralyzed.²⁰ (p 1492) Facial pain or altered facial sensation may occur.

See list of lesions that may produce cavernous sinus syndrome (p.1376).

Osteopetrosis

AKA “marble bone disease” (there is also some confusion with the term osteosclerosis; osteosclerosis fragilis generalisata is the obsolete term for osteopetrosis). A rare group of genetic disorders of defective osteoclastic resorption of bone resulting in increased bone density, may be transmitted either as autosomal dominant or recessive.²¹ The dominant form is usually benign and is seen in adults and adolescents. The recessive (“malignant”) form is often associated with consanguinity, and is similar to hyperostosis cranialis interna (see above), but in addition to the proclivity for the skull, also involves ribs, clavicles, long bones, and pelvis (long-bone involvement results in destruction of marrow and subsequent anemia). Cranial nerves involved primarily include optic (optic atrophy and blindness are the most common neurologic manifestation), facial, and vestibulo-acoustic (with deafness), trigeminal nerve may also be involved. There may also be extensive intracranial calcifications, hydrocephalus, intracranial hemorrhage and seizures.

Bilateral optic nerve decompression via a supraorbital approach may improve or stabilize vision.²¹

91.9 Binocular blindness

☐ Bilateral occipital lobe dysfunction:

1. bilateral posterior cerebral artery flow impairment
 - a) top of the basilar syndrome
 - b) increased intracranial pressure
 - hydrocephalus with shunt malfunction
 - pseudotumor cerebri (idiopathic intracranial hypertension) (p.768)
 - cryptococcal meningitis: decreased visual acuity (p.376)
2. trauma: bilateral occipital lobe injury (e.g. contrecoup injury)

☐ Seizures. Epileptic blindness

☐ Migraine. Cortical spreading depression

☐ Posterior ischemic optic neuropathy. Usually in the setting of shock

☐ Bilateral vitreous hemorrhage. E.g. with SAH (Terson’s syndrome)

☐ Functional. Conversion reaction, hysterical blindness...

91.10 Monocular blindness

Due to a lesion anterior to the optic chiasm.

1. Amaurosis fugax: often described as a “shade coming down” over one eye
 - a) TIA: usually due to occlusion of the retinal artery (p.1271)
 - b) giant cell arteritis (GCA) (p.195): usually due to ischemia of optic nerve or tracts (less commonly due to retinal artery occlusion)²²

2. trauma: optic nerve injury
3. ruptured carotid cavernous aneurysm: resultant carotid cavernous fistula increases intraocular pressure by impeding venous return
4. intraorbital pathology: tumors
5. injury within the globe: retinal detachment, ocular trauma
6. unilateral vitreous hemorrhage: e.g. with SAH (Terson's syndrome)

91.11 Exophthalmos

91.11.1 General information

Alternate spelling: exophthalmus.

Definition: abnormal protrusion of the eyeball. Some authors reserve the term exophthalmos for cases due to endocrinopathies and use proptosis (of the eye) for other causes, but these terms are widely used interchangeably.

Criteria: different criteria are proposed. Anterior displacement of 18 mm (Hertal exophthalmometry can be used to measure clinically – requires intact lateral orbital bone). CT criteria: for most accurate results, the patient's should have their eyes open and fixated on a point in the primary gaze position. position of equator of the globe (widest part) is distal to a line drawn from lateral orbit to medial canthus, >2/3 of the globe anterior to this line.

91.11.2 Pulsatile

1. carotid cavernous fistula (CCF) (p.1256)
2. transmitted intracranial pulsation due to defect in orbital roof
 - a) seen unilaterally e.g. in neurofibromatosis type 1 (p.604)
 - b) post-op following procedures that remove orbital roof or wall
3. vascular tumors

91.11.3 Non-pulsatile

1. tumor
 - a) intraorbital tumor: may be due to mass effect from tumor or to compromised venous drainage from the orbit
 - optic glioma (p.631)
 - optic sheath neuroma
 - lymphoma
 - optic sheath meningioma²³
 - orbital involvement with multiple myeloma (p.714)
 - orbital invasion by invasive pituitary adenoma (p.771)
 - in peds: metastatic neuroblastoma
 - in peds: Langerhans cell histiocytosis (p.777) as part of Hand-Schüller-Christian (triad: DI, exophthalmos and lytic bone lesions (particularly of cranium))
 - b) due to hyperostosis from a sphenoid ridge meningioma
2. Graves' disease (hyperthyroidism + exophthalmos) (p.1375): even though the exophthalmos is usually bilateral with this (80%), thyroid disease is still the most common cause of unilateral proptosis²⁴
3. enlargement of periorbital fat²⁵
4. infection: orbital cellulitis (usually has concomitant sinusitis)
5. inflammatory: orbital pseudotumor. Usually unilateral (p.569)
6. hemorrhage
 - a) traumatic
 - b) spontaneous
7. 3rd nerve palsy: can cause up to 3 mm proptosis from relaxation of the rectus muscles
8. cavernous sinus occlusion (may affect both eyes)
 - a) cavernous sinus thrombosis (p.1309)
 - b) cavernous sinus tumor obstructing venous outflow
9. pseudo-exophthalmos
 - a) congenital macrophthalmos (bull's eye)
 - b) lid retraction: e.g. in Graves' disease (p.1403)
 - c) coronal craniosynostosis can cause a "relative" proptosis (p.253)

91.12 Ptosis

AKA blepharoptosis. Drooping of the upper eyelid.

Distinguished from pseudoptosis (lid droop not resulting from weakness of levator palpebrae superioris (LPS)) which can be due to enophthalmos (globe displaced posteriorly, e.g. with orbital floor blow-out fracture), microphthalmia, blepharospasm, Duane syndrome.

Etiologies of ptosis:

1. congenital: most are simple (autosomal dominant inheritance), complicated ptosis is associated with other findings (e.g. ptosis with ophthalmoplegia)
2. traumatic: injury to eyelid, orbital roof fracture...
3. neurogenic:
 - a) third nerve palsy (p.565)
 - involvement of main trunk of third nerve: can occur intradurally or within cavernous sinus. Ptosis may be an early sign of pituitary tumor expansion (apoplexy) (p.720)
 - involvement of the superior division of the third nerve within the orbit
 - b) Horner's syndrome (p.564): ptosis here is partial (may be a pseudoptosis since weakness is in tarsal muscles, not LPS), and the lower eyelid will higher than the uninvolved contralateral lower eyelid
4. myogenic ptosis
 - a) botulinum toxin injection (e.g. Botox®)
 - b) myasthenia gravis
5. mechanical ptosis
 - a) tumors: neurofibroma, hemangioma, malignant melanoma, mets...
 - b) extension of mucocele of frontal sinus
6. pharmacologic (drugs). Partial list:
 - a) corticosteroids: including topical
 - b) alcohol
 - c) opium

91.13 Pathologic lid retraction

1. hyperthyroidism (p.1376)
2. psychiatric: schizophrenia...
3. steroids
4. Parinaud's syndrome (p.99)

91.14 Macrocephaly

Macrocephaly means increased size of the head.²⁶ Although sometimes used synonymously, some contend that the term macrocrania by convention refers to a head circumference >98th percentile.²⁷ (p.203) Also, not to be confused with macrencephaly AKA megalencephaly (see below). In a pediatric practice the 3 most common etiologies in decreasing order of frequency: familial (parents have big heads), benign subdural fluid collections of infancy (p.904), and hydrocephalus.

1. with ventricular enlargement
 - a) (hydrostatic) hydrocephalus (HCP), see for etiologies (p.394)
 - communicating
 - obstructive
 - b) hydranencephaly (p.288)
 - c) constitutional ventriculomegaly: ventricular enlargement of no known etiology with normal neurologic function
 - d) hydrocephalus ex vacuo: loss of cerebral tissue (more often associated with microcephaly, e.g. with TORCH infections)
 - e) vein of Galen aneurysms: see below
2. with normal or mildly enlarged ventricles
 - a) "external hydrocephalus": prominent subarachnoid spaces and basal cisterns; see External hydrocephalus (AKA benign external hydrocephalus) (p.400)
 - b) subdural fluid
 - hematoma
 - hygroma
 - effusion benign and symptomatic
 - benign subdural collections of infancy (p.904)

- c) cerebral edema: some consider this to be a form of pseudotumor cerebri²⁶
 - toxic: e.g. lead encephalopathy (from chronic lead poisoning)
 - endocrine: hypoparathyroidism, galactosemia, hypophosphatasia, hypervitaminosis A, adrenal insufficiency...
 - d) familial (hereditary) macrocrania: parents also have large heads, the brains eventually “catch up”
 - e) idiopathic
 - f) megalencephaly (AKA macrencephaly): an enlarged brain (p.289)
 - g) neurocutaneous syndromes: usually due to increased volume of brain tissue (megalencephaly, see above).²⁶ Seen especially in neurofibromatosis and congenital hypermelanosis (Ito’s syndrome). Less common in tuberous sclerosis and Sturge-Weber. Also seen in the rare hemimegalencephaly syndrome
 - h) arachnoid cyst (AKA subependymal or subarachnoid cyst)²⁶: a duplication of the ependyma or arachnoid layer filled with CSF. Usually reach maximal size by 1 month of age and do not enlarge further. Treatment is required in $\approx 30\%$ due to rapid enlargement or growth beyond first month. Cyst may be shunted or fenestrated. Prognosis with true arachnoid cyst is generally good (unlike porencephalic cyst) if no increased ICP or progressive macrocephaly during 1st year of life
 - i) arteriovenous malformation: especially vein of Galen “aneurysm” (p.1255). Auscultate for cranial bruit. With vein of Galen aneurysms, macrocephaly may be due to HCP from obstruction of the sylvian aqueduct.²⁶ With other malformations, macrocrania may be due to increased pressure in venous system without HCP
 - j) brain tumors without hydrocephalus: brain tumors are rare in infancy, and most cause obstructive HCP. Tumors that occasionally present without HCP includes astrocytomas. May also be seen in the rare diencephalic syndrome, see tumor of anterior hypothalamus (p.632)
 - k) “gigantism syndromes”
 - Soto’s syndrome: associated with advanced bone age on x-ray, and multiple dysplastic features of face, skin and bones
 - exomphalomacroglossia-gigantism (EMG) syndrome: hypoglycemia (from abnormalities in islets of Langerhans), large birth weight, large umbilicus or umbilical hernia and macroglossia
 - l) “craniocerebral disproportion” (p.903) ²⁶: may be the same as benign extra-axial fluid of infancy
 - m) achondroplastic dwarf: cranial structures are enlarged but the skull base is small, giving rise to a prominent forehead and an OFC ≥ 97 th percentile for age, hypoplasia of midface, and stenosis at foramen magnum. Head growth follows different curve than normal (OFC ≥ 97 th percentile for age is not unusual and does not necessitate shunting)
 - n) Canavan’s disease: AKA spongy degeneration of the brain, an autosomal recessive disease of infancy prevalent among Ashkenazi Jews. Produces symmetrical low attenuation of hemispheric white matter on CT²⁸ and macrocephaly
 - o) neurometabolic diseases: usually due to deposition of metabolic substances in the brain. Seen in Tay-Sachs gangliosidosis, Krabbe disease...
3. due to thickening of the skull
- a) anemia: e.g. thalassemia
 - b) skull dysplasia: e.g. osteopetrosis (p.1401)

91.15 Tinnitus

91.15.1 General information

May be either subjective (heard only by patient) or objective (e.g. cranial bruit, can be heard by examiner as well, usually with a stethoscope place over the cranium, orbit, or carotid arteries in the neck). Objective tinnitus is almost always due to vascular turbulence (from increased flow or partial obstruction).

91.15.2 Pulsatile tinnitus

Most cases are due to vascular lesions.

1. pulse synchronous:
 - a) carotid cavernous fistula (p.1256)

- b) AVM:
 - cerebral (pial) AVM
 - dural AVM (p.1251)
 - c) glomus jugulare tumor (p.654)
 - d) cerebral aneurysm: (rare) possibly with turbulent flow in giant aneurysm
 - e) hypertension
 - f) hyperthyroidism
 - g) idiopathic intracranial hypertension (pseudotumor cerebri) (p.766)
 - h) transmitted bruit: from heart (e.g. aortic stenosis), carotid artery stenosis (especially external carotid)
 - i) dehiscent jugular bulb or high riding jugular bulb: normal venous variant
 - j) rarely with posterior fossa tumors: CP-angle tumors e.g. vestibular schwannoma or meningioma, vascular intraparenchymal tumors e.g. hemangioblastoma (especially in CPA)
 - k) lesions that can present with a red tympanic membrane
 - aberrant carotid artery in middle ear
 - persistent stapedial artery: rare. Arises from aberrant ICA or from junction of horizontal and vertical petrous ICA. Foramen spinosum is absent on the affected side. Enlargement of anterior tympanic segment of seventh nerve canal
 - glomus tympanicum tumor (p.654)
 - l) sigmoid sinus diverticulum
2. non pulse-synchronous: asymmetrical enlargement of sigmoid sinus and jugular vein may produce a low grade hum

Workup for pulsatile tinnitus:

1. MRI without and with enhancement: to look for tumors, e.g. glomus jugulare
2. angiogram: include internal and external carotid injections
3. tests that are usually not helpful and should not be ordered routinely
 - a) carotid ultrasound: nonspecific, not sensitive
 - b) MRI/MRV: may miss small dural fistulas and do not give details needed for treatment for large ones

91.15.3 Non-pulsatile tinnitus

1. occlusion of external ear: cerumen, foreign body
2. middle ear infection (otitis media)
3. otosclerosis
4. stapedial muscle spasms: as occurs in hemifacial spasm
5. CP-angle tumors: including vestibular schwannoma (p.670)
6. Meniere's disease (p.573)
7. labyrinthitis
8. endolymphatic sac tumors: e.g. as in von Hippel-Lindau disease (p.705)
9. drugs
 - a) salicylates: aspirin, bismuth subsalicylate (Pepto Bismol®)
 - b) quinine
 - c) aminoglycoside toxicity: streptomycin, tobramycin (tinnitus precedes hearing loss)

91.16 Facial sensory changes

1. circumoral paresthesias
 - a) hypocalcemia
 - b) syringobulbia
2. unilateral facial sensory changes
 - a) trigeminal nerve neuroma
 - b) vestibular schwannoma (VS): to involve Cr. N. V a VS has to be >2 cm diameter; see symptoms from 5th nerve compression under vestibular schwannoma (p.671)
 - c) compression of the spinal trigeminal tract (large compressive lesions may cause bilateral alteration of facial sensation) that chiefly manifests in diminution of pain and temperature sense with little effect on touch sense.²⁹ The tract usually extends down into the spinal cord as far as ≈ the C2 vertebral level (although it may occasionally extend down to C4)

91.17 Language disturbance

1. aphasia:
 - a) injury to speech areas of brain
 - Wernicke’s aphasia (p.98): classically produces fluent aphasia (normal sentence length & intonation, devoid of meaning)
 - Broca’s aphasia (p.98): faltering, dysarthric
 - conduction aphasia (p.98): fluent spontaneous speech and paraphasias, but patients understand spoken or written words, and are aware of their deficit
 - b) transitory aphasia following a seizure; see Todd’s paralysis (p.443)
 - c) primary progressive aphasia of adulthood: idiopathic & degenerative
2. akinetic mutism: seen with bilateral frontal lobe dysfunction (e.g. with bilateral ACA distribution infarction due to vasospasm from a-comm aneurysm rupture or with large bilateral frontal lesions; may actually be abulia) or with bilateral cingulate gyrus lesions
3. muteness of cerebellar origin^{30,31}
4. following transcallosal surgery: as a result of bilateral cingulate gyrus retraction or thalamic injury together with section of the midportion of the corpus callosum³²

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92 Differential Diagnosis (DDx) by Signs and Symptoms – Primarily Spine and Other

92.1 Diagnoses covered outside this chapter

Table 92.1 Differential diagnoses by signs and symptoms, spine and other – covered outside this chapter	
DDx	
ankylosing spondylitis (p. 1125)	
bladder dysfunction (p. 92)	
brachial plexopathy (p. 542)	
carpal tunnel syndrome (p. 521)	
cervical stenosis (p. 1086)	
lateral disc herniation (p. 1059)	
meralgia paresthetica (p. 534)	
myopathy	
spinal cord tumors (p. 783)	
spinal epidural abscess (p. 349)	
spinal stenosis	
• lumbar (p. 1101)	
synovial cyst (spinal) (p. 1144)	
thoracic outlet syndrome (p. 554)	
torticollis (p. 1533)	
urinary retention (p. 94)	

92.2 Myelopathy

Items marked with a dagger (†) may present as a spinal epidural mass.

- Congenital:
1. (Arnold)-Chiari malformation (p.284): Type I often presents in early adulthood

2. tethered cord: often may not present until after some trauma

3. syringomyelia: may be congenital or post-traumatic in quadriplegics, usually presents with a central cord syndrome – see Syringomyelia (p. 1144) – or progressive myelopathy

4. neurenteric cyst (p.290)

5. cord compression that occurs with some mucopolysaccharidoses: e.g. Morquio syndrome (due to atlanto-axial subluxation), Hurler syndrome

6. hereditary spastic paraplegia: family history is key. Diagnosis of exclusion¹
- Acquired:
1. cervical or thoracic spinal stenosis: often degenerative disease superimposed on congenitally narrow canal (congenital narrowing is frequent in achondroplastic dwarfs)

2. traumatic: including spinal shock, hematomyelia, spinal epidural hematoma (see vascular below), barotrauma, electrical injuries, compression by bone fracture†. May follow minor trauma in the setting of spinal stenosis

3. herniated intervertebral disc†: myelopathy more common in thoracic region, radiculopathy more common in cervical region (long tract signs are rare with herniated cervical disc)

4. kyphosis

5. extramedullary hematopoiesis† (p.171): hypertrophy of marrow → cord compression. Primarily in chronic anemias (e.g. thalassemia major)
6. bony compression secondary to incompetence of odontoid process or transverse atlantal ligament†. May be congenital, traumatic (p.978), neoplastic, or inflammatory (especially rheumatoid arthritis)
7. epidural lipomatosis† (p.1150): hypertrophy of epidural fat most often due to years of exogenous steroid therapy²
8. ossification of the posterior longitudinal ligament (OPLL) (p.1127) ³
9. arachnoiditis ossificans: a rare condition (only ≈ 43 case reports as of 1998⁴) involving calcification of the arachnoid membrane. In the T-spine, may occur as ossified plaques or in a cylindrical form surrounding the spinal cord. May be difficult to detect on MRI and myelography. Plain unenhanced CT may be optimal for diagnosis
10. vertebral Paget's disease† (p.1120)
11. idiopathic spinal cord herniation (p.1150) ^{5,6}: rare. Thoracic spinal cord herniates through an anterior dural defect frequently producing a Brown-Séquard syndrome or spastic paraparesis

□ Neoplastic:

1. spine/spinal cord tumors† (p.783) (for details)
 - a) extradural (55%):
 - primary tumors (rare) include: neurofibromas, chordomas, osteoid osteoma, aneurysmal bone cyst, vertebral hemangioma⁷
 - if age >40 yrs, suspect extradural lymphoma (primary or secondary) or leukemic deposits (chloroma), especially with pre-existing diagnosis of hematopoietic or lymphatic disorder
 - epidural metastases (p.814) become increasingly common after age 50 yrs. Occurs in up to 10% of cancer patients. 5–10% of malignancies present initially with cord compression
 - b) intradural-extramedullary (40%): meningiomas, neurofibromas
 - c) intradural-intramedullary: primary cord tumors (p.784) (ependymoma, astrocytoma) and rarely intramedullary mets
2. carcinomatous meningitis (p.811): neurologic deficit usually cannot be localized to a single level
3. paraneoplastic syndrome (p.542): including effects on spinal cord or on peripheral nerves

□ Vascular:

1. hematoma/hemorrhage
 - a) spinal epidural hematoma† (p.1131): usually associated with anticoagulation therapy⁸
 - traumatic: following LP or epidural anesthesia (p.1131)
 - spontaneous⁹: rare. Includes hemorrhage from spinal cord AVM (p.1131) or from vertebral hemangioma (p.1140)
 - b) spinal subarachnoid hemorrhage: as in spinal epidural hematoma (p.165), this may also be post-traumatic (e.g. following LP^{10,11}) or secondary to spinal cord AVM
 - c) spinal subdural hematoma
 - d) hematomyelia
2. spinal cord infarction: uncommon with the elimination of syphilitic endarteritis. Most often in the territory of the anterior spinal artery, sparing posterior columns. Most commonly ≈ T4 level (watershed zone)
 - a) atherosclerosis of radicular artery in elderly patient with hypotension is now the major cause of this rare condition
 - b) clamping aorta during surgery (e.g. for abdominal aortic aneurysm)
 - c) hypotension (relative or absolute) during surgery in the sitting position in the presence of spinal stenosis.¹² May be improved by avoiding absolute hypotension, using awake fiber-optic intubation and positioning, intraoperative SSEP monitoring and inducing hypertension if changes occur with positioning, avoidance of sitting position, and avoiding hyperflexion, hyperextension and traction
 - d) aortic dissection
 - e) embolization of spinal arteries
3. spinal cord vascular malformations† (p.1140): 10–20% present as sudden onset of myelopathy usually in patients <30 yrs,¹³ myelopathy may be secondary to:
 - a) mass effect from AVM: spinal AVMs account for <5% of lesions presenting as cord “tumors”
 - b) rupture → SAH, hematomyelia, or epidural hematoma
 - c) watershed infarction due to “steal”
 - d) spontaneous thrombosis (necrotizing myelopathy of Foix-Alajouanine disease (p.1141) ¹⁴): presents as spastic → flaccid paraplegia, with ascending sensory level
4. radiation myelopathy: due to microvascular occlusion (p.1563)

5. secondary to iodinated contrast material used for mesenteric or aortic angiography. Especially when angiogrammed in presence of hypotension, where cardiac output is shunted away from viscera and into spinal radicular arteries. Treatment: place patient sitting, remove ≥ 100 ml of CSF via LP and replace with equal amount of saline over 30 mins¹⁵

□ Autoimmune:

1. post-viral (or post-vaccination): may actually be etiology of auto-immune process (i.e. transverse myelitis). Viral prodrome present in $\approx 37\%$ of cases of ATM. Viral infection is usually most damaging to gray matter (e.g. poliomyelitis)

□ Demyelinating:

1. acute (idiopathic) transverse myelitis (ATM) (p. 187). Peak incidence during first 2 decades of life. Abrupt onset of LE weakness, sensory loss, back pain, and sphincter disturbance indistinguishable from spinal cord compression. Thoracic region most common. CT and myelogram are normal. MRI may demonstrate. CSF $\rightarrow$ pleocytosis and hyperproteinemia
2. multiple sclerosis (MS): diagnosed in only 7% of patients presenting as acute transverse myelopathy. Although more common in young adults, MS can occur at any time in life. Myelopathy of MS is usually insidious, and is usually incomplete (i.e. some sparing). Affects myelin, thus sparing gray matter. Abdominal cutaneous reflexes are almost always absent in MS
3. Devic syndrome (neuromyelitis optica (NMO)): a variant of MS characterized by acute bilateral optic neuritis and transverse myelitis (extending ≥ 3 levels,¹⁶ often causing cervical myelopathy). Spinal cord edema may become so severe as to cause complete block on myelography. More common in Asia and India than U.S. or Europe. Compared to classic MS: the myelopathy tends to be more severe (pathology: more necrosis as opposed to incomplete demyelination) and with less chance of recovery. Distinct serum IgG antibodies (NMO-IgG) may help differentiate from MS¹⁷

□ Metabolic/toxic:

1. (subacute) combined system disease (CSD) (AKA subacute combined columnar degeneration): due to vitamin B12 (cyanocobalamin) deficiency.
 - a) etiologies:
 - dietary deficiency of B12 (B12 is a water soluble vitamin present in meats and animal products)
 - pernicious anemia: intestinal malabsorption of B12 in the distal ileum due to lack of secretion of intrinsic factor (a small polypeptide) by gastric parietal cells¹⁸)
 - other gastric disorders: low gastric pH e.g. in Zollinger-Ellison syndrome can inhibit attachment of intrinsic factor to ileal receptors
 2. clinical: onset is gradual and uniform. Begins with symmetrical paresthesias in feet or hands (posterior column involvement) $\rightarrow$ leg stiffness, weakness, and proprioceptive deficits with unsteadiness that is worse in the dark $\rightarrow$ spasticity $\rightarrow$ paraplegia $\rightarrow$ bowel and bladder dysfunction. Dementia (confusion, memory impairment, irritability...) occurs in advanced cases due to cerebral white matter changes. Visual disturbances with or without optic atrophy may be due to optic nerve demyelination
 3. labs:
 - a) serum B12: the most sensitive test. However, normal B12 levels do not R/O B12 deficiency. If there are neurologic symptoms then check malonic acid or other markers of B12 deficiency such as methylmalonic acid (also check homocysteine to R/O folate deficiency)
 - b) CBC: most (but not all) patients will have a macrocytic (megalocytic) anemia (folic acid deficiency also produces megaloblastic anemia. Folic acid corrects the anemia, even with CSD, but not the neurologic deficits which may actually worsen)
 - c) Schilling test: determines the cause of the B12 deficiency even if B12 injections have already been given (radiolabeled cyanocobalamin is given orally, followed by a parenteral flushing dose of nonradioactive vitamin, and the percentage of radioactivity is measured in the urine over 24 hours, performed once without and then once with added intrinsic factor, and then once following antibiotic therapy)
 4. imaging: T2WI MRI may demonstrate increased signal within the white matter of the spinal cord, predominantly in the posterior columns but may also be seen in spinothalamic tracts
 5. treatment: B12 injections q 1–3 months or large doses of oral preparations¹⁹ (other transport systems independent of intrinsic factor results in absorption of $\approx 1\%$ of orally administered B12, doses of 300–100,000 mcg result in absorption of more than the daily requirement of 1–2.5 mcg)
 6. toxins: e.g. local anesthetics used for spinal anesthesia rarely cause myelopathy

□ Infectious:

1. (para) spinal abscess, AKA spinal epidural abscess or epidural empyema (p.349) †: often history of staphylococcus infection, usually a skin furuncle. Vertebral osteomyelitis often accompanies.²⁰ Produces local tenderness, back pain, fever, elevated ESR
2. vertebral osteitis/osteomyelitis (p.353) †
3. pyogenic discitis†: spontaneous or following procedures (p.356)
4. HIV or AIDS related myelopathy: similar to B12 deficiency. Spastic weakness & ataxia. Can cause vacuolization of spinal cord. “Tropical (spastic) paraparesis of AIDS” also seen in HTLV-I infection²¹
5. tuberculosis: Pott’s disease, see Tuberculous vertebral osteomyelitis (p.354)
6. spinal meningitis with pachymeningitis
7. viral:
 - a) herpes varicella-zoster: rarely causes necrotizing myelopathy
 - b) Herpes simplex type 2: may cause ascending myelitis
 - c) cytomegalovirus: may cause transverse myelitis
8. syphilitic involvement: may cause tabes dorsalis, syphilitic meningomyelitis, or spinal vascular syphilis. Diagnosed by serum and CSF serology
9. parasitic cysts†
10. some forms of Creutzfeldt-Jakob disease (CJD) with predominant initial muscle wasting may mimic spinal cord disease or ALS (p.367)

□ Peripheral neuromuscular disorder:

1. Guillain-Barré syndrome (GBS) (p.184): rapidly ascending weakness (mimics cord compression) with areflexia and near normal sensation
2. chronic dysimmune neuropathies: presumed to be immune mediated²²
 - a) chronic immune demyelinating polyradiculoneuropathy (CIDP) (p.186): similar to GBS but can progress over a longer period of time
 - b) multifocal motor neuropathy (MMN): characterized by asymmetric muscle wasting, cramping & LE twitching. May mimic ALS, but is treatable (with IVIg or immunosuppression)
3. myopathies: including steroid myopathy (usually affects proximal > distal muscles)

□ Motor neuron diseases:

1. amyotrophic lateral sclerosis (ALS) (p.183): upper and lower motor neuron disease. Slight spasticity of LEs (extreme spasticity is rare), atrophic weakness of the hands and forearms, fasciculations in the UE, absence of sensory changes (including lack of pain), sphincter control usually preserved
2. primary lateral sclerosis: age > 50. No LMN signs. Slower progression than ALS (yrs to decades). Pseudobulbar palsy (p.178) is common²³

† items with dagger may also present as a spinal epidural mass

92.3 Sciatica

92.3.1 General information

Definition: pain in the distribution of the sciatic nerve. The sciatic nerve is comprised of components of nerve roots of L4-S3. The nerve passes out of the pelvis through the greater sciatic foramen along the back of the thigh. In the lower third of the thigh it divides into the tibial and common peroneal nerves.

92.3.2 Etiologies

The most common cause of sciatica is radiculopathy due to a herniated lumbar disc.²⁴ The differential diagnosis is similar to that for myelopathy (see above) but also includes:

1. congenital:
 - a) meningeal cyst (perineural cyst); see Spinal meningeal cysts (p.1142)
 - b) conjoined nerve root (p.275): initially dismissed as a possible cause of radiculopathy, but current thinking recognizes that these may be symptomatic possibly by tethering
2. acquired:
 - a) spinal stenosis/spondylosis/spondylolysis/spondylolisthesis
 - b) juxtafacet cyst: includes synovial cyst and ganglion cyst (p.1143)²⁵: detection is increasing with the use of MRI

- c) nerve root sheath cyst: may be congenital or acquired. May arise near axilla of nerve root and cause compression of adjacent roots. Treatment: excise cyst and oversew the ostium
 - d) arachnoiditis ossificans (p.1408): rare. In the lumbar region may occur as columnar, cylindrical, or irregularly shaped masses.²⁶ May produce low back pain, radiculopathy, or cauda equina syndrome
 - e) heterotopic ossification around the hip²⁷
 - f) injection injuries from misplaced IM injections
 - g) compartment syndrome of the posterior thigh
 - h) injury complicating total hip arthroplasty²⁸
 - i) radiation injury following treatment of nearby tumors
3. infectious:
- a) discitis (p.356): usually causes excruciating pain with any movement
 - b) Lyme disease (p.334)
 - c) herpes zoster: a rare cause of radiculopathy.²⁹ Lumbosacral dermatomes are involved in $\approx 10\text{--}15\%$ of zoster cases. Pain is usually independent of position. Typical herpetic skin lesions usually follow onset of pain by 3–5 days. 1–5% develop motor weakness (usually in arms or trunk). Sacral zoster can cause detrusor paralysis, producing urinary retention. 55% of those with motor symptoms have good recovery, 30% have fair to good recovery
4. neoplastic:
- a) spine tumors: multiple myeloma (p.714), metastases (p.821)
 - b) bone or soft-tissue tumors along the course of the sciatic nerve: may result in erroneous laminectomy for herniated lumbar disc.³⁰ Pain is usually insidious in onset, and not positional (see below)
 - intra-abdominal or pelvic neoplasm
 - tumors of the thigh
 - tumors in the popliteal fossa or calf
5. inflammatory:
- a) trochanteric bursitis (p.1101): may produce pseudoradiculopathy. Rarely extends to the posterior thigh or as far distally as the knee
 - b) myositis ossificans of the biceps femoris muscle³¹
6. vascular:
- a) sciatica may be mimicked by intermittent (i.e. vascular) claudication
 - b) psoas hematoma: usually in patient on anticoagulant. Sometimes drainage is required
7. referred pain of nonspinal origin: not dermatomal. Nerve root tensions signs (p.1101) are usually negative. Includes:
- a) pyelonephritis
 - b) renolithiasis including ureteral obstruction
 - c) cholecystitis
 - d) appendicitis
 - e) endometritis/endometriosis
 - f) posterior perforating duodenal ulcer
 - g) inguinal hernia, especially if incarcerated
 - h) aortic dissection (p.1414)
8. piriformis syndrome (PS): controversial. Piriformis muscle originates on anterior S2–4 VBs, sacrotuberous ligament and passes through the greater sciatic notch to attach to the greater trochanter of the femur. It is innervated by L5-S1. It is the principle external rotator of the extended hip. It may irritate or compress the sciatic nerve (AKA pseudosciatica, can mimic symptoms of a herniated disc). The superior gluteal nerve is spared as it has a take-off proximal to the muscle. Conversely, PS may occur secondary to lower lumbar radiculopathy. Produces pain in the sciatic distribution and weakness of external rotation and abduction of the hip. Signs: Freiberg test (pain with forced internal rotation of the hip with thigh extension) or the Pace test (pain on resisted abduction/external rotation of the hip). No well designed studies of treatments. Advocated therapies include: PT, stretching, injection of the muscle localized by digital rectal exam taking care not to inject the sciatic nerve itself & piriformis muscle section. Sometimes long-lasting relief can follow injection with local anesthetic. Use of botulinum toxin (Botox®) injections has been described
9. more peripheral involvement (i.e. neuropathy) that may be confused with radiculopathy. Including:
- a) femoral neuropathy mistaken for L4 radiculopathy (see below)
 - b) proximal sacral plexus lesion mistaken for S1 radiculopathy (see below)
 - c) diabetic neuropathy (p.545) including diabetic amyotrophy
 - d) tumors (see below)

92.3.3 Extraspinal tumors causing sciatica

□ Pain characteristics: pain is almost always insidious in onset.³⁰ It may be intermittent initially, but eventually all patients develop pain that is constant, progressive and unaffected by position or rest.³⁰ Significant night pain is described in ≈ 80%

Straight leg raising was positive in most, but in more than half the pain was localized to a specific point along the course of the nerve, distal to the sciatic notch.³⁰ Conservative treatment brings either no or only temporary relief.

Approximately 20% will have a previous history of tumor (usually neurofibromatosis or previous malignancy). Malignancies include³⁰: metastatic lesions, primary bone sarcomas (chondrosarcoma...), soft-tissue sarcomas (liposarcoma...). Benign tumors include: lipoma, neurofibroma, schwannoma, aneurysmal bone cyst of the sacrum, giant cell tumor of the sacrum, tenosynovial giant cell tumor.

In two-thirds of cases, a detailed medical history and physical exam allowed localization and even determining the nature (bone tumor vs. soft-tissue) of the lesion.³⁰ Radiographs that show the entire pelvis and the proximal femur will demonstrate almost all tumors in these locations.^{30,32}

92.3.4 Features differentiating radiculopathy in sciatica

General information

Sciatica may result from nerve root involvement within the spinal canal (e.g. with lumbar disc herniation). Clinically this produces a nerve root syndrome; see Nerve root syndromes (p.1049). Spinal imaging studies (MRI, myelogram/CT) will usually detect nerve root compression here. More peripheral involvement may be difficult to image.

L4 involvement

Femoral neuropathy is often mistakenly identified as an L4 radiculopathy. Distinguishing features are shown in □ Table 92.2.

L5 involvement

Peroneal nerve palsy may be mistaken for L5 radiculopathy (p.1416).

S1 involvement

Outside the spinal canal, S1 can also be involved as it enters the sacral plexus, e.g. by a pelvic tumor. In plexus lesions, EMG will show sparing of the paraspinal muscles (nerves to paraspinal muscles exit in the region of the neural foramen) and the gluteus maximus and medius (superior and inferior gluteal nerves take-off just distal to the paraspinal nerves).

Table 92.2 Distinguishing femoral neuropathy from L4 radiculopathy		
Feature	Femoral neuropathy	L4 radiculopathy
Sensory loss		
distribution (□ Fig. 1.14)	anterior thigh	dermatome from ≈ knee to medial malleolus, spares anterior thigh
Muscle weakness		
iliopsoas	weak	normal
thigh adductors	normal (innervated by obturator nerve)	may be weak
quadriceps	weak	weak

92.4 Acute paraplegia or quadriplegia

92.4.1 General information

Entities causing spinal cord compression usually present as: paraplegia or -paresis (or quadriplegia/paresis), urinary retention (may require bladder ultrasound or checking a post-void residual to detect), and impaired sensation below level of compression. May develop over hours or days. Reflexes may be hyper- or hypo-active. There may or may not be a Babinski sign. Excluding trauma, the most common cause is compression by tumor or bone.

92.4.2 Etiologies

Some overlap with myelopathy.

1. in infancy (may produce “floppy infant syndrome”)
 - a) spinal muscular atrophy (the most severe form is called Werdnig-Hoffmann disease and is usually fatal within months): autosomal recessive congenital disease of childhood with degeneration of anterior horn cells. Only rarely evident at birth (where it presents as a paucity of movement), produces weakness, areflexia, muscle and tongue fasciculations with normal sensation. Severe cases progress over the first year or two to quadriplegia
 - b) spinal cord injury during parturition: a rare sequela of breech delivery
 - c) congenital myopathies: e.g. infantile acid maltase deficiency (Pompe disease)
 - d) infantile botulism: ileus, hypotonia, weakness, mydriasis, *Clostridium botulinum* bacteria and toxin in feces
2. traumatic spinal cord injury
 - a) major trauma: diagnosis is usually evident
 - b) minor trauma: may cause cord injury in setting of spinal stenosis, may → central cord syndrome; see Central cord syndrome (p.944)
 - c) atlantoaxial dislocation: from major trauma or due to instability from tumor or rheumatoid arthritis
3. congenital
 - a) extradural spinal cord compression by bone secondary to cervical hemivertebra (symptoms not present at birth, may develop decades later, occasionally after minor trauma)
 - b) cervical stenosis (p.1088) (usually with superimposed spondylosis): quadriplegia or central cord syndrome may follow minor trauma
 - c) achondroplastic dwarfism: spinal stenosis (animal model: dachshund)
 - d) syringomyelia: usually presents with central cord syndrome
4. metabolic
 - a) combined system disease* (p.1409)
 - b) thallium poisoning: usually causes sensory and autonomic symptoms, quadriplegia and dysarthria may be seen in severe cases
 - c) central pontine myelinolysis (p.115)
5. infectious
 - a) epidural spinal infection (abscess or empyema)*
 - b) post-viral (or post-vaccination): may be a transverse myelitis*
6. peripheral neuromuscular disorder*
 - a) Guillain-Barré syndrome (p.186): classically an ascending paralysis, but paraparesis mimicking a spinal cord lesion is an unusual variant³³
 - b) myopathies
7. neoplastic*: spinal cord tumors
8. autoimmune*
9. vascular
 - a) acute pontomedullary infarction: age usually >50 yrs. Patient is quadriplegic, alert, with bulbar palsies (eye movement abnormalities, impaired gag and speech)
 - b) spinal cord infarction*: including AVM, radiation myelopathy...
10. miscellaneous compressive*: including epidural hematoma, bony compression, epidural lipomatosis
11. functional: hysteria, malingering
12. bilateral cerebral hemisphere lesion (involving both motor strips): e.g. post-cerebral irradiation or parasagittal lesion. Will not have sensory level

* For items with asterisk, see Myelopathy (p.1407) for details.

92.5 Hemiparesis or hemiplegia

92.5.1 General information

May be produced by anything that interrupts the corticospinal tract from its origin in the pyramidal cells of Betz in the motor strip down to the cervical spine. This results an upper motor neuron paralysis (see □ Table 29.4) which should also produce long tract findings, including Babinski sign ipsilateral to hemiplegia.

92.5.2 Etiologies

□ Lesions of the cerebral hemisphere in the region of the contralateral motor strip. Large lesions may also involve sensory cortex producing reduced sensation ipsilateral to the hemiparesis

1. tumor (neoplasm): primary or metastatic
2. traumatic: epidural or subdural hematoma, hemorrhagic contusion of the brain, compression by depressed skull fracture
3. vascular:
 - a) infarction
 - ischemic: embolic, low flow (due to atherosclerosis, arterial dissection...)
 - hemorrhagic: intracerebral hemorrhage, aneurysmal SAH...
 - b) TIA (p.1264)
4. infection: cerebritis, abscess

□ Lesions of the contralateral internal capsule. Produces pure motor hemiplegia without sensory loss. Most common etiology is ischemic lacunar infarct

□ Lesions of the brainstem. Ischemic infarct, hemorrhage, tumor

□ Lesions of cervicomedullary junction. Foramen magnum lesions (p.1367).

□ Unilateral spinal cord lesions. Above $\approx$ C5 ipsilateral to the weakness producing a Brown-Séquard syndrome (p.947) with contralateral sensory loss to pain and temperature. See etiologies (p.947).

□ Hypoglycemia. Can sometimes be associated with hemiparesis that clears after administration of glucose

□ Note. In a patient with unexplained hemiparesis/hemiplegia, especially after trauma, consider carotid dissection.

92.6 Low back pain

92.6.1 General information

The following considers primarily low back pain (LBP) without radiculopathy or myelopathy, although some overlap occurs. Trauma is usually obvious and is not discussed. See Sciatica (p.1410) for differential diagnosis of that and also Low back pain and radiculopathy (p.1414) for evaluation.

92.6.2 Acute low back pain

Similar to list for myelopathy (p.1407). Most cases are non-specific (e.g. lumbosacral sprain), only 10–20% can be given a precise pathoanatomical diagnosis³⁴:

□ Patients writhing in pain. Should be evaluated for an intraabdominal or vascular condition (e.g. pain of aortic dissection is typically described as a “tearing” pain): patients with neurogenic LBP tend to remain as still as possible, possibly needing to change positions at intervals

□ Unrelenting pain at rest:

1. spinal tumor (intradural or extradural) (p.1032)
 - a) primary or metastatic spine tumor: suspected in patients with pain duration > 1 month, unrelieved by bed rest, failure to improve with conservative therapy, unexplained weight loss, age > 50 yrs³⁵

- b) nocturnal back pain relieved by aspirin is suggestive of osteoid osteoma or benign osteoblastoma (p.792)³⁶
- 2. infection (especially in IV drug abusers, diabetics, post spinal surgery, immunosuppressed patients, or those with pyelonephritis or UTI post-GU surgery). Fever is somewhat insensitive for spinal infections. Spine tenderness to percussion has 86% sensitivity with bacterial infections, but a low specificity of 60%³⁵ Types of infections include:
 - a) discitis
 - b) spinal epidural abscess: should be considered in patients with back pain, fever, spine tenderness, or skin infection (furuncle)
 - c) vertebral osteomyelitis
- 3. inflammatory
- 4. sacroiliitis: may produce pain and tenderness over one or both SI joints. Pelvic x-rays may show sclerosis of one or both sacroiliac joints.
 - a) bilateral & symmetric
 - ankylosing spondylitis (p.1123): morning back stiffness, no relief at rest, improvement with exercise.³⁷ Usually seen in males with symptom onset before age 40 yrs. Positive Patrick's test (p.1048) and pain on compressing the pelvis with the patient in the lateral decubitus position
 - Reiter syndrome (after Hans Reiter, a German bacteriologist): a reactive arthritis (usually 1–3 weeks following certain bacterial infections) with involvement of at least one other non-joint area (urethritis, uveitis/conjunctivitis, skin lesions, mucosal ulcerations...). 75% are HLA-B27 positive
 - may occur in Crohn's disease
 - b) bilateral & asymmetric
 - psoriatic arthritis
 - rheumatoid arthritis: adult & juvenile forms
 - c) unilateral
 - gout
 - osteoarthritis
 - infection

□ Evolving neurologic deficit. (Cauda equina syndrome: perineal anesthesia, urinary incontinence or urgency or retention, progressive weakness) all require emergent diagnostic evaluation to rule-out treatable conditions such as:

1. spinal epidural abscess (p.349)
2. spinal epidural hematoma (p.1131)
3. spinal tumor (intradural or extradural) (p.783)
4. massive central disc herniation (p.1051)

□ Pathologic fracture. Acute pain in patients at risk for osteoporosis or with known Ca should prompt evaluation for pathologic fractures

1. lumbar compression fracture: see Osteoporotic spine fractures (p.1008)
2. sacral insufficiency fracture³⁸: especially in rheumatoid arthritis patients on chronic steroids, often with no antecedent history of trauma. May cause back pain and/or radiculopathy. Often missed on plain films, best seen on CT, but may also be detected on bone scan

□ Coccydynia (p.1038) . Pain and tenderness around the coccyx

□ Tears in the annulus fibrosus. ("Anular tears")³⁹ (NB: also present in 40% of asymptomatic patients between 50–60 yrs age, and 75% between 60–70 yrs⁴⁰)

□ Rarely following subarachnoid hemorrhage. (SAH) due to irritation of lumbar nerve roots and dura: usually accompanied by other signs of SAH (p.1156)

□ Myalgia. May be a side-effect of "statins" (drugs used to lower serum concentration of LDL cholesterol) with or without elevation of serum creatine phosphokinase, sometimes with accompanying weakness and rarely with severe rhabdomyolysis and myoglobinuria leading to renal failure (risk may be increased with renal or hepatic dysfunction, advanced age, hypothyroidism, or serious infection)⁴¹

□ Drug induced:

1. statins: see above under myalgia

- 2. phosphodiesterase type 5 (PDE5) inhibitors used for erectile dysfunction: all may be associated with LBP, but the incidence is higher with tadalafil,⁴² etiology unknown. Usually occurs 12–24 hours post-dose and resolves by 48 hours. Most respond to simple analgesics

92.6.3 Subacute low back pain

- 10%of patients with LBP have symptoms that persists >6 weeks.
Differential diagnosis includes causes of acute LBP (above) and also:
- 1. continued pain at rest should prompt evaluation for spinal osteomyelitis (especially with fever and elevated ESR) or neoplasm if not already done
 - 2. plain spine x-rays may show possibly causative conditions, although many or all of the following may also be seen in asymptomatic patients
 - a) spondylolisthesis (p.1098)
 - b) spinal osteophytes
 - c) lumbar stenosis
 - d) Schmorl’s node or nodule (p.1060): disc herniation through cartilaginous end-plate into vertebral body (NB: may also be seen in 19%of asymptomatic patients⁴³)

Chronic low back pain

After 3 months, only ≈ 5%of patients with LBP will continue to have persistent symptoms. A structural diagnosis is possible in only ≈ 50%of these patients. These patients account for 85%of the cost in lost work and compensation.³⁴

- Differential diagnosis includes causes of acute and subacute LBP listed above, as well as:
- 1. degenerative conditions
 - a) degenerative spondylolisthesis (p.1099)
 - b) spinal stenosis (affecting the spinal canal)
 - c) lateral recess syndrome
 - 2. spondyloarthropathies
 - a) ankylosing spondylitis: look for erosive changes adjacent to SI joint and positive test for HLA-B27 antigen
 - b) Paget’s disease of the spine: vertebral involvement is very common in a patients with Paget’s disease
 - 3. osteitis condensans ilii: increased density in ilium, usually asymptomatic (incidental) finding. Occasionally may produce low back pain or tenderness. Usually found in women who have been pregnant
 - 4. psychological overlay: including secondary gain (financial, emotional...)

92.7 Foot drop

92.7.1 General information

Key concepts

- weak anterior tibialis (foot extension) innervated by deep peroneal nerve (L4, 5)
- most common etiologies: L4/L5 radiculopathy, common peroneal nerve palsy
- in a patient with foot drop, check posterior tibialis (foot inversion) and gluteus medius (internal rotation of flexed hip) – both are spared in peroneal nerve palsy and both should be involved with L4/5 radiculopathy
- EMG can assist in localization and prognostication

Definition: weakness of anterior tibialis (primarily L4 and to a lesser extent L5), often accompanied by a weak extensor digitorum longus and extensor hallicus longus (primarily L5 with some S1 contribution), all of which are innervated by the deep peroneal nerve.

92.7.2 Underlying substrates of foot drop

The most common dilemma is to distinguish foot drop due to radiculopathy from that due to peroneal nerve palsy (usually common peroneal nerve). With common peroneal nerve (CPN) palsy, there

is sparing of posterior tibialis (foot inversion, innervated by posterior tibial nerve) and gluteus medius (internal rotation of the thigh with the hip flexed, innervated by superior gluteal nerve, primarily L5 with some L4, the takeoff is shortly after the roots exit from neural foramen). With L4 or L5 root lesions these muscles will also be weak, see □ Fig. 92.1.

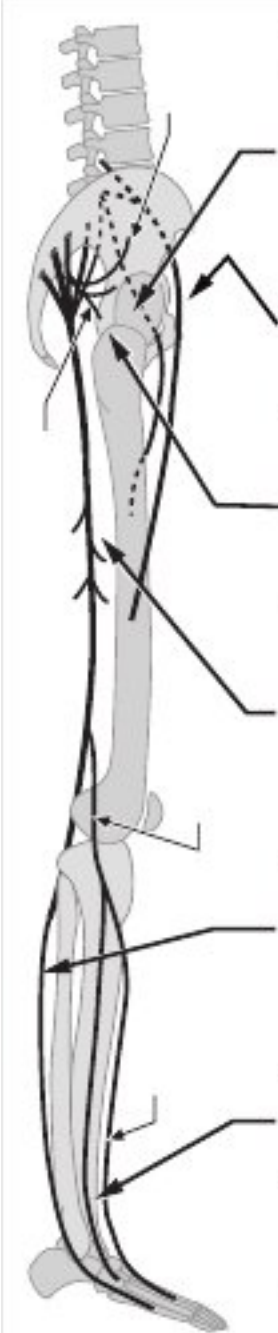
Flail foot results from paralysis of dorsiflexors plus plantarflexors, e.g. in sciatic nerve dysfunction as can occur during surgery for hip fracture/dislocation⁴⁴ or injection injuries (IM injections should be give superiorly and laterally to a line drawn between the posterior superior iliac spine and the greater trochanter of the hip). NB: the peroneal division of the sciatic nerve tends to be more vulnerable to injury than the tibial division.

92.7.3 Etiologies of foot drop

Three major categories: 1) muscular, 2) neurologic, 3) anatomic.

- 1. peripheral nerve palsies (more common). See □ Table 92.3 and □ Fig. 92.1.
 - a) peroneal nerve injury (also, see Common peroneal nerve palsy (p.535) for details including etiologies). Branches that may be involved:
 - deep peroneal nerve: isolated foot drop with minimal sensory loss (except possibly in great toe web space)
 - superficial peroneal nerve: weakness of peroneus longus and brevis (foot eversion) with no foot drop. Sensory loss: lateral aspect of lower half of leg and foot
 - common peroneal nerve: combination of above (i.e. foot drop + weak foot eversion, with sparing of tibialis posterior (foot inversion). Sensory loss: lateral aspect of lower half of leg and foot)
 - b) L5 radiculopathy: (or, less commonly, L4). The most common cause is HLD at L4–5, other etiologies include: lumbar spinal stenosis at L4–5, sacral ala fracture (p.1014)
 - results in pain and/or sensory changes in L5 (or L4) dermatome
 - weakness with radiculopathy tends to be more pronounced in distal muscles (e.g. anterior tibialis) than in proximal (e.g. gluteus maximus)
 - painless foot drop is unlikely to be due to radiculopathy; consider peroneal neuropathy, diabetic neuropathy, lesion anywhere along pyramidal tract, motor neuron disease...
 - c) lumbar plexus injury
 - d) lumbosacral plexus neuropathy (p.544)
 - e) injury to lateral trunk of sciatic nerve

Table 92.3 Localization of lesion with foot drop						
Lesion	Motor deficit ^a					Sensory changes
	anterior tibialis (L4, 5 ankle dor-siflexion)	peroneus longus/brevis (L5, S1 foot eversion)	tibialis posterior (L4, 5 foot inversion)	biceps femoris (L5, S1, 2 knee flexion)	gastro-cnemius (S1, 2 plantar-flexion)	
deep peroneal nerve	x					minimal, or great toe web space
superficial peroneal nerve		x				lateral distal leg and dorsum of foot
common peroneal nerve (CPN)	x	x				all of the above
L4 or L5 radiculopathy	x	x	x			dermatomal (□ Fig. 1.14)
peroneal division of sciatic nerve ^b	x	x	x	x		as with common peroneal
main trunk of sciatic nerve	x	x	X	x	x	lateral distal leg and entire foot
^a x denotes that the indicated muscle is involved (i.e. weak)						
^b see footnote (b) under □ Fig. 92.1						



Muscle to check	Rationale	Finding: exam
thigh adductors (adductors longus, brevis and magnus) (L2, L3): innervated by the obturator nerve	involvement indicates that the lesion includes more than sciatic nerve/L5 root (e.g. paravertebral mass, cauda equina lesion if bilateral findings)	weakness when adducting thigh while supine with knee extended
quadriceps femoris innervated by the femoral nerve	(same as above)	weakness of knee extension (L2, L3, L4)
muscles innervated by L5 branches that exit the lumbar plexus very close to the neural foramina	involvement indicates very proximal lesion (e.g. nerve root or very proximal (paravertebral) lumbar plexus) ^a	weakness of: 1. gluteus medius (L4, L5, S1): internally rotate thigh 2. gluteus maximus (L5, S1, 2): dig heel into bed while supine
L5 innervated muscles (via sciatic nerve) proximal to the take off of the common peroneal nerve ^b	if muscles listed above are intact, involvement of muscles to the right localizes the lesion to sciatic nerve above mid thigh (e.g injury to sciatic nerve at the greater sciatic notch)	1. slight weakness of biceps femoris (lateral hamstrings) (L5, S1, 2): flex the knee (with thigh flexed) 2. gastrocnemius weakness (foot plantarflexion) unless injury only to peroneal division of sciatic nerve ^b
Muscles innervated by (posterior) tibial nerve	sparing of these with foot drop indicates lesion distal to take off of common peroneal nerve (foot drop with weak foot inversion may be L4 or 5 radiculopathy)	weakness of tibialis posterior (L4, L5): invert foot (foot should be plantarflexed to eliminate anterior tibialis)
anterior tibialis innervated by deep peroneal nerve	involvement does not narrow down etiology until other muscles are examined	weakness of ankle dorsiflexion (foot drop)
muscles innervated by superficial peroneal nerve	preservation of these with foot drop localizes the lesion to the deep peroneal nerve	weakness of the peroneus longus and brevis (L5, S1): evert the foot
Abbreviations: CPN = common peroneal nerve DPN = Deep Peroneal nerve IGN = inferior gluteal nerve SGN = superior gluteal nerve		
^a : Note: EMG can differentiate root lesion from plexus by detecting involvement of paraspinal muscles which occurs in root but not in plexus lesions since dorsal rami exit proximal to plexus ^b : The peroneal of the sciatic nerve is more vulnerable to injury than the tibial division for several reasons. It is thus not unusual to see isolated peroneal nerve injuries above the knee e.g from hip dislocation or fractures, stab wounds, injection injuries...		

Fig. 92.1 Physical exam to localize the lesion in a patient with IE weakness

- f) peripheral neuropathy: weakness tends to be greater distally, producing wrist or foot drop. Classic example: Charcot-Marie-Tooth (p. 541), findings tend to be rather dramatic in spite of the fact that it often doesn't seem to bother the patient very much
- g) early in the course of motor neuron disease (ALS)
- h) heavy metal poisoning
- 2. central nervous system causes (foot drop here is usually painless)
 - a) cortical lesion (UMN): parasagittal lesions in region of motor strip (sensation will be spared if the lesion does not extend posteriorly to the sensory cortex).⁴⁵ There may be a Babinski sign or hyperactive Achilles reflex (so-called "spastic foot drop"). Usually painless
 - b) spinal cord injury: including cervical spinal myelopathy
- 3. non-neurogenic causes
 - a) muscular dystrophy
 - b) lead toxicity: in children may cause foot drop with no sensory loss
 - c) anterior compartment syndrome

92.7.4 Clinical

Loss of dorsiflexion causes foot slap with the front of the foot when the heel strikes the ground while walking. Also during the swing phase of gait the front of the foot may snag the ground (especially on

uneven surfaces) which may cause tripping; thus patients develop steppage gait (exaggerated thigh & knee flexion) on the affected side. Associated weakness of tibialis posterior, when present (e.g. with L5 radiculopathy) destabilizes the ankle permitting eversion which also predisposes to falls and to ankle fractures. Chronic foot drop may produce achilles tendon contracture with talipes equinus.

Wasting of the extensor digitorum brevis may be seen.

92.7.5 Evaluation

1. bloodwork: glucose, ESR
2. EMG: can help differentiate L5 radiculopathy from peroneal nerve palsy, plexus lesion (□ Fig. 92.1) or motor neuron disease (p. 182) (for details). EMG is not reliable until symptoms have been present at least $\approx$ 3 weeks
3. For suspected radiculopathy: MRI (or CT/myelogram if MRI is not possible)

92.8 Weakness/atrophy of the hands/UEs

92.8.1 Hand/UE weakness or atrophy with relatively preserved function in the LEs

1. cervical spondylosis (p. 1083): often causes sensory disturbance
2. cervical radiculopathy (p. 1069)
3. amyotrophic lateral sclerosis (ALS): no sensory involvement. One of the few causes of clinically prominent fasciculations. See details of ALS (p. 1410), other distinguishing features (p. 1410), fibrillations (p. 505)
4. spinal cord pathology
 - a) central cord syndrome (p. 944): typically causes more involvement (weakness, sensory disturbance) in the UE than the LE
 - b) syringomyelia (p. 1144): usually burning dysesthesias of the hands with dissociated sensory loss
5. brachial plexus injury (p. 550)
6. brachial plexus neuropathy (includes Parsonage-Turner syndrome) (p. 542)
7. peripheral nerve problems, including
 - a) carpal tunnel syndrome (p. 519)
 - b) ulnar neuropathy (p. 526)
 - c) other peripheral nerve entrapment syndromes (p. 515)
8. foramen magnum lesions (p. 1367): can cause (Bell's) cruciate paralysis⁴⁶ due to compression above the pyramidal decussation which produces bilateral UE weakness and possibly atrophy of the hands with sparing of the LEs⁴⁷ (in the differential diagnosis for central cord syndrome). Compression on one side may produce the similarly named but clinically different hemiplegia cruciata (spastic palsy of one UE and the contralateral LE)⁴⁷
9. thoracic outlet syndrome (p. 554)
10. botulism (p. 1399)
11. pharyngeal-cervical-brachial variant of Guillain-Barré syndrome (p. 186)

92.8.2 Atrophy of the first dorsal interosseous muscle

Etiologies: either C8/T1 nerve root or ulnar nerve involvement (either may be focal or diffuse). There are 4 main differential diagnoses:

1. ulnar neuropathy: check median nerve to see if findings extend to a nearby but separate nerve
 - a) at the elbow (p. 531)
 - b) at Guyon's canal (p. 531)
2. nerve root involvement:
 - a) cervical radiculopathy: C8 or T1
 - b) nerve root avulsion: weakness + sensory loss with normal SNAP on EMG (p. 243) usually with a history of precipitating trauma
3. lower brachial plexus involvement
 - a) thoracic outlet syndrome (p. 554)
 - b) Pancoast tumor (p. 542)
4. neurodegenerative disorders

- a) amyotrophic lateral sclerosis (ALS) (p.1086)
- b) multifocal motor neuropathy (MMN) (p.1410): a chronic dysimmune neuropathy with asymmetric muscle wasting, cramping & LE twitching

92.9 Radiculopathy, upper extremity (cervical)

See Weakness/atrophy of the hands/UEs (p.1420). In addition to those items:

1. primary shoulder pathology: characteristically, pain is aggravated by active and/or passive shoulder movement. In general, shoulder pathology does not produce pain referred to the neck
 - a) rotator cuff tear
 - b) bicipital tendonitis: tenderness over biceps tendon
 - c) subacromial bursitis: there may be tenderness over the AC joint
 - d) adhesive capsulitis
 - e) impingement syndrome: the “empty can test” is usually positive (each arm held out in front, 30° lateral to straight forward, thumbs pointing down, as in emptying out a soda can. Examiner pushes down on the patient’s hands while the patient resists. Test is positive if it reproduces pain)
2. shoulder pain is very common in polymyalgia rheumatica (p.198), typically worsens with movement
3. interscapular pain: a common location for referred pain with cervical radiculopathy, may also occur with cholecystitis or some shoulder pathologies
4. MI: some cases of cervical radiculopathy (especially left C6) may present with symptoms that are suggestive of an acute myocardial infarction
5. complex regional pain syndrome AKA reflex sympathetic dystrophy (p.497): may be difficult to distinguish from cervical radiculopathy. Stellate ganglion blocks may help.⁴⁸

92.10 Neck pain (cervical pain)

This section deals primarily with axial neck-pain without radicular features. For radicular features, see Radiculopathy, upper extremity (cervical) above.

1. cervical spondylosis (including facet arthritis)
2. cervical sprain: including whiplash associated disorder
3. fracture of the cervical spine: with upper cervical spine fractures (e.g. odontoid), patients characteristically hold their head in their hands, especially when going from recumbent to upright position
 - a) traumatic
 - b) pathologic (tumor invasion, rheumatoid arthritis)
4. occipital neuralgia (p.515)
5. herniated cervical disc:
 - a) lateral herniated disc: if symptomatic, tends to produce more radicular symptoms in the UE than actual neck pain
 - b) central disc herniation: if symptomatic, tends to produce myelopathy, does not produce any neck pain whatsoever in many cases
6. abnormalities of the craniocervical junction:
 - a) Chiari 1 malformation (p.277)
 - b) atlantoaxial subluxation
7. fibromyalgia: idiopathic chronic pain syndrome characterized by widespread nonarticular musculoskeletal pain, nodularity and stiffness^{49,50} without pathologic inflammation. Possible link to neuroendocrine dysfunction.⁵¹ Afflicts 2% of the population,⁵⁰ female: male ratio is 7:1. No diagnostic laboratory study. May be associated with psychiatric illness and multiple non-specific somatic complaints including malaise, fatigue, impaired sleep, GI complaints and cognitive impairment
8. Eagle’s syndrome: elongation of the styloid process. Surgical resection can ameliorate the pain. Two variants:
 - a) typical variant: history of tonsillectomy. Pharyngeal pain, dysphagia and otalgia
 - b) second variant: AKA carotid artery-styloid process syndrome. Carotidynia radiating into ipsilateral eye and vertex
9. crystal deposition diseases: gout, pseudogout, hydroxyapatite (HA) or calcium pyrophosphate dihydrate (CPPD) crystal deposition diseases. May appear as a crown-like density encompassing the odontoid process (crowned-dens syndrome)⁵² representing calcifications in the transverse ligament, seen best on cervical CT. May be treated with a short course of prednisolone (e.g. 15 mg/d) followed by NSAIDs

92.11 Burning hands/feet

1. spinal cord syndromes:
 - a) central cord syndrome (CCS) (p.944)
 - b) burning hands syndrome (p.937): a possible variant of CCS, described in football-related cervical spine injury
 - c) numb-clumsy hand syndrome (p.1085): seen in cervical myelopathy
2. complex regional pain syndrome (CRPS) AKA reflex sympathetic dystrophy (p.497)
3. peripheral neuropathy
 - a) diabetic amyotrophy AKA Bruns-Garland syndrome (p.545)
4. erythermalgia AKA erythromelalgia: rare disorder characterized by erythema, edema, increased skin temperature, and burning pain of the hands and/or feet. Usually refractory to medical management, some success reported with epidural bupivacaine,⁵³ lidocaine patches,⁵⁴ or cold soaks
 - a) primary erythermalgia: etiology is idiopathic
 - b) secondary erythermalgia: associated with autoimmune and rheumatologic factors
5. vascular:
 - a) occlusive arterial disease: atherosclerosis, Raynaud's syndrome
 - b) venous insufficiency

92.12 Muscle pain/tenderness

1. fibromyalgia: see above
2. myopathy
3. statin induced myopathy: can vary from mild (with symptoms of muscles aches, symptoms usually abate rapidly after discontinuation of the statin although, occasionally up to 2 months may be required) to severe (with rhabdomyolysis which can →nephropathy)
4. diffuse severe sensitivity to light touch is a marker of nonorganic pain⁵⁵

92.13 Lhermitte's sign

92.13.1 General information

Technically, a symptom (not a sign). Electrical shock-like sensation radiating down the spine usually provoked by neck flexion (shocks radiating up the spine are sometimes referred to as reverse Lhermitte's sign). Classically attributed to MS, but may occur in any process involving primarily the posterior columns of the spinal cord.

92.13.2 Etiologies

1. multiple sclerosis (MS) (p.179)
2. cervical spondylosis
3. subacute combined degeneration (p.1409): check for vitamin B12 deficiency
4. cervical cord tumor
5. cervical disc herniation
6. radiation myelopathy (p.1563)
7. Chiari type I malformation (p.277)
8. central cord syndrome (p.944)
9. SCIWORA (spinal cord injury without radiographic abnormality) (p.999)

92.14 Swallowing difficulties

1. mechanical: the term globus describes a sensation of a lump in the throat
 - a) ossification of the anterior longitudinal ligament (OALL) (p.1129)
 - b) diffuse idiopathic skeletal hyperostosis (DISH) (AKA Forestier disease) (p.1129): An enthesopathy
 - c) post-op following ACDF
 - it is normal to have a little swelling and fullness early post-op
 - may be increased with multiple levels and with anterior plating
 - as a complication from post-op hematoma
2. neurologic

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Part XXIII

Procedures, Interventions,
Operations

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93 General Information

93.1 Introduction

This section provides information useful in the O.R. that apply to a number of different topics. Some items that are pertinent to only one topic will be found in that section instead (e.g. transsphenoidal tumor removal is found in the section on pituitary tumors).

REMEMBER: before performing any invasive procedure, know the patient's coagulation status (history of anticoagulant and/or antiplatelet function medications, and if indicated: PT, PTT, INR, platelet count and platelet function assay, FDP...).

93.2 Intraoperative dyes

This section covers visible dyes that may be useful in the operating room. For radio-opaque dyes, see Contrast agents in neuroradiology (p.219). There is little information available in the literature regarding the intrathecal (IT) use of the following agents.

Indigo carmine: a blue dye which has been used intrathecally to locate CSF leaks. There are few published reports, and no accounts of adverse effects. In 1933 a report¹ of IT injection of 5 ml of 0.6% indigo carmine solution produced blue-green discoloration of the CSF draining through a fistula into the nose within 15 minutes, lasting for 5 hours, with no indication of toxicity. It is excreted in the urine (and not in mucous membranes). The consensus is that it should be relatively safe for IT use, but the manufacturer did not recommend this application.

□ Methylene blue: methylene blue is probably cytotoxic and appears to become fixed to neural tissue. It should therefore not be used as a stain in neurosurgical operations or diagnostic tests. CNS damage (some permanent) occurred in 14 patients given an IT injection of a 1% solution. Symptoms included: paraparesis, quadriplegia, multiple cranial nerve involvement (including anosmia and optic atrophy), dementia and hydrocephalus.²

Fluorescein: although intrathecal injection (e.g. to look for CSF leak) has been used by ENT surgeons with apparently acceptable results, there is a risk of seizures. 2.5% fluorescein is diluted 1:10 with CSF or saline and ≈ 6 ml is injected into the spinal subarachnoid space (or 0.5 ml of 5% fluorescein mixed with 5–10 ml of CSF³).

Fluorescein has also been used IV (adult dose: 1 amp IV) to help mark areas of the brain where there is breakdown of the blood brain barrier (BBB), e.g. in tumors (p.596), however, fluorescein is eventually excreted in mucus, urine, etc., and just about everything turns orange. It has also been used to perform intraoperative “visible angiograms” during the removal of AVMs.

Indigocyanine green (ICG): used for intraoperative angiography (p.1597).

93.3 Operating room equipment

93.3.1 Operating microscope – observer's eyepiece

For spine cases, the ideal location of the observer eyepiece is usually directly opposite the surgeon. For intracranial work, the observer's (assistant's) eyepiece is placed to right of operator's except in the following cases where it is placed to the left:

1. transsphenoidal surgery (when the surgeon stands to the patient's right)
2. right posterior fossa craniotomy in the lateral oblique (suboccipital) position

93.3.2 Head stabilization

General information

Options include

1. Non-pin head stabilization: When absolute head fixation is not indicated, the following options avoid some of the complications associated with pin-based head fixation (see below):
 - a) Horse-shoe head rest
 - b) Doughnut fashioned out of stockinette
 - c) Prone-view® for prone position e.g. during posterior spine surgery
2. Pin-based head-fixation: The most common system is the Mayfield head-holder.

Pin-based head-fixation

Indications for pin stabilization:

1. ☐ not recommended for use in age < 3 years, use pediatric pins with care in ages 3–10 years (this age cutoff is not based on scientific evidence, most reported complications occur in age > 3 years)
2. craniotomies:
 - a) most intracranial vascular operations: a radiolucent head-holder should be used if intraoperative angiogram is to be performed
 - b) often for tumor operations, especially if there is need for a self-retaining retractor system that attaches to the Mayfield head-holder (e.g. Budde halo)
 - c) when intraoperative image-guidance (IG) systems are used (an alternative here is to use mask-based or strap-on registration array holders)
3. cervical spine: often used for posterior cervical operations (laminectomies, instrumentation, fusions...)

Application of pin head holder:

1. “skull block” (p.1434) (blockade of scalp innervation) may be administered prior to pin placement. Critical for awake craniotomies (for the wake-up), and for vascular cases and cases with increased ICP where the block blunts the precipitous increase in blood pressure^{4,5} that may otherwise accompany pinning which may → increased ICP. For technique, see Typical sequence of anesthesia (p.1434).
2. plan pin placement
 - a) the manufacturer recommends that the pins be placed within a band-like area similar to a “sweatband” worn just above the orbits and the pinna
 - b) avoid placing pins on the thin temporal squamosa, and use with caution over the frontal sinus⁶
 - c) the single pin is typically placed anteriorly when the patient is in the supine position (☐ Fig. 94.5), for a posterior fossa approach in the prone position, if the craniotomy is on one side the single pin is placed on that side
 - d) the pins in the dual pin “rocker” should be equidistant from the centerline for maximal stability
3. appropriately sized sterile pins are placed in the head-holder:
 - a) ☐ children < 3 yrs age A: increased risk of calvarial penetration or depressed skull fractures. A padded cerebellar head rest should be used instead^{6,7}
 - b) children ≥ 3 yrs age and < 10 yrs age: special pediatric pins should be employed (these pins have a “shoulder” to insure shorter depth of penetration)
4. the pins are often coated with appropriate antibiotic ointment
5. the clamp is squeezed together, allowing the ratchet gears to slide, until the pins are initially seated in the skull
6. the knob housing the tension spring and gauge is tightened (each ring = 20 lbs)
 - a) adults: tighten until the third ring (60 lbs) is visible; up to 80 lbs has been described
 - b) pediatrics: less pressure should be applied, 30–40 lbs has been suggested.⁶ 6 Even with pediatric pins & decreased pressure, complications may occur, consider horseshoe headrest

Complications of pin head-fixation:

1. malposition of pins:
 - a) through unintended anatomic structures: pinna, orbit, superficial temporal artery, shunt hardware, prior craniotomy defect
 - b) poor fixation by not properly placing pins close to “equator” resulting in movement during surgery (with risk of cervical spine injury, injury to structures being operated on due to the sudden movement, loss of image guided registration) and possible skin laceration
2. skin penetration by pins: can cause injury to intracranial structures, infection including delayed abscess, epidural hematoma⁶...
 - a) overtightening of pins
 - b) incorrect pin selection: see above
 - c) soft skull: in elderly patients, poorly calcified skulls, pediatric patients⁶
3. skin necrosis: especially with pediatric pins due to the “shoulder” of the pins
4. skull fracture: including “ping-pong ball” fractures in young children
5. slippage of any of the joints or connections to the O.R. table
6. clamp breakage^{8,9}: inspect the head-holder for cracks prior to each application, store properly and maintain per the manufacturer’s specifications

7. bleeding from pin site: usually when the head-holder is removed. If bleeding does not stop after a minute or so of pressure, a suture or surgical staple may be used

93.4 Surgical hemostasis

93.4.1 Basic options

1. thermocoagulation
 - a) electrical coagulation:
 - monopolar (Bovie) cautery: electric current passes through the patient to a grounding pad. Because of possible transmission through electrically and thermally sensitive neural structures, this modality is not used directly on the brain or in proximity to named nerves (including cranial nerves) & nerve roots
 - bipolar cautery: current passes only between the tips of the cautery device. Used for precise coagulation. When used directly on or near to the brain or nerves, the current setting is typically reduced from that employed for general use
 - b) thermal units: e.g. AccuTemp® disposable eye cautery units (particularly useful to coagulate dura when inserting a ventriculostomy in the ICU)
 - c) laser: especially neodymium:yttrium-aluminum-garnet (Nd:YAG) laser
2. mechanical
 - a) bone wax: originated by Sir Victor Horsely. Inhibits bone formation
 - b) ligature: less commonly used in neurosurgery than other specialties
 - c) “silver clips” (e.g. HemoClips®)
3. chemical hemostasis: see below

93.4.2 Chemical hemostasis

See review¹⁰ for more information. Some key points:

1. gelatin sponge (Gelfoam®): no intrinsic coagulating effect. Absorbs 45 times its weight in blood which causes it to expand and tamponade bleeding. Absorbable. May be combined with thrombin as patties or as powder (e.g. FLOSEAL®, SurgiFlo®)
2. oxidized cellulose (Oxycel®) and oxidized regenerated cellulose (Surgicel®): absorbable. Acidic material that reacts with blood to form a reddish brown “pseudoclot.” Bactericidal to over 20 different organisms. May retard bone growth. Oxycel® interferes with epithelialization more than Surgicel®
3. microfibrillar collagen (Avitene®): promotes adhesion and aggregation of platelets. Loses effectiveness in severe thrombocytopenia (<10,000/ml). May be used on bone bleeding. Remove excess material to reduce risk of infection
4. thrombin (Thrombostat®): does not depend on any intermediate physiological agent. Caution: although thrombin may cause significant edema when placed on brain where the pia has been disrupted, practical experience indicates this is uncommon

93.5 Craniotomy general information

93.5.1 Cranial perforators

Perforators are one means used to create burr holes which enable the surgeon to access the brain for intracranial procedures. Many brands of perforators are designed with a slip clutch that disengages the drill and stops the outer shaft from spinning and entering the skull once the central part of the drill penetrates the inner table. While these clutches are generally reliable and immensely helpful, there is a possibility of malfunction such that the outer shaft continues to rotate and if it penetrates the inner table, the entire drill can plunge into the brain. In 8 months of 2005 the FDA received reports of 200 injuries as a result of the drill failing to disengage.¹¹ The FDA released a number of recommendations to reduce the risk of injury,¹¹ excerpts of which are shown here:

- select the appropriate perforator based on the skull thickness (pediatric vs. adult)
- keep the perforator perpendicular to the skull throughout the drilling process
- do not rock, rotate, or change the angle of the device during drilling
- avoid using excessive pressure on the drill. Brace the hand holding the drill on the other hand which should rest on the patient’s skull, and thereby prevent plunging if the drill completely penetrates the skull
- use caution when:
 - drilling through areas of irregular bone contour, curvature or variations in thickness

- drilling on skulls of infants/children, the elderly, or any other patient who might have softer bone consistency (including osteogenesis imperfecta...)
- perforating bone in an area where the bone might be diseased or incompetent or have loose fragments

93.5.2 Intraparenchymal cyst aspiration

When a cystic tumor or intracerebral hemorrhage is being operated on, an attempt should be made to insert a ventricular needle into the lesion and aspirate some but not all of the cyst contents. This often produces significant decompression. Avoid evacuating all of the contents otherwise the lesion might be difficult to find. The needle may then be left in place to allow localization of the lesion (or the needle track can be followed, which may occasionally be difficult).

93.5.3 Intra-operative brain swelling

Background

Under certain circumstances during surgery, the brain may start to severely swell out of the craniotomy wound. Etiologies of this emergency situation include:

1. extraparenchymal bleeding: from a vessel or intraoperative aneurysm rupture, remotely situated epidural/subdural hematoma
2. intracerebral hemorrhage
3. venous outflow obstruction
4. vasodilatation induced by hypercarbia
5. severe diffuse cerebral edema following stroke or traumatic brain injury (TBI)

Management

First efforts should be aimed at ruling out and correcting the aforementioned causes as well as some adjunctive measures. Most maneuvers are similar to those used in controlling an ICP crisis. During the process, it is critical to try to avoid having the brain compress itself against the craniotomy bone edges which can lacerate the cortex and can also further compromise cortical veins which impairs venous outflow causing more brain edema and swelling which accelerates the vicious cycle.

1. elevate the head of the patient (e.g. with reverse Trendelenburg of the O.R. table)
2. make sure the jugular veins are not kinked: this may require rotating the head by loosening the pivot that connects the table adapter to the Mayfield head holder and rotating the head to a more neutral position
3. rule-out hypercarbia: make sure the endotracheal tube is not kinked, check the patient's end-tidal $p\text{CO}_2$
4. measures to lower ICP and protect the brain
 - a) give Mannitol 1 gm/kg IV bolus
 - b) drain CSF if an option: from adjacent cistern or lumbar drain
 - c) have anesthesiologist hyperventilate to a PCO_2 of 30–35 mm Hg
 - d) have anesthesiologist induce burst suppression
5. emergently intubate patients who are undergoing awake craniotomy
6. consider intraoperative ultrasound if rapidly available to rule-out hematoma (intracerebral, EDH, SDH) which could potentially be immediately evacuated
7. during the above steps, place a moist sponge on the surface of the brain and gently but firmly apply evenly distributed pressure to push the brain back into the wound
8. if all else is failing, the craniotomy flap can be enlarged as much as possible to create a decompressive craniotomy. Enlarging the skin incision to do so is preferable to having too small a bony opening which risks brain compression/laceration against the edges. The skin is closed without the bone flap and without dural closure as in a decompressive craniectomy (p.1467)
9. a last ditch life-saving measure for continued uncontrollable swelling which is to be taken under advisement with eloquent cortex: use a gloved hand to sweep the herniating brain out of the wound (i.e. remove it from the patient)

93.5.4 Craniotomy pre- and post-op management

Risks

Many risks cannot be generalized for all craniotomies and are specific to various tumors, aneurysms, etc. General information:

1. post-operative hemorrhage
 - a) overall risk of post-operative hemorrhage^{12,13}: 0.8–1.1% The most common indication for craniotomy in these series was for meningioma, followed by trauma, aneurysm, and then intrinsic supratentorial tumors. 43–60% of the hematomas were intraparenchymal, 28–33% epidural, 5–7% subdural, 5% intrasellar, 8% mixed, 11% confined to superficial wound. Overall mortality was 32%
 - b) hematoma may occur at the surgical site or in remote locations, e.g. intracerebellar hemorrhage after pterional¹⁴ and temporal¹⁵ craniotomies
2. in craniotomy for brain tumor¹⁶:
 - a) risk of anesthetic complications: 0.2%
 - b) increased neurologic deficit in 1st 24 hours post-op: $\approx$ 10%
 - c) wound infection: 2%
3. postoperative headache (p. 1431)

Pre-op orders

1. for tumor: if patient on steroids, give $\approx$ 50% higher dose 6 hrs before and on-call to O.R. (stress doses); if not on steroids give dexamethasone 10 mg PO 6 hrs before and on call to O.R. (in A.M., give with sip water)
2. antiepileptic medication
 - a) if there is a history of seizures:
 - if already on antiepileptic drugs (AEDs) continue same doses
 - if not on AEDs, load with Keppra 500 mg or oral PHT (may give 300 mg PO q 4 hrs $\times$ 3 doses (total 900 mg) to load orally)
 - b) if no history of seizures
 - if surgery does not require a cortical incision (e.g. aneurysm) then AEDs are generally not used
 - if a cortical incision is anticipated, option to load with AEDs as above
3. prophylactic antibiotics: (optional) ideally 30–60 minutes before incision. For most antibiotics, it is given in the O.R. before the skin incision. For antibiotics that take a long time to infuse (e.g. vancomycin) it may help to order it to be given “on call to O.R.”
4. DVT prophylaxis: pneumatic compression boots or knee-high TED® hose

Post-op orders

Guidelines (individualize as appropriate) for patient to be extubated

1. admit PACU, transfer to ICU (neuro unit if available) when stable
2. VS: q 15 min $\times$ 4 hrs, then q 1 hr. Temperature q 4 hrs $\times$ 3 d, then q 8 hrs. Neuro check q 1 hr
3. activity: bed rest (BR) with HOB elevated 20–30°
4. knee high TED hose or pneumatic compression boots
5. I & O q 1 hr (if no Foley: straight cath q 4 hrs PRN bladder distension)
6. incentive spirometry q 2 hrs while awake (do not use following transsphenoidal surgery)
7. diet: NPO except minimal ice chips and meds as ordered
8. IVF: NS+20 mEq KCl/L @90 ml/hr
9. O₂: 2 L per NC
10. meds:
 - a) dexamethasone (Decadron®): if not on chronic steroids, give 4 mg IV q 6 hrs. Otherwise give stress doses based on patients current dose and length of treatment
 - b) H₂ antagonist, e.g. ranitidine 50 mg IVPB q 8 hrs
 - c) Anti-epileptic drug (AED) especially when cerebral cortex is violated: typically Keppra® (levetiracetam): 500 mg PO or IV q 12 hours. If there is no prior history of seizure, typically discontinue after $\approx$ 1 week
 - d) Cardene® drip: titrate to keep SBP < 160 mm Hg and/or DBP < 100 mm Hg (use cuff pressures, may use A-line pressures if they correlate with cuff pressures)
 - e) codeine 30–60 mg IM q 3–4 hrs PRN H/A
 - f) acetaminophen (Tylenol®) 650 mg PO/PR q 4 hrs PRN temperature > 100.5° F (38 C)
 - g) continue prophylactic antibiotics if used: (e.g. cefazolin (Kefzol®) 500–1000 mg IVPB q 6 hrs $\times$ 24 hrs, then D/C)
11. labs:
 - a) CBC once stabilized in ICU and q d thereafter
 - b) renal profile once stabilized in ICU and q 12 hrs thereafter
 - c) ABG once stabilized in ICU and q 12 hrs $\times$ 2 days, then D/C (also check ABG after any ventilator change if patient on ventilator)

12. call M.D. if any deterioration in crani checks, for $T > 101^{\circ}$ (38.5°C), sudden increase in SBP, SBP < 120 , U.O. $< 60\text{ ml}/2\text{-hrs}$
13. post-op CT: non-contrast post-op head CT is performed if the patient does not return to baseline neurologic function within a reasonable amount of time, also, performed routinely at many institutions following all craniotomies

93.5.5 Postoperative deterioration

General information

When the postoperative neurologic status is worse than pre-op, especially in a patient who deteriorates after initially doing well following surgery, emergency evaluation and treatment is indicated.

Possible etiologies:

1. hematoma (p.1429)
 - a) intracerebral hemorrhage (ICH)
 - b) epidural hematoma: at or remote from surgical site
 - c) subdural hematoma
2. cerebral infarction
 - a) arterial
 - b) venous infarction: especially with surgery on or around the venous sinuses (p.1466)
3. postoperative seizure: may be due to inadequate anticonvulsant levels, and may be exacerbated by any of the above (see below for management)
4. acute hydrocephalus
5. pneumocephalus; also see Pneumocephalus (p.887):
 - a) tension pneumocephalus: see Tension pneumocephalus (p.888)
 - b) simple pneumocephalus: the simple presence of air in the cranium can cause neurologic symptoms even if not under tension (as would commonly occur following the now outdated pneumoencephalogram). Symptoms include: lethargy, confusion, severe headache, nausea & vomiting, seizures. Air may be located over the cerebral convexities, in the p-fossa, and/or in the ventricles and usually resorbs with symptomatic improvement in 1–3 days
6. edema: may improve with steroids
 - a) worsening of cerebral edema: moderate post-op worsening of cortical function of immediately adjacent brain is not unexpected in many operations, and is usually transient. However, reversible etiologies (such as subdural hematoma (SDH)) must be ruled out
 - b) traction or manipulation of cranial nerves may cause dysfunction that may be temporary. Division of cranial nerves can cause permanent dysfunction
7. persistent anesthetic effect (including paralytics): unlikely in a patient who deteriorates after initially doing well post-op. Consider reversing medication given during surgery (caution re hypertension and agitation), e.g. naloxone, flumazenil (p.298), or reversal of pharmacologic muscle block (p.136)
8. vasospasm: following SAH or may be due to manipulation of blood vessels

Postoperative seizure management

1. intubate if patient does not rapidly regain consciousness, is not protecting airway, or has labored respirations
2. CTscan: rule out hematoma (intracerebral or extra-axial) or hydrocephalus
3. anticonvulsants:
 - a) draw blood for appropriate anticonvulsant level
 - b) bolus with additional anticonvulsants: do not wait for levels

93.5.6 Postoperative headache

General information

Persistent headache (H/A) is well described following posterior fossa craniectomy (incidence range: 0–83%⁷). The time course in one series¹⁸ was: 23% at 3 mos, 16% at 1 yr, and 9% at 2 yrs.

Persistent H/A may also be observed following supratentorial craniotomy¹⁹ (prevalence 1 year after anterior temporal lobectomy for seizures: 12%⁹). The “syndrome of the trephined” was first described in the French literature during World War I, and consisted of: headache and sometimes pulsatile pain (usually localized to the area of the skull defect), amnesia, inability to concentrate, insomnia... similar in some ways to postconcussive syndrome (p.923).

These H/A have been attributed to: traction on the dura when the bone is not replaced, tension on the dura due to tight dural closure, temporalis or nuchal muscle dissection, nerve entrapment in the closing sutures or in the healing scar, intradural blood and/or bone dust, CSF leak.¹⁹

Prevention

No single method or group of methods has been successful in completely eliminating the complaint of post-op H/A.^{20,21} Until further research can further advance the understanding of the cause and prevention of these H/A, it seems reasonable to employ the following measures as much as possible in an attempt to minimize these debilitating symptoms: restoring function of the temporalis or suboccipital musculature, rigid fixation of bone flaps, cranioplasty for large craniectomies, meticulous tension-free dural closure (using duraplasty when necessary), and keeping intradural blood clot and bone dust to the minimum possible.²² Cranioplasty following posterior fossa surgery for vestibular schwannoma reduced the incidence of post-op H/A from 17% to 4%.²³

Treatment

Initially, symptomatic treatment is indicated. Referral to a H/A specialist may be appropriate when it becomes apparent that the H/A are not resolving spontaneously after ≈ 3 months.²²

93.6 Intraoperative cortical mapping (brain mapping)

93.6.1 General information

Indications: typically used to locate motor strip, sensory cortex, or speech centers intraoperatively for surgery in and around these eloquent areas. Localization of these areas based on visible anatomy alone is unreliable. These techniques are typically employed in seizure surgery as well as in treating lesions in areas of eloquent brain.

Some techniques require an awake patient, with the surgery being done under local anesthesia with sedation. Motor and sensory cortex can also be localized in anesthetized patients using SSEPs (see below).

93.6.2 Phase reversal method for localizing primary sensory and motor cortex

General information

Utilizes intra-operative SSEPs to localize primary sensory and motor cortex in patients under general anesthesia (as opposed to using brain mapping techniques in awake patients).^{24,25}

Technique

See anesthesia requirements for intraoperative EP monitoring (p.107). A strip grid is placed on the surface of the brain perpendicular to the anticipated orientation of the central sulcus. SSEP stimulation is performed while recording through the strip grid. Phase reversal of the N20/P20 peak between a pair of electrodes in the strip grid indicates that those electrodes straddle the central sulcus ($\square$ Fig. 93.1) with primary motor cortex located anteriorly, and sensory cortex posteriorly. The grid is then repositioned and the test is run again to verify the findings.

93.6.3 Awake craniotomy

General information

Usually employed for brain mapping, especially for speech areas. Numerous techniques and protocols have been described. Typically, the patient is temporarily anesthetized with short acting agents (inhalational and/or injectable). This is supplemented with local anesthetic. The craniotomy is then performed and the patient is allowed to wake up while the brain is exposed to permit neurophysiologic testing during surgery. If (short-acting) paralytics are used, it is critical to reverse these agents 15–30 minutes prior to applying the electrical stimulation and that a train-of-four muscle twitch can be elicited.

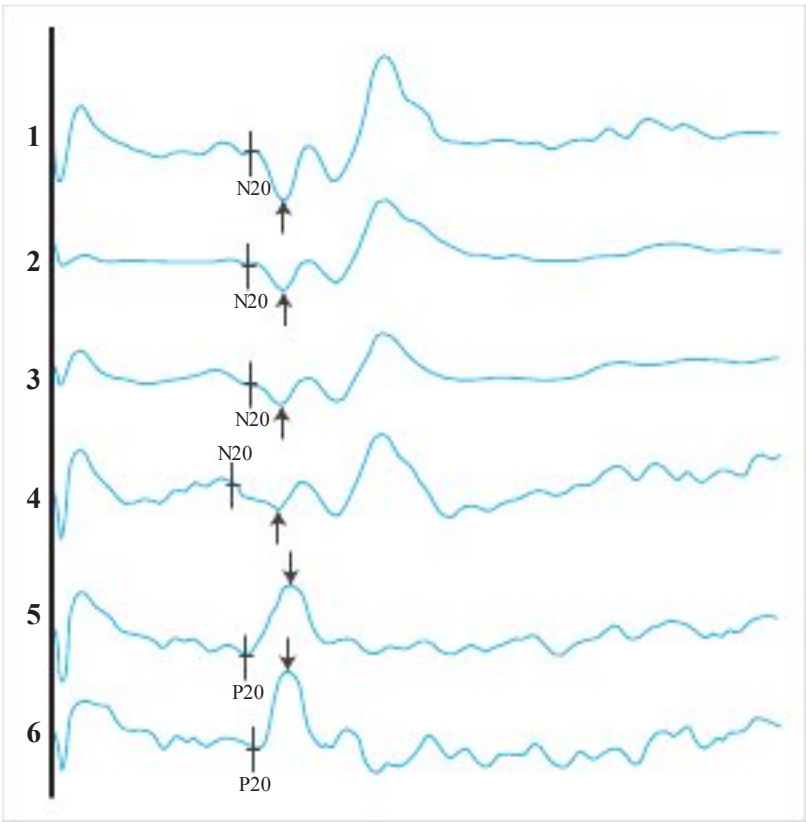


Fig. 93.1 Phase reversal
Intra-op 6-electrode recording strip placed on the brain during SSEP recording. Phase reversal of the negative N20 peak (arrows) to a positive P20 peak between electrodes #4 & 5 indicates that electrodes #4 & 5 straddle the central sulcus.

Booking the case: Awake craniotomy

Also see defaults & disclaimers (p. 27) and pre-op counselling (see below).

1. position: depends on lesion location, with pin headholder (for image guided navigation if used)
2. equipment:
 - a) microscope if needed e.g. for tumor dissection
 - b) image guided navigation system (if used)
 - c) ultrasonic aspirator (for tumors)
3. anesthesia: pre-op consult for “awake craniotomy” & skull block
4. consult neurology or neuropsychology to be available during surgery for intra-op neurologic testing for “awake craniotomy”
5. EEG techs to perform intra-op EEG and provide brain stimulator
6. post-op: ICU
7. consent (in lay terms for the patient – not all-inclusive):
 - a) procedure: surgery on the brain to be performed with periods where the patient will be woken up for testing, (plus whatever else is planned, e.g. removal of tumor, removal of seizure source...)
 - b) alternatives: the same surgery under general anesthesia, nonsurgical management, (for some diagnoses, e.g. tumor, radiation therapy)
 - c) complications: (usual craniotomy complications: stroke, bleeding, coma, death, infection, seizures), difficulty accurately mapping the desired areas of the brain

Indications

1. surgery in eloquent brain (near motor strip (Brodmann’s area 4 in □ Fig. 1.1) or speech/language centers (Wernicke’s & Broca’s areas)) or thalamus, including tumors and epileptic foci
2. removal of brainstem tumors
3. some seizure surgery to look for seizure focus

Contraindications to awake craniotomy

1. patient unlikely to be able to cooperate: very young or very elderly patients, confused patients, those with significant speech deficits already present or language barrier

Patient counselling pre-op

Patients need to be aware of what the sequence of events will be and what will be expected of them. It may be helpful to have them practice reading some typical material that will be used in the O.R.

Patients over age ≈ 40 usually need reading glasses to see written material, and they should have their own available in the O.R., although the temples (earpieces) usually can't be accommodated. The patient should be advised that there may be some pain involved.

Patient positioning for surgery

Significantly more time must be spent on patient positioning to ensure that they will be as comfortable as possible without moving. Extra padding is employed. Access to the patient's face is necessary for the anesthesiologist and the neurophysiologist.

Typical sequence for anesthesia

See reference.²⁶

- 1. in the pre-op holding area, load with Precedex® (dexmedetomidine) 0.5 mcg/kg IV over 20 minutes followed by intra-op infusion at 0.4–1.0 mcg/kg/hr
- 2. induction of anesthesia utilizes propofol 3 mg/kg IV followed by laryngeal mask airway (LMA) placement
- 3. skull block⁴: injection of local anesthetic (e.g. 30 ml of 0.5% bupivacaine) to permit the skin incision and also rigid head fixation with pins (as required for image navigation devices, and situations where no head movement can be tolerated during surgery) without pain at the time of the wake-up. Injection at 4 regions on each side as shown in □ Fig. 93.2:

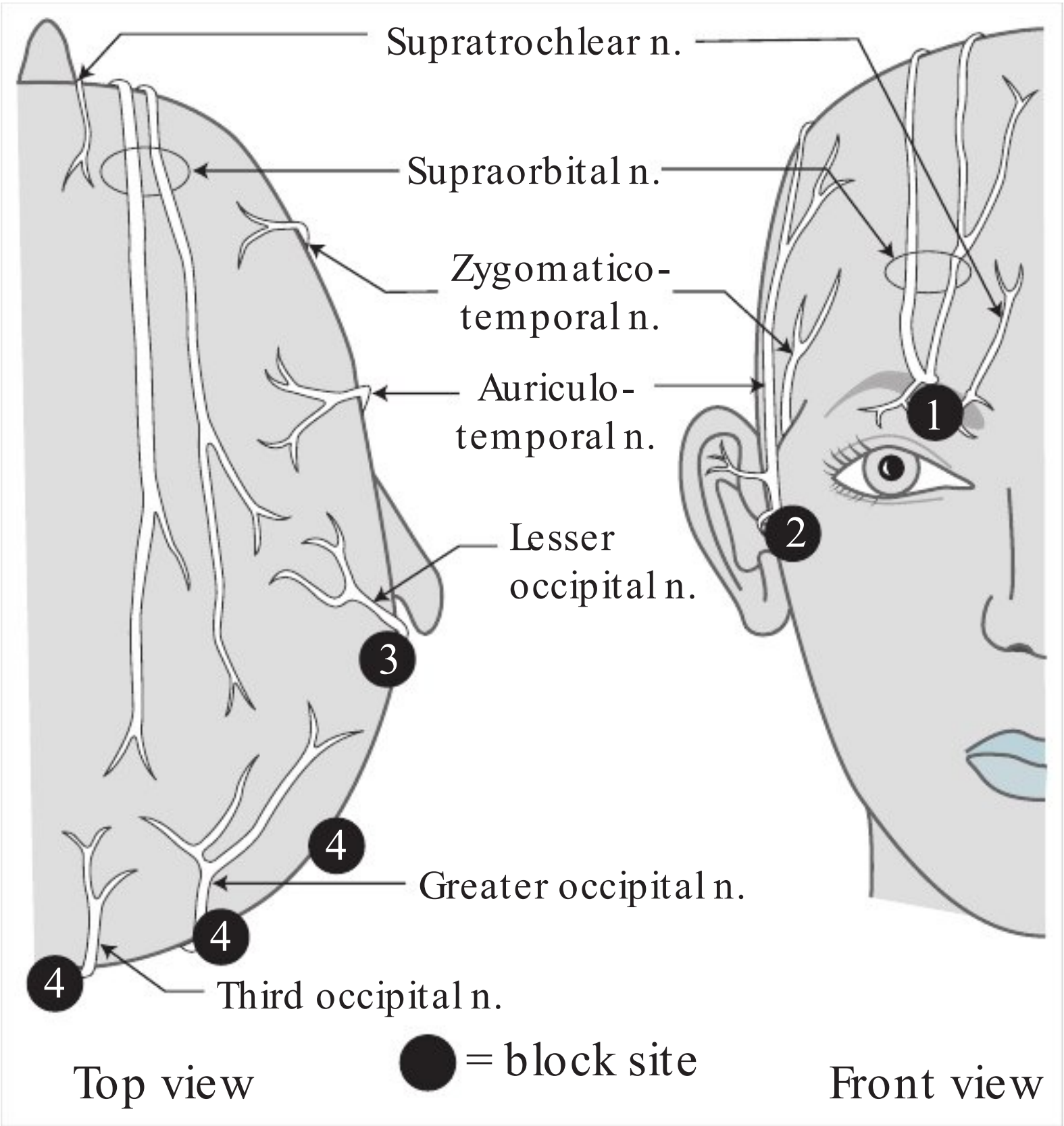


Fig. 93.2 Infiltration sites for skull block

- supraorbital & supratrochlear nerves: 2 ml injected 1.5 cm above the supraorbital foramen above the medial third of the orbit. NB: if you are going to use surface matching to register the patient for image guidance (e.g. BrainLab or Stealth), injecting here may deform the skin and affect the registration accuracy. Consider injecting a lower volume of higher concentration agent (e.g. 2%lidocaine)
 - auriculotemporal nerve: 5 ml injected 1.5 cm anterior to the tragus. □ Caution: to avoid anesthetizing the facial nerve, inject just deep to the subcutaneous tissue
 - postauricular branches of the greater auricular nerve: 2 ml 1.5 cm posterior to the antitragus
 - greater, lesser & third occipital nerves: inject 5 ml with a 22 gauge spinal needle at the mastoid process and proceed along the nuchal ridge until the midline is reached
4. start inhalational anesthesia with 0.5 MAC desflurane with the patient breathing spontaneously while the scalp incision, craniotomy, and dural opening are performed (the dura is pain sensitive, the brain is not)
 5. as the dural opening is begun, the desflurane is turned off and a remifentanyl infusion of 0.1–0.2 mcg/kg/min IV is started
 6. by the time the dural opening is completed, the desflurane has usually worn off and the LMA can be removed
 7. remifentanyl is then titrated for pain control
 8. neurophysiologic testing can usually be performed at this time (e.g. below)
 9. the operation may often be carried to completion with the patient awake, although once the intracranial part of the operation is completed, more pain relief may be desired and general anesthesia may be needed for pain control or agitation (LMA may suffice here)

93.6.4 Speech mapping

General information

Typical settings for a constant current generator using a bipolar electrode are shown in □ Table 93.1. If a voltage based unit is used, start at 1 volt and increase.

Techniques for speech mapping

There are numerous methodologies. One protocol:

1. requires awake craniotomy
2. once temporal lobe is exposed, a recording electrode strip is place on the brain surface
3. using a bipolar stimulator, start with a low current (e.g. 2 mA) and begin stimulate an area of cortex for 3–5 seconds, and observe for afterdischarges (akin to a focal seizure) on the recording strip. If no afterdischarges, increase the current in 2 mA increments up to a maximum of ≈ 10 mA. If afterdischarges occur, back off by 1–2 mA and then test that area for speech changes as follows
4. stimulate cortex while patient names objects shown on picture cards (automatic verbalization, such as counting, is robust and may persist). Observe for effects ranging from total speech arrest to paraphasic errors
5. repeat the above steps at the next area (first finding threshold for afterdischarges and then stimulating while testing)

Table 93.1 Settings for constant current generator	
Control	Setting ^a
frequency	50–60 Hz
waveform	biphasic square wave
duration	2–4 mS peak-to-peak
mode	repeat
polarity	normal
current	varies between 2–16 mA
^a not all settings are present on all models	

93.7 Cranioplasty

93.7.1 Indications

1. cosmetic restoration of external skull symmetry
2. relief of symptoms due to craniotomy defect (p.1431)
3. protection from trauma (blunt or penetrating) in area of post-craniotomy or posttraumatic skull defect

93.7.2 Timing

In certain cases, no consensus is possible. To avoid the risk of infection, some authors recommend delaying cranioplasty with a “foreign body” (e.g. tantalum mesh) at least 6 months after an open (i.e. contaminated) wound or one that traverses the nasal sinuses. Others perform primary closure at the time of repair of the skull fracture. In “clean” cases (e.g. repair of defect after removing hemangioma of the skull) there is little argument against immediate cranioplasty.

93.7.3 Material

Options for material include:

1. the patient’s own bone if it was previously removed in a sterile fashion and saved either in a sterile manner under refrigeration, or in an abdominal pocket at the time of craniectomy
2. materials that can be formed by the surgeon
 - a) methylmethacrylate: mixed in the O.R., molded to desired shape, and allowed to set before being attached to the skull usually with plates (alternatively, sutures or wire may be used). Setting is an exothermic reaction and ample irrigation and preferably removal from the operative site during the process avoids undesirable transmission of heat to the brain
 - b) mesh: may be made of titanium or tantalum.
3. pre-fabricated custom flaps may be made by a number of commercial vendors using thin-cut CT scans to generate computer models of the defect and, if available, to use the contralateral side as a “mirror image” model for the desired shape
 - a) methylmethacrylate
 - b) PEEK (poly-ether-ether-ketone)
 - c) titanium
4. split thickness calvaria

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When foreign material is used, some recommend perforating the flap with a dozen or so drill holes to prevent the accumulation of fluid (either underneath the flap, or between the flap and the skull). This cannot be done with tantalum plates.

Localizing levels in spine surgery

Identifying the correct level in spine surgery may be extremely challenging in certain situations. With the proliferation of minimally invasive spine techniques and the associated reduction in the structures that are directly visualized, the reliance on intraoperative imaging to determine the spinal level has increased.

Potential pitfalls which increase the chances of error, including:

1. pre-op, pathology is usually identified on MRI, and there are problems in translating MRI images to imaging available in the O.R.
 - a) thoracic lesions: pre-op MRI usually counts from the top (C2) down, and in surgery it is often necessary to count from the bottom up, the count could be off if there are not 5 lumbar vertebra and 12 pairs of ribs
 - b) lumbar spine: a well developed S1–2 disc (so-called lumbarized S1) or an L5 vertebra fused to the sacrum (sacralized L5) can confuse the count
2. not all patients have 12 ribs, or 5 “lumbar” vertebrae. In the modal (most common) human spine, there are 24 presacral vertebrae, however some individuals have 23 and others have 25 (variations include: 11 or 13 rib bearing vertebrae, or a lumbosacral transitional vertebrae; the terminology of a “lumbarized S1 vertebrae” or a “sacralized L5 vertebrae” is imprecise and confusing). A HLD at the ultimate disc space (usually L5-S1) most often impinges on the 25th nerve root (however, in the variant cases, it may actually impinge on the 24th or 26th root)²⁷

3. patients may have variant or ambiguous anatomy (e.g. a well-developed S1–2 disc space, an enlarged L1 transverse process which mimics a rib)
4. some “landmarks” used for localizing levels are unreliable or changeable
5. plain radiographs (and fluoroscopy) has difficult imaging the upper thoracic and sometimes the lower cervical spine
 - a) on lateral imaging, the shoulder often obscure lower cervical/upper thoracic levels
 - b) on AP imaging, the pronounced kyphosis of this region requires cranio-caudal angulation of the x-ray beam, which throws off imaging at other levels
6. spinous processes of lumbar and especially thoracic levels are below the corresponding VB
7. changes may occur between the time of the pre-op imaging and the surgery

Aids in determining spinal levels:

1. image guided systems, when available
2. some O.R. suites have intra-op MRI or CT (or CT-like modalities, e.g. O-arm™ by Medtronic or ARCADIS® C-arm by Siemens) or image guided spine technology
3. □ pre-op plain x-rays: lumbar (for lumbar pathology) and lumbar + thoracic (for thoracic pathology) to verify there are 12 thoracic vertebrae and 5 lumbar
4. on lateral lumbar spine x-rays, the top of the iliac crests are even with the L4 spinous process or the L4–5 interspinous space
5. on sagittal MRI there are generally no numbering cues, but on axial MRI, the sacral ala are reliably identifiable and this can be used to identify L5-S1 disc space
6. counting methods (if possible, using more than 1 method is highly recommended)
 - a) counting up from T12 or L5 on fluoro: be sure there are 12 ribs. You can “bridge” from lumbar or lower thoracic spine to higher thoracic levels using an instrument on the patient as a marker, or you can count up to one level (e.g. T9) and then while a hemostat is placed at this level on the fluoro screen, under real-time fluoro the machine is slowly moved up the spine and the hemostat is moved along with T9 (to count up from L5, verify there are 5 lumbar (i.e. non-rib bearing) presacral vertebrae). Radiation safety: avoid live fluoro as much as possible
 - b) AP view: starting at T12 (lowest rib) or from L5
 - c) lateral view: starting at the L5 and counting up
 - d) counting down from T1 (first rib) on AP fluoro: the fluoro machine may need to be angled caudally from the anterior position because of the thoracic kyphosis. Sometimes counting pedicles helps
 - e) by palpation: with thoracotomy, in the upper T-spine you can palpate the ribs from the inside from T1 and count down. The rib inserts at the upper end of the thoracic vertebra near the junction with the VB above (e.g. the T5 rib joins T5 close to the T4–5 disc space)

93.8 Bone graft

93.8.1 Use of bone graft extenders/substitutes as an adjunct to fusion

Practice guideline: Bone graft extenders and substitutes

Level I²⁸: autologous bone or recombinant human bone morphogenic protein (rhBMP-2) bone graft substitute is recommended in the setting of an ALIF in conjunction with a threaded titanium cage

Level III²⁸:

- rhBMP-2 in conjunction with hydroxyapatite and tricalcium phosphate may be substituted for autograft in some cases of posterolateral fusion
- calcium phosphate is recommended as a bone graft extender, especially when combined with autologous bone

93.8.2 Assessing surgical lumbar fusion

See Practice guideline: Radiographic assessment of fusion (p.1438).

Practice guideline: Radiographic assessment of fusion

Level I²⁹: static x-rays alone are not recommended

Level II²⁹:

- in the absence of rigid instrumentation, lack of motion between vertebrae on lateral flexion/extension x-rays is highly suggestive of successful fusion
- □ technetium-99 bone scanning is not recommended

Level III²⁹: radiographic techniques, often in combination, may be used when failed lumbar fusion is suspected, including: static and flexion/extension x-rays, CTscan

Practice guideline: Correlation between fusion and outcome

Level III³⁰: the correlation between fusion and clinical outcome is not strong, and in any given situation fusion status may be unrelated to outcome

93.8.3 Bone graft properties

General information

For spine fusions, components of bone graft that are important for fusion:

1. osteoinduction: recruitment of mesenchymal cells and the stimulation of these cells to develop into osteoblasts and osteoclasts
2. osteogenesis: formation of new bone by host or graft mesenchymal stem cells transformed into osteoblasts
3. osteoconduction: the structure of the graft that acts as a scaffold upon which new bone and blood vessels form
4. mechanical stability: the structural anatomical biomechanical support provided, e.g. following discectomy, corpectomy or resection of vertebral tumor

□ Table 93.2 summarizes the properties of various bone graft materials (adapted^{31,32,33}). See the sections that follow for more details.

Table 93.2 Characteristics of bone graft materials ^a (see text for details)				
Material	Mechanical stability	Osteogenic	Osteo-inductive	Osteo-conductive
Cancellous autograft	±	++++	++	++++
Cortical autograft	+++	+	+	+
Vascularized autograft	+++	+++	++	+++
Allograft	+	–	±	+
Bone marrow aspirate	–	+	±	+
Demineralized bone matrix (DBM)	–	–	+	+
Bone morphogenic protein (BMP)	–	–	++++	–
Collagen	–	–	–	–
Ceramics	+	–	–	+++
^a – no effect, ± minimal or no effect, + mild, ++ moderate, +++ strong, ++++ very strong effect				

Autograft

Common donor sites: iliac crest, rib,³⁴ fibula, bone removed during decompression. Characteristics:

1. PROS: no histocompatibility or disease transmission issues
2. CONS:
 - a) persistent post-op donor site pain: occurs in as many as 34% of patients (the severity of which was graded as “unacceptable” in 3%)³⁵
 - b) increased surgical risks of:
 - blood loss
 - wound infection
 - fracture
 - cosmetic deformity
 - increased operative time to procure
 - numbness from nerve injury (e.g. cluneal nerves see below)
 - hematoma
3. subtypes
 - a) cancellous bone: provides all graft components except mechanical stability
 - b) cortical bone:
 - provides superior and immediate mechanical strength
 - has diminished osteoinduction and osteoconduction capacity
 - c) corticocancellous bone: e.g. tricortical iliac crest wedge. Contains all bone graft components
 - d) vascularized autograft:
 - technically challenging
 - best suited for areas that are scarred, irradiated, or that span long segments
 - e) autologous bone marrow:
 - source of osteoprogenitor cells and osteoinductive substrates
 - diminished donor site risks
 - no osteoconductive nor structural properties

Allograft

Acquired through organ procurement agencies. Primarily frozen or freeze dried. Donor sites include: ilium, tibia, fibula, femur, rib.

1. PROS: eliminates risks associated with harvesting autograft
2. CONS:
 - a) small but real risk of disease transmission
 - b) provide only osteoconduction (lacks osteoinduction and osteogenesis)
 - c) availability may vary from time-to-time
3. subtypes
 - a) tricortical block, bicortical plug, or unicortical dowel
 - b) corticocancellous: matchsticks, crushed
 - c) cancellous: cubes, block, crushed, bone powder
4. uses: allografts are acceptable for structural grafts such as in anterior spinal interbody fusion, where compressive forces are applied to the graft. However, for onlay grafts such as for posterior cervical fusion, the lack of osteoinductive and osteogenetic properties is a critical shortcoming

Deminerlized bone matrix (DBM)

Prepared by acid extraction, reducing antigenicity, but preserving some osteoconductive and variable osteoinductive properties.

1. available as putty, gel, chips, granules or powder
2. primarily used as an adjunct to other grafting materials
3. CONS:
 - a) increased cost
 - b) variable efficacy between preparations and batches of the same preparation
 - c) no mechanical or structural properties

Bone morphogenic proteins (BMP)

(AKA bone morphogenetic proteins). Biological compounds that induce the transformation of mesenchymal stem cells into osteoblasts (osteoinduction) with the potential to induce ectopic bone formation. There are ≈ 20 different proteins from the transforming growth factor- β family. Produced using recombinant DNA technology.

1. a carrier matrix is required to retain the soluble factor at the graft site (i.e. to prevent the BMP from diffusing into adjacent tissues, thereby reducing the desired effect and possibly inducing bone growth at undesired foci)
2. FDA approved in U.S. only for ALIF. Other uses are “off label”
3. available preparations: rhBMP-2 (Infuse® by Medtronic)
4. PROS: increases fusion rates
5. CONS:
 - a) expensive
 - b) ectopic bone formation, bone resorption (so-called osteolysis) or remodelling at the graft site³⁶
 - c) in anterior cervical spine surgery: neck swelling with airway compromise, hematoma, painful seroma³⁶

Collagen

Used primarily as a carrier for other osteoinductive, osteoconductive, or osteogenetic materials and as a composite with other graft extenders

1. PROS: contributes to vascular ingrowth, mineral deposition, and growth factor binding
2. CONS:
 - a) minimal structural support
 - b) potential immunogenicity

Ceramics

Includes tricalcium phosphate, calcium carbonate & hydroxyapatite.

1. PROS: no risk of disease transmission
2. CONS: only recommended for use as bone graft extenders (i.e. must be combined with autograft, bone marrow aspirate, BMP...)

93.8.4 Bone graft procurement

Iliac crest

Anterior iliac bone graft

Should be obtained at least $\approx 3\text{--}4$ cm lateral to the anterior superior iliac spine (ASIS) to avoid the lateral femoral cutaneous nerve and to reduce the risk of avulsion fractures of the remaining ilium. When a tricortical graft is taken, keep the dissection in the subperiosteal plane and avoid electrocautery on the medial (inner) surface when detaching the iliacus muscle to avoid injury to the ilioinguinal, iliohypogastric and lateral femoral cutaneous nerves.

Posterior iliac crest bone graft

May be used to obtain corticocancellous strips or plates for onlay bone grafts, or tricortical grafts which may be used as strut grafts or for C1–2 arthrodesis.

Posterior bone grafts (□ Fig. 93.3) are taken from the medial 6–8 cm of the iliac crest to avoid the superior cluneal nerves (which cross the posterior iliac crest ≈ 8 cm lateral to the posterior superior iliac spine) with resultant buttock numbness or the development of painful neuromas. A vertical incision just medial to the posterior superior iliac spine usually works well.

The spine may sometimes be found on corpulent patients by locating the “dimple of Venus” (fossae lumbales laterales – indentation sometimes visible superior to the gluteal cleft, directly superficial to the sacroiliac joint) and incising slightly lateral to it. Avoid mistaking the sacrum for the iliac spine.

The gluteus maximus is dissected off the lateral surface subperiosteally. To avoid fractures extending into the iliac crest, a wide osteotome should be used to create a “stop cut”; alternatively, a sagittal saw may be used. Avoid penetration through the inner cortical surface of the crest so as not to enter the pelvis and possibly cause an intra-abdominal hematoma. Another potential complication is fracture extension into the greater sciatic notch with possible injury to the gluteal arteries and sciatic nerve among others. Once the graft is removed and cancellous bone is gouged out, the exposed bone surfaces should be waxed and closed system drainage should be used to reduce the risk of local hematoma formation.

Fibula

Autogenous fibular graft provides a high arthrodesis rate,³⁷ but may be associated with significant morbidity, and so may be best reserved for salvage procedures.³⁸ Preserve the proximal fibular head

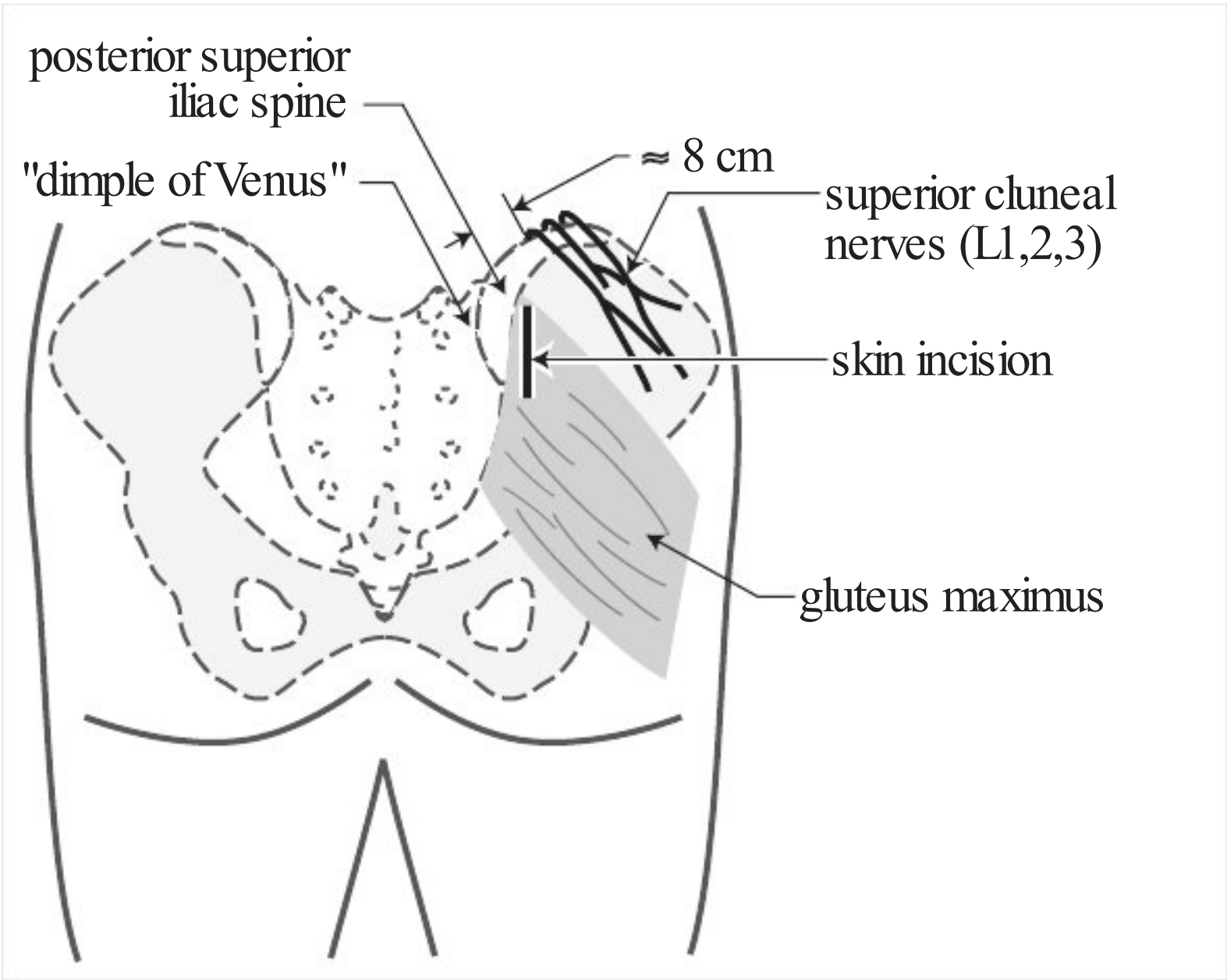


Fig. 93.3 Posterior iliac crest bone donor site

to avoid injury to the peroneal nerve. At least 7 cm of distal fibular should be maintained to preserve ankle stability.³⁴

93.9 Stereotactic surgery

93.9.1 General information

The term stereotactic (Greek: stereo=3-dimensional, tactic=to touch) surgery was initially used in animals, and was based on atlases of three dimensional coordinates compiled from dissections. The term was then used for surgery performed in humans, usually for thalamic lesioning to treat Parkinsonism (see Surgical treatment of Parkinson's disease (p.1524)), where the target site to be lesioned was located relative to landmarks with intraoperative pneumoencephalography or contrast ventriculography. Use of this procedure fell off dramatically in the late 1960's with the introduction of L-dopa for Parkinsonism.³⁹

Current techniques would be more appropriately termed image-guided stereotactic surgery. In the first part of the procedure, a CT scan or MRI (or occasionally, angiogram) is performed. For increased precision, some sort of "fiucial" markers or frame is attached to the patient's head during this image acquisition phase. Acceptable accuracy can often be obtained using high resolution thin cut imaging slices (usually with a 0 angle of the gantry) and then surface matching algorithms in the guidance system will match the pre-op CT/MRI to the patient's head.

The second part of the procedure usually takes place in an operating room. The patient is "reiistered" with the pre-op images, and then tracking cameras follow the movement of instruments with appropriate attachments to show in "real-time" the location of the instrument with respect to the pre-op image. An important limitation to be aware of is the fact that the pre-op images are "historical" and are not updated as the surgical procedure alters the anatomy of the patient. For example, even the administration of mannitol can cause brain shifts that may cause the target of the surgery to move away from its pre-op location by several millimeters.

93.9.2 Indications for stereotactic surgery

1. biopsy (also, see below)
 - a) deeply located cerebral lesions: especially near eloquent brain
 - b) brain stem lesions: may be approached through the cerebral hemisphere⁴⁰
 - c) multiple small lesions (p.333) (e.g. in some AIDS patients)
 - d) patient medically unable to tolerate general anesthesia for open biopsy
2. catheter placement
 - a) drainage of deep lesions: colloid cyst, abscess
 - b) indwelling catheter placement for intratumoral chemotherapy
 - c) radioactive implants for interstitial radiation brachytherapy⁴¹
 - d) shunt placement: for hydrocephalus (rarely used) or to drain cyst
3. electrode placement
 - a) depth electrodes for epilepsy
 - b) “deep brain stimulation” for chronic pain (requires electrophysiologic stimulation)
4. lesion generation
 - a) movement disorders: Parkinsonism (p.1526), dystonia, hemiballismus
 - b) treatment of chronic pain
 - c) treatment of epilepsy (rarely used)
5. evacuation of intracerebral hemorrhage
 - a) using an Archimedes’ screw device^{42,43}
 - b) with adjunctive urokinase^{44,45} or recombinant tissue-plasminogen activator (p.1344)⁴⁶
6. stereotactic “radiosurgery” (p.1564) (see Stereotactic radiosurgery & radiotherapy)
7. to localize a lesion for open craniotomy (e.g. AVM,⁴⁷ deep tumor)
 - a) using a ventricular-type catheter
 - b) using a blunt biopsy needle or introducer⁴⁸
 - c) systems using visible light laser beam for guidance
8. transoral biopsy of C2 (axis) vertebral body lesions⁴⁹
9. “experimental” or unconventional applications
 - a) stereotactic clipping of aneurysms⁵⁰
 - b) stereotactic laser surgery
 - c) CNS transplantation⁵¹: e.g. for Parkinsonism (p.1525)
 - d) foreign body removal⁵²

93.9.3 Stereotactic biopsy

This section presents information regarding stereotactic brain biopsy (SBB) in general. For SBB in specific conditions, see the index entry for that condition. Biopsy may be through a small “cookie cutter” craniotomy, or through a smaller burr hole with a needle. Although it may be performed under local or general anesthesia, it is more commonly done under general.

Contraindications

1. coagulation disorders
 - a) coagulopathies: bleeding diatheses, iatrogenic (heparin or coumadin)
 - b) low platelet count (PC): PC<50,000/ml is an absolute contraindication, it is desirable to get the PC≥100,000
2. inability to tolerate general anesthesia and to cooperate for local anesthesia

Yield

The yield rate (i.e. the ability to make a diagnosis from a SBB) reported in large series in the literature ranges from 82–99% in nonimmunocompromised (NIC) patients, and is slightly lower in AIDS patients at 56–96% Higher yield rates in AIDS may result from improved surgical technique and histologic evaluation.⁵³

The yield rate is higher for lesions that enhance with contrast on CT or MRI (99% in NIC patients) than with lesions that do not enhance (74%).⁵⁴

Complications

The most frequent complication is hemorrhage, although most are too small to have clinical impact. The risk of a major complication (mostly due to hemorrhage) in NIC patients ranges from 0–3% (with most <1%), and 0–12% in AIDS.⁵⁴ Higher complication rates seen in AIDS patients in some series may

be due to reduced platelet count or function, and to vessel fragility in primary CNS lymphoma. In NIC patients, multifocal high grade gliomas had the highest complication rate.
Infection is an infrequent complication with needle biopsy.

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94 Specific Craniotomies

94.1 Posterior fossa (suboccipital) craniectomy

94.1.1 Indications

To gain access to the cerebellum, cerebellopontine angle (CPA), to one vertebral artery, posterior brainstem, fourth ventricle, pineal region, or, using extreme lateral posterior fossa approach to the antero-lateral brain stem. See paramedian (p.1447) and midline (p.1450) suboccipital craniectomies for details.

94.1.2 Position

Options

Position options include:

1. sitting position: see below
2. lateral oblique (p.1446): patient three-quarters oblique (almost prone),
3. semi-sitting
4. supine with shoulder roll, head almost horizontal
5. prone
6. Concorde position: prone, thorax elevated, neck flexed and tilted away from the side on which the surgeon will be standing

Sitting position

Used less frequently than in the past because of associated complications and acceptable alternative positions (except for some specific circumstances). However, some experts feel that the risks of the sitting position have been greatly overstated.¹

Advantages

1. improved drainage of blood and CSF out of surgical site
2. enhanced venous drainage which helps reduce venous bleeding and also ICP
3. easy ventilation due to unencumbered chest
4. patient's head may be kept exactly midline, aiding operator orientation, and reducing risk of kinking of vertebral arteries

Disadvantages/risks

1. possible air embolism (see below)
2. fatigue of operators hands
3. increased surgical risks from placement of CVP catheter (required to treat possible AE): e.g. pneumothorax with subclavian vein catheterization, thrombosis
4. risk of post-op hematoma at the operative site may be increased since potential venous bleeders may remain occult while the patient is sitting, but may manifest when patient returns to a horizontal position post-op. However, one study found no such increased incidence²
5. risk of post-op subdural hematoma: 1.3% of p-fossa cases³
6. possible brachial plexus injury: prevent this by not allowing patient's arms to hang at the side. Instead, fold them across abdomen
7. midcervical quadriplegia^{4,5}: presumably due to flexion myelopathy.^{6,7,8} The combination of the sitting position with hypotension⁹ or neck flexion with possible compression of the anterior spinal artery, ± cervical bar, and elevation of the head thus reducing the arterial pressure may all contribute
8. sciatic nerve injury (piriformis syndrome)¹⁰: prevent this by flexing patient's knees (reduces tension on sciatic nerve)
9. extent of post-op pneumocephalus is more pronounced, and may increase the risk of tension pneumocephalus¹¹; see Pneumocephalus (p.887)
10. venous pooling of blood in the LEs under anesthesia may cause relative hypovolemia and should be counteracted by binding the LEs prior to positioning
11. decreased cerebral blood flow due to lower hemodynamic arterial pressure¹²

Air embolism (AE): A potentially fatal complication of any operation when an opening to air occurs in a non-collapsible vein (e.g. diploic vein or a dural sinus) when there is a negative pressure in the

vein (e.g. when the head is elevated above the heart).¹³ Air is entrained in the vein and can become trapped in the right atrium of the heart which may impair venous return causing hypotension. May also produce cardiac arrhythmias. Paradoxical air embolism can occur in the presence of a patent foramen ovale¹⁴ or pulmonary AV fistula, and may produce ischemic cerebral infarction.

Greater negative pressures occur in the sitting position due to the extreme elevation of the head, but AE can occur in any operation with the head elevated higher than the heart. Incidence: a wide range has been quoted in the literature, and depends on the monitoring method used: $\approx 7\text{--}25\%$ incidence with the sitting position using Doppler monitoring is an estimate.³

For operations with a significant risk of AE, a right atrial CVP line is recommended (to aspirate air), and monitoring for air embolism; options include: transesophageal echo (the most sensitive), precordial Doppler monitoring. (Although technically the risk of air embolism includes any case where the head is higher than the right cardiac atrium, practically it is limited to cases where the head of the bed is $\approx >30^\circ$ which is mostly limited to the sitting position for posterior fossa tumors.)

Diagnosis and treatment:

AE should be suspected in any operative case in which the surgical site is higher than the heart when there is any unexplained hypotension or decrease in EtCO₂.¹⁶

- transesophageal echocardiography (TEE). Bubbles can be seen on the 2D echo display
pros: considered the most sensitive monitoring modality
cons: significant false positive rate, expensive, invasive, requires experience and vigilance
- precordial doppler U/S: probe may be placed over 2nd to 4th intercostal space either to right or left of sternum, or posteriorly between the scapula and spine. AE is heralded by a change in sonic intensity and character at first by a superimposed irregular high-pitched swishing sound, and then as more air is entrained so called “mill wheel” or machinery sounds dominate
pros: the most sensitive of the non-invasive techniques
cons: difficult in morbidly obese patients and in certain patient positions (e.g. prone or lateral), interference from other sounds in the OR, requires vigilance

The earliest indication of AE may be a rise in the end-tidal nitrogen (requires mass-spectrometer on monitor), then a fall in the end tidal pCO₂ occurs. Machinery sounds in the precordial Doppler also suggest AE. Hypotension may develop. Measures shown in □ Table 94.1 should be immediately instituted.

Lateral oblique position

AKA “park bench” position.

- Axillary roll for the down side arm (see □ Fig. 94.1) (or, position the patient so that the down side arm extends over the edge of the table and is held in place by a sling formed by the Mayfield table attachment with copious padding)
- Upper arm supported on pillows or towels (avoid using a Mayo stand which restricts the ability to laterally tilt the OR table during surgery)
- Adhesive tape to gently pull down on the upper shoulder
- Bring the patient’s back as close to the side edge of the table as possible (usually limited by the travel of the head clamp) to bring the patient closer to the surgeon
- Elevate thorax 10-15°
- Tilt the vertex of the head towards the floor (see below)
- Optional spinal drainage (usually for large tumors)
- Pillow between the legs
- Secure patient with adhesive tape over pads so the table can be “airplaned” (rolled) during the operation

Table 94.1 Treatment for air embolism	
1.	find and occlude site of air entry, or else rapidly pack wound with sopping wet sponges/laps and wax bone edges
2.	lower patient’s head if at all possible (30° or less from horizontal)
3.	jugular venous compression (bilateral best; second choice: right only)
4.	rotate patient LEFT side down (attempt to trap air in right atrium)
5.	aspirate air from right atrium via CVP catheter
6.	ventilate patient with 100% O ₂
7.	discontinue nitrous oxide if used (may expand AE) ¹⁵
8.	use pressors and volume expanders to maintain BP
9.	PEEP is ineffective in preventing or treating AE; may increase the risk of paradoxical AE ¹³

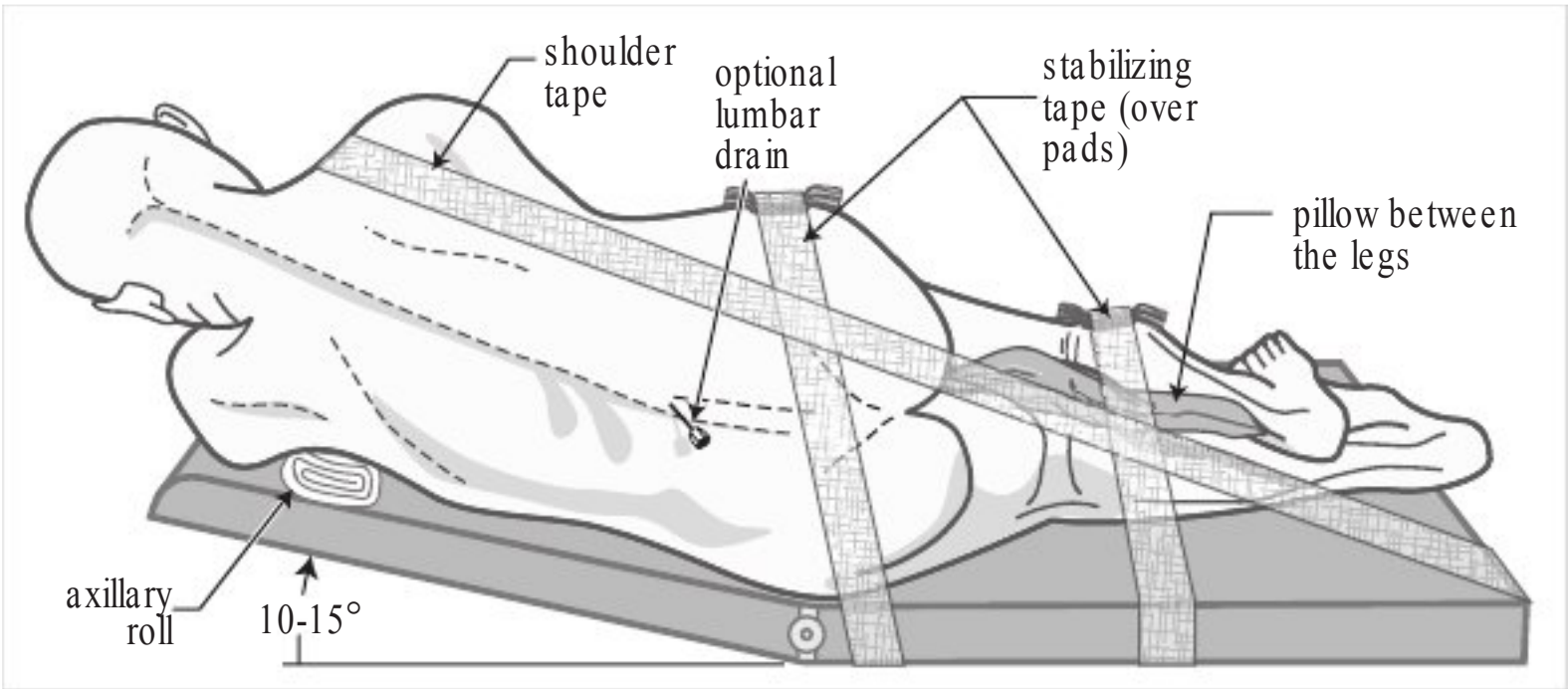


Fig. 94.1 Lateral oblique (“park bench”) position

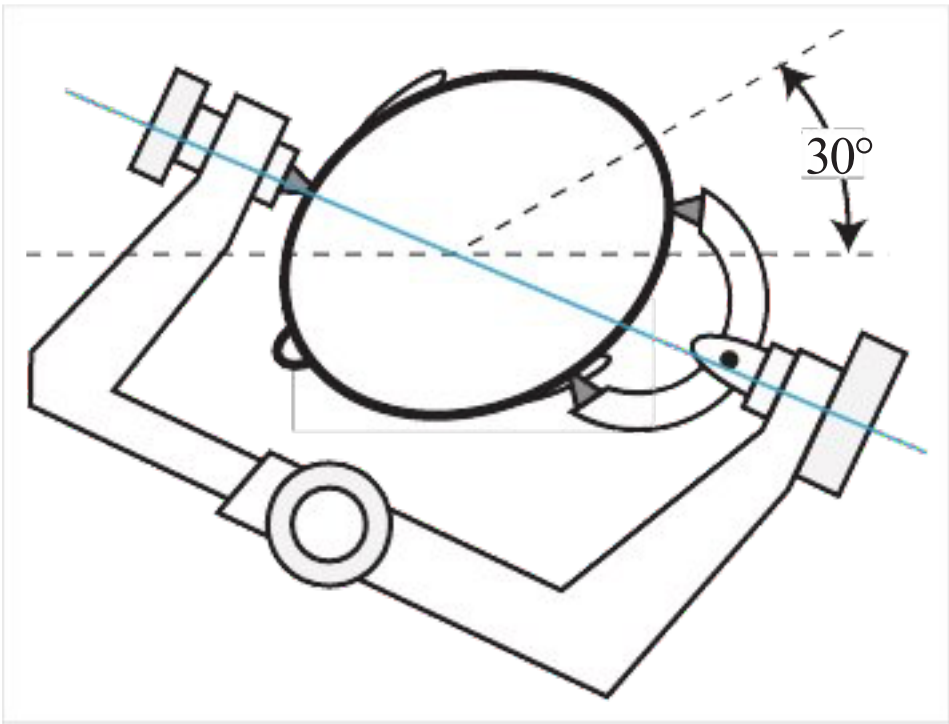


Fig. 94.2 Position of head and headholder for right suboccipital craniectomy (looking down on top of patient’s head)

For access to the porus acusticus or more caudally (e.g. for vestibular schwannomas; not necessary for microvascular decompression for trigeminal neuralgia).

Get the shoulders out of the way by flexing the neck as much as possible while maintaining patient airway (aided by use of non-kinking wire-reinforced ET tube, so-called “armored tube”). The upper shoulder is retracted caudally by adhesive tape (avoid excess traction which may injure brachial plexus).

Head positioning

A Mayfield head-clamp is placed with the single pin on the side of the lesion, slightly anterior to a true-lateral on the skull (□ Fig. 94.2.). The head is then rotated 20–30° face-down from the horizontal.

94.1.3 Paramedian suboccipital craniectomy

Indications

- 1. access to the cerebellopontine angle (CPA)
 - a) CPA tumors, including:
 - vestibular schwannoma
 - CPA meningioma
 - epidermoid

- b) microvascular decompression
 - trigeminal neuralgia
 - hemifacial spasm
 - miscellaneous: geniculate neuralgia, glossopharyngeal neuralgia
- 2. lesions of one cerebellar hemisphere:
 - a) tumors: metastases, hemangioblastomas...
 - b) hemorrhage within cerebellar hemisphere
- 3. access to vertebral artery
 - a) aneurysms: PICA, vertebrobasilar junction
 - b) vertebral endarterectomy
- 4. access to antero-lateral brainstem tumors (extreme lateral p-fossa approach)
 - a) foramen magnum tumors, including: chordomas, meningiomas

Position, skin incision, craniectomy, approach...

See list of alternatives (p.1445). See lateral oblique position (p.1446).

Skin incision

Linear (paramedian) incisions

Access to CPA. For microvascular decompressions and small CPA tumors, a linear incision provides adequate exposure and involves less trauma to overlying muscles, and may be easier to get water-tight closure than with midline incision. For all of the following, the linear skin incision is located 5 mm medial to the mastoid notch (a palpable landmark, □ Fig. 94.3):

1. “5–6–4” incision (incision placed 5 mm medial to mastoid notch, extending from 6 cm above notch to 4 cm below). High enough to expose transverse sinus:
 - a) for approach to fifth nerve: microvascular decompression for trigeminal neuralgia
2. “5–5–5” incision (5 mm medial, extending 5 cm up to 5 cm down), used for approach to seventh/eighth nerve complex:
 - a) microvascular decompression for hemifacial spasm
 - b) small vestibular schwannoma
3. “5–4–6” incision (5 mm medial, extending 4 cm up to 6 cm down): used for approach to lower cranial nerves:
 - a) glossopharyngeal neuralgia

“Hockey-stick” incision

Useful for cerebellar hemispheric lesions as well as for larger CPA lesions where getting the muscles out of the way will facilitate maneuvering instruments about the posterior fossa.

Incision is made in the midline starting at $\approx$ C2 spinous process, proceeding superiorly to just above the inion, and then laterally to just beyond the mastoid tip (□ Fig. 94.4). A short optional caudal curve may be made laterally to further remove the muscle from the operative field.

Craniectomy

Landmarks

The location of the inferior margin of the transverse sinus is quite accurately estimated at two finger-breadths above the upper limit of the mastoid notch (usually just above the superior nuchal line). This should be the upper limit of the skull opening.

For microvascular decompression

Craniectomy $\approx$ 2 cm diameter placed in the angle between transverse and sigmoid sinuses.

For small tumors (<2.5 cm)

Craniectomy $\approx$ 4 cm diameter placed in the angle between transverse and sigmoid sinuses.

For large tumors

A larger craniectomy may be needed, the size of which is limited by:

1. transverse sinus superiorly
2. foramen magnum inferiorly (which may be opened as prophylaxis against tonsillar herniation in the event of p-fossa edema post-op)

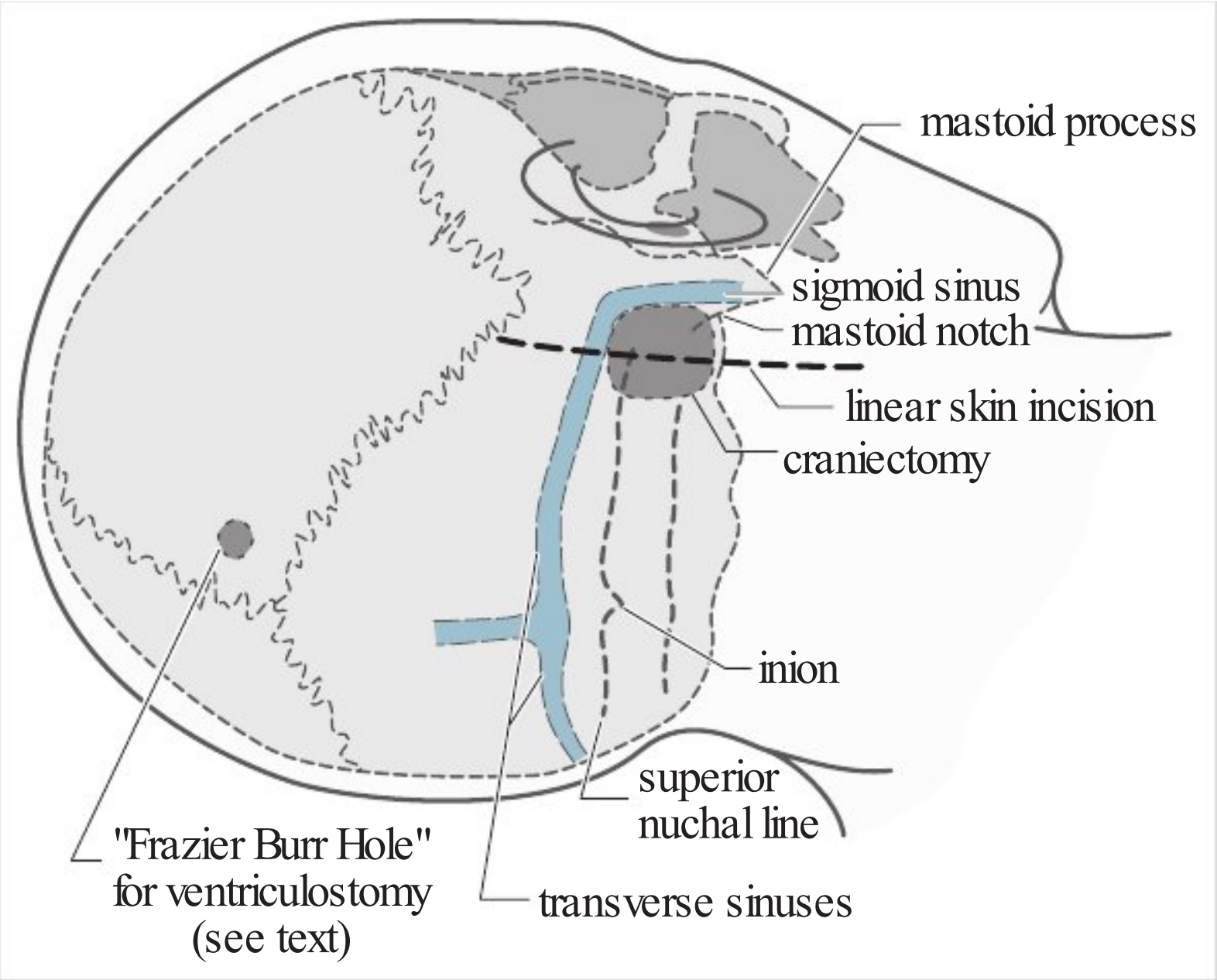


Fig. 94.3 Paramedian suboccipital craniectomy

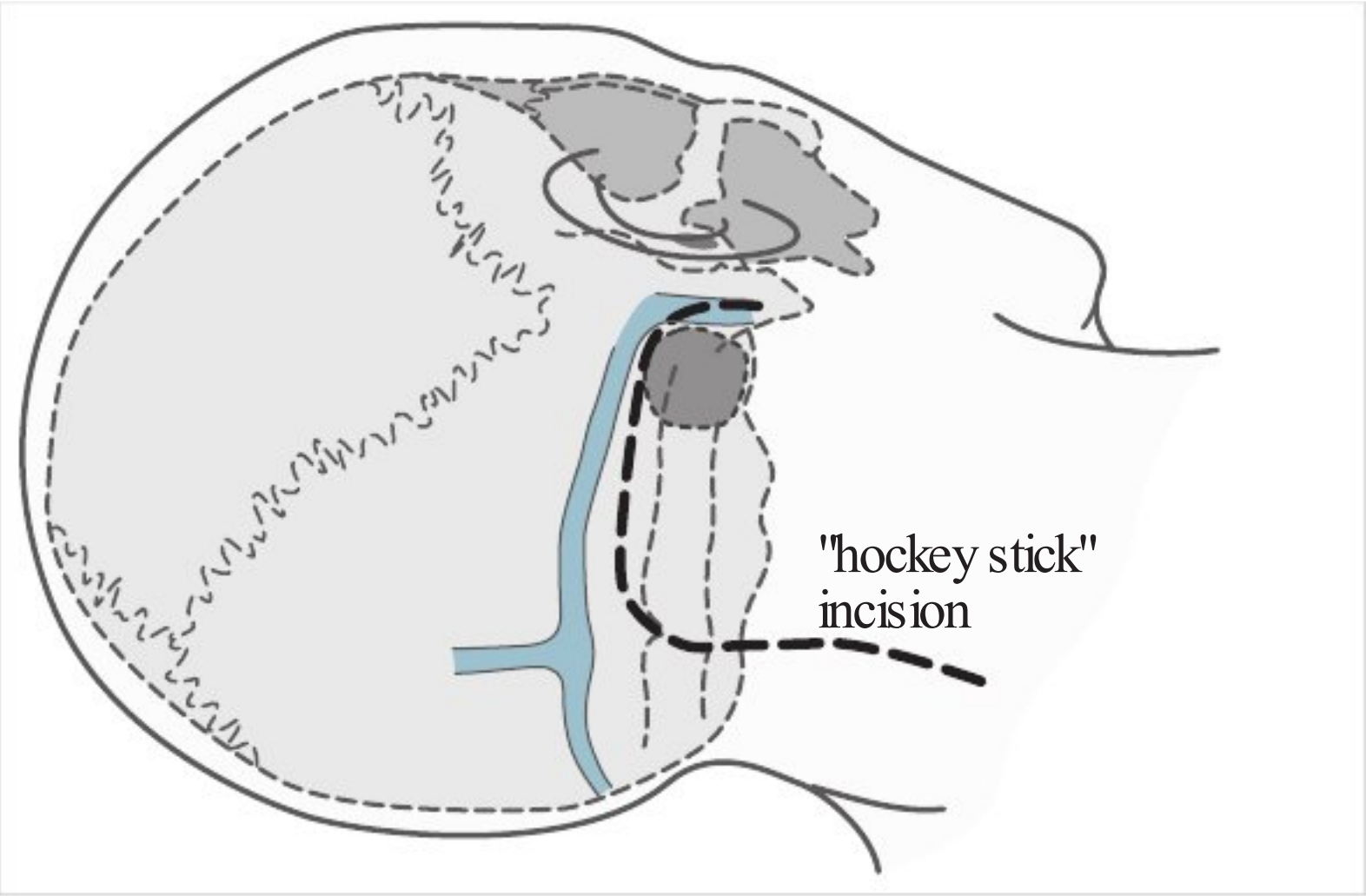


Fig. 94.4 "Hockey-stick" skin incision

3. sigmoid sinus laterally (opening mastoid air cells is acceptable, but to prevent CSF leak, these must be packed with bone wax and muscle (or bone dust from craniectomy¹⁷), and may be covered with reflected dura or fascia)
4. midline medially (unless the tumor extends across the midline)

For approach to lower cranial nerves

(e.g. for glossopharyngeal neuralgia).

Craniectomy is extended inferiorly to $\approx 1/2$ cm above foramen magnum.

Burr hole for emergency ventriculostomy

Optionally placed prophylactic occipital burr hole (Frazier burr hole) usually for intraparenchymal cerebellar tumors or any situation where post-op swelling or hydrocephalus is likely (not commonly used for microvascular decompression or small vestibular schwannomas).

Location: 3–4 cm from midline. In adults, 6–7 cm above the inion¹⁸; in pediatrics, 2–3 cm above the transverse sinus¹⁹ (p. 429) (i.e. ≈ 3 –4 cm above the inion).

See Post-op management (p. 1451) for use.

Approach to the CPA

The angle of approach determines which portion of the posterior fossa is visualized.

1. retracting the cerebellum inferiorly (working in the junction of the tentorium and petrous bone) gives access to the region of the trigeminal nerve, e.g. for microvascular decompression for trigeminal neuralgia
2. medial retraction gains access to the region of the porus acusticus, e.g. for vestibular schwannomas
3. superior retraction gains access to the lower cranial nerves, e.g. for geniculate neuralgia

94.1.4 Midline suboccipital craniectomy

Indications

Access to the midline or both sides of the posterior fossa

1. midline posterior fossa lesions
 - a) cerebellar vermis and paravermian lesions, including: vermis AVM, cerebellar astrocytoma near the midline
 - b) tumors of the fourth ventricle: ependymoma, medulloblastoma
 - c) pineal region tumors
 - d) brainstem lesions: brainstem vascular lesions (e.g. cavernous angioma)
2. decompressive craniectomies
 - a) for Chiari malformation
3. cerebellar tumors: metastases, hemangioblastoma, pilocytic astrocytoma...

Position, skin incision, craniectomy, approach...

See positioning (p. 1446).

Skin/fascia incision

Midline incision from ≈ 6 cm above inion to $\approx$ C2 spinous process. Take the incision a little higher if a Frazier burr hole is to be done (can then utilize the same skin incision). The skin incision should leave the muscles and fascia intact. It is often difficult to place Raney clips on the skin in this region. To facilitate water-tight closure, the fascia is “T’d” at the top, leaving a cuff of tissue on the occiput just above the superior nuchal line.

Craniectomy

Craniectomy implies removal of bone (often piecemeal) with no intention of replacing it. Although craniotomy with replacement of bone flap at end of procedure has been used successfully, there is some concern that if there is post-op swelling, the inelastic bone flap may cause more pressure to be transmitted to the brain stem.

Usually taken down to foramen magnum. For cerebellar hemisphere tumors, many remove the posterior arch of C1 (caution re vertebral arteries on superior aspect of C1).

Approach

A “Y” shaped durotomy is often used. If the lesion has a cystic component, aspiration through a ventricular needle is used to partially decompress it.

94.1.5 Extreme lateral posterior fossa approach

Allows access to antero-lateral region of brain stem. Differs from above in that the skin incision is designed to get the bulk of the skin and muscle flap out of the way.

Key: remove the lip of the foramen magnum as far laterally as possible, best done with a diamond drill.

94.1.6 Cranioplasty for suboccipital craniectomy

Methylmethacrylate cranioplasty as part of the closure following suboccipital crani for vestibular schwannoma reduced the incidence of post-op H/A from 17%to 4%²⁰

94.1.7 Post-op considerations for p-fossa cranis

Post-op check

In addition to routine, the following should be checked:

- 1. respirations: rate, pattern (see Intubation below)
- 2. follow closely for hypertension (see below)
- 3. evidence of CSF leak through wound

Post-op management

Intubation

Post-op intubation for 24–48 hours is sometimes maintained on a precautionary basis: many complications often have respiratory arrest as the initial manifestation (see below), and the patient may deteriorate precipitously from this point. There is a trade-off as the stimulus of the endotracheal tube may exacerbate hypertension and patient agitation, and so sedation is often required, which may obscure the neuro exam and depress respirations. If the patient wakes up extremely well from an uncomplicated p-fossa crani and it is not late at night, most surgeons will extubate.

Hypertension

Hypertension should be avoided at all costs to prevent bleeding from tenuous vessels (e.g. nitropruside should be prepared prior to termination of the operation, and should be hanging and ready to titrate to keep SBP ≥ 160 mm Hg during the reversal of anesthesia and post-op).

Physician should be called for any sudden changes in BP post-op (may indicate elevated pressure in posterior fossa, see below).

Post-op complications

Posterior fossa edema and/or hematoma

In the posterior fossa, a small amount of mass effect can be rapidly fatal due to the paucity of room and the immediate transmission of pressure directly to the brain stem. It can also occlude CSF circulation through the aqueduct and cause acute hydrocephalus with the attendant risk of tonsillar herniation. Increased pressure in the p-fossa is usually heralded by sudden increases in BP or changes in respiratory pattern (pupillary reflexes, level of consciousness and ICP are not affected until late). See

□ Table 94.2 for emergency treatment measures.

Table 94.2 Emergency treatment for p-fossa swelling
□ Rapid intubation, ventricular tap (through previously placed burr hole, if possible, see below), and reoperation is indicated. The wound should be opened immediately wherever patient is (recovery room, ICU, floor...). CT scanning may cost valuable minutes; it is rarely appropriate to delay treatment for this (must be judged on an individual basis).

To expedite ventricular taps, a prophylactic occipital burr hole (Frazier burr hole) is often placed during posterior fossa surgery to permit drainage of CSF from the lateral ventricles in the event of acute hydrocephalus from blockage of the 4th ventricle or aqueduct. If acute hydrocephalus develops (e.g. from a hematoma), an emergent percutaneous ventricular tap with ventricular needle (or, if not available, spinal needle) is performed, passing the needle through the burr hole aiming for the middle of the forehead. In the presence of acute hydrocephalus, CSF should be encountered at a depth of 3–5 cm. NB: this maneuver may provide a few more minutes while preparing for the definitive treatment of re-opening the wound; however, hydrocephalus may not initially be present since it takes some time to develop.

Suboccipital pseudomeningocele

An “internal” CSF fistula. Incidence following suboccipital craniectomy: 8²¹-28%²²

May be asymptomatic, but also may be associated with H/A, nausea/vomiting, local pain/tenderness. Some are soft and compressible, others may be tense.

Indications for operation:

1. external leak (CSF fistula, see below)
2. threatening integrity of incision
3. cosmetic deformity
4. causing symptoms

Treatment options (up to 67% require permanent CSF drainage²³):

1. noninvasive measures: expectant management, fluid restriction, head wrapping, keeping HOB elevated, acetazolamide. Steroids may be used if aseptic meningitis is suspected
2. percutaneous aspiration: “tap and wrap”.^{19 (p 436), 24} Risks introducing bacteria, causing infection
3. direct surgical exploration with multilayer re-closure^{19 (p 436)}
4. lumbar drain: effective only if pseudomeningocele communicates with the subarachnoid space.
 - May produce acute posterior fossa syndrome (H/A, nausea, vomiting, ataxia...) ²¹ especially if the pseudomeningocele doesn’t communicate. Symptoms usually resolve with prompt discontinuation of lumbar drainage.^{21, 22} Other potential complications: vagal nerve palsy, tonsillar herniation, subdural hematoma, kinking of PCA → stroke. Drainage options:
 - a) external drain (temporary)
 - b) lumboperitoneal shunt (permanent)
5. ventricular drainage
 - a) EVD (temporary)
 - b) shunt (permanent)

CSF fistula

Occurs in 5–17% of cases. A potential source of meningitis, thus CSF leak must be treated immediately.

Etiologies: controversial. May include:

1. abnormal CSF hydrodynamics (i.e. hydrocephalus). Maneuvers to stem the leak will likely fail until the CSF is shunted or hydrodynamics normalize
2. poor wound closure: probably blamed more often than it is the actual cause
3. subarachnoid scarring

May be associated with meningitis (aseptic or infectious), multiple operations. Formation may be facilitated by coughing/sneezing, postural changes, one-way ball-valve mechanism due to a tissue flap.

An external CSF leak may occur through:

1. the skin incision
2. via the eustachian tube; see possible routes of egress following suboccipital vestibular schwannoma removal (p.683):
 - a) through the nose (CSF rhinorrhea)
 - b) down the back of the throat
3. the ear (CSF otorrhea) in cases with perforated TM

Treatment:

Initial treatment measures to temporize in the hope that CSF hydrodynamics will normalize and/or that the leak site will scar closed within a few days:

1. elevate the HOB

2. lumbar subarachnoid drainage
3. if the leak occurs through the skin incision:
 - a) reinforce the incision with sutures, e.g. running locked 3–0 nylon after preparation of the skin with antimicrobial and local anesthetic
 - b) alternatively, the incision may be painted with several coats of collodion

If persistent, a CSF fistula requires surgical correction, see CSF fistula (cranial), for general information (p.384), see CSF fistula following suboccipital removal of vestibular schwannoma (p.684).

Fifth or seventh nerve injuries

Causes diminished corneal reflex with potential corneal ulceration; initially managed with isotonic eye drops (e.g. Natural Tears®) q 2–4 hrs & PRN, or with a moisturizing insert (e.g. Lacricert®) q day, and at night with an eye patch or taping eyelid shut.

Miscellaneous

Supratentorial intracerebral hemorrhage has been described, and may result from transient hypertension.²⁵

94.2 Pterional craniotomy

94.2.1 Indications

1. aneurysms
 - a) all aneurysms of anterior circulation
 - b) basilar tip aneurysms
2. direct surgical approach to cavernous sinus
3. suprasellar tumors
 - a) pituitary adenoma (when there is a large suprasellar component)
 - b) craniopharyngioma

94.2.2 Technique

Position, skin incision, craniectomy, approach...

1. supine, ipsilateral shoulder roll if head turned >30° (see below)
2. elevate thorax 10–15°: reduces venous distension
3. flex knees
4. Mayfield 3 pin head-holder: applied between true AP and true lateral (so that it is ≈ horizontal when head is rotated to the necessary position, see □ Fig. 94.5)
5. neck extended 15°: allows gravity to retract frontal lobe away from skull base
6. head rotated from vertical as shown in □ Fig. 94.5

Room arrangement

1. microscope: observer tube to operator's right for either right or left pterional crani

Skin incision

See □ Fig. 94.6. From zygomatic arch 1 cm in front of tragus (to avoid frontalis branch of facial nerve and frontal branch of superficial temporal artery), curving slightly anteriorly, staying behind hairline to widow's peak, optional additional curve beyond midline to aid in skin retraction. Over temporalis muscle, incise skin down to but not through temporalis fascia.

The temporalis muscle may be incised caudal to the skin incision (i.e. closer to zygomatic arch): this minimizes the muscle mass that needs to be retracted inferiorly and yet keeps the scar behind hairline (note: there is an increased risk of frontalis weakness with this technique than if the temporalis muscle is incised in-line with the skin incision).

Craniotomy

There are numerous ways to cross the pterion (the lesser wing of the sphenoid makes this difficult). One method is outlined here, □ Fig. 94.7.

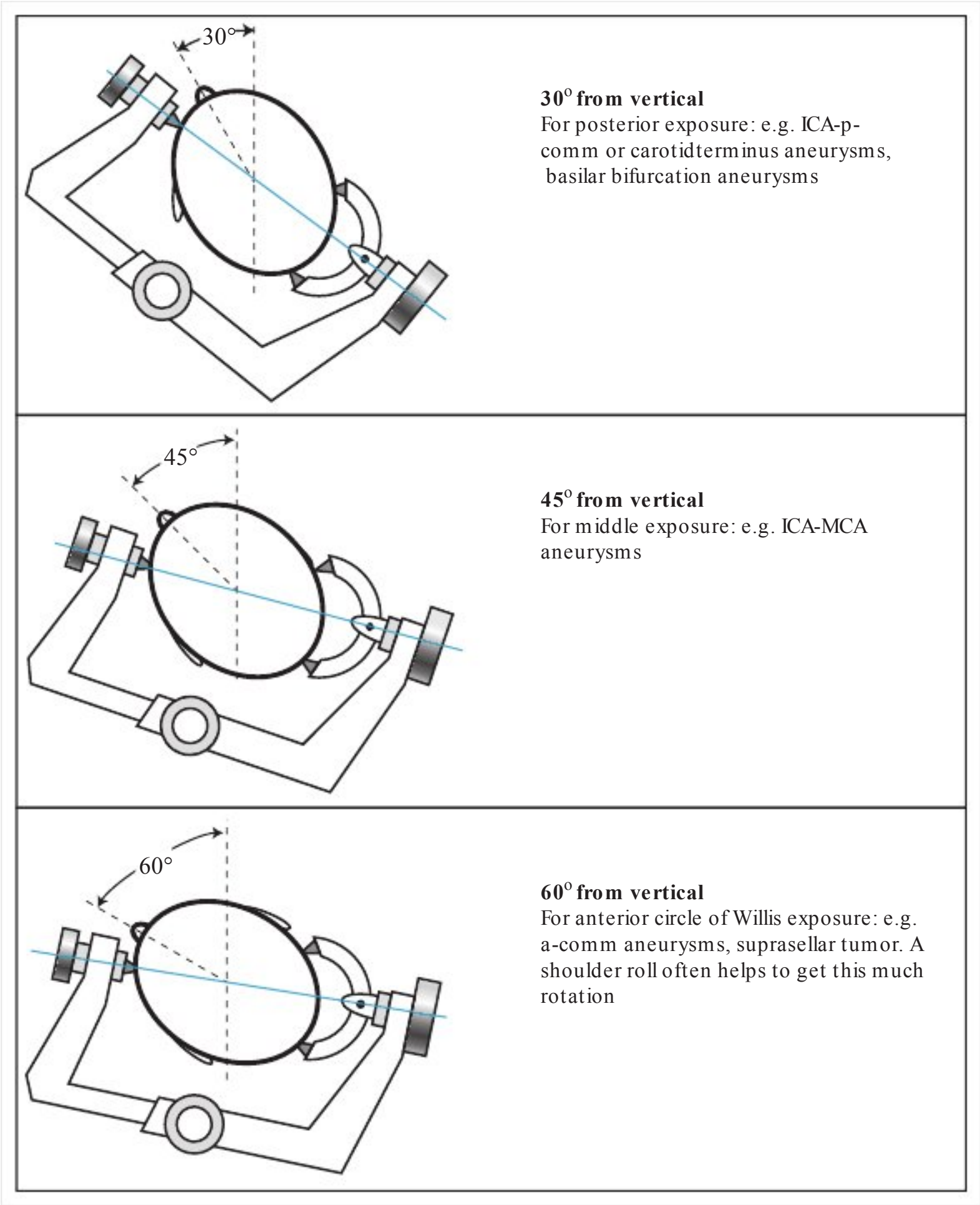


Fig. 94.5 Head position for pterional craniotomy depending on exposure required. The blue line indicates the approximate centerline.

Burr holes

Two burr holes are sufficient; made as far caudally as possible to minimize the amount of bone to be rongeured off to gain access to the floor of the middle cranial fossa. One burr hole is made at the posterior insertion of the zygomatic arch (“A” in □ Fig. 94.7); this burr hole may be placed slightly forward when exposure is centered over structures around the ACoA (e.g. suprasellar tumor). The second burr hole (“Z”) is made at the intersection of the zygomatic bone (near the frontozygomatic suture), the superior temporal line and the supraorbital ridge. The hole should be as low as possible on the orbit; aim the drill slightly superiorly to avoid actually entering the orbit. The dura is dissected off the inner table with a Penfield #3 dissector.

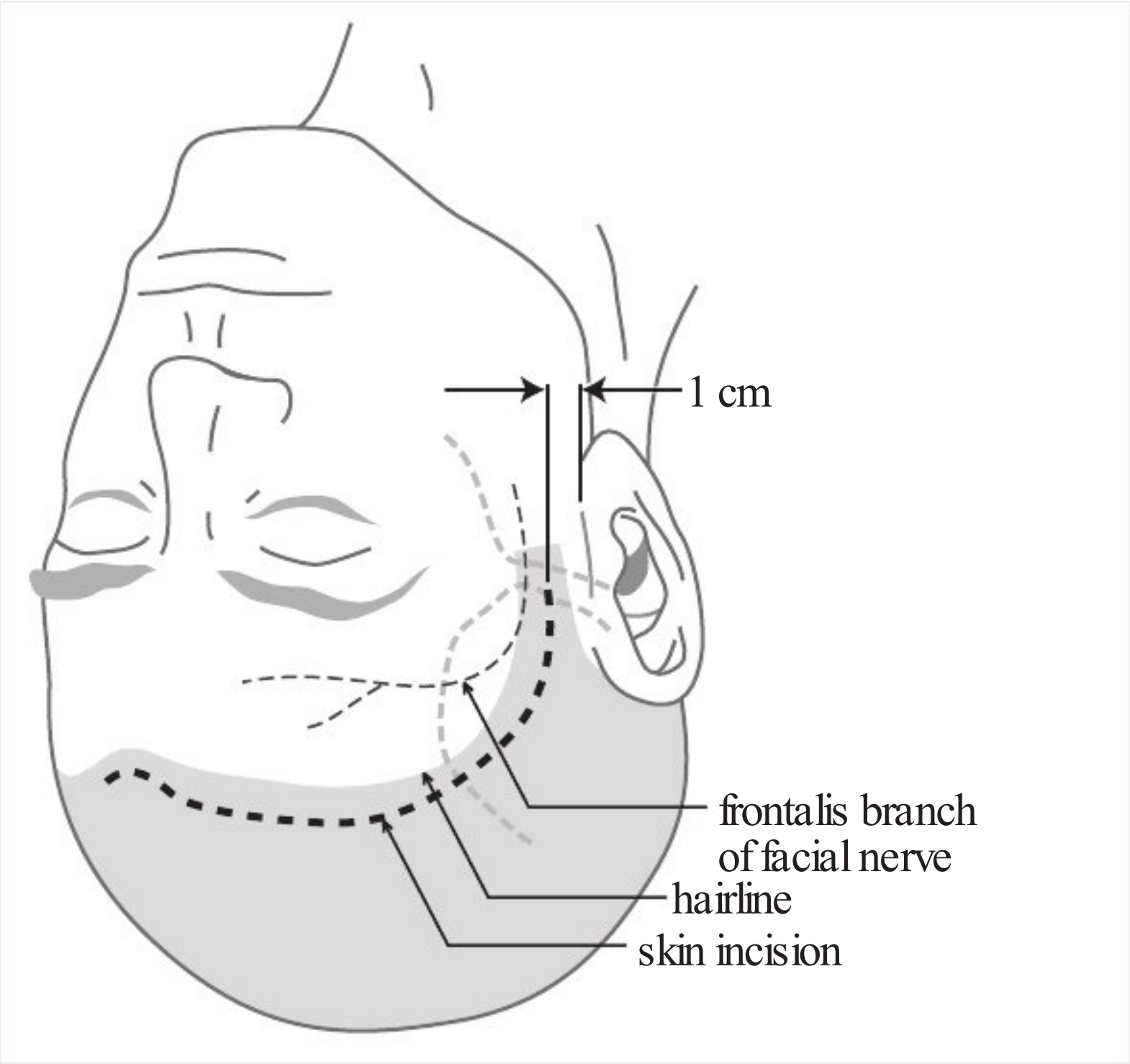


Fig. 94.6 Skin incision for pterional craniotomy

Craniotomy

The resulting bone flap is centered over the depression of the sphenoid ridge. Approximately 33% of the craniotomy is anterior to the anterior margin of temporalis muscle insertion, $\approx 66\%$ is posterior.

With the craniotome, starting at the frontal burr hole the craniotomy is taken anteriorly across the anterior margin of the superior temporal line staying as low as possible on the orbit (to obviate having to rongeur bone, which is unsightly on the forehead). The distance “B” from the medial extent of the craniotomy to the frontal burr hole is 3 cm for anterior circulation aneurysms. For the approaches to skull base (e.g. Dolenc approach), distance “B” is larger and takes the opening to $\approx$ the mid orbit. Then from point “B”, a sharp superior turn is made and the opening is taken back to point “A.” The height (“H”) of the craniotomy needs to be only ≈ 3 cm for aneurysms of the Circle of Willis, and slightly larger (≈ 5 cm) for middle cerebral artery aneurysms. Minimal exposure of temporal cortex is necessary for aneurysms of the skull base region. For large flaps (e.g. for tumors), “H” is made larger to expose more temporal lobe.

From the frontal burr hole, the craniotomy is then taken posteriorly towards the depression corresponding to the sphenoid wing until the drill hangs up.

The craniotomy from the posterior burr hole is taken forward towards the depression corresponding to the sphenoid wing until the drill hangs up.

The bone between the two points where the drill hangs up is scored with the craniotome, and then the bone is fractured at this point. A rongeur is used to remove as much sphenoid wing as possible.

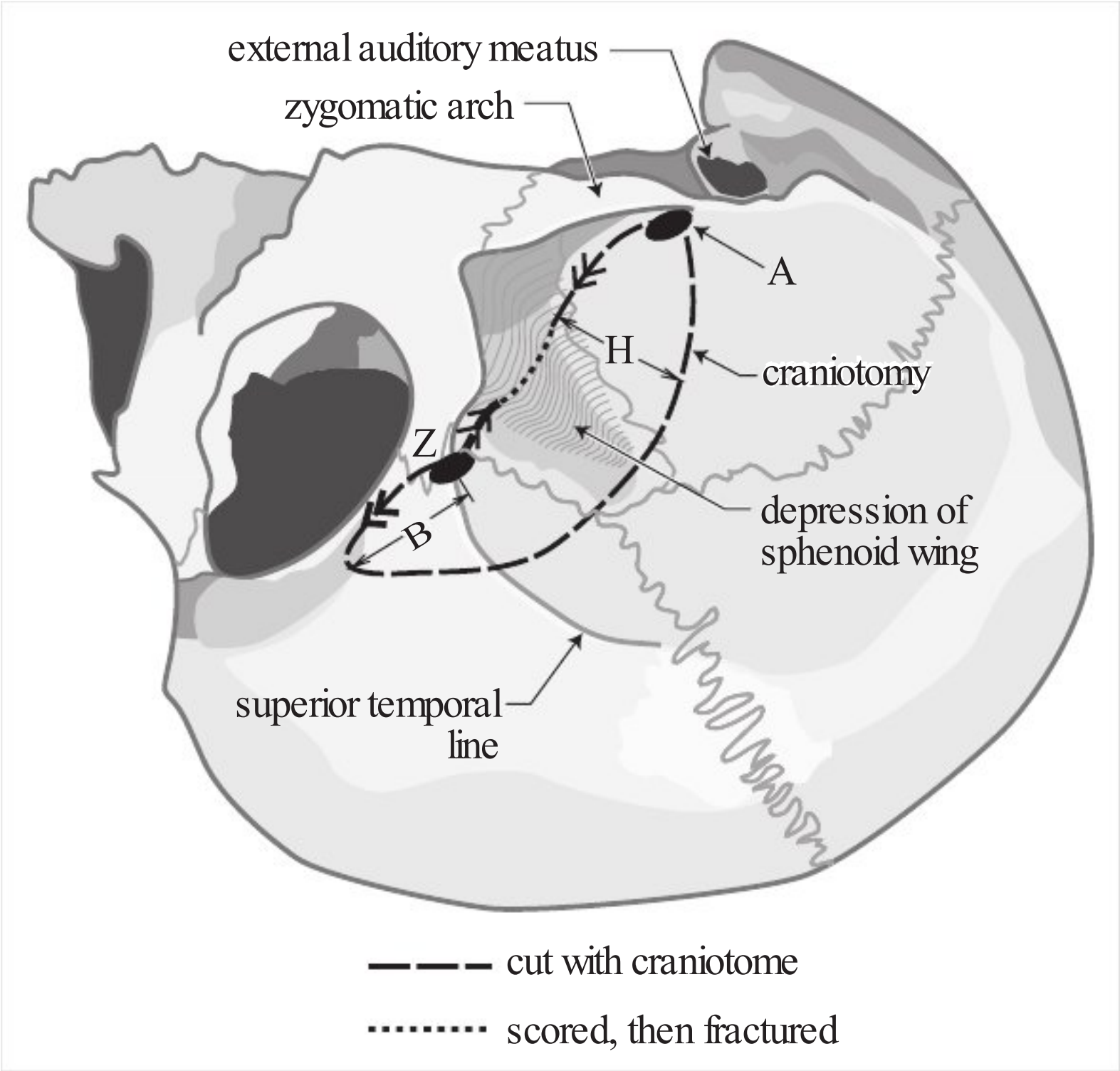


Fig. 94.7 Skull landmarks for right pterional craniotomy

Dural flap

Curvilinear, centered over sphenoid wing, retracted inferiorly with dural stitch.

Dissection

For some anterior circulation aneurysms (e.g. MCA aneurysms) and for the Yasargil approach to basilar tip aneurysms, the sylvian fissure needs to be split. This can be accomplished by working from the lateral aspect of the fissure medially, or, by starting at the point where the carotid artery penetrates the fissure and working laterally. The latter method may be easier when prolific veins overly the junction of the frontal and temporal lobe. There are no arteries that cross the sylvian fissure, and so if the correct plane is maintained, no arteries need to be sacrificed.

□ Fig. 94.8 shows a theoretical exposure of the circle of Willis possible through a pterional craniotomy. This diagram is semi-schematic, and in reality dissection would be directed either anteriorly (e.g. to expose ACoA) or posteriorly (e.g. for basilar tip aneurysms) but not both.

94.3 Temporal craniotomy

94.3.1 Indications

- 1. temporal lobe biopsy: herpes simplex encephalitis
- 2. temporal lobectomy: for resection of seizure focus, decompression post-trauma...

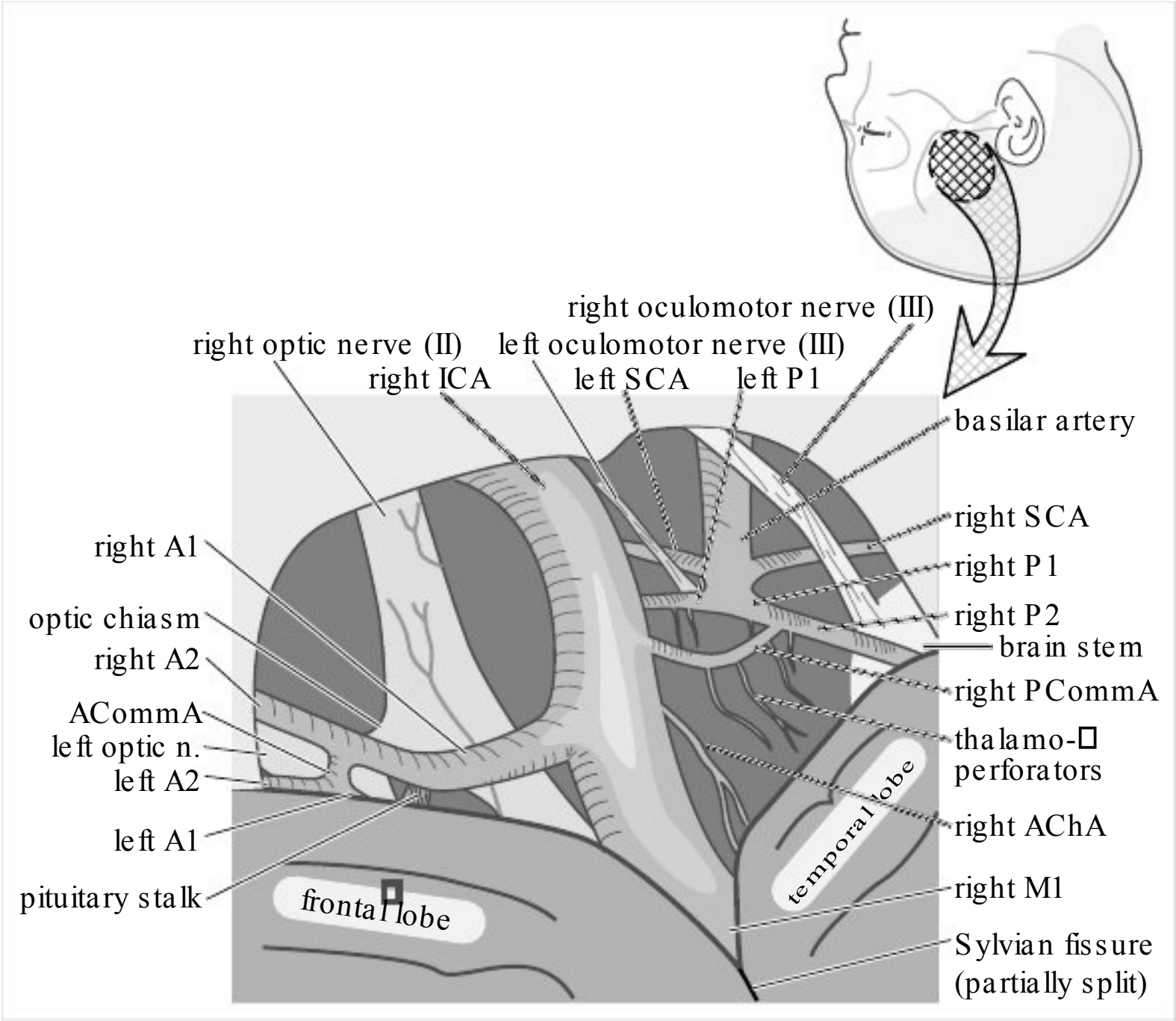


Fig. 94.8 Right sided sylvian fissure dissection, surgical view

3. hematoma (epidural or subdural) overlying temporal lobe
4. tumors of the temporal lobe
5. small, laterally located vestibular schwannomas²⁶
6. access to the floor of the middle cranial fossa (including foramen ovale/Meckel's cave, the labyrinthine and upper tympanic portion of the facial nerve)
7. access to medial temporal lobe e.g. for amygdalo-hippocampectomy (p.1556) or for mesial temporal sclerosis (p.441)

94.3.2 Technique

See □ Fig. 94.9. Two basic methods for temporal craniotomy:

1. small craniotomy or craniectomy through a linear skin incision: good for cortical biopsy or draining chronic subdural hematoma. Also permits access to floor of middle fossa. Simple quick closure
2. question-mark skin incision with standard craniotomy flap: useful for temporal lobe exposure for tumor or acute hematoma

94.3.3 Position, skin incision, craniectomy, approach...

1. patient supine with shoulder roll (to assist in rotating neck to get head almost horizontal)
2. elevate thorax 10–15°: reduces venous distension
3. flex knees slightly
4. Mayfield 3 pin head-holder: true AP with single pin anteriorly
5. head rotated almost horizontal to floor: avoid over-extending to prevent kinking neck veins

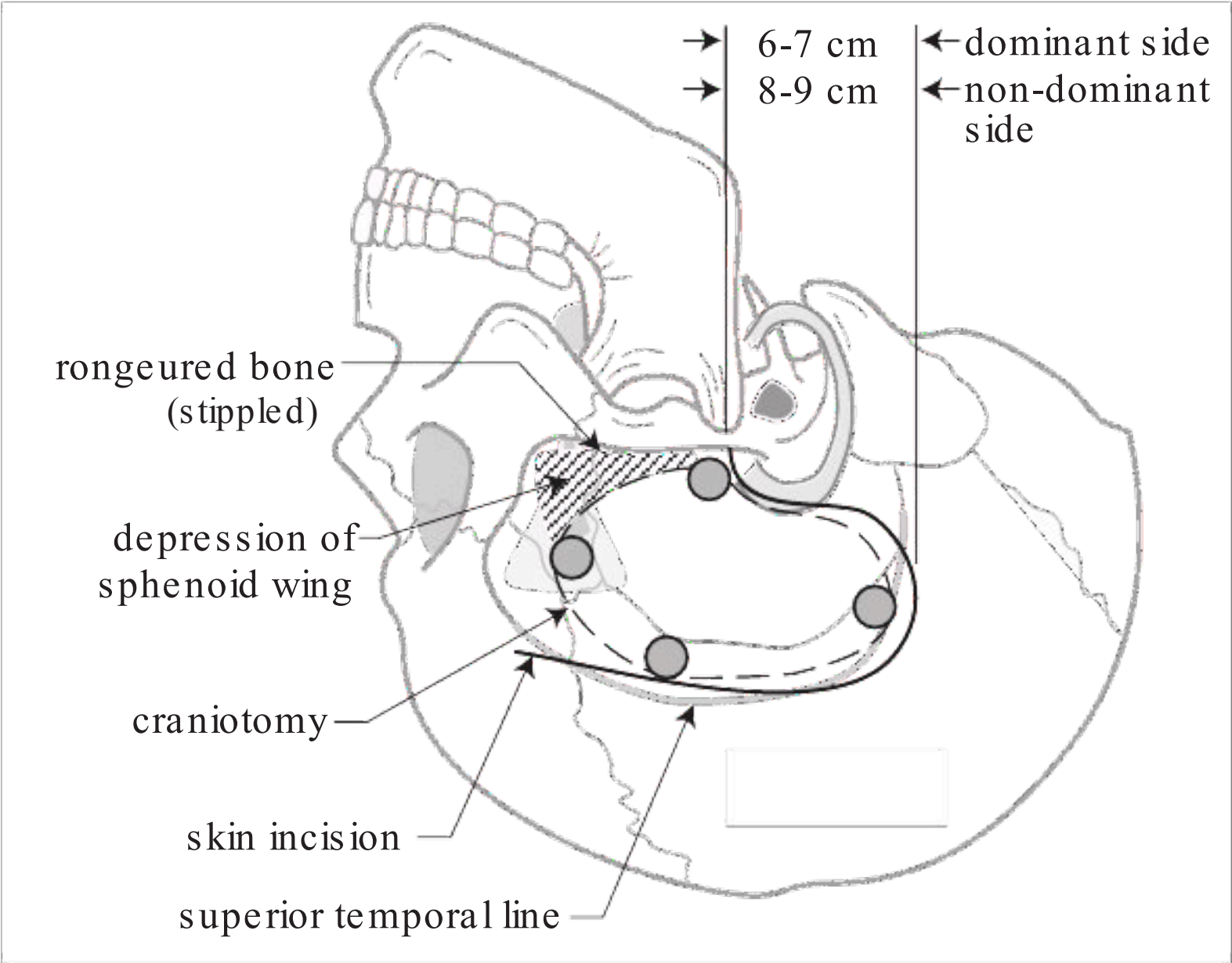


Fig. 94.9 Temporal craniotomy (exposing entire temporal lobe)

94.3.4 Craniotomies

Small craniectomy

Linear skin incision completely within the extent of the temporalis muscle. To access the temporal tip: place the incision midway between the lateral canthus and external auditory canal (EAC); extend it from the zygomatic arch upward for ≈ 6 cm. For small, laterally located vestibular schwannomas, the incision is made 0.5 cm anterior to the EAC extending $\approx 7\text{--}8$ cm above the zygomatic arch.²⁶ To drain a subdural, place the incision just anterior to the tragus and start it 1–2 cm above the zygomatic arch for ≈ 6 cm (modified based on the location of the epicenter of the subdural). Take the incision down to temporalis fascia with the knife, and incise the fascia and muscle with Bovie cautery. Spread with self-retaining retractors, and make a burr hole. Enlarge with rongeurs and/or Kerrison punches.

Standard craniotomy

Question-mark skin incision

See □ Fig. 94.9. Used for access to the temporal lobe including tip (a reverse question mark incision may be used to gain access to the middle and posterior temporal lobe).

1. the pinna is either sutured inferiorly out of the way before draping, or it can be folded under the drapes which may be stapled to the skin
2. the lower limb extends from the zygomatic arch just anterior to the tragus (to avoid the superficial temporal artery)
3. curve as far posteriorly as $\approx 6\text{--}7$ cm on the non-dominant side, or $\approx 8\text{--}9$ cm on the dominant side at the level of the top of the pinna (these dimensions allow access to the “safe” area of temporal tip for lobectomy)
4. then superiorly to the level of the superior temporal line
5. then anteriorly towards the forehead, stopping at the hair line

Burr hole placement

1. at the posterior insertion of the zygomatic arch
2. at upper anterior junction of zygomatic arch
3. one or two burr holes along posterior and superior aspect of the skin incision

Craniotomy

Connect the burr holes with the craniotome, keeping as low as possible in the middle fossa to minimize the amount of bone that must be rongeured. The remaining bone is rongeured down to the floor of the middle fossa (cross-hatched area in □ Fig. 94.9).

Temporal lobectomy

□ Danger points:

1. dominant hemisphere: Wernicke's speech area. Although variable (see Temporal lobectomy (p.1556)), one can usually safely resect up to 4–5 cm from temporal tip without use of mapping techniques to localize speech
2. non-dominant hemisphere: one can resect up to 6–7 cm before running the risk of injuring the optic radiation
3. sylvian fissure (middle cerebral artery): it is best to amputate the temporal lobe backward from the tip for the extent of the desired resection, and then work deep
4. medially, the incisura should be identified to avoid injury to the brain stem which lies just medial to this

94.4 Frontal craniotomy

94.4.1 Indications

1. access to frontal lobe: e.g. for infiltrating tumor
2. approach to third ventricle or to sellar region tumors in some situations, including craniopharyngiomas, planum sphenoidale meningiomas
3. repair of ethmoidal CSF fistula

94.4.2 □ Danger points

1. anterior cerebral arteries in the midline (deep)
2. superior sagittal sinus (SSS) in the midline (note: the SSS may be sacrificed in its anterior third without engendering venous infarction in most cases, whereas venous infarction will almost always occur with division of the SSS posterior to that)
3. avoid inadvertently crossing the midline into the contralateral hemisphere through the corpus callosum
4. dominant hemisphere: Broca's (motor speech) area is located in the inferior frontal gyrus

94.4.3 Technique

Craniotomy options

Two basic choices for craniotomy:

1. unilateral craniotomy through a curved skin incision taken anteriorly up to the hairline: used when one does not need to be low in the frontal fossa in the midline (otherwise the skin incision would have to be taken far into forehead) and when there is no need to cross the midline
2. large bifrontal skin incision from "ear-to-ear" (souttar skin incision²⁷) allowing low approach to one or both frontal fossa

Unilateral frontal craniotomy

□ Fig. 94.10. Skin incision starts <1 cm anterior to the tragus, and does not need to go all the way down to the zygomatic arch. It curves superiorly and slightly posteriorly before being taken to the midline frontally.

Burr holes

1. at the junction of the superior temporal line and the orbital rim

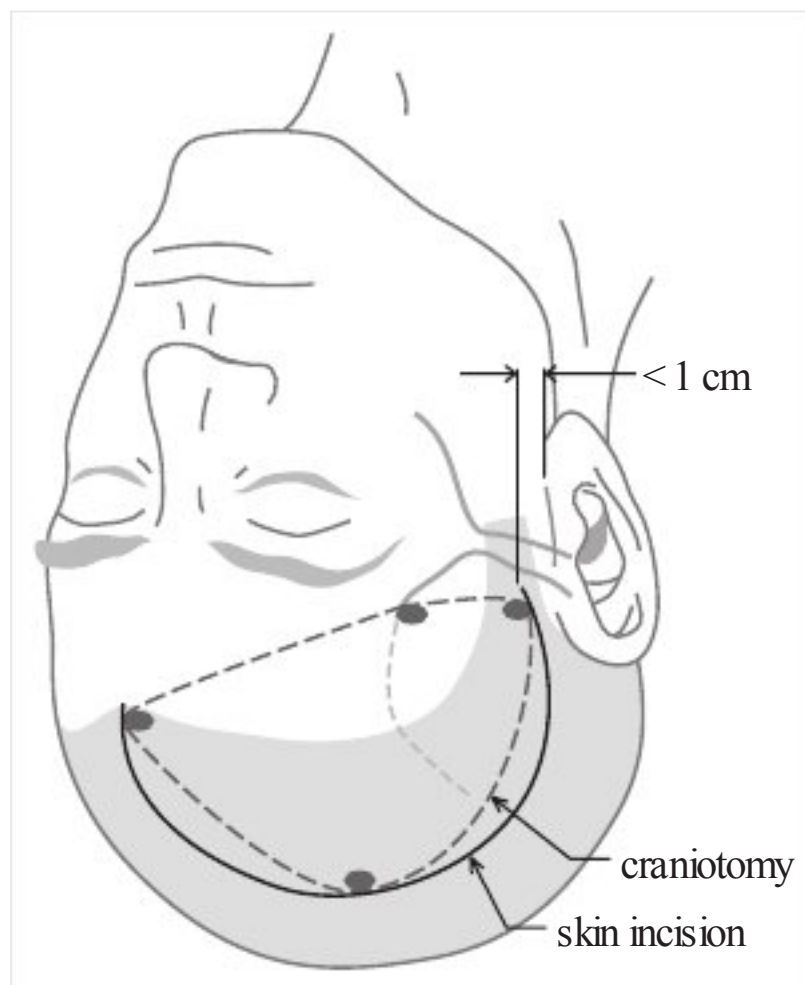


Fig. 94.10 Unilateral frontal craniotomy

2. just posterior to the depression of the sphenoid wing (behind the pterion)
3. anteriorly just behind the hairline to avoid having a burr hole under the forehead (which causes an unsightly depression)
4. superiorly

Bilateral frontal craniotomy

1. “ear-to-ear” or souttar skin incision
 - a) just behind hairline with a slight widow’s peak at the front
 - b) does not need to go all the way to the zygomatic arch, it just needs to be $\approx$ as low as the orbital roof
 - c) unlike pterional craniotomy, usually do not need to incise the temporalis muscle and fascia. Dissect the flap off the muscle/fascia
 - d) if a periosteal flap is likely to be needed, it sometimes helps not to incise the periosteum at the same time as the skin incision. Then, the periosteum can be incised behind the skin incision to yield a longer periosteal graft than would have otherwise been obtained
2. burr holes: to avoid burr hole defects on the forehead, the bone flap can be created with two burr holes straddling the superior sagittal sinus (SSS) close to the skin incision, and two burr holes laterally
3. the SSS may be divided low, near the orbital roof, with little risk
4. if the frontal sinus is entered, it is dealt with as outlined under Frontal sinus fractures (p.886)

94.5 Petrosal craniotomy

94.5.1 Indications

1. lesions of the petrous apex (e.g. petroclival meningiomas)
2. lesions of the clivus (e.g. chordomas) with both posterior fossa and supratentorial components

94.5.2 Advantages

Spares sinus and otologic apparatuses. Minimizes cerebellar and temporal lobe retraction.

94.5.3 Technique

See reference.²⁸

Position

1. patient supine, ipsilateral shoulder roll
2. elevate thorax 10°: reduces venous distension
3. flex knees
4. Mayfield 3 pin head-holder: close to true AP with single pin on forehead
5. head positioned to place petrous base at highest point of field:
 - a) head rotated 40–60° from vertical
 - b) head abducted towards contralateral shoulder
 - c) neck extended 15°: allows gravity to retract frontal lobe away from skull base

Skin incision

Reverse question mark starting from zygomatic arch 1 cm anterior to tragus, arcing posteriorly over ear, descending to 0.5–1 cm medial to mastoid notch.

Temporalis muscle and periosteum reflected anteriorly and inferiorly.

Craniotomy

Four burr holes are utilized, two on each side of the sinuses (near the junction of the transverse and sigmoid sinuses).

94.6 Approaches to the lateral ventricle

Surgical approaches to the trigone – Contemporary Neurosurgery Vol. 27 No.5²⁹

Classic review¹⁸ (p 561–74) summarized:

1. atrium (AKA trigone); numerous approaches include²⁹:
 - a) middle temporal gyrus: through the dilated temporal horn
 - b) lateral temporal parietal
 - c) superior parietal occipital
 - d) transcallosal (see below)
 - e) transtemporal horn: access to temporal horn is via lobectomy of the temporal tip
 - f) occipital lobe incision or occipital lobectomy: recommended only if patient has homonymous hemianopsia pre-op
2. frontal horn
 - a) middle frontal gyrus
3. midventricular body
 - a) transcallosal
 - b) middle frontal gyrus: usually prevents access to vascular supply until most of the tumor is removed (especially for tumors supplied primarily by posterior choroidal artery)
4. temporal horn
 - a) middle temporal gyrus
 - b) transtemporal horn

94.7 Approaches to the third ventricle

94.7.1 General information

Classic references review the microsurgical anatomy³⁰ and surgical approaches,³¹ and are briefly summarized below.

Alternative approaches for lesions of the anterior third ventricle³²:

1. transcortical: approach is through the lateral ventricle and is feasible only in the presence of hydrocephalus; especially useful if the tumor extends from the third ventricle into one of the lateral ventricles. Risk of seizures is 5% (higher than with transcallosal) (p. 1466).
2. transcallosal: may be preferable in the absence of hydrocephalus (see below)
 - a) anterior transcallosal: good visualization of both walls of third ventricle; risk of bilateral fornix damage
 - b) posterior transcallosal: allows approach to quadrigeminal plate or pineal region; risk of damage to deep veins
3. subfrontal: allows four different approaches
 - a) subchiasmatic: between optic nerve and optic chiasm
 - b) optico-carotid: through the triangular space bordered by optic nerve medially, carotid artery laterally, and ACA posteriorly

- c) lamina terminalis: above the optic chiasm³³
- d) transsphenoidal: requires removal of tuberculum sellae, planum sphenoidale, and anterior wall of the sella turcica
- 4. transsphenoidal
- 5. subtemporal
- 6. stereotactic: may be useful for aspiration of colloid cysts; see Stereotactic drainage of colloid cysts (p.759)

94.7.2 General principles of tumor removal

Summarized.³¹ During the approach, deep veins should be preserved at all costs, even if it means stretching them to the point that they may rupture.

It is helpful to place a suture through the tumor capsule to act as a tether.

The tumor should first be removed from within the capsule; techniques include aspiration, and then opening the capsule and debulking from within. The capsule may then be collapsed and dissected from adherent structures. If the capsule adhesions seem unyielding, the most likely cause is incomplete intracapsular evacuation.

Vessels on the surface of the tumor should be presumed to be supplying normal brain, and should be dissected off the capsule once it is completely emptied.

94.7.3 Transcallosal approach to lateral or third ventricle

General information

Performed through an interhemispheric approach to the corpus callosum (CC) via a parietal craniotomy, usually right sided in a left-hemisphere dominant patient.

Indications

Primarily for tumors or lesions of the lateral or third ventricle, including:

- 1. colloid cysts
- 2. craniopharyngiomas
- 3. cysticercosis cysts
- 4. thalamic glioma
- 5. AVM

Booking the case: Transcallosal surgery

Also see defaults & disclaimers (p.27).

- 1. position: supine with pin headholder
- 2. equipment:
 - a) microscope
 - b) image guided navigation system
- 3. post-op: ICU
- 4. consent (in lay terms for the patient – not all-inclusive):
 - a) procedure: operation between the two halves of the brain to remove lesion
 - b) alternatives: non-surgical management, surgery through the surface of the brain (transcortical), radiation therapy for some diagnoses
 - c) complications: stroke, “disconnection syndrome” (uncommon) (p. 1556), hydrocephalus with possible need for a shunt, memory deficits

Technique

See references.^{30,31,34}

General information

□ Fig. 94.11. Image-guided navigation is very helpful in ascertaining the correct trajectory which permits minimizing the size of the callosotomy, and helps distinguish the corpus callosum from the cingulate gyri.

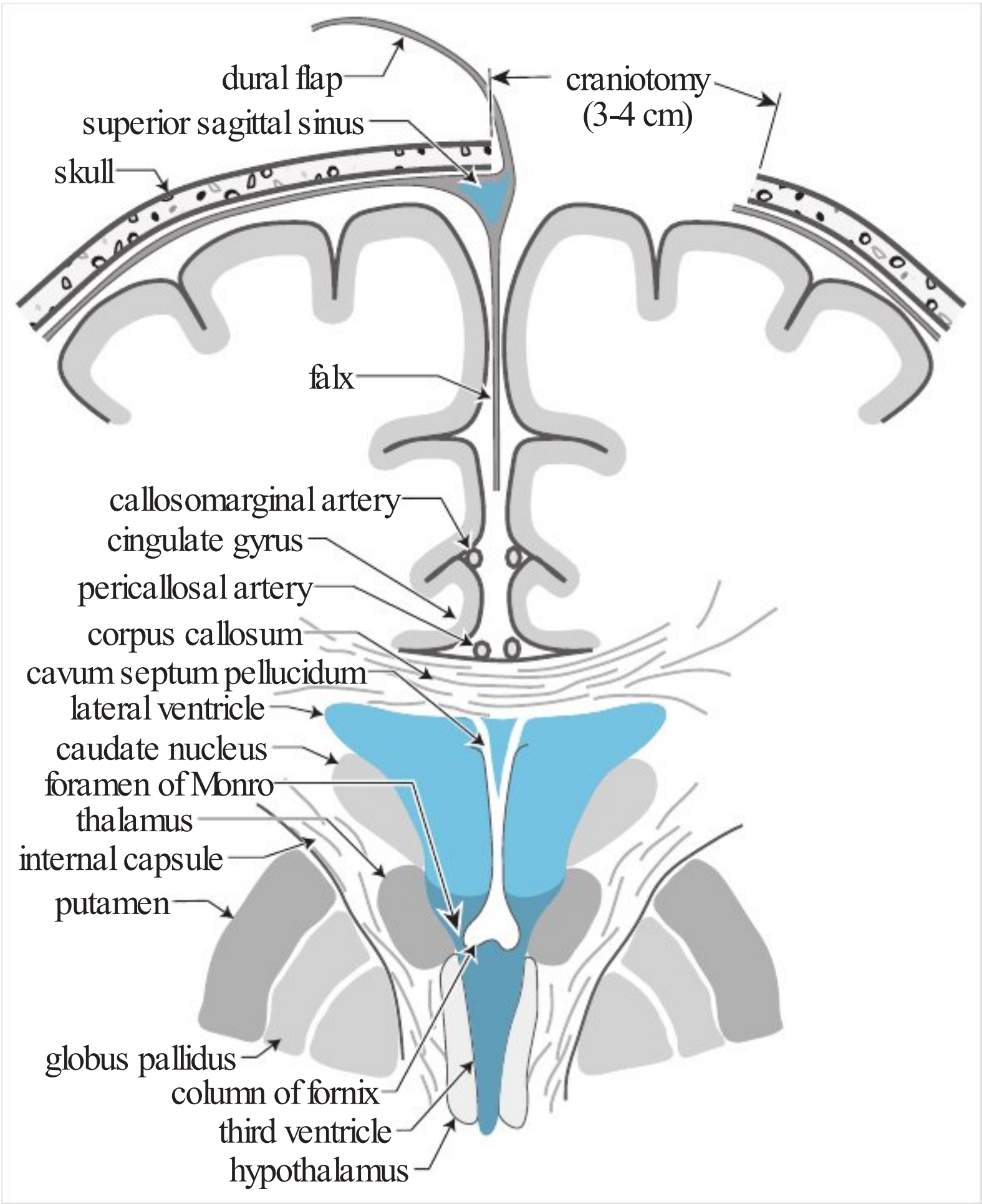


Fig. 94.11 Transcallosal approach to the third ventricle: frontal view

Position
Supine with neck flexed. Thorax elevated 20°. Spinal drain not used. Keep the head perfectly vertical to minimize disorientation that can easily occur with this approach. Alternatively, gravity retraction may be employed either by tilting the head slightly to the right (causing the right hemisphere to fall away) or by using the lateral position.

Skin incision
Either of the following may be used:

1. inverted “U” with the top just left of midline, extending from 6 cm anterior to the coronal suture to 2 cm behind the coronal suture, taking the sides for 7–8 cm
2. souttar skin incision

Craniotomy

Pre-op angiography is recommended to plan the position of the flap to avoid sacrificing large cortical veins. MRI may also suffice for this.³⁵ There tends to be fewer veins bridging from the cortex to the superior sagittal sinus anterior to the coronal suture, therefore this is often a good location to enter the interhemispheric fissure. The bone flap is either trapezoidal or triangular in shape, for adequate exposure it is critical to go all the way to the superior sagittal sinus (SSS). Several techniques may be used. NB: the SSS is often to the right of the sagittal suture (p.61).

1. to expose the SSS, straddle the SSS with paired burr holes anteriorly and posteriorly, dissect the dura from the inner table between pairs, and make the longitudinal cut on the left of midline.
Disadvantage: removing the midline bone puts the SSS at greater risk of injury and makes it more difficult to control lacerations
2. one can make the long cut well to the right of midline, and then under direct vision rongeur off the bone to the SSS. Safe, but leaves a large bone gap that may need to be filled (e.g. with methyl methacrylate) and is time consuming
3. most risky for causing a sagittal sinus laceration is to make the long cut just right of the midline (over the edge of the SSS, which may lacerate it)

To stay away from the motor strip and to keep the sagittal sinus exposure as anterior as possible, 2/3 of the opening should lie anterior to coronal suture, 1/3 posterior (generally: 6 cm total with 4 cm anterior and 2 cm posterior). The craniotomy extends laterally to 3–4 cm to right of midline. The last cut with the craniotome should connect the burr holes along the sinus (midline); leaving this cut for last permits rapid access to the sinus in case it is torn. The dural flap is based towards the sagittal sinus.

Approach to corpus callosum

None, or at most, only one bridging vein from the cortex to the sagittal sinus may be sacrificed (and then, only if it is not a large draining vein). Gently retract the right hemisphere. Avoid retractors on the sagittal sinus to prevent injury to the SSS which may lead to sinus thrombosis (once CSF is released (with the callosotomy) retraction will be easier). Enter the interhemispheric fissure and follow the falx deep. Open the arachnoid membrane beyond the deep edge of the falx.

The two cingulate gyri may be adherent in the midline, and can easily be mistaken for the corpus callosum (CC). This error may be compounded by mistaking the callosomarginal arteries for the pericallosal arteries. Erroneously entering the cingulate gyrus disorients the surgeon and could cause injury to the pericallosal arteries. To differentiate: the CC is a pure white structure, is usually deeper than one anticipates, and is appreciated beneath the paired pericallosal arteries. Image guided surgery or measuring the depth to the CC on the pre-op MRI may help.

Callosotomy

The callosotomy is usually performed between the two pericallosal arteries. Some arterial branches may cross the midline, occasionally it is necessary to sacrifice some. Trajectory: a line drawn from the coronal suture (in the midline) to the external auditory canal (the foramen of Monroe lies along this line); this helps avoid the tendency to tunnel posteriorly through the CC. Either the bipolar cautery, suction and sharp knife, or the laser is used to make the callosotomy. In hydrocephalus, the callosum will be thin. Entering the lateral ventricle releases CSF which aids retraction. When the foramen of Monroe is occluded (e.g. with colloid cyst), it helps to fenestrate the septum pellucidum to prevent it from bulging into the ventricle in which one is operating (otherwise, as CSF is aspirated from the ipsilateral lateral ventricle, it cannot escape from the other).

Disconnection syndrome (p.1556): more common with posterior callosotomy (near the splenium) where more visual information crosses. The risk is reduced by creating a callosotomy <2.5 cm in length extending posteriorly from a point 1–2 cm behind the tip of the genu.³⁶ For an interforaminal approach, the callosotomy must be perfectly midline.

Approach to third ventricle

Usually, the callosotomy will not be exactly midline, and one of the lateral ventricles will be entered. Great care must be taken to correctly identify which lateral ventricle has been entered, another potentially disorienting pitfall. For orientation (□ Fig. 94.12), the choroid plexus passes forward in the choroidal fissure to the foramen of Monroe (which is medial) where it converges with the thalamostriate vein approaching from a more lateral position in the groove between the thalamus and caudate. The septal and caudate veins approach the foramen from anterior. With colloid cysts, the

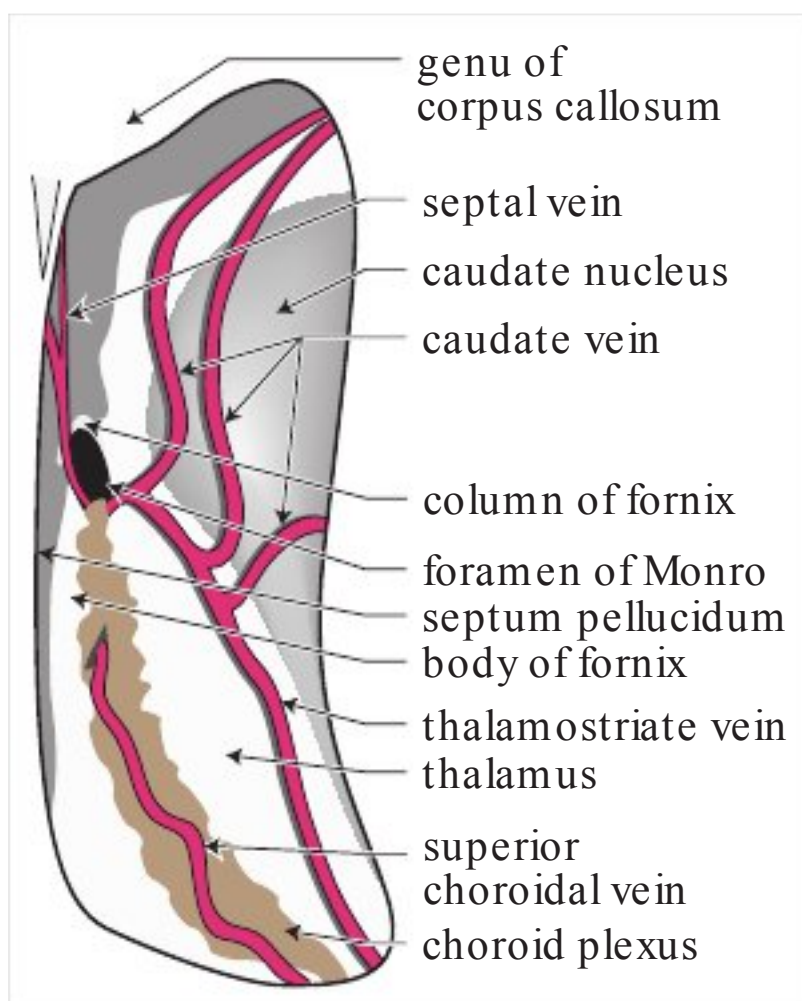


Fig. 94.12 Right foramen of Monro viewed from above through right lateral ventricle³⁰⁾

foramen of Monro may be hard to recognize initially as it will be plugged with the cyst which can resemble the ependymal lining of the ventricle, but on close inspection is usually slightly grayer (the choroid plexus enters the posterior aspect of the foramen).

Another possible pitfall upon incising the CC is entering a cavum septum pellucidum (p.1375). The give-away here is that no landmarks will be visible.

Alternative approaches to third ventricle

1. interforaminal³⁶: go above body of fornix, approaches roof of third ventricle. Well suited for lesions of the mid and posterior third ventricle. Callosotomy should be as close to midline as possible
2. from lateral ventricle through the foramen of Monro: with hydrocephalus, the foramen of Monro is usually dilated. If the foramen is too small for adequate access to the third ventricle, one can:
 - a) return to the interforaminal approach (see above) or
 - b) enlarge the foramen of Monro only if absolutely necessary. Either by:
 - opening the foramen laterally
 - the “subchoroidal” approach, making the incision posteriorly (sacrificing the thalamostriate vein) which is reportedly well tolerated^{31,37}
 - last resort: incising the antero-superior margin of the foramen through the column of one fornix.³⁶ Caution: if the other fornix is non-functional for any reason, this would produce a bilateral forniceal lesion and may (but not definitely³⁶) result in loss of short term memory and ability for new learning

Colloid cyst removal

It is critical to debulk and empty a cystic lesion, such as a colloid cyst, before delivering it from the third ventricle through the foramen of Monro. This will minimize the retraction and manipulation of the fornix. Inserting a needle and aspirating may work. The partially emptied cyst is grasped with a micro-pituitary and is delivered into the lateral ventricle through the foramen of Monro. One should only attempt to deliver the empty capsule through the foramen of Monro (p.1462). There is usually a stalk attaching the lesion to the roof of the third ventricle, this is coagulated with bipolar cautery and divided.

For other tumors, if the tumor is too large to fit through the foramen of Monro, it should be gutted from within.

Complications

1. venous infarction, may be due to:
 - a) sacrifice of critical cortical draining veins: plan the flap to avoid this with preoperative angiography, or with sagittal T2WI MRI images³⁸
 - b) superior sagittal sinus (SSS) thrombosis.³⁹ Factors that may contribute to sinus injury include³⁵:
 - injury from retractor: avoid placing retractor on sinus (deformation of midline should not exceed 5 mm)
 - over-retraction of the dural sinus flap or on SSS itself (lateral deformation should be <2 cm)
 - injury during the opening of the bone in the region of the sinus
 - over-use of bipolar coagulation in the region of the SSS
 - hypercoagulable state of the patient, including dehydration
2. transient mutism as a result of bilateral cingulate gyrus retraction or thalamic injury in conjunction with section of the midportion of the callosum³⁸

94.7.4 Transcortical approach to lateral or third ventricle

Indications

In the absence of hydrocephalus, it is difficult to navigate through the ventricular system. Thus, with normal sized ventricles, the third ventricle and region of the foramen of Monroe are better approached transcallosally (p.1462).

1. tumors of the atrium of the lateral ventricle
2. tumors of the roof of the third ventricle
3. third ventricular tumors with significant extension into one lateral ventricle

Approaches

1. posterior parietal
2. middle temporal gyrus: useful when temporal horn of lateral ventricle is dilated due to hydrocephalus caused by the tumor; access is through the temporal horn
3. middle frontal gyrus approach: a 4 cm incision is made parallel to the axis of the middle frontal gyrus, above and anterior to the expressive speech center (Broca's area) and anterior to the motor strip³¹; about the same point as used for frontal ventriculostomy, see Kocher's point (p.1512)

94.8 Interhemispheric approach

94

94.8.1 Indications

For lesions abutting on midline, deep to surface, but superficial to corpus callosum (lesions that can "fall away" from midline). Similar to transcallosal approach above, except that the pathology can be placed on the down side, which allows gravity to retract the hemisphere and thus minimizes pressure necrosis injury from mechanical retractors.

94.8.2 Technique

Position

True lateral (prevents getting lost from unusual angles). Head tilted slightly up.

Approach

Similar to transcallosal (p.1462). Need to be sure that lateral portion of craniotomy extends at least 4 cm from midline to minimize the necessity of retraction of brain against bone.

94.9 Occipital craniotomy

94.9.1 Indications

Occipital lobe tumors including posterior falx meningiomas or tentorial meningiomas with only supratentorial component. Occipital lobe intracerebral hemorrhages.

94.9.2 Positions

Supine

Shoulder roll on affected side; elevate thorax 15°. Mayfield 3 pin headholder with single pin in forehead off to the side of the crani, double pin just over midline on opposite side.

Lateral oblique

1. affected side up, can operate either
 - a) from behind patient similar to p-fossa crani for CPA lesion
OR
 - b) from top of table
2. alternative approach: affected side down. Useful in lesions adjacent to the falx; see Interhemispheric approach (p.1466)

94.10 Decompressive craniectomy

94.10.1 Indications

Indications (controversial) include:

1. malignant middle cerebral artery occlusion syndrome (p.1303) primarily for nondominant hemisphere. Use on dominant side is more controversial
2. traumatic intracranial hypertension
 - a) as an adjunct for persistent intracranial hypertension when other ICP control measures fail (p.914)⁴⁰
 - b) early in the management: may be considered for patients undergoing emergent surgery (for fracture, EDH, SDH...)⁴¹
3. uncontrollable brain swelling during craniotomy (p.1429)
4. reported in children with refractory nontraumatic intracranial hypertension⁴² (e.g. infection, infarction, Reye's syndrome...)

94.10.2 Potential complications

1. bleeding
2. herniation of the brain through the opening, compressing and lacerating the brain on the bone edges (risk may be reduced by making generous craniectomy)
3. post-op injury to the brain from inadvertent external pressure applied to the now relatively less protected brain
4. post-op fluid collections: hygromas or hematomas at the operative site, on the contralateral side, or interhemispheric

94.10.3 Techniques

General considerations

1. it is necessary to open the dura
2. options for the removed bone flap
 - a) discard it: this may be the best option when the bone flap has been contaminated as a result of an open traumatic scalp laceration
 - b) place it in a separate subcutaneous pouch in the patient's abdomen for later retrieval and reimplantation into the skull. This is especially helpful if the patient's own skull is preferred and the patient does not live in the area where he/she is having the surgery
 - c) store it for future implantation: saturate with sterile solution (e.g. RPMI medium 1640 www.invitrogen.com/GIBCO) and then place within sterile storage (e.g. intestinal bags which are then placed in a sterile plastic container) and store in a bone freezer at -80° C
 - d) for non-contaminated situations (e.g. stroke): reimplantation can be considered after 6–12 weeks
3. bone openings need to be large (e.g. > 12 cm diameter,⁴³ often > 15 cm)

Hemicraniectomy

1. some prefer use of a Mayfield headholder placed low (□ Fig. 94.13) to give greater access⁴¹ (not feasible with severe comminuted skull fractures)

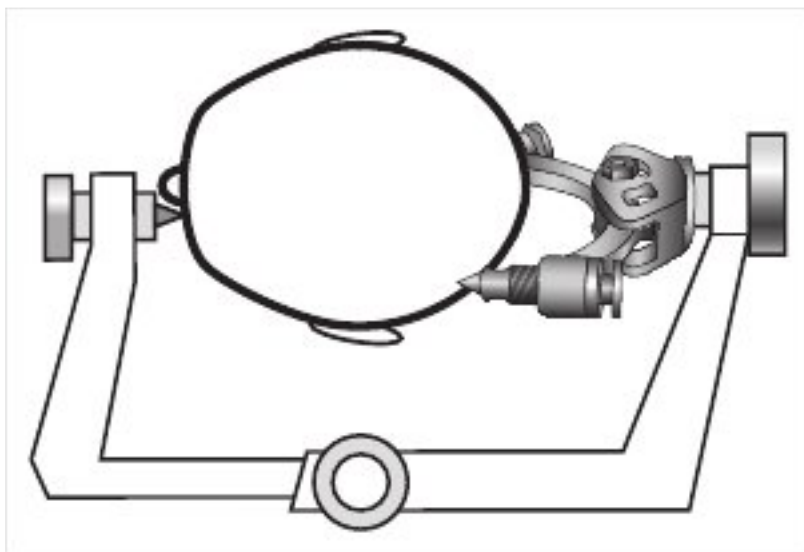


Fig. 94.13 Position of head and headholder for right hemispheric craniectomy (looking down on top of patient's head)

2. AP axis of head is placed horizontal to floor (unless C-spine not cleared or if neck too immobile – one may compensate for this by rotating table)
3. skin incision: two options
 - a) ☐ Fig. 94.14A starts at widow's peak, similar to trauma flap (p.837), but with increased exposure by taking it posteriorly close to theinion then turning sharply anteriorly and hugging the ear to preserve blood supply
 - b) ☐ Fig. 94.14B "T" incision. Less risk of flap ischemia. The "T" joins the midline incision behind the coronal suture to preserve the STA⁴¹
 - c) burr holes (☐ Fig. 94.15): a burr hole is made just above the posterior root of the zygomatic arch, a second one may be made just behind the frontal insertion of the zygomatic arch, inferior to the superior temporal line
 - d) bone flap: proceed posteriorly from the posterior zygomatic arch using the footplated craniotome. Posteriorly, stay ≈ 1 cm superior to asterion to avoid the transverse sinus. The flap is taken 1 cm beyond the lambdoid suture, and then up towards the sagittal suture, crossing the lambdoid suture again (this leaves a small amount of bone posteriorly on which the head can rest post-op). An anterior turn is made 1 cm short of the sagittal suture to avoid the superior sagittal sinus, and the sagittal suture is paralleled. The coronal suture is crossed and the drill is taken as low as possible in the frontal fossa near the midline. Staying as low as possible, the orbital roof is followed posteriorly towards the second burr hole. The burr holes are then connected
 - e) some bone may need to be rongeuired to expose the floor of the middle fossa (stippled area in ☐ Fig. 94.15)
 - f) dural opening: based inferiorly, taken to 1 cm short of the craniotomy edge. Dural releasing incisions may be made at intervals up to the bone margin to avoid strangulation of the brain on the dural edge
 - g) duraplasty
 - onlay: 2 cm wide strips of dural substitute can then be placed partway under the dural edge around the periphery to isolate the brain from the undersurface of the skin flap where there will be a gap in the dura
 - some authors suture a dural draft in place
 - h) the dural flap is then replaced on top of the brain and dural substitute strips, and is not sutured

Bilateral craniectomy

The above procedure can be performed bilaterally, however, it is difficult to position the head to do this. Alternatively, a bifrontal craniectomy can be performed.

1. skin incision: bicoronal, posterior to the coronal suture (☐ Fig. 94.16)
2. burr holes: may use the same ones as for hemispheric craniectomy (see above) bilaterally. Addition burr holes to straddle the superior sagittal sinus may be made if a large single bone flap is planned
3. bone flap (☐ Fig. 94.17): two options, both are taken back to the coronal suture
 - a) a single large bone flap⁴⁴ extending back to the coronal sutures, or
 - b) two frontal flaps leaving a thin strip of bone in the midline overlying the superior sagittal sinus (if this strip is too wide, it can damage the brain)
4. dural opening: bilateral, based against the midline (superior sagittal sinus)

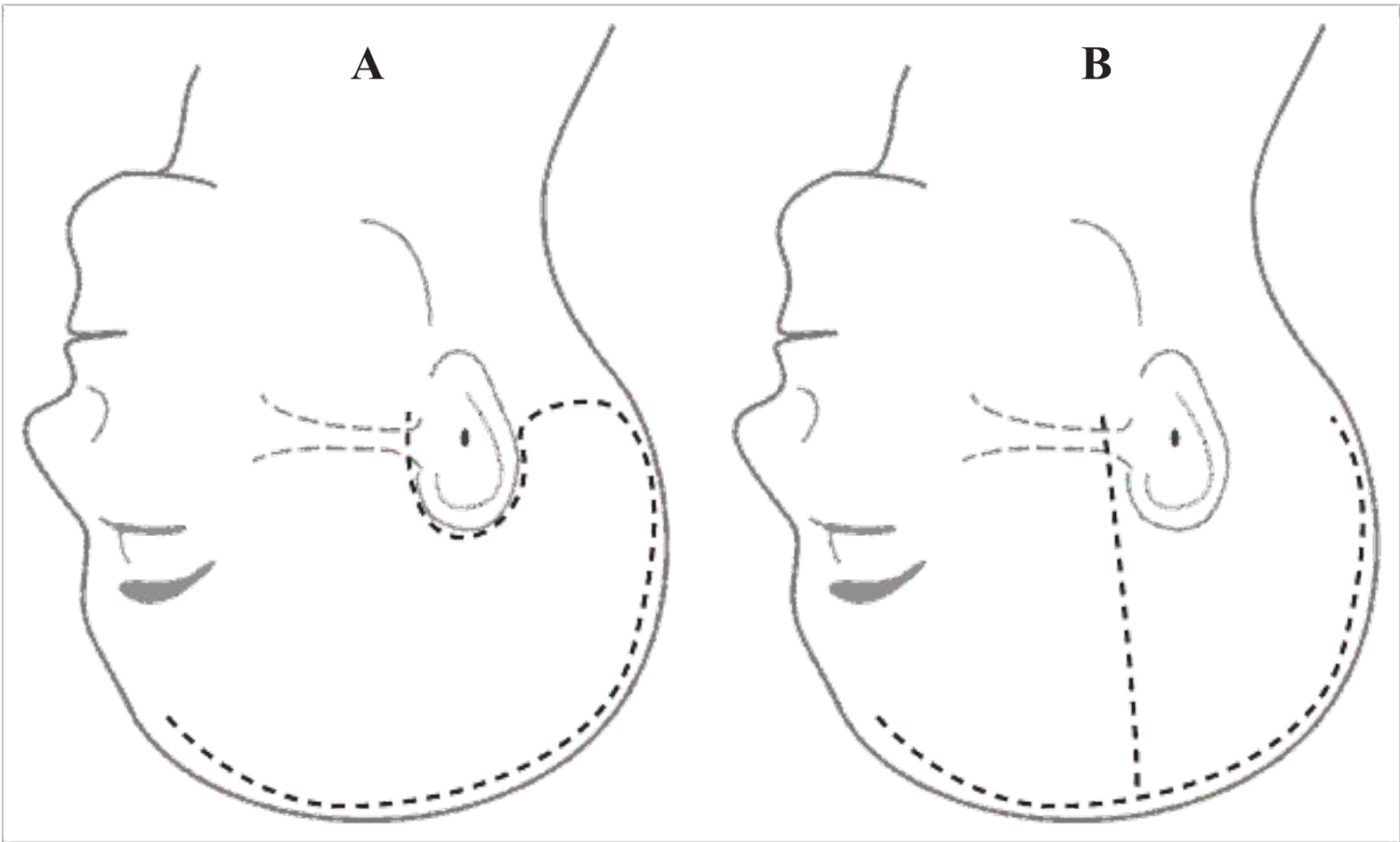


Fig. 94.14 Two options for skin incision for hemicraniectomy (see text)

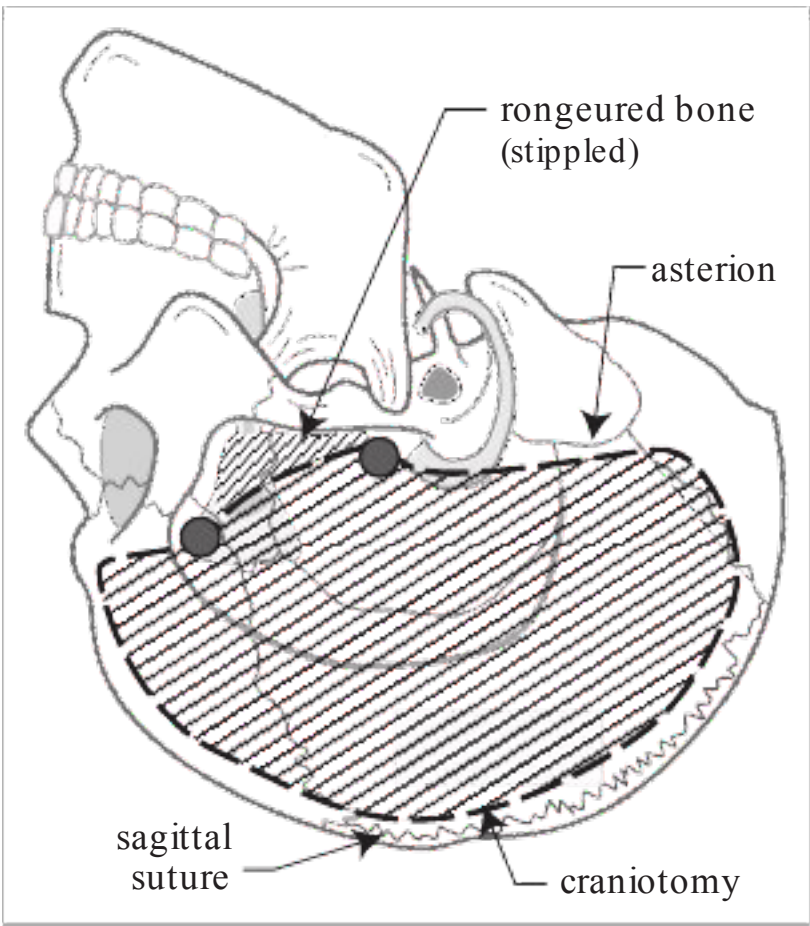


Fig. 94.15 Hemicraniectomy bone flap

Posterior fossa decompressive craniectomy

- 1. skin incision: midline skin incision from above inion to $\approx$ C2 spinous process
- 2. bone opening: laterally to sigmoid sinuses, superiorly to transverse sinus. C1 laminectomy is typically performed as well⁴²
- 3. dural opening: “Y” shaped incision

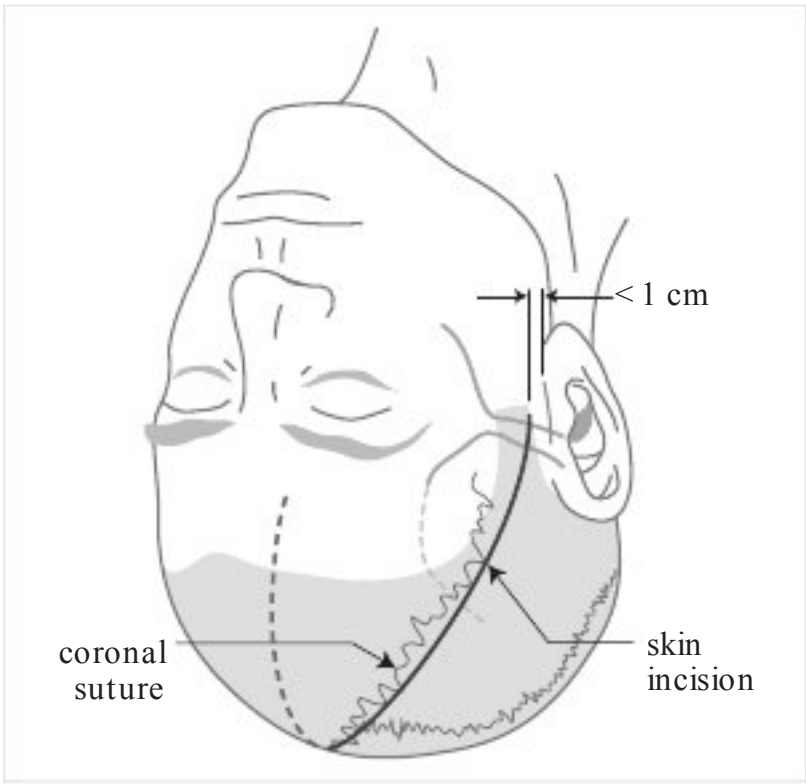


Fig. 94.16 Bilateral craniectomy skin incision

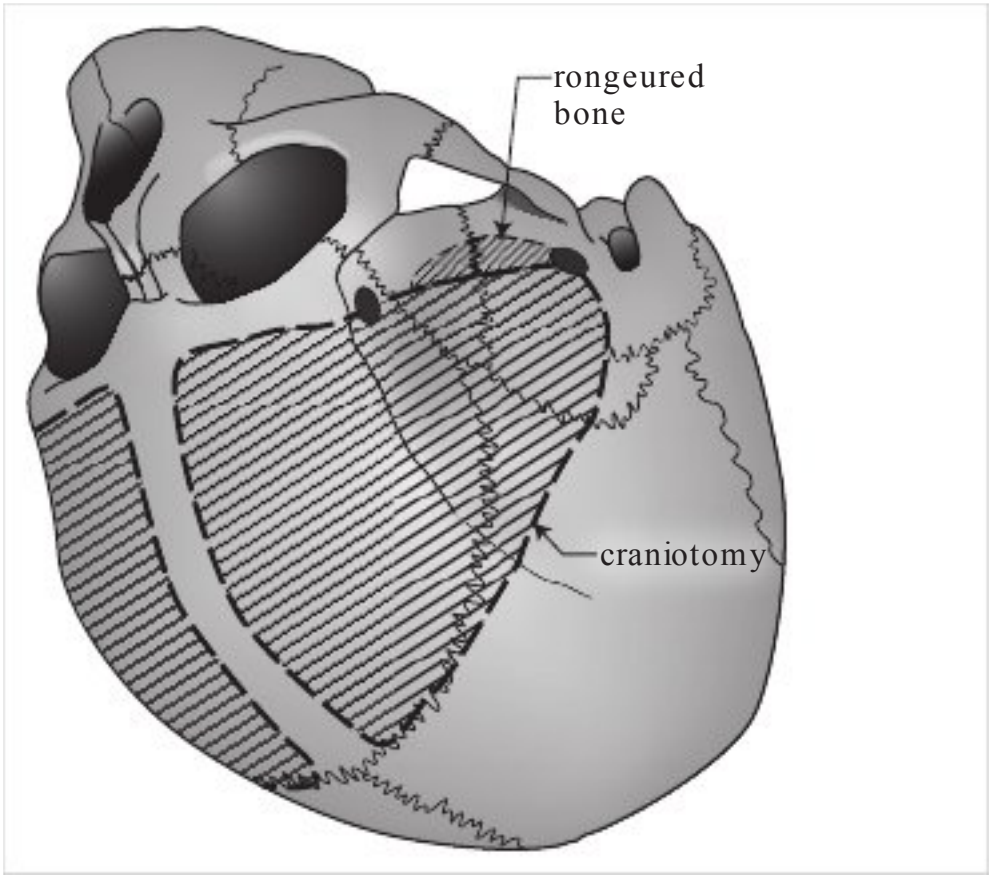


Fig. 94.17 Bilateral craniectomy skull flap Shown with 2 separate frontal flaps (the preserved midline bone strip over the superior sagittal sinus is optional)

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95 Spine, Cervical

95.1 Anterior approaches to the cervical spine

1. anterior odontoid screw (p.1476)
2. C1–3 (upper cervical spine):
 - a) transoral approach: including odontoidectomy (p.1472)
 - b) extrapharyngeal approaches: use nasotracheal intubation (so that the mandible can be completely closed) through the contralateral nares. The head is slightly extended and is rotated 15° to the contralateral side. Avoid any oral tubes
 - medial extrapharyngeal approach: medial to carotid sheath. Provides a more anterior position than the lateral retropharyngeal approach. Structures encountered: branches of external carotid artery, upper laryngeal nerves, hypoglossal nerve
 - lateral retropharyngeal approach: only the spinal accessory nerve is encountered
3. C3-C7: standard anterior cervical discectomy approach
 - a) For 1 or 2 level ACDF or 1 level corpectomy, a horizontal incision is usually employed
 - b) For more levels, a vertically oriented incision may be preferred to facilitate access

95.2 Transoral approach to anterior craniocervical junction

95.2.1 General information

Primarily useful for midline extradural lesions (approach to intradural lesions has been described,¹ but the use has been extremely limited because of difficulties obtaining watertight closure and increased risk of meningitis). Refinements in techniques and equipment (e.g. flexible reinforced oral endotracheal tube, McGarver or Crockard retractor, operating microscope, and suturing transnasal red-rubber catheters to the uvula to aid in retraction) allows access from as high as the inferior third of the clivus to as low as C3 (and sometimes C4²) vertebral body without need for tracheostomy or splitting of the tongue. Additional access can be achieved with use of extended techniques including splitting of the hard & soft palate, tongue splitting, and transmandibular approach.

95.2.2 Transoral odontoidectomy

Indications

Anterior extradural compression of the cervicomedullary junction as with pannus from rheumatoid arthritis, irreducible basilar invagination, tumors of C2, infection.

95

Stabilization

75% of patients undergoing transoral removal of the odontoid process required posterior fusion afterwards³ due to ligamentous instability.^{4,5} While the stabilization intuitively seems like it should be done first, it is often done following the decompression at the same sitting or at a soon to follow date. Some reasons for decompressing before stabilization:

1. positioning the patient for fusion may cause neurologic compromise if there is cord compression
2. a post-op MRI can be done to determine if enough decompression was achieved from the odontoidectomy. If not, a laminectomy can be done at the same time as the posterior stabilization
3. the amount of destabilization may not be known until after the odontoidectomy – in some cases a C1–2 fusion may suffice⁵

Stabilization usually entails posterior occipitocervical fusion. Occasionally fusion may be limited to C1–2 or C1–3 without the occiput. It is also possible to place an anterior strut between the body of C2 and the clivus, or between C2 and C1. Fibula is recommended. Metal instrumentation should be avoided.

Pre-op preparation

1. make sure that the patient can open the mouth at least 25 mm. If not, then other approaches such as translabiomandibular should be considered

2. for conditions resulting in malalignment or basilar invagination, cervical traction for 1 or more days is sometimes required
3. radiographic evaluation
 - a) cervical MRI without and with contrast to define the soft tissue pathology
 - b) CT of the craniocervical junction with sagittal and coronal reconstruction
 - c) CTA to assess the position and involvement of the vertebral arteries. Measuring the distance between the VAs provides useful information

Booking the case: Transoral approach

Also see defaults & disclaimers (p. 27).

1. position: supine with pin headholder
2. equipment
 - a) microscope
 - b) high-speed drill with long bits
 - c) C-arm
 - d) image guided navigation system (if used)
3. instruments
 - a) transoral set (usually includes oral retractor such as Crockard, Dingman, Dickman-Sonntag...)
 - b) long instruments: microdiscectomy instruments often work
4. anesthesia: awake, fiberoptic endotracheal intubation
5. some surgeons use ENT to perform the approach and closure and for follow-up
6. consent (in lay terms for the patient – not all-inclusive):
 - a) procedure: transoral resection of odontoid, placement of halo-vest immobilization, MEP monitoring (MEP should be consented specifically due to risk of seizures). Need for posterior stabilization at the same setting or in the immediate future
 - b) alternatives: nonsurgical management, radiation therapy for some diagnoses
 - c) complications: CSF leak with possible meningitis, spinal cord injury, wound breakdown, swallowing difficulties (may require PEG tube), breathing problems (may require tracheostomy), seizures with MEP

Technical considerations

For details, see references.^{2,3,6} Some key points:

Awake fiberoptic orotracheal intubation is employed. Nasotracheal (NT) is used by some, but at the narrow upper part of the exposure the NT tube tends to get in the way.

SSEP and MEP monitoring are used in appropriate cases.

Positioning: 3-point fixation with a Mayfield headholder is typically used. The patient is supine with no neck rotation (distorts the anatomical relationships and may bring one VA closer to the midline). Tilt the whole patient or the table towards the surgeon. 10–15° neck extension improves the exposure. Alternatively, the surgeon can stand above the patient who is kept perfectly supine.

A specialized retractor (e.g. Crockard transoral retractor) or a conventional Dingman retractor is placed. Verify that the tongue is not being compressed against the teeth.

Landmark: the tubercle of the atlas can be palpated through the posterior pharynx to locate the midline and for craniocaudal orientation.

The mucosa of the posterior pharynx is infiltrated with 1% lidocaine with epinephrine. Some authors culture the oropharynx to obtain drug sensitivities of organisms for use in the event of infection. Some advocate liberal topical use of 1% hydrocortisone ointment to the mucosa of the oropharynx and posterior tongue at the beginning and also during the operation to reduce intra-op and post-op swelling. Others feel it has no effect, and IV Decadron is used by some.

A 3 cm long vertical midline incision is made.

To reduce the risk of C1 spreading and allowing basilar invagination, C1 ring-sparing surgery may be attempted by removing only the inferior half to two-thirds of the anterior C1. When C1 ring-sparing is not done, the central 3 cm of the atlas is removed with a high-speed drill.

There is ≈ about 20–25 mm working distance between the two vertebral arteries at their point of closest approximation where they enter the foramen transversarium at the inferior aspect of the lateral mass of C2.

The odontoid is hollowed out “like a canoe”) using a high speed drill, checking progress on lateral fluoro at frequent intervals. Once the bone has been reduced to a thin shell, it can be fractured in

towards the hollowed out portion using curettes. The superior tip of the odontoid is particularly challenging due to the apical ligament.

Closure: a two layer closure is preferred by some. Others recommend a single layer closure incorporating deep muscle, superficial muscle, and mucosa.² If the dura has been violated, a fascial patch is secured with tissue adhesive and a lumbar subarachnoid drain is placed in the O.R. and maintained at low pressure for 3–4 days. An NG tube is placed under direct visualization to avoid injury or penetration of the mucosal closure.

Posterior stabilization

Transoral odontoidectomy produces instability in most cases (sometimes delayed).^{4,5}

For basilar invagination or occipitocervical instability, an occipitocervical fusion is recommended (p. 1474).⁶

For C1–2 instability alone, a posterior C1–2 arthrodesis may be performed (p. 1479).⁶

Possible complications

1. dural tear with CSF leak and risk of meningitis
2. vertebral artery injury
3. spinal cord injury

Post-op care

1. NG feeding or IV hyperalimentation is used initially (in anticipation of oropharyngeal swelling (lasts 2–3 days) and to avoid disruption of the mucosal closure)
2. intubation is maintained until the swelling subsides. Initial removal of the endotracheal tube over a tube changer facilitates reintubation if needed; the tube changer can be removed if no problem develops after 1 hour⁶
3. if the NG tube comes out, it should only be replaced under direct vision (usually by ENT physician) to avoid injury/penetration of the mucosal incision
4. halo-vest immobilization is maintained until the posterior fusion is performed
5. for staged procedures when the fusion is being done at a later date, a post-op MRI should be done to assess the degree of decompression. If further decompression is needed, then a laminectomy can be added to the posterior fusion

95.3 Occipitocervical fusion

The patient will lose about 30% of neck flexion mobility with an occipital to C1 fusion.

Indications for occipitocervical fusion⁷:

1. traumatic occipitoatlantal dislocation
2. absence of a complete arch of C1. Note: Alternatively, C1–2 lateral mass fusion (with or without lateral mass screws) (p. 1479) may be used in cases b and c if only the posterior arch of C1 is compromised
 - a) congenital
 - b) post-decompression
 - c) posttraumatic: “bursting” C1 fracture (bilateral or multiple C1 ring fractures). NB: some feel that this may be satisfactorily treated with halo immobilization until the atlas fracture heals (as they almost all do) followed by C1 to C2 wiring/fusion.⁸
3. congenital anomalies of the occipitocervical joints
4. upward migration of the odontoid into the foramen magnum
5. marked irreducible shifts of C1 or C2

Disadvantages of occipitocervical fusion:

1. loss of movement at the occipitoatlantal junction further reduces the range of motion as follows⁹:
 - a) flexion/extension: reduced by $\approx 30\%$ (13° occurs at occiput-C1 junction)
 - b) lateral rotation: 10° is lost
 - c) lateral bending: 8° is lost
2. non-union rate is higher than with C1–2 fusion alone¹⁰

Options:

1. keel plate (placed centrally over the thickest portion of the occipital bone) connected via rods to cervical screws (C2 pedicle screws and C3 lateral mass screws): reduced range of motion (ROM) to 17% of normal in a cadaver study¹¹ (for technique, see below)
2. occipital condyle (OC)-C1 polyaxial screws¹²: see below
3. occipital-C1 (AKA atlantooccipital) transarticular screws (see below)
4. looped rod wired to the occiput via wire cables placed through holes drilled in the occiput. Reduced ROM only to 31% of normal¹¹

95.3.1 Keel plate occipital-cervical fusion

Pre-op planning:

1. CT scan through C2
 - a) to rule-out aberrant position of foramen transversarium
 - b) to measure diameter of pedicles (may be best done on coronal sections due to the fact that the axial images are not usually oriented along path of pedicle) and estimate length of screws to be used
 - c) to verify trajectory of screws
2. measure thickness of occipital bone to determine screw-length for occipital screws

Technique:

1. occipital keel screws/plate
 - a) a drill, tap and screwdriver with flexible shafts or universal joints are usually needed because of interference from patient's skin
 - b) midline holes are preferred since occipital bone is thickest here
 - c) drill with drill guide to 8 mm, check depth with probe, if the inner cortex has not been breached, drill to 10 mm, check depth again, keep drilling 2 mm at a time until the inner cortex is breached, use that screw length
 - d) **SCREWS** 4.5 diameter blunt screws, 8–12 mm length
2. C2 pedicle screws (p. 1485)
3. C3 lateral mass screws (p. 1485) (if used)

95.3.2 Occipital condyle to C1 polyaxial screw fusion

See references.^{12,13}

Utilizes polyaxial screws placed in the occipital condyles that are then connected to screws placed at lower levels (see below) via connecting rods.

1. PROS compared to occipital plate/keel instrumentation:
 - a) circumvents problem of poor occipital bone purchase which may occur with keel plates
 - b) can be used even following posterior fossa craniectomy
 - c) greater surface area for fusion
 - d) avoids risk of intracranial injury from occipital screws
2. CONS: due to condylar variability, not all patients are candidates
3. biomechanics: compared to occipital plate, similar stiffness in flexion-extension and axial rotation, increased stiffness to lateral bending¹⁴
4. clinical: 1 patient, 2 years F/U with solid OC fusion¹⁵

Pre-op planning: CT scan occiput through C2.

Technique:

□ Structures to avoid include: hypoglossal nerve in hypoglossal canal (just above the occipital condyles (OC)), carotid and vertebral arteries, jugular bulb. IG may be helpful

1. occipital condyle screws
 - a) **ENTRY** 4–5 mm lateral to the foramen magnum, 1–2 mm rostral to the atlanto-occipital joint (do not need to or want to expose the entire condyle – there is an emissary vein laterally which is best left alone)
 - b) **TRAJ** 12–22° medial (mean: 17°), 5° maximal superior angulation
 - c) **SCREWS** 3.5 mm diameter polyaxial screws; bicortical purchase obtained using 20–24 mm length (mean: 22 mm)
2. condyle screws are connected with 3 mm diameter rods to either:
 - a) screws in C1 lateral mass and C2 pedicles (p. 1483), or
 - b) C1–2 transarticular screws (p. 1480)

95.3.3 Occipital-C1 (AKA atlantooccipital) transarticular screws

See references.^{16,17}

1. PROS: no compromise of C1–2 joint
2. CONS: steep trajectory requires additional incision at level of C-T junction
3. ENTRY midpoint of posterior C1 lateral mass
4. TRAJ 10–20° medially, aiming cranially to TARGET middle of occipital condyle
5. SCREWS 28–32 mm cannulated lag screws
6. biomechanics: $\approx$ equal to occipital plate-C1 lateral mass fusion¹⁸
7. clinical data: 2 cases reported, 2 year F/U, no complication

95.3.4 Post-op immobilization/bracing

1. for severe C1 fractures, or those with impaired bone healing capacity (elderly or unreliable patients, smokers...) a halo-vest is recommended $\times$ 8–12 weeks
2. otherwise, if C1 is not badly damaged, a collar that limits flexion (e.g. Miami-J collar) suffices $\times$ 8–12 weeks

95.4 Anterior odontoid screw fixation

95.4.1 Introduction

50% of axial rotation of the head occurs at the C1–C2 complex. Treatment of odontoid fractures by C1–2 fusion significantly reduces this mobility (although subaxial articulations will compensate to some degree over time). Odontoid screw fixation (OSF) attempts to treat odontoid fractures by restoring the structural integrity of the odontoid process (osteosynthesis) without sacrificing the normal mobility.

Stability of the C1–2 joint depends primarily on the integrity of the odontoid process and the atlantal transverse ligament (p.70) (which is the most important structure holding the odontoid process in position against the anterior arch of C1).

95.4.2 Evaluation

A full set of C-spine x-rays is needed, including an open-mouth odontoid view. MRI is recommended to rule-out disruption of atlantal transverse ligament. Cervical axial CT with coronal and sagittal reconstructions are also recommended to demonstrate the orientation of the fracture path and to verify the integrity of the posterior elements. Complex motion polytomography may be used if the CT findings are unclear.

95.4.3 Indications

Reducible odontoid Type II fracture (and Type III fractures where the fracture line is in the cephalad portion of the body of C2 in an elderly patient who may not fuse as well with immobilization as a younger patient¹⁹). The transverse ligament must be intact.

95.4.4 Contraindications

1. fractures of the C2 vertebral body (except cephalad Type III fracture)
2. disruption of atlantal transverse ligament: see Transverse atlantal ligament (TAL) injuries (p.1479). May be directly demonstrated on MRI. Indirect evidence: if the sum of the overhang of the lateral masses of C1 on C2 exceeds 7 mm (rule of Spence (p.970))
3. large odontoid fracture gap
4. irreducible fracture
5. age of fracture: controversial. Fusion rates in fracture $>$ 18 months old was 25%²⁰ Fracture $<$ 6 months old have $\approx$ 90% fusion rate²⁰
6. patients with short, thick necks and/or barrel chest: makes it difficult to achieve the proper angle. May be circumvented by the instrumentation distributed by Richard-Nephew which utilize a cannulated flexible drill, tap and screwdriver
7. pathologic odontoid fracture
8. fracture line in oblique orientation to frontal plane (shearing forces can cause malalignment during screw tightening)

Booking the case: Odontoid screw fixation

See defaults & disclaimers (p. 27).

1. position: supine head on horseshoe headrest, halter traction
2. anesthesia: awake fiberoptic or nasotracheal intubation. Do NOT use a wire reinforced endotracheal tube
3. equipment: 2 C-arms for biplane fluoro, or O-arm image guidance
4. instrumentation:
 - a) ACDF surgical set
 - b) tube retractor (e.g. METRx® by Medtronic)
 - c) some surgeons use specialized instrumentation (e.g. Apfelbaum set)
5. implants: cannulated screw set
6. consent (in lay terms for the patient – not all-inclusive):
 - a) procedure: surgery to place screw(s) from the front of the neck across the fractured odontoid bone. Possible posterior approach in case the anterior approach cannot be completed
 - b) alternatives: nonsurgical management in a collar, fusion
 - c) complications: screw breakage/pullout, failure to fuse which might require addition surgery (which will reduce neck motion)

95.4.5 Technique summary

Preparation

Various instrumentation systems have been developed to facilitate the procedure. The following describes some of the basic elements that are not specific to any one instrumentation (see reference by Apfelbaum²¹ for details of his instrumentation distributed by Aesculap Instrument Corporation, South San Francisco, CA).

Two C-arm fluoroscopy machines are mandatory for bi-plane imaging (simultaneous AP and lateral views). Some surgeons prefer placing 2 screws if there is enough bone to accommodate them, however, this may also reduce the amount of bone surface that can heal and the fusion rate appears to be the same.²²

Anesthetic considerations

The anesthesiologist is positioned at the foot of the table. Awake fiberoptic or nasotracheal intubation is recommended, especially for easily dislocatable fractures. Do NOT use a wire reinforced endotracheal tube since the wire interferes with the AP imaging.

Position

Supine. The neck is placed in extension (critical to performing the procedure) either with Holter traction and a small shoulder roll with the head on a doughnut (a strip of tape across the forehead stabilizes the head), or a radiolucent head holder may be used. Place the lateral fluoro unit first, then the AP unit slides into the “C” of the lateral unit. Lateral fluoroscopy is used to assess reduction of the fracture fragment, and the head is repositioned to try and achieve reduction. If there is retrolisthesis of the odontoid, the neck may need to be slightly less extended. A radiolucent mouth gag is placed to hold the mouth open for AP transoral imaging (a small tape roll works well). Abort the procedure if AP and lateral fluoroscopic views do not adequately image the odontoid.

Approach

A Cloward-type of horizontal skin incision at $\approx$ C5–6 (the entry site can be localized by placing a guidewire adjacent to the patients neck and taking a lateral fluoro) and approach identical to anterior cervical discectomy is used (all the way to exposing the longus coli muscles (p. 1073)). A Kittner is used to dissect superiorly anterior to the longus coli muscles in the loose areolar tissue up to C2. A self-retaining retractor (e.g. Caspar retractor – not distractor) with a superior retractor blade is attached (or a hand-held retractor, preferably radiolucent, may be used). Alternatively, a retractor tube system²³ (e.g. METRx® by Medtronic) may be used. The bovie is used to remove the soft tissue over the inferior front of C2.

Procedure

Localization: lateral fluoro is used to place the tip of an awl as far anteriorly as possible on the inferior endplate of C2 (□ Fig. 95.1; a common error is to go too far back along the inferior margin of C2, then the guidewire end up towards the back of the dens). AP fluoro is used place the awl in the exact center of the C2 body in the lateral dimension. The awl is used to make a pilot hole at this location.

Guidewire placement, drilling, tapping and ultimately screw placement are performed while monitoring the progress on frequent fluoro images, aiming for the exact middle of the dens on AP fluoro, and aiming towards the apex of the odontoid fracture fragment (skimming just within the anterior part of the C2 vertebral body) on lateral fluoro.

Drilling is performed under fluoro all the way through the apical cortex of the dens to avoid cracking the dens with the screw (the area just distal to the apex of the dens is safe).

A titanium partially threaded (lag) screw is placed. If an appropriately sized lag screw is not available, one can overdrill the part of the path through the body of C2 up to the fracture. In this way, a fully threaded screws can be used which will slip through the overdrilled hole and still have a lag effect on the fracture fragment. If a second side-by-side screw is used, it may be fully threaded. In chronic nonunion cases, prior to advancing the screw a bifaced curette may be inserted within the fracture space to freshen the fracture site. The screw(s) should be drawn up tightly to the inferior edge of C2. □ Fig. 95.1 shows the final position of an anterior odontoid screw.

At the end of the procedure confirm integrity of the transverse ligament by carefully flexing the neck under lateral fluoro.

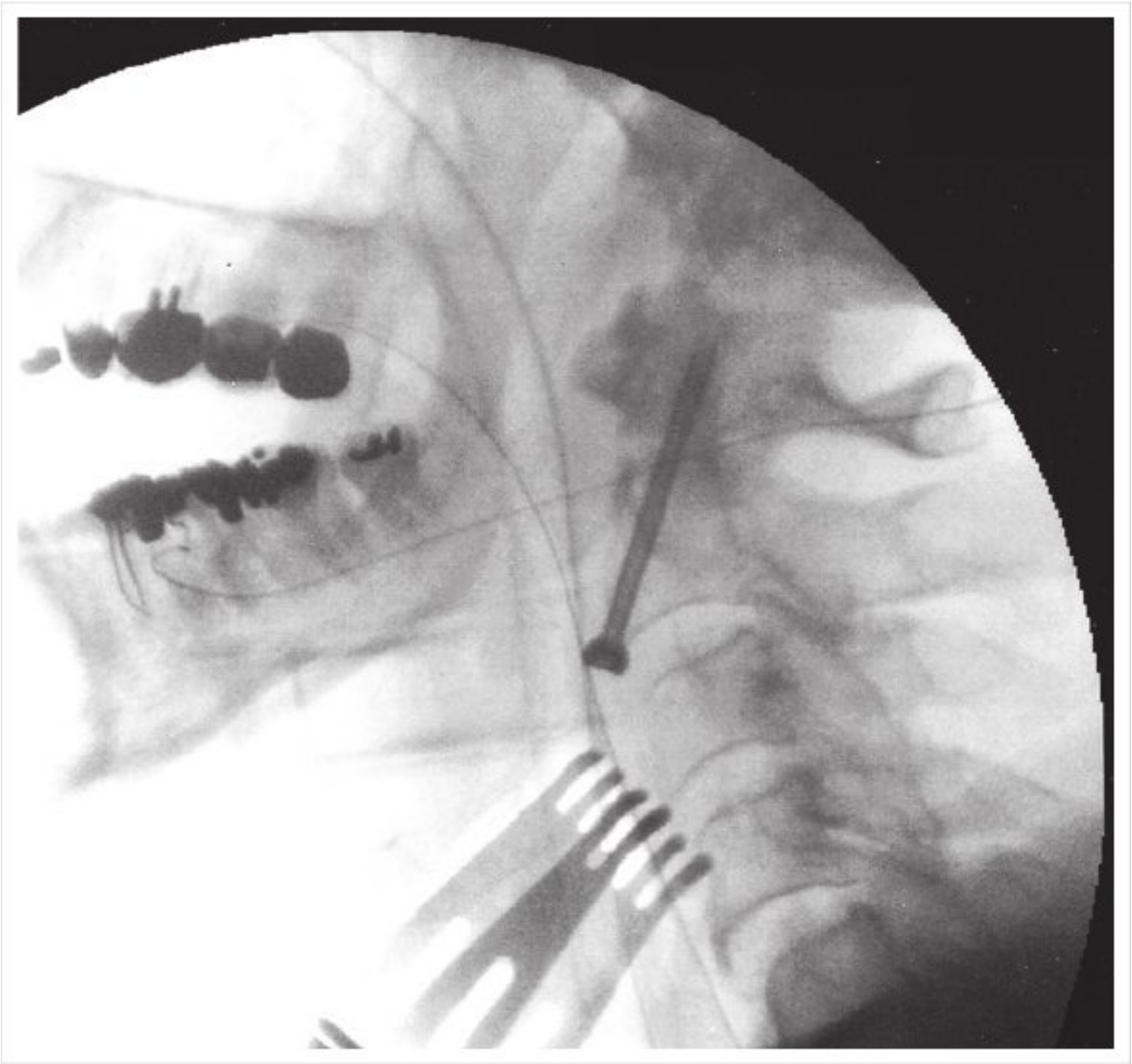


Fig. 95.1 Final position of anterior odontoid screw

Postoperative immobilization

The immediate post-op strength of the odontoid+screw is only $\approx 50\%$ of the normal odontoid. Therefore, a cervical brace is recommended for 6 weeks¹⁹ (although some authors don't use one²¹). If the patient has significant osteoporosis, a halo brace is recommended.

Results

Healing takes ≈ 3 months (or longer with chronic nonunion). With fractures <6 months old, the union rate was 95%. Chronic nonunions >6 months old have a significant risk of hardware failure (screw breakage or pull-out), with a bony union rate of 31% and 38% rate of presumed fibrous union.²¹ Thus, in cases of chronic nonunion >6 months old, C1–2 arthrodesis is probably a better choice unless the need to maintain motion is worth the risk of needing a second operation if this one fails.

The average technical complication rate is $\approx 6\%$ (2% screw malposition, 1.5% screw breakout).

95.5 Atlantoaxial fusion (C1–2 arthrodesis)

95.5.1 Indications

NB: The patient will lose $\approx 50\%$ of head rotation with C1–2 fusion.

- Instability of the C1–2 joints, including:
 1. atlantoaxial dislocation due to incompetence of the transverse atlantal ligament (TAL):
 - a) rheumatoid arthritis (p. 1136): symptomatic patients, or asymptomatic patients with subluxation ≥ 8 mm
 - b) local infection
 - c) trauma
 - d) Down syndrome (p. 1138): due to laxity of the TAL
 2. incompetence of the odontoid process
 - a) odontoid fractures meeting surgical criteria, including
 - Type II fractures with >6 mm displacement
 - instability at the fracture site in halo-vest traction
 - chronic nonunion of odontoid fractures
 - disruption of the transverse ligament
 - b) following transoral odontoidectomy
 - c) tumors destroying the odontoid process
- Vertebrobasilar insufficiency with head turning (p. 1307) (bow hunter's sign)

95.5.2 Technical considerations

Come cases require incorporation of the occiput in addition to C1–2.

Surgical options include:

- Rigid instrumentation:
 1. C1–C2 fusion using polyaxial screws connected by rods:
 - a) C1: screws placed in lateral masses. May be used in cases where the posterior arch of C1 is compromised
 - b) C2 screw options:
 - screws may be placed in pedicles (pars)
 - screws may be placed in lateral masses
 - crossed C2 laminar screws²⁴
 2. C1–2 posterior transarticular facet screws (TAS)^{25,26,27}

□ Posterior cervical wiring and fusion. With the development of rigid fixation, these techniques are used less frequently. While they are poor in limiting rotation, they are effective in limiting flexion. And since the Dickman & Sonntag technique is effective in limiting extension, it has recently been used to offload C1 lateral mass screws which have a tendency to break at the point of entry to the bone of C1

1. interspinous fusion technique of Dickman and Sonntag (p. 1483)
2. not presented here:

- a) Brooks fusion²⁸ (the Smith-Robinson technique as modified by Griswold²⁹): C1 to C2 sublaminar wires with 2 wedge bone grafts
- b) Gallie fusion³⁰ (p 1477–93) and its modifications: midline wire under the arch of C1 with an “H” bone graft

□ Halifax clamps with fusion.³¹ These clamps are effective in minimizing movement in flexion, but are less stable in extension or with rotation

□ Odontoid compression screw fixation (p.1476). Essentially only for odontoid Type II fractures <6 months old with intact transverse ligament (p.1476).³² Preserves more mobility than C1–2 fusion

□ Combined anterolateral and posterior bone grafting.³²

□ Combining anterior (transoral) decompression with posterior fusion. Indicated when a significant anterior mass is present causing neural compression and/or making passage of sublaminar wires at C1 unsafe

95.5.3 Techniques of atlantoaxial fusion

Positioning

The patient is placed in a halo ring (with a gap in the back and secured to the table using a Mayfield adapter) or Mayfield pin fixation and is then placed prone on the O.R. table on chest rolls. The table will usually need to be positioned in a maximal reverse-Trendelenburg position to bring up the surgical area. The patient's feet are allowed to rest on a padded footplate on the table to prevent the patient from sliding down. Lateral intraoperative x-rays are taken after patient positioning.

Incision and approach

A midline skin incision is made from just below the inion to the spinous process of C5 or C6.

C1–2 transarticular facet screws (TAS)

May be used as an adjunct to posterior C1–2 wiring and bone graft – e.g. technique of Dickman and Sonntag (p.1483) – to achieve immediate stabilization without the need for postoperative external orthosis, or in cases where the posterior arch of C1 is fractured or absent. □ A major risk of the procedure is vertebral artery (VA) injury. Therefore many practitioners have adopted C1 lateral mass screws (p.1481).

Selection of candidates

May be appropriate in elderly patients or those with rheumatoid arthritis, in whom there may be slow fusion, or for those who have failed a previous attempt at C1–2 wiring/fusion. Also in young individuals who have ligamentous laxity.

All patients must have thin cut CT scans from the occipital condyles through C3 with sagittal reconstruction through the C1–2 facet on both sides to look for the presence of a vertebral artery in the intended path of the screw. Also, risk of VA injury can be reduced using CT scans reconstructed along the planned trajectory of the screw (aiming from a point 4 mm above the inferior C2 facet to a point in the anterior C1 button on CT³³).

Technique summary

A number of instrumentation sets are available for the procedure, and each has its own nuances. The following is intended to primarily cover the basic procedure common to most or all (see reference by Apfelbaum²¹ for details with that system).

Position:

Patient supine, with the head clamped in a Mayfield head-holder with a slight military tuck of the chin. Lateral C-arm fluoroscopy is used for the procedure, and some have advocated biplane fluoro.

Approach:

Utilize a standard midline posterior laminectomy approach from occiput to the C3 spinous process. The lamina of C2 and the posterior arch of C1 are exposed to the lateral aspect of the C2 inferior articular facet. The lateral extent of the spinal canal is defined using a small angled curette. The C1–2 facet is curetted to facilitate arthrodesis and permits observation of the drill as it crosses the joint.

ENTRY 1–2 mm superior to the C2–3 facet on the midline axis of the pars interarticularis. The trajectory is determined fluoroscopically using a K-wire placed on the side of the neck as a guide, aiming it through the C2 inferior articular process, pars interarticularis, superior articular process and across the C1–2 articulation into the lateral mass of C1. This helps establish the appropriate entry site for the drill guide through a separate stab wound, usually around the T1–2 level, 2–3 cm off the midline.

TRAJ A pilot hole is then drilled using visual guidance to maintain a straight parasagittal course (it helps to stand on 1 or 2 footstools to eliminate some of the parallax error) and fluoroscopic guidance to maintain the trajectory towards the C1 lateral mass. An assistant can reduce any atlanto-axial translational malalignment using a towel clip on C1 or C2 just prior to the drill crossing the C1–2 facet joint. To minimize the risk of VA injury, keep the drill as far dorsally as possible within the pars interarticularis. The pilot hole is then tapped and a fully threaded titanium screw is placed. If brisk arterial bleeding (not bone bleeding) occurs after drilling or tapping the first side, the VA may have been injured. The screw may still be placed but the contralateral hole and screw should not be placed. A post-op arteriogram is then performed to assess for propagating thrombus or dissection. Barring any contraindications, the procedure is repeated on the contralateral side. After screw placement, then posterior bone fusion – e.g. technique of Dickman and Sonntag (p.1483) – is performed. External immobilization is usually not employed post-op (the screws are considered to supply adequate internal immobilization).

Results

A fusion rate of up to 99% with no complications has been reported.²⁵ Injury to the vertebral artery is the main potential complication.

C1–2 lateral mass screws

Placement of polyaxial mini screws in C1 lateral mass and C2 pedicle with rod fixation. Originated by Goel and Laheri³⁴ in 1994 and promulgated in 2001 by Harms and Melcher.³⁵

Advantages over C1–2 transarticular screws (see above):

1. the more superior and medial trajectory should reduce the risk of VA injury³⁵
2. may be used in the presence of C1–2 subluxation
3. may be usable in certain cases of aberrant VA course
4. in selected cases, this can be used for temporary fixation without fusion (since joint spaces remain intact) and the hardware may be removed after an appropriate time to reclaim motion in the C1–2 articulation

Booking the case: C1–2 lateral mass fusion

Also see defaults & disclaimers (p.27) and pre-op assessment (see below).

1. position: prone, pin headholder
2. anesthesia: awake fiberoptic or nasotracheal intubation
3. equipment: C-arm or O-arm image guidance
4. implants:
 - a) mini-polyaxial screws (smooth shank screws needed for C1)
 - b) cable required for interspinous graft (optional, but recommended)
 - c) have rep bring in occipital plates and instrumentation in case of inability to place C1 screws therefore enabling occipital-cervical fusion as a bail-out option
5. consent (in lay terms for the patient – not all-inclusive):
 - a) procedure: surgery to place screws & rods from the back of the neck to stabilize, and usually to fuse the top 2 bones of the neck
 - b) alternatives: nonsurgical management in a collar, in some cases screws may be temporary and no fusion would be done
 - c) complications: screw breakage/pullout, failure to fuse which might require addition surgery, loss of some neck bending motion is expected ($\approx 20\%$ is typical)

Surgical technique (excerpted highlights)

See references.^{35,36}

NB: if fusion is to accompany screw placement (i.e. permanent screw placement), strong consideration should be given to supplemental interspinous fusion, if not contraindicated (p.1483) to prevent fatigue breakage of C1 screws.

Applied anatomy: there is no true neural foramen at C1–2, the C2 nerve root lies on the posterior surface of the capsule of the C1–2 articular joint.

Pre-op assessment

It is mandatory to know the position of the VA on both sides (and in particular, the location of both foramina transversarium of C1), and the following bony information (requires thin-cut CTscan):

- 1. cranio-caudal thickness (height) of the posterior arch of C1 (in case the arch needs to be drilled to facilitate screw placement)
- 2. to determine screw length: distance from the planned entry point (see below) to the planned exit target (midposition of the anterior part of the superior C1 VB)
- 3. to estimate medio-lateral angle for screws

Approach

Completely expose the C1–C2 complex. Dissect over the superior surface of the C2 pars interarticularis to expose the C1–C2 joint to accurately locate the entry point for the C1 lateral mass screws. Bleeding is controlled with bipolar cautery and/or Gelfoam-thrombin. Complete exposure of the posterior face of the inferior C1 facet also mobilizes the C2 root from the underlying attachments and facilitates its inferior mobilization.

- 1. C1 lateral mass screws **ENTRY** visualization commonly requires caudal retraction of the C2 dorsal root ganglion (occasionally this may not be feasible³⁶; sacrificing the C2 root may be required but this can lead to post-op pain and numbness³⁷; technique is to divide the preganglionic nerve fibers and to close the dural defect³⁶). The screw entry point is the midpoint of the inferior part of the C1 lateral mass (for both mediolateral and cranio-caudal directions). An awl or a 1- to 2-mm high-speed drill is used to mark the position to prevent slippage while drilling the hole. Drilling a portion of the inferior arch of C1 is sometimes needed to allow screw placement (caution: the thickness of the arch in the cranio-caudal dimension varies widely, and the horizontal segment of the VA lies immediately above – use pre-op CT for planning)
- 2. C1 screw **TRAJ** averages $\approx 17^\circ$ medially, $\approx 22^\circ$ rostrally, **TARGET** the superior aspect of the anterior tubercle of C1 on lateral fluoro (see □ Fig. 95.2)
- 3. C1 **SCREWS** 3.5 or 4 mm diameter, length is determined from pre-op fine-cut CT to obtain bicortical purchase (□ CAUTION: the ICA may be as close as 1 mm to the ideal exit site of the screw³⁸ □ some authors use only unicortical purchase). The screw needs to be proud to bring it up to the level of the C2 screw (it may actually be necessary to have the C1 screw protruding 1–2 mm more than the C2 screw in order to allow rod attachment³⁶), and it should have an ≈ 8 mm unthreaded superficial portion to minimize irritation of the C2 nerve which could produce occipital neuralgia
- 4. C2 pedicle (pars) screws, see description of placement (p.1483)
- 5. if a fusion is to be performed: the posterior arch of C1 and the C2 lamina are decorticated with a drill. Onlay fusion substrate is then placed, taking care not to compress the dura. Optional adjunct: intra-articular decortication and packing bone within the C1–2 joint

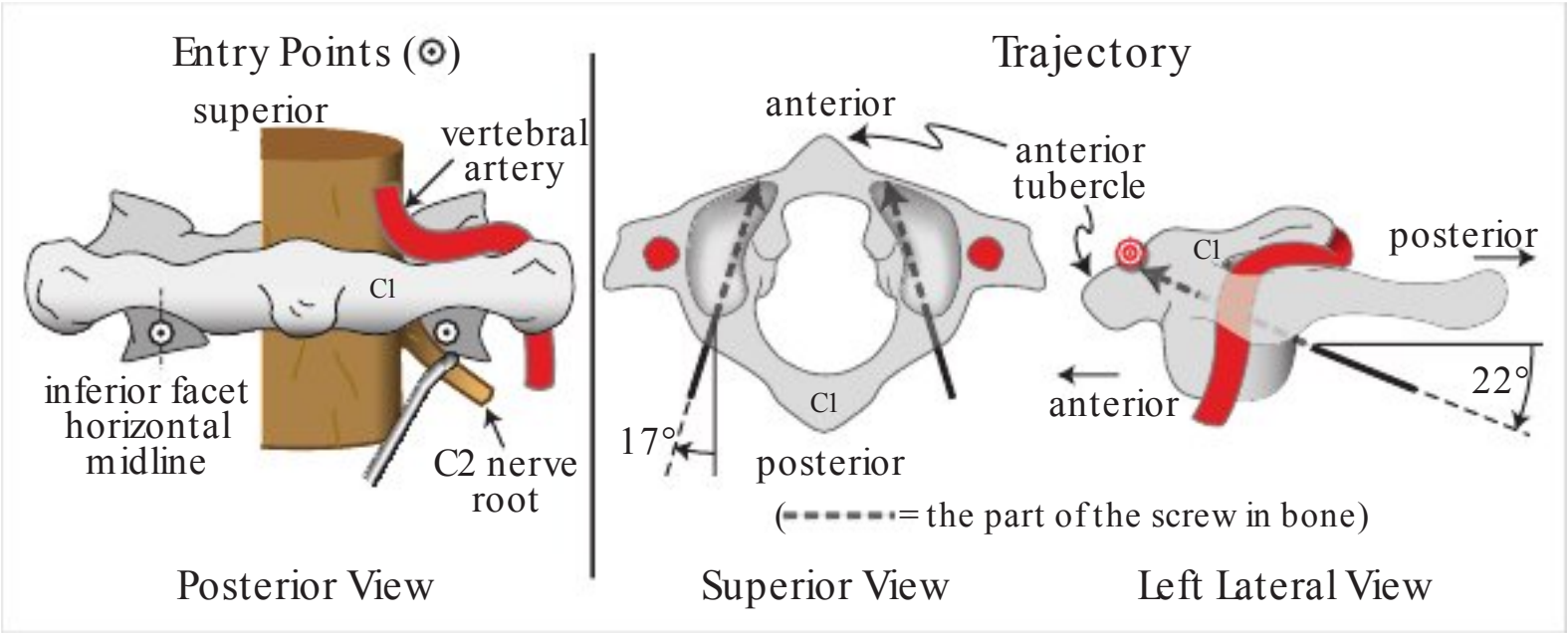


Fig. 95.2 Screw entry point and trajectory for C1 lateral mass screws

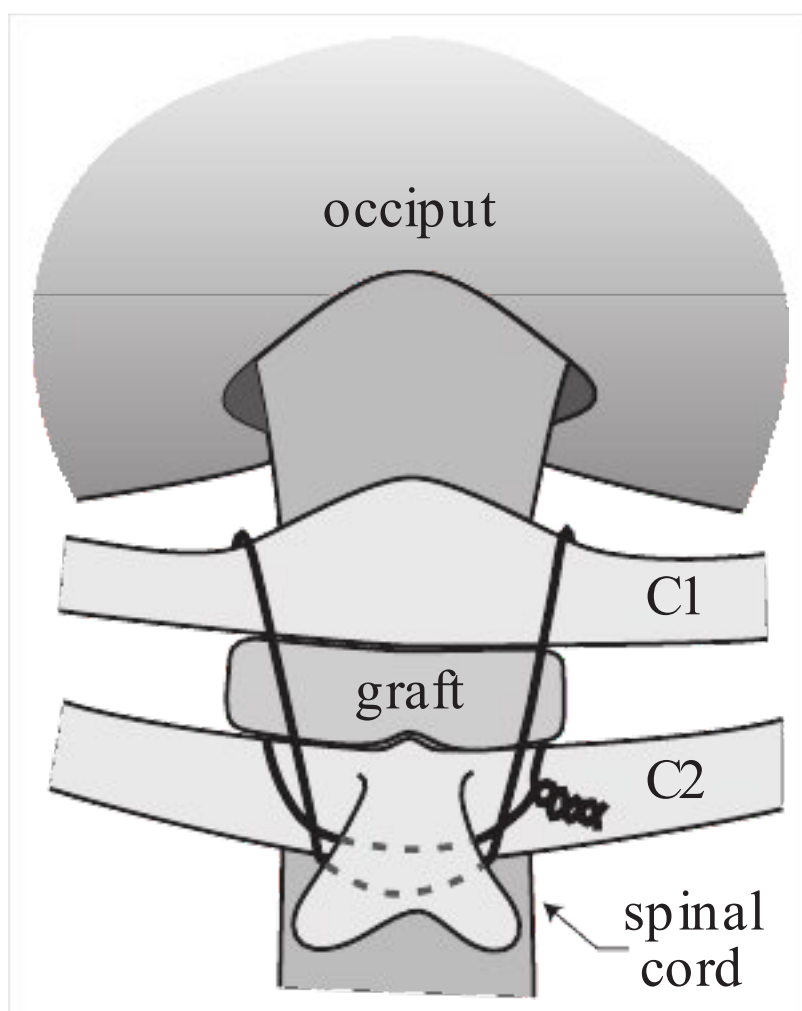


Fig. 95.3 Dickman and Sonntag C1–2 interspinous fusion

Post-op care

A cervical collar (soft or rigid, as preferred) for 4–6 weeks suffices.

Interspinous fusion technique of Dickman and Sonntag

A single bicortical graft is used, with multistranded cable passed sublaminar to C1 only. The bone graft is wedged between C1 and C2 (trapping it between loops of cable),^{39,40} see □ Fig. 95.3. Currently, this technique is infrequently used as the primary fixation for C1–2 fusion (unless technical difficulties prevent e.g. C1–2 lateral mass fusion). However, it may be most useful for limiting extension to offload C1 lateral mass screws to reduce the risk of screw breakage.⁴¹

Cannot be used if the posterior ring of C1 or C2 is fractured.

Bone graft

Autologous bone is preferred. Bone is often taken from the posterior iliac crest (p.1440). A tricortical graft of ≈ 4 cm length and >1 cm height is obtained. The top edged is removed to create a bicortical graft of ≈ 1 cm height.

95.6 C2 screws

95.6.1 Options

1. pedicle screws (pars interarticularis screws): directed medially (see below)
2. lateral mass screws: directed laterally. Length is sized to fall short of foramen transversarium
3. C1–2 transarticular screws (p.1480): associated with more risk of VA injury
4. translaminar screws^{42,43}: 1 year stability appears to be less than C2 pedicle screws when used for subaxial fusions, but was $\approx$ as effective for axial fusions (C1–2 or C1–3).⁴⁴ May be useful as a “bail-out” for subaxial fusions when the C2 pars diameter is too small for pedicle screws⁴⁵

95.6.2 C2 pedicle (pars) screws

Check CTscan or MRI to rule-out aberrant location of vertebral artery or unusual location of foramen transversarium before placing C2 pedicle screws. Some find image guided navigation systems to be helpful.

- Technique:
1. **ENTRY** palpate the medial and superior aspect of the pars with a Penfield 4 dissector (□ Fig. 95.4). Enter at the estimated center of the surface projection of the C2 pars at the midpoint medio-laterally³⁵ in the supero-medial quadrant of the surface of the C2 isthmus
 2. **TRAJ** 20–30° medially (through the central axis of the C2 pedicle),⁴⁶ 25° superiorly (on lateral fluoroscopy, place the screw parallel to the pars) (□ Fig. 95.5). To assist with trajectory, expose the proximal upper and medial border of the C2 pars interarticularis, and use a Penfield 4 to palpate during drilling (□ Fig. 95.4)
 3. drill a shallow entry point, then drill with drill-stop set at 12 mm, monitoring progress at intervals under fluoro and palpating with probe, and if no breakout, then complete drilling by gradually increasing drilling depth by 2 mm increments either up to 15–20 mm to stay in the pedicles, or up to ≈ 30 mm depth to perform osteosynthesis for a hangman's fracture. □ If withdrawal of the drill is followed by brisk bleeding, the screw should be inserted immediately to stop the bleeding. This bleeding may be from the vertebral artery; however, it is usually due to injury to the venous plexuses, and will not have any ill effects. In such cases it is best to not place the contralateral screw and to obtain an angiogram very soon post-op
 4. **SCREWS** 3.5 mm dia. Screw length is not critical except when attempting to bridge a fracture gap (osteosynthesis) e.g. with a hangman's fracture in which case screws of 20–30 mm length are placed to avoid penetrating anterior C2 cortex (lag screws are used for this, or the proximal bone can be overdrilled); for most purposes screw lengths of 15–20 mm length are used. Shorter screws (15–16 mm length) can still grip the pedicle with lower risk of VA injury

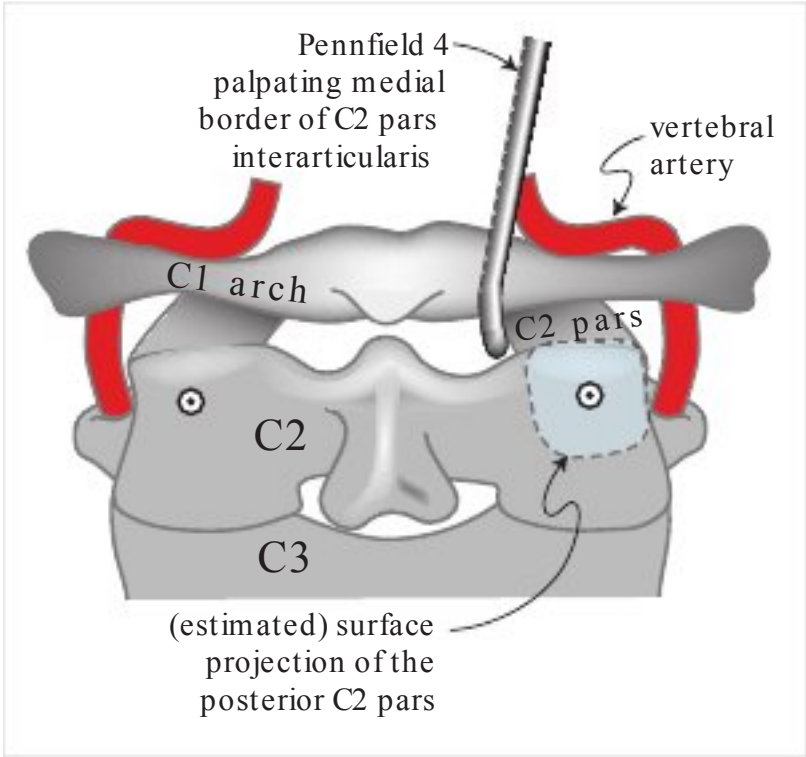


Fig. 95.4 Entry point for C2 pedicle screw placement (posterior view)

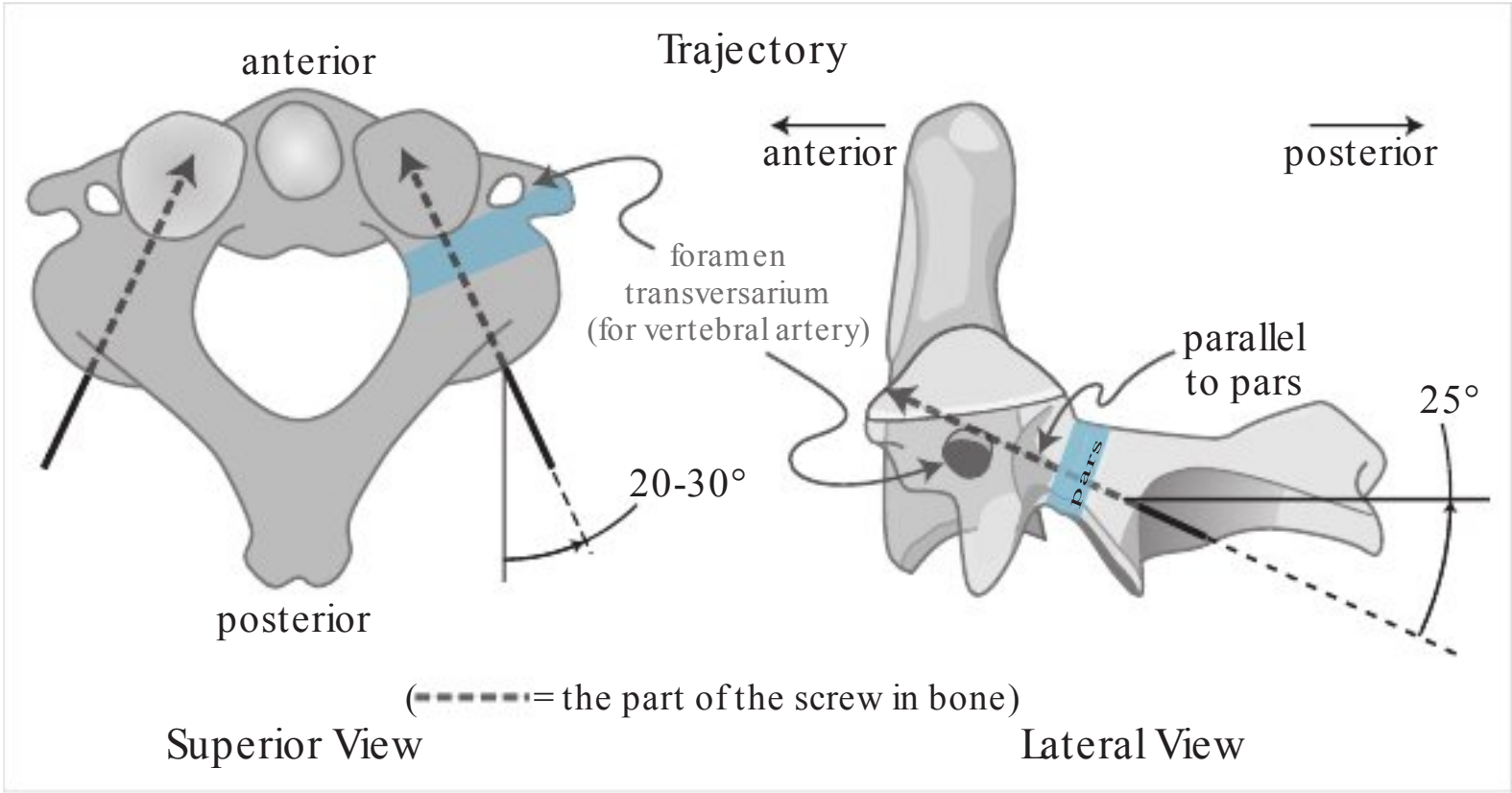


Fig. 95.5 Screw trajectory for C2 pedicle screws

95.6.3 C3–6 fixation

Lateral mass screws

Generally applicable to C3–6. The lateral masses of the thoracic spine are usually too small and not strong enough⁴⁷ for these screws. C7 is a transitional level, and lateral mass screws may sometimes be used. Occasionally even T1 may be amenable (see below).

Technique:

A number of methods have been promulgated with various screw entry points and trajectories (some are shown in □ Table 95.1). Comparing 3 techniques⁴⁸ there was a lower risk of nerve injury with the following (method of An⁴⁹):

- 1. **ENTRY**⁴⁹ 1 mm medial to the midpoint of the lateral mass (□ Fig. 95.6). In the cranial-caudal direction, the midpoint is used. A Penfield 4 may be used to palpate the medial wall of the pars to help determine entry point and trajectory
- 2. **TRAJ** 30° laterally, 15° cephalad (editor’s note: for upper cervical levels more cephalad trajectory is used, for caudal cervical levels 15° or less may be closer) (□ Fig. 95.6).⁴⁹ To get the lateral angulation, the holes are best drilled from the contralateral side of the patient, holding the drill shaft almost up against the spinous processes (if they are still present)
 - a) **SCREWS** 3.5 mm diameter, 14–16 mm length (for C3–6)
 - b) rod size: 3.5 mm diameter rods are usually used, and can be placed as far caudally as T3 as long as there is not gross instability (below T3, 5.5 mm diameter rods are used either via transitional rods or with rod connectors, e.g. “domino” connector)

Spinous process wiring may be used with intact spinous processes to help secure the bone graft.⁴⁷

Transarticular screw fixation

An alternative to lateral mass fusion. First described in 1972 by Roy Camille. May be used alone or as an anchor point.

Table 95.1 Comparison of methods for lateral mass screw placement for C3–6				
Method	Entry point		Trajectory angle	
	Medio-lateral	Cranio-caudal	Medio-lateral	Cranio-caudal
An	1 mm medial to mid-point	midpoint	30° lateral	15° cephalad
Magerl	2 mm medial to midpoint	2 mm cranial to midpoint	20–25° lateral	parallel to facet joint ^a
Roy-Camille	midpoint	midpoint	0–10° lateral	0°

^aangle can be determined by inserting probe into the joint

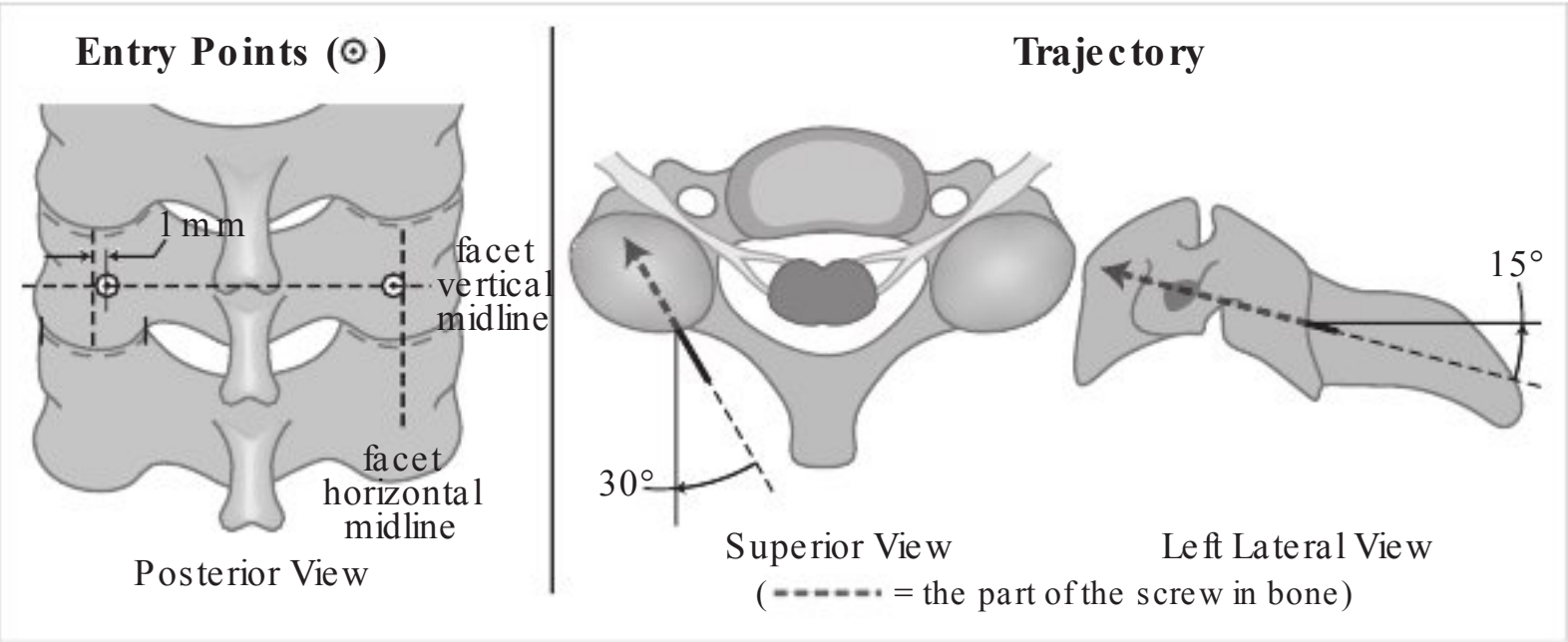


Fig. 95.6 Screw entry point and trajectory for C3–6 lateral mass screws (method of An)

1. PROS:
 - a) screws cross 4 cortical surfaces for better purchase
 - b) compresses across the joint to promote fusion
 - c) useful at cervico-thoracic junction where trajectory preserves facet capsule
 - d) lower implant profile
2. CONS: cannot correct deformity
3. ENTRY midpoint of lateral mass
4. TRAJ perpendicular to joint, neutral to 5° lateral (to avoid VA and exiting root)
5. biomechanics: stability equivalent to lateral mass screws⁵⁰
6. clinical: 25 patients (81 screws), 71 anchor, 10 fixation, 3.5 years F/U: solid fusion, no complications⁵¹

Translaminar cervical screw fixation

May be used in cervical or thoracic spine.^{52,53}

1. indications: salvage technique when anatomy precludes pedicle screws
2. PROS:
 - a) avoids complications related to pedicle screws
 - b) no need for fluoroscopy (reduces radiation exposure)
3. CONS: requires intact posterior elements (cannot do with laminectomy)
4. ENTRY contralateral spinolaminar junction (at base of spinous process)
5. TARGET junction of the transverse process and the superior facet contralateral to the entry point
6. SCREWS 3.5–4.5 mm × 26 mm polyaxial screw
7. biomechanics: no data
8. clinical: 7 patients (C-T fixation), 14 months F/U, no hardware complications. Inconsequential ventral penetration in 5%⁵²

C7 screws

C7 is a transitional level, and as a result either the lateral masses or the pedicles or both may be relatively small.

Screw fixation options:

1. pedicle screws (p.1489): recommended especially when the C7 lateral mass is of inadequate size for lateral mass screws.⁴⁹ Placement with fluoroscopy may be difficult due to shoulder artifact on lateral fluoro, and direct visualization of the medial wall of the pedicle may be required as in the thoracic spine
2. lateral mass screws⁴⁶:
 - a) ENTRY as for C3–6 (see above)
 - b) TRAJ compared to C3–6 screws, slightly less lateral at ≈ 15° and a little less cephalad at ≈ 10°
 - c) SCREWS 3.5 mm diameter, 14 mm length
 - d) biomechanics: lab studies indicate that C7 lateral mass screws are biomechanically equivalent to C7 pedicle screws in constructs extending down to C7⁵⁴
3. C7 transfacet screw⁵⁵:
 - a) PROS: reduced risk to spinal cord and nerve roots
 - b) CONS: disrupts C7-T1 facet capsule, so T1 must be included in fusion. short screws result in low pullout strength □ may be best used as an intermediate anchor point and not an construct endpoint
 - c) ENTRY 1–2 mm medial and superior to center of facet
 - d) TRAJ 30° inferiorly and 20° laterally, TARGET goal is bicortical purchase
 - e) SCREWS 3.5 mm diameter × 8–10 mm polyaxial screws
 - f) biomechanics: equivalent to C7-T1 pedicle screws⁵⁶
 - g) clinical: 10 patients, long cervico-thoracic fixation, 6 months F/U, 3 patients with solid fusion

95.7 Anterior vertebral body screw-plate fixation

Plate should be contoured to directly contact the front of the VBs being instrumented.

Typically, fixed screws are used in the lowest level, and variable screws are used at levels above. However, practices vary.

For the most rostral and most cranial screw holes, the plate should be sized such that the edge of the VB would just be visible when looking through the hole.

Screw length: Typical lengths vary from 12 mm (usually in women) to 16 mm. As a guide, if Caspar pins are used during the operation, most have 14 mm of threads in the VB, the screw length can

be chosen based on how close these Caspar pins come to the posterior VB. The screws in the plate are usually angled, so an additional 1–2 mm can be accommodated beyond this length.

95.8 Zero profile interbody devices

These devices incorporate holders for screw placement into an interbody cage without need for a separate anterior plate. Most often used in cervical spine.

1. PROS:
 - a) Are often easier to place adjacent to a previously placed anterior plate (because the plate will often cover too much of the VB to provide enough room for another plate on the same VB)
 - b) Avoids plates that are not parallel to the long axis of the spine
 - c) Posterior migration of the cage is prevented once the screws are placed
2. CONS: Biomechanical stability is less than that with a plate (4-screw devices are more stable than 3 screw devices)

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96 Spine, Thoracic and Lumbar

96.1 Anterior access to the cervico-thoracic junction/upper thoracic spine

96.1.1 Sternal splitting procedure

Permits access down to T3 (occasionally as far as T5) from an anterior midline approach (access to this region with a lateral (transthoracic) approach is poor due to the small volume of the pulmonary apices).

The neck and thorax are prepped down to the umbilicus. A hockey stick incision may be used, the horizontal portion is the usual for an ACDF. The vertical limb is centered over the sternum. In most cases, the services of a CV surgeon are employed to split the sternum and divide the sternocleidomastoid. This approach does not violate the pericardium or pleura, and a chest tube is not required (but is often used as a large bore drain to prevent hemomediastinum, and also as a precaution in case the parietal pleura is cut during the exposure). Because of the depth of the approach, longer instruments than the routine 7 inch length instruments used for an ACDF are required.

The exposed edges of the sternum may also be used to obtain cancellous bone for the graft.

96.2 Anterior access to mid and lower thoracic spine

96.2.1 Transthoracic approach

Position: lateral position on a bean-bag on the O.R. table with the break in the table under the level of pathology (remember to unbreak the table prior to instrumenting). Stabilize the patient using adhesive tape over surgical towels. An axillary roll is placed. A double-lumen endotracheal tube is used to permit deflating the lung on the side of the thoracotomy. If the patient does not tolerate completely deflating one lung, it is often adequate to just partially deflate the lung.

To increase the exposure, a rib may be resected. Generally, the level opened and the rib removed are one or two levels above the level of pathology (e.g. for T7 VB tumor, the T6 or T5 rib are removed).

If a compression plate is going to be used, the goal is to be lateral on the VB, to achieve this try to position it as far posterior as possible (rongeur off a little bit of the rib heads to facilitate this).

96.2.2 Anterior access to mid thoracic spine

Laterality of approach: if the pathology does not dictate use of one side over another:

1. advantages of right-sided thoracotomy: the heart, mediastinum and brachiocephalic vein do not impede access
2. advantages of left-sided thoracotomy: aorta is easier to mobilize and retract

Determining the level in the upper T-spine can be quite difficult intraoperatively. Counting up from the sacrum on an AP view using live fluoro sometimes will work when lateral spine x-rays cannot penetrate the lower c-spine due to the shoulders.

96.2.3 Anterior access to lower thoracic spine

Unless pathology is predominantly right-sided, a left-sided thoracotomy is preferred (easier to mobilize aorta than vena cava).

At about T10 and below, the attachment of the diaphragm increases the difficulty of the approach. In this area, a retrocoelmic approach (outside the pleural cavity) can facilitate surgery.

96.3 Thoracic pedicle screws

96.3.1 General information

Preferable to lateral mass screws because the transverse processes (which are analogous to the lateral masses in the cervical spine) of most thoracic vertebrae are not as strong.¹ Thoracic pedicles are

usually very narrow in the lateral dimension (the width is a little larger at the cranial end) and are very tall in cranio-rostral direction. Image guidance systems may also be helpful.

Accurate placement of thoracic pedicle screws is generally more challenging than the lumbar spine. There are at least 4 methods to place these screws, and a combination of methods may be used.

1. Intra-operative fluoro: for parts of the thoracic spine, biplane fluoro may be used as in the lumbar spine (see above)
 - a) PROS:
 - Allows percutaneous screw placement
 - generally good accuracy in screw placement
 - b) CONS:
 - due to the dense bone of the shoulders, the thoracic spine is usually difficult to image from T1 to about T4 on lateral fluoroscopy. For non-percutaneous cases, Steinman pins may be placed at the estimated entry points for the screws, and AP fluoro is used to fine-tune the position so that the screw enters the pedicle at the desired location
 - may increase radiation exposure to surgery team and patient
2. “Free hand” placement based on anatomic landmarks. X-rays are usually still obtained at the end, and any screws with unsatisfactory placement must be revised
 - a) PROS:
 - As the number of levels (and therefore, screws) placed in a patient increases, may save time over other methods
 - The small facetectomies that are needed to visualize the facet joint of the level below can facilitate correction of spinal curves and provide a good fusion surface
 - likely reduces radiation to the surgical team and the patient
 - b) CONS: Steep learning curve: this method probably takes the most practice to perfect
3. Performing small laminotomies at each level where pedicles are not exposed by a laminectomy, and use position of pedicle either by visualization or by palpate of the medial and superior aspect of the pedicles with a dissector to get an approximation of the entry point and pedicle trajectory
 - a) PROS: can permit accurate placement at essentially any level with potentially less radiation (depending on how often the surgeon checks screw position)
 - b) CONS: takes a little time at each level, but is over all comparable to other methods
4. Image guidance using instruments that are fitted with specialized markers that are tracked real-time by “cameras” that project the drill and/or screw location on a CT or x-ray image viewed in the OR.
 - a) PROS:
 - reduces intra-operative radiation to surgical team, and to a lesser extent to the patient
 - Allows percutaneous screw placement
 - b) CONS: accuracy may be compromised by movement of spinal segments relative to the registration array, or by technical errors. The surgeon must be vigilant for screw placement that does not look appropriate based on the anatomy

96.3.2 Fluoro and laminotomy thoracic pedicle screw placement technique:

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1. **ENTRY:** See Freehand thoracic pedicle screw placement technique (p.1491) for entry points using landmarks. Alternatives: making a small laminotomy to palpate the medial and cephalad edges of the pedicle with a Penfield #4 dissector, using Steinman pins at estimated entry points and taking AP fluoro to fine tune (see □ Fig. 96.1).
2. **TRAJ**
 - a) below T1: 5–10° medially and 10–20° caudad¹ (□ Fig. 96.2). A thoracic Lenke probe may be used as a pedicle finder.
 - b) T1: if a lateral mass screw is placed at T1 (instead of a pedicle screw), aim almost straight down at the floor (with patient positioned horizontally, i.e. without Trendelenberg or reverse-Trendelenberg)
3. **SCREWS** Smaller pedicles (usually T1–4, especially in females) usually require the smallest screw diameter (typically 4.5 mm). Others may accommodate 5.5 mm. Typical length: 20–25 mm
4. Rod size: when connecting to a cervical rod, down to ≈ T3 you can use a 3.5 mm diameter cervical rod throughout (here, the stiffer cobalt-chrome rod may be advantageous over titanium) with some systems (e.g. Mountaineer by DePuy using a 4.35 diameter screw). Below T3, ≈ 5.5 mm diameter rods (or 6.35 mm for scoliosis surgery) are usually used either via a transitional rod or using a domino connector to mate the two rods

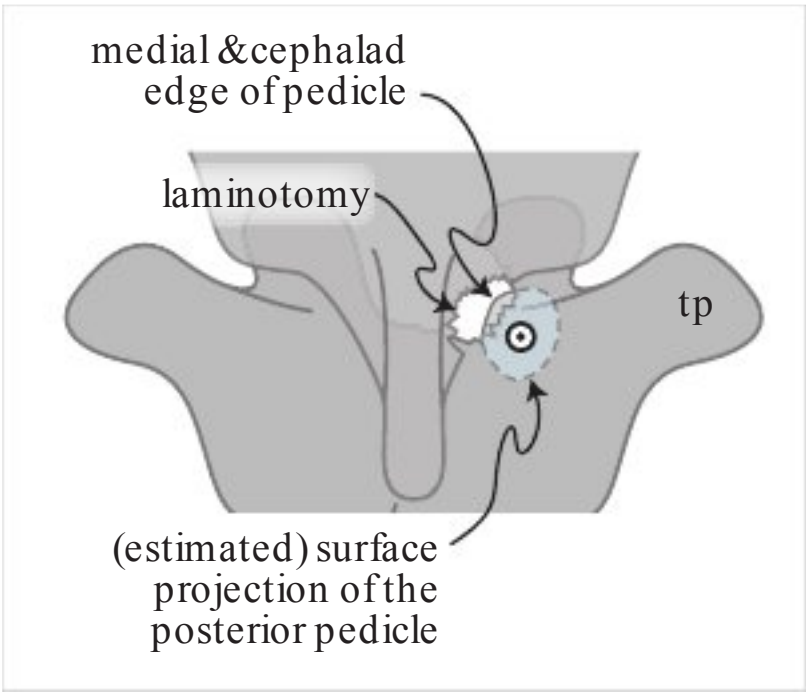


Fig. 96.1 Entry point for thoracic pedicle screws (posterior view) tp = transverse process

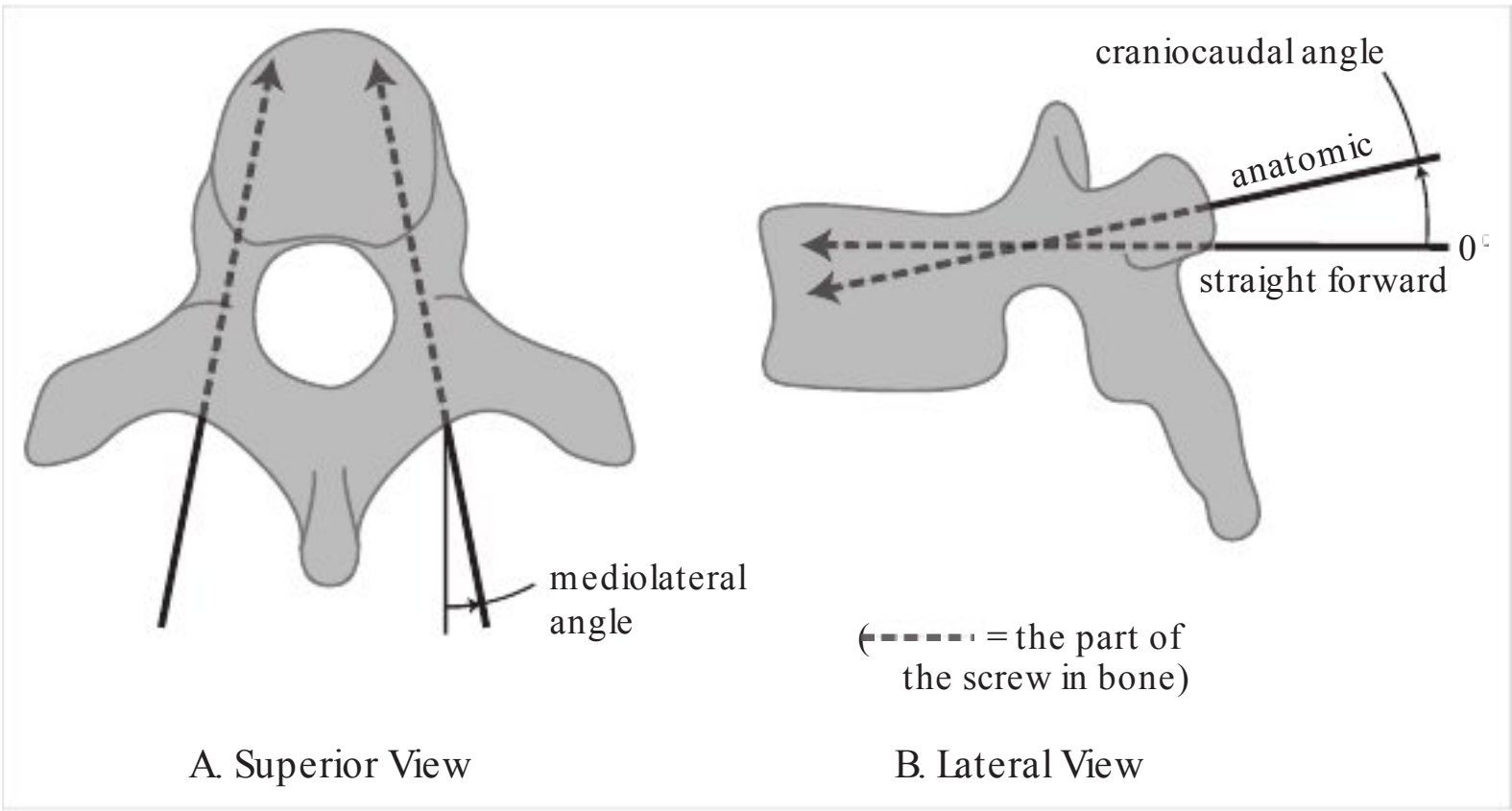


Fig. 96.2 Screw trajectory for thoracic pedicle screws

96.3.3 Freehand thoracic pedicle screw placement technique

Advantages and disadvantages

Possible advantages

- May speed up surgery especially when instrumenting a large number of levels
- Reduces the amount of x-rays/fluoroscopy needed during surgery
- Avoids the challenges of getting fluoro aligned for each level in a scoliotic spine especially when a rotational component is present
- Not impeded in areas that are difficult to image on fluoroscopy (typically upper thoracic spine)
- Accuracy as good or better than other techniques
- Obtains bone for use in graft (from facetectomies)
- Exposes bone (of facet joints) to assist in fusion
- Releases joints which facilitates reduction of scoliosis

Possible disadvantages

- Cannot be used if anatomy distorted by previous fusions, congenital anomalies...
- Steep learning curve: requires learning new aspects of anatomy, necessitates performing numerous screw placements, usually with a mentor, to become proficient
- Possible increased blood loss from increased exposure and facetectomies

Procedure

The details of the procedure are beyond the scope of this book. A summary of key points/information²:

- Complete exposure of entire posterior elements (lamina, pars interarticularis, facets, transverse process (tp)...) at every level is mandatory
- Intraoperative electrophysiologic monitoring: SSEP, MEP, triggered EMG
- Facetectomies: performed bilaterally at every level except the most superior (that one is not being fused to the level above). Make 2 cuts: 1st is parallel to canal. Caution: spinal cord is at risk on side of concavity
- It is essential to develop a series of steps that are repeated without deviation with every screw placement:
 - make pilot hole at entry point (as indicated below)
 - insert curved pedicle finder with the tip directed laterally until 2 cm deep (do not force)
 - completely remove the pedicle finder and rotate it 180° and re-insert it with the curve directed medially
 - palpate all 4 walls and the deep end with a ball probe, push it gently through the cancellous bone until the tip strikes the cortical bone of the anterior vertebral body (sound and feel). Place a clamp where the probe enters the bone and...
 - measure the depth that the probe was inserted to determine maximum screw length; if the length is much shorter than expected, the probe may be hitting the lateral VB wall and consideration should be given to redirecting the screw more medially
 - tap the pedicle (i.e. cut the threads for the screw using a tap)
 - probe 2nd time with ball probe
 - place screw
 - run a set of motor evoked potentials on electrophysiologic monitoring
 - after all screws are placed, stimulate each screw with triggered EMG: stimulation at <6 mA (some surgeons use cutoff of 8 mA instead) or a threshold that is 65% or more decreased from the average of the other screws indicates a possible medial pedicle breach and should prompt a re-evaluation of the screw (remove screw & re-probe the hole or obtain a fluoro image)²

Entry points²

- Craniocaudal (□ Fig. 96.3)
 - T1, 2, 3 & 12: even with the middle of the transverse process (mnemonic: T1–2–3 “mid tp”) same for T12 (like lumbar spine below it)
 - T7, 8 & 9: even with the top of the transverse process (mnemonic: T7–8–9 “top of the line”)
 - For levels in between these 2 groups (T4, 5 & 6), gradually move slightly superior from mid tp at T1–2–3 to the top of the tp at T7–8–9
- Mediolateral (□ Fig. 96.3)
 - T1, 2 & 3: even with the lateral edge of the pars
 - T7, 8 & 9: just lateral to midposition of the base of the superior articular facet (the most medial starting points)
 - T11, 12: at or just medial to lateral edge of pars
 - Levels in between these groups: gradually transition the position

Trajectories

- Trajectories using landmarks
 - Practitioners of freehand screw placement use landmarks more than angles. The advantage is that angles are difficult to estimate, and using landmarks allows screws to be placed even in rotated, scoliotic spines.
 - The screw is inserted perpendicular to the surface of the superior articular facet (that is exposed during the facetectomies) while also “aiming” at the contralateral pedicle.
- Trajectories using angles: may be helpful to conceptualize
 - Craniocaudal angle (□ Fig. 96.2 – B): either of 2 trajectories are used³
 - Straightforward screw placement: 0° to horizontal (parallel to the superior endplate – allows fixed-head screws for spine derotation)
 - Anatomic screw placement: 10–15° caudal⁴ (parallel to the pedicle – provides a longer path for screw contact with bone but requires a multiaxial screw head)
 - Mediolateral trajectory⁴ (□ Fig. 96.2 – A)
 - the angle gradually becomes more medial as you progress from
 - T12 where the angle is slightly lateral (≈ –5°) to

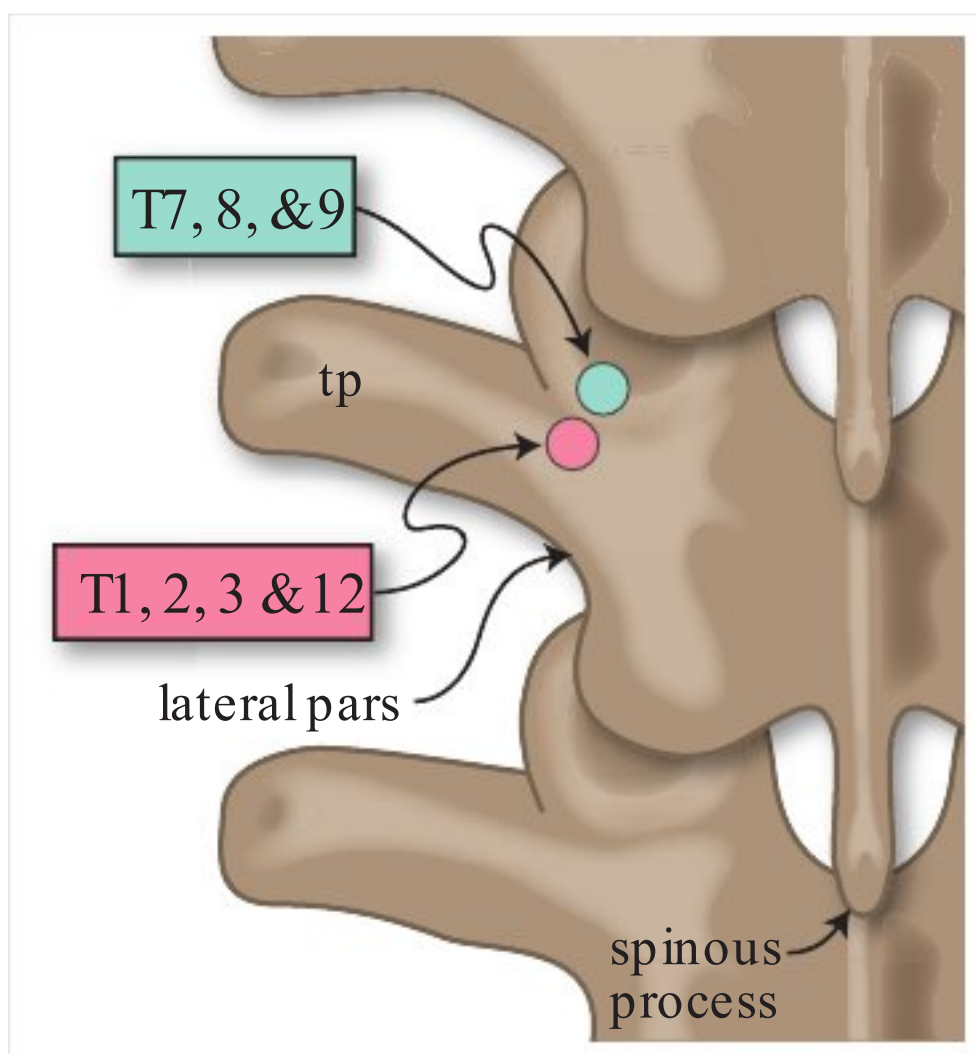


Fig. 96.3 Entry points for freehand thoracic pedicle screw placement on left side. Posterior view, stylized representation of generic thoracic level.

- T1 where the angle is $\approx 27^\circ$ medial
- (at T11 it is close to 0° and as a rough approximation it increases by 2° per level above this)

Screw size

- Length: the depth to the anterior cortex varies from 40–45 mm when measured along axis of pedicle (or 30–42 mm when measured parallel to sagittal plane).⁴ Typical thoracic screw length is 35–40 mm.
- Diameter: the narrowest pedicles in the mediolateral dimension are typically T4–7.⁴ Screw diameter should be approximately 80% of the pedicle diameter.

96.4 Anterior access to thoracolumbar junction

96.4.1 Retroperitoneal approach

Unless the pathology is predominantly left-sided, a left-sided approach is preferred because the spleen is easier to retract than the liver, and the aorta is easier to mobilize than the inferior vena cava. It is important to flex the ipsilateral leg to relax the psoas muscle, permitting safer retraction of the ipsilateral lumbosacral plexus.

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96.5 Anterior access to the lumbar spine

96.5.1 Anterior lumbar interbody fusion (ALIF)

Retroperitoneal usually through a Pfannenstiel's abdominal incision.

Relatively contraindicated in males because of risk of retrograde ejaculation in 1–2% (as high as 45% in some reviews). Other risks: injury to great vessels, especially with calcified arteries, especially at L4–5.

Bowel prep the day before surgery for complex cases.

Position: Trendelenburg position, place the level of the iliac crests over the kidney rest or use a sacral bump to increase lordosis.

As a result of the bifurcation of the great vessels (aorta and inferior vena cava) which ranges from just above to just below the L4–5 disc space, this approach is best suited for access to L5–S1.

At L5–S1, the anterior sacral artery runs down the midline of the VB and has to be sacrificed to do an ALIF.

96.6 Instrumentation/fusion pearls for the lumbar and lumbosacral spine

1. a lumbar fusion that includes L1 should not be terminated at L1 or T12
2. the taller the disc space the less likely that interbody grafts are well suited:
 - a) the disc may not be significantly degenerated to require discectomy
 - b) tall disc space means larger interbody implants which requires more retraction of nerve to insert (using a PLIF technique)
3. a long fusion should not be terminated at or near a vertebral level that is at the apex of scoliosis^{5 (p 382)}
4. laminectomy without fusion should be avoided at the apex of scoliosis
5. posterior midline fusion: early experience with midline fusions resulted in lumbar spinal stenosis as a late complication. Therefore, current fusion techniques include postero-lateral fusion, interbody fusion (from anterior or posterior approach), facet fusion...

96.7 Lumbosacral pedicle screws

96.7.1 General information

Pedicle screw pull out strength is determined in part by the major screw diameter which should be 70–80% of pedicle diameter (larger screws can break through the pedicle wall or can burst the pedicle). The minor diameter determines the strength of the screw and should be ≥ 5.5 mm in the adult lumbar spine. The length should allow penetration of 70–80% of the VB. Bicortical purchase or anterior VB penetration should be avoided to reduce the risk of injury to great vessels or abdominal viscera.

96.7.2 Placement techniques

There are at least 4 screw placement techniques

1. Intra-operative fluoro: biplane fluoro facilitates this technique
 - a) PROS:
 - allows percutaneous screw placement
 - generally good accuracy in screw placement
 - b) CONS:
 - imaging may be difficult in some parts of the lumbar spine, especially in larger patients. In these cases, the Steinman pin method below can be used to supplement
 - may increase radiation exposure to surgical team and patient
2. Steinman pin method: Steinman pins are placed at the estimated entry points for the screws, and AP and lateral fluoro is used to fine-tune the position so that the screw enters the pedicle at the desired location
3. Freehand placement based on anatomic landmarks. Usually with X-ray assistance. Greatly facilitated at levels where a laminectomy has been performed since the medial pedicle is exposed and is easily palpated
 - a) PROS: likely reduces radiation to the surgical team and the patient
 - b) CONS: requires somewhat more experience than the other methods
4. Image guidance using instruments that are fitted with specialized markers that are tracked real-time by “cameras” that project the drill and/or screw location on a CT or x-ray image viewed in the O.R.
 - a) PROS:
 - reduces intra-operative radiation to surgical team, and to a lesser extent to the patient
 - allows percutaneous screw placement
 - b) CONS: accuracy may be compromised by movement of spinal segments relative to the registration array, or by technical errors. The surgeon must be vigilant for screw placement that does not look appropriate based on the anatomy

96.7.3 Open lumbar pedicle screw technique (see below for percutaneous placement)

1. **ENTRY** at the base of the transverse process, at the intersection of the center of the transverse process (in the rostral-caudal direction) and the sagittal plane through the lateral aspect of the

superior facet. If a laminectomy has been performed at that level, the location of the pedicle is then verified by palpation using a probe within the spinal canal, otherwise fluoroscopy is used

- 2. **TRAJ**
 - a) the approximate mediolateral trajectory is shown in □ Table 96.1, and equals the lumbar vertebral number multiplied by 5° for each level from L1 to L5.⁶ The angle of the screw in the rostral-caudal direction is determined by fluoroscopy, maintaining a course that is parallel to the vertebral end plate. A frameless stereotactic navigation system may help orient screw trajectory
 - b) S2 screws are oriented laterally and superiorly and can be as long as 60 mm
- 3. **SCREWS** goal is for screw to cross ≈ 2/3 of the VB (typical screw lengths: 40-55 mm except for S1 which are usually only 35-40 mm long)
- 4. rod diameter: typically 5–6.5 mm

X-ray verification once pedicle screws are placed: on AP view if the screw tip crosses the midline to the contralateral side, there is likely to be a breach of the medial pedicle (sensitivity 0.87, specificity 0.97, accuracy 0.98),⁷ and if the screw does not pass medial to the medial pedicle wall there is likely to be lateral pedicle/VB violation (sensitivity 0.94, specificity 0.90, accuracy 0.96)⁷).

96.7.4 Percutaneous pedicle screws

The principles here are also employed in accessing the pedicles for e.g. vertebroplasty/kyphoplasty, percutaneous biopsy of pathology in the pedicle and/or vertebral body.

Basic principles:

- 1. requires AP and lateral fluoroscopy, or “O-arm” imaging (essentially an image guided intra-operative CTscan). With fluoroscopy, biplane fluoro (1 C-arm dedicated for AP view, another for lateral) greatly expedites the procedure
- 2. this method can be employed essentially from T1 through S1 as long as adequate AP & lateral imaging of the involved level is possible. Using fluoro for upper thoracic placement (e.g. above ≈ T5) is challenging (small pedicles, and the shoulders interfere with lateral x-ray)
- 3. the skin entry site is lateral to the lateral edge of the pedicle. This permits the needle to pass through the pedicle in a medial direction into the VB. The degree of angulation and therefore the distance off the midline for the entry site depends on the vertebral level being accessed (thoracic pedicles are oriented in a more AP direction, lumbar pedicles angulate medially inward) as well as the amount of overlying muscle/fat

Procedure:

- 1. a needle, typically a Jamshidi needle, is placed such that the tip is just short of entering the pedicle on the lateral fluoro (on left in □ Fig. 96.4)
- 2. at this point, on the AP view the needle tip should be at or just barely lateral to the lateral edge of the pedicle near the equator of the pedicle (on the right side, this would be at the “3:00” position, on the left side this would be at the “9:00” position)
- 3. the needle is advanced to just enter the pedicle on lateral fluoro, at this point it should be just within the pedicle margin at the 3:00 or 9:00 position on the AP view (as shown on the right in □ Fig. 96.4)

Table 96.1 Medial angles for lumbar pedicle screw	
Level	Medial angle
L1	5° medially
L2	10° medially
L3	15° medially
L4	20° medially
L5 & S1 ^a	25° medially
S2	40–45° laterally
^a aim for sacral promontory	

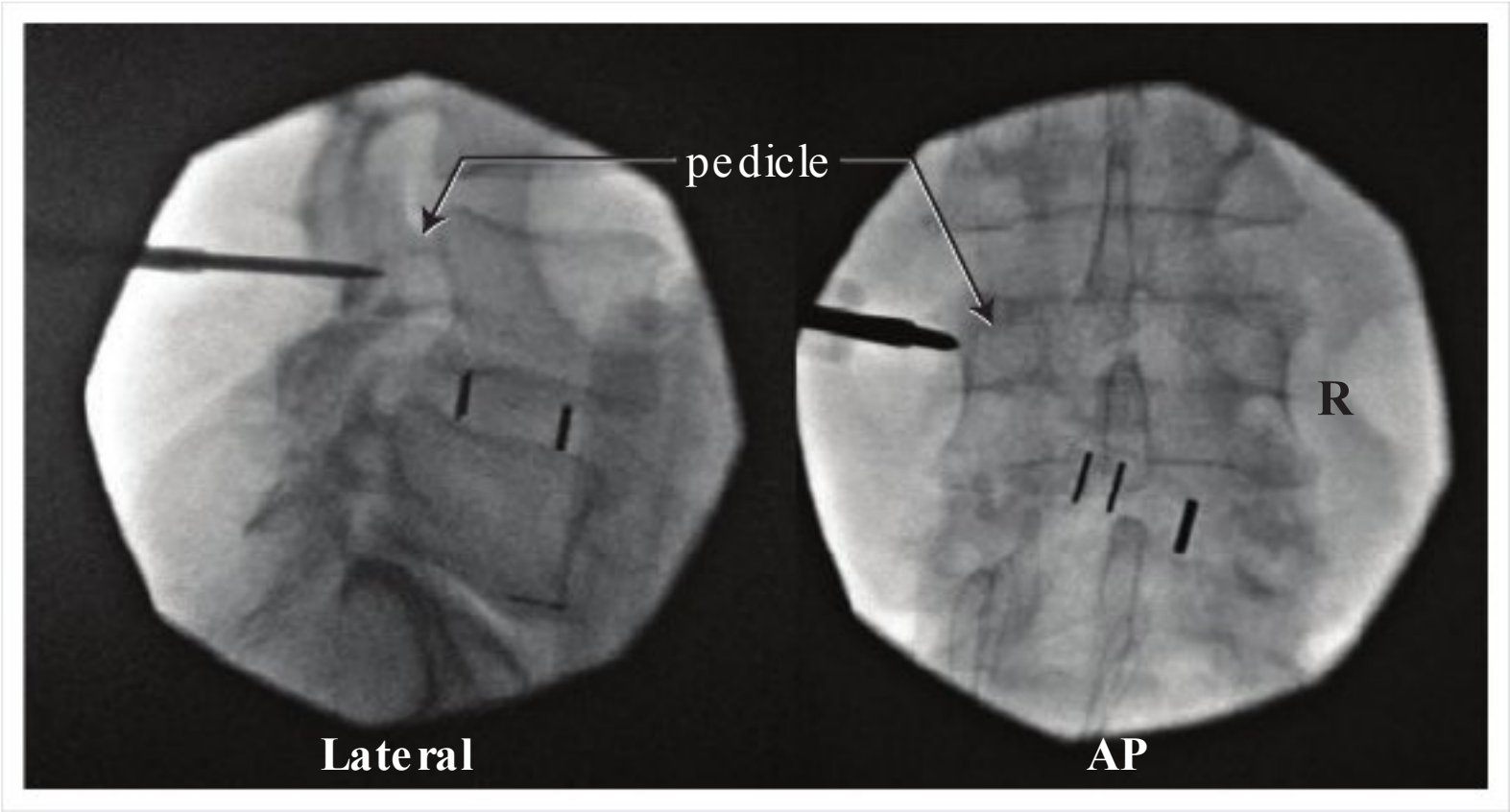


Fig. 96.4 Pedicle cannulation – entering pedicle

- 4. continue advancing the needle into the pedicle. Intermediate fluoro images can be obtained (e.g. to monitor trajectory on the lateral fluoro), but the next critical landmark is when the needle tip is just traversing the junction of the pedicle and the VB on the lateral fluoro (i.e. just entering the VB, as shown on the left in □ Fig. 96.5). it should be close to but never more medial than the medial border of the pedicle on the AP view (as shown on right in □ Fig. 96.5). If this criteria is maintained, the needle cannot breach the medial wall of the pedicle where it can threaten neural structures or compromise the purchase of the pedicle screw
- 5. subsequent steps differ for various procedures and manufacturers

96.7.5 Pedicle-screw rod diameters

Approximate weight guidelines for pedicle-screw rod diameters are shown in □ Table 96.2.

96.7.6 Lumbar lateral mass screw fixation

Trademarked names for equipment related to this procedure include “Cortical Screws.”

These screws enter the bone of the lumbar spine near the infero-medial border, and are directed superiorly and laterally, thereby passing close to 3 cortical margins, which provides a pull-out strength close to (but probably slightly less) than pedicle screws.

- 1. PROS: placement through a midline incision is somewhat easier than trying to get a more laterally situate entry point for a pedicle screw, and the supero-lateral trajecdtory does not require fighting the paraspinal muscles as occurs in trying to get a medial trajectory with pedicle screws
- 2. CONS:
 - a) pull-out strength slightly less than pedicle screws
 - b) the bone that has to be left unviolated at the screw entry site may interfere with some of the decompression

96.7.7 Translaminar lumbar screw fixation

- 1. indications:
 - a) short segment lumbar fusion
 - b) posterior component in a 360° fixation combined with interbody fusion
- 2. PROS:
 - a) small incision, minimal soft tissue disruption
 - b) decreased cost (fewer screws implanted)
 - c) decreased blood loss
 - d) adjacent facet joint spared

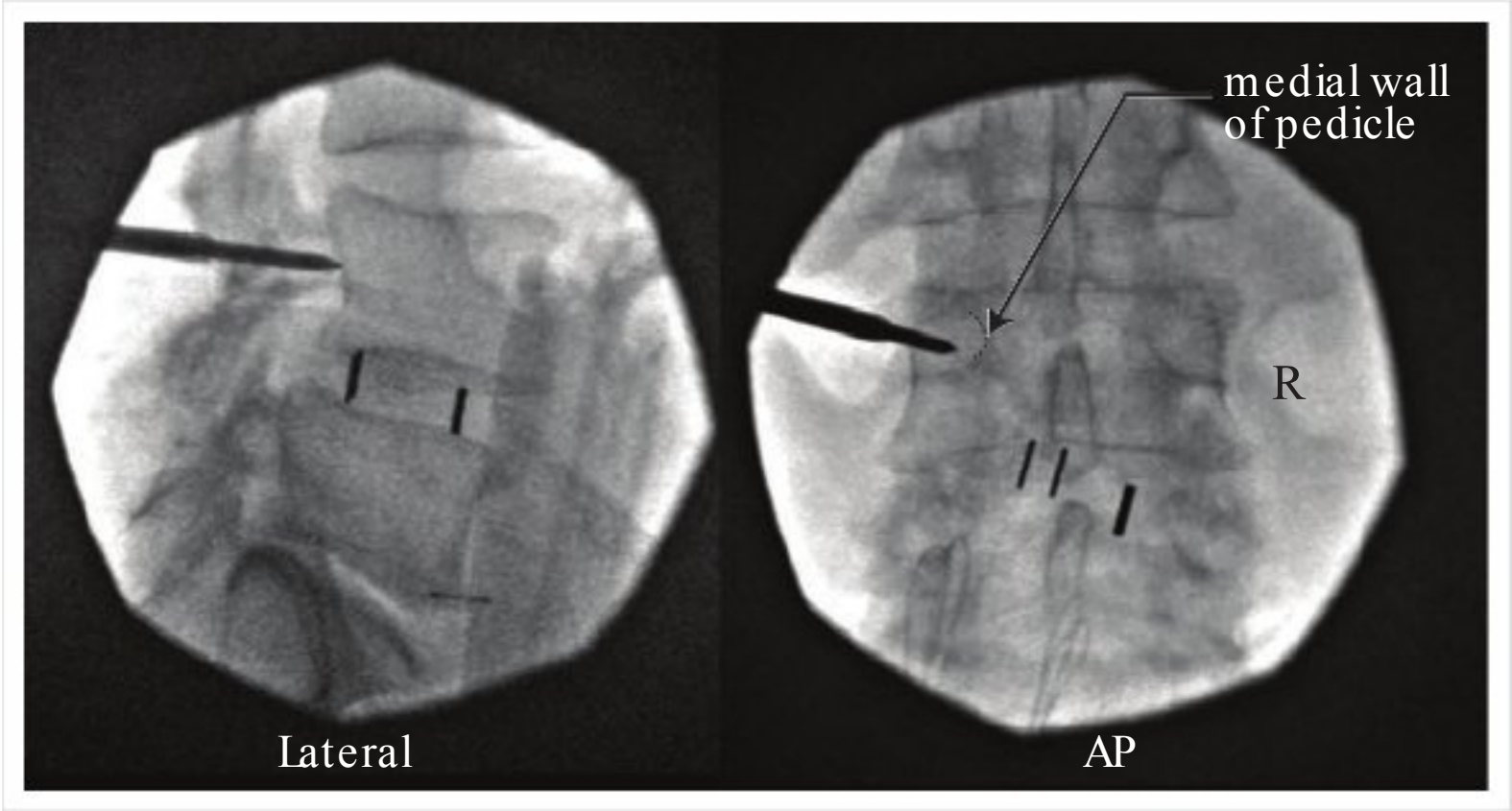


Fig. 96.5 Pedicle cannulation – entering the vertebral body

Table 96.2 Minimum recommended titanium rod diameter for lumbar pedicle-screw fixation		
Patient weight		Rod diameter (mm)
(lbs)	(kg)	
30–90	12–40	4.5
90–225	40–100	5.5
>225	>100	6.35 (1/4 inch)

3. CONS:

a) requires intact posterior elements (cannot use with laminectomy)

b) cannot reduce
4. ENTRY skin incision 5–7 off midline, screw entry into bone in contralateral spinous process. Can be placed bilaterally
5. drill between the tables of the lamina across the center of the facet joint, terminating at the base of the transverse process
6. SCREWS 4.5 mm diameter fully threaded screws (no polyaxial head)
7. biomechanics: equivalent to bilateral pedicle screws.⁸ Limited in extension⁹
8. clinical: 476 patients, 10 years mean F/U, 74%good outcomes⁹

96.7.8 Posterior lumbar interbody fusion (PLIF and TLIF)

Originally developed by Cloward¹⁰ in 1943. Bilateral laminectomy and aggressive discectomy followed by the placement of bone grafts into the decorticated disc space. It has been advocated to reduce the movement in an abnormal “motion segment” (defined as the area between two vertebra). Relatively contraindicated with well preserved disc-space height.

Many PLIFs when studied ≈ 1 year later show re-collapse of the disc space, which raises the question as to whether the PLIF has any benefit over simple discectomy. Areas of concern include: concerns regarding injury to nerves at the time of surgery or later due to retropulsion of bone graft.

Transforaminal lumbar interbody fusion (TLIF): a variation on a PLIF where the graft is placed from one side (via the “neural foramen) after complete removal of the facet joint on that side. Requires much less nerve root retraction than PLIF, and is often advantageous for re-operations with primarily unilateral pathology where going through the foramen avoids the scar tissue.

Stand-alone PLIFs or TLIFs may be associated with progressive spondylolisthesis at that level and are usually supplemented with pedicle screws/rods.

96.8 Minimally invasive lateral retroperitoneal transpsoas interbody fusion

96.8.1 General information

First introduced by Luiz Pimenta in 2001^{11,12} as an adaptation of an endoscopic lateral transpsoas approach to lumbar fusion described by Bergey et al.¹³ Trademarked names include “extreme-lateral” (XLIF™, NuVasive, San Diego, CA) or “direct-lateral” lumbar interbody fusion (DLIF™, Medtronic, Memphis, TN); the generic term lateral lumbar interbody fusion (LLIF) will be used here. Variants to the approach include the oblique lumbar interbody fusion (OLIF™, Medtronic, Memphis, TN) which utilizes a pre-psoas approach (at L5-S1 the OLIF is halfway between an ALIF and a LLIF). A retroperitoneal approach. Indirectly decompresses nerves by distracting the disc space and fuses the spine with an interbody cage having a large cross-sectional area. Access is best from L1–5. For L1–2, one can retract the 12th rib, or go between 11th & 12th rib, or excise the 12th rib. Iliac crest prevents access to L5-S1 (axial-LIF may be used here) and occasionally to L4–5 (see below). A similar retropleural approach may be employed in the thoracic spine up to T4. □ With thoracic lateral interbody fusions, DO NOT penetrate the contralateral anulus. Intra-operative EMG monitoring is critical, so the anesthesiologist needs to use only short-acting neuromuscular blockade at beginning of case. In males, implants are typically 55-60 mm in length (oriented along patient’s lateral axis) if placed in the midposition of the VB, or 45 mm in the anterior portion (lengths are 10% shorter in females).¹⁴ Potential advantages include less tissue trauma, minimal blood loss, shorter operation time, less wound issues, placement of a larger cage, early patient mobilization,^{15,16} no risk of durotomy with CSF leak.

96.8.2 Indications

- central lumbar spinal stenosis (mild to moderate only) with neurogenic claudication
- foraminal stenosis (indirect decompression)
- spondylolisthesis grade 1 or 2
- axial low back pain associated with degenerative disc disease
- total disc replacement
- for correction of sagittal or coronal imbalance
- adjacent segment failure: LLIF is particularly attractive here because it obviates dealing with scar tissue and (often) hardware from previous surgery and which also reduces risk of durotomy
- burst fractures and tumors in the thoracolumbar area (corpectomy)
- to retrieve damaged or malpositioned lumbar disc replacement devices¹⁷
- adult spinal deformity: can be used to correct scoliosis and to increase lumbar lordosis especially when combined with release of the anterior longitudinal ligament (ALL release)¹⁶

96.8.3 Contraindications

1. cases requiring direct decompression: includes
 - a) pathology within the spinal canal e.g. herniated disc, synovial cyst, where simply distracting the disc space is less likely to correct the pathology
 - b) somewhat imprecisely defined “pinpoint” central canal stenosis (some cases may still respond)
2. tall disc spaces: disc space height > 12 mm usually implies that further distraction may be difficult to achieve. However, the interbody cage can still prevent compression at these levels when the patient stands up
3. prior retroperitoneal surgery on the planned side of LLIF (can still be done on contralateral side, sometimes may still be feasible on ipsilateral side)
4. pathology at L5-S1: the technique cannot reliably access L5-S1 due to the interference from the ilium of the pelvis
5. may not be able to access L4–5 if the iliac crest extends more than ≈ halfway up the L4 VB. Sometimes it is necessary to position the patient on the O.R. table with the table flexed and a bump under the hip to see if the space will “open up” and permit access. Angled instruments can usually permit acceptable access to the space of the lumbar plexus is not located too far forward on the VB
6. anomalous vascular anatomy interfering with approach: check position of great vessels on pre
7. relative contraindications:
 - a) osteoporosis: may also be contraindication to lateral plates
 - b) active infection (relatively contraindicated with any fusion technique)

96.8.4 Surgical Technique (MIS retroperitoneal transposas approach)

1. Position
 - a) lateral decubitus position with the top of the iliac crest just superior to the table break
 - b) choice of side: If there is no reason not to do so, the left side is usually up. Factors that might influence which side is to be up:
 - right side up if access is needed to L4-5 and the iliac crest is higher on left and would interfere (use AP x-rays, lateral x-rays and/or lumbar CT to evaluate)
 - previous retroperitoneal surgery would cause one to consider placing the contralateral side up
 - if scoliosis is present and the intent is to correct this, the concave side is usually up: this usually provides better access to L4/5 if it is an operative level. Also usually allows access to multiple levels through potentially fewer and smaller incisions because the corridors to each disc space tend to converge
 - if ACR is planned, a posterior location of the great vessels and especially the lack of any soft tissue between the vertebral body/osteophyte and the vessel would speak for using the contralateral side if that is more favorable (if not, ACR may not be advisable)
 - c) true orthogonal position: the c-arm fluoroscopy is placed horizontal with 0° tilt, the patient position is then fine-tuned until the spinous process is exactly in the center between the pedicles on the AP view. If this is not possible due to rotation between levels, a relatively neutral level is chosen to start, and the OR table will need to be rotated slightly for each level as it is being worked on in order to make that level true lateral (centering spinous process). Adhesive tape is applied to maintain the patient in this position
2. lateral fluoroscopy is used to mark the index disc space transversely and the posterior third of the disc space vertically. An exception is at L4/5, where the vertical mark is at the middle of the disc space based on the anatomic safe zones¹⁸
3. retroperitoneal access through a single lateral skin incision and blunt dissection through abdominal muscles and fascia (external oblique, internal oblique, transversalis)
4. transposas approach and retractor placement is achieved using sequential tubular dilators that are placed under the guidance of fluoroscopic imaging (or navigation) and using directional EMG monitoring (Neurovision™, NuVasive, San Diego, CA) allowing the dilator placed anterior to the main lumbar plexus
5. discectomy and preparation of the disc space without violation of the endplates (to reduce the risk of subsidence) are performed. Work straight up-and-down in the ventral-dorsal plane (to avoid injuring the ALL anteriorly or entering the spinal canal posteriorly)
6. interbody spacer is placed usually with the posterior edge in the posterior third of the disc space (as the EMG monitoring allows)
7. for anterior column release (MIS-ACR), additional steps have to be performed after discectomy, which include dissection/section of the ALL, and placement of hyperlordotic cages (20 or 30 degrees) usually more anteriorly in the disc space than a routine cage. This is a technique for the advanced lateral access surgeon

96.8.5 Instrumented augmentation (pedicle screws or lateral plate)

96

Standalone cage

May be feasible in the following circumstances:

- no osteoporosis
- no instability on pre-op lateral flexion/extension x-rays
- the ALL was not disrupted during LLIF surgery
- a cage width of at least 22 mm (26 mm or larger is optimal) has been placed: the large surface area reduces the risk of subsidence

When these conditions are not met, additional instrumentation should be considered.

Lateral plate and screws

May be applied through the same exposure. May not be optimal in patients with osteoporosis or age > 55 due to increased risk of subsidence from weaker bone. Not as practical for multilevel LLIFs.

Posterior instrumentation

For cases with instability, posterior instrumentation (e.g. pedicle screws, including percutaneously placement) may be superior. Also, may be indicated if laminectomy is needed for direct decompression.

96.8.6 Complications

1. thigh numbness: incidence is $\approx 10\text{-}12\%$ ^{16,19} Due to injury to the genitofemoral nerve. Risk of direct injury in the quadrant anterior to the midline at L2-3 and in the anterior quadrant at L3-4 and L4-5.¹⁸ A sensory nerve – cannot be monitored with EMG. Usually transient, resolves in ≈ 2 weeks
2. thigh flexion weakness: due to psoas injury. Risk increases if >2 levels are done. Usually transient, resolves in 1–8 weeks
3. femoral nerve injury with quadriceps weakness: due to nerve root/plexus injury²⁰ (probably from psoas retraction – shorter operations may have lower risk) or psoas hematoma. Onset may be delayed 1–2 d post-op. Incidence $<1\%$ May improve slowly over a long time (>9 months)
4. Lumbar plexus injuries: the risk of direct injury to the lumbar plexus is reduced by staying at or anterior to “safe” working zones defined by Uribe et al.¹⁸ as follows (NB: it is often possible to work posterior to these zones using EMG monitoring to ascertain the proximity of the plexus, and conversely injuries can occur even in these safe zones):
 - L1-2 through L3-4: middle of the posterior quarter of the VB
 - L4-5: midpoint of the VB
5. contralateral femoral nerve injury
6. genitofemoral neuralgia
7. abdominal viscera perforation
8. vascular injury²¹ including common iliac artery (at the L4-5 level) or aortic injury (above L4-5), common iliac vein or inferior vena cava
9. kidney-ureter injury
10. graft subsidence
11. unintended rupture of anterior longitudinal ligament
12. psoas/retroperitoneal hematoma ipsilateral/contralateral
13. abdominal wall paresis or hernia²²
14. rhabdomyolysis
15. retrograde ejaculation (primarily with ACR and pre-psoas approaches)

96.8.7 Postoperative care

- for single-level lumbar LLIF: mobilize the patient in the immediate post-op period without a brace
- hip flexion pain on the approach side is anticipated during the immediate postoperative period
- transient hip flexion weakness related to psoas muscle manipulation is usually self-limited and improves by 8 weeks post-op
- in case of significant leg weakness (femoral nerve injury), a lumbar CT or MRI is indicated to rule out compression by a psoas hematoma, disc extrusion or malposition of cage or screw. If compression is ruled out, patients can be followed by postoperative EMG at 6 weeks to define extent of injury (neuropraxia, axonotomesis, neuronotomesis), again at 3 months to evaluate if the expected interval improvement from neuropraxia has occurred and 5 months to follow up axonal growth²⁰

96.8.8 Outcomes

- fusion rates range from 91-100%²³
- outcomes scales (ODI and VAS leg and back) are significantly improved at follow up²⁴

Booking the case: Lateral interbody fusion

Also see defaults & disclaimers (p. 27) and pre-op orders (see below).

1. position: lateral decubitus, typically left-side up unless specified otherwise
2. equipment: C-arm
3. implants:
 - a) interbody graft
 - b) some stabilizing hardware is usually needed, especially if spondylolisthesis is present. Options:
 - pedicle screws: bilateral, or unilateral
 - interspinous clamps
4. consent (in lay terms for the patient – not all-inclusive):
 - a) procedure: surgery through the side to place a spacer between two of the vertebrae (back-bones) to make more space for the nerves and to stop painful movement, screws/plates etc. will then need to be placed either from the side through the same opening or sometimes from the back. In case the procedure cannot be done from the side due to the position of the lumbar plexus (uncommon, when it occurs it is usually an issue at L4-5) determine if patient wants to have a posterior procedure (e.g. TLIF) and put that on the consent. Be sure to notify the vendor of this possibility
 - b) alternatives: nonsurgical management, open surgery through the back
 - c) complications: thigh weakness (usually temporary), knee weakness (uncommon), thigh numbness, graft subsidence/migration, failure to achieve desired relief

96.9 Transfacet pedicle screws

96.9.1 General information

Screws place directly across the lumbar facet joint into the pedicle of the level below. No rod is needed. Immobilizes only, does not provide any decompression, distraction or fusion. Therefore not intended for use as a standalone. Can be placed percutaneously.

96.9.2 Indications

Placement is optimal for L3–4, L4–5 or L5-S1. Difficulty increases in upper lumbar levels.

May be used as adjunct to:

1. ALIF
2. LLIF (when lateral plate not used)
3. contralateral to TLIF (pedicle screws could be used on the side of the TLIF, or a spinous process clamp could be used)
4. axial-LIF (Ax-LIF)

96.9.3 Contraindications

A transfacet pedicle screw cannot be used where the facet has been removed (e.g. for a TLIF) or with a pars defect in the upper of the two levels to be fused.

96.9.4 Technique

1. placed percutaneously or via an open procedure usually in the prone position
2. approximate skin incision site: a single midline ≈ 1.5 cm vertical incision is used
 - a) for L5-S1 or L4–5: incision at L3 spinous process
 - b) for L3–4: incision at L2 spinous process

3. use AP & lateral fluoro to guide trajectory
- a) AP fluoro: lay a guidewire on the patient’s back and orient it to pass through the desired pedicle. Use a skin marker on the patient’s back to mark the guidewire’s trajectory

b) lateral fluoro: initial bony target is the midpoint of the inferior facet of the upper level. The tip of the guidewire should contact the bone directly posterior to the inferior endplate of the upper level

96.10 Facet fusion

A bone dowel (e.g. TruFUSE® by MinSURG) is placed into a predrilled opening in the facet joint to promote fusion across the joint. Marketed as a possible standalone.

96.11 S2 screws

May be directed medially (analogous to pedicle screws), or more commonly, directed laterally and superiorly into the ala. In either case, bicortical purchase is necessary.
The main structure to avoid is penetrating the sacroiliac (SI) joint with the screw.

96.12 Iliac screws

Wide exposure is needed. On the initial few cases, the surgeon may be better served by exposing all the way to the posterosuperior aspect of the sciatic notch so that the screw trajectory can be aimed using a palpating finger.
A small amount of bone is removed just below and medial to the posterior superior iliac spine. This prevents the head of the screw from being too superficial which may cause discomfort or skin breakdown. The screw trajectory aims towards the acetabulum to pass approximately 1 cm superior to the sciatic notch on AP fluoro. Avoid penetrating the cortex, especially in the sciatic notch.
An offset adapter is usually required to connect to rods passing through pedicle screws in the levels above.
SCREWS Length 50–70 mm (the screw should end just above midpoint of the sciatic notch, or slightly medial to that). Diameter: 6–8 mm.

96.13 Post-op clinic visits – lumbar and/or thoracic spine fusion

96.13.1 Visit schedule

Patients are seen in the clinic at intervals depending on the preference of the surgeon. A typical follow-up schedule with studies routinely performed is shown in □ Table 96.3. For specific problems, additional investigations are usually needed.

Table 96.3 Sample post-op lumbar fusion clinic visit schedule ^a	
Time post-op	Agenda
7–10 d	wound check, D/C sutures/staples if used
6 wks	AP & lateral LS-spine x-ray in brace
10–12 wks	• AP & lateral LS-spine x-rays with flexion/extension views out of brace • if x-rays look good and patient is doing well, begin weaning brace
6 months	• AP & lateral LS-spine x-rays with flexion/extension views • some surgeons release patients at this time if they are doing well
1 year (optional)	• AP & lateral LS-spine x-rays with flexion/extension views • release patient if they are doing well
^a the same schedule can be used for thoracic fusions with the difference that standing AP & lateral x-rays are done in place of flexion/extension views	

96.13.2 Post-op x-rays

Items to check on post-op x-rays include:

1. alignment
2. position of grafts if used (e.g. interbody grafts)
3. integrity of hardware (look for screw or rod breakage, screw pullout, rod disconnection)
4. lucencies around screws which may indicate motion and implies nonunion
5. any evidence of fusion (may be difficult, e.g. with synthetic interbody cages)
6. on flexion/extension films: look for motion across fused segments (sometimes absence of motion may be the only evidence of fusion on plain x-rays) and the development of abnormal motion at adjacent segments.

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97 Miscellaneous Surgical Procedures

97.1 Percutaneous ventricular puncture

97.1.1 Indications

In pediatrics, may be used to remove hemorrhagic ventricular fluid following intraventricular hemorrhage, or to obtain CSF specimen in cases of suspected ventriculitis. May be used emergently in pediatrics or adult as a temporizing measure in patient herniating from obstructive hydrocephalus.

97.1.2 Peds

Shave hair. 5 minute Betadine® prep.

The right side is preferred. Enter through coronal suture just lateral to anterior fontanelle (AF) using a 20–22 Ga. spinal needle. If a CT scan has been done, it may be used to help judge angulation (usually varies between contra- and ipsi-lateral medial canthus and intersection with EAM).

97.1.3 Adult

See reference.¹

Only used emergently. Takes advantage of thin orbital roof in adult.

Prep conjunctiva and skin with antiseptic (e.g. ophthalmic betadine). Elevate the eyelid and depress the globe. Using a 16–18 Ga. spinal needle, penetrate the anterior third of orbital roof (1–2 cm behind orbital rim) with firm pressure (may need gentle tapping). Aim at coronal suture in the midline. The frontal horn should be about 3–4 cm deep.

97.2 Percutaneous subdural tap

97.2.1 Indications

Utilized in pediatrics. Used to be done for diagnostic purposes, but this has been supplanted by CT, MRI & ultrasound. Currently, this procedure may be used emergently for decompression, to drain subdural collections and to obtain fluid for diagnostic tests, such as culture (repeat taps may be used, but surgery should be considered after ≈ 5 –6).

97.2.2 Technique

Shave hair. Prep 5 minutes with povidone iodine (Betadine®). Using a short 20–21 Ga. spinal needle (spinal needle is recommended because the stylet may reduce the risk of implanting epidermal cells into the CNS), penetrate the lateral margin of the anterior fontanelle (AF) or coronal suture at least 2 cm off midline. Remove the stylet and aspirate. With bilateral fluid collections, bilateral taps should be done.

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97.3 Lumbar puncture

97.3.1 Contraindications

1. risk of tonsillar herniation (see below)
 - a) known or suspected intracranial mass
 - b) non-communicating hydrocephalus
2. infection in region desired for puncture: choose another site if possible
3. coagulopathy
 - a) platelet count should be $>50,000/\text{mm}^3$ (p.154)
 - b) patient should not be on anticoagulants because of risk of epidural hematoma (p.1131) or subarachnoid hemorrhage² with secondary cord compression
4. use caution in suspected aneurysmal SAH: excessive lowering of the CSF pressure increases the transmural pressure (pressure across the aneurysm wall) and may precipitate rerupture
5. caution in patients with complete spinal block: 14% will deteriorate after LP³

6. relative contraindication: Chiari malformation. There is some evidence that draining CSF may precipitate herniation. This is less of a concern with successfully surgically treated Chiari malformation.

Elevated ICP and/or papilledema by themselves are NOT contraindications (e.g. LP is actually used diagnostically and as a treatment in idiopathic intracranial hypertension, see below).

97.3.2 Technique

Background and anatomy

The spinal cord and column are the same length in a 3-month fetus. After that, the spinal column grows faster than the cord. As a result, the conus medullaris is located rostral to the termination of the thecal sac in the adult, situated between the middle thirds of the vertebral bodies of L1 and L2 in 51–68% of adults (the most common location), T12–L1 in $\approx 30\%$ and L2–3 in $\approx 10\%$ (with 94% of cords terminating within the territory of L1 and L2 vertebral bodies).⁴ The thecal sac ends $\approx$ S2. The tips of the spinous processes as palpated on the surface are located caudal to the corresponding VB. The intercrystal line (connecting the superior border of the iliac crests) crosses the spine at the L4 spinous process or between the L4 and L5 spinous processes in most adults.

Procedure

Position: the procedure is usually performed in the lateral decubitus position. As the needle is advanced, it is helpful to have the patient bring the knees up and to flex the neck in order to open up the spaces between the posterior elements of the spine.

For diagnostic LP, a 20 Ga spinal needle is often selected. Larger needles (e.g. 18 Ga) may be used e.g. with pseudotumor cerebri to encourage post-procedure drainage of CSF into the soft tissues of the back.

The back is prepped and draped to create a sterile working area.

Entry point: in an adult, use the L4–5 interspace in most cases (located at or just below the intercrystal line) or 1 level higher (L3–4). Peds: L4–5 is preferred over L3–4.

The needle is always advanced with the stylet in place at least through the skin and some subcutaneous tissue to avoid introducing epidermal cells which may cause iatrogenic epidermoid tumors; see Complications following LP (p.1507). The needle is aimed slightly cranially (to parallel the spinous processes) and usually a little down towards the bed (aiming towards the umbilicus). If a Quincke LP (standard) needle is used, the bevel is turned parallel to the length of the spinal column to reduce the risk of post-LP H/A (p.1509). In general, if bone is encountered it is more often due to deviation from a true midline trajectory rather than a failure to aim correctly in the rostral-caudal direction. The needle should be withdrawn to just below the skin surface before attempting a new trajectory.

If during insertion of the needle the patient experiences pain radiating down one LE, this usually indicates that a nerve root has been encountered. The needle should be withdrawn immediately and reinserted aiming more towards the side contralateral to the extremity that experienced the pain. The stylet is removed at intervals during the insertion to look for CSF (a distinct pop is sometimes felt as the needle penetrates the dura).

Once CSF flows, the needle is connected to a manometer through a 3-way stopcock, the pressure is measured and recorded (see below), and CSF is drained into sterile tubes (1–2 ml for each tube) for laboratory analysis (see below). The practitioner should also note the color of the fluid (clear, blood tinged, xanthochromic...) and the clarity (clear, cloudy, purulent...).

At the end of the procedure, the stylet should be replaced before the needle is withdrawn (to reduce post-LP H/A, see below).

Opening pressure: The opening pressure (OP) should be measured and recorded for every LP. To be meaningful, the patient should be lying down and as relaxed as possible (should not be in forced fetal position), with the bed flat. The variation of pressure with respirations is usually a good indication of a communicating fluid column (the fluctuation is in-phase with the respiratory pressures in the inferior vena cava, rising with inspiration and falling with expiration⁵). Normal values: in the left lateral decubitus position, average OP = 12.2 ± 3.4 cm H₂O (8.8 ± 0.9 mm Hg).⁶ Also, see □ Table 23.1 for peds.

Queckenstedt's test: if a subarachnoid block is suspected (e.g. from spinal tumor), compress the jugular vein (JV) first on one side then on both (do not compress carotid arteries). If there is no block, the pressure will rise to 10–20 cm of fluid, and will drop to the original level within 10 seconds of release of the JV.^{7(p 11)} Do not do JV compression if intracranial disease is suspected.

97.3.3 Laboratory analysis

Routinely, three tubes are sent for analysis as shown in □ Table 97.1. See □ Table 23.4 for interpreting the results of the laboratory analysis.

If the tap is possibly traumatic (i.e. bloody), or if having an accurate cell count is essential (e.g. to R/O SAH) then 4 tubes are collected, and the first and last are sent for cell counts and are compared; see Traumatic tap (p.1506).

If special cultures are required (e.g. acid fast, fungal, viral) they are also specified on the tube for culture & sensitivity (C & S).

If CSF for cytology is desired (e.g. to R/O carcinomatous meningitis or CNS lymphoma), then at least 10 ml of CSF must be sent in one tube to pathology (where it is spun down, and examined for cells).

97.3.4 Mining useful data with a traumatic tap

General information

A traumatic tap (TT) occurs when the spinal needle damages a blood vessel with the result that (usually venous) blood either alone or admixed with CSF will be obtained.

Estimating true WBC count in CSF with a traumatic tap

When many RBCs and WBCs are present in the CSF due to a traumatic tap (TT), it is difficult to know if there is a true leukocytosis on the CSF. It may help to determine if the WBCs are elevated or if they are present in the same ratio as in the peripheral blood. In non-anemic patients, there should be ≈ 1–2 WBCs for every 1000 RBCs (as a correction^{8 (p 176)}: subtract 1 WBC for every 700 RBCs^{8 (p 176)}). In the presence of anemia or peripheral leukocytosis, use Fishman’s formula^{8 (p 176)} shown in Eq (97.1) to estimate the original WBC count in the CSF before the TT,

$$WBC_{CSF\;Original} = WBC_{CSF} \times \frac{WBC_{Blood} \times RBC_{CSF}}{RBC_{Blood}}$$

97:16

where WBCCSF original=WBC count in the CSF before the TT, WBCCSF & RBCCSF=WBC & RBC counts measured in the CSF, and WBCblood & RBCblood =WBC & RBC per mm³ in the peripheral blood.

Estimating true total CSF protein content with a traumatic tap

If the hemogram and peripheral protein are normal, then have the cell count and protein content run on the same tube, and the correction is^{8 (p 176)}:

- subtract 1 mg per 100 ml of protein for every 1000 RBC per mm³

Differentiating SAH from traumatic tap

See typical findings in SAH (p.1160). Some features helpful in differentiating SAH from TT are shown in □ Table 97.2.

Table 97.1 Routine tests for CSF		
Test	If there is no concern about possible traumatic tap	If there is concern about traumatic tap
cell count		Tube 1
gram stain + C & S (culture & sensitivity)	Tube 1	Tube 2
protein and glucose	Tube 2	Tube 3
cell count	Tube 3	Tube 4

Table 97.2 Features distinguishing traumatic tap from SAH		
Feature	Traumatic tap (TT)	SAH
RBC count (and gross appearance of bloodiness)	declines as CSF drains (compare first tube to last tube)	usually >100,000 RBCs/mm ³ , changes little as CSF drains
ratio of WBC:RBC	similar to the ratio in peripheral blood (above)	usually promotes a leukocytosis (elevated WBC count)
supernatant	clear	xanthochromic ^a (rarely in <2 hrs, present in 70% by 6 hrs, and >90% by 12 hrs after SAH)
clotting of fluid	usually clots if erythrocyte count >200,000/mm ³	usually does not clot
protein concentration	fresh bleeding elevates CSF protein from normal by only ≈ 1 mg per 1000 RBC	blood breakdown products elevate this more than TT (measured protein exceeds the sum of normal protein + 1 mg protein/1000 RBC)
repeat LP at higher level	usually clear	remains bloody
opening pressure	usually normal	usually elevated
^a NB: other conditions can cause xanthochromia		

97.3.5 Complications following LP

General information

The overall risk of disabling or persistent symptoms (defined as severe H/A lasting >7 days, cranial nerve palsies, major exacerbation of pre-existing neurological disease, prolonged back pain, aseptic meningitis, and nerve root or peripheral nerve injuries) has been estimated at 0.1–0.5%⁹ Severe side effects, which include brainstem herniation, infection, subdural hematoma or effusion, and SAH, are rare.¹⁰ (p 171–2)

Possible complications

1. tonsillar herniation
 - a) acute herniation in the presence of mass lesion (see below)
 - b) chronic tonsillar herniation (acquired Chiari 1 malformation): this has been reported after multiple traumatic LPs with presumed post-LP CSF leak¹¹
2. infection (spinal meningitis)
3. “spinal headache”: usually positional (diminishes with recumbency) (see below)
4. spinal epidural hematoma (p. 1131): usually seen only with coagulopathy
5. spinal epidural CSF collection: may be fairly common in patients with post-LP H/A. Usually resolves spontaneously
6. epidermoid tumor: risk may be increased by advancing LP needle without stylet (transplanting a core of epidermal tissue)^{12,13,14}
7. impinging nerve root with needle: usually causes transient radicular pain, may cause permanent radiculopathy in some
8. intracranial subdural hygroma or hematoma^{15,16} (rare)
9. vestibulocochlear dysfunction¹⁷:
 - a) subclinical (demonstrated on audiogram) or moderate reduction in hearing may occur, and seems to correlate with post-procedure CSF leakage. Most studies show reduction is at frequencies <1000 Hz
 - b) sudden hearing loss may occur. Perform audiogram to quantify loss. Treat with bedrest for several days, prednisone 60 mg/d tapered over 2–3 weeks
 - c) pathogenesis: reduced CSF pressure may reduce perilymph pressure through the cochlear aqueduct (may be especially pronounced with a patent aqueduct),¹⁸ producing endolymphatic hydrops
10. ocular abnormalities
 - a) abducens palsy: almost invariably unilateral. Often delayed 5–14 days post-LP, usually recovers after 4–6 wks¹⁹
11. dural sinus thrombosis²⁰ (usually with underlying thrombophilia)

Risk of acute tonsillar herniation following lumbar puncture

The question of when to do LP first (to save time) and when to obtain CTscan first to R/O intracranial mass (for safety) before performing an LP is controversial.

Issues

The time delay to initiating antibiotics is the most important variable in the outcome of meningitis. Time may be more crucial in community acquired meningitis, where a virulent organism infects a sometimes immune compromised host (e.g. children or the elderly), than in post-op neurosurgical meningitis (usually a low virulence organism, e.g. Staph. aureus, in an immune intact host where the BBB has been disrupted).

The theoretical risk in performing an LP with an intracranial mass is that the resultant shift in pressure may precipitate tonsillar herniation.

Starting antibiotics without first having a CSF specimen from an LP may prevent cultures from growing in the lab which then risks the difficulties inherent in managing partially treated meningitis, or a suboptimal choice of antibiotic medication.

Clinical evaluation for possible contraindication to LP is unreliable. Papilledema is one possible indication of increased ICP. Papilledema takes a minimum of 6 hrs to develop after the onset of increased ICP, and in most cases it requires up to 24 hrs to develop. Therefore, its absence does not insure normal intracranial pressure. Furthermore, papilledema may be seen in conditions where there is not a contraindication to LP, e.g. idiopathic intracranial hypertension, where LP is one of the accepted treatments (p.766).

The ready availability of CT scans, often within the emergency department itself, may involve a delay of only a few minutes, if qualified personnel to interpret the study are also immediately available. However, in a busy E/R, there may be delays related to other patients also requiring emergency CT, the need to transmit scans to an off-site radiologist...

Historical information

Herniation following LP was more common prior to ≈ 1950, long before CT scans were available, where the procedure was performed even when some patients had clear evidence of ↑ ICP, large bore spinal needles (12–16 gauge) were more commonly employed, and large quantities of CSF were removed for therapeutic purposes. In a 1969 report of 30 patients who deteriorated after LP,²¹ 73% had localizing signs (hemiparesis, anisocoria...) and 30% had papilledema. None of 5 patients with cerebral abscess deteriorated after the first of multiple LPs.

In a series of 129 patients with ↑ ICP,²² the complication rate reported was 6% however, some of these complications were probably unrelated to LP, and many of these patients were in extremis. In 7 series totalling 418 patients, a complication rate of 1.2% was calculated.²²

Σ

Herniation as a result of LP is consistently reported only in patients with severe non-infectious processes, often with accompanying signs of mass effect (localizing signs, papilledema...). Thus, in cases of suspected meningitis in the absence of focal findings and papilledema, if a CTscan cannot be performed and interpreted within a few minutes, the benefits of performing an LP with a needle 20 gauge or smaller and removing only a few ml of CSF and starting empiric antibiotics probably outweigh the small risk of herniation. In the unlikely event that there is acute deterioration associated with the withdrawal of a few ml of CSF, the (anecdotal) recommendation is to immediately replace the fluid through the LP needle.

Post LP (myelogram) H/A

General information

AKA “postspinal headache” or “spinal headache.” May also follow procedures other than LP/myelogram, such as dural opening (p.1055). Can also occur with spontaneous intracranial hypotension (p.389) and following decompressive craniectomy.²³

Clinical features

Important distinctive characteristic: H/A occurs when patient is erect, and is completely or partially (but significantly) relieved when recumbent. May be associated with nausea, vomiting, dizziness, or visual disturbances.

□ Time course. Most post-LP headaches (PLPHA) have a delayed onset 24–48 hrs after the LP, and although they may occur weeks post-LP, most also develop within 3 days. The duration of PLPHA varies, with a mean of 4 days,²⁴ and reports of duration of months²⁵ and even > 1 year.²⁶

Pathophysiology

Thought to be due to continued CSF leakage through the hole in the dura,²⁷ which reduces the CSF “cushion” of the brain. In the upright position, the pull of gravity on the brain produces traction on the blood vessels and any structures tethering the brain to the pain-sensitive dura. CSF may sometimes be demonstrable in the epidural space.

Epidemiology following LP

Reported incidence range is 2–40% (typically $\approx$ 20%), higher after diagnostic LP than for epidural anesthesia.²⁴

See also variables in LP that impact upon the risk of PLPHA (p.1508) (e.g. incidence is lower with smaller gauge spinal needle).

1. Risk factors for post-LP H/A outside the control of the physician:
 - a) age: incidence $\uparrow$ in younger patients
 - b) sex: incidence $\uparrow$ in females
 - c) prior headache history (including previous PLPHA)
 - d) body size: $\uparrow$ with small body mass index = weight/height²²⁸
 - e) pregnancy
2. Variables that have been shown to influence the incidence of PLPHA:
 - a) needle size: larger needles carry increased risk²⁹
 - b) bevel orientation: orienting the bevel parallel to the longitudinally running fibers of the dura reduces the risk of PLPHA³⁰
 - c) replacing the stylet prior to needle removal lowers the incidence²⁸
 - d) the number of dural punctures (may not be totally under the physician’s control)
3. Variables that may or may not influence the incidence of PLPHA: needle type
 - a) Quincke needle: bevelled edge with cutting tip (the standard LP needle). Incidence of PLPHA with 20 and 22 gauge Quincke needles: 36%³¹
 - b) atraumatic needles: a number of types are available (Sprotte, Whitacre...). Most are “pencil pointed” and may produce a hole with a lower incidence of transdural leak.³² Unproven²⁸
4. Factors found not to affect the incidence of PLPHA:
 - a) the position of the patient after LP (does not seem to prevent PLPHA, but may delay the onset of symptoms^{33,34})
 - b) volume of fluid removed at the time of LP
 - c) hydration following LP²⁸

Treatment for H/A following LP

Initial “conservative” measures include:

1. flat in bed for at least 24 hrs
2. hydration (PO or IV)
3. analgesics for H/A
4. tight abdominal binder
5. desoxycortisone acetate 5 mg IM q 8 hrs²⁴
6. caffeine sodium benzoate 500 mg in 2 cc IV q 8 hrs up to 3 d max (70% of patients had relief with 1 or 2 injections)³⁵
7. high-dose steroids: report of success in a case of intracranial hypotension associated with spontaneous slit ventricles tapering down from a starting dose of dexamethasone 20 mg/day³⁶
8. blood patch if refractory

□ Epidural blood patch. For refractory post-lumbar puncture or post-myelogram H/A. Works in one application in over 90% of cases, may be repeated if ineffective.²⁵ Theoretical risks: infection, cauda equina compression, failure to relieve H/A.

□ Technique. Summary: 10 ml of non-heparinized autologous blood injected into epidural space.

Accessing epidural space (one of several techniques): proceed as routine LP. When ligaments are traversed, and needle tip is nearing spinal canal, stylet is removed. Then, either place drop of sterile saline in hub (hanging drop technique) and advance while watching for it to be drawn into needle as epidural space is entered, or gently try injecting air with small syringe (preferably glass, lower

resistance) while advancing, when the epidural space is entered, resistance to injection disappears, but CSF cannot be aspirated.

A venipuncture site is prepared aseptically. 10 ml of the patient's blood is withdrawn. After verifying CSF cannot be aspirated through the spinal needle, the blood is injected into the epidural space. After 30 minutes supine, patient may ambulate ad lib.

97.4 Lumbar catheter CSF drainage

97.4.1 General information

Insertion of a catheter into the lumbar subarachnoid space for the purpose of draining CSF. Usually connected to a closed drainage system similar to that for an EVD. Generally used for periods of only a few days or so.

97.4.2 Indications

1. to reduce CSF pressure on a site of CSF leak/fistula. Example situations:
 - a) dural breach during spine surgery or craniotomy (especially posterior fossa)
 - b) for spontaneous CSF fistula (rare) (p.386)
2. to reduce intracranial pressure in cases of communicating hydrocephalus: e.g. drain test for NPH, or when an infected shunt has been removed
3. to reduce CSF pressure to attempt to increase perfusion of the spinal cord: e.g. during surgery for abdominal aortic aneurysm, or following spinal cord injury

97.4.3 Contraindications

As with lumbar puncture (see above).

97.4.4 Insertion technique

Positioning, entry site, and trajectory are all similar to lumbar puncture (see above). Instead of a spinal needle, a Tuohy needle is used. The bevel is inserted parallel to the fibers of the dura (rostral-caudal). The needle is then rotated 90° (usually pointing rostrally) and the catheter is threaded into the needle. □ If the catheter does not thread, the needle must be withdrawn together with the catheter – attempting to withdraw the catheter through the needle will shear off the catheter at the tip of the needle.

97.4.5 Management

The orders for the nursing staff to maintain the catheter include:

1. instructions as to how to regulate the CSF drainage. Most commonly one of:
 - a) by pressure: accomplished by specifying a height of the drip chamber, usually at the level of the tragus or shoulder
 - b) by withdrawing a specified amount of CSF per hour: usually 10–20 cc. This method reduces the risk of overdrainage if the drip chamber is too low
2. instructions for the exit site: usually treated as an arterial-line

97.4.6 Complications

1. infection
2. overdrainage: usually as a result of the drainage bag being too low when using the pressure drainage method described above (either from falling to the floor, or not being raised when the patient sits or stands up) or from catheter disconnection. Can cause:
 - a) subdural hematoma from tearing of bridging veins from downward displacement of the brain
 - b) headache
3. pneumocephalus: usually from placing the drain height below the site of a fistula, and air is drawn in through the fistula tract
 - a) tension pneumocephalus: usually with a ball-valve effect at the fistula site
4. catheter pull out: frequently occurs simply as a result of patient movement in bed or with patient transfers

97.5 C1–2 puncture and cisternal tap

97.5.1 Indications

Situations where CSF specimen is required but access via LP is difficult or contraindicated (lumbar arachnoiditis, superficial infection, marked obesity, patients who cannot be turned on their sides...), or to instill contrast to demonstrate the rostral extent of a block documented by dye injected via LP. Spinal headache is less common with these procedures than with LP. C1–2 puncture is safer than cisternal tap.

□ Contraindicated: in patient with Chiari malformation (often present in myelomeningocele) due to low lying cerebellar tonsils and medullary kink.

Normal CSF values for glucose and protein differ only slightly from CSF obtained by lumbar puncture. Opening pressures averaged 18 cm of fluid with lateral puncture.

97.5.2 C1–2 puncture

AKA lateral cervical puncture. Equipment: LP tray (useful for the specimen tubes, extension tube for contrast injection under fluoroscopy, lidocaine, and spinal needle) with a standard 20 Ga spinal needle, contrast if needed (e.g. Iohexol®). It is preferred to perform the procedure under fluoro, however it has also been described without fluoroscopic guidance with a completely cooperative patient.³⁷

Patient position: supine in bed without a pillow, with the head straight up. Avoid any head rotation which may bring the vertebral artery (VA) into the needle path.³⁸ Place head within lateral fluoroscopy unit (since this is rarely available, a C-arm fluoro positioned horizontally may be used).

If iodinated dye is being injected for myelography, the head should be elevated to prevent contrast from running into posterior fossa; in cases with cervical spine injury, one can put the entire bed in reverse Trendelenburg.

Entry point: 1 cm caudal and 1 cm posterior (dorsal) to the tip of the mastoid process. Needle insertion: use a 25 Ga. needle to anesthetize the skin at the entry point. Under fluoro, advance a larger needle (e.g. 21 Ga.) towards the C1–2 interspace while injecting local anesthetic: aim for a target in the middle of the posterior third of the bony spinal canal (or, alternatively, 2–3 mm anterior to the posterior margin of the bony canal) (“X” in □ Fig. 97.1). Leave this needle in as a marker. Insert the 20 Ga. spinal needle parallel to the marker needle. Verify the course with fluoro. If fluoro is not used, insert the spinal needle at the entry point, and advance it parallel to the plane of the bed, perpendicular to the neck.³⁷ If the needle penetrates deeply without encountering bone or CSF, it is most likely that the tip is too par posterior. If bone is encountered, redirect the needle in the rostro-caudal plane.

Several “pops” may be felt, and the stylet should be removed after each to check for CSF return. The subarachnoid space is $\approx$ 5–6 cm deep to the skin surface in most adults.³⁹ The needle must be supported more than with a lumbar puncture.

To inject iodinated contrast, use e.g. $\approx$ 5 ml of 180 mg% Iohexol® for cervical myelogram, watch dye on fluoro (should be able to see it in subarachnoid space).

Risks

Case report of a death from subdural hematoma due to puncture of an anomalous vertebral artery⁴⁰ (found in $\approx$ 0.4% of population). If the VA is penetrated, the needle is withdrawn and local pressure is applied. Penetration of the upper spinal cord/lower medulla (risk of serious neurologic sequelae, even from this, is small). Herniation (as with LP) when there is increased ICP.

97.5.3 Cisternal tap

Suboccipital access to the cisterna magna. Usually done with patient sitting, with neck slightly flexed.⁴¹ Overlying hair should be shaved. Local anesthetic is infiltrated. A 22 gauge spinal needle is inserted exactly in the midline between the inion and the C2 spinous process, directed superiorly towards the glabella until the needle strikes the occiput or enters the cisterna magna. If the occiput is encountered, the needle is withdrawn slightly and reinserted directed slightly inferiorly, the process is repeated (“walking down the occiput”) until the cisterna magna is entered (a “pop” will be felt).

The distance from the skin surface to the cisterna magna is 4–6 cm, and from the dura to the medulla is $\approx$ 2.5 cm. However, due to tenting of the dura, the needle may be very close to the medulla before entering the subarachnoid space.

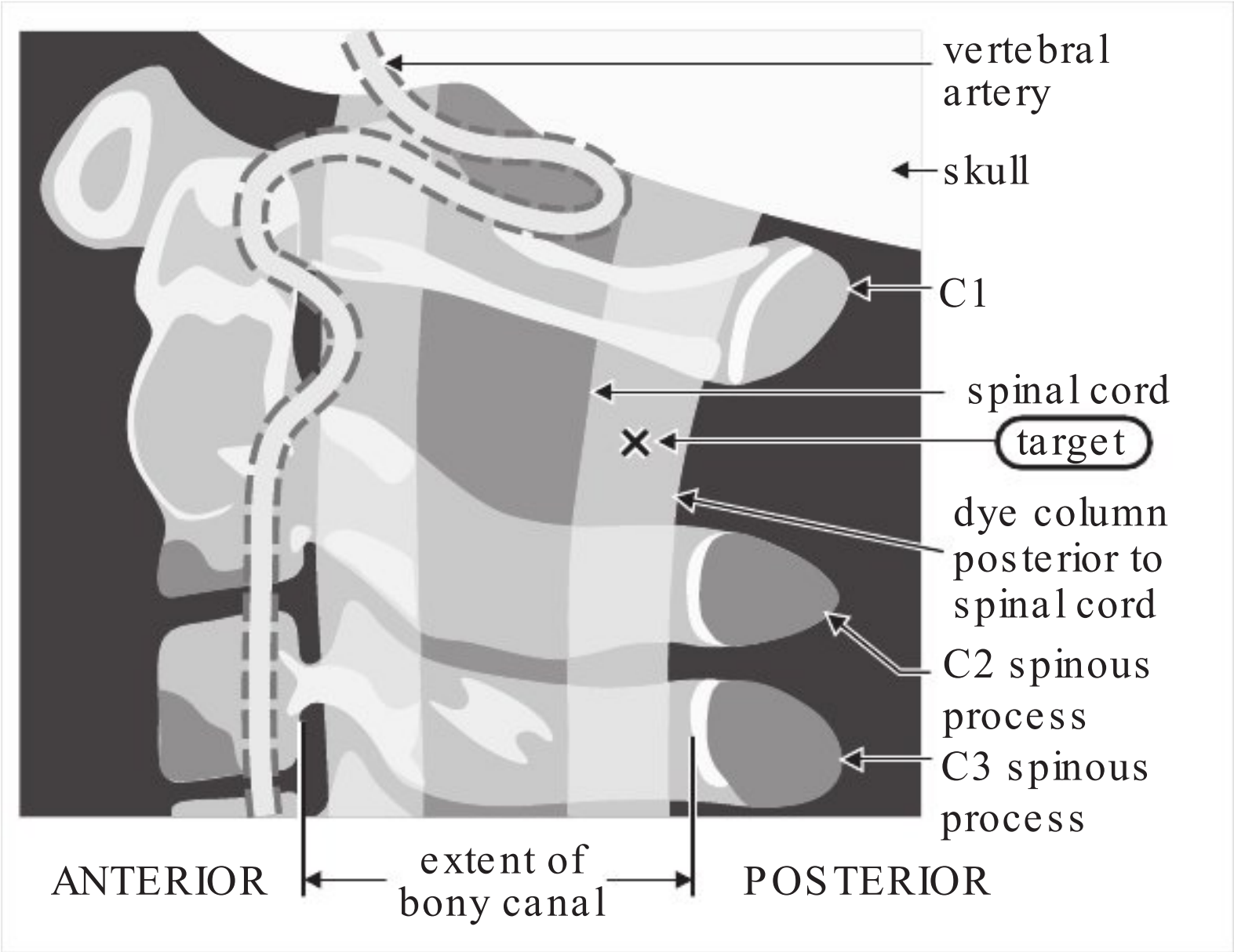


Fig. 97.1 C1–2 puncture target*
* left lateral view through upper C-spine: composite diagram of a myelogram and vertebral arteriogram illustrating the relative location of the spinal cord, CSF space, and VA. Only bony landmarks will be visible with fluoroscopy

Risks

- 1. hemorrhage in the cisterna magna: may be due to perforation of a large vessel³⁷
- 2. piercing the medulla oblongata: may cause vomiting, respiratory arrest...
- 3. positioning may compromise blood flow in the vertebral artery in elderly patients

97.6 CSF diversionary procedures

97.6.1 Ventricular catheterization

Most common insertion sites⁴² (p 151–3):

- 1. □ Kocher’s point (coronal): localizes an entry point into the frontal horn of the lateral ventricle that passes anterior to the motor strip. The right side is usually used. Often employed for ICP monitors, EVDs, shunts, ventriculoscopes.... Originally described as, “about 2 cm from the central line and 3 cm from the precentral fissure.”⁴³ A number of surface landmarks have been described to locate a point anterior to the motor strip that is in this general vicinity, and the author will often refer to the position as “Kocher’s point.” Commonly cited landmarks:
 - a) entry site: 2–3 cm from midline which is approximately the mid-pupillary line with forward gaze, 1 cm anterior to coronal suture which is approximately 11 cm up from the nasion (to avoid the motor strip)
 - b) trajectory: direct catheter perpendicular to surface of brain, which can be approximated by aiming in coronal plane towards medial canthus of ipsilateral eye and in AP plane towards EAM
 - c) insertion length: advance catheter with stylet until CSF is obtained (should be <5–7 cm depth; this may be 3–4 cm with markedly dilated ventricles). Advance catheter without stylet 1 cm deeper. □ CAUTION: if CSF is not obtained until very long insertion length (e.g. ≥ 8 cm) the tip is probably in a cistern (e.g. prepontine cistern) which is undesirable

2. □ occipital-parietal region: commonly used for CSF shunt
 - a) entry site: a number of means have been described, including:
 - Frazier burr hole: placed prophylactically before p-fossa crani for emergency ventriculostomy in event of post-op swelling. Location: 3–4 cm from midline, 6–7 cm above inion⁴⁴ (p 520) (caution: an error in locating the inion could put the catheter in an undesirable location if this method alone is used)
 - parietal boss: flat portion of parietal bone
 - follow point from mid-pupillary line parallel to sagittal suture until it intersects line extending posteriorly from the top of the pinna
 - ≈ 3 cm above and ≈ 3 cm posterior to top of pinna
 - b) trajectory: insert the catheter parallel to skull base:
 - initially aim for middle of forehead
 - if this fails, aim for ipsilateral medial canthus
 - c) insertion length: ideally, the tip should be just anterior to the foramen of Monroe in the frontal horn.⁴⁵ Ventriculoscopic guidance (if available) increases the accuracy to a significant degree. In the absence of this:
 - intracranial length should be $\approx$ two-thirds of the length of the skull (this is short enough to prevent penetration of frontal brain parenchyma, but long enough to take tip beyond the foramen of Monroe to prevent catheter from ending up in the temporal horn where choroid plexus increases the chance of obstruction)
 - in adults without macrocrania the inserted length is usually ≈ 12 cm when the burr hole is in line with the axis of the lateral ventricle⁴⁶ (lengths > 12 cm are rarely required). In hydrocephalic infants usually ≈ 7 –8 cm is required
 - use the stylet for the initial ≈ 6 cm of insertion, then remove it and insert the remaining length (keeps the catheter straight during penetration of occipital parenchyma and prevents the tip from dropping into the temporal horn where there is choroid plexus, also the temporal horn may collapse and occlude the catheter when the HCP is resolved)
3. Keen's point (posterior parietal): (placement in trigone) 2.5–3 cm posterior and 2.5–3 cm superior to pinna (was the usual site of occurrence of cerebral abscesses arising from otitis media, and was often used to tap these)
4. Dandy's point: 2 cm from midline, 3 cm above inion (may be more prone to damage visual pathways than above)

97.6.2 Ventriculostomy/ICP monitor

General information

AKA intraventricular catheter (IVC) or external ventricular drainage (EVD).

Elevated INR: for patients with an elevated INR (such as patients on warfarin), it has generally been recommended that placement of intraparenchymal catheters be delayed until the INR be ≤ 1.6 in order to reduce the risk of hemorrhage to an acceptable level.⁴⁷ However, intraparenchymal ICP monitors were placed without hemorrhagic or thrombotic complications in 11 patients with grade III/IV hepatic encephalopathy associated with fulminant hepatic failure who had an average INR of 3 when it was performed within 15–120 minutes after receiving 36.7 micrograms/kg IV of recombinant activated factor VII (rFVIIa).⁴⁸

Insertion technique

Unless contraindicated (e.g. right ventricular bleed), the right (non-dominant) side is preferred. Shave entire ipsilateral hemicranium and contralateral frontal portion. Clip the hair around the planned incision site and the exit site for the tunneled catheter (avoid shaving which degrades the skin's barrier function against infection). Prep with any acceptable surgical prep (betadine prep for 5 minutes, Chloroprep, Duraprep...)

Site: approximately Kocher's point (see above). To avoid motor strip, enter 1–2 cm anterior to coronal suture (estimated position of coronal suture: follow line up midway between lateral canthus and EAM, or measure 11 cm up from the nasion), and to avoid the sagittal sinus, 2–3 cm lateral to midline (2 fingerbreadths or ≈ 3 cm is commonly employed as an approximation). Incision oriented in sagittal plane (in case it needs to be incorporated in flap); elevate periosteum; place self-retaining retractor; make twist drill hole. Bone-wax edges to stop bone bleeding; cauterize dura with bipolar coagulator; incise dura in cruciate fashion with #11 scalpel blade; cauterize incised dural edges and then pia/arachnoid with bipolar.

For ventriculostomy: insert catheter perpendicular to brain surface⁴⁹ to a depth of 5–7 cm (most catheters are marked at 5 and 10 cm). With any ventricular enlargement, CSF should flow at least by

3–4 cm depth (with normal ventricles, this may be 4–5 cm). If no CSF is encountered here and the catheter is passed further until CSF is obtained, it is unlikely to be due to catheterization of frontal horn of lateral ventricle (in this case, at $\approx 9\text{--}11$ cm the tip will often be in the pre-pontine cistern, a subarachnoid space, which is undesirable). If unsuccessful after a maximum of three attempted passes, then place a subarachnoid bolt or intraparenchymal monitor.

For (Richmond) subarachnoid bolt: screw in until tip is flush with inner table.

Removal

Patients receiving anticoagulants need to have normal coagulation and platelet function before discontinuing the catheter to reduce the risk of intracranial hemorrhage. For heparin and LMW heparin, stop the drug 24 hours prior to discontinuing the drain.

“Sump drainage”

The tip of a 25 gauge butterfly may be bent at a 90° angle, and inserted into a subcutaneous reservoir for prolonged ventricular drainage.⁵⁰ In one series OF 34 patients, this was used for prolonged periods (up to 44 days) with acceptably low infection rate.⁵¹ The use of a one-way valve, continuous antibiotics (ampicillin and cloxacillin) and meticulous technique was credited for the lack of infection.

97.6.3 Ventricular shunts

Booking the case: Ventricular shunt

Also see defaults & disclaimers (p. 27).

1. position: supine with shoulder roll
2. implants: need to specify shunt manufacturer and valve type (e.g. programmable, low profile...) Uncommon components (e.g. ultra-low pressure, tumor filter) may need special order
3. equipment:
 - a) C-arm for ventriculo-atrial shunts
 - b) endoscopic display (e.g. if NeuroPen is used)
 - c) image guided navigation system (infrequently used)
4. consent (in lay terms for the patient – not all-inclusive):
 - a) procedure: surgery to insert a permanent drainage tube from the brain to the abdomen, outside of the lungs, vein near the heart (as appropriate) to drain excess cerebrospinal fluid
 - b) alternatives: nonsurgical management (rarely effective for hydrocephalus), third ventriculostomy (for certain cases)
 - c) complications: infection, suboptimal position which might require re-operation, failure to relieve hydrocephalus/symptoms, subdural hematoma, bleeding in the brain, shunts are mechanical devices and will eventually fail (break, block up, move...) and need repair/replacement (sometimes sooner rather than later). Abdominal shunts: risk of bowel injury (which could require further surgery)

97 Ventricular catheter

Kocher’s point is currently used in most cases for insertion site of ventricular catheter; see Ventricular catheterization (p.1512) for technique. An alternative is an occipital burr hole aiming toward the frontal horn of the lateral ventricle.

An inverted “J” shaped incision is used to keep hardware from lying directly under the skin incision (minimizes risk of skin breakdown and also creates additional barrier to infection of subjacent hardware). CSF should be sent for culture at the time of insertion since it has been estimated that in $\approx 3\%$ of patients the CSF is already infected. 4 mg of preservative-free gentamicin may instilled into the ventricular catheter by the technique of barbotage, (a technique to administer a drug while reducing the amount of drug lost in the dead space of the catheter: a portion of the antibiotic solution is injected into the CSF, then a lesser amount of CSF is aspirated, a second portion is then injected and the process is repeated until all of the medication is administered).

If you think the catheter is in the ventricle, but you don’t get CSF flow, it may be due to low pressure, you can compress the jugular veins or lower the head of the bed to try and induce CSF flow.

Connectors

If a connector must be used near the clavicle, place it rostral to (above) the clavicle. □ Avoid placing it caudal to the clavicle which increases the risk of disconnection.

Distal catheter placement options

All things being equal, the general order of preference for distal catheter placement:

1. peritoneal cavity: see below
2. pleural space (p.416): not for age ≤ 7 years. For technique, see below
3. right atrium or superior vena cava (p.1516)
4. infrequently used distal shunt sites
 - a) gall bladder
 - b) internal jugular vein (with the catheter pointing “upstream”)
 - c) superior sagittal sinus

Ventriculoperitoneal shunt

Peritoneal catheter

For small children, use at least 30 cm length of intraperitoneal tubing to allow for continued growth (120 cm total length of peritoneal tubing was associated with a lower revision rate for growth without significant increase in other complications⁵²). A silver clip is placed at the point where the catheter enters the peritoneum so that the amount of residual intraperitoneal catheter can be determined on later films (more important in growing children).

Distal slits on the peritoneal catheter may increase the risk of distal obstruction,⁵³ and some authors recommend that they be trimmed off. Wire reinforced catheters should not be used because of excessively high rate of viscus perforation, and this tubing was designed to prevent kinking which is not a problem with modern shunts.

Open technique

A vertical incision lateral and superior to the umbilicus is one of several choices. The following layers should be identified as they are traversed to avoid confusing preperitoneal fat with omentum and erroneously placing the tip in the preperitoneal space:

1. subcutaneous fat
2. anterior sheath of the abdominis rectus muscle (anterior rectus sheath)
3. abdominis rectus muscle fibers: should be split longitudinally
4. posterior rectus sheath
5. preperitoneal fat (may be very well developed in a few individuals, but is essentially nonexistent in most)
6. peritoneum (usually closely adherent to the posterior rectus sheath)

Trocar technique

1. place a Foley catheter to decompress the bladder
2. 1 cm skin incision above and lateral to the umbilicus
3. pull abdominal skin anteriorly (away from patient)
4. insert trocar aiming toward the ipsilateral iliac crest
5. feel 2 “pops” of penetration: 1 = anterior rectus sheath, 2 = posterior rectus sheath/peritoneum
6. peritoneal catheter should feed easily through trocar
7. □ contraindications: prior abdominal surgery, extremely overweight patients

VP shunt, post-op orders (adult)

1. flat in bed (to avoid overshunting and possible subdural hematoma) with gradual mobilization
2. if peritoneal end is new or revised, do not feed until bowel sounds resume (usually at least 24 hrs, due to ileus from manipulation of peritoneum)
3. shunt series (AP & lateral skull, and chest/abdominal x-ray) as baseline for future comparison (some surgeons obtain these films immediately post-op in case some immediate revision is indicated, e.g. ventricular catheter tip in temporal horn)

Ventriculopleural shunt insertion

See reference.⁵⁴

See also more details (p.416). Not for age ≤ 7 years. A 3 cm horizontal incision is made just below the level of the breast either in the midclavicular line or in the anterior axillary line. Divide the subcutaneous tissue, deep fascia, and pectoralis muscle. The external and internal intercostal muscles are divided along the superior margin of the inferior of the two ribs exposed (to avoid the neurovascular bundle running along the inferior margin of each rib). A self retaining retractor between the ribs aids the exposure. The parietal pleura is visualized with the visceral pleura sliding underneath with each respiration. The pleura is not opened until the catheter is brought out subcutaneously at this incision. Have the anesthesiologist hold respirations, and nick the parietal pleura (or use a blunt-tip hemostat to pop through) to admit the catheter. Allow the lung to drop away and insert 20–40 cm of tubing into the pleural cavity. If the pleural opening is lax around the catheter, it can be snugged with a 4–0 absorbable suture. Have the anesthesiologist provide a valsalva maneuver before cinching down the pleural suture, and again before closing the deep muscle layer. A chest tube is usually not required. A maneuver that may sometimes be helpful is to place a red-rubber catheter next to the shunt tube at the same time (to permit the escape of air from the pleural space). Begin closing, but prior to placing the last deep suture, have the anesthesiologist perform a valsalva maneuver and allow air to escape through the red-rubber catheter (you can place the end in saline to see the bubbles). Once the bubbles stop, pull the red-rubber catheter and close the last stitch. If the bubbles don't stop, there is an air leak in the visceral pleura and a pigtail catheter or a chest tube connected to a Pleur-evac® should be used.

Ventriculo-atrial shunt insertion

Open method

The common facial vein (CFV) is located by making a diagonal cervical incision across the anterior border of the sternocleidomastoid at or just below the level of the angle of the mandible (the CFV may be as far as ≈ 2 cm below this point). The platysma is divided, and the CFV is located as it joins the internal jugular vein (IJV) at the level of the hyoid bone. The CFV is cannulated with the atrial tubing, and is secured with a snug ligature close to the junction with the IJV. If the CFV is not suitable, a purse string suture is placed directly in the IJV, and the IJV is then opened in the center of the purse string and cannulated.

Percutaneous method

May be utilized in adults (and possibly peds). The IJV is catheterized using the Seldinger technique⁵⁵ with a guide-wire and needle through a stab wound at the anterior margin of the SCM. Fluoroscopy is used to place the tip of the wire at the desired location (see below). A No. 13 French peel-away introducer and dilator are then inserted over the wire, which is then bent at the skin edge and withdrawn⁵⁶ (for a pediatric case: may use a No. 7 French introducer with a 1.5 mm O.D. lumboperitoneal catheter for the distal atrial catheter). The atrial catheter is cut to the length of the wire distal to the bend, and the catheter is then threaded into the introducer. The position of the catheter tip should again be confirmed (e.g. with radio-opaque contrast under fluoroscopy). A short skin incision is then made starting at the point where the catheter penetrates the skin to permit subcutaneous tunneling of the tubing.

Location of distal tip

Placement pointer: if the catheter repeatedly goes down the wrong vessel (e.g. the subclavian vein), a “J” guidewire may help. Also, rotating the head to a more neutral position sometimes works.

The ideal location of the distal tip is in the right atrium (unlike the location for central catheters in the superior vena cava (SVC)) so that the turbulent blood flow will reduce the risk of thrombus formation. The tip may enter the right atrium, but must not penetrate the tricuspid valve. A number of methods for optimal placement of the distal shunt tip may be employed, and include:

1. using an intraoperative chest x-ray to locate the tip between the level of T6-T8 vertebra in an adult. In a growing child, initially insert to $\approx$ T10 level. This method is subject to error due to malalignment of the x-ray beam (parallax error)
2. locating the tip near the level described above, then inject iodinated contrast, e.g. 20 ml of Omnipaque 180 (iohexol) (p.219) under intraoperative fluoroscopy to locate the tip in SVC
3. fill the catheter with normal or 3% saline and use the catheter as an EKG electrode. The P-wave changes from a downward to a biphasic morphology as the tip enters the atrium. A sharp upward deflection occurs as the tricuspid valve is approached.⁵⁷ Some recommend advancing the tip to maximal P-wave amplitude and then backing off a centimeter or two
4. fill the catheter with heparinized saline (1–5 U per cc NS) and measure the pressure as the tip is advanced,⁵⁸ leave tip just short of where atrial pressure tracing occurs
5. utilizing intraoperative echocardiography⁵⁹

A growing patient is followed with annual CXRs. When the catheter tip is above $\approx$ T4, the catheter must be lengthened or converted to a VP shunt.

97.6.4 Third ventriculostomy

General information

See also indications and complications (p.415).

Older techniques include a subfrontal approach, opening the chiasmatic and lamina terminalis cisterns, and making a 5–10 mm opening in the lamina terminalis. Stereotactic third ventriculostomy (using contrast ventriculography⁶⁰ or CT guided) has also been described. Current technique, endoscopic third ventriculostomy (ETV) often with assistance of image-guidance consists of fenestrating the floor of the third ventricle using a ventriculoscope.

Ventriculoscopic technique

1. equipment: requires a rigid endoscope (does not work well with flexible)
2. image guided stereotactic technology helps immensely with the trajectory, but once you've entered the third ventricle, you must navigate by visual landmarks and cannot rely on image guidance because of the limitations of the accuracy
3. burr hole: 2–3 cm lateral to the midline just anterior to the coronal suture (Kocher's point)
4. pass through the foramen of Monro and fixate the sheath just within the third ventricle
5. the floor of the third ventricle is inspected and must be thin enough and translucent enough to permit visualization of the basilar artery and mammillary bodies. If these structures cannot be visualized then the procedure should be aborted
6. the location of the opening is chosen:
 - a) in the midline (avoids p-comm and PCA)
 - b) in the region of the tuber cinereum (prominence of the base of the hypothalamus, extending ventrally into the infundibulum and pituitary stalk)
 - c) posterior to the infundibular recess
 - d) anterior to the mammillary bodies
 - e) anterior to the tip of the basilar artery
7. an effective technique consists of "rubbing through" the floor of the third ventricle either with a probe or Decq forceps. Alternatively, hydrodissection or bipolar electrocautery may be used to thin down the lamina. ☐ Do not use laser due to possibility of injury to basilar artery!⁶¹
8. the opening can be enlarged with the Decq forceps, or a 3 french use of a Fogarty balloon or a double balloon (Fogarty or Neuroballoon™ catheter (Integra LifeSciences 7CBD10)). The balloon is inflated distal to the opening in the floor and is then withdrawn through the opening
9. the opening does not need to be large (unlike e.g. fenestrating an arachnoid cyst): $\approx$ 4–5 mm is usually adequate^{62,63}
10. after penetrating through the floor of the third ventricle, make certain that you can see vessels (sometimes the arachnoid is not perforated, or there is a second membrane or webs of membranes that need to be lysed)
11. consider injection of diluted iohexol or other intrathecal contrast agent into the lateral/third ventricle (see ventriculogram) prior to removal of scope. CT head 1 hour after surgery will show diffuse subarachnoid contrast in cisterns and over convexity if ETV successful
12. sagittal T2 weighted, thin-slice sequence will show drop out of T2 signal at stoma of ETV

97.6.5 LP shunt placement

Technique of insertion

See reference.⁶⁴

1. position: lateral decubitus position, both knees flexed (right side up preferred)
2. prep back, flank and abdomen
3. 1 cm skin incision over L4–5 or L5–S1 (in obese patients, use larger skin incision carried down to fascia overlying spinous processes. This may also be superficially incised between spinous processes to aid insertion)
4. tilt table to 30° reverse-Trendelenburg to expand lumbar subarachnoid space
5. insert 14 gauge Tuohy needle into subarachnoid space, with opening directed rostrally (caudal placement also acceptable). Confirm placement by CSF flow
6. remove trocar, insert shunt tubing such that $\leq$ 8 cm of catheter (for L4–5 insertion) lies within spinal canal (minimizes conus medullaris irritation)
7. needle then withdrawn over catheter

8. make flank incision, pass tunneler from flank to back incision. Feed catheter from back to flank. Withdraw tunneler over catheter
9. abdominal placement:
 - a) open: incision made through peritoneum. Catgut (or other absorbable suture) purse-string placed in peritoneal opening
 - b) trocar
10. pass tunneler from abdominal incision to flank incision. Feed catheter from flank to abdomen. Withdraw tunneler over catheter
11. verify CSF flow. Place catheter inside peritoneum. For open technique: cinch purse-string snugly, but loose enough to allow catheter to slide with gentle pushing
12. snug fitting retaining sleeve placed around catheter at all three incisions, and secured to subcutaneous tissue with non-absorbable suture

Lumboperitoneal (LP) shunt evaluation

When problems develop, evaluation of function may be more difficult than with VP shunt. Evaluation may include:

1. abdominal x-rays: AP & lateral x-rays can rule out breakage or migration of a shunt component
2. noncontrast brain CTscan: can rule-out complication such as subdural hematoma
3. LP: perform LP just above or below level of lumbar catheter. The pressure may be 0 or negative, and it may be necessary to aspirate CSF to confirm placement
 - a) can give indirect evidence of shunt function by measuring the CSF pressure, which should be low if the shunt is working (only helpful in cases where the shunt was placed for elevated CSF pressure, e.g. pseudotumor cerebri; not helpful in NPH).
 - b) “shunt-o-gram”: inject contrast into subarachnoid space through LP needle
 - radionuclide (p.421): inject radio-isotope via LP and look for subsequent tracer activity in peritoneal cavity
 - with water-soluble contrast⁶⁵: inject 10 ml of iohexol and monitor the flow of contrast fluoroscopically as the patient is brought vertical. Coughing or valsalva maneuver will accelerate the flow of contrast
4. shunt tap: if an antechamber has been installed, it is accessed after cleaning the skin with antiseptic using a 22 Ga. or smaller needle placed perpendicular to the dome to prevent leakage. If there is no access chamber, it is sometimes possible to tap the tubing itself with a 27 gauge butterfly needle

97.7 Ventricular access device

97.7.1 General information

An indwelling ventricular catheter connected to a reservoir that is situated under the scalp for the purpose of chronic access to the intrathecal space (usually the ventricular system), or sometimes other intracranial compartments such as tumor cysts. Sometimes referred to as an Ommaya® reservoir, this is actually a trade name.

97.7.2 Indications

1. administration of intrathecal (IT) antineoplastic chemotherapy:
 - a) for CNS neoplasms, including: carcinomatous meningitis, methotrexate for CNS lymphoma or leukemia (p.713)
 - b) IT chemotherapy is often used for the following even in the absence of CNS involvement because of the high relapse rate in the CNS: acute lymphoblastic leukemia, lymphoblastic lymphoma, Burkitt’s lymphoma
2. administration of intrathecal antibiotic for chronic meningitis
3. chronic removal of CSF from infants with intraventricular hemorrhage
4. for fluid aspiration from a chronic tumor cyst that is resistant to therapy (radiation or surgery)

Booking the case: Ventricular access device

Also see defaults & disclaimers (p. 27).

1. position: supine
2. equipment
 - a) endoscopic display (e.g. if NeuroPen is used)
 - b) C-arm (optional) to verify position of ventricular catheter
 - c) image guided navigation system (infrequently used)
3. implants: need to specify reservoir manufacturer
4. consent (in lay terms for the patient – not all-inclusive):
 - a) procedure: surgery to insert a tube into the fluid space within the brain (ventricle) which is connected to a port under the skin so that fluids can be removed or injected (usually medication)
 - b) alternatives: sometimes fluid can be removed and medication can be injected using a lumbar puncture (spinal tap). The effectiveness of this may not be the same as the operation being discussed here
 - c) complications: infection, suboptimal position which might require re-operation, subdural hematoma, bleeding in the brain, this is a mechanical device and may eventually fail (break, block up...) and need repair/replacement

97.7.3 Technique of insertion

See reference.⁶⁶

Preferably placed in the right frontal region, unless indicated otherwise (e.g. for tumor cyst). Usually placed under endotracheal general anesthesia, although local anesthesia occasionally may be used (e.g. for patients too ill to tolerate general anesthesia).

Patient position: supine, head midline, neck flexed 5°.

Incision: inverted “U”, slightly larger than the reservoir (the original Ommaya® reservoir is 3.4 cm diameter), with the center over the coronal suture approximately 3 cm from midline, roughly centered near Kocher’s point (p. 1512). A circle of pericranium of diameter equal to that of the reservoir is excised and saved. Alternatively, the pericranium may be flapped separately in the opposite direction (i.e. a right-side-up “U”), and closed over the reservoir to help secure it in position.

Make a burr hole over the coronal suture 3 cm off midline. A cruciate incision in the dura is made large enough to visualize the cortical surface, minimal cortical bipolar coagulation is used, and a pial/cortical incision is made to avoid surface vessels.

One may inject 15–20 cc of filtered air into the ventricles with a ventricular needle prior to the catheter insertion to guide the tip of the catheter with intra-operative lateral skull x-rays (intraoperative pneumoencephalogram). The trajectory is towards a point intersecting a plane 2 cm anterior to the EAM aiming minimally towards the midline (1–2°). Alternatively, one may aim perpendicular to the surface of the skull.⁴⁹ A total length of ≈ 7.25 cm of catheter is fixed to the base of the reservoir which allows the catheter to lie on the floor of the anterior horn of the lateral ventricle in most adults. This location can be verified with intraoperative pneumoencephalography⁶⁶ or with ventriculoscopy techniques.

The excised pericranium is placed over the dura, and the reservoir is sutured to the pericranium. Note: the dome of the original Ommaya® reservoir has a low resistance, and may be easily collapsed if too much tension is placed on the overlying scalp. If early use of the reservoir is desired (i.e. within 48 hrs post-op), the skin closure should be performed with a running nonabsorbable suture (e.g. nylon) and coated with collodion, and the surgical site can then be left without a gauze dressing for easier access to the reservoir. A skin tattoo can be created over the center of the reservoir (to assist in localizing the reservoir for injection) using India ink and a pricking the skin with a sterile needle.

97.7.4 Reservoir puncture

The scalp is prepped with antimicrobial scrub, and using sterile technique, a 25 gauge or smaller butterfly needle is introduced at an oblique angle, preferably with a non-coring needle. The original (Ommaya®) reservoir has firm plastic bottom surface which can be penetrated if too much force is applied.

97.8 Sural nerve biopsy

97.8.1 General information

Although a number of peripheral nerves may be biopsied, the sural nerve fulfills the criteria of being well studied, expendable with minimal deficit, easily accessible, and often involved in the pathologic process in question.

97.8.2 Indications

Nerve biopsy plays a small role in diagnosing peripheral neuropathies, but may be very accurate for vasculitis, amyloidosis, Hansen's disease, metachromatic and globoid leukodystrophy, neoplastic infiltration of peripheral nerve, and relapsing polyneuritis.^{67(p 316)} May help distinguish the two types of Charcot-Marie-Tooth syndrome. May show demyelination in diabetic amyotrophy (p.545). The yield of sural nerve biopsy may be low if it is normal on nerve conduction testing. Disadvantages of the sural nerve is that it will usually be spared in purely motor neuropathies.

97.8.3 Risks of procedure

1. sensory loss in the sural nerve distribution is expected, but often does not persist for more than several weeks (unless the underlying disease process prevents this)
2. problems with wound healing: the ankle is a notorious region for poor circulation and the loss of sensation (from the disease or biopsy) may render the area subject to repeated trauma without the patient being aware. Furthermore, many patients with an undiagnosed systemic disease requiring a sural nerve biopsy will have poor wound healing (a significant number are also diabetic)
3. failure to make a diagnosis: although biopsy may be able to exclude some contingencies, it often does not make a specific diagnosis

97.8.4 Applied anatomy

The sural nerve is formed by the merging of the distal portion of the medial sural cutaneous nerve (one of the terminal branches of the tibial nerve) and the anastomotic ramus of the common peroneal nerve. It is entirely sensory except for some unmyelinated autonomic fibers. It supplies cutaneous sensation to the posterolateral third of the leg, the lateral heel and foot, and the little toe. At the level of the ankle it lies between the Achilles tendon and the lateral malleolus. This location is constant, superficial, and relatively protected from external trauma which might otherwise confuse the analysis.

97.8.5 Technique

Modified technique.^{68(p 771-2)} Usually done under local anesthesia with sedation. Side to biopsy: if one leg shows more involvement, that is often the preferred side to biopsy.

Position: patient prone. If patient is not under general anesthesia, use 3/4 oblique (not prone) pillow between the legs. The leg to be biopsied is uppermost and is flexed 90° at the knee to relax tension on the nerve, the ankle is slightly everted. Compressing the calf (can be done using a sterile Penrose drain as a temporary tourniquet during surgery) distends the lesser saphenous vein (LSV) at the lateral malleolus (LM), which (when visible) reliably locates the sural nerve usually deep and anterior to the vein.

After prepping, drape the limb with a sterile stockinette or similar drape, infiltrate local anesthetic subcutaneously just posterior to the LM and proximally, paralleling the Achilles tendon for ≈ 10 cm. A 7–10 cm incision is made overlying the course of the LSV beginning usually just posterior to and ≈ 1 cm proximal to the LM. The vein can be seen through the translucent Scarpa's fascia. The fascia is incised over the vein which is gently retracted to reveal the nerve usually deep to the vein. A common pitfall is to go too deep, the nerve is fairly superficial; it is not necessary to go through the thick fascia. If at any time you see tendons to the toes, you have gone too deep.

To differentiate the sural nerve from the LSV (which may resemble the nerve in some cases): the nerve has many branches at acute angles especially proximal to the LM, vs. the vein which has right angle branches. If in doubt, a frozen section may be helpful to verify that the biopsied structure is a nerve in order to avoid potentially embarrassing explanations and the possible need to repeat the procedure.

After exposing at least 3–5 cm of the nerve, anesthetize the proximal portion with 0.5% lidocaine using a 27 Ga. needle and cut it sharply just distal to the infiltration site (note: sometimes a biopsy

of only a portion of the nerve's fascicles may suffice: this is accomplished by opening the epineurium for the length of the exposure and teasing out a fascicle with minimal branching). Cut the nerve with slight tension on it to allow the ends to retract deep to the skin incision to prevent the formation of a scar neuroma. Some pathologists request that the proximal end of the nerve be marked, e.g. with a suture.

If it is desired to obtain a biopsy of the sural nerve higher up for comparison, it may be accessed in the mid upper calf between the heads of the gastrocnemius muscle. Here it may be as deep as $\approx$ 2 cm. Gently tugging on the exposed nerve in the ankle may help in localization.

A dissolvable subcuticular closure may be used. Generous padding should be placed over the incision to protect it from bumps (since many patients are numb from the pathology and/or surgery, there is increased risk of inadvertent surgical wound injuries). An elastic pressure dressing is applied after closure.

97.8.6 Nerve handling

For light microscopy which suffices in most cases, immerse the nerve in formalin. For electron microscopy, glutaraldehyde is used. For biochemical and immunofluorescence studies, use rapid freezing.

97.8.7 Post-op care

Pressure dressing should be worn for protection for two weeks. The patient is allowed to walk but should restrict their activity for 2–3 days. If nonabsorbable sutures are used instead of subcuticular closure, they should be left in place 10–14 days.

97.9 Nerve blocks

Also see Occipital nerve block (p.516).

97.9.1 Stellate ganglion block

General information

□ Do not perform stellate ganglion block bilaterally (can cause bilateral laryngeal paralysis → respiratory compromise). Stellate ganglion is actually closer to C7 than C6, but risks at C7 are much higher (closer to pleura → pneumothorax, vertebral artery → arterial injection → seizures and/or hematoma, recurrent laryngeal nerve → unilateral vocal cord paralysis → hoarseness (common), brachial plexus → UE weakness). Other complications: intradural injection → spinal anesthesia, phrenic nerve block.

Technique

Patient supine; interscapular roll; head tilted backward, mouth slightly open to relax strap muscles. Displace SCM and carotid sheath laterally, insert 1.5 inch 22 Ga. needle to contact Chassaignac's tubercle (anterior tubercle of transverse process of C6) AKA carotid tubercle (the most prominent in the C-spine) usually at the level of the cricoid cartilage, approximately 1.5–2 inches above clavicle.

Withdraw needle 1–2 mm and aspirate (do not inject intravascularly). Inject small test dose, then full 10 ml of 0.5% bupivacaine (Marcaine®) or 20 ml of 1% lidocaine. Remove needle and elevate patient's head on pillow to facilitate spread.

Verify block by Horner's syndrome, and anhidrosis and increased warmth of ipsilateral hand.

97.9.2 Lumbar sympathetic block

Technique

Patient prone on fluoro table. Use local anesthetic to allow insertion of 20–22 gauge spinal needles (10 to 12.5 cm long) at L2, L3 and L4 levels. Needle inserted 4.5–5 cm lateral to spinous process until transverse process contacted, then redirected caudally and inserted to a depth 3.5–4 cm deeper than transverse process. Final needle tip position should be just anterolateral to vertebral bodies. At each level, instill $\approx$ 8 ml of 1% lidocaine local after verifying that nothing can be aspirated.

Keep patient on bed rest for several hours, then ambulate with assist; watch for orthostatic hypotension due to vascular pooling in blocked lower extremity.

97.9.3 Intercostal nerve block

Indications

1. postthoracotomy pain
2. intercostal neuralgia
3. postherpetic neuralgia
4. pain from rib fractures

General principles

In order to obtain good anesthesia, the following should be noted:

1. a good site for injection is in the posterior axillary line (PAL) because
 - a) this is proximal to the origin of the lateral cutaneous nerve (which originates $\approx$ in the anterior axillary line)
 - b) this avoids the scapula for nerves above $\approx$ T7
 - c) this reduces the risk of pneumothorax from injecting closer to the spine (the latter requires a longer needle path and there is increased difficulty palpating landmarks)
2. due to overlap, at least 3 intercostal nerves usually need to be blocked to achieve at least some area of anesthesia; it is usually necessary to block 1–2 intercostal nerves both above and below the affected dermatome
3. the intercostal nerves lie on the undersurface of the corresponding rib in close proximity to the pleura; the order of structures from top down is: rib, vein, artery, nerve

Technique

1. after raising a skin wheal at the desired level in the PAL, insert a 22 Ga. or smaller needle directly against the rib
2. walk the needle down the rib millimeter by millimeter until the needle just slips under the rib; to avoid pleural puncture, do not advance the needle more than one-eighth inch deep to the anterior surface of the rib
3. aspirate to be certain that there is no air (from lung penetration) or blood (from entering the intercostal artery or vein)
4. if no air or blood returns, inject 3–5 ml of local anesthetic
5. if there is any question about lung penetration, obtain a portable CXR to R/O pneumothorax

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98 Functional Neurosurgery

For functional neurosurgery related to pain, see Pain procedures (p. 1541).

98.1 Deep brain stimulation

A variety of conditions may be treated including:

1. movement disorders
 - a) Parkinson's disease: STN stimulation may be superior to best medical management^{1,2} because of similar efficacy to levodopa with fewer side effects (primarily dyskinesias) (see below)
 - b) dystonia (p. 1528)
 - c) tremor (p. 1537)
2. epilepsy (p. 1554)
3. pain (p. 1550): response is variable, typically only 25–60% respond
4. potential uses
 - a) psychiatric disorders: mainly
 - Tourette syndrome: thalamic & pallidal DBS (case reports^{3,4})
 - obsessive compulsive disorders: anterior capsule and STN stimulation⁵ and, recently, targets more posterior and rostral⁶
 - depression: subgenual cingulate gyrus⁷ and anterior capsule stimulation⁸
 - b) obesity⁹
 - c) drug addiction¹⁰
 - d) hypertension (case report of lowering BP in a patient who was being treated for pain¹¹)

98.2 Typical targets used in functional brain surgery

□ Fig. 98.1 depicts relationships of typical targets to other structures and effects of DBS (or lesioning). This figure is intended for illustrative purposes only, and is not presented for purposes of performing surgical procedures.

98.3 Surgical treatment of Parkinson's disease

98.3.1 Historical background

Prior to the availability of effective medications, surgical procedures were developed to treat Parkinson's disease. An early procedure was ligation of the anterior choroidal artery. Due to variability in distribution, destruction often extended beyond the desired confines of the pallidum and the results were too unpredictable. Anterodorsal pallidotomy became an accepted procedure in the 1950's, but long-term improvement was mainly in rigidity, while tremor and bradykinesia did not improve.¹² The ventrolateral thalamus subsequently became the preferred target. Lesions there were most effective in diminishing tremor. In actuality, the tremor was often not the most debilitating symptom, particularly since it is a resting tremor at first (it may become more pervasive later). Bradykinesia and rigidity were frequently more problematic. Furthermore, the procedure only reduces tremor in the contralateral half of the body, and bilateral thalamotomies were not recommended due to an unacceptably high risk of post-op dysarthria and gait disturbance. Use of thalamotomy fell off dramatically in the late 1960's with the introduction of L-dopa.¹³

98.3.2 Current trends

However, at some point most patients will experience problematic side effects and/or resistance to treatment with antiparkinsonian drugs. This, together with recent refinements in surgical techniques resulting in improved outcomes have produced in a resurgence of interest in the operative treatment of Parkinson's disease. Tissue transplantation (e.g. with adrenal medullary tissue) appears to have only modest benefits (see below). Lesioning or stimulation techniques have therefore gained in popularity with renewed interest in the posteroventral pallidum as the target, which was actually pioneered by Leksell around the time that thalamotomy came into vogue.¹²

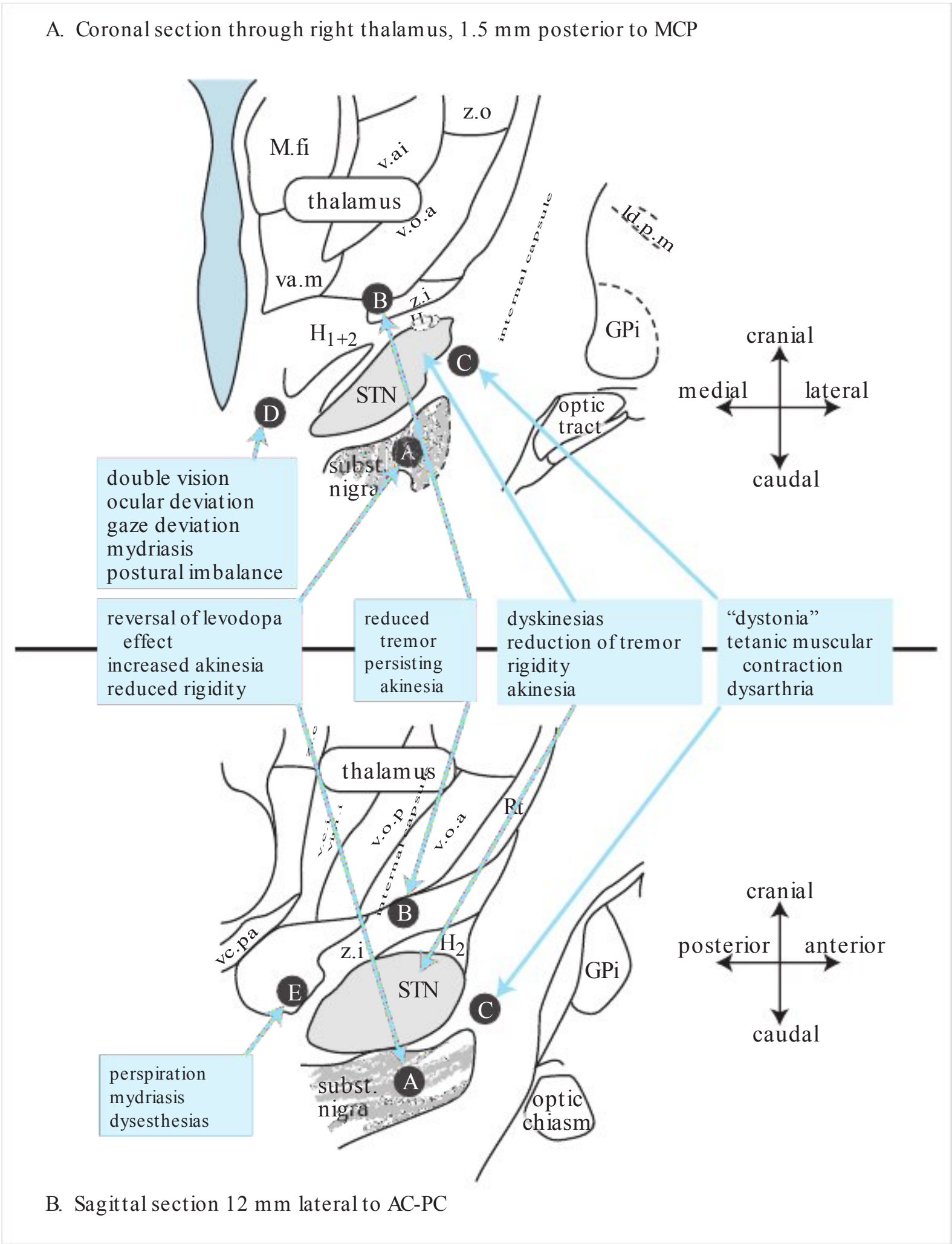


Fig. 98.1 Illustration of some targets for functional brain surgery
Abbreviations: AC=anterior commissure, GPi=globus pallidus interna, H₁=Forel’s H₁ field, MCP=midcommissural point (halfway between AC & PC), PC=posterior commissure, STN=subthalamic nucleus, subst.nigr=substantia nigra, z.i=zona incerta

98.3.3 Tissue transplantation

Tissue transplantation for Parkinson’s disease is generally limited to research centers. The present status of implantation of fetal dopaminergic brain cells into Parkinson’s disease patients is that it may reduce the severity of the illness and increase the effectiveness of levodopa.¹⁴ For ethical reasons, this procedure is rarely performed in the U.S.

Other transplanted tissues include cells from the patient's own adrenal medulla. After initial enthusiastic results,¹⁵ later studies failed to corroborate the dramatic outcomes, and benefits appear to be modest.^{16,17,18}

A double-blinded, randomized, placebo-controlled trial¹⁹ of 34 subjects with severe PD noted initial improvement at 6 and 9 months, but found no efficacy 2 years after fetal mesencephalic cell transplantation. Of note: immunosuppression was only used for six months. Further research is ongoing.²⁰

98.3.4 Ablative surgery and electrical stimulation

General information

Ablative surgery has largely given way to less destructive deep brain stimulation (DBS).

Pallidotomy

See references.^{21,22}

General information

Pallidotomy may work by one of the following mechanisms: directly destroying portions of the internal segment of the globus pallidus internus (GPi), interrupting pallidofugal pathways, or diminishing inputs to the medial pallidum (especially from the subthalamic nucleus; see Pathophysiology (p.177)). Although early methodologies included stereotactic radiosurgery,²³ modern techniques (excluding very select cases) rely primarily on radiofrequency or cryoprobe lesioning after confirming target location by electrical stimulation.

Electrical stimulation

Deep brain stimulation (DBS) in the area of the GPi²⁴ and subthalamic nucleus (STN) can also relieve parkinsonian symptoms²⁵ without irreversibly destroying tissue. A randomized study showed similar efficacy between thalamotomy and DBS, but fewer side-effects with DBS.²⁶ A more recent target of interest for DBS is the pedunculopontine nucleus (PPN).

Indications

1. patients refractory to medical therapy (including multiple agents). However, some investigators feel the response to surgery might be better if done early
2. primary indication (based on an opinion survey²⁷): patients with levodopa-induced dyskinesias (especially those with associated painful muscle spasms). Initial results indicate that these are very responsive to pallidotomy
3. gait and postural instability²⁸ as well as falls and freezing (non-human primate data)²⁹ may respond to DBS of the pedunculopontine nucleus (PPN)
4. patients primarily with rigidity or bradykinesia (unilateral or bilateral), on-off fluctuations or dystonia. Tremor may be present, but if it is the predominant symptom, then using the ventralis intermedius (VIM) nucleus of the thalamus as the target (for ablation (thalamotomy) or stimulation)³⁰ is a better procedure. VIM stimulation is also used to treat essential tremor³¹

Contraindications

1. patients with significant dementia: further cognitive impairment has been noted primarily in patients with cognitive deficits prior to treatment
2. patients with risk of intracerebral hemorrhage: those with coagulopathy, poorly controlled hypertension, those on anti-platelet drugs that cannot be withheld (may consider stereotactic radiosurgery lesions for these rare patients, see below)
3. patients with ipsilateral hemianopsia: due to the risk of post-op contralateral hemianopsia from optic tract injury which would make the patient blind
4. age ≥ 85 yrs
5. patients with secondary Parkinsonism (p.177) i.e. not idiopathic Parkinson's disease: respond poorly, presumably due to different pathophysiology. Look for:
 - a) signs of autonomic nervous system dysfunction (suggests Shy-Drager)
 - b) EOM abnormalities (may occur in progressive supranuclear palsy (PSNP))
 - c) long-tract signs
 - d) cerebellar findings (as in olivo-ponto-cerebellar atrophy (OPCA))
 - e) failure to improve with levodopa

- f) MRI: lacunar infarcts in basal ganglia (as in arteriosclerotic Parkinsonism), or tumor in region of substantia nigra
- g) PET scanning (if available): decreased striatal metabolism detected by deoxyglucose PET scan (suggests striato-nigral degeneration (SND))

Technique

General information

Antiparkinsonian medications are withheld the morning of the procedure to bring out symptoms. A stereotactic frame is applied under local anesthetic parallel to the orbitomeatal line (which aligns with the anterior-posterior commissural (AC-PC) line).

Radiologic target localization

May utilize: MRI, CT, and/or ventriculography. MRI is the most common imaging modality, and may best demonstrate the desired anatomy, but is susceptible to geometric distortion. Therefore many centers also utilize CT and/or ventriculography to supplement MRI. T1WI images are commonly employed, however some feel that optimal MRI imaging may be performed with gadolinium-enhanced axial and coronal projections using 1 mm slice intervals and a STIR or spoiled GRASS volume acquisition protocol.

The posterior commissure is the white-matter band at the level of the pineal that crosses at the posterior third ventricle.

The typical initial target²⁷ is shown in □ Table 98.1. Avoid encroachment on internal capsule (medial to GPi) and optic tract (inferior to GPi). Lesions of the subthalamic nucleus are associated with hemiballismus. An entry site is chosen from imaging studies, and is usually just anterior to the coronal suture and 15–20 mm lateral to the midline. A 4 mm twist drill hole is used. The trajectory should avoid midline venous structures, arterioles within sulci (therefore enter through a gyrus), and passage through the lateral ventricle.

Electrophysiologic target localization

Stimulation:

The patient must be awake for the procedure. For patients with dyskinesias that occur only following a dose of medication, their normal dose of medicine is given after imaging to bring out the symptoms for the procedure. Stimulation is required to verify the neurophysiologic target which varies between individuals. Macroelectrode stimulation may be done with the lesioning electrode. Impedance typically drops when a white matter tract is encountered. The impedance of the desired target is usually >600 Ω. Stimulate with square wave at 1, 5, 50 and 100 Hz with voltage range of 0.5–3 volts (NB: above ≈ 2 V you may be seeing wide-field stimulation). Pallidum stimulation usually increases (but occasionally decreases) contralateral muscle tone. Also look for reduction of tremor or dyskinesia. Contralateral weakness or hypotonia indicates proximity to the internal capsule. Visual scotomata suggests stimulation of the optic tract.

Micro-electrode recording:

About half the institutions surveyed perform microelectrode recording, and half the remaining centers were considering starting it.

Lesioning

Kondziolka et al.²² use a 1.1 mm diameter probe with a 3 mm exposed tip. A small lesion is made at 45°C for 30 seconds, before making the definitive lesion at 70–80°C for 60 seconds. The probe is withdrawn 3–4 mm and a second lesion is made. Lesions with cryoprobes may be associated with a higher incidence of intracerebral hemorrhage.

For the very rare patient in whom insertion of an electrode is contraindicated (e.g. refractory coagulopathy), lesioning may be done with stereotactic radiosurgery, however, this eliminates the

Table 98.1 Target for pallidotomy	
Typical initial target	Median
1–3 mm anterior to the midpoint of the AC-PC line	2 mm
18–23 mm lateral ^a	21 mm
2–6 mm inferior	5 mm
^a may be decreased in women (start at ≈ 19 mm), or increased when the 3rd ventricle is dilated	

critical ability to verify the site of the planned lesion electrophysiologically before a permanent lesion is made.

Unilateral pallidotomy produces primarily contralateral effects, although some ipsilateral changes occur. Bilateral procedures are usually staged separately with a 3–12 hiatus between sides. Although they can be done in one sitting, bilateral pallidotomies may carry an increased risk of speech difficulties and cognitive decline.

Results

At present, the major focus of therapy has been on improvement of motor symptoms. Although 97% of patients showed at least some improvement (some poor results may derive from inclusion of some patients with secondary Parkinsonism), in 17% the degree of improvement was graded as mild.

Significant reduction of levodopa induced dyskinesias occurred in 90%. Bradykinesia improved in 85%, rigidity in 75% and tremor in 57%. Other areas of improvement include: speech, gait, posture, and reduction of on-off phenomenon and freezing. Although symptoms may be ameliorated, overall functional improvement may not be remarkable.³²

Although dosages of antiparkinsonian medication are often reduced, continued medical therapy is usually required, and no change is made for at least 2 months following pallidotomy.

Indications are that beneficial surgical effects can last ≥ 5 years, with early failures possibly due to production of too small of a lesion, and late failures possibly due to progression of the disease.

Ongoing studies are investigating longer term results, microelectrode recording, alternate lesioning targets, the role of early surgery... Until more information is available, one cannot make any statements about the optimal target, localizing method, etc.

Complications

Visual field deficit occurs in 2.5% due to proximity of the optic tract to the globus pallidus. Hemiparesis may occur due to the nearby passage of the internal capsule. Intracerebral hemorrhage may also occur. Dysarthria occurs in $\approx 8\%$ but is usually temporary. Speech difficulties and also cognitive decline may be more risky when bilateral pallidotomies are performed at the same sitting.

Thalamic lesions

Lesioning the thalamic ventralis intermedius nucleus (VIM) nucleus reduces Parkinsonian tremor in $>85\%$. It can also be useful in the treatment of rigidity and drug induced dyskinesias by extending the lesioning anteriorly to include the ventral oralis. However thalamotomy does not improve symptoms of akinesia or bradykinesia and can result in worsening of gait symptoms or speech problems.

Subthalamotomy

Lesioning the subthalamic nucleus (STN) is classically associated with intractable hemiballism. There are few studies as a result, but the limited data suggests that selective lesioning in this region provides relief on par with pallidotomy. Postoperative hemichorea is a known complication but is generally transitory and mild.³³ DBS in this region may be a better option (p. 1524).

98.4 Dystonia

Pallidal stimulation is the primary surgical treatment for dystonia.³⁴ Response is better for primary dystonias, e.g. tardive dystonias than for secondary dystonias such as postanoxic, postencephalitic, perinatal and poststroke dystonias³⁴ (other targets need to be assessed).

For primary dystonias, the globus pallidus internus (GPi) is the most common primary target (□ Fig. 98.1). Good results have also been reported with STN DBS.

98.5 Spasticity

98.5.1 General information

Results from lesions in upper motor neuron pathway, causing absence of inhibitory influence on alpha motor-neurons (α MN) (alpha spasticity) as well as on gamma motor neurons (intrafusal fibers) (gamma spasticity). Causes uninhibited reflex arc between α MN and Ia afferents from muscle spindles resulting in a hypertonic state of muscles with clonus, and sometimes with involuntary movements. Etiologies include: injury to cerebrum (e.g. stroke) or spinal cord (spasticity is an

expected sequelae of spinal cord injury rostral to the conus medullaris), multiple sclerosis, and congenital abnormalities (e.g. cerebral palsy, spinal dysraphism).

98.5.2 Clinical

General information

Increased resistance to passive movement, hyperactive muscle stretch reflexes, simultaneous activation of antagonistic muscle groups, may occur spontaneously or in response to minimal stimuli. Characteristic postures include scissoring of legs or hyperflexion of thighs. May be painful, or may disrupt patient’s ability to sit in wheelchair, lay in bed, drive modified vehicles, sleep, etc. May also promote development of decubitus ulcers. A spastic bladder will have low capacity and will empty spontaneously.

Spasticity is often exacerbated by same type of stimuli that aggravate autonomic hyperreflexia; see Autonomic hyperreflexia (p.1020).

The onset of spasticity following spinal cord injury may be delayed for several days to months (the latency period is attributed to “spinal shock” (p.931), during which time there is decreased tone and reflexes).³⁵ Onset of spasticity following spinal shock starts with increasing flexor synergistic activity over 3–6 mos, with more gradual increases of extensor synergy which ultimately predominates in most cases.

Some “beneficial” aspects of mild spasticity:

- 1. maintains muscle tone and therefore bulk: provides support for patient when sitting in wheelchair, helps prevent decubitus ulcers over bony prominences
- 2. muscle contractions may help prevent DVTs
- 3. may be useful in bracing

Grading spasticity

Assessment should be performed with patient supine and relaxed. The Ashworth scale (□ Table 98.2) is commonly used for the clinical grading of the severity of spasticity. Many attempts have been made to quantitate spasticity electrodiagnostically, the most reliable has been H-reflex measurement.

98.5.3 Treatment

General information

Depends on extent of useful function (or potential for same) present in areas at and below the level where spasticity starts (complete spinal cord injuries usually have little function, whereas patients with MS may have significant function).

Medical treatment

- 1. “prevention”: measures to decrease inciting stimuli (physical therapy to prevent joint damage, good skin & bladder care... see Autonomic hyperreflexia (p.1020))
- 2. prolonged stretching (more than just range of motion): not only prevents joint and muscle contractures, but modulates spasticity
- 3. oral medications³⁷; (see Surgical treatment (p.1530) for intrathecal medications: few drugs are effective without significant undesirable side effects

Table 98.2 Ashworth scores ³⁶	
Ashworth score	Degree of muscle tone
1	no increase in tone (normal)
2	slight increase, a “catch” with flexion or extension
3	more marked increase, passive movements easy
4	considerable increase, passive movements difficult
5	affected part rigid in flexion or extension

- a) diazepam (Valium®): activates GABAA receptors, increases pre-synaptic inhibition of α MN. Most useful in patients with complete spinal cord injuries.
 - start with 2 mg PO BID-TID, increase by 2 mg per day q 3 days up to a max of 20 mg TID.

Side effects: may cause sedation, weakness, decreased stamina (most of which may be minimized by gradual increases in dosage). Abrupt discontinuation may cause depression, seizures, withdrawal syndrome

- a) baclofen (Lioresal®): activates presynaptic GABAB receptors of Ia muscle spindle afferents, causes pre-synaptic inhibition of α MN and decreases nociception. May be most useful in patients with spinal cord lesions (complete or incomplete).
 - start with 5 mg PO BID-TID, increase in 5 mg increments q 3 days up to max of 20 mg QID. Side effects: sedation, lowers seizure threshold. Must be tapered to discontinue (abrupt discontinuation may result in seizures, rebound hyper-spasticity or hallucinations).
- b) dantrolene (Dantrium®): reduces depolarization induced Ca^{++} influx into sarcoplasmic reticulum of skeletal muscle; acts on all skeletal muscle (with no preferential effect on spasmogenic reflex arc).
 - start with 25 mg PO q d, increase q 4–7 days to BID, TID, then QID, then by 25 mg per day up to max of ≈ 100 mg QID (may take 1 week at new steady state to see effect); Side effects: muscle weakness (may make ambulation impossible), sedation, idiosyncratic hepatitis (may be fatal; more common in patients on >300 mg/d $\times >2$ mos) that is often preceded by anorexia, abdominal pain, N/V; D/C if no benefit is seen by ≈ 45 days; follow LFTs (SGPT or SGOT)
- c) progabide: activates both GABAA and GABAB receptors. Useful for patients with severe flexor spasms
- d) theoretical benefits may be derived from other agents, but they have not been used for some practical reason in each case³⁵ (e.g. phenothiazines reduce gamma spasticity, but only at high PO doses or parenterally; clonidine; tetrahydrocannabinol...)

Surgical treatment

General information

Reserved for spasticity refractory to medical treatment, or where side effects of medications are intolerable. Generally either orthopedic (e.g. tendon release operations (tenotomies) of heel cord or hamstrings, iliopsoas myotomies, etc.) or neurosurgical (e.g. nerve blocks, neurectomies, myelotomy, etc.). Herz et al.³⁸ recommend percutaneous radiofrequency foraminal rhizotomy, usually with percutaneous radiofrequency sciatic neurectomy as the initial ablative procedure (see below).

1. nonablative procedures
 - a) intrathecal (IT) baclofen (see below)
 - b) intrathecal morphine (tolerance and dependence may develop)
 - c) electrical stimulation via percutaneously placed epidural electrodes³⁹
2. ablative procedures, with preservation of potential for ambulation
 - a) motor point block³⁵ (intramuscular phenol neurolysis): preserves sensation and existing voluntary function. Especially useful in patients with incomplete myelopathies; time consuming
 - b) phenol nerve block: similar to motor point block, but used when spasticity more severe and complete block of muscle desired. Open phenol block used instead of percutaneous when nerve is mixed and sensory preservation is desired (also reduces post-block dysesthesias)⁴⁰
 - c) selective neurectomies³⁵
 - sciatic neurectomy (may be done with RF lesion)³⁸
 - obturator neurectomy: useful if strong hip adductor spasticity that causes scissoring and wasted energy expenditure in ambulating
 - pudendal neurectomy: useful if excessive detrusor dyssynergy interferes with bladder retraining
 - d) percutaneous radiofrequency foraminal rhizotomy: small unmyelinated sensory fibers are more sensitive to RF lesions than larger myelinated A-alpha fibers of motor units. Technique: start at S1 on one side, and work up to T12, then repeat on other side. At each level: verify needle position by stimulating with 0.1–0.5 V and watch for movement in appropriate myotome (tip should be extradural, avoid subarachnoid placement), ablate with 70–80° C $\times$ 2 mins for S1, and 70° C $\times$ 2 mins for L5 to T12 (to preserve motor function). If symptoms recur, may repeat with lesions at 90° C $\times$ 2 mins
 - e) myelotomies⁴¹
 - Bischof's myelotomy: divides anterior and posterior horns via laterally placed incision, disrupts reflex arc. No effect on α -spasticity

- midline “T” myelotomy: interrupts reflex arc from sensory to motor units without disrupting connections from corticospinal tract to anterior motor neurons. Slightly higher risk of losing motor function.
Technique: laminectomy from T11 to L1. Mobilize midline dorsal longitudinal vein and incise cord in midline from T12 at a depth of 3 mm to S1 at a depth of 4 mm (preserving S2-S4 maintains bladder reflex pathways. Unilateral extension up to conus medullaris reduces bladder spasticity and increases capacity before reflex emptying occurs)
 - f) selective dorsal rhizotomy^{42,43}: uses intraoperative EMG and electrophysiological stimulation to eliminate sensory rootlets involved in “handicapping spasticity” (leaves rootlets subserving “useful spasticity” intact). Interrupts the afferent limb of pathologic reflex arc. May be temporary, but seems to persist at least ≈ 5 yrs. No effect on α-spasticity. Ambulatory children with cerebral palsy have improved gait, nonambulatory children are improved but are still not able to ambulate afterwards
 - g) stereotactic thalamotomy or dentatotomy: may be useful in cerebral palsy.⁴⁴ Useful for unilateral dystonia, but cannot be used for bilateral dystonia as bilateral lesions would be required which jeopardizes speech. Effective only for dystonia distal to shoulders or hips, and should not be used if the condition is rapidly progressive
3. ablative procedures, with sacrifice of potential for ambulation (in complete cord injuries, nonablative procedures are not indicated because there is no motor function to recover). Used after failure of percutaneous rhizotomy (see above) and “T” myelotomy (see above)
- a) intrathecal injection of 6 ml of 10%phenol (by weight) in glycerin mixed with 4 ml of iohexol (Omnipaque® 300) (p.219) for a final concentration of 6%phenol and ≈ 120 mg iodine/ml. Administered via LP at L2–3 interspace with patient in lateral decubitus position (most symptomatic side down) under fluoro until T12-S1 nerve root sleeves are filled (sparing S2–4 for bladder function). Patient is maintained in this position × 20–30 mins and then kept sitting upright × 4 hrs⁴⁵ (absolute alcohol provides more permanent blocks, but is hypobaric and more difficult to control)
 - b) selective anterior rhizotomy: results in flaccid paralysis with denervation atrophy of muscles
 - c) neurectomies, often combined with tenotomies³⁸
 - d) intramuscular neurolysis by phenol injection³⁸
 - e) corpectomy⁴⁶: most drastic measure, reserved for patients who do not respond to any other measure. Results in total flaccidity with loss of benefits from mild spasticity. Converts bladder from UMN to LMN control. Works well for progressive deficit from syringomyelia and for spasticity, but poor for “phantom” leg pain⁴⁷
 - f) cordotomy: rarely used

Intrathecal baclofen

See references.^{48,49,50,51}

Selection criteria used in one study⁵⁰ are shown in □ Table 98.3. Other indications include: stroke,⁵² cerebral palsy, TBI, dystonia, stiff-man syndrome.

Test doses: Incremental test doses of 50, 75, and then 100 mcg intrathecal baclofen (ITB) via lumbar puncture or temporary catheter were used,⁵⁰ randomly alternated with placebo, with dose escalation halted if a response to active drug occurred. The following parameters were evaluated at 0.5, 1, 2, 4, 8 & 24 hrs post injection: pulse and respiratory rate, BP, hypertonia (Ashworth score, □ Table 98.2), reflexes, spasm score, voluntary muscle movement, and adverse effects (if any, including seizures). Pump implantation was offered if there was a 2 point reduction in the Ashworth score and muscle spasm score for ≥4 hrs after bolus injection of active drug without intolerable side effects. Usual daily dose for ITB is twice the test dose, typically 200 micrograms/d.

Table 98.3 Selection criteria for baclofen pump

- age 18–65 yrs (older patients treated on compassionate use basis)
- able to give informed consent
- severe, chronic spasticity (≥ 12 mos duration) due to spinal cord lesion or MS
- spasticity refractory to oral drugs (including baclofen), or unacceptable side effects
- no CSF block (e.g. on myelography)
- positive response to IT baclofen at test dose ≤ 100 mcg and no response to placebo
- no implanted programmable device such as cardiac pacemaker^a
- females of childbearing potential: not pregnant & using adequate contraception
- no hypersensitivity (allergy) to baclofen
- no history of stroke, impaired renal function, or severe hepatic or GI disease

^athis study used a programmable IT pump

- Alternatively, give 25 mcg IT in the O.R., and if the patient improves, insert subcutaneous pump.³⁷
- Pump systems: Available programmable systems include N’Vision, manufactured by Medtronic, Inc., Minneapolis, MN.
- Insertion technique: IT catheter is typically inserted $\approx$ L2–3, and is threaded rostrally $\approx$ 3 levels, but should be no higher than T10 (risk of rostral progression of hypotonia).
- Post-op orders: Guidelines for post-operative orders following baclofen pump insertion
1. admit PACU, transfer to:
 - a) floor if insertion follows test dosing or if patient has just been transitioned from stable PO dose
 - b) ICU if there has been a hiatus in baclofen therapy
 2. neuro checks q 2 hrs for 1st 24 hours
 3. baclofen:
 - a) for patients on oral or IV baclofen: continue baclofen at the previous dose via the same route (oral or IV) until the ITB takes effect (usually 2–4 hrs, full effect is delayed up to 24 hrs). The IV/PO drug is then tapered
 - b) if there has been a hiatus in baclofen therapy: baclofen 20 mg PO QID
 4. have 2 vials of IV physostigmine available and labeled “FOR EMERGENCY USE ONLY” for possible baclofen overdose

- Baclofen overdose:
- ABCs (airway/breathing/circulation). Intubate if necessary
1. empty pump reservoir to stop drug flow (record amount withdrawn)
 2. administer physostigmine if not contraindicated:
 - a) ☐ adult: 0.5–1.0 mg IM or IV @rate $\leq$ 1 mg/min (may repeat q 10–30 minutes PRN)
 - b) ☐ peds: 0.02 mg/kg IM or IV @rate $\leq$ 0.5 mg/min (may repeat q 5–10 min up to 2 mg max)
 3. withdraw 30–40 ml CSF either via LP or through catheter access port
 4. notify the pump manufacturer

Complications: Device-related complications are shown in ☐ Table 98.4. The frequency of most is $\approx$ 1% except catheter-related problems which had a rate of $\approx$ 30%⁵⁰

Complications related to the drug therapy itself include: overdosage problems (rostral progression of hypotonia, respiratory depression, coma, and seizures).

Intrathecal baclofen withdrawal: Interruption of ITB therapy may occur as a result of: empty pump reservoir, pump battery failure, catheter migration/breakage/kinking/disconnection/occlusion, programming error. Steps in assessing the infusion system are shown in ☐ Table 98.5.

The severity of withdrawal syndrome depends on dose of drug used (increased with higher dose) and duration of therapy (increased with longer therapy).

Syndromes: with abrupt discontinuation of ITB

Table 98.4 ITB pump complications ^a
1. mechanical problems a) pump underinfusion b) catheter problems: occlusion, kink, dislodgment, cut, break or disconnection 2. wound complications a) pocket erosion b) incisional pain c) infection d) seroma (may require aspiration) e) CSF collection
^a device-related complications requiring a secondary invasive procedure

Table 98.5 Assessing ITB infusion systems
• interrogate the pump (with device programmer) • refill reservoir with drug if empty (by experienced ITB practitioner) • obtain AP & lateral x-rays to assess location of catheter tip and to look for breaks, kinks or migration

1. mild withdrawal symptoms: return of spasticity and rigidity, tachycardia, piloerection (goose-bumps) & pruritus
2. more significant withdrawal symptoms: seizures & hallucinations
3. severe withdrawal symptoms (estimated incidence: 3–5%)⁵³: increased rebound spasticity, rigidity, fever, labile BP, and reduced level of consciousness – resembling but distinct from malignant hyperthermia (p.108) or neuroleptic malignant syndrome. If untreated, the severe syndrome can progress over 24–72 hours to rhabdomyolysis (with elevated creatine kinase (CK) and transaminase), hepatic and renal failure, DIC, and occasionally death

Management of abrupt ITB withdrawal syndrome⁵³:

1. ABCs (airway/breathing/circulation). Intubate if necessary
2. primary goal is to reestablish ITB therapy at the same dose as soon as possible
3. assess pump/system as outlined in □ Table 98.5
4. early use of high-dose oral/enteral baclofen: ≥ 120 mg/d in 6–8 divided doses if the patient's condition permits (NB: PO baclofen is not reliable as the lone treatment for ITB withdrawal, and safety not established for age <12 yrs)
5. attempt to restore ITB therapy at or near pre-withdrawal dosage by experienced physician by one of the following:
 - a) using a programmed bolus through the pump
 - b) via the catheter access port
 - c) via LP
 - d) via new externalized catheter
6. if restoration of ITB therapy is delayed and if symptoms persist
 - a) move pt. to ICU if not already there
 - b) parenteral benzodiazepine infusion: diazepam or midazolam. Titrate dose to reduce muscle rigidity, hyperthermia, BP lability, seizures
 - c) cyproheptadine⁵⁴: a serotonin antagonist. Start with 4 mg po q 6 hrs
 - d) diphenhydramine 50 mg PO or IM, may repeat q 6 hrs for pruritis
 - e) dantrolene may not be as effective as it is for malignant hyperthermia
 - f) rectal baclofen

If it is necessary to electively (or semi-electively) remove a pump system, the optimal scenario is to gradually taper the drug by reprogramming the pump and/or by filling the reservoir with solution of decreased baclofen concentration.

98.6 Torticollis

98.6.1 General information

AKA wry neck. A form of dystonia resulting in a failure to control head position (if shoulders or trunk are also involved, dystonia is a more proper label).

98.6.2 Etiologies

A symptom of diverse causes. Differential diagnosis includes:

1. congenital torticollis (may be the initial presentation of dystonia musculorum deformans)
2. spasmodic torticollis, AKA wry neck: a specific subtype of torticollis that is idiopathic by definition. The shortened sternocleidomastoid (SCM) muscle is usually in spasm
3. extrapyramidal lesions (including degenerative): often alleviated by lying down; EMG shows abnormal grouped activity
4. psychogenic (often mentioned, seldom verified)
5. torticollis from atlantoaxial rotatory subluxation (p.968): the elongated SCM may be in spasm (opposite of that in spasmodic torticollis)
6. neurovascular compression of the 11th nerve (see below)
7. hemorrhage into sternocleidomastoid muscle (with subsequent contracture)
8. infection of the cervical spine
9. cervical adenitis
10. syringomyelia
11. cerebellar tumors in children
12. bulbar palsies

13. “pseudotorticollis” may develop as an unconscious correction to reduce diplopia that occurs with imbalance of extraocular eye musculature

98.6.3 Non-surgical treatment of torticollis

Should be attempted first, and includes:

- 1. relaxation training, including biofeedback
- 2. thorough neuropsychiatric evaluation
- 3. trans-epidermal neuro-stimulation (TENS) to the neck

98.6.4 Surgical procedures

Reserved for disabling, refractory cases. Includes:

- 1. dorsal cord stimulation
- 2. local injection of botulinum toxin: may work for retrocollis, is poor for lateral torticollis (must inject posterior cervicals and both SCM, and may cause temporary pharyngeal muscle dysfunction resulting in dysphagia), and is totally ineffective for anterocollis
- 3. selective rhizotomy and spinal accessory nerve section

98.6.5 Other treatments for torticollis

Stereotactic electrocoagulation of Forel’s H1 field.

98.6.6 Torticollis of 11th nerve origin

- 1. usually a horizontal type (manifests as horizontal head movement) which may be exacerbated when supine (unlike extrapyramidal torticollis)
- 2. contraction of SCM is usually accompanied by activity in contralateral agonist muscles
- 3. may be treated surgically. Procedures include
 - a) sectioning of the anastomotic branches between the 11th nerve and the upper cervical posterior root (C1 anastomotic branch is sensory only)
 - b) microvascular decompression of the 11th nerve (most cases caused by vertebral artery, but PICA compression is also described⁵⁵). Relief takes several weeks post-op

98.7 Neurovascular compression syndromes

98.7.1 General information

Syndromes due to compression of cranial nerves at the root entry zone (REZ) (or in the case of motor nerves, root exit zone). The REZ (AKA Obersteiner-Redlich zone) is the point where central myelin (from oligodendroglial cells) changes to peripheral myelin (from Schwann cells).

Syndromes include:

- 1. trigeminal neuralgia, see Trigeminal neuralgia (p.479)
- 2. hemifacial spasm: see below
- 3. disabling positional vertigo
- 4. some forms of torticollis of 11th nerve origin (above)

98.7.2 Hemifacial spasm

General information

Key concepts

- intermittent unilateral painless contractions of facial muscles
- typically caused by compression of VII nerve by AICA
- along with palatal myoclonus: the only movement disorder that persists in sleep
- responds well to microvascular decompression, but risk of hearing loss is ≈ 20%

Hemifacial spasm (HFS) is a condition of intermittent, painless, involuntary, spasmodic contractions of muscles innervated by the facial nerve in one side of the face only. May be limited to the upper or lower half of the face only, and excess lacrimation may be present. HFS usually begins with rare contractions of the orbicularis oculi, and slowly progresses to involve the entire half of the face and increases in frequency until the ability to see out of the affected eye is impaired.

HFS may be associated with trigeminal neuralgia, geniculate neuralgia; see Tic convulsif (p.493), or vestibular and/or cochlear⁵⁶ nerve dysfunction.

HFS is more common in women, is seen more often on the left, and usually presents after the teenages. Auditory function testing reveals abnormal acoustic middle ear reflex in almost half of patients, indicating some degree of VIII compromise.⁵⁶

Meige's syndrome: hemifacial spasm with oral movements.

□ Note. HFS and palatal myoclonus are the only involuntary movement disorders that persist during sleep.⁵⁷

Etiologies

1. vascular compression syndrome (see below): the most common etiology (much more common than with trigeminal neuralgia)
2. idiopathic
3. tumor compressing the nerve
4. can follow some cases of Bell's palsy
5. conditions that can mimic HFS
 - a) blepharospasm (bilateral spasmodic closure of the orbicularis oculi muscles) which is more common in the elderly, and may be associated with organic brain syndrome. Blepharospasm is notorious for disappearing when the patient presents for medical evaluation (an effect of alerting), but may be elicited by asking patient to gently close the eyes and then rapidly open them, following which a blepharospasm may occur. HFS usually involves more than the ocular muscles
 - b) facial myokymia: continuous facial spasm which may be a manifestation of an intrinsic brain-stem glioma or of multiple sclerosis. Often associated with other findings

Vascular compression

HFS is usually caused by compression of the facial nerve at the root exit zone (REZ) by a vessel, which is most often an artery (most commonly AICA⁵⁸ (either pre- or postmeatal⁵⁹), but other vascular possibilities include an elongated PICA, SCA, a tortuous VA, the cochlear artery, a dolichoectatic basilar artery, AICA branches...), aneurysm, a vascular malformation, and rarely, veins have been implicated. In typical HFS (onset in the orbicularis oculi, and progressing downward over the face), the vessel impinges on the antero-caudal aspect of the VII/VIII nerve complex, in atypical HFS (beginning in the buccal muscles and progressing upward over the face) the compression is rostral or posterior to VII.⁶⁰

Vessels contacting the REZ of the vestibular nerve may cause vertigo, whereas tinnitus or hearing loss may result from cochlear nerve REZ compression.

Infrequently, benign tumors or a cyst in the cerebellopontine angle, multiple sclerosis, adhesions, or osseous skull deformities will be the cause of HFS.

Evidence indicates that there is not cross (ephaptic) conduction at the compressed REZ, but that the facial motonucleus is involved secondarily as a result of the REZ compression, via a phenomenon similar to kindling.⁶¹ In addition to the spasm, a 2nd electrophysiological phenomenon associated with HFS is synkinesis, where stimulation of one branch of the facial nerve results in delayed discharges through another branch (average latency: 11 mSec⁶²).

Evaluation

In typical cases of HFS, the diagnostic work-up is negative.

Most patients should have MRI of the posterior fossa (CT scan is less sensitive here) to R/O tumors or AVMs.

Vertebral angiography is usually not performed if imaging is normal. The neurovascular compression responsible for HFS usually cannot be identified on angiography.

Treatment

Medical management

HFS is generally a surgical condition. Early, mild cases may be managed expectantly. Carbamazepine and phenytoin are generally ineffective, unlike the situation with the causally similar condition of trigeminal neuralgia. Local injection of botulinum toxin (Oculinum®) may be effective in treating HFS and/or blepharospasm.^{63,64} Baclofen has been advocated but is not very effective.

Surgical management

General information

Many ablative procedures are effective for HFS (including sectioning of divisions of the facial nerve), however, this leaves the patient with some degree of facial paresis. The current procedure of choice for HFS is microvascular decompression (MVD) wherein the offending vessel is physically moved off of the nerve, and a sponge (e.g. Ivalon®, polyvinyl formyl alcohol foam) is interposed as a cushion. Other cushions may not prove to be as satisfactory (muscle may disappear, and Teflon felt may thin⁶⁵).

Most often, the offending vessel approaches the nerve at a right angle, and causes grooving in the nerve. Compression must occur at the root exit zone; decompression of vessels impinging distal to this area is usually ineffective.

Operative risks: see below.

Post-operatively, there may be episodes of mild HFS, however they usually begin to diminish 2–3 days following MVD. Severe spasm that does not abate suggests failure to achieve adequate decompression, and reoperation should be considered.

Surgical results of MVD depends on the duration of symptoms (shorter duration has better prognosis) as well as on the age of the patient (elderly patients do less well). Complete resolution of HFS occurred in 44 (81%) of 54 patients undergoing MVD, however, 6 of these patients had relapse.⁶⁶ 5 patients (9%) had partial improvement, and 5 (9%) had no relief.

Technique of MVD

Intra-operative brainstem auditory evoked potential (BAER),⁶⁷ or more applicable, direct VIII nerve monitoring⁶⁸ may help prevent hearing loss during MVD for 7th or 8th nerve dysfunction. Furthermore, monitoring for the disappearance of the (delayed) synkinetic response may aid in determining when adequate decompression has been achieved (generally reserved for teaching institutions).⁶¹

For a diagram of the normal anatomy of the CPA, see □ Fig. 1.9. The facial nerve should not be manipulated, and one should avoid dissection around the VII and VIII nerves near the IAC.⁶⁹ Vessels must be preserved, especially the cochlear artery and small perforators.

Place gentle medial traction on the cerebellum (<1 cm is recommended⁶⁹), and incise the arachnoid membrane between the flocculus and the eighth nerve (to avoid tension on nerves that could cause post-op deficit). The IX nerve may be followed medially from the jugular foramen to locate the origin of the VII nerve (the origin of VII is 4 mm cephalad and 2 mm anterior to that of the IX nerve⁷⁰).

Surgical results

Complete resolution of spasm occurs in $\approx 85\text{--}93\%$ ^{65,71,72,73,74} Spasm is diminished in 9% and unchanged in 6%⁷⁴ Of 29 patients with complete relief, 25 (86%) had immediate post-op resolution, and the remaining 4 patients took from 3 mos to 3 yrs to attain quiescence.

Recurrence

Return of symptoms after a period of complete resolution of HFS occurs in up to 10% of patients, 86% of recurrences happen within 2 yrs of surgery, and the risk of developing recurrence after 2 yrs of post-op relief is only $\approx 1\%$ ⁷⁴

Surgical complications

1. ipsilateral hearing loss: may occur from traction injury or a vasospasm
 - a) total hearing loss occurs in $\approx 13\%$ (range: 1.6–15%) (2.8% in one series,⁵⁶ 15% in another series⁶⁶)
 - b) partial hearing loss: 6%
2. facial weakness
 - a) transient: 18%
 - b) permanent facial weakness: 6%⁷²

- 3. ataxia in 1–6%
- 4. other complications that are minor or temporary include:
 - a) aseptic meningitis (AKA hemogenic meningitis) in 8.2%
 - b) hoarseness or dysphagia in 14%
 - c) CSF rhinorrhea in 0.3%
 - d) perioral herpes in 3%⁶⁹

98.8 Hyperhidrosis

98.8.1 General information

Either essential (primary, or idiopathic) or secondary (etiologies include: hyperthyroidism, diabetes mellitus, pheochromocytoma, acromegaly, parkinsonism, CNS trauma, syringomyelia, hypothalamic tumors, menopause).⁷⁵

Due to overactivity of eccrine sweat glands (found over entire body, highest concentration in palms and soles of feet). They produce a hypotonic secretion with saline as the primary constituent. These glands are under control of the sympathetic nervous system, however, the neurotransmitter is paradoxically acetylcholine (i.e. they are cholinergic, unlike most sympathetic end organs which are adrenergic). Most eccrine sweat glands serve a thermoregulatory function, however, those on the palms and soles respond primarily to emotional stress.⁷⁵

Essential hyperhidrosis is a generalized condition that usually manifests mostly in the palms. The incidence is unknown, although it was $\approx 1\%$ in an Israeli study (probably high).

98.8.2 Treatment

Mild cases are treated medically with:

- 1. topical agents: astringents (potassium permanganate, tannic acid...) or antiperspirants (contact dermatitis usually limits use of these agents)
- 2. or systemically with anticholinergics: including atropine, propantheline bromide... (side effects of dry mouth and blurred vision usually limits use of these)
- 3. tap water iontophoresis: may produce keratinization of palmar epithelium

Severe cases refractory to medical therapy may be candidates for surgical sympathectomy (see below).

98.9 Tremor

Thalamotomy or thalamic stimulation may be useful for tremors that are refractory to medical treatment, including parkinsonian (p.1526), essential,^{76,77} cerebellar and post-traumatic.³⁰

98.10 Sympathectomy

98.10.1 Cardiac sympathectomy

With the advances in percutaneous coronary artery techniques, cardiovascular surgery and drugs, cardiac sympathectomy for angina pectoris has found less application. However, it may still be useful in patients who have no further treatment options. Bilateral sympathectomy from the stellate ganglion through the T7 ganglia is required. Newer thoracoscopic techniques may revive some interest in this.

98.10.2 Upper extremity sympathectomy

Various pathologies that may be indications for upper extremity sympathectomy are shown in □ Table 98.6.

Table 98.6 Indications for UE sympathectomy
• essential hyperhidrosis • primary Raynaud’s disease • shoulder-hand syndrome • intractable angina • $\pm$ causalgia major (p.497)

Removal of the only second thoracic ganglion is probably adequate, and avoids a Horner’s syndrome in most. Techniques used include: anterior transthoracic, thoracic endoscopic,⁷⁸ percutaneous radiofrequency, and supraclavicular. An approach via a midline posterior incision with a T3 costo-transversectomy allows bilateral access.^{75,79} The risk of significant complications is ≈ 5% and include pneumothorax, intercostal neuralgia, spinal cord injury, and Horner’s syndrome.

98.10.3 Upper thoracic sympathectomy

Approaches include:

- 1. posterior paravertebral approach
- 2. axillary thoracotomy with transthoracic exposure of the sympathetic chain
- 3. supraclavicular, retropleural exposure
- 4. percutaneous radiofrequency technique^{80,81}
- 5. video endoscopic approach⁸²

98.10.4 Lumbar sympathectomy

Primary indication is for causalgia major of the lower extremity. Preoperative lumbar sympathetic blocks may be utilized to evaluate patient for response.

Removal of the L2 and L3 sympathetic ganglion is usually adequate to remove sympathetic tone from the lower extremities (occasionally L1 and sometimes T12 are also removed for causalgia of the thigh).

The most common approach is a retroperitoneal approach through a flank incision. The patient is placed in a lateral oblique position, and the skin incision is made from the anterior superior iliac spine to the tip of the 12th rib. The peritoneum is dissected from the muscular wall and is retracted anteriorly. The kidney and ureter are retracted anteriorly; injury to the ureter being a major risk of the operation. The sympathetic chain is identified on the lateral aspect of the vertebral bodies. The vena cava makes a right-sided approach more difficult as the aorta is easier to deal with on left-sided approaches.

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99 Pain Procedures

99.1 General information

Medical therapy must be maximized before a patient is a candidate for a pain procedure. Usually this requires escalating the dose of oral narcotic pain medications until the point that the pain is relieved or the side effects (usually somnolence or hallucinations) are intolerable (e.g. up to 300–400 mg/day of MS Contin may sometimes be necessary).

99.2 Choice of pain procedure

□ Table 99.1 shows some pain procedures that may be used for various indications (the list is not intended to be all inclusive, but is rather to serve as a starting point for organizing pain procedures). In general, nonablative procedures are exhausted before resorting to ablative procedures.

99.3 Types of pain procedures

See pain procedures particular to trigeminal neuralgia (p.482).

Techniques for other conditions include:

- 1. electrical stimulation
 - a) deep brain stimulation (p.1524) ¹: targets include thalamus and periaqueductal or periventricular gray matter
 - b) spinal cord stimulation (p.1547)
- 2. direct drug administration into the CNS:
 - a) different routes: spinal (p.1545), epidural or intrathecal, intraventricular (p.1547)
 - b) different agents: local anesthetics, narcotics (without motor, sensory, or sympathetic impairment seen with local anesthetics) (p.1545)
- 3. intracranial ablative procedures:
 - a) cingulotomy: theoretically reduces the unpleasant affect of pain without eliminating the pain. Must be done bilaterally, recently with MRI. Intolerable pain usually recurs after ≈ 3 mos. 10–30%develop flattened affect
 - b) medial thalamotomy: no longer used (presented for historical reasons). Controversial. Was used for some for nociceptive cancer pain. Performed stereotactically
 - c) stereotactic mesencephalotomy²: for unilateral head, neck, face and/or UE pain. Use MRI to create lesion 5 mm lateral to sylvian aqueduct at the level of the inferior colliculus. Unlike spinal cordotomy, the lesion is not near any motor tracts. Main complication is diplopia due to interference with vertical eye movement, often transient
- 4. spinal ablative surgical procedures
 - a) cordotomy: see below
 - open
 - percutaneous
 - b) cordectomy
 - c) commissural myelotomy: for bilateral pain (p.1544)
 - d) punctate midline myelotomy: for relief of visceral cancer pain
 - e) dorsal root entry zone lesion (p.1550)
 - f) dorsal rhizotomy: not useful for large areas of involvement

Table 99.1 Choice of pain procedures ^a			
Unilateral pain		Bilateral or midline pain	
Head, face, neck, UE	Pain at or below C5 dermatome	Below diaphragm	Above diaphragm
DBS	cordotomy ^b	spinal IT narcotics	intraventricular narcotics
stereotactic mesencephalotomy		↓ commissural myelotomy	
^a abbreviations: IT=intrathecal, UE or LE=upper or lower extremity			
^b cordotomy (open or percutaneous) if pain is unresponsive to or too high for spinal IT narcotics			

- g) dorsal root ganglionectomy (an extraspinal procedure)
- h) sacral cordotomy: for patients with pelvic pain who have colostomy and ileostomy. A ligature is tied around the dural sac below S1 nerve roots
- 5. sympathectomy: possibly for causalgia major; see Sympathectomy (p.1537) and Complex regional pain syndrome (CRPS) (p.497)
- 6. peripheral nerve procedures
 - a) nerve block³:
 - neurolytic: injection neurodestructive agents (e.g. phenol or absolute alcohol) on or near the target nerve
 - nonneurolytic: using local anesthetics, sometimes in combination with corticosteroids
 - b) neurectomy: (e.g. intercostal neurectomy for pain due to infiltration of chest wall by malignancy). Performed open or percutaneously with radiofrequency lesion. May sacrifice motor function with mixed nerves
 - c) peripheral nerve stimulators: rarely discussed

99.4 Cordotomy

99.4.1 General information

Interruption of the lateral spinothalamic tract fibers in the spinal cord. Cordotomy is the procedure of choice for unilateral pain below the C5 dermatomal level ($\approx$ nipple; occasionally pain as high up as the mandible may be treated), in a terminally ill patient. Better for aching pain, poor for central pain, dysesthesias, causalgia (deafferentation pain) midline visceral pain. May be performed as an open procedure, but is more easily performed percutaneously at the C1–2 interspace (which limits the procedure to the cervical region). If there is any contralateral pain, it will tend to be magnified following the procedure and often leads to dissatisfaction with cordotomy. If there is any bladder dysfunction, it will usually be worse following cordotomy. Bilateral cervical cordotomies carries a risk of the loss of automaticity of breathing⁴ (one form of sleep apnea, so-called Ondine's curse⁵). Therefore, if bilateral cordotomies are desired, the second should be staged after normal respiratory function and CO₂ responsiveness are verified following the first procedure, or the second stage may be done as an open procedure in the thoracic region.

Review the cross sectional spinal cord anatomy for relationships of the critical tracts (spinothalamic and corticospinal) to the dentate ligament, the anterior spinal artery, respiratory ( Fig. 1.13), and bladder areas ( Fig. 3.1).

99.4.2 Pre-op evaluation

Spirometric measurement of minute volume before and after breathing a mixture of 5%CO₂ and 95% O₂ for 5 minutes. If the MV decreases, these patients are at increased risk of having sleep apnea (usually transient), no increased risk if MV increases or stays the same. Also, patients with <50% of predicted values on PFTs are not candidates.

In patients with pulmonary cancer contralateral to the planned side of cordotomy, check that the contralateral diaphragm is functioning with fluoroscopy, otherwise if the ipsilateral diaphragm is lost due to cordotomy, the patient may be hypopneic.

99.4.3 Percutaneous cordotomy

General information

Indicated for unilateral pain below $\approx$ C4–5 in a terminally ill patient. Radiofrequency current is used to lesion the lateral spinothalamic tract.

99

Technique

Patient does not need to be NPO. Usual pain medications should be given. The patient must be awake and cooperative (any movement with the needle in the cord may lacerate the cord), however one may give e.g. hydroxyzine (Vistaril®) 50 mg IM on call to procedure for relaxation.

The procedure is performed in the x-ray department with either fluoroscopic or CT guidance. For fluoroscopy, the head is placed in a Rosomoff headholder with the height adjusted to keep the mastoid process in the same horizontal plane as the acromioclavicular joint. Working on the side contralateral to the pain, local anesthetic without epinephrine is infiltrated 1 cm caudal to the mastoid tip. An 18 gauge lumbar puncture needle is inserted perfectly horizontal aiming halfway between

the posterior margin of the body of C2 and the anterior portion of the C2 spinous process. Stay rostral to the C2 lamina to avoid the nerve (which is painful).

The dura will be penetrated at about the time that the tip of the needle is approximately even with the midline of the odontoid process on AP fluoro. A few ml of CSF are aspirated and shaken in a syringe together with a few ml of Pantopaque®, and several ml of the mixture are injected into the subarachnoid space under lateral fluoro guidance (note: Pantopaque is no longer available, and water soluble agents are less effective). A needle endoscopic technique may be able to localize the spinal cord anterior to the dentate ligaments. Some dye will layer on the anterior cord, some on the dentate ligament, and most in the posterior thecal space. The dye will only stay momentarily on the dentate ligament, thus be ready to immediately advance the needle just barely anterior to this while monitoring the tip impedance which will jump from $\approx 300\text{--}500\ \Omega$ (ohms) in the CSF to $\approx 1200\text{--}1500\ \Omega$ as the spinal cord is penetrated.

Stimulation at 100 Hz. should produce contralateral tingling at a threshold of ≤ 1 volt. No motor response should be elicited with 100 Hz. in the spinothalamic tract, and if muscle tetany occurs, lesioning must not be performed. If tingling is in the arm, lesioning will usually render from the arm and below analgesic. If tingling is in the lower extremity it will render only that limb analgesic. Stimulation at 2 Hz. should produce ipsilateral twitching of the arm or neck at $\approx 1\text{--}3$ volts.

Radiofrequency lesioning is performed for 30 seconds while the patient sustains contraction of the ipsilateral hand and the voltage is gradually increased from zero. Any twitching of the hand is indication to back down on the voltage. A second lesion is performed in the same region and is usually less painful. The appropriate body area is then checked for analgesia to pinprick.

If the procedure is performed satisfactorily, an ipsilateral Horner’s syndrome usually occurs.

Complications

For complications, □ Table 99.2.

Outcome

In experienced hands, 94% will achieve at least significant pain relief at the time of hospital discharge. The level of analgesia falls with time. At 1 year 60% will be pain free, and at 2 years this will be only 40%

Post-procedure management

CSF leakage will cease spontaneously. Patient is kept supine for 24 hrs to prevent “spinal” (post-LP) headache. Pain medication appropriate to post-operative management is prescribed. If successful, one can rapidly stop the narcotics for the primary pain, withdrawal syndromes occur only rarely.

99.4.4 Open cervical cordotomy (Schwartz technique)

General information

A relatively quick method for open cervical cordotomy.⁶ Can theoretically be done under local for patients who cannot tolerate general anesthesia.

Table 99.2 Post-cordotomy complications	
Complication	Frequency
ataxia	20%
ipsilateral paresis	5% total 3% permanent
bladder dysfunction	10% total 2% permanent
postcordotomy dysesthesia	8%
sleep induced apnea	0.3% unilateral cordotomy 3% bilateral cordotomy
death (respiratory failure)	0.3% unilateral cordotomy 1.6% bilateral cordotomy

Technique

Position: prone; face carefully placed on padded horseshoe headrest, neck slightly flexed to open the interlaminar spaces and to lower the head to prevent accumulation of intracranial air.

Skin incision: midline from occiput to C3. Working only on the side contralateral to the pain, muscles are stripped off the posterior lip of the foramen magnum, and from the lamina of C1 and C2. A Schwartz or Gelpi retractor is engaged between the occiput and C2. To increase exposure, the inferior half of C1 and superior half of C2 lamina are removed with a punch.

Dural incision: the ligamentum flavum is thin between C1–2, and can usually be opened with the dura in a linear incision from the lamina of C1 to C2 placed in the lateral third of the exposure, taking care to avoid bleeding from epidural veins. An angle is cut in the incision at either end to allow increased dural retraction. Tack-up sutures are placed in the dura, the arachnoid is opened, the dentate ligament is located and is gripped with a hemostat and divided between the hemostat and the dura.

Cordotomy: the dentate ligament is used to slightly rotate the spinal cord. A cordotomy knife (or 11 blade) with bone wax placed at 5 mm, is inserted into the cord in an avascular area just anterior to the dentate ligament, sharp side down. The anterolateral quadrant of the cord is cut with the following caveats:

- do not go posterior to the dentate ligament (to avoid corticospinal tract)
- do not cross the midline of the spinal cord
- do not injure the anterior spinal artery
- for patients with lower extremity pain, be sure to start exactly at the dentate ligament (to avoid missing lumbar and sacral fibers).

99.5 Commissural myelotomy

99.5.1 General information

AKA mediolongitudinal myelotomy. Interrupts pain fibers crossing in the anterior commissure on their way to the lateral spinothalamic tract.

99.5.2 Indications

Bilateral or midline pain, primarily below the thoracic levels (including abdomen, pelvis, perineum and lower extremities).

99.5.3 Technique

Laminectomy must extend at least 3 levels above the highest dermatome involved in pain. The dura is opened longitudinally and the operating microscope is then used to identify the midline sulcus (this is usually very difficult to see, and is then estimated as being halfway between where the dorsal roots enter the cord). Veins in the midline are sacrificed for the length of the proposed incision. A number 11 scalpel blade is then placed in a hemostat with 6–7 mm of the tip exposed. The blade is inserted in the midline at the upper end of the desired incision and is then passed caudally for the length of the planned incision (usually 3–4 cm).

99.5.4 Outcome

60% of patients have complete pain relief, 28% have partial, and 8% have none.

99.5.5 Complications

Weakness in the lower extremities occurs in $\approx 8\%$ (usually lower motor neuron, presumably due to injury to anterior horn motor neurons). Dysesthesias occur in almost all patients, but persists $>$ a few days in $\approx 16\%$ (these patients also have impaired joint position sense, all of which are presumably due to posterior column injury). Bladder dysfunction is seen in $\approx 12\%$. Sexual dysfunction may also occur. There is a risk of injury to the anterior spinal artery (rare).

99.6 Punctate midline myelotomy

99.6.1 Indications

Pelvic and visceral pain refractory to other therapies.⁷

99.6.2 Technique

Interruption of a midline posterior column pathway.

99.7 CNS narcotic administration

99.7.1 Intraspinal narcotics

General information

Spinal narcotics may be administered epidurally or intrathecally for pain relief. Satisfactory pain control can usually be achieved for pain below the neck, although for pain above the diaphragm/umbilicus some recommend intraventricular morphine (p.1547).⁸ May also be performed on a “one-time” basis e.g. injection into epidural space following a lumbar laminectomy. Or, it may be given on a short term continuous basis, via an external epidural or intrathecal catheter. It may also be performed on an intermediate-term basis (<60 days) with the use of a subcutaneous reservoir⁹ or on a long-term basis with an implantable drug infusion pump¹⁰ (e.g. Infusaid® or Medtronic® pump). Advantages over systemic narcotics include less sedation and/or confusion, less interference with GI motility (constipation), and possibly less N/V. The effectiveness is usually limited to ≈ 1 year and is thus not indicated for chronic benign pain. With time, increased doses are required because of the development of tolerance and/or progression of disease¹¹ with the concomitant development of the usual narcotic side effects.

Agents to use

Must be preservative-free (for either intrathecal or epidural use). This may be prepared by a pharmacist (e.g. add enough preservative free 0.9% saline to 1 or 3 gm morphine sulfate powder to yield a total of 100 ml produces 10 or 30 mg/ml solution respectively, and then filter this through a 0.22 μ m filter¹²). Alternatively, commercially available preparations include Duramorph® (available as 0.5 or 1 mg/ml) and Infumorph® (available in 20 ml ampules of 10 or 25 mg/ml), any of which may be diluted to a lower strength with preservative free diluent (normal saline). Cross tolerance to systemic narcotics does occur, and spinal narcotics are more effective in patients who have not been on continuous high dose IV opiates (patients on high-dose IV narcotics need higher initial intraspinal narcotic doses).

□ Side effects. Include pruritus (often diffuse, and may be experienced most intensely in the nose), respiratory depression (the respiratory depression with spinal narcotics is usually very gradual, and is often easily detected by monitoring respiratory rate q 1 hr and taking action if the rate decreases), urinary retention, and N/V.

Trial injection

Before implanting a permanent delivery system a test injection should be performed to verify pain relief and tolerance for medication. Administered via percutaneously inserted epidural or intrathecal catheter connected to an external pump. Doses required for intrathecal catheters are usually ≈ 5 –10 times lower than those for epidural catheters.

Sample post-injection orders after a one-time injection:

1. use no other narcotics for ≈ 24 hrs (with a continuous infusion additional narcotics should be withheld until the effect of the spinal narcotics has been determined)
2. 2 ampules (0.4 mg each) of naloxone (Narcan®) and syringe taped to patients bed (for the first 24 hrs after a single injection; at all times with continuous infusion)
3. head of bed elevated $\geq 10^\circ$ for 24 hrs
4. record respiratory rate q 1 hr for 24 hrs; if asleep and respiratory rate < 10 breaths/min, awaken patient. If unable to awaken, administer naloxone 0.4 mg IV and notify physician. Repeat naloxone 0.4 mg IV q 2 min PRN

5. optional: pulse oximeter for 24 hrs
6. diphenhydramine (Benadryl®) 25 mg IV q 1 hr PRN itching
7. droperidol (Inapsine®) 0.625 mg (which is 0.25 ml of the 2.5 mg/ml standard concentration available) IV q 30–60 mins PRN nausea
8. PRN supplemental pain medication:
 - a) narcotic agonist/antagonist: e.g. nalbuphine (Nubain®) 1–4 mg IV q 3 hrs,
OR
 - b) ketorolac tromethamine (Toradol®) 15 mg IV or IM or 30 mg IM q 6 hrs (use lower dose for weight < 50 kg, age > 65 yrs, or reduced renal function)

Implantable drug delivery pumps

Although satisfactory pain control can be achieved with either epidural or intrathecal narcotics (morphine diffuses easily through the dura to the CSF where it gains access to pain receptors), epidural catheters commonly develop problems with scarring and may become less effective sooner than intrathecal catheters. Pumps should only be implanted if patients have successful pain control with test injection of spinal epidural (5–10 mg) or intrathecal (0.5–2 mg) morphine. A life expectancy of > 3 months is recommended for implantable pumps (if shorter longevity is anticipated, an external pump may be used).

One such series of commonly used implantable drug delivery pumps is manufactured by Infusaid Inc.. The only needle that should be used with their devices are special 22 gauge Huber (non-coring) needles. Delivery rates increase with body temperature 10–13% per °C above 37° C, they decrease by the same amount for every °C below 37°C, and also they become inaccurate at ≤ 4 ml of reservoir fluid. These pumps should never be allowed to run until empty, as this may permanently affect accuracy and reliability of drug delivery. In addition to the pump reservoir port, most models have one or more side “bolus” ports that delivers injected fluid directly to the outlet tubing. One should not aspirate when accessing either port.

Medtronic produces a programmable pump.

Surgical insertion

Similar to the insertion of a lumbar-peritoneal shunt (p.1517). The patient is placed in the lateral position, such as on a bean-bag device. The pump is inserted into a subcutaneous pocket, created with a slightly curved 8–10 cm skin incision. The pump may be sutured to the fascia of the abdomen (in obese patients, it may be sutured to the subcutaneous tissue). Excess tubing should be coiled underneath the pump to prevent inadvertent puncture when accessing either reservoir.

The spinal catheter is inserted through a Tuohy needle inserted between lumbar spinous processes either percutaneously or via a small incision 2–3 mm lateral to the spinous processes. Alternatively, it may be inserted directly via a hemilaminectomy. Fluoroscopy may be used intraoperatively to verify rostral placement of the catheter; radiographic visualization of the catheter may be aided by filling it with iodinated contrast, e.g. Omnipaque-300 (p.219). All bends in the tubing should be very gradual to avoid kinking.

Post-op pain management

Although the pump will be infusing when the patient leaves the operating room, unless they have been on intraspinal narcotics up until the time of surgery, it will usually take several days for the drug to reach equilibrium in the CSF before the level of pain control will be adequate. This can be mitigated by a bolus infusion (3–4 mg morphine for epidural catheters, or 0.2–0.4 mg for intrathecal catheters).

Complications

Meningitis and respiratory failure are rare complications. CSF fistula and spinal H/A may occur. Disconnection or dislodgment of catheter tip may result in failure to control pain, but can usually be surgically corrected.

Outcome

Cancer pain is significantly improved in up to 90% Success rate for neuropathic pain (e.g. postherpetic neuralgia, painful diabetic sensory neuropathy): 25–50%

99.7.2 Intraventricular narcotics

Indications

May be used for cancer pain (especially head and neck)¹³ unresponsive to other methods in patients with a life expectancy < 6 mos.

Technique

An intraventricular catheter is connected to a ventricular access device (p. 1518). 0.5–1 mg of intrathecal morphine is injected via the VAD and usually provides ≈ 24 hrs of analgesia.

Complications

□ Side effects. Common ones include dizziness, N/V. The risk of respiratory depression is minimized by using correct dosing. Complications in a series of 52 patients¹³: bacterial colonization of reservoir (4%), dislodged catheter (2%), blocked catheter (6%), postoperative meningitis (2%).

Outcome

Pain is successfully controlled in 70% at 2 mos, but thereafter the effectiveness diminishes as a result of tolerance to the narcotics.

99.8 Spinal cord stimulation (SCS)

99.8.1 General information

Originally developed as dorsal column stimulation (DCS), it has since been determined that pain relief also occurs with ventral stimulation (without stimulation induced paresthesias seen with DCS). Pain relief in humans persists beyond the stimulation time, and is not reversed by naloxone. The exact mechanism of action is undetermined, but probably involves some combination of neurohumoral (i.e. endorphin), antidromic stimulation of a spinal pain “gate”, and supraspinal center stimulation. GABA and serotonin levels have been shown to be increased with SCS.

99.8.2 Indications

Possible indications include:

1. pain¹⁴: postlaminectomy pain syndrome (the most common indication, especially if LE pain > back pain; see below), complex regional pain syndrome (CRPS, formerly “reflex sympathetic dystrophy”) (p. 1548), postthoracotomy pain (intercostal neuralgia), multiple sclerosis, diabetic neuropathy (p. 1549) and sometimes postherpetic neuralgia
2. refractory angina pectoris (p. 1549)
3. painful limb ischemia from inoperable peripheral vascular disease (p. 1549)
4. functional: spastic hemiparesis, dystonia, bladder dysfunction
5. □ generally not used for cancer pain or for patients with limited life expectant.

99.8.3 Technique

In order for SCS to be effective, it is necessary for the patient to feel the stimulation in the areas of pain.¹⁵ Two techniques are used to place electrodes in the epidural space:

1. plate-like electrodes placed via hemilaminectomy
2. wire-like electrodes placed percutaneously with a Tuohy needle.

Following electrode placement, a trial with an external generator over several days determines if SCS is effective. The electrodes are removed unless clear improvement occurs, in which case an implantable pulse generator is placed subcutaneously.

99.8.4 Complications

With plate electrodes, there is a 3.5% incidence of infection which respond to electrode removal and IV antibiotics. Less common complications: electrode migration (usually seen with first few weeks),

lead breakage (less common with present systems), CSF leak, radicular pain, intermittent interference with cardiac pacemakers, and weakness.

99.8.5 Outcome

Success rate in pain control is ≈ 50%improvement in 50%of patients in experienced hands at specialized centers where multidisciplinary approach is available.¹⁵ In a retrospective long-term follow-up study (ave =96 months) of 410 patients that had SCS implantation due to numerous indications, success rate was 74%¹⁶

Prognosticators of a poor response to SCS include: pain resulting from spinal cord injury, from lesions proximal to the ganglion (e.g. root avulsion), failed back syndrome with back pain >LE pain and multiple previous operations (see below), psychological factors such as litigation, workers compensation, familial/marital discord or drug seeking behavior.¹⁷

99.8.6 Specific syndromes treated

Failed back surgery syndrome

Σ
The addition of SCS improves pain control over either PTor medical management alone for failed back surgery syndrome. At 24 months, SCS is as effective as reoperation in treating radicular pain, with no difference in ADLs or work status.

In the PROCESS trial¹⁸ (Prospective Randomized Controlled Multicenter Trial of the Effectiveness of Spinal Cord Stimulation), 100 patients with failed back surgery syndrome were randomized to SCS placement plus conventional medical management (52 patients) vs. conventional medical management alone (48 patients). The health-related quality of life, measured using EuroQol-5D questionnaire, was greater in the SCS group despite a higher total health care cost at 6 months.

In long-term follow-up at 24 months, the primary outcome (>50%relief in leg pain) is achieved in 37%of patients that were randomized to SCS plus conventional medical management and 2%of patients that were randomized to conventional medical management only. Patients were allowed to cross-over. After cross-over, the primary outcome is achieved in 47%of patients (34 of 72) who had SCS plus medical management as final treatment versus 7%of patients (1 of 15) in the other group (P=0.02).¹⁹

In another randomized prospective study, patients with persistent or recurrent radicular pain after lumbosacral surgery were randomized to reoperation vs. SCS. On an average of 3-year follow-up, the SCS group required less opiate analgesics. 9 of 19 patients in the SCS group compared to only 3 of 26 patients reoperated had self reported pain relief and satisfaction (P<0.01), and there was no difference in ADLs and work status. Patients in the SCS group were less likely to crossover to undergo reoperation (5 of 24 patients from the SCS group versus 14 of 26 patients in the reoperation group, P=0.02).²⁰ In failed back-surgery syndrome SCSs are better for radicular pain than for low back pain.

Complex regional pain syndrome

Σ
SCS may be effective for treating CRPS during the first couple of years, however no significant benefit was evident at 5-year follow-up.

CRPS is a chronic pain condition marked by continuous disabling intense aching or burning pain. Type I has no known nerve injury, Type II follows a nerve injury (p.497). The exact mechanism of the condition leading to intense disproportional pain is unknown and treatment options are limited. In a randomized clinical trial,²¹ patients with CRPS Type I were randomized to receive SCS plus physical therapy (PT) (36 patients) or PT alone (18 patients). 24 of 36 patients had successful SCS trial and

underwent implantation. At 6 months, in the group that received SCS plus PT, pain intensity reduced by 2.4 cm on the visual-analogue scale as opposed to an increase of 0.2 cm in the PT only group ($P<0.001$). In addition, 39% of patients in the SCS group, had “much improved” globally perceived effect vs. 6% ($P=0.01$). The health-related quality of life only improved in the SCS group. At 2-year follow-up, pain intensity in the SCS group reduced by 2.1 vs. 0.0 cm in the PT group compared to baseline ($P<0.001$) and global perceived effect were “much improved” in 43% vs. 6% ($P=0.001$).²² However, these benefits were no longer significant in 5 years.²³

Peripheral vascular disease

Σ

SCS does help with pain due to inoperable limb ischemia. It may or may not improve healing of pressure ulcers.

In a retrospective, non-control study of 38 patients, $\approx 94\%$ experienced pain relief and $\approx 50\%$ experienced healing of ischemic ulcers.²⁴

In a recent review²⁵ of six controlled studies of nearly 450 patients, SCS+medical treatment was compared to medical treatment alone. Although there was no significant difference in ulcer healing, the use of analgesics was less and limb salvage after 12 months was significantly higher in the SCS group (relative risk = 0.71).

Angina pectoris

Σ

SCS was as effective as CABG in controlling refractory angina and protecting against MIs. SCS improves exercise capacity by an unknown mechanism.

SCS reduces anginal pain and improves exercise capacity by an unknown mechanism, which may be related to decrease in myocardial oxygen consumption²⁶ or altering myocardial blood flow²⁷ rather than just masking symptoms

In a multicenter, randomized, prospective clinical trial, comparing SCS to CABG in selected patient,²⁸ there was no significant difference in decreasing anginal attack and nitrates consumption between the groups. 5-year follow-up from this trial found that both CABG and SCS offered similar protection from angina pectoris and myocardial infarction.²⁹

In a prospective study of 104 patients who underwent SCS placement for refractory angina pectoris (average follow-up ≈ 13 months), 73% had $>50\%$ reduction of weekly anginal episodes compared to baseline.³⁰

Diabetic neuropathy

Σ

Available data is limited, but SCS may be a viable modality for refractory pain from diabetic neuropathy. Further study is needed.

No good clinical data are available. A few studies with small numbers of patients suggest SCS can provide significant pain relieve in most patients with diabetic neuropathy that failed conservative management.^{31,32,33}

A small prospective, open-label study reported 9 of 11 patients with diabetic neuropathy that failed conservative treatment had significant pain relieved after SCS implantation at 6 months. Pain score on visual analogue scale decreased from 77 to 34. Microcirculatory perfusion did not change significantly from baseline.³³

99.9 Deep brain stimulation (DBS)

Deafferentation pain syndromes (anesthesia dolorosa, pain from spinal cord injury, or thalamic pain syndromes) may benefit from stimulation of sensory thalamus (ventral posteromedial (VPM) or ventral posterolateral (VPL)). DBS for chronic neuropathic pain produces a 40–50% reduction in pain in about 25–60% of patients.³⁴

Nociceptive pain syndromes are more likely to benefit from stimulation of periventricular gray matter (PVG) or periaqueductal gray matter (PAG) although PAG stimulation is rarely used because it often produces unpleasant side effects. Still, response rate has been only $\approx 20\%$ ³⁵ resulting in failure of the FDA to approve these devices for pain.

Cluster headaches: may respond to hypothalamic stimulation, but larger trials with longer follow-up are needed.³⁴

99.10 Dorsal root entry zone (DREZ) lesions

99.10.1 General information

Although use has been reported for a variety of indications, DREZ lesions appear to be most effective in treating the following:

DREZ lesions appear to be most effective in treating the following:

1. deafferentation pain resulting from nerve root avulsion.^{36,37,38} This most commonly occurs in motorcycle accidents
2. spinal cord injuries (SCI) with pain around the lowest spared dermatome with caudal extension of pain restricted to a few dermatomes (SCI with diffuse pain involving the entire body and limbs below the injury is less responsive)
3. post herpetic neuralgia (p.493): usually good initial response, but early recurrence in $\leq$ few months is common, and only 25% have long-term relief of pain
4. postamputation phantom limb pain: there is some support for this in the literature, but others feel this is not a good indication³⁹
5. ☐ generally not used for cancer pain

99.10.2 Technique

A laminectomy is performed over the involved segment(s) using radiographic localization. The dura is opened, and the DREZ is identified under microscope magnification using intact posterior rootlets above or below for orientation (contralateral rootlets may also be used to estimate the mirror-image location). Lesions are created ipsilateral to the avulsed nerve roots by radiofrequency current (approximately 50–60 lesions are required for several segments, each lesion is done at 75° for ≈ 15 seconds) or selective incisions extending from the last completely normal rootlet at the rostral end to the first normal rootlet caudally. The lesioning needle or knife blade is angled 30–45° medially and inserted to a depth of 2–3 mm. DREZ lesions may be combined with a cordectomy at the level of anatomic cord disruption in paraplegic patients.³⁹

99.10.3 Post-op management

Bed rest for 3 days may reduce the risk of CSF leakage. Analgesics appropriate for a multilevel laminectomy are administered.

99.10.4 Complications

Ipsilateral weakness (related to corticospinal tract) or loss of proprioception (dorsal columns) occurs in 10% of patients, and is permanent in $\approx$ half (i.e. 5%).

99.10.5 Outcome

In pain related to brachial plexus avulsion, 80–90% long-term significant improvement can be expected. Paraplegics with pain limited to the region of injury have an 80% rate of improvement, compared to 30% for those with pain involving the entire body below the lesion.

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100 Seizure Surgery

100.1 General information, indications

20% of patients continue to have seizures despite aggressive medical management with AEDs. Many of these patients may be candidates for surgical procedures to control their seizures.¹

Seizure disorder must be severe, medically refractory with satisfactory trials of tolerable medication for at least 1 year, and disabling to the patient. Medically refractory is usually considered two attempts of high-dose monotherapy with two distinct AEDs, and one attempt at polytherapy.

The three general categories of patients suitable for seizure surgery have²:

1. partial seizures
 - a) temporal origin: the largest group of surgical candidates (especially mesial temporal lobe epilepsy (MTLE) which is often medically refractory)
 - b) extratemporal origin
2. symptomatic generalized seizures: e.g. Lennox-Gastaut
3. unilateral, multifocal epilepsy associated with infantile hemiplegia syndrome

100.2 Pre-surgical evaluation

100.2.1 General information

All patients should undergo high resolution MRI study to rule out neoplasm, AVM, cavernous malformations, mesial temporal sclerosis or hippocampal lesion. Noninvasive techniques allow localization in the majority of cases.

100.2.2 Noninvasive evaluation techniques

□ Video-EEG monitoring. Pre-operative long-term inpatient video-EEG monitoring (surface electrodes) to correlate the clinically disabling seizure with appropriate electrical abnormalities and possibly to identify the seizure focus is required.

□ High-resolution MRI. The imaging modality of choice. Extremely good for detecting hippocampal asymmetry of mesial temporal sclerosis (MTS), and neuronal developmental abnormalities (e.g. cortical dysplasia) that may produce complex partial seizures (CPS).³

□ CAT scan. A seizure focus may enhance with IV contrast shortly following a seizure. Subtle enhancement may be present on the side of the focus on interictal CT scan.⁴

□ PET scan (positron emission tomography). Interictal PET scan using fluorine-18 deoxyglucose (18FDG) shows hypometabolism lateralized to the side of temporal lobe focus in 70% of patients with medically refractory CPS (does not show actual site of origin). Useful when MRI and EEG cannot localize.

□ SPECT scan (single photon emission tomography). Used to demonstrate increased blood flow during a seizure to help localize site of onset. [99m] Technetium (Tc) hexamethyl-propylene-amine-oxime (HMPAO) is usually administered immediately after onset of seizure, and the scan may be obtained within several hours.⁵

□ MEG (Magnetoencephalography). Functional imaging technique for mapping brain activity by recording magnetic fields created by neuronal activity (electrical current).⁶ Synchronized neuronal currents induce a weak magnetic field. Clinical uses include detecting and localizing pathological activity in patients with epilepsy and in localizing eloquent cortex for pre-operative surgical planning. Requires a magnetically shield room.

100.2.3 Mildly invasive techniques

□ Wada test.⁷ AKA intracarotid amytal test. Localizes dominant hemisphere (side of language function) and assesses ability of hemisphere without lesion to maintain memory when isolated. Usually reserved for candidates for large resections.⁸ Each cerebral hemisphere is individually anesthetized

via selective carotid catheterization (usually by a neurointerventionalist) and injection of short acting barbiturate.

Start with angiogram to assess cross flow and to R/O persistent trigeminal artery (p.83). Significant cross-flow is a relative contraindication to anesthetizing the side of dominant supply (patient goes to sleep).

Wada test may be grossly inaccurate with high flow AVM. Also, portions of hippocampus may be supplied by posterior circulation (not anesthetized by ICA injection).

EEG monitoring is usually performed during the test when it is being done for seizure surgery. Patient will show delta waves during deepest level of anesthesia.

Technique

- instruct patient as to what is expected
- catheterize ICA: usually start on side of lesion
- have patient hold contralateral arm in air, and instruct them to hold it there
- inject 100–125 mg sodium amobarbital (Amytal®) rapidly into internal carotid artery (effect starts almost instantaneously, begins to subside after ≈ 8 minutes; (may subside in ≈ 2 minutes with AVM where flow rates are high)
- determine adequacy of injection by assessing motor function in elevated arm (should be $\approx$ flaccid)
- assess language skills by showing patient pictures of objects and ask them to name each one out loud and remember each one
- assess memory function by asking patient to name as many of the pictures as they can ≈ 15 minutes after test: if they have difficulty, ask them to pick out pictures from a group that contains additional ones not shown to patient
- repeat procedure on other side (use lower Amytal doses with each subsequent injection)

100.2.4 Evaluation techniques requiring surgery

□ EEG obtained with invasive electrodes. Indications: Lack of lateralizing or localizing electrophysiology in pre-operative evaluation requires invasive electrodes for better definition of seizure focus.

Surgical options:

- Depth electrodes
 - Electrodes are placed stereotactically
 - stereoencephalography (sEEG): popularized in Europe by J. Talairach and J. Bancaud during the 1950s for invasive mapping of refractory focal epilepsy. The technique requires the placement of multiple depth electrodes in an orthogonal orientation to localize seizure onset.^{9,10,11}
 - 2-3% risk of intracerebral hemorrhage.⁸ Risk of infection with depth electrodes⁸: 2-10%
- Subdural grids or strips
 - Grids are frequently used for extra-operative functional mapping (helpful in children or in the mentally retarded). Subdural grid electrodes are placed with a craniotomy.
 - Surface strip electrodes may be placed through a burr hole
 - useful technique for intra-operative functional mapping

100.3 Surgical techniques

100.3.1 Basic procedures

Three basic types of procedure: resections, disconnections and stimulation.¹

1. resections

- a) resection of epileptic focus: higher chance of completely controlling seizures. Performed in noneloquent brain. Seizures must have focal onset (resection not encouraged if multifocal onset). Includes:
 - anterior temporal lobectomy or amygdalo-hippocampectomy for MTLE: see below
 - neocortical resections: especially with neuronal migration abnormalities
- b) resection of lesion in secondary epilepsy (lesional epilepsy e.g. tumor, AVM, cavernous malformation¹²...). In most cases the seizure focus is in or near the lesion, but some structural lesions are not responsible for the seizures. For seizure foci within the mesial temporal lobe, seizure control is better when lesionectomy is accompanied by amygdalo-hippocampectomy¹³

2. disconnections: used when eloquent brain is involved, or to separate the electrical activity of the two cerebral hemispheres

- a) section of corpus callosum (callosotomy): when drop attacks are the most disabling seizure type or for multiple bilateral foci (see below)

- b) hemispherectomy: for unilateral seizures with widespread hemispheric lesions and profound contralateral neurologic deficit. If any cortex is left, must make sure it is functionally deafferented (disconnected)
 - anatomic hemispherectomy
 - functional hemispherectomy: preservation of the basal ganglia isolates the abnormal side with $\approx 80\%$ seizure control rate (similar to anatomic hemispherectomy, but with lower complication rate).
- c) multiple subpial transection¹⁴: for partial seizure originating in eloquent cortical areas. The cortex is transected at 5 mm intervals, thus interrupting the horizontal spread of the seizure while sparing the vertically oriented functional fibers
- 3. stimulation: usually offered to non-resective candidates (poor localization or surgical failures). It is a reversible and adjustable mode of therapy.
- 4. Open Loop stimulation: blind stimulation continuously or intermittently
 - a) vagus nerve stimulation (p. 1558)
 - b) deep brain stimulation (DBS)
 - centromedian nucleus of the thalamus¹⁵: better for generalized tonic-clonic seizures
 - bilateral anterior nucleus of the thalamus: for partial seizures¹⁶
 - hippocampus¹⁵: for partial seizures
- 5. Closed Loop stimulation: react to a burst of stimulation (detection). Responsive cortical stimulation (RNS)¹⁷ requires a receive and a send electrode for electrical stimulation

Neurostimulation offers between 30-40% seizure reduction for most of the patients implanted. Risks include hardware failure, hemorrhage (not typically seen with VNS), infection and stimulation induced side-effects.

100.3.2 Anesthetic considerations

If intraoperative electrocorticography is to be performed:

- under local anesthesia: the only anesthetic agents that may be used are narcotics (usually fentanyl) and droperidol
- under general anesthesia: avoid benzodiazepines and barbiturates

100.3.3 Intraoperative electrocorticography (ECoG)

Subdural strips and/or grids are useful for ECoG and motor/speech mapping.

Methohexital (Brevitol®) may be given to try to provoke a seizure: observe for ↓ fast activity in suspected focus.

100.3.4 Intraoperative cortical mapping

See techniques of cortical mapping (p. 1432).

100.4 Surgical procedures

100.4.1 Corpus callosotomy

Indications and contraindications

Partial or total section may be most effective for generalized major motor seizures. Of little benefit for simple or complex seizures. Benefit has been supported for:

1. frequent episodes of atonic seizures (“drop attacks”) where loss of postural tone → falls and injuries¹⁸ (70% reduction with callosotomy), typically seen in Lennox-Gastaut syndrome
2. possibly for generalized seizure disorder with unilateral hemisphere damage (e.g. infantile hemiplegia syndrome); hemicortical resection may be better for this type, whereas callosotomy may promote partial seizures.

Note: a “functional hemispherectomy” is recommended over “anatomically complete” hemispherectomy to reduce morbidity and mortality²

3. some patients with generalized seizures without identifiable, resectable focus
4. ☐ Contraindication: major behavioral and/or language deficits may occur even with partial division in patients with speech and dominant handedness located in opposite hemispheres (“crossed dominance”). Thus, Wada test is recommended in all left handed patients

Technical details

Division of the anterior two-thirds of the corpus callosum (CC) (minimizes the risk of disconnection syndrome, see below) may be advantageous over complete callosotomy (controversial). Some advocate sectioning the CC with intraoperative EEG until the typical bisynchronous discharges that are usually seen become asynchronous.¹⁹ No need to section anterior commissure. Can usually be performed via a bifrontal craniotomy utilizing a bicoronal skin incision.

May produce post-op ↓ verbalization or akinetic mutism that usually resolves in weeks.

MRI sagittal cuts are ideal for assessing extent of division of the CC.²⁰

Disconnection syndrome

In a patient with a dominant left hemisphere, consists of left tactile anomia, left sided dyspraxia (may resemble hemiparesis), pseudohemianopsia, right sided anomia for smell, impaired spatial synthesis of right hand resulting in difficulty copying complex figures, decreased spontaneity of speech, incontinence.

More common with larger surgical sections of the CC. Risk is less if the anterior commissure is spared. Patients usually adapt after 2–3 months, with final function normal for most daily activities (deficits may show up on neuropsychological testing).

100.4.2 Mesial temporal lobe epilepsy (MTLE)

General information

80% of patients with medically intractable temporal seizures have a demonstrable focus in the anterior-mesial temporal lobe. Most patients have neuronal loss and gliosis of mesial temporal structures (mesial temporal sclerosis, MTS). Thus, a standard resection of temporal tip (with amygdalo-hippocampectomy) may be performed. In a randomized trial, anterior temporal lobectomy (ATL) was shown to be superior to medical management for treatment of medically resistant epilepsy. Results at one year show seizure reduction and improved quality of life in those patients who underwent ATL versus medical management alone for intractable epilepsy.²¹

Limits of resection (without significant neurologic deficit)

Note that these values are generally considered safe, however, variations occur from patient to patient and only intraoperative mapping can reliably determine the location of language centers.²² Most centers spare the superior temporal gyrus.²³ The following measurements are made along the middle temporal gyrus:

- dominant temporal lobe: up to 4–5 cm may be removed. Over-resection may injure speech centers, which cannot be reliably localized visually
- non-dominant temporal lobe: 6–7 cm may be resected. Slight over-resection may → partial contralateral upper quadrant homonymous hemianopsia; resection of 8–9 cm → complete quadrantanopsia

Alternatively, intraoperative electrocorticography may be used to guide resection of electrically abnormal areas.

Resection should be performed in subpial plane to prevent injury to vascular branches.

100.4.3 Selective amygdalo-hippocampectomy (SAH)

Surgery for temporal lobe epilepsy was initially treated with anterior temporal lobectomy as described by Penfield and Baldwin. In 1958, Niemeyer described a more selective approach to the hippocampus and amygdala through the middle temporal gyrus.²⁴ It would be almost thirty years until selective amygdalo-hippocampectomy was later modified. The goal of SAH is to remove the epileptogenic focus while minimizing disruption of nearby neurovascular structures and white matter tracts. Image guidance is very helpful for these techniques. More recent studies have compared standard ATL to SAH and found that both techniques have similar outcome with regard to seizure freedom but suggest an improved neuropsychological outcome with SAH versus ATL.^{25,26}

Three basic approaches:

1. transcortical: inferior temporal gyrus (ITG) approach. This technique uses a minimal access approach via a trephine craniotomy for SAH²⁷
2. transsylvian: approach requires a pterional craniotomy. More restrictive and greater risk of injury to M1 portion of MCA within sylvian fissure²⁸
3. subtemporal: uses a temporal fossa approach to access mesial structures^{29,30}

100.5 Risks of seizure surgery

- Major risks are related to³¹:
- 1. removal of essential areas of cortex
 - 2. injury to medullary core underlying cortical resection (projection fibers, association fibers, and/or commissural fibers): the most common deficit after temporal lobectomy is a contralateral (homonymous) superior quadrantanopsia (so-called “pie-in-the-sky” defect, due to an injury to Meyer’s loop wherein the fibers for the superior visual field of the optic radiation take a slight rostral “detour” towards the temporal tip)
 - 3. injury to vessels in area of resection → ischemic damage to areas supplied: especially sylvian branches during temporal lobectomy, anterior choroidal artery resulting in hemiparesis during mesial temporal lobe resection, or ACA branches with corpus callosotomy
 - 4. injury to nearby cranial nerves: especially third nerve during hippocampectomy where it lies medial to tentorium

100.6 MRI guided laser interstitial thermal therapy (MRGLITT)

Laser-induced thermal therapy uses thermal energy to induce cell death by damaging DNA and causing protein denaturation. The current therapy is performed with simultaneous MRI stereotactic guidance and real time feedback from the ablated lesion.^{32,33} It is considered less invasive than microsurgery. Main advantage is a shorter post-operative recovery period. Technique has been used for lesional and nonlesional epilepsy. Preliminary seizure control ≈60-70% No long-term data is yet available.

100.7 Postoperative management for seizure surgery (epilepsy surgery)

- 1. ICU for observation (24 hrs)
- 2. for seizures in the immediate post-op period (“honeymoon seizures”), not necessary to treat only one brief generalized seizure, otherwise load appropriately with IV Keppra or phenytoin
- 3. 10 mg dexamethasone (Decadron®) IV before surgery followed by q 8 hrs as necessary (longer taper for laser ablation and radiosurgery)
- 4. anticonvulsants are continued × 1–2 years even if no post-op seizures occur
- 5. after discharge: neuropsychiatric evaluation 6-12 months after surgery

100.8 Outcome

Modified Engel classification outcome is shown in □ Table 100.1.³⁴

100.8.1 Outcome with resection of seizure focus

The greatest effect of seizure surgery is reduction of seizure frequency,²³ however, any surgical procedure may fail to have a beneficial effect.

Seizure control is usually assessed at 1, 3 & 6 most post-op, and then annually. A post-op MRI is usually obtained at 3 most post-op to assess extent of surgical resection. Most patients take anti-epileptic drugs (AEDs) for 2 years post-op, and then may be discontinued in those free of seizures.

Recurrent seizures: although late seizures may occur, 90% of seizures that recur do so within 2 years.

Table 100.1 Modified Engel classification of seizure surgery outcome	
Class	Description
I	seizure free or residual auras
II	rare disabling seizures (<3 complex parital seizures per year)
III	worthwhile seizure reduction
IV	no worthwhile seizure improvement

2 years post-op in patients maintained on AEDs: 50%are seizure-free, and 80%have over 50% reduction of seizure frequency.

For temporal lobectomies in the dominant hemisphere without intraoperative monitoring, there is a 6%risk of mild dysphasia. Significant memory deficits occur in ≈ 2%

100.8.2 Radiosurgery for epilepsy

Stereotactic radiosurgery has been suggested to be an effective treatment for pharmacoresistant epilepsy with the potential for less morbidity than resection.^{35,36} Seizure free outcomes ≈ 65%for MTLE (delayed response to therapy ≈6-12 months). Potential for long-term complications remains a concern (radionecrosis).³⁷

100.8.3 Vagus nerve stimulation (VNS)

Electrodes wrapped around the vagus nerve in the neck are connected to an implanted programmable generator to stimulate the nerve to reduce seizure frequency. As is also true with many AEDs, the mechanism of action is not well understood.

Indications: Although it has been used (off label) for treatment resistant depression and other psychiatric conditions, the FDA approved indication is for adjunctive therapy for patients >12 years old with partial onset seizures refractory to medical treatment.

Complications: Main risk of surgery is transient or permanent vocal cord paralysis.

Outcome: In a 12 year retrospective review of VNS in 12 patients,³⁸ mean seizure frequency decreased by 26%at 1 year, by 30%at 5 years, and by 52%after 12 years.

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101 Radiation Therapy (XRT)

101.1 Introduction

Ionizing radiation comprises a portion of the electromagnetic spectrum and includes x-rays and gamma rays (both of which are electromagnetic radiation and transmit their energy via photons) and particulate radiation. The goal of XRT in treating tumors is to cause cell death or to stop cell replication. Photons impart critical energy to achieve this result by the photoelectric effect (at lower energies, <0.05 MeV), by Compton scattering (at higher energies of 0.1–10 MeV, e.g. in linear accelerators and Gamma knives), or by pair-production (at the highest energies).¹ In the Compton effect, the initial collision of the photon with an atom creates a free electron which then ionizes other atoms and breaks chemical bonds. The absorption of radiation by indirect ionization in the presence of water produces free radicals (containing an unpaired electron) which causes cellular injury (usually by damaging DNA) within the tumor.

See discussion of radiation dosage and units (p.223).

101.2 Conventional external beam radiation

101.2.1 Fractionation

The practice in which the total radiation dose is delivered in a series of smaller brief applications. This is one means of increasing the therapeutic ratio (the ratio of the effectiveness of XRT on tumor cells to that of normal cells). Radiation injury is a function of the dose, the exposure time, and the area exposed. Radiation oncologists refer to the four “R’s” of radiobiology²:

- 1. Repair of sublethal damage
- 2. Reoxygenation of tumor cells that were hypoxic before XRT: oxygenated cells are more sensitive than hypoxic cells because oxygen combines with unpaired electrons to form peroxides which are more stable and lethal than free radicals
- 3. Repopulation of tumor cells following treatment
- 4. Redistribution (or reassortment) of cells within the cell cycle: cells in the mitotic phase are the most sensitive

101.2.2 Dosing

The biologically effective dose of fractionated radiation is often modeled by the linear-quadratic equation (LQ-model) shown in Eq (101.1), where n = the number of doses, d = dose per fraction, and the factors α & β are used to describe the cell response to radiation. A high α/β ratio ≥ 10 is designated as early-responding tissue such as tumor cells, and a ratio ≤ 3 is considered late-responding tissue (mitotically quiescent), such as normal brain and also AVMs.

Linear quadratic equation:

biologically effective dose (BED) (Gy)^{1/4} n d $\hat{A}$ $\left[1 + \frac{d}{\alpha/\beta} \right]$ 101:10

101.2.3 Cranial radiation

General information

Following surgery for tumor (craniotomy or spinal surgery), most surgeons wait $\approx 7-10$ days before instituting XRT to the surgical site (allows initiation of healing from surgery).

Two CNS tumors that “melt away” with XRT but tend to recur later:

- 1. lymphoma
- 2. germ cell tumors

Radiation injury and necrosis

General information

Radiation necrosis (RN) may mimic recurrent (or denovo) tumor both clinically and radiographically. Differences in prognosis and treatment make it important to distinguish between tumor and RN.

Pathophysiology

As radiation is selectively toxic to more rapidly dividing cells, the two normal cell types within the CNS most vulnerable to RN are vascular endothelium (which have a turnover time of $\approx 6\text{--}10$ mos) and oligodendroglial cells. Vascular injury may be the primary limiting factor to the tolerance of cranial XRT.³ Injury from XRT occurs at lower doses when given concurrently with chemotherapy (especially methotrexate).

Etiology of side effects

The mechanism(s) by which XRT causes side effects is not known with certainty, but may be due to:

1. damage to vascular endothelium: effects on cerebral vasculature may differ substantially from effects on systemic vessels³
2. glial injury
3. immune system effect

Radiation effects are divided into 3 phases⁴:

1. acute: occur during treatment. Rare. Usually an exacerbation of symptoms already present. Probably secondary to edema. Treat with $\uparrow$ steroids
2. early delayed: few weeks to 2–3 mos following completion of XRT. In spinal cord $\rightarrow$ Lhermitte's sign. In brain $\rightarrow$ post-irradiation lethargy & memory difficulties
3. late delayed: 3 mos–12 yrs (most within 3 years). Due to small artery injury $\rightarrow$ thrombotic occlusion $\rightarrow$ white matter atrophy or frank coagulative necrosis

Manifestations of radiation effects:

1. decreased cognition
 - a) dementia may develop following XRT⁵ in as little as 1 year post-XRT. Incidence was higher when doses of 25–39 Gy were given in fractions >300 cGy⁶
 - b) children: may attain lower IQ by ≈ 25 points, especially with >40 Gy whole brain XRT. Measurable IQ differences occur in children radiated before age 7, but more subtle deficits occur even in older children⁷
2. radiation necrosis
3. injury to anterior optic pathways
4. injury to hypothalamic-pituitary axis $\rightarrow$ hypopituitarism $\rightarrow$ growth retardation in children; see radiation injury to pituitary (p.744)
5. primary hypothyroidism (especially in children)
6. may induce formation of new tumor: tumors most commonly identified as having increased incidence following radiation treatment are gliomas (including glioblastoma⁸), meningiomas,⁹ and nerve sheath tumors.¹⁰ Skull base tumors have been reported following EBRT¹¹
7. malignant transformation: e.g. after SRS for vestibular schwannomas (p.686)
8. leukoencephalopathy: profound demyelinating/necrotizing reaction 4–12 mos after combined RXT and methotrexate, especially in children with acute lymphoblastic leukemia (ALL) and adults with primary CNS tumors

Evaluation (differentiating RN from recurrent tumor)

Greenberg IMHO

Over the years many methods have been championed to differentiate radiation necrosis from recurrent high-grade glioma. Some are listed below. None have proven adequately reliable, and this may not even be a useful exercise. Tumor cells are frequently found on biopsy. The decision whether to reoperate is usually based on whether there is progressive mass effect (regardless of whether it is necrosis or tumor) taking into consideration the patient's neurologic condition, projected longevity, patient desires...

CT and MRI

Cannot reliably differentiate some cases of RN from tumor (especially astrocytoma; RN occasionally resembles glioblastoma).

MR spectroscopy (p.233) was reliable in distinguishing pure tumor (elevated choline) from pure RN (low choline), but was less definitive with mixed tumor/necrosis.¹²

DWI: mean ADCs were lower with recurrence ($1.18 \pm 0.13 \times 10^{-3}$ mm/s) vs. necrosis ($1.4 \pm 0.17 \times 10^{-3}$ mm/s)¹³ (not all cases biopsy proven).

Nuclear brain scan

Some reports of success with thallium-201 and technetium-99 m brain scans.

Computerized radionuclide studies

PET (positron emission tomography) scan: because positron emitting isotopes have short half-lives, PET scanning requires a nearby cyclotron to generate the radiopharmaceuticals at great expense. Utilizing [18F]-fluorodeoxyglucose (FDG), regional glucose metabolism is imaged and is generally increased with recurrent tumor, and is decreased with RN. Specificity for distinguishing RN from tumor recurrence is >90% but sensitivity may be too low to make it reliable.¹⁴ Amino acid tracers such as [11C]methionine and [18F]tyrosine are taken up by most brain tumors,¹⁵ especially gliomas, and may also be used to help differentiate tumor from necrosis. Accuracy may be increased by fusing PET scan with MRI.¹⁶

SPECT (single positron emission computed tomography): “poor man’s PET scan.” Uses radio-labeled amphetamine. Uptake depends on presence of intact neurons and the condition of cerebral blood vessels (including blood brain barrier). Decreased radionuclide uptake indicates necrosis, whereas tumor recurrence has no decreased uptake.

Treatment

Symptoms from any form of radiation toxicity often respond initially to steroids.

Reoperation and excision is appropriate if there is deterioration from mass effect, regardless of whether the mass effect is from recurrent tumor or RN; the decision to reoperate should be based on the patient’s Karnofsky rating (p.1358). Although some benefit has been shown, most reoperation studies are biased because they often select the patients who are doing better.

Other forms of therapy include: hyperbaric oxygen and anticoagulation.

Patients with documented tumor recurrence (as opposed to RN) may also be considered for additional radiation (external beam, interstitial brachytherapy, or stereotactic radiosurgery (SRS)) or chemotherapy.

Prevention

Injury is dependent on total radiation dose, number of treatments or fractions (less damage occurs with more frequent small treatments), and volume treated.

Various studies to determine the tolerance of normal brain to XRT have estimated that 65–75 Gy given over 6.5–8 wks in 5 fractions/week is usually tolerated (radiation necrosis will occur in $\approx$ 5% after 60 Gy fractionated in 30 treatments over 6 weeks). Other studies have shown tolerance to 45 Gy for 10 fractions, 60 Gy for 35 fractions, and 70 Gy for 60 fractions.⁴

101.2.4 Spinal radiation

General information

Most spine tumors are metastatic cancer. There is no proof that any treatment for spinal metastases will prolong survival. Treatment goals, regardless of the treatment modality, are pain relief and preservation of function.

Radiation therapy (XRT) is the main treatment modality for spinal metastases. Even tumors that are not considered “radio-sensitive” can respond to XRT.

Typical spinal radiation

For most metastatic spine tumors treated with conventional radiation therapy (i.e. not stereotactic radiosurgery), the usual fractionation is 30 Gy administered over 10 fractions.

Emergency spinal radiation

For acute spinal cord paralysis from tumor, if emergency surgery is not a consideration, 8 Gy can be administered in the 1st fraction for lymphoma, multiple myeloma, and (although this is not standard) it could be considered for small-cell (neuroendocrine) carcinoma since it is so radiosensitive.

Side effects

- 1. radiation myelopathy: see below
- 2. those due to overlap with GI tract: N/V, diarrhea
- 3. bone marrow suppression
- 4. growth retardation in children¹⁷
- 5. risk of developing cavernous malformations of the spinal cord (p.1247)

Radiation myelopathy

Radiation myelopathy (RM) typically occurs in patients with spinal cord included in radiation therapy (XRT) ports used to treat cancer outside the spinal cord, includes breast, lung, thyroid, and epidural mets. Radiation neuropathy may occur with irradiation in the region of the axilla for carcinoma of the breast (p.544). In the lower extremities, XRT for pelvic or bone tumors (e.g. of the femur) may produce lumbar plexopathy. In addition to permanent changes, radiation therapy may also produce spinal cord edema which may resolve after completion of radiation therapy.

Epidemiology

Incidence difficult to estimate due to the fact that the onset is typically delayed together with the poor survival of patients with malignant disease requiring XRT.

Most cases reported involve the cervical cord in spite of the higher frequency with which the thoracic cord is exposed to XRT (perhaps due to higher XRT doses to the head and neck and longer survival than with lung Ca).¹⁸ Delay between completion of XRT and onset of symptoms is usually ≈ 1 yr (reported range: 1 mos-5 yrs).

Important factors relating to the occurrence of RM include¹⁸:

- 1. rate of application (probably the most important factor)
- 2. total radiation dose
- 3. extent of cord shielding
- 4. individual susceptibility and variability
- 5. amount of tissue radiated
- 6. vascular supply to the region radiated
- 7. source of radiation

Pathophysiology

Effects of XRT on the spinal cord that lead to RM are:

- 1. direct injury to cells (including neurons)
- 2. vascular changes, including endothelial cell proliferation →thrombosis
- 3. hyalinization of collagen fibers

Clinical

Clinical types of radiation myelopathy

Four clinical types have been described and are shown in □ Table 101.1.

Onset is usually insidious, but abruptness has also been described; the presentation often mimics epidural mets. First symptoms: usually paresthesias and hypesthesia of LEs, and Lhermitte’s sign. Then spastic weakness of LEs with hyperreflexia develops. A Brown-Séquard syndrome is not uncommon.

Table 101.1 Types of radiation myelopathy	
Type	Description
1	benign form; commonly several mos following XRT (reported as late as 1 yr). Usually resolves completely within several mos. Mild sensory symptoms (frequently limited to a Lhermitte’s sign) without objective neurological findings
2	injury to anterior horn cells→ lower motor neuron signs in arms or legs
3	described only in experimental animals after doses larger than normal XRT. Complete cord lesion within hours due to injury to blood vessels
4	the type commonly reported. Chronic, progressive myelopathy (see text)

Approximately 50% of patients developing RM also have dysphagia from esophageal strictures requiring dilatations (the dysphagia often predates the myelopathy).

Evaluation

Essentially a diagnosis of exclusion. Radiographic imaging (CT, myelography) will be normal. MRI may show spinal cord infarction. The history of previous radiation is key. The differential diagnosis is included in Acute paraplegia or quadriplegia (p.1413).

Prognosis

Prognosis for Type 4 RM is poor. Usually progresses to complete (or near complete) cord lesion. Paraplegia and/or sphincter involvement are poor signs.

Prevention

Maximum recommended cord radiation dose depends on size of port, and varies with investigator. With large field techniques (>10 cm of cord), the risk of RM is negligible with ≤ 3.3 Gy in 42 days (0.55 Gy/wk), and with small field techniques ≤ 4.3 Gy in 42 days (0.717 Gy/wk). Larger doses may possibly be given safely if fractionated over longer periods. Recommended upper limit: 0.2 Gy/fraction.

101.3 Stereotactic radiosurgery and radiotherapy

101.3.1 General information

Key concepts

- Stereotactic radiosurgery (SRS): the application of a single large dose of radiation to a stereotactically localized target usually ≤ 3 cm diameter with minimal radiation delivered to surrounding tissue. May be a single treatment or up to 5 treatment fractions
- Stereotactic radiotherapy (SRT) employs hypofractionated dosing (2-5 treatment fractions) and the target can be larger
- Both SRS and SRT may be performed with properly equipped linear accelerators (LINAC Scalpel, CyberKnife®...), collimated beams from multiple radioactive sources (Gamma Knife®), or less commonly, protons and heavy charged particle beams.

Stereotactic radiosurgery (SRS)

Lars Leksell coined the term “radiosurgery” in 1951.¹⁹ His concept was to replace the use of electrodes or “knives” (scalpels) with multiple intersecting beams of radiation to converge on an intracranial target through an intact skull. The dose of radiation at the point of intersection (the “isocenter”) is higher than outside of the isocenter where the dose falls off sharply such that adjacent tissues receive only minimal radiation from each individual incident beam. When combined with a reliable method for aiming the beams at an intracranial target (e.g., using a stereotactic frame system and three-dimensional imaging) the technique came to be known as stereotactic radiosurgery.

Initially developed to create a necrotic lesion in specific nuclei or pathways for functional disorders, it has subsequently been proven that subnecrotic doses could trigger cellular reactions in tumors and vasculature that would lead to tumor shrinkage or control and obliteration of vascular malformations.

With conventional radiation treatments, the “R’s” of radiobiology (p.1560) are exploited. In contrast, with SRS, precision and accuracy are used to affect the target (e.g. damage tumor or thrombose AVM) and spare normal tissue.

Stereotactic radiotherapy (SRT)

Methods of immobilization and point-of-delivery imaging technologies have since made it practical to deliver stereotactic radiation spread out over several individual treatment sessions (known as fractions) when desirable. Many authors refer to this as stereotactic radiation therapy (SRT). The precise definition of SRT has evolved over time, with some publications describing it as using a

conventional fractionation scheme (1.8-2 Gy / fraction). However the majority of authors refer to SRT in terms of a hypofractionated approach, generally limited to five treatment fractions (see below).

Fractionation capitalizes on the differential response of normal tissue from tumors to radiation insult; see the Four “R’s” of radiobiology (p.1560). The value of fractionation is more apparent for tissues with high rates of proliferation and less capability to repair sublethal DNA damage (high α/β ratio in Eq (101.1)).²⁰ However these models apply more directly in conventional fractionation schemes; their use in single- or hypo-fractionated schemes is a subject of current research.

Treatment over multiple fractions renders impractical the traditional stereotactic frame. SRT therefore employs various techniques for patient immobilization including thermoplastic masks, dental-impression-based bite-blocks, and other relocatable frame systems. Displacement errors can be as high as 2-8 mm with mask systems, however these uncertainties can be reduced with in-room imaging systems such as cone-beam CT (CBCT) which can assist in patient localization, as well as intra-fraction motion monitoring techniques such as orthogonal kilovoltage x-rays, surface tracking systems, and infrared marker tracking systems.

Blurred lines between SRS and SRT

Although some purists insist that SRS is performed in a single session, the current definition was expanded by AANS/CNS/ASTRO in 2007 to include radiosurgical procedures “using a rigidly attached stereotactic guiding device, other immobilization device and/or a stereotactic image-guidance system ... performed in a limited number of sessions, up to a maximum of five.”²¹

To further muddy the waters, for billing purposes in the U.S., the cpt codes used by Medicare for SRS (for brain and spine)^{22,23} describe treatment in one session, whereas SBRT (stereotactic body radiation therapy) allows treatment delivery not to exceed 5 fractions within the body. For 5 or more fractions, Medicare considers it to be intensity modulated radiation therapy (IMRT).

Comparison of SRS technologies

Various methods for delivering SRS/SRT are clinically available. Three main categories (based on sources of radiation) are Gamma Knife, linear accelerator based, and heavy charged particle radiosurgery. Fundamentally, there is no difference between a photon created by radioactive decay (gamma ray) from a photon created using electrical energy in a linear accelerator (x-ray).

□ Gamma knife radiosurgery. In the original Gamma Knife (GK), the source of radiation is gamma decay of 201 cobalt-60 sources that align with an inner collimator to direct the resulting photon beams. A treatment couch includes a mount for an external collimation “helmet”, each of which has either a 4-, 8-, 14-, or 18-mm diameter beam aperture for each source. The stereotactic frame affixed to the patient’s head is positioned within the collimator helmet so the area to be treated is at the focus point of treatment unit. Several dwell positions (also called “shots” or “isocenters”) may be defined to match dose distributions to irregularly shaped targets.

In the newer GK model Perfexion®, 192 cobalt-60 sources are distributed on 8 sectors attached to sector drive motors to move the sources along a fully internalized tungsten collimator. This allows each sector to move from “home”, 4-, 8-, 16-mm collimators or shielded positions. The design makes possible composite use of different beam diameters to help optimize dose distribution. The Gamma Knife is specifically designed for cranial and upper cervical lesions, and is best suited for smaller lesions (<3 cm diameter).

As the gamma sources age, output declines, and treatment times necessarily become longer. Eventually, the sources must be replaced, which is a time consuming and expensive process.

□ Linear accelerator-based radiosurgery. Linear accelerators (linacs) generate x-rays by accelerating electrons and directing them to strike a target of a substance with high atomic number. The x-ray source and beam collimation is mounted on a rotating gantry, creating a fixed isocenter in space. Beam convergence is achieved via rotating arcs with the isocenter fixated on the target. The target is aligned to the isocenter using a treatment table that is adjustable in up to 6 degrees of freedom (6 dof, 3 translations, 3 rotations). Beams are collimated using either narrow-aperture cones (SRS cones) or multileaf collimators (MLCs), the latter of which use banks of computerized leaves to shape the treatment field and may be modulated to achieve specific dose distributions. Linacs are more versatile than GK in that they can treat cranial and extracranial targets, often have built-in CBCT imaging to assist with patient setup and target localization, and can have a higher dose rate (and thus potentially faster treatment). However they are generally technically more complicated than GK and require substantially more quality assurance to maintain technical confidence. The

Cyberknife is an SRS-specific linac that uses a robotic arm rather than an isocentrically-mounted linac to achieve 6 dof targeting.

□ Heavy charged particle radiosurgery. Heavy charged particles (protons or helium ions) from a cyclotron can be used for radiosurgery.²⁴ Unlike high-energy photons (gamma and x-rays), which deposit the majority of their energy upon entrance into tissue and continue to deposit decreasing amount of energy as they travel through the body, heavy charged particle beams have a shorter, bounded range of penetration wherein particles sharply increase energy deposition near the terminal depth of penetration (Bragg peak effect). Particle radiosurgery achieves a well-localized volume of high dose radiation by taking advantage of cross firing of a number of beams as well as the Bragg peak. Due to the expense and increased complexity of heavy charged particle SRS, this therapy is only available in a few centers in the world.

101.3.2 Indications

In general, SRS is useful for well-circumscribed lesions less than approximately 3 cm diameter. For larger lesions, the radiation dose must be reduced because of anatomic and radiobiological constraints.

Published uses of SRS include:

- Vascular lesions
 - AVMs (including dural arteriovenous fistulas)
 - Cavernous malformations
- Tumors
 - Metastases
 - Vestibular schwannomas
 - Meningiomas
 - Pituitary adenomas
 - Gliomas
 - Others: craniopharyngioma, pineal tumors, etc.
- Functional disorders
 - Trigeminal neuralgia^{25,26}
 - Intractable chronic pain: thalamotomy²⁷
 - Movement disorders: pallidotomy for Parkinson's disease or thalamotomy for tremor (Usually not a technique of choice because of inability to perform physiologic stimulation prior to lesioning. May be a consideration for the rare patients who cannot undergo placement of a stimulator/lesioning needle)
 - Psychiatric diseases (e.g. obsessive compulsive disorder)
 - Epilepsies²⁸

101.3.3 Contraindications

Compressive tumors of the spinal cord, brainstem or optic structures: even with the sharp falloff of radiation dose, there remains radiation delivered within a few millimeters of the margins of the isocenter. This, together with post-radiation swelling, might create significant risk of neurologic injury. Surgical removal should be considered in these situations especially for benign lesions in young individuals.

101.3.4 Treatment procedure

The treatment procedure includes placement of stereotactic frame (in framed-based SRS), obtaining stereotactic images, target definition, treatment planning and execution of treatment.

Target localization

MRI is the predominant imaging modality for SRS procedures given its superior soft tissue and tumor contrast. Typical MRI protocols include T1-weighted pre- and post-contrast images using 3D pulse sequences. If visualization of structures in surrounding CSF (trigeminal nerves, CPA tumors) is required, special sequences such as constructive interference in steady state (CISS) is useful. Fat-saturation sequences are used in cases with previously resected skull base tumors. NB: there is a 1-2 mm shift due to spatial distortion artifact from the MRI magnet. This effect is more prominent in high field strength MRI.

CT accuracy is never better than 0.6 mm, which is the pixel size. Usually is used when MRI is contraindicated or in cases where distortion in MRI may be a concern (dental brace, shunt). CT does not suffer susceptibility distortion as MRI and one can use a stereotactic CT to merge with nonstereotactic MRI for treatment planning.

Stereotactic angiography remains the best method to define the AVM nidi and their arterial supply and venous drainage. Given that angiograms provide only two sets of orthogonal images (usually AP and lateral images), CTA and/or MRI/MRA can be used as an adjunct to provide vascular anatomy on axial plane.

Treatment planning

Treatment planning is the process by which a dose distribution is created with the goal of adequately treating the target areas while largely sparing normal surrounding structures. In GK treatment, a dose distribution is created by defining one or more isocenters (shots). Each isocenter can make use of full complement beams of various diameters, however, in certain cases one or more of beams can be blocked to shape the dose distribution for irregular targets and protect adjacent critical structures. For linac-based SRS, treatment planning is accomplished using computer simulation programs to help select the number of arcs or beams with certain orientation. Also, static and dynamic collimators have been developed. Intensity modulation is also a means of delivering the desired dose to a target while decreasing the dose given to surrounding structures.

Lesions that are not round or ellipsoid in shape are not a problem when using linac, but with older GKs, multiple isocenters must be used to conform to an irregular surface. This results in multiple “hotspots.” This problem is eliminated with the GK Perfexion, which can create a single isocenter using different beam diameters. In the case of multiple metastases, it can generate individually shaped isocenters for each tumor and thus avoid having to change the helmet or add isocenters during a treatment.²⁹

□ Normal tissue tolerance. Cranial nerves: Damage to small nutrient vessels and Schwann cells or oligodendroglia are the possible mechanisms of radiation injury to cranial nerves. Special sensory nerves (optic, vestibulocochlear) are the most radiosensitive. The precise dose tolerance of cranial nerve is unclear; the optic nerve can probably tolerate doses lower than 8-10 Gy. Nerves in the parasellar region, the facial nerve and lower cranial nerves tend to tolerate higher doses.

SRS treatment may also have a deleterious effect in structures sensitive to swelling, such as brain-stem. However, the structures at highest risk are those within the higher isodose lines immediately adjacent to the lesion.

□ Table 101.2 shows maximum recommended doses of various organs for a single fraction. In the brain, critical radiation sensitive structures include: optic vitreous, nerve, and chiasm, brain stem, pituitary gland, and cochlea.

□ Dose. Dose is usually prescribed as a particular dose (in units of Gray) to the periphery of target. The periphery of the target is usually defined as the isodose curve, which covers the substantial (usually 95-100%) of the target. The isodose curve is the curve of equal dose usually defined as a percentage of the maximum dose point. Traditionally, treatment planning for GK uses the 50%isodose line because for a single isocenter this is the location of steepest dose gradient. Linac-based SRS used higher isodose line (70-90%) to increase homogeneity of dose distribution.

Dose-volume relation: The dose of radiation that can be tolerated is highly dependent on the volume being treated (larger treatment volumes require lower doses to avoid radiation injury). Dose selection is made based on known information or is estimated from dose-volume relationship. If uncertain, err on the side of a lower dose or slightly underdose the margin of tumor. Previous radiotherapy must also be taken into account by the treatment team, as local structures are more sensitive.

Table 101.2 Maximum recommended radiation dose of critical organs (delivered in a single fraction)		
Structure	Maximum dose (cGy)	% of maximum (at a prescribed dose of 50 Gy)
eye lens (cataract induction begins at 500 cGy)	100	2%
optic nerve ³⁰	100	2%
skin in beam	50	1%
thyroid	10	0.2%

101.3.5 Lesion-specific issues

Arteriovenous malformations and other vascular lesions

SRS is best accepted for the treatment of small to moderate-sized (<3 cm) AVMs that are deep or border on eloquent brain and have a “compact” (i.e. sharply demarcated) nidus. The radiation induces endothelial cell damage, smooth muscle cell proliferation, thickening of the vascular wall, and ultimately obliteration of the lumen over a period of 2-3 years (latency period).³¹

Lesions other than AVMs: Dural arteriovenous fistulas (AVFs) have also shown promising response to SRS.³² However, dural AVFs with cortical drainage should not be treated with SRS as these AVFs pose high risk of hemorrhage. SRS is of no benefit for venous angiomas.³³ SRS for cavernous malformations remains controversial, as the lesions could not be precisely evaluated with MRI or angiography. However, published retrospective studies have demonstrated reduction of hemorrhage rates following SRS.^{34,35}

□ Dose. Early study from Karlsson et al. demonstrated increased obliteration rate with increased margin dose.³⁶ The effect reached the plateau at 25 Gy and a higher dose produced more complications without extra benefits. The optimal dose for AVMs generally ranges between 23-25 Gy. The dose might be reduced for nidi at critical locations or with a large volume. At McGill with linac SRS, they use 25-50 Gy delivered to the 90%isodose curve at the edge of the nidus. With Bragg-peak, complications occurred less frequently with doses <19.2 Gy compared to doses above that (this may reduce the obliteration rate or increase the latency period).³⁷

□ Results. SRS has an overall obliteration rate of 70-80%across all cases of AVMs treated.³⁸ At 1 year, 46-61%of AVMs were completely obliterated on angiography, and at 2 years 86%were obliterated. There was no reduction in size in <2%of cases. Smaller lesions have higher obliteration rates (with Bragg-peak in AVMs <2 cm diameter, 94%thrombosed at 2 years, and 100%at 3 years).³⁷) AVMs >25 mm in diameter have only ≈ 50%chance of obliteration with 1 SRS treatment.

□ Grading system. A few SRS-based grading systems have been developed to predict patient outcomes since scales developed for surgical resection (e.g. the Spetzler-Martin scale) have not proven applicable for AVM radiosurgery. Long-term outcomes of 1012 patients treated with GK at UVA were analyzed to create the Virginia Radiosurgery AVM Scale (□ Table 101.3 and □ Table 101.4).³⁸ Multi-variate analysis showed that AVM volume, non-eloquent location, and no history of hemorrhage, were independent predictors of AVM obliteration without hemorrhage or permanent neurologic deficits.

The radiosurgery-based grading system by Pollock and Flickinger³⁹ has been validated in both GK and linac treatment centers. This AVM score is calculated using the following formula:

$$\text{AVM score} = \frac{\text{AVM volume (cm}^3\text{)}}{10} + \frac{\text{Age (years)}}{10} + \frac{\text{Location}}{10}$$

Location: hemispheric=corpus callosum=cerebellar ¼ 0; basal ganglia=thalamus=brainstem ¼ 1

□ Embolization. Controversy exists as to whether embolization is helpful or harmful prior to SRS. Some experts find that target definition after embolization is extraordinarily difficult because of multiple small residual nidi. Experimental studies have shown attenuation of radiation effect by embolic material.⁴⁰ However, for large AVMs, embolization remains an effective adjunct to reduce

Table 101.3 Virginia Radiosurgery AVM Scale ³⁸ : variables and points		
Variable		Points
AVM volume (cm ³)	<2	0
	2-4	1
	>4	2
Eloquent location		1
History of hemorrhage		1

Table 101.4 Virginia Radiosurgery AVM Scale ³⁸ : total points and %favorable outcome	
Total points	Favorable outcome (%)
0	83
1	79
2	70
3	48
4	39

the size of nidi to make AVMs amenable to SRS. Additionally, pre-SRS embolization should be considered for high flow fistulas, which are more radioresistant, and intranidal/perinidal aneurysms, which are at risk for rupture.

□ Large AVMs. Large AVMs (>10 cm³) remain a significant challenge to treat with any modality. Single stage SRS has resulted in low obliteration and high complication rates. To maximize the dose/volume response, SRS can be performed in a volume-staged fashion. The Pittsburgh group reported overall rates of total obliteration of 18%, 45% and 56%at 5, 7, and 10 years, respectively in a group of 47 patients with large AVMs. Ten patients had hemorrhage after SRS, and 5 of these patients died.⁴¹

□ Residual lesions after SRS. Factors associated with treatment failures include: incomplete angiographic definition of the nidus (the most frequent factor, responsible for 57%of cases), recanalization of the nidus (7%), masking of nidus by hematoma, and a theorized “radiobiological resistance.”⁴² In some, no discernible reason for failure could be identified. In this series the complete obliteration rate was ≈ 64% If AVMs persist, retreatment with SRS 2-3 years later is an option.⁴²

The ultimate goal of radiosurgery for AVMs is elimination of risk of hemorrhage. An imperfect surrogate marker for this goal has been total obliteration of the nidus on imaging. Reduced, unchanged, or increased hemorrhagic rates during the latency period have been reported. Proton beam treatment of AVMs affords no protection against hemorrhage in the first 12-14 months following treatment.²⁴; this is similar to the 12-24 month latency for photon radiation.⁴³ Hemorrhages may occur during the latency period even in AVMs that had never bled before,³⁷ and the question has been raised whether a partially thrombosed AVM is more likely to bleed because of increased outflow resistance. Yen et al. reviewed a large group of AVM patients undergoing radiosurgery and reported that the hemorrhagic rates reduced from 6.6 %before SRS to 2.5%after SRS. The protective effect is more significant in those AVMs that had a prior hemorrhage (10.4%to 2.8%).⁴⁴

Metastases

The gold standard and level 1 recommendation for a single brain metastasis amenable to surgical resection is surgery followed by WBRT.^{45,46} This does not apply to extremely radiosensitive tumors such as lymphoma, small cell lung cancer, germ cell tumors, and multiple myeloma. There has not been a randomized study to compare surgery alone to SRS alone. A prospective randomized study by Muacevic et al. showed comparable survival and local tumor control between SRS vs. surgery plus WBRT in patients with a single brain metastasis but a higher incidence of distant recurrence in the SRS arm.⁴⁷ When tissue is required for diagnosis, surgery should be considered. Tumors with significant mass effect should also be resected if the patients are fit for surgery. In patients with ≤3 mets, single dose SRS provides superior survival advantage as compared to WBRT (level III recommendation). Radiographic local control rate of ≈ 88%(reported range: 82-100%) has been cited.⁴⁸

No significant difference has been found with SRS between tumors considered “radiosensitive” and those that are “radioresistant” as defined by standards developed for EBRT (see □ Table 52.7; however, histology may affect the rate of response). The lack of significance of “radioresistance” may be due in part to the fact that the sharp dose drop-off with SRS allows higher doses to be delivered to tumors than would be used with EBRT.

General guidelines for when to consider SRS for brain metastases are:

- total tumor number ≤10
- total tumor volume ≤15 cm³
- single tumor volume is <10 cm³, and
- no leptomeningeal disease present.

Recent studies on SRS for brain metastases have shown that total tumor volume is a better predictor of overall survival, local control, and even distant failure than tumor number.^{49,50,51}

□ Dose. RTOG 90-05 study recommended 24 Gy as the maximum tolerated doses of single fraction SRS for tumors ≤ 20 mm in maximum diameter; 18 Gy for 21-30 mm; and 15 Gy for 31-40 mm.⁵²

Vestibular schwannomas (VS)

Possible indications of SRS for VS are: poor surgical candidates (due to poor medical condition and/or advanced age, some use >65 or 70 years as a cutoff), patient refusing surgery, bilateral VS, post-operative treatment of incompletely removed VS that continue to grow on serial imaging, or recurrences following surgical removal. See the Management algorithm under Vestibular schwannomas for strategies that take into account the natural history.

A few nonrandomized prospective studies comparing GK SRS to microsurgery in small to middle sized VS have demonstrated superiority of SRS treatment.^{53,54} The largest series of 2336 VS with at least 3 years follow-up from Marseille showed a tumor control rate of up to 97.5% trigeminal nerve injury 0.5% facial palsy 0.5% and hearing preservation at 65%.⁵⁴ However long-term follow-up is needed. SRS following deliberate subtotal resection of large Koos stage IV VS reduced the risk of facial weakness compared with GTR.⁵⁵ Dose range for SRS of VS is usually 11-12 Gy. SRS tolerable dose to cochlea seems to be around 4.2 Gy, but also depends on age and pre-SRS hearing class.⁵⁶ Temporary increase of tumor size following SRS is a common finding and can be managed conservatively.

SRT or multisession SRS has been used to treat VS. Selch et al. reported a local control rate of 100% with a median follow-up of 36 months.⁵⁷ Their group showed the hearing preservation rate has been high (93%) and incidence of facial nerve injury has been low (2.2% new facial palsy, 2.2% facial numbness). SRT has not been compared head-to-head with SRS, but preliminary results suggest the results of both modalities might be similar.

□ Dose. Sample SRT protocol: 6-MV linac with a micromultileaf collimator used to deliver 54 Gy in 30 fractions of 1.8 Gy prescribed to the 90% isodose line via 7-22 noncoplanar static fields or 4-6 noncoplanar dynamic arcs to a target defined as the tumor volume plus a margin of 1-3 mm. Multisession SRS has been used to treat large VS. In a study of 33 patients with VS greater than 8 cm³ treated with Cyberknife, radiological growth control was 94% at a median follow-up of 48 months.⁵⁸ Hearing was retained in 7 of 8 patients with serviceable baseline hearing. Only 1 patient each had cranial nerve deficits including vertigo, tongue paresthesia and trigeminal neuralgia.

Meningiomas

In a large series of meningiomas treated with SRS by the Pittsburgh group,⁵⁹ among 942 patients with 1045 tumors treated, the control rate for patients with surgically verified WHO grade I meningiomas was 93%. Control rate for presumed meningiomas based on imaging (no prior histological confirmation) was 97%. The control rates for WHO grades II and III tumors were 50% and 17% respectively.

□ Dose. The authors used a mean dose of 14 Gy to the tumor margin.⁵⁹

Pituitary adenomas

Surgery is the mainstay of management for pituitary tumors especially for nonsecretory tumors with mass effect and secretory tumors that fail medical treatment. For residual/recurrent secretory or nonsecretory tumors, SRS is an appropriate tool to arrest tumor growth and/or normalize hormonal function. In a large series of 418 patients with pituitary adenomas residual/recurrence treated with GK at UVA, the overall tumor control rate was 90% in patients with imaging follow-up available. The endocrine remission rates are 53% for acromegaly, 54% for Cushing's disease, and 26% for prolactinomas.

□ Dose. The usual dose for nonsecretory tumors is ≈ 16 -18 Gy, and a higher dose is required for secretory tumor ≈ 25 Gy.

□ Results. Early studies of SRS showed limited numbers of hypopituitarism and cranial nerve injuries. Available long-term follow-up studies now demonstrate a 20-30% rate of new endocrinopathy after SRS and a low but not negligible risk of cranial nerve damage.⁶⁰

Infiltrating tumors

SRS is generally not indicated as a primary treatment for infiltrating tumors, e.g. gliomas, due to the lack of a definable capsule and very close relationship between target volume and tolerated radiation dose. SRS has been used for recurrent lesions following standard treatment (surgical excision and adjuvant EBRT 60 Gy and Temozolomide). One of the arguments for SRS in these tumors is the fact that 90% of recurrences are within the original radiographic solid tumor volume.⁶¹ However, support for SRS boost declined after a 2004 RTOG trial 9305 showing no survival benefit in newly diagnosed GBM patients with upfront use of SRS followed by EBRT and BCNU chemotherapy. SRS can be used as a salvage treatment for small ($<10\text{ cm}^3$), recurrent GBM after standard treatment. Kong et al. showed SRS salvage treatment in these cases prolonged OS to 23 from 12 months.⁶²

101.3.6 Treatment morbidity and mortality

Immediate morbidity and mortality

Immediate mortality from the actual treatments themselves is probably zero. Morbidity: all but $\approx 2.5\%$ of patients were discharged home within 24 hours. Many centers do not admit patients overnight. Some immediate adverse reactions include⁶³:

- 16% of patients require analgesics for post-procedural headaches and antiemetics for nausea/vomiting
- at least 10% of patients with subcortical AVMs had focal or generalized seizures within 24 hrs of treatment (only one was on subtherapeutic AEDs. All were controllable with additional AEDs.)

Premedication

The Pittsburgh Gamma Knife group gives methylprednisolone 40 mg IV and phenobarbital 90 mg IV immediately after the radiation dose to patients with tumors or AVMs to reduce these adverse effects.⁶³ For small lesions and patients without prior history of seizures, premedication with steroids or antiepileptics are probably not necessary.

Delayed morbidity

Long-term morbidity directly related to the radiation may occur, and just as with conventional XRT, is more frequent with larger doses and treatment volumes. Radiation complications include:

- radiation-induced changes, high intensity on MRI T2WI, or low density on CT: usually occur ≈ 13 months after SRS for AVMs. The incidence is 34% with 8.6% having imaging changes associated with neurological symptoms (focal deficits, seizures, or headache) and 1.8% developing radiation necrosis and permanent deficits.⁶⁴ The possible mechanisms of this side effect include glial cell damage, breakdown of blood brain barrier or early venous thrombosis. Premature venous thrombosis or occlusion before obliteration of AVM nidus can produce venous hyperemia or intracranial hemorrhage⁶⁵
- vasculopathy: diagnosed by narrowing seen on angiography or by ischemic changes on imaging in $\approx 5\%$ of cases
- cranial nerve deficits: occur in $\approx 1\%$ of all cases. Incidence is higher with tumors of CPA or skull base
- radiation induced tumors: only a few case reports of newly formed malignant tumors (glioblastomas) or malignant transformation of a benign tumor (vestibular Schwannoma). Loeffler reported 6 cases over 80,000 radiosurgical procedures for benign diseases.⁶⁶ Radiation-induced meningioma is a well-known complication of radiation therapy.⁶⁷ In a large series of AVM patients treated with SRS, the incidence of radiation-induced meningioma was estimated to be 0.7%⁶⁸

101.4 Interstitial brachytherapy

101.4.1 General information

Technique whereby radioactive implants are used to deliver locally high doses of radiation directly to tumors while exposing nearby normal brain to less toxic doses. At present, the numbers are too small and the follow-up too short to determine the efficacy of interstitial brachytherapy.⁶⁹

Interstitial brachytherapy (IB) may reduce the rate of tumor growth, but it rarely produces clinical improvement. Patients are generally not considered for IB unless their Karnofsky score is ≥ 70 .

101.4.2 Techniques

Techniques include:

- 1. insertion of high activity iodine-125 pellets which remain in place (either by conventional open surgery or by stereotactic technique)
- 2. insertion of catheters (so-called afterloading catheters) containing radioactive source (such as gold or I125) by stereotactic technique, which are then removed at a predetermined time (usually 1–7 days)
- 3. instillation of radioactive liquids (e.g. phosphorous isotope) into a cyst cavity

I125 has several characteristics that favor its use: it emits low-energy gamma rays which are absorbed by surrounding tissues minimizing radiation exposure of the normal brain, medical personnel and visitors. It is available as low-activity (<5 mCi) or high-activity (5–40 mCi) seeds.

Treatment planning is devised to deliver 60 Gy to the edge of a volume that extends 1 cm beyond the contrast-enhancing tumor, with variations included to spare radiosensitive structures (e.g. optic chiasm). Usual delivery rates are 40–50 cGy/hr to the tumor margin (30 cGy/hr is the critical dose for cessation of human tumor growth) requiring that the seeds stay in the afterloading catheter ≈ 6 days.

101.4.3 Radiation necrosis

Symptomatic radiation necrosis (RN) occurs in ≈ 40% of cases, and may occur as early as several months after IB. It may be impossible to differentiate from recurrent tumor in many cases. Symptomatic treatment is often achieved with increased corticosteroid dosages. Continued neurologic deterioration may require craniotomy.

101.4.4 Outcome

IB is often used as a “last ditch” effort in a patient with a recurrent malignant tumor who has received maximal external beam irradiation and who is not a candidate for reoperation (as expected, the results in patients with such poor prognoses are not good). However, patients eligible for IB are usually better than those who are not candidates, and this may bias the results towards a better outcome.⁷⁰ Some studies with early (primary treatment) use have shown possible benefit.⁷¹

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102 Endovascular Neurosurgery

102

102.1 General information

102.1.1 Introduction

Endovascular Neurosurgery, AKA Neuroendovascular surgery, endovascular and surgical Neuroradiology (ESNR), or interventional neuroradiology (INR), combines catheter based techniques and imaging for the diagnosis and treatment of specific cerebral and spine conditions.

102.1.2 Indications/Conditions treated

Endovascular neurosurgery include diagnosis as well as the following treatment of:

- Aneurysms: coiling ($\pm$ stent or balloon assistance), flow diverting stents e.g., Pipeline, parent vessel sacrifice
- Arteriovenous malformations (AVMs): embolization (pre-operative or curative)
- Dural arteriovenous fistula (DAVF): curative or palliative embolization
- Spinal AVMs: embolization
- Arteriovenous fistulas: e.g. Carotid Cavernous fistulas (CCF)
- Acute embolic stroke: intra-arterial clot thrombolysis or mechanical thrombectomy
- Cranial sinus thrombosis (CST): Thrombolysis or mechanical thrombectomy
- Cerebrovascular arterial dissections: stenting or parent artery sacrifice
- Cervical internal carotid artery stenosis: angioplasty/stenting
- Tumors: embolization. Primarily used before surgery as an adjunct to decrease vascularity e.g., with some meningiomas and hemangioblastomas
- Intracranial atherosclerosis
- Vasospasm
- Transverse sinus stenosis: stenting as in pseudotumor cerebri
- Inferior Petrosal sinus sampling for localizing pituitary macroadenomas
- Iatrogenic vascular injuries: stenting or embolization to achieve hemostasis
- Refractory epistaxis: embolization to achieve hemostasis
- Wada testing: evaluation for language and memory localization (e.g. epilepsy patients being considered for surgery)
- Intraarterial chemotherapy: e.g. retinoblastoma
- Intraoperative angiography: typically used in aneurysm surgery to confirm exclusion of the aneurysm and patency of parent vessels, and during AVM surgery to confirm elimination of nidus

102.1.3 Contraindications

- Uncorrected bleeding disorders
- Poor renal function (due to iodine dye load)
- Relative contraindication: connective tissue disorder that predisposes to vessel dissection
- For spinal angiography: thoracic aortic aneurysm (relative)

102.1.4 Risks of cerebral angiography

Risk varies with the nature of the pathology being investigated and with the experience of the angiography team. Overall risk of a complication resulting in a permanent neurologic deficit^{1,2}: 0.1% In ACAS, there was a 1.2% complication rate (p.1276).

102.1.5 Miscellaneous angiography

□ Tumors. While angiography is no longer used diagnostically for tumors, there are a few general principles worth knowing. Typically, non-vascular deep lesions cause changes in venous structures, whereas superficial lesions affect arterial structures. Malignant neoplasm (e.g. glioblastoma): the classic feature on angiography is an early draining vein. Meningiomas: the stain (contrast) “arrives early, stays late” (appears early in arterial phase, blush persists beyond venous phase); see also other angiographic findings with meningiomas (p.695).

□ Allcock test. Evaluates flow through the posterior communicating arteries by vertebral injection with simultaneous common carotid artery compression in the neck.

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102.2 Pharmacologic agents

102.2.1 General information

This section presents drugs as they are used in neuroendovascular procedures.³ The indications cited are specific to endovascular intervention.

102.2.2 Abciximab (ReoPro)

General information

The Fab fragment of an antibody. Prevents binding of fibrinogen to platelet GP IIb/IIIa receptors. Platelet inhibition lasts up to 48 hours.

Indications and case selection

- Acute endoarterial thrombus during endovascular intervention
- Dissection with thrombus adherent to intimal flap
- Prophylaxis for intracranial or extracranial stent implantation

Dosing

□: bolus with 0.25 mg/kg IV over 10–60 minutes (shorter duration for the acute complications during intervention) followed by infusion of 0.125 mcg/kg/min (max. 10mcg/min) for 12 hours.

Reversal

Discontinue abciximab infusion. Allow 10–30 minutes for clearance of the drug from plasma, followed by platelet transfusion. Surgical intervention should be delayed for 12 – 24 hours after discontinuation.

102.2.3 Aspirin

General information

Irreversibly inactivates cyclo-oxygenase, resulting in platelet inhibition by preventing formation of prostaglandins from arachidonic acid.

Indications and case selection

- Intra (short-term) and post-procedural (short + long-term) prophylaxis of thromboembolic events e.g., during
 - diagnostic cerebral angiography
 - coil embolization of aneurysms
 - stent implantation (typically with a second antiplatelet agent)
 - balloon test occlusions
 - therapeutic occlusion of large arteries
- Subacute management of procedural complications e.g.,
 - parent artery coil herniations
 - thrombus or clot on coil phenomena
 - in-stent thrombus (alone or in combination with a second agent)

Dosing

□: 325–1300 mg orally, daily

Uncoated Aspirin (ASA) achieves peak plasma concentrations within 30 – 40 minutes.^{4,5} Enteric-coated ASA achieves peak plasma concentrations in up to 6 hours.⁶ 60% of population is resistant to the antiplatelet effect of low dose (81 mg) ASA, while up to 30% are resistant to 325 mg/day.^{7,8} However, the assessment of ASA resistance is highly assay dependent. A dose related effect i.e., improved

response to increasing the dose, indicates that ASA also exert antiplatelet effects through non-cyclo-oxygenase pathways.⁹

Reversal

Reversal is achieved by platelet transfusion.

ASA causes an irreversible inactivation of the platelet cyclo-oxygenase that persists through the lifespan of the exposed platelets.

102.2.4 Clopidogrel (Plavix™)

General information

A platelet ADP receptor antagonist.

Indications and case selection

- Prevention of intra-procedural and short-term post-procedural (4–12 weeks) thromboembolic events related to endovascular procedures including
 - Coil embolization of wide-neck cerebral aneurysms where stent will be used
 - Stent implantation (with a second antiplatelet agent)
 - Therapeutic occlusion of large arteries (often with a second antiplatelet agent)
- Subacute management of procedural complications (alone or in combination with a second agent)
 - Parent artery coil herniations
 - Thrombus or clot on coil phenomena
 - In-stent thrombus (may be more effective than other agents)

Dosing

□: 75 mg PO daily. Start 5 days prior to the actual procedure because there is a 3–7 days latency period to full therapeutic effect

LD: 300 mg PO, if there was no time to achieve therapeutic effect over a course of days. A therapeutic effect can usually be achieved within 2 to 3 hours of LD.

Reversal

Platelet transfusion.

102.2.5 Eptifibatide (Integrilin®)

General information

A reversible inhibitor of platelet aggregation by preventing the binding of fibrinogen, von Willebrand factor, and other adhesive ligands to GP IIb/IIIa. Platelet aggregation inhibition appears dose and concentration dependent and is reversible following discontinuation of eptifibatide. It causes a 5-fold increase in bleeding time and has no measurable effect on PT or aPTT.

Indications and case selection

These are the same as for abciximab (see above).

Dosing

□: bolus 180 mcg/kg IV (max 22.6 mg) over 1–2 minutes followed by infusion of 2 mcg/kg/min .

Reversal

- Discontinue drug. Significant reduction of anti-platelet effects occurs in 2–4 hours.¹⁰
- For clinical signs or imaging evidence of hemorrhage:
 - For ICH, hemodynamic compromise, or decrease in Hb > 5 g/dl or decrease in HCT > 15% give platelet transfusion
 - For a decrease in Hb < 5 g/dl or decrease in HCT < 15% give desmopressin □: 0.3 mcg/kg × 1.

102.2.6 Heparin

General information

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A glycosaminoglycan that indirectly inhibits thrombin by modulating antithrombin III (AT III) activity. It also indirectly inactivates factors IXa, Xa, XIa and XIIa and restores electronegativity to endothelial surfaces. It prevents thrombin-platelet aggregation and inhibits von Willebrand factor. The anticoagulant effects are immediate. The half-life of IV heparin is approximately 1.5 hours.^{11,12}

Indications and case selection

- Prophylaxis during diagnostic angiography (used in flush solutions only)^{3,13}
- Neuroendovascular procedures including,
 - coil embolization of intracranial aneurysms,
 - therapeutic occlusion of carotid, vertebral, or other large cerebral artery
 - Transarterial embolization of brain AVM or dural AVF
 - Percutaneous transluminal angioplasty
 - Intra- or extracranial stent implantation
 - Balloon test occlusion of carotid, vertebral, or other large cerebral artery

Dosing

□: Flush systems: 6000 i.u. per liter of 0.9%normal saline (6 i.u. per cc) for flush systems used in neuroendovascular procedures. During intraoperative angiography: 2500 i.u. per liter of 0.9%normal saline (2.5 i.u. per cc). The lower dosage during surgery is a precaution against excessive bleeding from operative sites e.g., during craniotomy.

Endovascular interventions:

Coil embolization for unruptured aneurysm: 5000 units bolus. Check the activated clotting (ACT, □ Table 102.1) 20–30 minutes after bolus and then hourly. Administer heparin 0–5000 units hourly prn to maintain ACT between 250–300.

Coil embolization for ruptured aneurysm: Place the framing coil prior to heparin bolus.

Angioplasty with/without stenting: 5000 units bolus. Perform ACT 20–30 minutes after bolus and then hourly. Administer heparin up to 5000 units hourly prn to maintain ACT between 300–350.

Post procedure heparinization: For aneurysm embolization, the following regimens are recommended:

Heparin dosing for post coiling neurosurgery patients

- University of Cincinnati Protocol¹⁴: devised to administer a safe dose of heparin that does not require repeat blood draws and titration to ACT or aPTT
 - Weight-based dosing
 - Weight ≤75 kg: 900 units/hour, no bolus doses, for 12 hrs
 - Weight >75 kg: 1300 units/hour, no bolus doses, for 12 hrs
 - Labs: no ACT or other laboratory draws required.
- Alternative: The most commonly used heparin weight based protocol (NB: this differs from dosing e.g. for coronary indications or treatment of DVT or PE¹⁵): bolus dose of 60–70 units/kg followed by a maintenance IV infusion rate of 18 units/kg/hour. Titrate to ACT as shown in □ Table 102.1.

Reversal

IV protamine sulfate at 1 mg per 100 units of circulating heparin (not to exceed 50 mg total).

A preloaded syringe of 50 mg should be available at all times. Normally protamine is administered as an IV infusion over 10 – 30 minutes, to prevent idiosyncratic hypotension and anaphylactoid

Table 102.1 Recommended ACT for various endovascular procedures	
Indications	ACT (Seconds)
Procedures involving deep arterial injury e.g., percutaneous transluminal angioplasty with/without stenting. Procedures with significant stasis of blood flow e.g., balloon occlusion of parent vessels	High (300–350)
Procedures in which the above mentioned thrombogenic elements are absent e.g., embolization of an aneurysm or AVM	Moderate (250–300)

symptoms. In an emergency e.g., vessel or intracranial aneurysm perforation, anticoagulation must be immediately reversed by rapid IV bolus of 10 mg protamine over 1–3 minutes.

102.2.7 Nitroglycerine

General information

Produces strong and immediate vasodilatation by stimulation of cGMP, which results in vascular smooth muscle relaxation.

Indications and case selection

Vessel spasm during catheterization.

Dosing

□: 100 – 300mcg through the catheter.

102.2.8 Papaverine

General information

A benzyloquinolone alkaloid that vasodilates by inhibition of cAMP and cGMP phosphodiesterases in smooth muscle, leading to increased intracellular levels of cAMP and cGMP. It may also inhibit the release of calcium from the intracellular space by blocking calcium ion channels in the cell membrane. Papaverine is short acting, with half-life of less than 1 hour.

Indications and case selection

Pre-treatment for angioplasty. The resulting vasodilatation will assist with balloon catheter placement. Due to its short duration of action necessitating repeated administration, other agents e.g., verapamil, are generally preferred.

Dosing

□: 300 mg of 3%papaverine (30 mg/ml) at pH 3.3, is diluted in 100 ml of normal saline to obtain a 0.3%concentration. It is administered intra-arterially through the microcatheter, which is positioned just proximal to the affected vascular segment at a rate of 3 ml/minute.

□ Do not mix Papaverine with contrast agents or heparin, which may result in precipitation of crystals.

102.2.9 Sodium Amytal

General information

A barbiturate derivative that activates GABA_A receptors. When it is administered into cerebral vasculature during Wada test, it temporarily anesthetizes the perfused region, resulting in isolation of the contralateral hemisphere permitting assessment of cortical functions including language and memory. It blocks neuronal activity and in conjunction with lidocaine, which blocks axonal activity, it is administered as a test injection in procedures such as spinal AVM embolization, or embolization of spinal tumors, prior to the injection of the embolic agent.¹⁶

Indications and case selection

Wada Test

Test injection prior to embolization of feeder vessel of AVM. In order to suppress both neuronal and axonal activity, amytal injection is followed by xylocaine injection.

Dosing

□: 50–100 mg per test injection via catheter.

For administration, dilute 500 mg sodium amytal in 20 cc 0.9% NS to yield a concentration of 25 mg/cc of sodium amytal.

For Wada test, after catheterization of vessel of interest, 100 mg of Sodium Amytal (4 cc of the preparation described above) are injected through the catheter. Additional boluses of 25 mg (1 cc) or modification of the original bolus may be required, per the neurologist's request.

102.2.10 Tissue Plasminogen Activator (tPA)

General information

A fibrin specific thrombolytic protease that converts plasminogen to plasmin.

Indications and contraindications

Indications:

- Cranial sinus thrombosis (CST)
- Patients with acute ischemic stroke who are 18 years of age, or older; have clinical diagnosis of ischemic stroke with measurable neurologic deficit
 - symptom onset ≤ 4.5 hours prior to initiation of treatment: given IV
 - up to 6 hours after symptom onset: given by intra-arterial administration. May be up to 24 hours after onset for posterior circulation (where there is less likelihood of hemorrhagic conversion of infarct).

□ Absolute Exclusion criteria:

- Brain CT scan demonstrating ICH prior to treatment.
- Only minor or, rapidly improving symptoms of stroke.
- Clinical presentation suggestive of SAH, even with a normal CT.
- CT demonstrating ICH or SAH.
- Active internal bleeding.
- Known bleeding diathesis including platelet count $< 100 \times 10^3/\text{mm}^3$.
- Heparin within 48 hours with an elevated APTT.
- Current oral anticoagulation use e.g., warfarin, or recent use with an elevated PT (> 15 sec) or INR (> 1.7).
- Current use of direct thrombin inhibitors (desirudin, bivalirudin, argatroban)¹⁷ or direct factor Xa inhibitors (Rivaroxaban, Edoxaban, Betrixaban)¹⁸ with elevated aPTT, INR, platelet count, Ecarin Clotting Time (ECT), TT, or factor Xa activity assays.
- Intracranial or intraspinal surgery, severe TBI, or previous stroke within 3 months.
- Suspected aortic dissection associated with stroke.
- Suspected subacute bacterial endocarditis or vasculitis.
- Recent (within past 7 days) arterial puncture at a non-compressible site.
- Lumbar puncture within 7 days.
- History of ICH.
- Known AVM or aneurysm.
- Persistent SBP > 185 mm Hg or DBP > 110 mm Hg at time of treatment, or patient requires aggressive treatment to reduce BP within these limits.

Relative Exclusion criteria:

- Witnessed seizure at the same time as onset of symptoms of stroke
- NIHSS > 22 (severe deficit) or < 4 (mild deficit)
- Baseline CT demonstrating extensive ischemic changes e.g., sulcal effacement, mass effect, or edema, hypodensity in more than 1/3 of MCA territory distribution
- Major surgery or serious trauma within 14 days
- History of GI or urinary tract hemorrhage within 21 days
- Recent acute MI (past 3 months)
- Post myocardial infarction pericarditis
- Blood Glucose < 50 mg/dl (2.7 mmol/l) or > 400 mg/dl
- Age > 80 years
- Pregnancy
- History of ischemic stroke AND diabetes
- History of terminal cancer or other medical condition with limited life expectancy
- History of advanced dementia

Dosing

□: Intravenous: 0.9 mg/kg (max. 90 mg). The first 10% of the calculated dose is administered as an IV bolus over 1 minute and the remainder is infused over an hour, within the 4.5 hour window.

□: Intraarterial: The maximum intra-arterial dose is 22 mg. It is independent of any previously administered intravenous dose.

1–2 mg tPA is administered manually distal to the clot, then administer an infusion of 0.5 mg/ml at 20 ml/hr (10 mg/hr). The infusion is prepared by mixing 10 mg of tPA in 20 ml in normal saline, resulting in a concentration of 1 mg tPA per 2 ml saline (or 0.5 mg/ml). An infusion pump may be used for more precise administration.

Angiography is performed every 15 minutes (following infusion of 2.5 mg tPA) as the catheter is gradually drawn back through the clot. The lesion is re-crossed after each angiogram. If the artery is still occluded, inject 1 – 2 mg tPA manually and resume the tPA infusion.

Discontinue tPA if

- adequate recanalization is achieved
- extravasation of contrast material is noted on angiography
- The maximum dose has been administered, or the administered dose approaches the maximum dose without clinical or angiographic improvement.

In CST, usually 2–5 mg are administered through the thrombus and then an infusion started at a rate of 1 mg/hr, usually for 12 hours. If a clot burden is still present on angiography, a longer duration of administration until the clot resolves is a consideration.

For CST, the infusion is prepared in a concentration of 1 mg/10 ml (0.1 mg/ml), for a rate of 10 ml/hr.

Reversal

FFP transfusion.

102.2.11 Verapamil

General information

A nondihydropyridine calcium channel blocker that reduces the influx of calcium through L-type calcium channels in smooth-muscle cells, enabling vasodilation. Half-life is about 3–7 hours.

Indications and case selection

- Before balloon angioplasty: chemical vasodilatation prior to mechanical vasodilatation may enable a smoother and safer angioplasty.
- Mild vasospasm, that does not warrant angioplasty.
- Moderate vasospasm, that cannot be safely treated with angioplasty.

Dosing

□: 5–10 mg IA. It is slowly infiltrated (over 2–10 minutes) into the vasospastic vessel via a microcatheter into the intracranial vessel, and/or via diagnostic or guide catheter into larger vessels e.g., ICA or, VA.

Reversal

For clinically significant hypotension or high degree AV block: treat with vasopressors and cardiac pacing e.g., epinephrine, norepinephrine and vasopressin infusions. Calcium chloride infusion is given in large doses e.g., 1 gm/hr for more than 24 hours. 20% intralipid infusion (100 ml bolus followed by continuous infusion at 0.5 ml/kg/hr) may also be useful.¹⁹ Atropine may be administered for bradycardia. Hemodialysis is ineffective.

102.2.12 Xylocaine

General information

Blocks the fast voltage gated Na⁺ Channels in neuronal cell membranes. There may be inhibition of postsynaptic neurons and consequently, action potentials.

Indications and case selection

As a local anesthetic prior to arteriotomy. Preparations with or without epinephrine may be used.

Cardiac xylocaine is used to test axonal function. It may be used alone or in conjunction with amobarbital.

Used for functional testing in case of spinal vascular disorders.

Provocative functional testing in cerebral AVMs.

Dosing

□: for local anesthesia : Approximately 5 ml of 2% lidocaine (max. 4 mg/ml to 280 mg; 14 ml)

□: for neurophysiological testing (Wada test): 10 – 40 mg i.a. of cardiac lidocaine.

Reversal

Lidocaine overdose²⁰: intravenous bolus of 20% lipid emulsion 1.5 ml/kg over 1 min and start IV infusion at 15 ml/kg/hr. The bolus may be repeated twice at 5 min intervals, if cardiovascular stability is not restored. Additionally, the infusion may be doubled to 30 ml/kg/hr if instability persists after 5 minutes. Lipid emulsion is continued until cardiovascular stability is restored or maximum dose administered.²⁰ Do not exceed the maximum cumulative dose of 12 ml/kg.

Concurrent supportive care includes following ACLS protocols e.g., securing airway, hemodynamic support etc.

□ Propofol is not a substitute for lipid emulsion in such cases. It has only 10% lipids which is too low to be of benefit, and the cardio-depressant properties of propofol may be counterproductive in such situations.

102.3 Neuroendovascular Procedure Basics

102.3.1 Vascular access

General information

Most commonly, vascular access is obtained through femoral artery. If femoral access is not possible then the radial artery, brachial artery or carotid artery (least common) may be used.

Femoral artery access

After prepping and draping the groin, place the left little finger on anterior superior iliac spine and span with thumb to the pubic symphysis. This approximately demarcates the ilioinguinal ligament. Bisect this line with the other hand and palpate the femoral pulse. The site of access is 3 finger-breadths below the point of bisection, to ensure the vessel puncture site is below the ilioinguinal ligament and therefore, compressible. A marker (e.g. hemostat) can be placed over the pulse and a fluoro shot taken to confirm that the planned access site is situated over the middle of the femoral head.

A small, superficial stab incision in the skin is made at the selected point, after infiltration of local anesthesia. In an elective case, a micropuncture set with a 7 cm 21 or 23 gauge needle may be used for vessel puncture. In an emergency (e.g., stroke) a larger 18G single wall needle is used. The artery is palpated and immobilized between the index and middle finger of one hand, while the needle is introduced through the stab at 45°. Once the artery is accessed, blood will emanate through the needle hub. An exchange over wires is made using modified Seldinger technique to place the sheath of desired size. The sheath is connected to continuously running flush of heparinized saline and secured with adhesive or suture to prevent dislodgement.

Radial artery access

First perform Allen's test with pulse oximetry to ensure the hand has satisfactory vascular supply, in case the procedure results in radial artery occlusion.

Allen's Test: Palpate the radial and ulnar arteries and place a pulse oximeter on the thumb or index finger. Make the patient flex and extend their fingers repeatedly. With digital pressure, compress both the radial and ulnar arteries during finger extension and maintain the compression until the oximetry pulse is lost. The wrist is maintained in approximately 20° flexion, in order to avoid false positive test when the wrist is in hyperextension. Release the pressure on the ulnar artery. Measure the time taken to achieve visual capillary refill in finger pads and at least 92% oxygen saturation. Normal capillary refill time is <5 sec, refill times of 5–15 secs are considered equivocal. A refill time longer than 15 sec is abnormal. Allen's test can also be performed using ultrasonography.

Reverse Allen's Test: It should be performed when the radial artery is being subjected to a repeat procedure. Compress both the radial and ulnar arteries with digital pressure during finger extension and maintain the compression. Release the pressure on the radial artery. Measure the time taken to achieve visual capillary refill in finger pads and at least 92% oxygen saturation, as indicated above.

During access, the following modifications are made from the technique described above. A shorter needle 21 G (e.g., 3 cm, instead of 7 cm) is used. Avoid entering the artery at a steep angle, as this may cause difficulties in threading the wire through the artery.

Advance a 0.018" wire through the needle hub into the radial artery and remove the needle. Make a nick in the skin over the wire, to aid in smoother insertion of larger sheaths. A 4 Fr micropuncture sheath (with dilator) over the wire is advanced over the wire and then the wire and dilator are removed. A cocktail of heparin (5000 IU/ml), verapamil (2.5 mg), 2% lidocaine (1.0 ml), and nitroglycerin (0.1 mg) are administered through the introducer sheath to relieve and/or prevent vasospasm. The patient should be forewarned about a transient but uncomfortable sensation of severe burning, as the cocktail is injected into the artery. The standard 0.035" wire is advanced through the sheath. The desired sheath size is then placed.

Note: After completion of the procedure, a closure device is not used for radial artery. Only manual compression for 15–20 minutes is applied.

102.3.2 Sheath Management

Once placed, the sheath is connected to a continuously running heparinized saline solution at a rate of 30 ml/hr. The flush consists of 6000 units of heparin in 1000 ml of 0.9% saline bag (6 units/ml). The saline bag is placed in a pressure infuser which is inflated to 300 mmHg. It is important that the sheath be continuously irrigated and the saline bag be under a pressure greater than the patient's own arterial pressure. The patient's leg on the side of the sheath is kept straight to prevent kinking of the sheath. If the intention is to maintain the sheath for several hours or a few days, it should be secured by suturing to the patient's skin.

102.3.3 Arteriotomy closure

After completion of the procedure, if vascular access will not be required within next few days, the sheath is removed. To do this, the following options are available:

- **Pressure**
 - **Manual Pressure:** Palpate the artery proximal to site of arteriotomy, remove sheath and apply manual pressure for 15–30 minutes and gradually decreasing pressure every 5 minutes.
 - **Femostop™:** Prior to application, ensure ACT is <150 and BP is under control. The inner circle of the Femostop dome should be positioned 1 cm superior and 1 cm medial to the actual puncture site and over the femoral artery. Inflate the Femostop to 20–30 mm Hg above the patient's systolic pressure. If this does not result in hemostasis, inflate to higher pressures until distal pulses are occluded. Maintain distal pulse occlusion for 5–7 minutes, then readjust the manometer pressure until good pedal pulse and good color of extremities is achieved. Continue to progressively decrease the applied pressure over the course of several hours, until the device can be discontinued entirely. Usually, a Femostop is maintained for 6–12 hours.
- **Percutaneous closure devices**
 - **Angioseal™:** effects closure by deposition of fibrin plug in the vessel which is then pulled back into the arteriotomy defect of the vessel wall. Angioseal is available in 6 and 8 Fr sizes. Deployed by exchanging it with the sheath over a wire. After deployment, the patient remains supine for 2 hours (with pillow under the head) with the leg on the side of Angioseal kept straight. Patient is mobilized thereafter.
 - **Mynx™:** a bio-absorbable sealant that is deposited in the arteriotomy defect. Available in 5, 6, 7 Fr sizes. Tends to be the least painful
 - **Starclose™:** Also exchanged with the sheath over a wire. Deployment results in stitch closure of the arteriotomy defect. Tends to be more painful

102.4 Diagnostic angiography for cerebral subarachnoid hemorrhage

102.4.1 General information

Objective: diagnose source of hemorrhage, assess supplying blood vessels and collateral vessels (may require additional provocative maneuvers), assess vessels for possible bypass, assess for vasospasm.

The most common cause of nontraumatic SAH is ruptured cerebral aneurysms. Other causes include AVM (cerebral or spinal), vasculitis, pre-truncal nonaneurysmal subarachnoid hemorrhage, arterial dissection, dural sinus thrombosis, pituitary apoplexy, bleeding dyscrasias, sickle cell disease and cocaine abuse. In 14–22%a cause for SAH is not found on angiography.²¹

102.4.2 Setup

A sheath is placed in the femoral artery, connected to a continuously running flush of heparinized saline. A diagnostic catheter is attached to a continuously running flush of heparinized saline. It is introduced into the vasculature, via the sheath and advanced over the guidewire through the aorta to the target vessel. The catheter is advanced with the guidewire leading and pulled back with the guide wire completely retracted into the catheter. Once the catheter is in desired position, the guidewire is withdrawn completely and angiography performed, by using manual injections or the auto-injector.

102.4.3 Planning

Pre-planning the procedure will save time, amount of contrast administered and radiation exposure. Standard views (half Towne’s and lateral for anterior circulation; Towne’s and lateral for posterior circulation) are obtained. Additionally, the images in appropriate projections are obtained, keeping in mind the patient’s pathology. As needed, rotational angiography is performed and the 3D reconstruction manipulated for further examination, as well as, selection of further projections for angiography.

102.4.4 Additional views

In case of aneurysms, based on their location, the projections as shown in □ Table 102.2 may prove useful.

102.5 Disease-specific intervention

102.5.1 Aneurysms

General information

The coil vs. clip debate is a moving target, especially given the fact that as long-term data about endovascular treatment is becoming available, the technology continues to evolve which renders the data obsolete. While there will always be a role for surgical clipping, endovascular therapy has emerged as a first line therapy for most aneurysms (particularly ruptured aneurysms or in poor surgical candidates). Surgery still remains a strong option for MCA aneurysms and many believe for most PICA aneurysms.

Indications

The selection of aneurysm for treatment depends upon the following considerations:

- Ruptured vs. unruptured. A ruptured aneurysm needs to be treated urgently, as the risk of re-rupture is 2–3%/day in first few days and 20%in 2 weeks. The risk of morbidity and mortality of an untreated ruptured aneurysm is 45-50%^{22,23}
- Symptoms other than rupture. Aneurysms that present with symptoms e.g., cranial nerve palsy, loss of vision or ischemia, may be at a higher risk of rupture than asymptomatic aneurysms (some symptoms may be due to acute expansion).^{24,25,26,27,28,29}
- Size. Large aneurysms (>7–10 mm) are more likely to rupture than small (<7 mm) aneurysms.^{29,30}
- Shape. An irregularly shaped aneurysm may be at a greater risk of rupture then a spherical saccular aneurysm. Irregular shape includes morphological characteristics such as daughter blebs, or irregular borders.³¹

Table 102.2 Projection for angiography, based on aneurysm location		
Aneurysm location	Projection	Additional view
Anterior communicating	25° oblique away from injection side; center beam on lateral aspect of ipsilateral orbital rim; orient tube in Towne’s view.	Submental vertex, but the image may be degraded by the amount of interposed bone.
Posterior communicating	55° paraorbital oblique away from injection side; center beam 1 cm posterior to inferior portion of lateral rim of ipsilateral orbit; orient x-ray tube 12° cephalad	
Carotid terminus	25° oblique away from injection side; center beam 3–4 cm above lateral aspect of ipsilateral orbital rim; orient x-ray tube in Towne’s view.	Submentovertex view
Ophthalmic artery	25° oblique away from injection side; center beam 3–4 cm above lateral aspect of ipsilateral orbital rim; orient tube in Towne’s view.	Submentovertex view
Superior hypophyseal	25° oblique away from injection side; center beam 3–4 cm above lateral aspect of ipsilateral orbital rim; orient x-ray tube in Towne’s view.	Submentovertex view
Posterior inferior cerebellar artery (PICA)	55° paraorbital oblique away from injection side; center beam on foramen magnum; orient x-ray tube 12° cephalad	
Vertebrobasilar junction aneurysms	15° oblique away from injection side; center beam on foramen magnum; orient x-ray tube 25° Towne.	Submentovertex view
Anterior inferior cerebellar artery aneurysm (AICA)	AP or Submentovertex view, center beam on nasion, orient x-ray tube 15° caudad	
Basilar bifurcation	Oblique 25° away from or towards injection side, center beam 3–4 cm above lateral aspect of ipsilateral superior orbital rim, orient x-ray tube 25° Towne.	Submentovertex view

- Aspect ratio. In addition to size, the aspect ratio (aneurysm depth/neck width) may predict the aneurysms at risk of rupture. An aspect ratio greater than 1.6 may create low flow conditions in the dome of the aneurysm that lead to stasis, thrombosis and a fibrinolytic cascade that results in breakdown of the intima.³² A retrospective evaluation of 75 ruptured and 107 unruptured aneurysms, demonstrated the mean aspect ratio to be 2.7 for ruptured and 1.8 for unruptured aneurysms (p<0.001). The mean depth of aneurysm was also greater in ruptured aneurysms (7.7 ±4.9 mm vs 5.1 ± 4.5 mm). 75% of the ruptured aneurysms were<10 mm, and 62% of these had aspect ratio>1.6.³³
- Location. The ISUIA studies demonstrated posterior circulation aneurysms had a higher risk of rupture compared to anterior circulation.^{24,34} Conversely, cavernous segment aneurysms had a 0% risk of rupture until they reached a size of 13–24 mm, when the 5 year cumulative risk became 3%
- Choice of treatment. The choice of treatment should take all of above in addition to individual patient considerations e.g., age, and overall health.

Wide necked aneurysms were previously thought better suited for clipping, but the availability of stents that may act as scaffolding for coils e.g. Enterprise™ stent or act as vascular reconstruction devices e.g., Pipeline. has greatly increased the spectrum of aneurysms amenable to endovascular treatment. Additionally, small aneurysms (<4 mm) in size may be less favorable for coiling.

Endovascular Options

□ Coiling. This is the treatment of first choice for most narrow-necked aneurysms. The first coil (framing coil) used is equal to the size of aneurysm, or slightly oversized (especially in unruptured aneurysm). The coils are progressively downsized as embolization progresses. The goal is to maximally pack the aneurysm such that no contrast is seen entering it and without causing any loop to herniate into the vessel lumen. Even after maximal packing, the actual packing is about 20–30%³⁵

□ Coiling with stenting. For wide-necked aneurysms, a stent may be used to prevent the coils from herniating out of the aneurysm into the blood vessel. When a stent is used, the patient is required to be on ASA (indefinitely) and Plavix (for at least 1 month). Therefore, stent assisted coiling is generally avoided in ruptured aneurysms, in part due to the fact that if an EVD is needed it cannot be safely placed with the patient on Plavix. Therefore, stent assisted coiling is generally avoided in ruptured aneurysms. However, it has also been undertaken successfully in ruptured cases, with 93% technical success, clinically significant ICH in 8% (including 10% known to have EVDs), and significant thromboembolic events in 6%³⁶

□ Balloon-assisted coiling. This technique may be used for wide necked aneurysms where stenting is deemed less desirable e.g., ruptured aneurysm, since it bypasses the need for dual anti-platelet therapy. A balloon catheter is selected based on the diameter of the artery and the width of the aneurysm neck. The balloon segment of the catheter (indicated by radiopaque markers) is positioned across the neck and kept inflated during coil deposition into the aneurysm. It is deflated prior to coil detachment and the stability of the deposited coil assessed. If the coil appears stable, it is detached. This inflation-deflation technique is continued as required, until the aneurysm has been completely coiled.

□ Dual catheter technique. In this case two microcatheters are placed in the aneurysm. Coil loops are deposited alternatively from each.

□ Coil types. Bare platinum coils: Guglielmi detachable coils are bare platinum coils that come in various diameters (mm) and lengths (cm). They come as standard, soft and ultra soft, as well as 360° and helical configurations.

More recently, the same manufacturer has introduced Target coils. The advantage over GDC is “less kickback” when being deposited in the aneurysm especially during final stages when coiling is near completion.

Other bare platinum coils include Trufill/Orbit coils, Micrus, and Target.

□ Hydrocoils. Platinum coils coated with hydrogel which expands upon contact with blood filling the residual space between loops.

□ Liquid embolic agents. A less commonly used technique where a DMSO compatible catheter is placed into the aneurysm and a balloon catheter inflated across its neck, while a liquid embolic agent (e.g. Onyx 500) is injected into the aneurysm. Onyx may be used in sidewall aneurysms that have a wide neck (≥ 4 mm) or with dome-to-neck ratio < 2 , that are not amenable to surgical clipping. It may also be used in patients who are allergic to metal (e.g., cobalt, chromium, platinum or tungsten). It is important to keep the balloon catheter inflated until the Onyx solidifies, as otherwise embolization into the normal vessel may produce a stroke.

□ Pipeline Embolization Device (PED). The woven design of this particular stent makes it low porosity which attenuates blood entry into the aneurysm and therefore encourages stasis. If needed, two or more of these may be deployed within each other across the neck of the aneurysm, to cause adequate flow stasis in the aneurysm. Angiography immediately post deployment, demonstrates contrast stasis in the aneurysm. Six month follow up angiography usually demonstrates complete obliteration of the aneurysm, such that it is no longer visualized. Once an aneurysm is successfully treated with PED, there is a 0% recurrence rate. The most recent iteration of the device known as Pipeline Flex™ is a 48 braid device that is somewhat easier to deploy than the earlier version. It is available in diameters from 2.5 – 5 mm. Sometimes, coil deployment in the aneurysm may also be required to encourage aneurysm thrombosis.

A strategy used in case of ruptured aneurysms is to “undercoil” the aneurysm to address the risk of re-rupture and then complete treatment with PED when it becomes safe to use dual antiplatelet therapy

In case of ruptured blister aneurysms, Pipeline device has been used with success by administering an abciximab bolus (0.125mcg/kg) IV, approx. 10 minutes prior to device deployment.^{37,38}

Indications:

- Large or giant wide-necked ICA aneurysms, from petrous to the superior hypophyseal segments.
- Presently, patients must be 22 years of age or older.
- PED has been used in cases outside the established indications e.g. in MCA, vertebral and basilar arteries.^{39,40,41}

Contraindications:

- Ruptured aneurysm (due to requirement of pre-embolization dual anti-platelet therapy). However, it has been used in ruptured blister aneurysms.^{37,38}
- Patients in whom dual anti-platelet therapy is contraindicated.
- Patients who have not received dual anti-platelet therapy (ASA and clopidogrel) prior to procedure.
- Active bacterial infection.
- Metal allergy to cobalt, chromium, platinum or tungsten.
- Patients with pre-existing stent in the parent artery at the target aneurysm location (relative contraindication).

Treatment of aneurysmal rupture during coiling

- notify anesthesia: for assistance with critical care management and in case you need to go to the O.R.
- immediately lower the blood pressure
- inflate balloon if balloon-assisted coiling
- immediately reverse anticoagulation. Give 50 mg of protamine (protamine should always be available during the procedure)
- do not remove the coil that caused the perforation, continue deploying it and follow with additional coils in rapid succession
- insert an extraventricular drain (EVD)

102.5.2 Endovascular management of vasospasm

General information

A ruptured aneurysm needs to be secured by coiling or clipping prior to hyperdynamic therapy or endovascular intervention for vasospasm.

Indications

Failure of resolution, or worsening of symptoms of vasospasm with hyperdynamic therapy for 12 - 24 hours.

Patients with conditions such as congestive heart failure, cardiac ischemia or pulmonary edema, that limit the institution of hyperdynamic therapy.⁴²

Treatment options

In addition to pharmacological (hyperdynamic or triple-H) therapy, endovascular options for management of vasospasm include intra-arterial chemical spasmolysis, utilizing selective catheterization of the involved segment and, mechanical spasmolysis using angioplasty.

□ Chemical spasmolysis. Verapamil: Drug of first choice for spasmolysis in many centers. Its advantage is a relatively long half-life (6–12 hours).

Indications:

- Mild to moderate vasospasm that does not warrant angioplasty. Or, when angioplasty cannot be used safely.
- Vasospasm consequent to manipulation during endovascular intervention.
- Prior to performing angioplasty, so that the dilatation is performed on the relaxed dilated artery, rather than a relatively rigid vasoconstricted one.

Dose: 5-10 mg injected gradually (over 2–10 minutes) via microcatheter, as the microcatheter is withdrawn through the spasmodic segment. Up to 20 mg may be given into each arterial tree. It is injected gradually to prevent a significant drop in BP or bradycardia.

Other drugs used for chemical spasmolysis include nicardipine, papaverine and nitroglycerine. Similar to verapamil, nicardipine has a relatively long half-life (9 hours). Papaverine (<1 hour) and nitroglycerine (minutes) are comparatively short acting.

□ **Angioplasty.** A balloon mounted catheter is positioned across the arterial segment in spasm. The balloon is gradually inflated under fluoroscopic visualization to the desired width (equal to or less than the caliber of the adjacent non-spastic segment). The rate of balloon inflation is at a rate ≤ 1 atm/15 sec (to “stretch” not “crack” the vessel)

A combination of chemical and mechanical spasmolysis may also be used such that initially the spastic segment is dilated using verapamil infiltration, followed by balloon angioplasty.

Balloon dilatation produces durable results. However, even verapamil induced spasmolysis has been noted to persist on angiographies performed 48–72 hours post treatment.

Angioplasty is performed under direct fluoroscopic visualization. To this effect, contrast media is admixed with normal saline (50:50 to 2/3 -1/3 ratio) that is used to inflate the balloon

Precautions during angioplasty:

- Avoid inflating the balloon beyond the normal caliber of the vessel.
- Inflate gradually at a rate ≤ 1 atm/15 sec (to “stretch” not “crack” the vessel)
- Do not inflate the balloon beyond its indicated “burst pressure.” This can also be avoided by selecting a balloon catheter of appropriate width of the particular vessel and length of the segment in spasm.

□ **Complications and management.** Vessel rupture. This can be avoided by not inflating the balloon beyond the normal caliber of the vessel. Following rupture, reverse therapeutic heparinization by administration of protamine 1 mg. per 100 units of heparin (max. 50 mg). In such an emergency, rapid IV bolus of protamine (10 mg over 1–3 minutes) is delivered rather than over 10 – 30 minutes, which is usually done to prevent idiosyncratic hypotension and anaphylactoid symptoms.

Maintain wire (0.014”) access across the injured vessel. Exchange for a balloon catheter 1 mm smaller in size and inflate it for about 20 minutes. Then deflate and perform angiography to see if bleeding has been controlled. Where feasible, use a covered stent to address the perforation. If necessary, sacrifice the involved vessel using coils. With this approach e.g., in the M2 segment of MCA, the complication of stroke is accepted with the aim of saving the patient’s life. Additionally, decompressive craniotomy and evacuation of the clot may be required.

□ **Dissection.** If it is minor, non-flow limiting and without a raised intimal flap, no intervention may be necessary. The dissection may be re-examined after an interval to verify resolution. If managed conservatively, place the patient on aspirin, with or without Plavix.

In case of a significant dissection, place a stent across the affected segment and start patient on ASA and plavix. Plavix 75 mg PO daily is usually administered for a month while ASA is continued indefinitely.

□ **Thromboembolic complications.** A thrombus consequent to hardware, vessel injury blood stagnation or inadequate heparinization is apparent as a filling defect on control angiography. This can usually be readily addressed by administration of abciximab bolus of 0.25 mg/kg IV over 10–15 minutes, followed by infusion of 0.125 mcg/kg/min (max. 10mcg/min) for 12 hours. Repeat angiography 15 minutes following the initiation of amciximab. If the thrombus persists, an angioplasty may be performed to flatten the thrombus against the wall of the vessel. Once blood flow is restored, its lytic properties, as well as the abciximab may resolve the thrombus. Another consideration is to deploy a stent to restore lumen and blood flow.

□ **ICH.** This may happen due to causes including vessel injury, bleeding into a previous area of infarct, hyperperfusion in a previously compromised area, or hypertension. When detected, discontinue and reverse the heparinization using protamine. Monitor patient closely. A decompressive craniotomy or craniectomy may be required if the hemorrhage is significant.

□ **Surveillance and follow-up.** In addition to diligent monitoring in NICU, the progress or resolution of vasospasm can also be assessed by performing serial TCDs. Depending upon extent of availability, these can be daily, every other day or at least bi-weekly (e.g., q Monday and Thursday). The TCDs may be discontinued when resolution of vasospasm is apparent clinically and radiologically.

Angiography may be repeated approximately 3 days following initial intervention, or when there is significant change indicating worsening vasospasm. It need not be repeated, if the patient is

obviously improving. Alternatively, CTA may be used for surveillance. At the time of surveillance angiography, additional treatments with chemical spasmolysis and/or balloon angioplasty may be considered. Therefore, start off with a larger sheath (usu. 6 Fr) to enable intervention.

Usually, the patient will remain in the NICU for approximately 10–14 days considering that vasospasm is maximal by 7–8 days and usually resolves by 14 days.^{43,44,45,46}

102.5.3 Arteriovenous malformation

Indications for endovascular intervention

- The most common form of endovascular intervention for AVM is preoperative embolization to facilitate surgical AVM resection.
- Presence of associated lesions e.g., aneurysm or pseudoaneurysm on the feeding pedicle or nidus, venous thrombosis, venous outflow restriction, venous pouches or dilatations.
- A small surgically inaccessible AVM, or where surgery carries a high risk of morbidity and mortality. Curative AVM embolization is rare and limited to small lesions with simple angioarchitecture. The small surgically inaccessible AVM can also be treated with radiosurgery, which has a better track record than curative AVM embolization attempts
- As a palliative treatment in an AVM that is not completely treatable by any approach or their combination, due to location and/or diffuse morphology, but is symptomatic. Use with caution: data suggests that partial embolization of complex AVMs may increase rupture rate and worsen outcome.

AVM embolization

This can be performed using numerous agents (or their combination).

The agents include coils, onyx, NBCA and PVA.

□ Coils. These may be used to close down a vessel supplying the AVM, an AVM pouch or aneurysms on the arteries associated with AVM. However, it cannot be relied upon to completely obliterate the AVM nidus, or effect a cure.

□ Onyx™. A “lava-like” liquid embolic agent – Ethylene vinyl alcohol (EVOH) copolymer; ethylene and vinyl alcohol dissolved in dimethyl sulfoxide (DMSO) with micronize tantalum (for radioopacity) – that solidifies through the process of precipitation which is initiated when it comes into contact with an aqueous solution (e.g. blood, body fluids, normal saline, water) to form a cast. Not an adhesive. Amongst all agents currently available, it has the best and most controlled penetration of the AVM nidus. Therefore, Onyx has the greatest likelihood of achieving complete cure. Complete cure rates with onyx alone are possible in 20 – 51% of highly selected patients.^{47,48,49} Supplied in pre-mixed ready-to-use vials of Onyx-18, Onyx-34 and Onyx-500 in which the number (e.g. 18 for Onyx-18) corresponds to the nominal viscosity (as measured in centistokes) which the manufacturer controls by altering the EVOH concentration. Prior to use, the product must be shaken on a mixer for at least 20 minutes. Higher numbers indicate greater viscosity. Onyx 18 is most commonly used, Onyx 34 is used for very high flow AVMs, while 500 is used for aneurysm embolization.

Onyx is used with DMSO compatible microcatheters (marathon, echelon, rebar, ultra flow) that have been primed with approximately 0.3 to 0.8 ml of DMSO injected slowly through the microcatheter (depending on the deadspace of the microcatheter) prior to injecting Onyx itself. This is done to eliminate any contrast, saline or blood within the microcatheter that will cause the onyx to solidify in the catheter. The microcatheter tip is wedged in an arterial branch supplying the AVM, as close to the nidus as possible. It is ensured that the branch is exclusively supplying the AVM, by performing angiography through the microcatheter. The microcatheter is then primed with DMSO and Onyx embolization commenced under fluoroscopy. The onyx is visible due to admixed tantalum powder. It is injected slowly and continuously avoiding too much force, as this could cause the Onyx to reflux instead of flowing forward and prematurely close off access to the nidus. No more than 1 cm of Onyx reflux on the microcatheter should usually be allowed, as it may make the removal of catheter difficult later and may well cause disastrous complications. If reflux is noted, wait for 1–2 minutes and then continue with the injection. The already deposited Onyx will form a plug around the microcatheter causing the new deposit of Onyx to flow forward. The injection is stopped once Onyx is noted only to reflux and not flow into the nidus. Care is taken not to allow the Onyx to flow into the major draining vein or sinus, especially before the entire arterial supply is interrupted. The concept is the same as in surgical removal of AVMs, where major draining veins are occluded only after eliminating the arterial supply of AVM. Otherwise, the result is the same with ensuing venous

hypertension resulting in AVM rupture and catastrophic ICH. In case of a large AVM, it is sometimes best to stage the procedure, addressing one major arterial tree at a time.

Intravenous or combined IV, and IA embolization of AVM has been performed in a select group of small, deep seated AVMs, with single deep drainage.⁵⁰

□ NBCA. N-butyl cyanoacrylate is an embolic agent that is a glue which rapidly solidifies when it comes in contact with blood. Since the introduction of Onyx, its use has declined considerably due to the drawbacks of very short working time, greater risk of embolization into venous sinuses and, adhesion to catheters rendering their extraction following embolization difficult. Such adhesion may cause inadvertent catheter fractures with retained foreign body, or intracranial bleeds.

□ PVA. Polyvinyl alcohol (PVA) particles are available in sizes ranging from 50 to 1000 mcm. While it may be useful in temporarily devascularizing the AVM e.g., in preparation of craniotomy and surgical resection of AVM, the endovascular treatment with PVA alone is not durable.

Postoperative Management

Post-op orders:

- Admit the patient to NSICU.
- Keep leg on side used for procedure straight X 2 hrs (in case of angioseal closure) or 6–8 hours (in case manual compression was applied), with HOB elevated 15°.
- Post procedure heparinization is not required with AVMs because most ischemic events occur during the procedure and are related to the passage of embolic materials into normal blood vessels.¹¹
- Check groins, DP’s, vitals and neuro checks q 15 min X 4, q 30 min X 4, then q hr.
- Maintain mild hypotension for 12–72 hours especially in case of larger AVMs and monitor the patient for perfusion pressure break through bleeding, seizures and other possible complications.
- Review/Resume preprocedure medications (Hold metformin for 48 hours after intervention; hold all oral hypoglycemics, until good PO intake established).⁵¹

Follow-up:

- Outpatient appointment in 4 weeks.
- Follow-up angiography at 3 months. In case of staged AVM embolization, the interval between sessions is at the discretion of the endovascular neurosurgeon e.g., 1–4 weeks apart.

102.5.4 Dural arteriovenous fistulae (DAVF)

General information

These are abnormal direct arteriovenous shunts within the leaflets of dura.

Classification

There are numerous DAVF classifications. Borden (p.1252) and Cognard (p.1252) are amongst the more widely used.

Indications for endovascular intervention

- DAVF with “aggressive” features (□ Table 102.3) are always considered for treatment. Due to the high annual mortality rate (10.4%) and annual hemorrhage rate (8.1%), the treatment should be expeditious.⁵²

Table 102.3 Differentiating aggressive from benign DAVF	
“Aggressive” symptoms	“Benign” symptoms
• Cortical venous reflux (CVR): the hallmark of an aggressive DAVF • Intracerebral hemorrhage • Focal neurological deficit • Dementia • Papilledema • Increased intraocular pressure	• Pulsatile bruit • Orbital congestion (without increased intraocular pressure) • Cranial nerve palsy • Chronic headaches

- DAVF with angiographic findings including:
 - Selective contrast injection into ICA or VA demonstrating delayed cerebral circulation time. The is indicative of venous congestive encephalopathy.
 - Pseudophlebitic pattern: Brain surface demonstrating tortuous, dilated collateral veins in the venous phase of the angiogram. This finding is associated with a greater risk of hemorrhage, or non-hemorrhagic neurological deficit.
 - Cortical venous reflux (CVR). To ensure that this is not missed, always perform selective (rather than global, non-selective) angiography when assessing for DAVF. Venous stenosis or obstruction is commonly found in patients with CVR.

If a DAVF is detected, look for additional fistulae, as they are multiple in up to 8%

Those with “benign” features (see □ Table 102.3), may be considered for treatment if the symptoms are causing considerable discomfort to the patient, or are angiographically progressive. In many cases the treatment is palliative i.e., the symptoms are reduced but the fistula is not completely obliterated. The benign type of DAVF should continue to be followed, even when treatment is not indicated because of the risk of conversion to a more aggressive form with CVR. The follow-up may be clinical, with radiological assessment for any change in symptoms. A suggested protocol is to perform MRA with gadolinium annually, with follow-up conventional catheter angiography at 3 years. If there is any change in the patient’s clinical condition, whether it is worsening, improvement or resolution of the symptoms, perform standard angiography to assess for CVR.

Asymptomatic DAVF can usually be followed.

Contraindications to endovascular intervention

Most contraindications are relative, and a risk benefit assessment is performed on a case by case basis:

- Provocative tests demonstrating intolerance to occlusion
- Recent major surgery
- Pregnancy
- Contraindication to anticoagulants and/or thrombolytics.
- NBCA should not be used in those with allergy to cyanoacrylates, ethiodol or iodine. Premedication in those with iodine allergies is a consideration.
- PVA should not be used as a therapeutic option (other than in cases of epistaxis). It is usually indicated for pre-surgical devascularization of lesion.

DAVF embolization

The approach can be transarterial, transvenous or a combination. When feasible, a transvenous approach is preferred, as the probability of fistula obliteration is greater via the venous route. Very infrequently, it is possible to access the venous side from the arterial route because of a large connection between the dural artery and adjacent vein e.g., in traumatic DAVF. This is usually not possible in spontaneous DAVF, as the feeding artery is too small.

For transvenous route, consider the potential outcome of venous occlusion e.g., venous infarct, if the sinus being occluded is also the main source of drainage for normal veins. In such a situation, consider highly selective occlusion that will spare the normal drainage. Alternatively, instead of performing a complete occlusion, consider partial treatment only such that CVR is eliminated, converting the fistula into the benign Borden type I.

When using the venous route, ascertain that the venous channel is not tenuous (e.g. acute DAVF) rendering it prone to rupture during manipulation. The venous walls become sturdier when the fistula has been present for a while.

□ Coils. Measure the maximum width of the fistulous site to be occluded and select the appropriate sized coils. Deposit as many coils as necessary to occlude the fistula.

Sometimes a “combined” strategy may be used, where initially coils are deposited to slow the rapid blood flow through the fistula, followed by occlusion using a liquid embolic agent. If this strategy is used, start off with or exchange for a microcatheter compatible with the liquid embolic agent.

□ Onyx. During transarterial embolization with liquid embolic agents, wedge the catheter as close as possible to the fistula. The technique of deposition is the same as for AVM (see above). It is important to disrupt the fistulous connection to achieve cure. Therefore, Onyx must penetrate into the venous aspect of the DAVF.

- NBCA. It is used less frequently since the availability of Onyx, for reasons described above.
- PVA. It really has no significant role in the treatment of DAVF.

102.5.5 Carotid cavernous fistulae (CCF)

General information

CCF categorized as direct and indirect. The direct type are usually posttraumatic and are a high-flow single shunt between the ICA and the cavernous sinus. Indirect are low flow from meningeal branches. The section below describes management of direct type.

Indications for endovascular intervention

Direct fistulae usually require treatment as they frequently do not resolve spontaneously. Other indications: corneal exposure, diplopia, proptosis, intolerable bruits or headaches.

Timing of treatment

If the patient is stable, treatment may usually be performed within a couple of days of the diagnosis (i.e., treatment does not have to be emergent).

Indications for urgent treatment: ICH, epistaxis, increased IOP, decreased visual acuity, rapidly progressive proptosis, cerebral ischemia and enlargement of traumatic aneurysm beyond the cavernous sinus.

CCF embolization

The treatment goal is to eliminate the fistula.

Angiography is performed to identify the exact location and size of the fistula and its venous drainage. To address the high flow, consider angiography at 7.5 fps, instead of the usual 2–4 fps. In addition to the CCF, also look for other vascular injuries/anomalies.

Selective catheterization of both ECAs and ICAs are performed, to assess their contribution to the CCF. Angiography is also performed after manual compression of the CCA on the side of fistula, to better assess cross flow from the contralateral side. The digital compression will attenuate the high blood flow to the fistula, enabling its visualization. Do not compress both carotids simultaneously.

Rotational angiography with 3D reformatting, may be performed to study the fistula and select appropriate working views for intervention. It is important to be cognizant of the venous involvement including, cavernous sinuses, superior and inferior ophthalmic veins, sphenoparietal sinus, superior and inferior petrosal sinuses and the pterygoid plexi.

The following routes may be utilized for treating CCF: Transarterial; transvenous; and, via superior ophthalmic vein (if conventional routes are not available). Other indications: corneal exposure, diplopia, proptosis, intolerable bruits or headaches.⁵³

- Coils. Currently, the method of choice is transarterial coil embolization of the CCF.

Using roadmapping, the microcatheter is advanced over microwire into cavernous sinus via the fistula. This may take some effort. Coils are then deployed and detached, as previously described for aneurysms. Periodic angiography is performed and further coils placed, until the cavernous sinus (CS) is completely occluded. Complete occlusion is indicated by no further contrast entering the CS.

- Onyx. Refer to AVM section (p.1589) for details regarding usage. In case of a high flow fistula, prior to Onyx deposition, it may be advisable to initially deposit coils into the CS to slow the blood flow. This will prevent the inadvertent embolization of Onyx into the draining veins and sinuses. A balloon can be inflated within the parent ICA in order to protect it as Onyx is being injected into the cavernous sinus.

- NBCA. NBCA should be used with great caution, preferably after having slowed the flow through the CCF in order to prevent untoward deposition in the venous sinuses.

There is also potential for reflux into the carotid artery, which could cause a stroke. This may particularly occur when the CCF closure is near completion and the pressure gradient between the carotid artery and the CS is lowered. As with Onyx, a balloon can be inflated in the parent artery to protect it.

□ Detachable balloons. Balloons were initially utilized with success for endovascular CCF treatment and remain available outside of the US; however, these balloons are no longer available in the U.S. due to technical concerns relating to premature detachment and deflation over time.

□ Coil occlusion of the ICA. The desirable treatment for CCF is the occlusion of the fistula itself, essentially resulting in ICA reconstruction. This is frequently not possible. If the CCF is not amenable to treatment by any other route, sacrifice of the involved ICA is an option, especially if satisfactory supply has been confirmed from the contralateral ICA via anterior communicating and/or supply via the posterior communicating arteries.

Post-operative management and follow up

Post-op orders:

- Admit the patient to NSICU for overnight observation. Further ICU stay will depend upon the patient's clinical condition.
- Consult or arrange follow up with ophthalmic surgery.
- 0.9%NS+20 meq KCl @ 150 cc/hr X 2 hrs, then decrease to 100 cc/hr if patient is NPO overnight, then advance diet as tolerated.
- Keep the leg that was used for procedure straight for 2 hrs in case of Angioseal closure, or 6–8 hours when manual compression was applied. May elevate 15° HOB e.g., a pillow under the head.
- Check groin, DPs, vitals and neuro checks q 15 min X 4, q 30 min X 4, then q hr
- Review/Resume preprocedure medications (Hold metformin for 48 hours after intervention; hold all oral hypoglycemics, until good PO intake established).

Follow-up:

- D/C next morning after mobilizing, if there are no complications /other ongoing medical concerns requiring hospitalization.
- F/u on outpatient basis in 4 weeks.
- F/u angiography at 3 months.

102.5.6 Vertebrojugular fistulae (VJF)

Etiologies

- Iatrogenic e.g., during spine surgery or angiography, chiropractic manipulation, nerve block injection or radiation therapy^{54,55}
- Trauma e.g., penetrating injury, or GSW
- Vasculitis

Endovascular management of VJF

□ Stenting. A polytetrafluoroethylene (PTFE) covered stent, e.g., Jostent, may be used to cover the ostia of the fistula.⁵⁴

□ Coil occlusion. In the presence of adequate blood flow through contralateral healthy vertebral artery, the fistulous artery may be occluded with coils.⁵⁶ Verify that the arterial wall with the fistulous connection is part of the occluded segment.

□ NBCA occlusion. Rarely, NBCA occlusion has been performed when stenting or coils occlusion were not possible.⁵⁷ Onyx may also be used similarly.

102.5.7 Carotid dissection

General information

See general information (p.1324).

Angiographic features

Luminal stenosis (65%), occlusion (28%), pseudoaneurysm (28%), luminal irregularity (13%), embolic distal branch occlusion (13%), intimal flap (12%) and slow ICA-MCA flow (11%).⁵⁸

Management

The initial management in the absence of ICH is intravenous heparin for 7 days followed by warfarin.⁵⁹ The goal aPTT with heparin is 1.5 – 2.0 times the control value (50–80 sec). Warfarin is continued for 3–6 months with target INR range of 2.0 – 3.0. If anticoagulation is contraindicated, antiplatelet therapy is a consideration. In pregnant individuals, obtain obstetric consultation prior to initiating anticoagulation or anti-platelet therapy.

Indications for endovascular intervention

- Persistent ischemic symptoms despite anticoagulation therapy.
- Flow-limiting lesion with hemodynamic compromise

Contraindication to anticoagulation and/or anti-platelet therapy

- Impending risk of stroke
- Expanding pseudoaneurysm formation
- Iatrogenic dissection during endovascular procedure where flow compromise is apparent

Stenting with/without coiling

The endovascular treatment for carotid dissection is stenting. In case of intimal flap, the stent will appose the flap back to the arterial wall. Pseudoaneurysms have also been successfully occluded with stenting. Both uncovered and covered stents have been used successfully.⁶⁰ JoStent is a PTFE covered stent that is available in US. A vein covered stent has also been used.⁶¹

In case of a pseudoaneurysm that continues to show significant residual filling after stenting, coiling of the pseudoaneurysm will cause occlusion.⁶⁰

After stenting, the patient remains on dual antiplatelet therapy (ASA+Plavix) for at least a month and ASA alone indefinitely.

Follow-up

Follow-up should be arranged for patients on warfarin (e.g., “Coumadin clinic”).

Follow-up study in 3–6 months, which could be CTA, Doppler ultrasonography or catheter angiogram.

102.5.8 Subclavian artery stenosis

General information

Radiologically demonstrable stenosis of the subclavian or innominate artery is present in approximately 17%. Of these, 2.5% have angiographic flow reversal in vertebral artery. Only 5.3% of those with angiographic steal have neurologic symptoms.⁶²

□ Symptoms. The 5 D's of VBI i.e., diplopia, dysarthria, defective vision, dizziness and drop attacks.

Other symptoms include headache, nystagmus, hearing loss and focal seizures.^{63,64}

The arterial stenosis is proximal to the origin of VA. Symptoms are induced by exercise or exertion using the arm ipsilateral to the stenosis. The increased flow demand due to the exertion results in retrograde blood flow through the VA. The neurological symptoms may be because of continuous brainstem ischemia or more commonly, ischemia due to ipsilateral arm exercise or exertion.⁶⁵

Indications for endovascular intervention

Symptomatic subclavian artery stenosis i.e., stenosis resulting in subclavian steal syndrome.

Endovascular intervention

This includes angioplasty and stenting. A balloon mounted stent e.g., Express LD may be used, as the stent is deployed concurrently with angioplasty.⁶⁶ However, if the stenosis is particularly severe (e.g., <2 mm), pre-dilatation may be performed by a smaller balloon to achieve a caliber of 4 mm at site of stenosis. Normal antegrade blood flow is restored following successful angioplasty and stenting.

Postoperative management

The patient is monitored at least overnight in NSICU.

After stenting, the patient remains on dual antiplatelet therapy (ASA+Plavix) for at least 1 month, and ASA alone indefinitely.

Follow-up study in 3–6 months, which could be CTA, Doppler ultrasonography or catheter angiogram.

Complications of angioplasty and stenting

The frequency of complications is 17.8% (of 73 procedures) for innominate and VA angioplasty and stenting. These include access-site bleeding and distal embolization.⁶⁷

102.5.9 Ischemic Stroke

IV tPA

The target time for IV tPA administration (“door to needle time”) is within 60 minutes from time of arrival to hospital. However, IV tPA can be administered up to 4.5 hours after the onset of symptoms if the patient does not have any contraindications.⁶⁸

Dose: 0.9 mg/kg (max 90 mg) with 10% of the dose administered as a bolus over 1 minute and the rest infused over 60 minutes.

Endovascular intervention

General information

Recent studies have established the effectiveness and relative safety of endovascular intervention. These trials favor rapid endovascular intervention in acute ischemic stroke with proximal vessel occlusion, small infarct core and moderate to good collateral circulation.^{69,70,71,72}

Indications and case selection for endovascular intervention

Intra-arterial tPA may be indicated for the following situations:

- Persistent symptoms of stroke despite IV tPA and adequate medical management
- Where angiography may be performed and treatment administered within 3 and 6 hours after symptom onset with an NIHSS score greater than 4, or those with an NIHSS score of greater than 20 and the ability to be treated within 6 hours
- Posterior circulation strokes may be treated endovascularly for up to 24 hrs (due to a lesser likelihood of hemorrhagic conversion of infarct)
- CTA or MRA demonstrating a diffusion-perfusion mismatch. In face of significant penumbra, it may be worthwhile to perform endovascular intervention even outside the therapeutic window. Conversely, intervention may be abandoned even within the therapeutic window, if the stroke is complete. Centers are relying more and more on neuroimaging rather than the therapeutic window
- When IV tPA is contraindicated e.g., recent surgery

Contraindications to intervention

Most contraindications are relative and have to be weighed against the risk of not intervening. These contraindications include:

- Hemorrhagic infarct or, ICH
- CT demonstrating hypodensity or mass effect consistent with evolving infarct of more than one-third of middle cerebral artery territory
- Recent major surgery
- Pregnancy
- When considering stenting, contraindication to anticoagulants and/or thrombolytics

Preprocedural management

This may be under the supervision of a stroke neurologist, or the neurosurgeon. Ensure the following:

- Rapid transfer of patient to a stroke center/facility with endovascular capabilities.
- ABC's take precedence.

- Ensure patient has two intravenous lines, preferably 18G or larger. Start monitoring BP, pulse oximetry, ECG, O₂ saturation, cardiac rate and rhythm, respiratory rate. Insert a Foley catheter.
- Verify laboratory values including Platelet count, BUN, CR, APTT, PT/INR. β -HCG for females of reproductive age group.
- Maintain MAP $\geq$ 90 mm Hg.
- CT scan head: To rule out ICH.
- CTA: To assess location of the clot (hyperdense artery sign (p.1281)) and vascular tortuosity.
- MRI head (select cases).
- If available and can be done without delay, then perfusion studies e.g., CTP or MRP. These perfusion studies will demonstrate viable brain (penumbra) vs completed stroke.
- In centers where available, CT, CTA and CTP all are performed during the same session on CT scanner.
- Be cognizant of renal insufficiency, diabetes, congestive heart failure etc., in which case consider diluted non-ionic contrast agent and carefully pre-plan, to maintain contrast load to minimum.
- If the patient is not responding to IV tPA or it is contraindicated, then endovascular intervention is considered.
- The goal of intervention is to re-establish circulation, as soon as possible.

Techniques

□ Stent retrievers. Due to higher success rate, stent retrievers have become the method of first choice for clot removal in embolic stroke. The recanalization rate is 88.8 – 100%^{73,74,75,76} The two devices currently available in USA are Solitaire and Trevo. A 7 or 8 Fr sheath is placed in the femoral artery, through which a 6 Fr balloon guide catheter is positioned in the ICA (in case of anterior circulation strokes). Angiography is performed to identify site of occlusion. Using fluoroscopy and road mapping, a microcatheter is advanced over a microwire, across the site of occlusion. The microwire is removed and the stent retriever is advanced through the microcatheter such that it extends proximal and distal to the clot. The stent retriever is unsheathed by retracting the microcatheter as the retriever is maintained stationary. The stent retriever expands to its actual size and this results in restoration of flow in the occluded artery. After five minutes, the balloon on the guide catheter is inflated to arrest blood flow. Maintaining gentle aspiration on the guide catheter, the stent retriever and microcatheter are retracted simultaneously. Once both the microcatheter and retriever are within the guide catheter, vigorous aspiration is applied as the two devices are concurrently retracted and removed from the patient. Angiography is performed to confirm reconstitution of circulation.

Some surgeons administer a small amount of i.a. tPA as ‘mop up’ after mechanical thrombolysis, to address potential distal debris.

Vessel perforation during stent retriever withdrawal has been reported.⁷⁷

□ Penumbra aspiration. Until the introduction of stent retrievers, the penumbra device boasted the highest recanalization rate. A recanalization rate of >80% is quoted in literature.^{78,79}

This device includes a microcatheter that is advanced over a microwire, through the positioned guide catheter. The tip of the microcatheter is positioned adjacent to the proximal aspect of the clot. A separator is advanced through the microcatheter that is advanced back and forth through the clot to disrupt it. The proximal end of the microcatheter is connected to an aspiration pump that is turned on to aspirate the clot fragments.

Unlike stent retrievers that affect recanalization within minutes, Penumbra aspiration device takes longer, with median time of 49 min.⁸⁰ Its use is confined to the straight arterial segments because of risk of vessel perforation by the separator action.

□ Intra-arterial tPA. This may be the simplest endovascular technique to undertake, when compared to above. However, on its own, while the recanalization rates may be better than IV tPA, they are inferior to the above-mentioned mechanical techniques.^{81,82} Currently, i.a. tPA is used in conjunction with other techniques in ischemic stroke.

In addition to above, other techniques to extract thrombus have also been employed with mixed results including, aspiration with a simple syringe attached to a microcatheter, usage of snares, angioplasty at site of thrombus, stenting etc.

102.5.10 Cranial sinus thrombosis

General information

Also see Cerebrovascular venous thrombosis (p.1308).

Hydration with IV fluids and IV anticoagulation are part of the initial treatment for cranial sinus thrombosis (CST). Prior to initiation of treatment, blood for hypercoagulopathy tests is drawn.

Indications for endovascular intervention

- Persistent ischemic symptoms despite anticoagulation therapy.
- Contraindication to anticoagulation and/or anti-platelet therapy including hemorrhagic infarct.⁸³
- Impending risk of stroke.

Endovascular treatment

Chemical Thrombolysis: A catheter may be advanced to the involved sinus or close to it, through the femoral vein. The advantage of local administration is that, a larger amount of tPA actually reaches the clot vs systemic administration through a peripheral vein. Usually, 2–5 mg are administered through the thrombus and then an infusion started at a rate of 1 mg/hr, usually for 12 hours. If clot burden is still there on angiography, the infusion may be continued for longer, until the clot resolves.

For CST, the infusion may be prepared in a concentration of 1 mg/10 ml (0.1 mg/ml), for a rate of 10 ml/hr

Mechanical Thrombolysis: Similar to arterial embolic stroke, devices such as Stentriever or Penumbra may be used for clot extraction. Additionally, devices intended for other sites e.g., clot extraction from dialysis fistula, have also been used in cranial sinuses.⁸³ The challenge during endovascular intervention is negotiating the sigmoid-transverse sinus junction especially when using bulkier catheters e.g., AngioJet.

102.5.11 Tumor embolization

Indications

Preoperative devascularization of vascular tumors including

- meningiomas: embolization for meningiomas is location specific, size dependent, institution dependent, and potentially controversial. May be best reserved for large hypervascular convexity meningiomas. Usually performed 24 hours to 1 week preoperatively. The devascularization causes attenuation in intraoperative blood loss and the resultant necrosis frequently renders the tumor softer and easier to remove. However, tumor swelling may occur and occasionally an emergency craniotomy may be required
- hemangiopericytomas
- juvenile nasopharyngeal angiofibromas
- glomus jugulare tumors
- hemangioblastomas
- vascular metastases

Technique

A sheath is placed in the femoral artery and a guide catheter is positioned as close as possible to the vessels of interest e.g., in case of a meningioma the guide catheter tip is positioned in the proximal ECA. Angiography and roadmapping are performed through the guide catheter. Using fluoroscopy and road mapping, a microcatheter is advanced over wire into the branches supplying the tumor. Angiography is performed through the microcatheter to ascertain the branch supplies the tumor and no concerning collaterals with intracranial circulation exist. A blank road map is obtained and embolization commenced. PVA particles or Onyx may be used for embolization. In case of Onyx, a DMSO compatible catheter must be used. PVA may be cheaper and quicker to use for tumor embolization. However, the devascularization is not durable and the occluded vessels may recanalize; therefore, with PVA the surgery should be performed within a few days of the embolization.

102.5.12 Intraoperative angiography

Typically used in aneurysm surgery to confirm exclusion of the aneurysm from the circulation and to verify patency of critical adjacent vessels, and during AVM surgery to confirm total elimination of the nidus.

1. using traditional iodinated contrast and fluoroscopy. Requires use of radiolucent headholder. Typically the introducer sheath is placed in the femoral artery at the time of initial pre-op angio, and is left in place for intraoperative use
2. indocyanine green (ICG)^{84,85}: can be visualized under normal light, or sometimes to better advantage when illuminated with near-infrared light. Use is restricted to surface vessels. May be less reliable with giant or wide-neck aneurysms or with thickwalled atherosclerotic

102.5.13 Refractory epistaxis

Indications

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Epistaxis that has not responded to treatment including manual compression, nasal packing, local vasoconstrictors, endoscopic cauterization or surgical ligation of sphenopalatine arteries

Preoperative management

Verify lab values including Platelet count, BUN, CR, APTT, PT/INR, and β -HCG for females of reproductive age group. In renal insufficiency, diabetes, CHF etc., use diluted non-ionic contrast agent and pre-plan carefully to maintain contrast load to minimum.

Liquids only on morning of procedure. NPO (for ≈ 6 hours) when procedure performed under general anesthesia.

Obtain informed consent for angiography and embolization of ECA branches

Ensure two I.V. lines inserted. Insert Foley. The patient will be more comfortable and cooperative with an empty bladder, if the procedure becomes prolonged.

□ Technique. Position patient on the neuroangiography table. Attach pulse oximetry and ECG leads for monitoring O_2 saturation, HR, cardiac rhythm respiratory rate and BP.

A sheath is placed in the femoral artery. A guide catheter is positioned in the proximal ECA on the side of bleeding or pathology. Angiography and roadmapping are performed through the guide catheter. Using fluoroscopy and road mapping, a microcatheter is advanced over the wire into the sphenopalatine branches. Angiography is performed through the microcatheter to ascertain appropriate positioning and to ensure no concerning collaterals with intracranial circulation exist. Contrast extravasation, tumor blush or pseudoaneurysms may be detected. A blank road map is obtained and embolization of the offending vessel commenced. PVA particles (250–300 mcgm) or Onyx (18 or 34) may be used. In case of Onyx, a DMSO compatible catheter is used. PVA may be cheaper and quicker to use.

Postoperative management

Post-op orders:

- Admit to ICU for overnight observation. Typically, nasal packing is left intact overnight and removed for inspection for bleeding the next day
- IV: 0.9%NS+20 meq KCl @150 cc/hr X 2 hrs, then decrease to 100 cc/hr, if patient is NPO
- Activity: Keep right/left leg (whichever side was used for procedure) straight for 2 hrs (in case of Angioseal closure), or 6–8 hours (in case manual compression was applied), with HOB elevated 15°. This is achieved by placing a pillow under the patient's head. There should be no flexion in the femoral region. If more head elevation is required, place bed in reverse-Trendelenberg position.
- Check groins, DP's, vitals and neuro checks q 15 min X 4, q 30 min X 4, then q hr.
- Advance diet as tolerated. Review/resume preprocedure medications (except oral hypoglycemics, until good PO intake established).

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Part XXIV

Appendix

103 Quick Reference
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XXIV

103 Quick Reference Tables and Figures

103

Table 103.1 Summary of findings in brain death (□ Table 19.1), see text for details (p. 308)	
Vital signs & general criteria	
• Core temp >36° C (96.8° F), SBP>100 mm Hg, no complicating drugs (BAC<0.8%)	
Absence of brainstem reflexes	
• Fixed pupils	
• Absent corneal reflexes	
• Absent oculovestibular reflex (calorics)	
• Absent oculoccephalic reflex: “Doll’s eyes” (p.301)	
• Absent gag reflex	
• Absent cough reflex	
No response to deep central pain	
Failed apnea challenge	

Table 103.2 Karnofsky performance status scale (□ Table 88.1), see text for details (p. 1358)		
Score	Criteria	Prognosis with malignant glioma
100	normal: no complaints, no evidence of disease	Better prognosis with malignant glioma (p. 624)
90	able to carry on normal activity: minor signs or symptoms	
80	normal activity with effort: some signs or symptoms	
70	cares for self: unable to carry on normal activity or to do active work	
60	requires occasional assistance: cares for most of needs	
50	requires considerable assistance and frequent care	
40	disabled: requires special care and assistance	
30	severely disabled: hospitalized; death not imminent	
20	very sick: hospitalized; active supportive care needed	
10	moribund: fatal processes are progressing rapidly	
0	dead	

Table 103.3 Gardener and Robertson modified hearing classification (□ Table 41.5), see text for details (p.674)				
Class	Generally considered	Description	Pure tone audiogram(dB)	Speech discrimination
I	serviceable	good-excellent	0–30	70–100%
II	serviceable	serviceable	31–50	50–59%
III	non-serviceable	non-serviceable	51–90	5–49%
IV	non-serviceable	poor	91-max	1–4%
V	non-serviceable	none	not testable	0

Table 103.4 American Academy of Otolaryngology-Head and Neck Surgery Foundation hearing classification system (□ Table 41.6), see text for details (p.674)				
Class	Generally considered	Pure tone threshold (dB)	Speech discrimination score (%)	
A	“useful”	≤ 30	AND	≥ 70
B	“useful”	> 30 AND ≤ 50	AND	≥ 50
C	“aidable”	> 50	AND	≥ 50
D	“nonfunctional”	any level		< 50

Table 103.5 Clinical grading of facial nerve function(House and Brackmann (□ Table 41.3), see text for details (p.672)		
Grade	Function	Description
1	normal	normal facial function in all areas
2	mild	slight weakness on close inspection
3	moderate	obvious but not disfiguring
4	moderate-severe	obvious weakness and/or disfiguring asymmetry
5	severe	barely perceptible motion
6	total paralysis	no movement

Table 103.6 Lumbar disc syndromes (□ Table 69.3), see text for details (p.1050)			
Syndrome	Level of herniated lumbar disc		
	L3–4	L4–5	L5-S1
root usually compressed	L4	L5	S1
% of lumbar discs	3–10%(5% average)	40–45%	45–50%
reflex diminished	knee jerk (Westphal’s sign)	medial hamstring	Achilles (ankle jerk)
motor weakness	quadriceps femoris (knee extension)	tibialis anterior (foot drop) & EHL	gastrocnemius (plantarflexion), ± EHL
decreased sensation	medial malleolus & medial foot	large toe web & dorsum of foot	lateral malleolus & lateral foot
pain distribution	anterior thigh	posterior LE	posterior IE, often to ankle

Table 103.7 Cervical disc syndromes (□ Table 70.1), see text for details (p. 1069)				
Syndrome	Cervical disc syndromes			
	C4–5	C5–6	C6–7	C7-T1
% of cervical discs	2%	19%	69%	10%
compressed root	C5	C6	C7	C8
reflex diminished	deltoid & pectoralis	biceps & bra-chioradialis	triceps	finger-jerk
motor weakness	deltoid	forearm flexion	forearm ext (wrist drop)	hand intrinsics
paresthesia & hypesthesia	shoulder	upper arm, thumb, radial forearm	fingers 2 & 3, all fingertips	fingers 4 & 5

Table 103.8 Muscle grading (modified Medical Research Council system, □ Table 29.2), see text for details (p. 505)		
Grade	Strength	
0	no contraction (total paralysis)	
1	flicker or trace contraction (palpable or visible)	
2	active movement with gravity eliminated	
3	active movement through full ROM against gravity	
4	active movement against resistance; subdivisions →	4– Slight resistance 4 Moderate resistance 4 + Slight resistance
5	normal strength (against full resistance)	
NT	not testable	

Table 103.9 Clinical criteria for spine stability (no C-spine imaging/x-rays needed), see Practice guideline for details (p. 953)	
• awake, alert, oriented (no mental status changes, including no alcohol or drug intoxication) • no neck pain (with no distracting pain) • no neurologic deficits	

Table 103.10 Normal prevertebral soft tissue (□ Table 12.2), see text for details (p. 214)				
Space	Level	Maximum normal width (mm)		
		Adults		Peds
		MDCT	Lateral X-Ray	
retropharyngeal	C1	8.5	10	unreliable
	C2–4	6–7 ^a	5–7	
retrotracheal	C5–7	18	22	14
^a CT data was deemed unreliable at C4				

Table 103.11 ASIA impairment scale (□ Table 62.13), see text for details (p. 944)	
Class	Description
A	Complete: no motor or sensory function preserved
B	Incomplete: sensory but no motor function preserved below the neurologic level (includes sacral segments S4–5)
C	Incomplete: motor function preserved below the neurologic level (more than half of key muscles below the neurologic level have a muscle strength grade <3)
D	Incomplete: motor function preserved below the neurologic level (more than half of key muscles below the neurologic level have a muscle strength grade ≥3)
E	Normal: Sensory & motor function normal

Table 103.12 Thoracolumbar injury classification & severity score (TLICS, □ Table 66.3), see text for details (p. 1007). Refer to □ Table 103.14 for management based on point total		
Category	Finding	Points
Radiographic findings	compression fx	1
	burst component or lateral angulation >15°	1
	distraction injury	2
	translational/rotational injury	3
Neurologic status	intact	0
	root injury	2
	complete SCI	2
	incomplete SCI	3
	cauda equina syndrome	3
Integrity of posterior ligamentous complex	intact	0
	undetermined	2
	definite injury	3
TLICS = Total Points →		

Table 103.13 Subaxial injury classification (SLIC, □ Table 65.1), see text for details (p.986). Refer to □ Table 103.14 for management based on point total	
Injury (rate the most severe injury at that level)	Points
Morphology	
No abnormality	0
Simple compression (compression fx, endplate disruption, sagittal or coronal plane VB fx.)	1
Burst fracture	2
Distraction (perched facet, posterior element fx.)	3
Rotation/translation (facet dislocation, teardrop fx., advanced compression injury, bilateral pedicle fx., floating lateral mass (p.994). Guidelines: relative axial rotation ≥ 11° or any translation not related to degenerative causes	4
Discoligamentous complex (DLC)	
Intact	0
Indeterminate (isolated interspinous widening with < 11° relative angulation & no abnormal facet alignment, ↑ signal on T2WI MRI in ligaments...)	1
Disrupted (perched or dislocated facet, <50% articular apposition, facet diastasis > 2 mm, widened anterior disc space, ↑ signal on T2WI MRI through entire disc...)	2
Neurologic status	
Intact	0
Root injury	1
Complete spinal cord injury	2
Incomplete spinal cord injury	3
● Continuous cord compression with neuro deficit	+1

Table 103.14 Management based on TLICS or SLIC (□ Table 66.4, □ Table 65.2), see text for details (p.1007)	
TLICS or SLIC	Management
≤ 3	nonoperative candidate
4	“grey zone”
≥ 5	surgical candidate

Table 103.15 Glasgow coma scale, recommended for age ≥ 4 yrs (□ Table 18.1), see text for details (p.296)			
Points	Best eye opening	Best verbal	Best motor
6	–	–	obeys
5	–	oriented	localizes pain
4	spontaneous	confused	withdraws to pain
3	to speech	inappropriate	flexion (decorticate)
2	to pain	incomprehensible	extensor (decerebrate)
1	none	none	none

Table 103.16 Children’s coma scale, for age <4 yrs (□ Table 18.2), see text for details (p.296)				
Points	Best eye	Best verbal		Best motor
6	–	–		obeys
5	–	smiles, oriented to sound, follows objects, interacts		localizes pain
		Crying	Interaction	
4	spontaneous	consolable	inappropriate	withdraws to pain
3	to speech	inconsistently consolable	moaning	flexion (decorticate)
2	to pain	inconsolable	restless	extensor (decerebrate)
1	none	none	none	none

Table 103.17 Measures to treat an acute ICP crisis (□ Table 56.6), see text for details (p.869)	
Check airway, neck position... For resistant or sudden IC-HTN, consider STAT unenhanced head CT	
Be sure patient is sedated and paralyzed	
Drain 3–5 ml CSF if IVC present	
Osmotic therapy: mannitol 1 gm/kg IV bolus or 10–20 ml of 23.4% saline	
Hyperventilate: to PaCO ₂ 30-35 mm Hg	
Pentobarbital 100 mg slow IV or thiopental 2.5 mg/kg IV over 10 minutes	

Table 103.18 Summary of initial steps for status epilepticus: adults and children >13 kg (□ Table 27.5), see text for details (p.469)	
O ₂ . Turn patient on their side. Check VS. Neuro exam	
Monitor/labs: Pulse oximetry. EKG/telemetry. □ Fingerstick glucose. Blood tests (do not wait for results to begin □): □ electrolytes, □ CBC, □ ABG, □ AED levels, □ LFTs, □ Mg ⁺⁺ , □ Ca ⁺⁺ , □ head CT	
• thiamine 100 mg IV and/or 50 ml of 50% dextrose (if needed)	
First-line AED: • lorazepam (Ativan®) 4 mg IV for adults, 2 mg IV for children >13 kg @<2 mg/min Repeat dose if necessary	
Second-line AED: given with failure of (or simultaneously with administration of) repeat dose of benzodiazepine • fosphenytoin: 15-20 mg PE/kg IV @ 150 mg PE/min (preferred drug: faster infusion rate, less irritation) OR • phenytoin: 15-20 mg/kg IV @ 50 mg/min (less expensive) If no response to loading dose, an additional 10 mg/kg IV may be given after 20 min. NB: following infusion rate guidelines is imperative	
□ phenytoin level ≈ 10 min after PHT loading dose; repeat 10 min later additional dose if required	
Alternative second-line AEDs: • sodium valproate: 20-30 mg/kg IV bolus (max rate: 100 mg/min) OR • phenobarbital: 20 mg/kg IV (start infusing @ 50-100 mg/min). A repeat dose of 25-30 mg/kg can be given 10 min after first dose. OR • Levetiracetam (Keppra®): 20 mg/kg IV bolus of over 15 minutes – evidence less clear	
If seizures continue >30 mins and are refractory to 1st and 2nd line AEDs: intubate in ICU and begin continuous infusion therapy (CIT) of: • Midazolam: 0.2 mg/kg IV loading dose followed by 0.2-0.6 mg/kg/hr OR • Propofol: 2 mg/kg IV loading dose followed by 2-5 mg/kg/hr	

Table 103.19 Hunt and Hess classification of SAH (□ Table 77.2), see text for details (p. 1162)	
Grade	Description
1	asymptomatic, or mild H/A and slight nuchal rigidity
2	Cr. N. palsy (e.g. III, VI), moderate to severe H/A, nuchal rigidity
3	mild focal deficit, lethargy, or confusion
4	stupor, moderate to severe hemiparesis, early decerebrate rigidity
5	deep coma, decerebrate rigidity, moribund appearance
Add one grade for serious systemic disease (e.g. HTN, DM, severe atherosclerosis, COPD) or severe vasospasm on arteriography.	

Table 103.20 Modified grading system of Fisher: correlation between the amount of blood on CT and the risk of vasospasm (□ Table 78.2), see text for details (p. 1180)		
Modified Fisher scale group	Blood on CT	Symptomatic vasospasm
	No SAH or IVH	
1	focal or diffuse thin SAH, no IVH	24%
2	focal or diffuse thin SAH, with IVH	33%
3	focal or diffuse thick SAH, no IVH	33%
4	focal or diffuse thick SAH, with IVH	40%

Table 103.21 Spetzler-Martin AVM grading system (□ Table 82.6), see text for details (p. 1243)	
Graded feature	Points
Size	
small (<3 cm)	1
medium (3–6 cm)	2
large (>6 cm)	3
Eloquence of adjacent brain	
non-eloquent	0
eloquent	1
Pattern of venous drainage	
superficial only	0
deep	1

Table 103.22 WFNS SAH grade (□ Table 77.4), see text for details (p. 1163)		
WFNS grade	GCS score	Major focal deficit
0		
1	15	–
2	13–14	–
3	13–14	+
4	7–12	+ or –
5	3–6	+ or –

Table 103.23 ICH Score (□ Table 87.5), see text for details (p. 1339)		
Feature	Finding	Points
GCS score	3–4	2
	5–12	1
	13–15	0
Age	≥80 years	1
	<80	0
Location	infratentorial	1
	supratentorial	0
ICH volume	≥30 cc	1
	<30 cc	0
Intraventricular blood	yes	1
	no	0
“ICH Score”= Total Points		0–6

Table 103.24 Mortality based on ICH Score (□ Table 87.6), see text for details (p. 1340)	
ICH Score	30 day mortality
0	0%
1	13%
2	26%
3	72%
4	97%
5	100%
6	? 100%

Table 103.25 Spinal nerve root motor distribution (□ Table 62.10), see text for details (p. 941)			
Segment	Muscle	Action to test	Reflex
C1–4	neck muscles		
C3, 4, 5	diaphragm	inspiration, TV, FEV1, VC	
C5, 6	deltoid	abduct arm >90°	
C5, 6	biceps	elbow flexion	biceps
C6, 7	extensor carpi radialis	wrist extension	supinator
C7, 8	triceps, extensor digitorum	elbow and finger extension	triceps
C8, T1	flexor digitorum profundus	grasp (flex distal phalanges)	
C8, T1	hand intrinsic	abduct little finger, adduct thumb	
T2–9	intercostals		
T9,10	upper abdominal	Beever’s sign	abdominal cutaneous reflex
T11,12	lower abdominal		
L2, 3	iliopsoas, adductors	hip flexion	cremasteric reflex
L3, 4	quadriceps	knee extension	infrapatellar (knee jerk)
L4, 5	medial hamstrings, tibialis anterior	ankle dorsiflexion	medial hamstrings
L5, S1	lateral hamstrings, posterior tibialis, peroneals	knee flexion	
L5, S1	extensor digitorum, EHL	great toe extension	
S1, 2	gastrocs, soleus	ankle plantarflexion	achilles (ankle jerk)
S2, 3	flex digitorum, flex hallucis		
S2, 3, 4	bladder, lower bowel, anal sphincter	clamp down during rectal exam	anal cutaneous reflex, bulbocavernosus & priapism

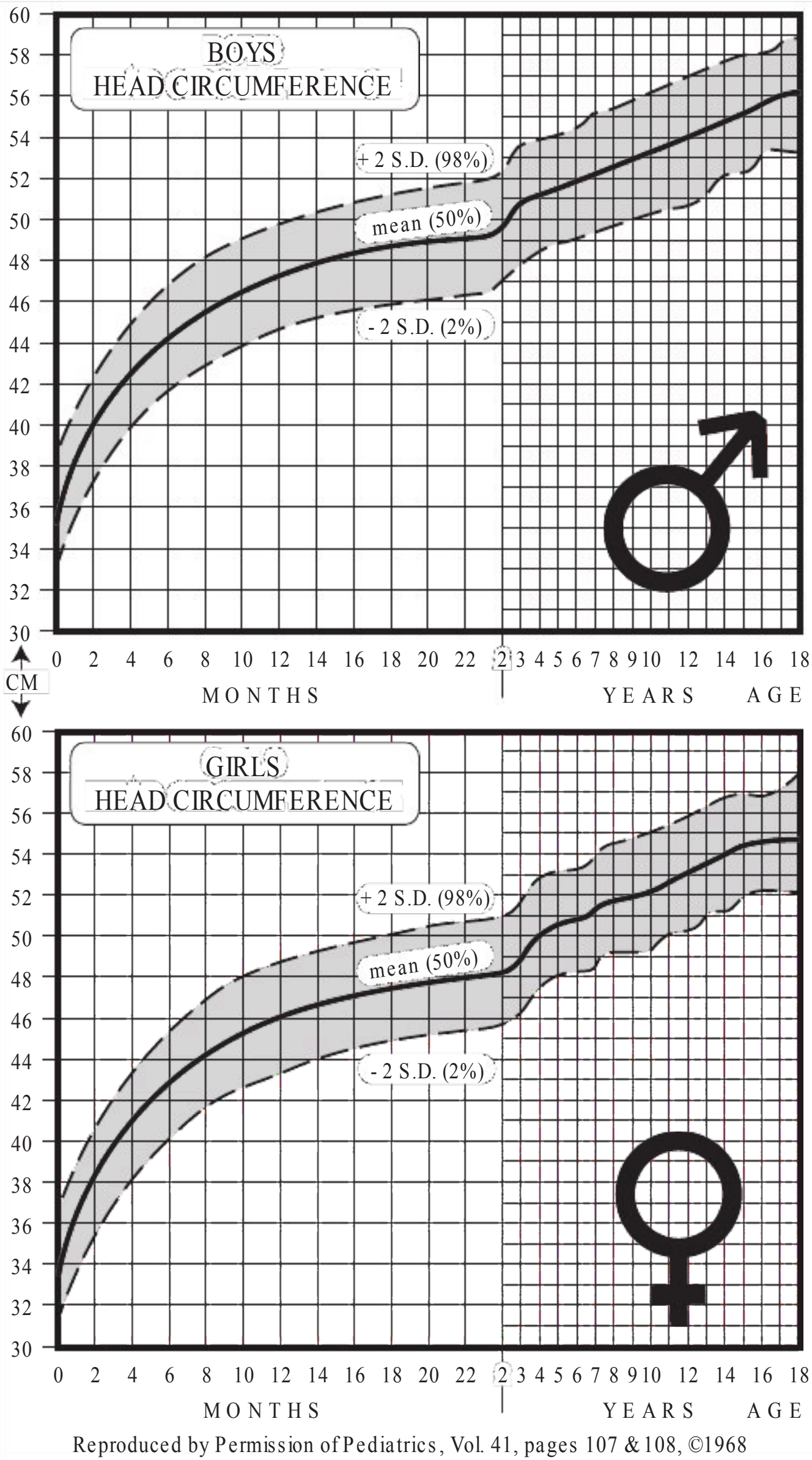


Fig. 103.1 Boys & girls head circumference (Reproduced by permission of Pediatrics 1968, Vol 41, page 107-108)

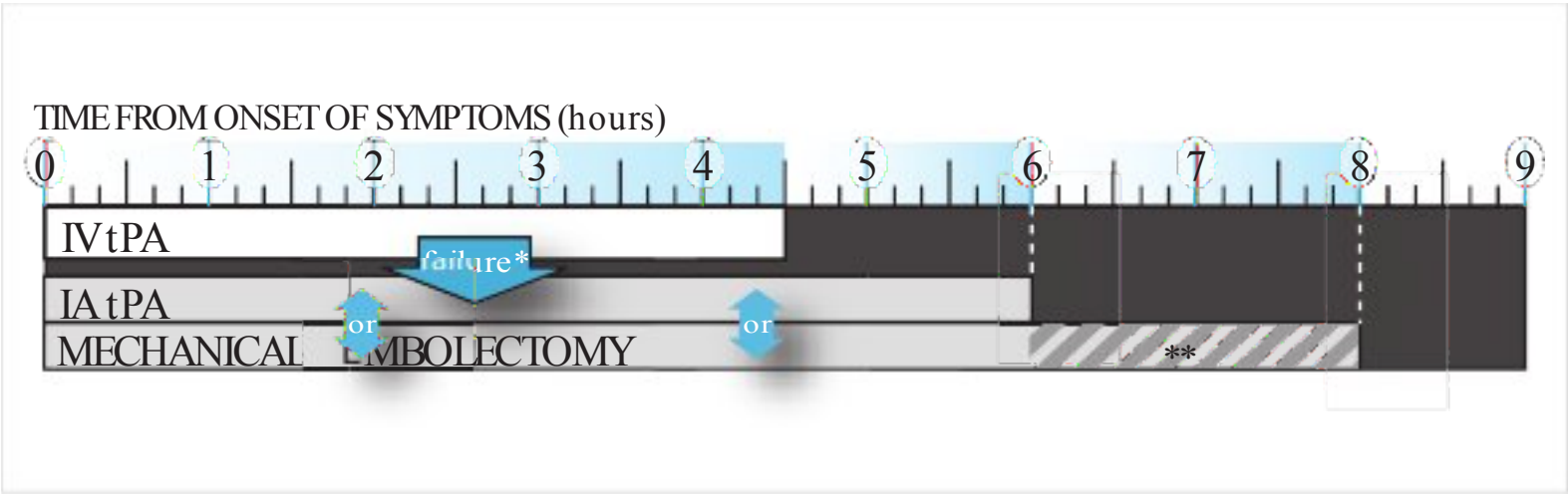


Fig. 103.2 Treatment options timeline (□ Fig. 84.1), see text for details (p.1286)
* option for failures with NIHSS>8-10
** from 6-8 hrs, check perfusion before mechanical embolectomy

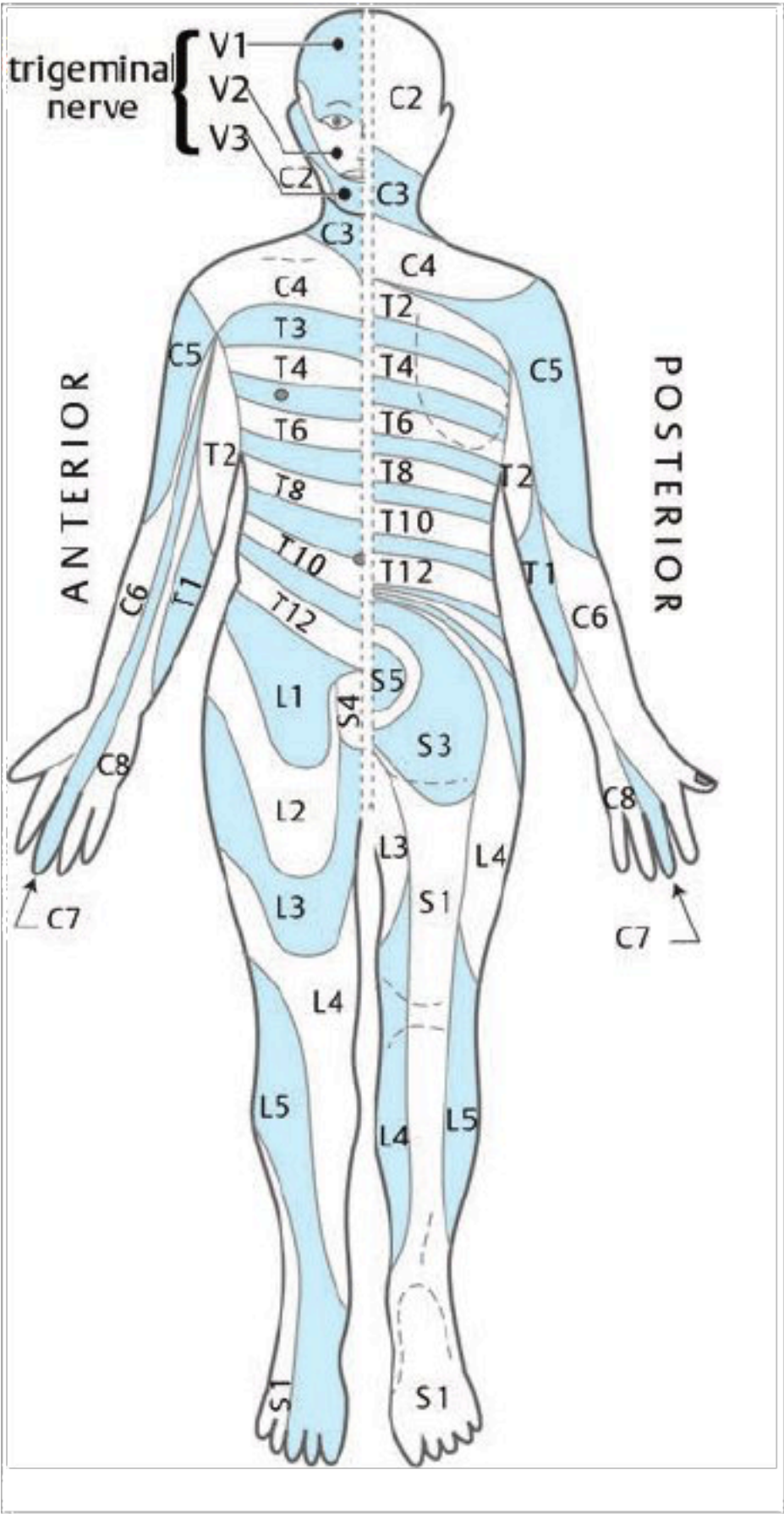


Fig. 103.3 Spinal nerve root sensory derma-
tomes (□ Fig. 1.14), see text for details
(p.73) (Redrawn from “Introduction to Basic
Neurology”, by Harry D. Patton, John W.
Sundsten, Wayne E. Crill and Phillip D.
Swanson, © 1976, pp 173, W. B. Saunders
Co., Philadelphia, PA, with permission)

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